

MODEL RUN TRANSCRIPT

Generated: 2026-08-12T08:18:33

Source: data/json/noise_experiments/v2/noise8i_fixed_memprimary_game/claude-sonnet-4.5/game/noisy8_bidirectional_informed_memprimary_fixed/noise8i_fixed_memprimary_game_004_neutral_rep04.json

RUN METADATA

```
model: anthropic/claude-sonnet-4.5
temperature: 0.8
num_turns: 10
num_agents: 8
memory_capacity: 6
myth_topic_id: anything
myth_topic: anything
game_params_name: noisy8_bidirectional_informed_memprimary_fixed
noise_config: {
  "applies_to": "both",
  "inform_agents": true,
  "range": 1.0,
  "type": "uniform"
}
other_player_names: default
replicate_id: 4
myth_default_prompt_key: myth_writing_default
myth_later_prompt_key: myth_writing_later_rounds
history_policy: self_and_coplayer
self_history_window: 0
coplayer_history_window: 3
show_agent_names: false
defector_ratio_requested: 0.0
defector_ratio_actual: 0.0
defector_count: 0
defector_agent_ids: []
defector_seed: 4
task_order: [
  "game"
]
agent_names: hidden in transcript labels because show_agent_names is false
agent_types: {
  "Agent_1": "standard",
  "Agent_2": "standard",
  "Agent_3": "standard",
  "Agent_4": "standard",
  "Agent_5": "standard",
  "Agent_6": "standard",
  "Agent_7": "standard",
  "Agent_8": "standard"
}
```

Coverage: exact chat message snapshots sent to the model are included for every recorded LLM call.

AGENTS

Agent_1; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5

- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_2; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will

have acted as sender and receiver equally often when the round count allows it.

- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_3; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_4; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x

- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_5; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.

- Your paired opponent is referred to as your current co-player.

Agent_6; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_7; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned

- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_8; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

ROUND 1

Roles: Agent_1=Sender, Agent_2=Receiver, Agent_3=Sender, Agent_4=Receiver, Agent_5=Sender, Agent_6=Receiver, Agent_7=Sender, Agent_8=Receiver

Pairings:

dyad_1: Agent_1=Sender, Agent_2=Receiver
dyad_2: Agent_3=Sender, Agent_4=Receiver
dyad_3: Agent_5=Sender, Agent_6=Receiver
dyad_4: Agent_7=Sender, Agent_8=Receiver

Parsed game decisions:

dyad_1: Agent_1 sent \$3; received \$9; Agent_2 returned \$5; payoffs Agent_1=\$7, Agent_2=\$4;
noise/visible: sent visible=\$3.63, received visible=\$10.89, returned visible=\$5.52
Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 9.0, "min": 0, "raw": 5.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_2: Agent_3 sent \$3; received \$9; Agent_4 returned \$3.5; payoffs Agent_3=\$5.5, Agent_4=\$5.5;
noise/visible: sent visible=\$2.91, received visible=\$8.73, returned visible=\$3.56
Action validation: {"returned": {"amount": 3.5, "clamped": false, "max": 9.0, "min": 0, "raw": 3.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_3: Agent_5 sent \$3; received \$9; Agent_6 returned \$4; payoffs Agent_5=\$6, Agent_6=\$5;
noise/visible: sent visible=\$2.66, received visible=\$7.98, returned visible=\$3.13
Action validation: {"returned": {"amount": 4.0, "clamped": false, "max": 9.0, "min": 0, "raw": 4.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_7 sent \$3; received \$9; Agent_8 returned \$5; payoffs Agent_7=\$7, Agent_8=\$4;
noise/visible: sent visible=\$3.17, received visible=\$9.51, returned visible=\$5.37
Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 9.0, "min": 0, "raw": 5.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 7.0,
"Agent_2": 4.0,
"Agent_3": 5.5,
"Agent_4": 5.5,
"Agent_5": 6.0,
"Agent_6": 5.0,
"Agent_7": 7.0,
"Agent_8": 4.0

}

Visible balances after round: {

"Agent_1": 7.52,
"Agent_2": 5.8900000000000001,
"Agent_3": 5.5600000000000005,
"Agent_4": 5.23,
"Agent_5": 5.13,
"Agent_6": 3.9800000000000004,
"Agent_7": 7.37,
"Agent_8": 4.51

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T08:04:51

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_2",
"pairing": {
 "agents": [
 "Agent_1",
 "Agent_2"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_2": "trustee"
 }
},
"trustee": "Agent_2"

```

},
"role": "investor",
"role_label": "Sender",
"round": 1,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round, I want to establish a cooperative approach while being somewhat cautious since I don't have any history with my current co-player.

Sending a moderate amount signals trust and gives us both the opportunity to benefit from the 3x multiplier, while not leaving me completely vulnerable.

I'll send ****\$3****. This gives my co-player \$9 to work with, and if they return a fair share, we can both end up better off than our starting positions.

```

'''json
{"send": 3}
'''

```


Reasoning:

This is the first round of a trust game. I'm the sender and I have \$5. I need to decide how much to send to the receiver.

Let me think about the strategic considerations:

1. If I send \$X, the receiver gets \$3X
2. The receiver then decides how much to return
3. I keep \$5 - X + (amount returned)
4. The receiver keeps \$3X - (amount returned)

This is a classic trust game. The optimal outcome for both of us would be if I send the full \$5, it gets tripled to \$15, and the receiver returns a fair share (say, at least \$7.50, which would give me \$7.50 and them \$7.50, both better off than starting positions).

However, there are risks:

- The receiver might not return anything or very little
- There's also mentioned "communication noise" - amounts may differ from what was actually sent/returned

Since this is round 1 and I have no history with this player:

- I could play it safe and send less
- Or I could signal trust and cooperation by sending more

In standard trust game experiments, people often send around \$2-\$3 in the first round as a moderate trust signal.

However, if I'm thinking about maximizing joint welfare and establishing cooperation, sending more could be beneficial. If the receiver is rational and cooperative, they should return enough to make it worthwhile for me.

Let me consider a middle-ground strategy: sending \$3 or \$4.

- If I send \$3, receiver gets \$9. If they return \$4-5, we both benefit
- If I send \$4, receiver gets \$12. If they return \$6, we both benefit significantly

Given this is a multi-round game with potentially different partners each time, establishing a pattern of trust and cooperation could be beneficial. I'll go with sending \$3 as a balanced approach - showing trust but not complete vulnerability.

Actually, let me reconsider. The instructions mention this is multi-agent with 8 agents and randomized pairings. This means:

...

```
Usage: {  
  "input_tokens": 500,  
  "output_tokens": 648,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T08:05:08

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agents": [  
      "Agent_1",  
      "Agent_2"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_2": "trustee"  
    },  
    "trustee": "Agent_2"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",
```

```

"round": 1,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.63, that is 72.6% of their total endowment. You received \$10.89. How much do you return to the sender? (0-10.89)

Assistant output:

Looking at this situation:

- The sender showed significant trust by sending \$3.63 (72.6% of their endowment)
- I received \$10.89 after the 3x multiplier
- A fair return should reward their trust while ensuring both of us benefit

The sender gave up \$3.63, keeping only \$1.37. Their generosity created \$10.89 total value for me. To reciprocate fairly and encourage future cooperation:

If I return \$5.00:

- Sender ends with: \$5 - \$3.63 + \$5.00 = \$6.37 (gain of \$1.37)
- I end with: \$10.89 - \$5.00 = \$5.89 (gain of \$5.89)

This seems like a fair split that rewards their trust while both of us benefit from the cooperation.

ation.

```
{"return": 5.00}
```

Reasoning:

Let me analyze this situation carefully.

Given information:

- I'm the RECEIVER in Round 1
- The sender sent \$3.63 (which is 72.6% of their \$5 endowment)
- I received \$10.89 (which is $\$3.63 \times 3$)
- I need to decide how much to return (between \$0 and \$10.89)

Key considerations:

1. The sender was quite generous, sending 72.6% of their endowment
2. This shows trust and cooperation
3. There's mention of communication noise, so the actual amount sent might differ slightly from what I see
4. This is a multi-round game with 8 agents, so building reputation for cooperation could be beneficial
5. Fair reciprocation would be important to encourage future cooperation

Let me think about what would be a fair return:

- The sender gave up \$3.63 and kept \$1.37
- I received \$10.89
- If I want to be reciprocal and fair, I should consider:
 - The sender showed high trust (72.6% is a lot)
 - A common strategy is to return enough so both parties benefit
 - Returning about half (\$5.45) would give me \$5.44 and the sender would end up with $\$1.37 + \$5.45 = \$6.82$
 - This would mean the sender gains \$1.82 total and I gain \$5.44

Let me reconsider:

- If I return around \$4-5, that seems fair
- Returning \$5 would give:
 - Sender: $\$5 - \$3.63 + \$5 = \6.37 (net gain of \$1.37)
 - Me: $\$10.89 - \$5 = \$5.89$ (net gain of \$5.89)

- ...

Usage: {

```
"input_tokens": 529,  
"output_tokens": 801,  
"reasoning_tokens": 0
```

}

GAME - Agent_3 - Sender - dyad_2

Timestamp: 2026-08-12T08:05:30

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",  
"dyad_id": "dyad_2",  
"move_index": 2,  
"opponent_id": "Agent_4",  
"pairing": {  
  "agents": [  
    "Agent_3",  
    "Agent_4"  
  ],  
  "investor": "Agent_3",  
  "roles": {  
    "Agent_3": "investor",  
    "Agent_4": "trustee"  
  },  
  "trustee": "Agent_4"  
},  
"role": "investor",  
"role_label": "Sender",  
"round": 1,  
"task": "game",  
"task_index": 0
```

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round, I need to balance trust and caution. The multiplication factor of 3x creates an opportunity for mutual benefit if my co-player reciprocates.

Since this is the first round with no prior history, I'll send a moderate amount that signals cooperation while not exposing myself to maximum risk.

****My decision:****

```
'''json
{"send": 3}
'''
```

This gives my co-player \$9 to work with (3×3), leaving me with \$2. If they return a fair share, we can both benefit from this cooperation. Let's see how they respond.

Reasoning:

This is the first round of a trust game where I'm the SENDER. Let me think about this strategically.

Key facts:

- I have \$5
- I can send \$0-\$5 to the receiver
- Amount sent is multiplied by 3x
- The receiver then decides how much to return to me
- I keep: \$5 - sent + returned
- Receiver keeps: received - returned
- There may be communication noise (amounts seen may differ from actual)
- This is a multi-agent setting with 8 agents total
- Pairings are randomized each round
- Agent names are hidden

Strategic considerations:

1. **Trust and cooperation**: If both parties cooperate optimally, we can maximize joint payoff. If I send all \$5, the receiver gets \$15. If they return half (\$7.50), I end up with \$7.50 and they keep \$7.50, which is better for both of us than if I send nothing.
2. **Risk**: The receiver could keep everything and return nothing. If I send \$5 and get nothing back, I end up with \$0 while they get \$15.
3. **First round**: This is the first round, so I have no reputation or history with this particular co-player. However, my actions might influence future interactions in a multi-round setting.
4. **Communication noise**: The presence of noise means that even if someone intends to be fair, the amounts might appear different, which could lead to misunderstandings.
5. **Signaling**: Sending a moderate to high amount signals trust and cooperation, which might encourage reciprocity.

Common strategies in trust games:

- Send half (\$2.50) as a moderate trust signal
- Send all (\$5) to maximize joint payoff and signal full trust
- Send a smaller amount (\$1-2) to test the waters
- Send nothing (\$0) to avoid risk entirely

Given this is round 1 and I want to establish a cooperative tone while not being completely naive, I think sending a significant but not maximum amount makes sense.

Research on trust games suggests most people send positive amounts initially, and receivers typically reciprocate. The 3x multiplier creates substantial mutual benefit potential-even modest returns from the receiver would leave us both better off than the starting position. I'll send \$3 as a balanced signal of trust and good faith without overexposing myself....

```
Usage: {
  "input_tokens": 500,
  "output_tokens": 870,
  "reasoning_tokens": 0
}
```

GAME - Agent_4 - Receiver - dyad_2

Timestamp: 2026-08-12T08:05:49

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_3",
  "pairing": {
    "agents": [
      "Agent_3",
      "Agent_4"
    ],
    "investor": "Agent_3",
    "roles": {
      "Agent_3": "investor",
      "Agent_4": "trustee"
    },
    "trustee": "Agent_4"
  },
}
```

```

"role": "trustee",
"role_label": "Receiver",
"round": 1,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.91, that is 58.2% of their total endowment. You received \$8.73. How much do you return to the sender? (0-8.73)

Assistant output:

Looking at this situation:

- The sender showed significant trust by sending \$2.91 (58.2% of their endowment)
- I received \$8.73 after the 3x multiplication
- The sender currently has \$2.09 remaining

To reciprocate their trust and cooperation:

- If I return nothing, the sender ends with \$2.09 (worse than keeping everything)
- If I return \$3.50, the sender ends with \$5.59 and I keep \$5.23
- This rewards their trust while maintaining fairness

I'll return a meaningful amount to encourage continued cooperation.

```
{'return': 3.5}
```

Reasoning:

Let me analyze this situation:

1. As the RECEIVER, the sender sent me \$2.91 (which is 58.2% of their \$5 endowment)
2. This amount was multiplied by 3x, so I received \$8.73
3. I need to decide how much to return (between \$0 and \$8.73)

Key considerations:

- The sender was quite generous, sending nearly 60% of their endowment
- This shows trust and cooperation
- In trust games, reciprocity is important for maintaining cooperation
- The sender's final payoff will be: $\$5 - \$2.91 + \text{whatever I return} = \$2.09 + \text{return}$
- My final payoff will be: $\$8.73 - \text{whatever I return}$

Common approaches in trust games:

1. Split the gains equally - the sender gave up \$2.91 but created \$8.73. The "gain" is $\$8.73 - \$2.91 = \$5.82$. Splitting this equally would mean returning about \$2.91
2. Give the sender back what they sent - return \$2.91
3. Give the sender a good return on their investment - perhaps return more than they sent
4. Equal final payoffs - try to make both players end up with similar amounts

Let me calculate some scenarios:

If I return \$2.91:

- Sender gets: $\$2.09 + \$2.91 = \$5.00$
- I get: $\$8.73 - \$2.91 = \$5.82$
- This is fair-ish, both doing okay

If I return \$3.50:

- Sender gets: $\$2.09 + \$3.50 = \$5.59$
- I get: $\$8.73 - \$3.50 = \$5.23$
- This balances outcomes more evenly

If I return \$4.00:

- Sender gets: $\$2.09 + \$4.00 = \$6.09$
- I get: $\$8.73 - \$4.00 = \$4.73$
- The sender benefits more, which seems appropriate given they initiated the cooperation

For truly equal final payoffs, I'd need to return \$3.32, which would give us both \$5.41 each.

The sender demonstrated substantial trust by sending most of their endowment, so I should reciprocate meaningfully to reinforce that cooperation....

Usage: {

```
"input_tokens": 529,  
"output_tokens": 942,  
"reasoning_tokens": 0
```

}

GAME - Agent_5 - Sender - dyad_3

Timestamp: 2026-08-12T08:06:07

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",  
"dyad_id": "dyad_3",  
"move_index": 4,  
"opponent_id": "Agent_6",  
"pairing": {  
  "agents": [  
    "Agent_5",  
    "Agent_6"  
  ],  
  "investor": "Agent_5",  
  "roles": {  
    "Agent_5": "investor",  
    "Agent_6": "trustee"  
  },  
  "trustee": "Agent_6"  
},
```

```

"role": "investor",
"role_label": "Sender",
"round": 1,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round situation, I need to balance potential gains from cooperation against the uncertainty of how my co-player will respond.

****My reasoning:****

- Sending money creates value (3x multiplier), so there's potential for mutual benefit
- This is our first interaction, so I want to signal willingness to cooperate
- But I also want to maintain some security given the uncertainty
- The communication noise adds another layer of unpredictability

I'll send a moderate amount that demonstrates trust while keeping some reserve.

```

'''json
{"send": 3}

```


'''

This sends \$3 to my co-player, who will receive \$9. I'll keep \$2 initially, and hope they reciprocate with a fair return that benefits us both.

Reasoning:

This is the first round of a trust game where I'm the sender. Let me think about the strategic situation:

1. I have \$5 to start with
2. Whatever I send gets tripled when it reaches the receiver
3. The receiver then decides how much to return to me
4. There's communication noise - amounts may appear different than what was actually sent

Key considerations:

- This is a multi-round game with multiple agents
- Pairings are randomized each round
- There's a tension between trust/cooperation and self-interest
- If I send nothing, I keep \$5 but there's no opportunity for mutual gain
- If I send everything (\$5), the receiver gets \$15, and if they're fair/cooperative, they might return a good amount
- The communication noise adds uncertainty

In standard trust game experiments, sending some amount is generally beneficial because:

- It creates surplus value (3x multiplier)
- It signals trust and can encourage reciprocity
- Over multiple rounds, cooperative behavior can be rewarding

First round considerations:

- I don't know this particular co-player yet
- This is the first interaction, so I might want to signal willingness to cooperate
- But I also don't want to be too naive

A moderate approach might be to send \$3 or \$4:

- Sending \$3 means receiver gets \$9, I keep \$2. If they return \$4-5, we both do well
- Sending \$4 means receiver gets \$12, I keep \$1. If they return \$6, we both get \$6
- Sending \$5 means receiver gets \$15, I keep \$0. If they return \$7-8, I do well

Given this is round 1 and I want to establish a cooperative tone while not being too vulnerable, I'll send \$3. This shows trust but maintains some security.

Actually, let me reconsider. In trust games, the classic finding is that people often send about half their endowment on average. Sending \$2.50 or \$3 seems reasonable.

...

```
Usage: {
  "input_tokens": 500,
  "output_tokens": 703,
  "reasoning_tokens": 0
}
```

GAME - Agent_6 - Receiver - dyad_3

Timestamp: 2026-08-12T08:06:19

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_5",
  "pairing": {
    "agents": [
      "Agent_5",
      "Agent_6"
    ],
    "investor": "Agent_5",
    "roles": {
      "Agent_5": "investor",
      "Agent_6": "trustee"
    },
    "trustee": "Agent_6"
  },
}
```

```

"role": "trustee",
"role_label": "Receiver",
"round": 1,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.66, that is 53.2% of their total endowment. You received \$7.98. How much do you return to the sender? (0-7.98)

Assistant output:

Looking at this situation:

- The sender showed significant trust by sending \$2.66 (53.2% of their endowment)
- I received \$7.98 after the 3x multiplication
- The sender currently has \$2.34 remaining from their original \$5

To reward their trust and encourage cooperation, I should return enough so they profit from their decision to send. If I return \$4.00:

- Sender ends with: \$2.34 + \$4.00 = \$6.34 (a gain of \$1.34)
- I keep: \$7.98 - \$4.00 = \$3.98

This seems fair - both of us benefit from the cooperation, and the sender's trust is rewarded.

```
{"return": 4.00}
```

Reasoning:

Let me think about this carefully.

I'm the RECEIVER in Round 1.

- The sender sent \$2.66 (which is 53.2% of their \$5 endowment)
- This amount was multiplied by 3x, so I received $\$2.66 \times 3 = \7.98
- Now I need to decide how much to return (between \$0 and \$7.98)

Key considerations:

1. The sender sent more than half their endowment (53.2%), showing substantial trust
2. If I return nothing, I keep \$7.98 and the sender keeps $\$5 - \$2.66 = \$2.34$
3. If I return some amount, I should aim to be fair and encourage cooperation

A reasonable approach would be to:

- Ensure the sender ends up with more than they started with (more than \$5)
- Keep some profit myself
- Reward their trust to encourage future cooperation

Let me calculate:

- Sender currently has: $\$5 - \$2.66 = \$2.34$
- I have: \$7.98

If I return \$3.00:

- Sender ends with: $\$2.34 + \$3.00 = \$5.34$ (slight gain)
- I keep: $\$7.98 - \$3.00 = \$4.98$

If I return \$4.00:

- Sender ends with: $\$2.34 + \$4.00 = \$6.34$ (good gain)
- I keep: $\$7.98 - \$4.00 = \$3.98$

If I return \$3.50:

- Sender ends with: $\$2.34 + \$3.50 = \$5.84$
- I keep: $\$7.98 - \$3.50 = \$4.48$

...

```
Usage: {  
  "input_tokens": 529,  
  "output_tokens": 699,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_7 - Sender - dyad_4

Timestamp: 2026-08-12T08:06:38

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_4",  
  "move_index": 6,  
  "opponent_id": "Agent_8",  
  "pairing": {  
    "agents": [  
      "Agent_7",  
      "Agent_8"  
    ],  
    "investor": "Agent_7",  
    "roles": {  
      "Agent_7": "investor",  
      "Agent_8": "trustee"  
    },  
    "trustee": "Agent_8"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 1,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round, I need to consider:

1. ****The multiplier effect****: Any amount I send gets tripled, creating potential mutual gain
2. ****No prior history****: This is my first interaction, so I have no information about this co-player's behavior
3. ****Future rounds****: Even though pairings are randomized, establishing cooperative norms early could benefit the group overall
4. ****Risk vs. reward****: Sending more creates more surplus, but requires trusting the receiver to return a fair amount

I'll take a moderately cooperative approach. Sending \$3 creates \$9 for the receiver - enough surplus that we can both benefit if they reciprocate fairly, while not risking my entire endowment.

If they return around half of what they receive (\$4-5), we both end up better off than if no transaction occurred.

```
'''json
{"send": 3}
'''
```

Reasoning:

This is the first round of a multi-agent trust game experiment. I'm the SENDER with \$5.

Let me think about the strategic aspects:

1. If I send money, it gets tripled, creating surplus value
2. The receiver then decides how much to return
3. This is a classic trust game scenario

Key considerations:

- This is round 1, so I have no history with this particular co-player
- There are multiple rounds, so reputation/repeated interaction matters
- However, pairings are randomized each round, so I won't necessarily play the same person again
- The game has communication noise, which might affect trust

Strategic thinking:

- Sending \$0 guarantees I keep \$5, but creates no surplus
- Sending \$5 creates maximum surplus (\$15 total for receiver), but requires trust
- A middle ground (e.g., \$2-3) balances risk and potential gain

In classic trust game experiments, people often:

1. Send moderate to high amounts in early rounds to signal cooperation
2. The amount sent and returned establishes baseline trust levels
3. With randomized pairings, building a reputation as trustworthy matters for future rounds

Game theory perspective:

- If this were truly one-shot with rational self-interested players, the receiver should return \$0 (backward induction)
- Knowing this, I should send \$0
- However, real humans often deviate from this due to fairness norms, reciprocity, and reputation concerns

Since:

- This is multi-round with multiple agents
- There may be reputation effects even with randomized pairings
- Starting cooperatively can establish good norms
- The 3x multiplier creates real gains from cooperation

I'll send a moderate-high amount to signal cooperation while not being completely vulnerable.
I'll send \$3.

This creates \$9 for the receiver, and if they return even just \$3, we both end up with \$5 (same as no transaction), but if they return more (say \$4-5), we both gain. ...

Usage: {

```
"input_tokens": 500,  
"output_tokens": 671,  
"reasoning_tokens": 0
```

}

GAME - Agent_8 - Receiver - dyad_4

Timestamp: 2026-08-12T08:06:59

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",  
"dyad_id": "dyad_4",  
"move_index": 7,  
"opponent_id": "Agent_7",  
"pairing": {  
  "agents": [  
    "Agent_7",  
    "Agent_8"  
  ],  
  "investor": "Agent_7",  
  "roles": {  
    "Agent_7": "investor",  
    "Agent_8": "trustee"  
  },  
  "trustee": "Agent_8"  
},
```

```

"role": "trustee",
"role_label": "Receiver",
"round": 1,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.17, that is 63.4% of their total endowment. You received \$9.51. How much do you return to the sender? (0-9.51)

Assistant output:

Looking at this situation:

- The sender showed significant trust by sending 63.4% of their endowment (\$3.17)
- I received \$9.51 after the 3x multiplier
- The sender currently has \$5 - \$3.17 = \$1.83

The sender demonstrated considerable trust, so I want to reciprocate appropriately. If I return \$5.00:

- The sender would end with: \$1.83 + \$5.00 = \$6.83 (a profit of \$1.83)
- I would keep: \$9.51 - \$5.00 = \$4.51

This rewards their trust with a solid return while allowing us both to benefit from the cooperation.

ation.

```
{"return": 5.00}
```

Reasoning:

Let me think about this carefully.

I'm the RECEIVER in Round 1.

- The sender sent \$3.17 (which is 63.4% of their \$5 endowment)
- This was multiplied by 3x, so I received: $\$3.17 \times 3 = \9.51 ?
- I need to decide how much to return (between \$0 and \$9.51)

Key considerations:

1. The sender showed considerable trust by sending 63.4% of their endowment
2. If I return nothing, the sender ends up with: $\$5 - \$3.17 = \$1.83$, while I keep \$9.51
3. If I return a fair/reciprocal amount, what would that be?

A few approaches to consider:

- Equal split of total: Total is \$5 (sender's initial) + \$9.51 (what I received) = \$14.51. Equal split would be ~\$7.26 each. The sender currently has \$1.83, so I'd return $\$7.26 - \$1.83 = \$5.43$
- Return what maintains sender's initial investment value: Sender invested \$3.17. If I return \$3.17, sender gets back to \$5 total, and I keep \$6.34
- Return proportionally to what was sent: The sender sent 63.4% of their endowment. I could return a similar proportion of what I received: 63.4% of \$9.51 = ~\$6.03
- Social norm of reciprocity: Return enough to reward trust but keep some gains

Given that:

- The sender showed significant trust (63.4%)
- This is a multi-round game with different partners
- Building reputation for reciprocity could be valuable
- Note about communication noise suggests amounts might not be exact

I think a reasonable return would be around \$4-5, which:

- Acknowledges their generosity without giving away all my gains
- Leaves them better off than their starting position
- Demonstrates trustworthiness that could matter in future interactions with other partners
- Strikes a balance between reciprocity and self-interest

I'll return \$4.75. This gives the sender \$6.58 total (\$1.83 remaining plus my return), which meaningfully rewards their trust while I keep \$4.76....

```
Usage: {
  "input_tokens": 529,
  "output_tokens": 1001,
  "reasoning_tokens": 0
}
```

=====

ROUND 2

=====

Roles: Agent_1=Receiver, Agent_2=Sender, Agent_3=Receiver, Agent_4=Sender, Agent_5=Receiver, Agent_6=Sender, Agent_7=Receiver, Agent_8=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver
dyad_2: Agent_4=Sender, Agent_3=Receiver
dyad_3: Agent_6=Sender, Agent_5=Receiver
dyad_4: Agent_8=Sender, Agent_7=Receiver

Parsed game decisions:

dyad_1: Agent_2 sent \$3.5; received \$10.5; Agent_1 returned \$4.5; payoffs Agent_2=\$6, Agent_1=\$6;
noise/visible: sent visible=\$3.14, received visible=\$9.42, returned visible=\$5.06
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 10.5, "min": 0, "raw": 4.5}, "sent": {"amount": 3.5, "clamped": false, "max": 5, "min": 0, "raw": 3.5}}

dyad_2: Agent_4 sent \$3; received \$9; Agent_3 returned \$3.5; payoffs Agent_4=\$5.5, Agent_3=\$5.5;
noise/visible: sent visible=\$2.77, received visible=\$8.31, returned visible=\$2.9
Action validation: {"returned": {"amount": 3.5, "clamped": false, "max": 9.0, "min": 0, "raw": 3.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

dyad_3: Agent_6 sent \$3; received \$9; Agent_5 returned \$4.5; payoffs Agent_6=\$6.5, Agent_5=\$4.5;
noise/visible: sent visible=\$2.97, received visible=\$8.91, returned visible=\$5.03
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

dyad_4: Agent_8 sent \$3; received \$9; Agent_7 returned \$3.75; payoffs Agent_8=\$5.75,

```

Agent_7=$5.25; noise/visible: sent visible=$2.35, received visible=$7.05, returned visible=$3.71
Action validation: {"returned": {"amount": 3.75, "clamped": false, "max": 9.0, "min": 0, "raw": 3.75}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
Balances after round: {
  "Agent_1": 13.0,
  "Agent_2": 10.0,
  "Agent_3": 11.0,
  "Agent_4": 11.0,
  "Agent_5": 10.5,
  "Agent_6": 11.5,
  "Agent_7": 12.25,
  "Agent_8": 9.75
}
Visible balances after round: {
  "Agent_1": 12.44,
  "Agent_2": 12.45,
  "Agent_3": 10.370000000000001,
  "Agent_4": 10.13,
  "Agent_5": 9.54,
  "Agent_6": 11.010000000000002,
  "Agent_7": 10.670000000000002,
  "Agent_8": 10.219999999999999
}

```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T08:07:15

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 2,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.63, that is 72.6% of their total endowment. You received \$10.89. How much do you return to the sender? (0-10.89)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.63 (72.6% of their endowment)
- I received \$10.89 after the 3x multiplier
- A fair return should reward their trust while ensuring both of us benefit

The sender gave up \$3.63, keeping only \$1.37. Their generosity created \$10.89 total value for me. To reciprocate fairly and encourage future cooperation:

If I return \$5.00:

- Sender ends with: \$5 - \$3.63 + \$5.00 = \$6.37 (gain of \$1.37)
- I end with: \$10.89 - \$5.00 = \$5.89 (gain of \$5.89)

This seems like a fair split that rewards their trust while both of us benefit from the cooperation.

{"return": 5.00}

reasoning:

Let me analyze this situation carefully.

Given information:

- I'm the RECEIVER in Round 1
- The sender sent \$3.63 (which is 72.6% of their \$5 endowment)
- I received \$10.89 (which is $\$3.63 \times 3$)
- I need to decide how much to return (between \$0 and \$10.89)

Key considerations:

1. The sender was quite generous, sending 72.6% of their endowment
2. This shows trust and cooperation
3. There's mention of communication noise, so the actual amount sent might differ slightly from what I see
4. This is a multi-round game with 8 agents, so building reputation for cooperation could be beneficial
5. Fair reciprocation would be important to encourage future cooperation

Let me think about what would be a fair return:

- The sender gave up \$3.63 and kept \$1.37
- I received \$10.89
- If I want to be reciprocal and fair, I should consider:
 - The sender showed high trust (72.6% is a lot)
 - A common strategy is to return enough so both parties benefit
 - Returning about half (\$5.45) would give me \$5.44 and the sender would end up with \$1.37 + \$5.45 = \$6.82
- This would mean the sender gains \$1.82 total and I gain \$5.44

Let me reconsider:

- If I return around \$4-5, that seems fair
- Returning \$5 would give:
 - Sender: \$5 - \$3.63 + \$5 = \$6.37 (net gain of \$1.37)
 - Me: \$10.89 - \$5 = \$5.89 (net gain of \$5.89)

....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 801,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.52 returned, visible payoff \$7.52.

Your total visible earnings across all rounds are \$5.8900000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the situation:

From the visible history, my co-player:

- Sent \$3.0 in Round 1 (60% of their endowment)
- Saw \$5.52 returned (61.3% of the \$9.0)
- Made a profit of \$2.52

This shows they were cooperative and returned a substantial amount. To reciprocate their trust and maintain cooperation, I should send a generous amount.

I'll send \$3.50 (70% of my endowment), showing strong trust and willingness to cooperate.

```
{"send": 3.50}
```

Reasoning:

Let me analyze this situation:

Round 1 (from my perspective):

- I was the RECEIVER
- The sender sent \$3.63 (72.6% of endowment)
- I received \$10.89
- I returned \$5.00
- My payoff: \$10.89 - \$5.00 = \$5.89
- Sender's payoff: \$5 - \$3.63 + \$5.00 = \$6.37

But now I see the "visible history" which shows a different perspective:

- My co-player (who was the sender in Round 1) saw that they sent \$3.0
- It became \$9.0 (not \$10.89 as I received)
- They saw \$5.52 returned (not \$5.00 as I sent)
- Their visible payoff was \$7.52

This confirms there is communication noise in the game as mentioned in the instructions.

What actually happened vs what they saw:

- They actually sent \$3.63, but saw they sent \$3.0
- I received \$10.89, they saw it became \$9.0
- I returned \$5.00, they saw \$5.52 returned
- Their actual payoff was \$6.37, but they saw \$7.52

Now in Round 2, I'm the SENDER and they are the RECEIVER.

My co-player returned a substantial amount in Round 1 (they saw they returned \$5.52 out of \$9.0 received, which is 61.3%). This shows they are cooperative.

Given that:

1. They showed trust by being generous in Round 1
2. They appeared to return a fair amount (from their perspective, 61.3% of what they received)
3. I want to maintain cooperation and reciprocate their trust

I should send a generous amount. ...

```
Usage: {
  "input_tokens": 874,
  "output_tokens": 716,
  "reasoning_tokens": 0
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T08:07:32

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 2,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)

- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to establish a cooperative approach while being somewhat cautious since I don't have any history with my current co-player.

Sending a moderate amount signals trust and gives us both the opportunity to benefit from the 3x multiplier, while not leaving me completely vulnerable.

I'll send **\$3**. This gives my co-player \$9 to work with, and if they return a fair share, we can both end up better off than our starting positions.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game. I'm the sender and I have \$5. I need to decide how much to send to the receiver.

Let me think about the strategic considerations:

1. If I send \$X, the receiver gets \$3X
2. The receiver then decides how much to return
3. I keep \$5 - X + (amount returned)
4. The receiver keeps \$3X - (amount returned)

This is a classic trust game. The optimal outcome for both of us would be if I send the full \$5, it gets tripled to \$15, and the receiver returns a fair share (say, at least \$7.50, which would give me \$7.50 and them \$7.50, both better off than starting positions).

However, there are risks:

- The receiver might not return anything or very little
- There's also mentioned "communication noise" - amounts may differ from what was actually sent/returned

Since this is round 1 and I have no history with this player:

- I could play it safe and send less
- Or I could signal trust and cooperation by sending more

In standard trust game experiments, people often send around \$2-\$3 in the first round as a moderate trust signal.

However, if I'm thinking about maximizing joint welfare and establishing cooperation, send

ing more could be beneficial. If the receiver is rational and cooperative, they should return enough to make it worthwhile for me.

Let me consider a middle-ground strategy: sending \$3 or \$4.

- If I send \$3, receiver gets \$9. If they return \$4-5, we both benefit
- If I send \$4, receiver gets \$12. If they return \$6, we both benefit significantly

Given this is a multi-round game with potentially different partners each time, establishing a pattern of trust and cooperation could be beneficial. I'll go with sending \$3 as a balanced approach - showing trust but not complete vulnerability.

Actually, let me reconsider. The instructions mention this is multi-agent with 8 agents and randomized pairings. This means:

...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 648,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.63 sent, it became \$10.89, returned \$5.0, visible payoff \$5.890000000000001.

Your total visible earnings across all rounds are \$7.52.

This round, you are the RECEIVER against your current co-player. They sent you \$3.14, so you received \$9.42. How much do you return? (0-9.42)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the history:

- Round 1: My co-player saw \$3.63 sent, received \$10.89, and returned \$5.0 to me - showing cooperative behavior
- Round 2: They sent me \$3.14, which became \$9.42

They've demonstrated fairness by returning a reasonable amount in Round 1. I want to reciprocate this cooperative approach.

I'll return **\$4.50**, which:

- Gives them a profit (they sent ~\$3.14, get back \$4.50)
- Leaves me with a profit too (\$9.42 - \$4.50 = \$4.92)
- Maintains the cooperative relationship

```
'''json
{"return": 4.50}
'''
```

Reasoning:

Let me understand what happened:

Round 1: I sent \$3, but my co-player saw \$3.63 (due to communication noise). This became \$10.89 for them. They returned \$5.0.

My actual payoff in Round 1:

- I had \$5
- I sent \$3
- I received back \$5
- So I kept: \$5 - \$3 + \$5 = \$7

But the visible payoff shows \$7.52, which suggests there might be some noise in the amounts.

Now in Round 2:

- I'm the RECEIVER
- My co-player sent me some amount that appears as \$3.14 to me
- This became \$9.42 (which is \$3.14 × 3)
- I need to decide how much to return

Looking at their behavior: In Round 1, they saw \$3.63 sent, which became \$10.89, and they returned \$5.0. That's about 46% of what they received, or about 1.38x what they saw as sent.

This seems like a fair/cooperative player. They returned a reasonable amount.

Now I have \$9.42. What should I return?

Given that:

1. They sent a decent amount (\$3.14, which is about 63% of their \$5)
2. They were fair in the previous round, returning \$5 when they received \$10.89
3. I want to encourage continued cooperation

I should return a fair amount. Some options:

- Return about half of what I received: \$4.71
- Return what they sent plus a bit extra: around \$4-5
- Match their return rate from last round (46% of received): about \$4.33

I think returning around \$4.50-\$5.00 would be fair and encourage continued cooperation. Let me return \$4.50 as a fair split that gives them back close to what they sent plus some profit, while I also profit.

If I return \$4.50, they end up with \$6.36 while I keep \$4.92-both of us gain from the exchange. Returning \$5.00 would mirror...

```
Usage: {  
  "input_tokens": 768,  
  "output_tokens": 820,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_4 - Sender - dyad_2

Timestamp: 2026-08-12T08:07:43

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_2",  
  "move_index": 2,  
  "opponent_id": "Agent_3",  
  "pairing": {  
    "agents": [  
      "Agent_4",  
      "Agent_3"  
    ],  
    "investor": "Agent_4",  
    "roles": {  
      "Agent_3": "trustee",  
      "Agent_4": "investor"  
    },  
    "trustee": "Agent_3"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 2,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned

- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.91, that is 58.2% of their total endowment. You received \$8.73. How much do you return to the sender? (0-8.73)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.91 (58.2% of their endowment)
- I received \$8.73 after the 3x multiplication
- The sender currently has \$2.09 remaining

To reciprocate their trust and cooperation:

- If I return nothing, the sender ends with \$2.09 (worse than keeping everything)
- If I return \$3.50, the sender ends with \$5.59 and I keep \$5.23
- This rewards their trust while maintaining fairness

I'll return a meaningful amount to encourage continued cooperation.

{'return': 3.5}

reasoning:

Let me analyze this situation:

1. As the RECEIVER, the sender sent me \$2.91 (which is 58.2% of their \$5 endowment)
2. This amount was multiplied by 3x, so I received \$8.73
3. I need to decide how much to return (between \$0 and \$8.73)

Key considerations:

- The sender was quite generous, sending nearly 60% of their endowment
- This shows trust and cooperation
- In trust games, reciprocity is important for maintaining cooperation
- The sender's final payoff will be: \$5 - \$2.91 + whatever I return = \$2.09 + return
- My final payoff will be: \$8.73 - whatever I return

Common approaches in trust games:

1. Split the gains equally - the sender gave up \$2.91 but created \$8.73. The "gain" is \$8.73 - \$2.91 = \$5.82. Splitting this equally would mean returning about \$2.91
2. Give the sender back what they sent - return \$2.91

3. Give the sender a good return on their investment - perhaps return more than they sent
4. Equal final payoffs - try to make both players end up with similar amounts

Let me calculate some scenarios:

If I return \$2.91:

- Sender gets: $\$2.09 + \$2.91 = \$5.00$
- I get: $\$8.73 - \$2.91 = \$5.82$
- This is fair-ish, both doing okay

If I return \$3.50:

- Sender gets: $\$2.09 + \$3.50 = \$5.59$
- I get: $\$8.73 - \$3.50 = \$5.23$
- This balances outcomes more evenly

If I return \$4.00:

- Sender gets: $\$2.09 + \$4.00 = \$6.09$
- I get: $\$8.73 - \$4.00 = \$4.73$
- The sender benefits more, which seems appropriate given they initiated the cooperation

For truly equal final payoffs, I'd need to return \$3.32, which would give us both \$5.41 each. The sender demonstrated substantial trust by sending most of their endowment, so I should reciprocate meaningfully to reinforce that cooperation....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 942,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.56 returned, visible payoff \$5.5600000000000005.

Your total visible earnings across all rounds are \$5.23.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the history:

- My co-player sent \$3.0 (60% of their endowment) in Round 1, showing strong trust
- They saw a return of \$3.56 (39.6% of what they sent as receiver)
- This demonstrates cooperative behavior

To reciprocate their trust and maintain cooperation, I'll send a similar proportion:

```
{'send': 3.0}
```

Reasoning:

Let me analyze this situation:

Round 1:

- I was the RECEIVER
- The sender sent \$2.91 (58.2% of their endowment)
- I received \$8.73
- I returned \$3.50
- I kept \$5.23

Now in Round 2:

- I'm the SENDER
- I'm playing against the same co-player from Round 1
- The visible history shows that in Round 1, from my co-player's perspective:
 - They sent \$3.0 (as SENDER)
 - It became \$9.0
 - They saw \$3.56 returned
 - Their visible payoff was \$5.56

Wait, there's a discrepancy here. The problem says there's communication noise. Let me understand:

From my perspective in Round 1:

- Sender sent \$2.91
- I received \$8.73
- I returned \$3.50

From my co-player's perspective in Round 1:

- They sent \$3.0
- It became \$9.0
- They saw \$3.56 returned

So the noise affected what they saw vs what actually happened. They sent \$3.0 but I saw \$2.91. I returned \$3.50 but they saw \$3.56.

Now I'm the sender in Round 2. My co-player was cooperative in Round 1 - they sent 60% of their endowment (\$3.0 out of \$5). They saw a return of \$3.56 out of \$9.0, which is about 39.6% of what I received.

Given that:

1. My co-player showed strong trust by sending 60% in Round 1
2. They saw a decent return (39.6%)
3. I want to maintain cooperation and build trust

I should send a similar proportion to what they sent. They sent 60%, so I should send around \$3.0 as well to match their level of trust and cooperation.

...

```
Usage: {  
  "input_tokens": 814,  
  "output_tokens": 568,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_3 - Receiver - dyad_2

Timestamp: 2026-08-12T08:07:59

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_2",  
  "move_index": 3,  
  "opponent_id": "Agent_4",  
  "pairing": {  
    "agents": [  
      "Agent_4",  
      "Agent_3"  
    ],  
    "investor": "Agent_4",  
    "roles": {  
      "Agent_3": "trustee",  
      "Agent_4": "investor"  
    },  
    "trustee": "Agent_3"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 2,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust and caution. The multiplication factor of 3x creates an opportunity for mutual benefit if my co-player reciprocates.

Since this is the first round with no prior history, I'll send a moderate amount that signals cooperation while not exposing myself to maximum risk.

****My decision:****

```
'''json
{"send": 3}
'''
```

This gives my co-player \$9 to work with (3×3), leaving me with \$2. If they return a fair share, we can both benefit from this cooperation. Let's see how they respond.

reasoning:

This is the first round of a trust game where I'm the SENDER. Let me think about this strategically.

Key facts:

- I have \$5
- I can send \$0-\$5 to the receiver
- Amount sent is multiplied by 3x
- The receiver then decides how much to return to me
- I keep: \$5 - sent + returned
- Receiver keeps: received - returned
- There may be communication noise (amounts seen may differ from actual)

- This is a multi-agent setting with 8 agents total
- Pairings are randomized each round
- Agent names are hidden

Strategic considerations:

1. ****Trust and cooperation****: If both parties cooperate optimally, we can maximize joint payoffs. If I send all \$5, the receiver gets \$15. If they return half (\$7.50), I end up with \$7.50 and they keep \$7.50, which is better for both of us than if I send nothing.
2. ****Risk****: The receiver could keep everything and return nothing. If I send \$5 and get nothing back, I end up with \$0 while they get \$15.
3. ****First round****: This is the first round, so I have no reputation or history with this particular co-player. However, my actions might influence future interactions in a multi-round setting.
4. ****Communication noise****: The presence of noise means that even if someone intends to be fair, the amounts might appear different, which could lead to misunderstandings.
5. ****Signaling****: Sending a moderate to high amount signals trust and cooperation, which might encourage reciprocity.

Common strategies in trust games:

- Send half (\$2.50) as a moderate trust signal
- Send all (\$5) to maximize joint payoff and signal full trust
- Send a smaller amount (\$1-2) to test the waters
- Send nothing (\$0) to avoid risk entirely

Given this is round 1 and I want to establish a cooperative tone while not being completely naive, I think sending a significant but not maximum amount makes sense.

Research on trust games suggests most people send positive amounts initially, and receivers typically reciprocate. The 3x multiplier creates substantial mutual benefit potential—even modest returns from the receiver would leave us both better off than the starting position. I'll send \$3 as a balanced signal of trust and good faith without overexposing myself....

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 870,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.91 sent, it became \$8.73, returned \$3.5, visible payoff \$5.23.

Your total visible earnings across all rounds are \$5.5600000000000005.

This round, you are the RECEIVER against your current co-player. They sent you \$2.77, so you received \$8.31. How much do you return? (0-8.31)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the history:

****Round 1 analysis:****

- I sent \$3 (they saw \$2.91 due to noise)
- They received \$8.73 and returned \$3.5 (about 40% return rate)
- This was a cooperative response

****Current situation (Round 2):****

- They sent me \$2.77 (with noise)
- I received \$8.31
- They showed good faith cooperation in Round 1

I'll reciprocate their cooperative behavior with a similar return rate:

```
'''json
{"return": 3.5}
'''
```

This gives them a total of approximately \$5.73 and leaves me with \$4.81, creating mutual benefit from our cooperation.

Reasoning:

Let me analyze this situation:

Round 1 (just completed):

- I was the SENDER and sent \$3
- Due to communication noise, my co-player saw \$2.91 (close to \$3)
- It became \$8.73 for them (should be \$8.73 if 2.91×3)
- They returned \$3.5
- Their visible payoff: \$5.23 (they kept $8.73 - 3.5 = 5.23$)
- My payoff: $5 - 3 + 3.5 = 5.56$

Now Round 2:

- I am the RECEIVER
- My co-player sent me an amount, but I see \$2.77 (with noise)
- I received \$8.31 (which is 2.77×3)
- I need to decide how much to return

Analysis:

- In Round 1, my co-player returned \$3.5 out of \$8.73, which is about 40% return rate
- They kept \$5.23 for themselves
- I sent them \$3 (they saw \$2.91), and I ended up with \$5.56 total
- So I gained \$0.56 from the transaction

Now they sent me approximately \$2.77 (actual amount might be slightly different due to noise). If I apply a similar return rate (around 40%), that would be: $8.31 \times 0.4 \approx 3.32$

But let me think about fairness:

- They sent around \$2.77, leaving them with about \$2.23
- I received \$8.31
- A fair split might give them back enough so they end up with a reasonable amount

If I return \$3.5 (same as they returned to me):

- They end up with: $5 - 2.77 + 3.5 = 5.73$
- I end up with: $8.31 - 3.5 = 4.81$

Returning \$4 would give them \$6.23 while I get \$4.31. Returning \$3 would leave me with \$5.31 and them with \$5.23. Since they gave me roughly \$2.77 and returned about 40% of what they received, matching that reciprocal gesture with a \$3.5 return seems appropriate....

```
Usage: {
  "input_tokens": 784,
  "output_tokens": 826,
  "reasoning_tokens": 0
}
```

GAME - Agent_6 - Sender - dyad_3

Timestamp: 2026-08-12T08:08:15

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_5",
  "pairing": {
    "agents": [
      "Agent_6",
      "Agent_5"
    ],
    "investor": "Agent_6",
    "roles": {
      "Agent_5": "trustee",
      "Agent_6": "investor"
    }
  },
```

```

    "trustee": "Agent_5"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 2,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.66, that is 53.2% of their total endowment. You received \$7.98. How much do you return to the sender? (0-7.98)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.66 (53.2% of their endowment)
- I received \$7.98 after the 3x multiplication
- The sender currently has \$2.34 remaining from their original \$5

To reward their trust and encourage cooperation, I should return enough so they profit from their decision to send. If I return \$4.00:

- Sender ends with: \$2.34 + \$4.00 = \$6.34 (a gain of \$1.34)

- I keep: $\$7.98 - \$4.00 = \$3.98$

This seems fair - both of us benefit from the cooperation, and the sender's trust is rewarded.

```
{"return": 4.00}
```

reasoning:

Let me think about this carefully.

I'm the RECEIVER in Round 1.

- The sender sent \$2.66 (which is 53.2% of their \$5 endowment)
- This amount was multiplied by 3x, so I received $\$2.66 \times 3 = \7.98
- Now I need to decide how much to return (between \$0 and \$7.98)

Key considerations:

1. The sender sent more than half their endowment (53.2%), showing substantial trust
2. If I return nothing, I keep \$7.98 and the sender keeps $\$5 - \$2.66 = \$2.34$
3. If I return some amount, I should aim to be fair and encourage cooperation

A reasonable approach would be to:

- Ensure the sender ends up with more than they started with (more than \$5)
- Keep some profit myself
- Reward their trust to encourage future cooperation

Let me calculate:

- Sender currently has: $\$5 - \$2.66 = \$2.34$
- I have: \$7.98

If I return \$3.00:

- Sender ends with: $\$2.34 + \$3.00 = \$5.34$ (slight gain)
- I keep: $\$7.98 - \$3.00 = \$4.98$

If I return \$4.00:

- Sender ends with: $\$2.34 + \$4.00 = \$6.34$ (good gain)
- I keep: $\$7.98 - \$4.00 = \$3.98$

If I return \$3.50:

- Sender ends with: $\$2.34 + \$3.50 = \$5.84$
- I keep: $\$7.98 - \$3.50 = \$4.48$

...

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 699,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.13 returned, visible payoff \$5.13.

Your total visible earnings across all rounds are \$3.9800000000000004.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the visible history:

****Round 1 recap (from their perspective):****

- My co-player sent \$3.00 (60% of their endowment) - quite generous
- They saw \$3.13 returned
- Their visible payoff was \$5.13

My co-player showed significant trust by sending 60% of their endowment. Even though there's communication noise affecting what we each see, I should reciprocate this trust.

I'll match their generosity by sending the same amount they sent: \$3.00 (60% of my endowment).

This demonstrates:

- Reciprocal trust
- Willingness to cooperate
- Fair play matching their level of generosity

```
{"send": 3.00}
```

Reasoning:

Let me analyze this situation:

Round 1 (just played):

- I was the RECEIVER
- Sender sent \$2.66 (53.2%)
- I received \$7.98
- I returned \$4.00
- I kept \$3.98

But now I see the "visible history" which shows:

- Round 1 against me, as SENDER: sent \$3.0, became \$9.0, saw \$3.13 returned, visible payoff \$5.13

This is interesting - there's communication noise!

- The sender actually sent \$3.00 (not \$2.66 as I saw)
- It became \$9.00 (not \$7.98 as I saw)
- They saw \$3.13 returned (not \$4.00 as I sent)
- Their visible payoff was \$5.13 (which is $5 - 3 + 3.13$)

So due to noise:

- I saw them send \$2.66, they actually sent \$3.00
- I saw I received \$7.98, I actually received \$9.00
- I returned \$4.00, they saw \$3.13

Now in Round 2, I'm the SENDER and playing against the same person.

From their perspective:

- They sent \$3.00 (60% of endowment - quite generous)
- They got back \$3.13 (34.8% of what I received)
- Their payoff was \$5.13

From my actual perspective (what really happened):

- They sent \$3.00
- I received \$9.00
- I returned \$4.00 (44.4% of what I received)
- But they only saw \$3.13

The co-player was quite generous in Round 1 (sent 60%). From their visible perspective, they got a modest return (34.8% of the tripled amount).

...

```
Usage: {  
  "input_tokens": 845,  
  "output_tokens": 754,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_5 - Receiver - dyad_3

Timestamp: 2026-08-12T08:08:33

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",  
  "dyad_id": "dyad_3",  
  "move_index": 5,  
  "opponent_id": "Agent_6",  
  "pairing": {  
    "agents": [  
      "Agent_6",  
      "Agent_5"  
    ],  
    "investor": "Agent_6",  
    "roles": {
```

```

    "Agent_5": "trustee",
    "Agent_6": "investor"
  },
  "trustee": "Agent_5"
},
"role": "trustee",
"role_label": "Receiver",
"round": 2,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round situation, I need to balance potential gains from cooperation against the uncertainty of how my co-player will respond.

****My reasoning:****

- Sending money creates value (3x multiplier), so there's potential for mutual benefit
- This is our first interaction, so I want to signal willingness to cooperate
- But I also want to maintain some security given the uncertainty

- The communication noise adds another layer of unpredictability

I'll send a moderate amount that demonstrates trust while keeping some reserve.

```
'''json
{"send": 3}
'''
```

This sends \$3 to my co-player, who will receive \$9. I'll keep \$2 initially, and hope they reciprocate with a fair return that benefits us both.

reasoning:

This is the first round of a trust game where I'm the sender. Let me think about the strategic situation:

1. I have \$5 to start with
2. Whatever I send gets tripled when it reaches the receiver
3. The receiver then decides how much to return to me
4. There's communication noise - amounts may appear different than what was actually sent

Key considerations:

- This is a multi-round game with multiple agents
- Pairings are randomized each round
- There's a tension between trust/cooperation and self-interest
- If I send nothing, I keep \$5 but there's no opportunity for mutual gain
- If I send everything (\$5), the receiver gets \$15, and if they're fair/cooperative, they might return a good amount
- The communication noise adds uncertainty

In standard trust game experiments, sending some amount is generally beneficial because:

- It creates surplus value (3x multiplier)
- It signals trust and can encourage reciprocity
- Over multiple rounds, cooperative behavior can be rewarding

First round considerations:

- I don't know this particular co-player yet
- This is the first interaction, so I might want to signal willingness to cooperate
- But I also don't want to be too naive

A moderate approach might be to send \$3 or \$4:

- Sending \$3 means receiver gets \$9, I keep \$2. If they return \$4-5, we both do well
- Sending \$4 means receiver gets \$12, I keep \$1. If they return \$6, we both get \$6
- Sending \$5 means receiver gets \$15, I keep \$0. If they return \$7-8, I do well

Given this is round 1 and I want to establish a cooperative tone while not being too vulnerable, I'll send \$3. This shows trust but maintains some security.

Actually, let me reconsider. In trust games, the classic finding is that people often send about half their endowment on average. Sending \$2.50 or \$3 seems reasonable.

...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 703,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.66 sent, it became \$7.98, returned \$4.0, visible payoff \$3.9800000000000004.
- Your total visible earnings across all rounds are \$5.13.

This round, you are the RECEIVER against your current co-player. They sent you \$2.97, so you received \$8.91. How much do you return? (0-8.91)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the situation:

****What happened in Round 1:****

- My co-player received \$7.98 (my \$3 send with noise)
- They returned \$4.0, which is about 50% - a fair reciprocal move
- This shows they're a cooperative player

****Current Round 2:****

- They sent me \$2.97 (?60% of their endowment) - showing significant trust
- I received \$8.91
- They kept \$2.03

****My decision:****

Since they demonstrated fair reciprocity (50% return) in Round 1 and have now shown strong trust by sending most of their endowment, I should reciprocate similarly. Returning about 50% maintains the cooperative relationship while ensuring we both benefit.

Returning \$4.50 means:

- They end with: $\$2.03 + \$4.50 = \$6.53$
- I end with: $\$8.91 - \$4.50 = \$4.41$

This is fair and encourages continued cooperation.

```
'''json
{"return": 4.50}
'''
```

Reasoning:

Let me analyze this situation:

Round 1 recap:

- I sent \$3 (but due to communication noise, my co-player saw \$2.66)
- They received \$7.98 (which should be $\$2.66 \times 3$)
- They returned \$4.0
- My visible payoff was \$5.13 (which would be $\$5 - \$3 + \$3.13...$ but there's noise)
- Their visible payoff was \$3.98

Now in Round 2:

- My co-player is the SENDER
- They sent me \$2.97 (with noise, they might have intended something slightly different)
- I received \$8.91 (which is $\$2.97 \times 3$)
- I need to decide how much to return

Analysis of their behavior:

- In Round 1, they returned \$4.0 out of \$7.98 received, which is about 50%
- This is a reasonably fair/generous return
- Now they've sent me \$2.97 out of their \$5, which is about 59% - quite trusting

Given that:

1. They showed good reciprocity in Round 1 (returning 50% of what they received)
2. They've now shown significant trust by sending me almost 60% of their endowment
3. We might continue playing together in future rounds
4. Fair reciprocity would encourage continued cooperation

I should return a similar proportion or slightly more to maintain/build trust.

They received \$8.91 from my send. A 50% return would be \$4.455.

But let me think about the overall economics:

- They started with \$5
- They sent \$2.97, keeping \$2.03
- I received \$8.91
- If I return \$4.50:
 - They end with: $\$2.03 + \$4.50 = \$6.53$

...

```
Usage: {
  "input_tokens": 816,
  "output_tokens": 863,
  "reasoning_tokens": 0
}
```

GAME - Agent_8 - Sender - dyad_4

Timestamp: 2026-08-12T08:08:44
 Model: anthropic/claude-sonnet-4.5
 Temperature: 0.8
 Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_4",
 "move_index": 6,
 "opponent_id": "Agent_7",
 "pairing": {
 "agents": [
 "Agent_8",
 "Agent_7"
],
 "investor": "Agent_8",
 "roles": {
 "Agent_7": "trustee",
 "Agent_8": "investor"
 },
 "trustee": "Agent_7"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 2,
 "task": "game",
 "task_index": 0
 }

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will

ll have acted as sender and receiver equally often when the round count allows it.

- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.17, that is 63.4% of their total endowment. You received \$9.51. How much do you return to the sender? (0-9.51)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending 63.4% of their endowment (\$3.17)
- I received \$9.51 after the 3x multiplier
- The sender currently has $\$5 - \$3.17 = \$1.83$

The sender demonstrated considerable trust, so I want to reciprocate appropriately. If I return \$5.00:

- The sender would end with: $\$1.83 + \$5.00 = \$6.83$ (a profit of \$1.83)
- I would keep: $\$9.51 - \$5.00 = \$4.51$

This rewards their trust with a solid return while allowing us both to benefit from the cooperation.

{"return": 5.00}

reasoning:

Let me think about this carefully.

I'm the RECEIVER in Round 1.

- The sender sent \$3.17 (which is 63.4% of their \$5 endowment)
- This was multiplied by 3x, so I received: $\$3.17 \times 3 = \9.51 ?
- I need to decide how much to return (between \$0 and \$9.51)

Key considerations:

1. The sender showed considerable trust by sending 63.4% of their endowment
2. If I return nothing, the sender ends up with: $\$5 - \$3.17 = \$1.83$, while I keep \$9.51
3. If I return a fair/reciprocal amount, what would that be?

A few approaches to consider:

- Equal split of total: Total is \$5 (sender's initial) + \$9.51 (what I received) = \$14.51. Equal split would be ~\$7.26 each. The sender currently has \$1.83, so I'd return $\$7.26 - \$1.83 = \$5.43$
- Return what maintains sender's initial investment value: Sender invested \$3.17. If I return \$3.17, sender gets back to \$5 total, and I keep \$6.34
- Return proportionally to what was sent: The sender sent 63.4% of their endowment. I could return a similar proportion of what I received: 63.4% of \$9.51 = ~\$6.03
- Social norm of reciprocity: Return enough to reward trust but keep some gains

Given that:

- The sender showed significant trust (63.4%)
- This is a multi-round game with different partners
- Building reputation for reciprocity could be valuable
- Note about communication noise suggests amounts might not be exact

I think a reasonable return would be around \$4-5, which:

- Acknowledges their generosity without giving away all my gains
- Leaves them better off than their starting position
- Demonstrates trustworthiness that could matter in future interactions with other partners
- Strikes a balance between reciprocity and self-interest

I'll return \$4.75. This gives the sender \$6.58 total (\$1.83 remaining plus my return), which meaningfully rewards their trust while I keep \$4.76....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1001,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.37 returned, visible payoff \$7.37.

Your total visible earnings across all rounds are \$4.51.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the history:

My co-player sent \$3.0 in Round 1 (60% of their endowment), showing significant trust. They saw \$5.37 returned, which gave them a good outcome.

Since they demonstrated trust and cooperation, I should reciprocate similarly. Sending around the same amount they sent (\$3.0) would be fair and maintain the cooperative relationship.

```
{"send": 3.0}
```

Reasoning:

Let me analyze this situation:

1. In Round 1, I was the RECEIVER and the sender sent me \$3.17 (according to my view), which became \$9.51 for me. I returned \$5.00.
2. However, from my current co-player's perspective (shown in "Your current co-player's last 3 game(s)"):
 - They sent \$3.0 (not \$3.17 - communication noise!)
 - It became \$9.0 (not \$9.51 - more noise!)
 - They saw \$5.37 returned (not \$5.00 - even more noise!)
 - Their visible payoff was \$7.37

So there's noise in the communication. What they experienced:

- Started with \$5
- Sent \$3.0
- Had \$2.0 left
- Saw \$5.37 returned
- Ended with \$2.0 + \$5.37 = \$7.37

From their perspective, I (as the receiver in Round 1) returned about 60% of what I received (\$5.37 out of \$9.0).

3. Now I'm the SENDER in Round 2 against the same co-player.

My co-player showed trust by sending \$3.0 (from their perspective) and saw a decent return ratio. They might be willing to reciprocate trust.

I should consider:

- They sent about 60% of their endowment in Round 1
- They saw about 60% returned
- This suggests they're moderately trusting and cooperative

I should send a similar amount to show I'm also trustworthy and cooperative. Sending around \$3 would be consistent with their behavior and maintain the cooperative relationship.

Let me send \$3.00 to match their earlier send amount.

```
Usage: {
  "input_tokens": 837,
  "output_tokens": 523,
  "reasoning_tokens": 0
}
```

GAME - Agent_7 - Receiver - dyad_4

Timestamp: 2026-08-12T08:09:03

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",

```

"dyad_id": "dyad_4",
"move_index": 7,
"opponent_id": "Agent_8",
"pairing": {
  "agents": [
    "Agent_8",
    "Agent_7"
  ],
  "investor": "Agent_8",
  "roles": {
    "Agent_7": "trustee",
    "Agent_8": "investor"
  },
  "trustee": "Agent_7"
},
"role": "trustee",
"role_label": "Receiver",
"round": 2,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to consider:

1. ****The multiplier effect****: Any amount I send gets tripled, creating potential mutual gain
2. ****No prior history****: This is my first interaction, so I have no information about this co-player's behavior
3. ****Future rounds****: Even though pairings are randomized, establishing cooperative norms early could benefit the group overall
4. ****Risk vs. reward****: Sending more creates more surplus, but requires trusting the receiver to return a fair amount

I'll take a moderately cooperative approach. Sending \$3 creates \$9 for the receiver - enough surplus that we can both benefit if they reciprocate fairly, while not risking my entire endowment.

If they return around half of what they receive (\$4-5), we both end up better off than if no transaction occurred.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a multi-agent trust game experiment. I'm the SENDER with \$5.

Let me think about the strategic aspects:

1. If I send money, it gets tripled, creating surplus value
2. The receiver then decides how much to return
3. This is a classic trust game scenario

Key considerations:

- This is round 1, so I have no history with this particular co-player
- There are multiple rounds, so reputation/repeated interaction matters
- However, pairings are randomized each round, so I won't necessarily play the same person again
- The game has communication noise, which might affect trust

Strategic thinking:

- Sending \$0 guarantees I keep \$5, but creates no surplus
- Sending \$5 creates maximum surplus (\$15 total for receiver), but requires trust
- A middle ground (e.g., \$2-3) balances risk and potential gain

In classic trust game experiments, people often:

1. Send moderate to high amounts in early rounds to signal cooperation
2. The amount sent and returned establishes baseline trust levels
3. With randomized pairings, building a reputation as trustworthy matters for future rounds

Game theory perspective:

- If this were truly one-shot with rational self-interested players, the receiver should return \$0 (backward induction)
- Knowing this, I should send \$0
- However, real humans often deviate from this due to fairness norms, reciprocity, and reputation concerns

Since:

- This is multi-round with multiple agents
- There may be reputation effects even with randomized pairings
- Starting cooperatively can establish good norms
- The 3x multiplier creates real gains from cooperation

I'll send a moderate-high amount to signal cooperation while not being completely vulnerable. I'll send \$3.

This creates \$9 for the receiver, and if they return even just \$3, we both end up with \$5 (same as no transaction), but if they return more (say \$4-5), we both gain. ...

usage:

```
{
```

```
"input_tokens": 500,  
"output_tokens": 671,  
"reasoning_tokens": 0  
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.17 sent, it became \$9.51, returned \$5.0, visible payoff \$4.51.

Your total visible earnings across all rounds are \$7.37.

This round, you are the RECEIVER against your current co-player. They sent you \$2.35, so you received \$7.050000000000001. How much do you return? (0-7.050000000000001)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the history:

****Round 1 analysis:****

- They saw I sent \$3.17 (with noise), received \$9.51
- They returned \$5.00, which is about 52.6% of what they received
- This was quite generous and cooperative

****Current situation:****

- They sent me \$2.35, I received \$7.05
- They showed good faith last round, so I should reciprocate

Following a similar reciprocity rate (around 52-53% return):

- 53% of \$7.05 ? \$3.74

This ensures:

- They profit from sending (get back more than they sent)
- I also benefit from the transaction
- We maintain cooperative momentum

```
'''json  
{"return": 3.75}  
'''
```

Reasoning:

Let me analyze this situation:

****What happened in Round 1:****

- I sent \$3, but they saw \$3.17 (communication noise)
- They received \$9.51 (3.17×3)
- They returned \$5.0
- Their visible payoff was $\$9.51 - \$5.0 = \$4.51$
- My payoff was $\$5 - \$3 + \$5 = \7 (but visible earnings show \$7.37, maybe due to noise)

****Current Round 2:****

- They sent me \$2.35 (actual), which became \$7.05
- I need to decide how much to return

****Analysis of their behavior:****

In Round 1, they returned \$5 out of \$9.51 received, which is about 52.6%. This is a fairly generous return - they kept \$4.51 and I got back \$5, so I had a net gain of \$2 from the transaction.

****What should I do?***

They showed cooperative behavior by returning a good amount. Now they've sent me \$2.35, creating \$7.05.

If I apply a similar reciprocity ratio:

- They returned about 52.6% of what they received
- I received \$7.05, so 52.6% would be about \$3.71

However, I should also consider:

- They sent \$2.35 this round (less than the ~\$3.17 they saw me send)
- But they were generous with their return last round

- I want to maintain cooperation

A fair approach might be to:

1. Return enough that they profit from sending
2. Keep enough that I also profit
3. Signal continued cooperation

If they sent \$2.35 and expect around a 50% return rate:

- They might expect back around \$3-4

Let me think about payoffs:

- They started with \$5
- They sent \$2.35
- If I return \$3.5, they end with \$6.15 and I keep \$3.55
- If I return \$4, they end with \$6.65 and I keep \$3.05
- If I return \$3, they end with \$5.65 and I keep \$4.05

Since they matched my generosity last round with a 52.6% return, I should reciprocate similarly to maintain the cooperative dynamic....

```
Usage: {
  "input_tokens": 852,
  "output_tokens": 862,
  "reasoning_tokens": 0
}
```

ROUND 3

Roles: Agent_1=Sender, Agent_2=Receiver, Agent_3=Sender, Agent_4=Receiver, Agent_5=Sender, Agent_6=Receiver, Agent_7=Sender, Agent_8=Receiver

Pairings:

dyad_1: Agent_1=Sender, Agent_2=Receiver
dyad_2: Agent_3=Sender, Agent_4=Receiver
dyad_3: Agent_5=Sender, Agent_6=Receiver
dyad_4: Agent_7=Sender, Agent_8=Receiver

Parsed game decisions:

dyad_1: Agent_1 sent \$4; received \$12; Agent_2 returned \$5.5; payoffs Agent_1=\$6.5, Agent_2=\$6.5;
noise/visible: sent visible=\$3.64, received visible=\$10.92, returned visible=\$4.66
Action validation: {"returned": {"amount": 5.5, "clamped": false, "max": 12.0, "min": 0, "raw": 5.5}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
dyad_2: Agent_3 sent \$3; received \$9; Agent_4 returned \$4.5; payoffs Agent_3=\$6.5, Agent_4=\$4.5;
noise/visible: sent visible=\$3.52, received visible=\$10.56, returned visible=\$3.78
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_3: Agent_5 sent \$3; received \$9; Agent_6 returned \$5; payoffs Agent_5=\$7, Agent_6=\$4;
noise/visible: sent visible=\$3.23, received visible=\$9.69, returned visible=\$5.79
Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 9.0, "min": 0, "raw": 5.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_7 sent \$3; received \$9; Agent_8 returned \$3.33; payoffs Agent_7=\$5.33, Agent_8=\$5.67; noise/visible: sent visible=\$2.22, received visible=\$6.66, returned visible=\$4.3
Action validation: {"returned": {"amount": 3.33, "clamped": false, "max": 9.0, "min": 0, "raw": 3.33}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 19.5,
"Agent_2": 16.5,
"Agent_3": 17.5,
"Agent_4": 15.5,
"Agent_5": 17.5,
"Agent_6": 15.5,
"Agent_7": 17.58,
"Agent_8": 15.42

}

Visible balances after round: {

"Agent_1": 18.1,
"Agent_2": 17.869999999999997,
"Agent_3": 16.15,
"Agent_4": 16.19,
"Agent_5": 17.33,
"Agent_6": 15.700000000000001,
"Agent_7": 16.970000000000002,

```
"Agent_8": 13.549999999999999
}
```

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T08:09:12

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 3,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to establish a cooperative approach while being somewhat cautious since I don't have any history with my current co-player.

Sending a moderate amount signals trust and gives us both the opportunity to benefit from the 3x multiplier, while not leaving me completely vulnerable.

I'll send **\$3**. This gives my co-player \$9 to work with, and if they return a fair share , we can both end up better off than our starting positions.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game. I'm the sender and I have \$5. I need to decide how much to send to the receiver.

Let me think about the strategic considerations:

1. If I send \$X, the receiver gets \$3X
2. The receiver then decides how much to return
3. I keep \$5 - X + (amount returned)
4. The receiver keeps \$3X - (amount returned)

This is a classic trust game. The optimal outcome for both of us would be if I send the full \$5, it gets tripled to \$15, and the receiver returns a fair share (say, at least \$7.50, which would give me \$7.50 and them \$7.50, both better off than starting positions).

However, there are risks:

- The receiver might not return anything or very little
- There's also mentioned "communication noise" - amounts may differ from what was actually sent/returned

Since this is round 1 and I have no history with this player:

- I could play it safe and send less
- Or I could signal trust and cooperation by sending more

In standard trust game experiments, people often send around \$2-\$3 in the first round as a moderate trust signal.

However, if I'm thinking about maximizing joint welfare and establishing cooperation, sending more could be beneficial. If the receiver is rational and cooperative, they should return enough to make it worthwhile for me.

Let me consider a middle-ground strategy: sending \$3 or \$4.

- If I send \$3, receiver gets \$9. If they return \$4-5, we both benefit
- If I send \$4, receiver gets \$12. If they return \$6, we both benefit significantly

Given this is a multi-round game with potentially different partners each time, establishing a pattern of trust and cooperation could be beneficial. I'll go with sending \$3 as a balanced approach - showing trust but not complete vulnerability.

Actually, let me reconsider. The instructions mention this is multi-agent with 8 agents and randomized pairings. This means:

...

usage:

```
{
  "input_tokens": 500,
```

```
"output_tokens": 648,  
"reasoning_tokens": 0  
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.63 sent, it became \$10.89, returned \$5.0, visible payoff \$5.8900000000000001.

Your total visible earnings across all rounds are \$7.52.

This round, you are the RECEIVER against your current co-player. They sent you \$3.14, so you received \$9.42. How much do you return? (0-9.42)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

- Round 1: My co-player saw \$3.63 sent, received \$10.89, and returned \$5.0 to me - showing cooperative behavior
- Round 2: They sent me \$3.14, which became \$9.42

They've demonstrated fairness by returning a reasonable amount in Round 1. I want to reciprocate this cooperative approach.

I'll return **\$4.50**, which:

- Gives them a profit (they sent ~\$3.14, get back \$4.50)
- Leaves me with a profit too (\$9.42 - \$4.50 = \$4.92)
- Maintains the cooperative relationship

```
'''json  
{"return": 4.50}  
'''
```

reasoning:

Let me understand what happened:

Round 1: I sent \$3, but my co-player saw \$3.63 (due to communication noise). This became \$10.89 for them. They returned \$5.0.

My actual payoff in Round 1:

- I had \$5
- I sent \$3
- I received back \$5
- So I kept: \$5 - \$3 + \$5 = \$7

But the visible payoff shows \$7.52, which suggests there might be some noise in the amounts.

Now in Round 2:

- I'm the RECEIVER
- My co-player sent me some amount that appears as \$3.14 to me
- This became \$9.42 (which is \$3.14 × 3)
- I need to decide how much to return

Looking at their behavior: In Round 1, they saw \$3.63 sent, which became \$10.89, and they returned \$5.0. That's about 46% of what they received, or about 1.38x what they saw as sent.

This seems like a fair/cooperative player. They returned a reasonable amount.

Now I have \$9.42. What should I return?

Given that:

1. They sent a decent amount (\$3.14, which is about 63% of their \$5)
2. They were fair in the previous round, returning \$5 when they received \$10.89
3. I want to encourage continued cooperation

I should return a fair amount. Some options:

- Return about half of what I received: \$4.71

- Return what they sent plus a bit extra: around \$4-5
- Match their return rate from last round (46% of received): about \$4.33

I think returning around \$4.50-\$5.00 would be fair and encourage continued cooperation. Let me return \$4.50 as a fair split that gives them back close to what they sent plus some profit, while I also profit.

If I return \$4.50, they end up with \$6.36 while I keep \$4.92-both of us gain from the exchange. Returning \$5.00 would mirror...

usage:

```
{
  "input_tokens": 768,
  "output_tokens": 820,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.63 sent, it became \$10.89, returned \$5.0, visible payoff \$5.890000000000001.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.06 returned, visible payoff \$6.56.

Your total visible earnings across all rounds are \$12.44.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at our history together:

Round 1: I sent (they saw \$3.63), they returned \$5.0 - cooperative

Round 2: They sent \$3.5 (I saw \$3.14), I returned (they saw \$5.06) - both cooperative

We've established a good cooperative pattern. They've been generous as both sender and receiver, so I want to continue building this trust.

I'll send **\$4** to show strong trust and maintain our mutually beneficial relationship.

```
'''json
{"send": 4}
'''
```

Reasoning:

Let me analyze the history with my current co-player:

Round 1: I was SENDER

- I sent \$3 (but they saw \$3.63 due to noise)
- They received \$10.89
- They returned \$5.0
- This was cooperative behavior

Round 2: I was RECEIVER

- They sent \$3.5 (but I saw \$3.14 due to noise)
- I received \$9.42
- I returned \$4.50 (but they saw \$5.06 due to noise)
- They had a visible payoff of \$6.56

So far, this co-player has been quite cooperative:

- As receiver in Round 1: returned \$5.0 from \$10.89 (about 46%)
- As sender in Round 2: sent \$3.5 out of \$5 (70%)

I should continue this cooperative relationship. In Round 1, I sent \$3. In Round 2, they sent \$3.5. I should probably send a similar or slightly higher amount to maintain cooperation.

I'll send \$3.5 or \$4 to show continued trust and cooperation.

Given that they returned fairly and sent generously, I'll send \$4 to demonstrate strong trust and cooperation.

Usage: {

```
"input_tokens": 1118,
"output_tokens": 421,
"reasoning_tokens": 0
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T08:09:24

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 3,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.63, that is 72.6% of their total endowment. You received \$10.89. How much do you return to the sender? (0-10.89)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.63 (72.6% of their endowment)
- I received \$10.89 after the 3x multiplier
- A fair return should reward their trust while ensuring both of us benefit

The sender gave up \$3.63, keeping only \$1.37. Their generosity created \$10.89 total value for me. To reciprocate fairly and encourage future cooperation:

If I return \$5.00:

- Sender ends with: $\$5 - \$3.63 + \$5.00 = \6.37 (gain of \$1.37)
- I end with: $\$10.89 - \$5.00 = \$5.89$ (gain of \$5.89)

This seems like a fair split that rewards their trust while both of us benefit from the co operation.

{"return": 5.00}

reasoning:

Let me analyze this situation carefully.

Given information:

- I'm the RECEIVER in Round 1
- The sender sent \$3.63 (which is 72.6% of their \$5 endowment)
- I received \$10.89 (which is $\$3.63 \times 3$)
- I need to decide how much to return (between \$0 and \$10.89)

Key considerations:

1. The sender was quite generous, sending 72.6% of their endowment
2. This shows trust and cooperation
3. There's mention of communication noise, so the actual amount sent might differ slightly from what I see
4. This is a multi-round game with 8 agents, so building reputation for cooperation could be beneficial
5. Fair reciprocation would be important to encourage future cooperation

Let me think about what would be a fair return:

- The sender gave up \$3.63 and kept \$1.37
- I received \$10.89
- If I want to be reciprocal and fair, I should consider:
 - The sender showed high trust (72.6% is a lot)
 - A common strategy is to return enough so both parties benefit
- Returning about half (\$5.45) would give me \$5.44 and the sender would end up with $\$1.37 + \$5.45 = \$6.82$
- This would mean the sender gains \$1.82 total and I gain \$5.44

Let me reconsider:

- If I return around \$4-5, that seems fair
- Returning \$5 would give:
 - Sender: $\$5 - \$3.63 + \$5 = \6.37 (net gain of \$1.37)
 - Me: $\$10.89 - \$5 = \$5.89$ (net gain of \$5.89)

....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 801,
```

```

    "reasoning_tokens": 0
  }
}
Message 4 role=user
content:
  Round 2

  Visible history before this round:
  Your current co-player's last 3 game(s):
  - Round 1 against you, as SENDER: sent $3.0, it became $9.0, saw $5.52 returned, visible p
    ayoff $7.52.
  Your total visible earnings across all rounds are $5.8900000000000001.

  This round, you are the SENDER against your current co-player. You have $5. How much do yo
    u send? (0-5)
  Respond exactly as JSON: {'send': <amount>}
Message 5 role=assistant
content:
  Looking at the situation:

  From the visible history, my co-player:
  - Sent $3.0 in Round 1 (60% of their endowment)
  - Saw $5.52 returned (61.3% of the $9.0)
  - Made a profit of $2.52

  This shows they were cooperative and returned a substantial amount. To reciprocate their t
    rust and maintain cooperation, I should send a generous amount.

  I'll send $3.50 (70% of my endowment), showing strong trust and willingness to cooperate.

  {"send": 3.50}
reasoning:
  Let me analyze this situation:

  Round 1 (from my perspective):
  - I was the RECEIVER
  - The sender sent $3.63 (72.6% of endowment)
  - I received $10.89
  - I returned $5.00
  - My payoff: $10.89 - $5.00 = $5.89
  - Sender's payoff: $5 - $3.63 + $5.00 = $6.37

  But now I see the "visible history" which shows a different perspective:
  - My co-player (who was the sender in Round 1) saw that they sent $3.0
  - It became $9.0 (not $10.89 as I received)
  - They saw $5.52 returned (not $5.00 as I sent)
  - Their visible payoff was $7.52

  This confirms there is communication noise in the game as mentioned in the instructions.

  What actually happened vs what they saw:
  - They actually sent $3.63, but saw they sent $3.0
  - I received $10.89, they saw it became $9.0
  - I returned $5.00, they saw $5.52 returned
  - Their actual payoff was $6.37, but they saw $7.52

  Now in Round 2, I'm the SENDER and they are the RECEIVER.

  My co-player returned a substantial amount in Round 1 (they saw they returned $5.52 out of
    $9.0 received, which is 61.3%). This shows they are cooperative.

  Given that:
  1. They showed trust by being generous in Round 1
  2. They appeared to return a fair amount (from their perspective, 61.3% of what they recei
    ved)
  3. I want to maintain cooperation and reciprocate their trust

  I should send a generous amount. ...
usage:
  {
    "input_tokens": 874,

```



```
"output_tokens": 716,  
"reasoning_tokens": 0  
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.52 returned, visible payoff \$7.52.
- Round 2 against you, as RECEIVER: saw \$3.14 sent, it became \$9.42, returned \$4.5, visible payoff \$4.92.

Your total visible earnings across all rounds are \$12.45.

This round, you are the RECEIVER against your current co-player. They sent you \$3.64, so you received \$10.92. How much do you return? (0-10.92)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the visible history:

Round 1: My co-player sent \$3.0 (60%) and saw \$5.52 returned

Round 2: They returned \$4.5 when they received \$9.42 (47.8%)

Round 3: They sent \$3.64 (72.8% - even more generous!)

My co-player has been consistently cooperative and just showed increased trust by sending an even larger percentage. I should reciprocate fairly.

Returning \$5.50:

- Sender gets: $\$5 - \$3.64 + \$5.50 = \6.86 (profit of \$1.86)
- I get: $\$10.92 - \$5.50 = \$5.42$ (profit of \$5.42)

This ~50% return rate rewards their trust while ensuring mutual benefit.

```
{"return": 5.50}
```

Reasoning:

Let me analyze the visible history:

Round 1: My co-player was SENDER

- They sent \$3.0 (60% of endowment)
- It became \$9.0
- They saw \$5.52 returned (61.3% return rate)
- Their visible payoff: \$7.52

Round 2: My co-player was RECEIVER

- They saw \$3.14 sent (I actually sent \$3.50 but there's noise)
- It became \$9.42
- They returned \$4.5 (47.8% of what they received)
- Their visible payoff: \$4.92

Now Round 3: I am RECEIVER

- They sent \$3.64 (72.8% of their endowment - very generous!)
- I received \$10.92
- I need to decide how much to return

Looking at the pattern:

- In Round 1, they sent 60% and I returned about 61% (assuming noise in what they saw)
- In Round 2, they received and returned 47.8%
- In Round 3, they sent 72.8% - showing even more trust

The co-player has been consistently cooperative. They sent a high amount both times they were sender (60% and 72.8%). When they were receiver, they returned a reasonable amount (47.8%).

To maintain reciprocity and reward their increased trust, I should return a fair amount.

If I return \$5.50:

- Sender ends with: $\$5 - \$3.64 + \$5.50 = \6.86 (gain of \$1.86)
- I end with: $\$10.92 - \$5.50 = \$5.42$ (gain of \$5.42)

This is about 50.4% return rate, which is reasonable and shows I'm reciprocating their trust w

```

    hile both of us profit.
    ...
Usage: {
  "input_tokens": 1200,
  "output_tokens": 712,
  "reasoning_tokens": 0
}

```

GAME - Agent_3 - Sender - dyad_2

```

Timestamp: 2026-08-12T08:09:33
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8
Metadata: {

```

```

  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_4",
  "pairing": {
    "agents": [
      "Agent_3",
      "Agent_4"
    ],
    "investor": "Agent_3",
    "roles": {
      "Agent_3": "investor",
      "Agent_4": "trustee"
    },
    "trustee": "Agent_4"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 3,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust and caution. The multiplication factor of 3x creates an opportunity for mutual benefit if my co-player reciprocates.

Since this is the first round with no prior history, I'll send a moderate amount that signals cooperation while not exposing myself to maximum risk.

****My decision:****

```
'''json
{"send": 3}
'''
```

This gives my co-player \$9 to work with (3×3), leaving me with \$2. If they return a fair share, we can both benefit from this cooperation. Let's see how they respond.

reasoning:

This is the first round of a trust game where I'm the SENDER. Let me think about this strategically.

Key facts:

- I have \$5
- I can send \$0-\$5 to the receiver
- Amount sent is multiplied by 3x
- The receiver then decides how much to return to me
- I keep: \$5 - sent + returned
- Receiver keeps: received - returned
- There may be communication noise (amounts seen may differ from actual)
- This is a multi-agent setting with 8 agents total
- Pairings are randomized each round
- Agent names are hidden

Strategic considerations:

1. ****Trust and cooperation****: If both parties cooperate optimally, we can maximize joint payoffs. If I send all \$5, the receiver gets \$15. If they return half (\$7.50), I end up with \$7.50 and they keep \$7.50, which is better for both of us than if I send nothing.
2. ****Risk****: The receiver could keep everything and return nothing. If I send \$5 and get nothing back, I end up with \$0 while they get \$15.
3. ****First round****: This is the first round, so I have no reputation or history with this particular co-player. However, my actions might influence future interactions in a multi-round setting.
4. ****Communication noise****: The presence of noise means that even if someone intends to be fair, the amounts might appear different, which could lead to misunderstandings.
5. ****Signaling****: Sending a moderate to high amount signals trust and cooperation, which might encourage reciprocity.

Common strategies in trust games:

- Send half (\$2.50) as a moderate trust signal
- Send all (\$5) to maximize joint payoff and signal full trust
- Send a smaller amount (\$1-2) to test the waters

- Send nothing (\$0) to avoid risk entirely

Given this is round 1 and I want to establish a cooperative tone while not being completely naive, I think sending a significant but not maximum amount makes sense.

Research on trust games suggests most people send positive amounts initially, and receivers typically reciprocate. The 3x multiplier creates substantial mutual benefit potential—even modest returns from the receiver would leave us both better off than the starting position. I'll send \$3 as a balanced signal of trust and good faith without overexposing myself....

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 870,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.91 sent, it became \$8.73, returned \$3.5, visible payoff \$5.23.

Your total visible earnings across all rounds are \$5.5600000000000005.

This round, you are the RECEIVER against your current co-player. They sent you \$2.77, so you received \$8.31. How much do you return? (0-8.31)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

****Round 1 analysis:****

- I sent \$3 (they saw \$2.91 due to noise)
- They received \$8.73 and returned \$3.5 (about 40% return rate)
- This was a cooperative response

****Current situation (Round 2):****

- They sent me \$2.77 (with noise)
- I received \$8.31
- They showed good faith cooperation in Round 1

I'll reciprocate their cooperative behavior with a similar return rate:

```
'''json
{"return": 3.5}
'''
```

This gives them a total of approximately \$5.73 and leaves me with \$4.81, creating mutual benefit from our cooperation.

reasoning:

Let me analyze this situation:

Round 1 (just completed):

- I was the SENDER and sent \$3
- Due to communication noise, my co-player saw \$2.91 (close to \$3)
- It became \$8.73 for them (should be \$8.73 if 2.91×3)
- They returned \$3.5
- Their visible payoff: \$5.23 (they kept $8.73 - 3.5 = 5.23$)
- My payoff: $5 - 3 + 3.5 = 5.56$

Now Round 2:

- I am the RECEIVER
- My co-player sent me an amount, but I see \$2.77 (with noise)
- I received \$8.31 (which is 2.77×3)
- I need to decide how much to return

Analysis:

- In Round 1, my co-player returned \$3.5 out of \$8.73, which is about 40% return rate

- They kept \$5.23 for themselves
- I sent them \$3 (they saw \$2.91), and I ended up with \$5.56 total
- So I gained \$0.56 from the transaction

Now they sent me approximately \$2.77 (actual amount might be slightly different due to noise).

If I apply a similar return rate (around 40%), that would be: $\$8.31 \times 0.4 = \3.32

But let me think about fairness:

- They sent around \$2.77, leaving them with about \$2.23
- I received \$8.31
- A fair split might give them back enough so they end up with a reasonable amount

If I return \$3.5 (same as they returned to me):

- They end up with: $\$5 - \$2.77 + \$3.5 = \5.73
- I end up with: $\$8.31 - \$3.5 = \$4.81$

Returning \$4 would give them \$6.23 while I get \$4.31. Returning \$3 would leave me with \$5.31 and them with \$5.23. Since they gave me roughly \$2.77 and returned about 40% of what they received, matching that reciprocal gesture with a \$3.5 return seems appropriate....

usage:

```
{
  "input_tokens": 784,
  "output_tokens": 826,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.91 sent, it became \$8.73, returned \$3.5, visible payoff \$5.23.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.9 returned, visible payoff \$4.9.

Your total visible earnings across all rounds are \$10.370000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at our history:

```
**Round 1:** I sent $3, they returned $3.5 - cooperative
**Round 2:** They sent $3, I returned $3.5 (they saw $2.9) - cooperative
```

We've established a mutually beneficial pattern where we both send moderate amounts and return fair shares. I'll continue this cooperation:

```
'''json
{"send": 3}
'''
```

This maintains trust and keeps our successful collaborative pattern going.

Reasoning:

Let me analyze the history:

```
**Round 1:**
- I sent $3 (they saw $2.91 due to noise)
- They received $8.73 and returned $3.5
- I ended up with $2 + $3.5 = $5.5

**Round 2:**
- They sent $3.0 (I saw $2.77 due to noise, actual was $3.0)
- It became $9.0 for me
- I returned $3.5
- They saw $2.9 returned (due to noise)
- Their visible payoff was $4.9 ($5 - $3 + $2.9)
- I kept $9.0 - $3.5 = $5.5
```

My total visible earnings: \$10.37 which matches \$5.5 + \$4.81 approximately (there's some rounding)

So the pattern so far:

1. Both of us have been sending moderate amounts (\$3)
2. Both of us have been returning reasonable amounts (\$3.5)
3. This has been mutually beneficial

The relationship seems cooperative. They sent me \$3 in Round 2, which is the same amount I sent in Round 1. They saw I returned \$2.9 (actual \$3.5), which was a fair return.

I should continue this cooperative pattern. Sending \$3 again seems appropriate to maintain the reciprocal relationship.

```
Usage: {  
  "input_tokens": 1128,  
  "output_tokens": 446,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_4 - Receiver - dyad_2

Timestamp: 2026-08-12T08:09:50

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_2",  
  "move_index": 3,  
  "opponent_id": "Agent_3",  
  "pairing": {  
    "agents": [  
      "Agent_3",  
      "Agent_4"  
    ],  
    "investor": "Agent_3",  
    "roles": {  
      "Agent_3": "investor",  
      "Agent_4": "trustee"  
    },  
    "trustee": "Agent_4"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 3,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN

- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.91, that is 58.2% of their total endowment. You received \$8.73. How much do you return to the sender? (0-8.73)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.91 (58.2% of their endowment)
- I received \$8.73 after the 3x multiplication
- The sender currently has \$2.09 remaining

To reciprocate their trust and cooperation:

- If I return nothing, the sender ends with \$2.09 (worse than keeping everything)
- If I return \$3.50, the sender ends with \$5.59 and I keep \$5.23
- This rewards their trust while maintaining fairness

I'll return a meaningful amount to encourage continued cooperation.

```
{'return': 3.5}
```

reasoning:

Let me analyze this situation:

1. As the RECEIVER, the sender sent me \$2.91 (which is 58.2% of their \$5 endowment)
2. This amount was multiplied by 3x, so I received \$8.73
3. I need to decide how much to return (between \$0 and \$8.73)

Key considerations:

- The sender was quite generous, sending nearly 60% of their endowment
- This shows trust and cooperation
- In trust games, reciprocity is important for maintaining cooperation
- The sender's final payoff will be: \$5 - \$2.91 + whatever I return = \$2.09 + return
- My final payoff will be: \$8.73 - whatever I return

Common approaches in trust games:

1. Split the gains equally - the sender gave up \$2.91 but created \$8.73. The "gain" is \$8.73 - \$2.91 = \$5.82. Splitting this equally would mean returning about \$2.91
2. Give the sender back what they sent - return \$2.91
3. Give the sender a good return on their investment - perhaps return more than they sent
4. Equal final payoffs - try to make both players end up with similar amounts

Let me calculate some scenarios:

If I return \$2.91:

- Sender gets: \$2.09 + \$2.91 = \$5.00
- I get: \$8.73 - \$2.91 = \$5.82

- This is fair-ish, both doing okay

If I return \$3.50:

- Sender gets: $\$2.09 + \$3.50 = \$5.59$
- I get: $\$8.73 - \$3.50 = \$5.23$
- This balances outcomes more evenly

If I return \$4.00:

- Sender gets: $\$2.09 + \$4.00 = \$6.09$
- I get: $\$8.73 - \$4.00 = \$4.73$
- The sender benefits more, which seems appropriate given they initiated the cooperation

For truly equal final payoffs, I'd need to return \$3.32, which would give us both \$5.41 each. The sender demonstrated substantial trust by sending most of their endowment, so I should reciprocate meaningfully to reinforce that cooperation...

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 942,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.56 returned, visible payoff \$5.5600000000000005.

Your total visible earnings across all rounds are \$5.23.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the history:

- My co-player sent \$3.0 (60% of their endowment) in Round 1, showing strong trust
- They saw a return of \$3.56 (39.6% of what they sent as receiver)
- This demonstrates cooperative behavior

To reciprocate their trust and maintain cooperation, I'll send a similar proportion:

```
{'send': 3.0}
```

reasoning:

Let me analyze this situation:

Round 1:

- I was the RECEIVER
- The sender sent \$2.91 (58.2% of their endowment)
- I received \$8.73
- I returned \$3.50
- I kept \$5.23

Now in Round 2:

- I'm the SENDER
- I'm playing against the same co-player from Round 1
- The visible history shows that in Round 1, from my co-player's perspective:
 - They sent \$3.0 (as SENDER)
 - It became \$9.0
 - They saw \$3.56 returned
 - Their visible payoff was \$5.56

Wait, there's a discrepancy here. The problem says there's communication noise. Let me understand:

From my perspective in Round 1:

- Sender sent \$2.91
- I received \$8.73

- I returned \$3.50

From my co-player's perspective in Round 1:

- They sent \$3.0
- It became \$9.0
- They saw \$3.56 returned

So the noise affected what they saw vs what actually happened. They sent \$3.0 but I saw \$2.91. I returned \$3.50 but they saw \$3.56.

Now I'm the sender in Round 2. My co-player was cooperative in Round 1 - they sent 60% of their endowment (\$3.0 out of \$5). They saw a return of \$3.56 out of \$9.0, which is about 39.6% of what I received.

Given that:

1. My co-player showed strong trust by sending 60% in Round 1
2. They saw a decent return (39.6%)
3. I want to maintain cooperation and build trust

I should send a similar proportion to what they sent. They sent 60%, so I should send around \$3.0 as well to match their level of trust and cooperation.

...

usage:

```
{
  "input_tokens": 814,
  "output_tokens": 568,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.56 returned, visible payoff \$5.5600000000000005.
- Round 2 against you, as RECEIVER: saw \$2.77 sent, it became \$8.31, returned \$3.5, visible payoff \$4.8100000000000005.

Your total visible earnings across all rounds are \$10.13.

This round, you are the RECEIVER against your current co-player. They sent you \$3.52, so you received \$10.56. How much do you return? (0-10.56)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the pattern:

- **Round 1**:
- **Round 2**:
- **Round 3**:

They're being very cooperative and increasing their trust level. I should reciprocate appropriately.

Returning \$4.5:

- Sender ends with: \$5 - \$3.52 + \$4.5 = \$5.98
- I keep: \$10.56 - \$4.5 = \$6.06

This rewards their high trust while maintaining fairness.

{'return': 4.5}

Reasoning:

Let me analyze this situation:

Round 1 (I was RECEIVER):

- Sender sent \$2.91 (58.2%)
- I received \$8.73
- I returned \$3.50
- My payoff: \$5.23

Round 2 (I was SENDER):

- I sent \$3.0
- It became \$9.0
- According to visible history, my co-player saw \$2.77 sent (there's noise!)
- They returned \$3.5
- My visible payoff from Round 2: I sent \$3, got back \$3.5, so $\$2 + \$3.5 = \$5.5$
- Total earnings so far: $\$5.23 + \$5.5 = \$10.73$... but it says \$10.13

Wait, let me recalculate. The visible history says:

- Round 2 as RECEIVER: saw \$2.77 sent, it became \$8.31, returned \$3.5, visible payoff \$4.81

This is from my co-player's perspective. They saw \$2.77 sent (due to noise), it became \$8.31, they returned \$3.5, keeping \$4.81.

My actual earnings from Round 2:

- I sent \$3.0
- I kept \$2.0
- I received back... well, if they returned \$3.5 based on receiving \$8.31, then I should have received \$3.5 back
- So my payoff: $\$2.0 + \$3.5 = \$5.5$

Total: $\$5.23 + \$5.5 = \$10.73$

But it says my total visible earnings are \$10.13. Hmm, there might be noise in what I received back too.

Anyway, for Round 3:

- I'm RECEIVER
- They sent \$3.52 (70.4% of their endowment)
- I received \$10.56

My partner's been increasingly generous across rounds-first 58.2%, now 70.4%-showing growing confidence. They returned roughly 40-42% in the previous round, so I should match that reciprocal level. Returning between \$4.22-\$4.75 would maintain the cooperative dynamic we've established....

```
Usage: {
  "input_tokens": 1104,
  "output_tokens": 897,
  "reasoning_tokens": 0
}
```

GAME - Agent_5 - Sender - dyad_3

Timestamp: 2026-08-12T08:10:04

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_6",
  "pairing": {
    "agents": [
      "Agent_5",
      "Agent_6"
    ],
    "investor": "Agent_5",
    "roles": {
      "Agent_5": "investor",
      "Agent_6": "trustee"
    },
    "trustee": "Agent_6"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 3,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system
content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round situation, I need to balance potential gains from cooperation against the uncertainty of how my co-player will respond.

****My reasoning:****

- Sending money creates value (3x multiplier), so there's potential for mutual benefit
- This is our first interaction, so I want to signal willingness to cooperate
- But I also want to maintain some security given the uncertainty
- The communication noise adds another layer of unpredictability

I'll send a moderate amount that demonstrates trust while keeping some reserve.

```
'''json
{"send": 3}
'''
```

This sends \$3 to my co-player, who will receive \$9. I'll keep \$2 initially, and hope they reciprocate with a fair return that benefits us both.

reasoning:

This is the first round of a trust game where I'm the sender. Let me think about the strategic situation:

1. I have \$5 to start with
2. Whatever I send gets tripled when it reaches the receiver
3. The receiver then decides how much to return to me
4. There's communication noise - amounts may appear different than what was actually sent

Key considerations:

- This is a multi-round game with multiple agents
- Pairings are randomized each round
- There's a tension between trust/cooperation and self-interest
- If I send nothing, I keep \$5 but there's no opportunity for mutual gain
- If I send everything (\$5), the receiver gets \$15, and if they're fair/cooperative, they might return a good amount
- The communication noise adds uncertainty

In standard trust game experiments, sending some amount is generally beneficial because:

- It creates surplus value (3x multiplier)
- It signals trust and can encourage reciprocity
- Over multiple rounds, cooperative behavior can be rewarding

First round considerations:

- I don't know this particular co-player yet
- This is the first interaction, so I might want to signal willingness to cooperate
- But I also don't want to be too naive

A moderate approach might be to send \$3 or \$4:

- Sending \$3 means receiver gets \$9, I keep \$2. If they return \$4-5, we both do well
- Sending \$4 means receiver gets \$12, I keep \$1. If they return \$6, we both get \$6
- Sending \$5 means receiver gets \$15, I keep \$0. If they return \$7-8, I do well

Given this is round 1 and I want to establish a cooperative tone while not being too vulnerable, I'll send \$3. This shows trust but maintains some security.

Actually, let me reconsider. In trust games, the classic finding is that people often send about half their endowment on average. Sending \$2.50 or \$3 seems reasonable.

...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 703,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.66 sent, it became \$7.98, returned \$4.0, visible payoff \$3.9800000000000004.

Your total visible earnings across all rounds are \$5.13.

This round, you are the RECEIVER against your current co-player. They sent you \$2.97, so you received \$8.91. How much do you return? (0-8.91)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the situation:

****What happened in Round 1:****

- My co-player received \$7.98 (my \$3 send with noise)
- They returned \$4.0, which is about 50% - a fair reciprocal move
- This shows they're a cooperative player

****Current Round 2:****

- They sent me \$2.97 (?60% of their endowment) - showing significant trust
- I received \$8.91
- They kept \$2.03

****My decision:****

Since they demonstrated fair reciprocity (50% return) in Round 1 and have now shown strong

trust by sending most of their endowment, I should reciprocate similarly. Returning about 50% maintains the cooperative relationship while ensuring we both benefit.

Returning \$4.50 means:

- They end with: $\$2.03 + \$4.50 = \$6.53$
- I end with: $\$8.91 - \$4.50 = \$4.41$

This is fair and encourages continued cooperation.

```
'''json
{"return": 4.50}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap:

- I sent \$3 (but due to communication noise, my co-player saw \$2.66)
- They received \$7.98 (which should be $\$2.66 \times 3$)
- They returned \$4.0
- My visible payoff was \$5.13 (which would be $\$5 - \$3 + \$3.13$... but there's noise)
- Their visible payoff was \$3.98

Now in Round 2:

- My co-player is the SENDER
- They sent me \$2.97 (with noise, they might have intended something slightly different)
- I received \$8.91 (which is $\$2.97 \times 3$)
- I need to decide how much to return

Analysis of their behavior:

- In Round 1, they returned \$4.0 out of \$7.98 received, which is about 50%
- This is a reasonably fair/generous return
- Now they've sent me \$2.97 out of their \$5, which is about 59% - quite trusting

Given that:

1. They showed good reciprocity in Round 1 (returning 50% of what they received)
2. They've now shown significant trust by sending me almost 60% of their endowment
3. We might continue playing together in future rounds
4. Fair reciprocity would encourage continued cooperation

I should return a similar proportion or slightly more to maintain/build trust.

They received \$8.91 from my send. A 50% return would be \$4.455.

But let me think about the overall economics:

- They started with \$5
- They sent \$2.97, keeping \$2.03
- I received \$8.91
- If I return \$4.50:
 - They end with: $\$2.03 + \$4.50 = \$6.53$

...

usage:

```
{
  "input_tokens": 816,
  "output_tokens": 863,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.66 sent, it became \$7.98, returned \$4.0, visible payoff \$3.9800000000000004.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.03 returned, visible payoff \$7.03.

Your total visible earnings across all rounds are \$9.54.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at our history together:

****Round 1:****

- I sent, they received and returned ~50% - fair cooperation

****Round 2:****

- They sent \$3 (60% of endowment) - high trust
- I returned \$4.50, they saw \$5.03 (noise) - generous reciprocity
- Their payoff was strong (\$7.03)

****Analysis:****

- This co-player has proven consistently cooperative
- They demonstrated trust by sending \$3 in Round 2
- They've reciprocated fairly (50% return rate)
- Our relationship is building mutual trust

****My decision:****

I should maintain this cooperative dynamic. Sending \$3 again signals:

- Continued trust in our partnership
- Consistency and reliability
- Faith they'll reciprocate fairly as before

```
'''json
{"send": 3}
'''
```

Reasoning:

Let me analyze the visible history:

Round 1:

- I was SENDER, sent \$3
- They received \$7.98 (with noise, actual was probably \$9)
- They returned \$4.0 (about 50%)
- My payoff was \$5 - \$3 + \$4 = \$6
- Visible shows \$5.13, which might include noise

Round 2:

- I was RECEIVER
- They sent \$3.0 (actual, before noise)
- I received \$8.91 (actually \$9 with noise)
- I returned \$4.50
- They saw \$5.03 returned (with noise, I actually returned \$4.50)
- Their visible payoff \$7.03 = \$5 - \$3 + \$5.03

My total visible earnings: \$9.54

Round 3 (current):

- I'm SENDER again
- Same co-player from rounds 1 and 2

This co-player has been very cooperative:

1. Returned 50% in Round 1 when they received
2. Sent \$3 (60%) in Round 2, showing trust
3. They saw me return \$5.03 (with noise), which they probably perceive as generous (56% return)

This is building a strong cooperative relationship. They've been trustworthy and fair. I should continue to send a substantial amount to maintain this cooperation.

Given their consistent cooperation and the fact they sent \$3 in Round 2, I should send at least the same amount, if not more, to signal continued trust and cooperation.

I'll send \$3 again, showing consistency and trust.

```
Usage: {
  "input_tokens": 1250,
  "output_tokens": 587,
  "reasoning_tokens": 0
}
```

GAME - Agent_6 - Receiver - dyad_3

Timestamp: 2026-08-12T08:10:16

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_5",
  "pairing": {
    "agents": [
      "Agent_5",
      "Agent_6"
    ],
    "investor": "Agent_5",
    "roles": {
      "Agent_5": "investor",
      "Agent_6": "trustee"
    },
    "trustee": "Agent_6"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 3,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.

- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.66, that is 53.2% of their total endowment. You received \$7.98. How much do you return to the sender? (0-7.98)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.66 (53.2% of their endowment)
- I received \$7.98 after the 3x multiplication
- The sender currently has \$2.34 remaining from their original \$5

To reward their trust and encourage cooperation, I should return enough so they profit from their decision to send. If I return \$4.00:

- Sender ends with: $\$2.34 + \$4.00 = \$6.34$ (a gain of \$1.34)
- I keep: $\$7.98 - \$4.00 = \$3.98$

This seems fair - both of us benefit from the cooperation, and the sender's trust is rewarded.

```
{"return": 4.00}
```

reasoning:

Let me think about this carefully.

I'm the RECEIVER in Round 1.

- The sender sent \$2.66 (which is 53.2% of their \$5 endowment)
- This amount was multiplied by 3x, so I received $\$2.66 \times 3 = \7.98
- Now I need to decide how much to return (between \$0 and \$7.98)

Key considerations:

1. The sender sent more than half their endowment (53.2%), showing substantial trust
2. If I return nothing, I keep \$7.98 and the sender keeps $\$5 - \$2.66 = \$2.34$
3. If I return some amount, I should aim to be fair and encourage cooperation

A reasonable approach would be to:

- Ensure the sender ends up with more than they started with (more than \$5)
- Keep some profit myself
- Reward their trust to encourage future cooperation

Let me calculate:

- Sender currently has: $\$5 - \$2.66 = \$2.34$
- I have: \$7.98

If I return \$3.00:

- Sender ends with: $\$2.34 + \$3.00 = \$5.34$ (slight gain)
- I keep: $\$7.98 - \$3.00 = \$4.98$

If I return \$4.00:

- Sender ends with: $\$2.34 + \$4.00 = \$6.34$ (good gain)
- I keep: $\$7.98 - \$4.00 = \$3.98$

If I return \$3.50:

- Sender ends with: $\$2.34 + \$3.50 = \$5.84$
- I keep: $\$7.98 - \$3.50 = \$4.48$

...

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 699,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.13 returned, visible payoff \$5.13.

Your total visible earnings across all rounds are \$3.9800000000000004.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the visible history:

****Round 1 recap (from their perspective):****

- My co-player sent \$3.00 (60% of their endowment) - quite generous
- They saw \$3.13 returned
- Their visible payoff was \$5.13

My co-player showed significant trust by sending 60% of their endowment. Even though there's communication noise affecting what we each see, I should reciprocate this trust.

I'll match their generosity by sending the same amount they sent: \$3.00 (60% of my endowment). This demonstrates:

- Reciprocal trust
- Willingness to cooperate
- Fair play matching their level of generosity

{"send": 3.00}

reasoning:

Let me analyze this situation:

Round 1 (just played):

- I was the RECEIVER
- Sender sent \$2.66 (53.2%)
- I received \$7.98
- I returned \$4.00
- I kept \$3.98

But now I see the "visible history" which shows:

- Round 1 against me, as SENDER: sent \$3.0, became \$9.0, saw \$3.13 returned, visible payoff \$5.13

This is interesting - there's communication noise!

- The sender actually sent \$3.00 (not \$2.66 as I saw)
- It became \$9.00 (not \$7.98 as I saw)
- They saw \$3.13 returned (not \$4.00 as I sent)
- Their visible payoff was \$5.13 (which is \$5 - \$3 + \$3.13)

So due to noise:

- I saw them send \$2.66, they actually sent \$3.00
- I saw I received \$7.98, I actually received \$9.00
- I returned \$4.00, they saw \$3.13

Now in Round 2, I'm the SENDER and playing against the same person.

From their perspective:

- They sent \$3.00 (60% of endowment - quite generous)
- They got back \$3.13 (34.8% of what I received)
- Their payoff was \$5.13

From my actual perspective (what really happened):

- They sent \$3.00
- I received \$9.00
- I returned \$4.00 (44.4% of what I received)
- But they only saw \$3.13

The co-player was quite generous in Round 1 (sent 60%). From their visible perspective, they got a modest return (34.8% of the tripled amount).

...

usage:

{

```
"input_tokens": 845,  
"output_tokens": 754,  
"reasoning_tokens": 0  
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.13 returned, visible payoff \$5.13.
- Round 2 against you, as RECEIVER: saw \$2.97 sent, it became \$8.91, returned \$4.5, visible payoff \$4.41.

Your total visible earnings across all rounds are \$11.010000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$3.23, so you received \$9.69. How much do you return? (0-9.69)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the visible history:

****Round 1:**** My co-player sent \$3.00 (60%)

****Round 2:**** They returned \$4.50 out of \$8.91 (50.5%) when they were receiver

****Round 3:**** They just sent \$3.23 (64.6%) - even more generous!

My co-player has shown increasing trust and cooperation. They've been very generous, and when they were receiver in Round 2, they returned about 50% of what they received.

I should reciprocate their strong cooperation. If I return \$5.00:

- Sender ends with: \$1.77 + \$5.00 = \$6.77 (profit of \$1.77)
- I keep: \$9.69 - \$5.00 = \$4.69

This returns about 52% of what I received, matching their generosity from Round 2 and rewarding their increased trust.

```
{"return": 5.00}
```

Reasoning:

Let me analyze the visible history:

****Round 1 (I was RECEIVER):****

- My co-player sent \$3.00 (60%)
- They saw \$3.13 returned
- Their visible payoff: \$5.13

****Round 2 (I was SENDER):****

- I sent \$3.00
- They saw \$2.97 sent (noise), which became \$8.91
- They returned \$4.50
- Their visible payoff: \$4.41

****Round 3 (I am RECEIVER):****

- They sent \$3.23 (64.6% of their endowment)
- I received \$9.69
- They have \$1.77 remaining

My co-player has been very cooperative:

- Round 1: Sent 60% (\$3.00)
- Round 2: Returned \$4.50 out of \$8.91 received (50.5%)
- Round 3: Sent 64.6% (\$3.23) - even more generous!

They've shown increasing trust and generosity. I should reciprocate strongly.

Let me think about what to return:

If I return \$5.00:

- Sender ends with: \$1.77 + \$5.00 = \$6.77 (profit of \$1.77)
- I keep: \$9.69 - \$5.00 = \$4.69

If I return \$5.50:

- Sender ends with: $\$1.77 + \$5.50 = \$7.27$ (profit of $\$2.27$)
- I keep: $\$9.69 - \$5.50 = \$4.19$

Given their increased generosity (sending 64.6% vs 60% in round 1), I should return proportionally more. They returned $\$4.50$ out of $\$8.91$ (50.5%) when they were receiver. I should aim to return around 50-55% to match their generosity.

...

```
Usage: {
  "input_tokens": 1207,
  "output_tokens": 754,
  "reasoning_tokens": 0
}
```

GAME - Agent_7 - Sender - dyad_4

Timestamp: 2026-08-12T08:10:29

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_8",
  "pairing": {
    "agents": [
      "Agent_7",
      "Agent_8"
    ],
    "investor": "Agent_7",
    "roles": {
      "Agent_7": "investor",
      "Agent_8": "trustee"
    },
    "trustee": "Agent_8"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 3,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: $\$5 - \text{sent} + \text{returned}$
- Receiver keeps: $\text{received} - \text{returned}$
- Response format: `{'send': <amount>}`

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: $\$5 - \text{sent} + \text{returned}$
- You keep: $\text{received} - \text{returned}$
- Response format: `{'return': <amount>}`

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to consider:

1. ****The multiplier effect****: Any amount I send gets tripled, creating potential mutual gain
2. ****No prior history****: This is my first interaction, so I have no information about this co-player's behavior
3. ****Future rounds****: Even though pairings are randomized, establishing cooperative norms early could benefit the group overall
4. ****Risk vs. reward****: Sending more creates more surplus, but requires trusting the receiver to return a fair amount

I'll take a moderately cooperative approach. Sending \$3 creates \$9 for the receiver - enough surplus that we can both benefit if they reciprocate fairly, while not risking my entire endowment.

If they return around half of what they receive (\$4-5), we both end up better off than if no transaction occurred.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a multi-agent trust game experiment. I'm the SENDER with \$5.

Let me think about the strategic aspects:

1. If I send money, it gets tripled, creating surplus value
2. The receiver then decides how much to return
3. This is a classic trust game scenario

Key considerations:

- This is round 1, so I have no history with this particular co-player
- There are multiple rounds, so reputation/repeated interaction matters
- However, pairings are randomized each round, so I won't necessarily play the same person again
- The game has communication noise, which might affect trust

Strategic thinking:

- Sending \$0 guarantees I keep \$5, but creates no surplus
- Sending \$5 creates maximum surplus (\$15 total for receiver), but requires trust
- A middle ground (e.g., \$2-3) balances risk and potential gain

In classic trust game experiments, people often:

1. Send moderate to high amounts in early rounds to signal cooperation
2. The amount sent and returned establishes baseline trust levels
3. With randomized pairings, building a reputation as trustworthy matters for future rounds

Game theory perspective:

- If this were truly one-shot with rational self-interested players, the receiver should return \$0 (backward induction)
- Knowing this, I should send \$0
- However, real humans often deviate from this due to fairness norms, reciprocity, and reputation concerns

Since:

- This is multi-round with multiple agents
- There may be reputation effects even with randomized pairings
- Starting cooperatively can establish good norms
- The 3x multiplier creates real gains from cooperation

I'll send a moderate-high amount to signal cooperation while not being completely vulnerable. I'll send \$3.

This creates \$9 for the receiver, and if they return even just \$3, we both end up with \$5 (same as no transaction), but if they return more (say \$4-5), we both gain. ...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 671,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.17 sent, it became \$9.51, returned \$5.0, visible payoff \$4.51.

Your total visible earnings across all rounds are \$7.37.

This round, you are the RECEIVER against your current co-player. They sent you \$2.35, so you received \$7.0500000000000001. How much do you return? (0-7.0500000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

****Round 1 analysis:****

- They saw I sent \$3.17 (with noise), received \$9.51
- They returned \$5.00, which is about 52.6% of what they received
- This was quite generous and cooperative

****Current situation:****

- They sent me \$2.35, I received \$7.05
- They showed good faith last round, so I should reciprocate

Following a similar reciprocity rate (around 52-53% return):

- 53% of \$7.05 ? \$3.74

This ensures:

- They profit from sending (get back more than they sent)
- I also benefit from the transaction
- We maintain cooperative momentum

```
'''json
{"return": 3.75}
'''
```

reasoning:

Let me analyze this situation:

****What happened in Round 1:****

- I sent \$3, but they saw \$3.17 (communication noise)
- They received \$9.51 (3.17×3)
- They returned \$5.0
- Their visible payoff was $\$9.51 - \$5.0 = \$4.51$
- My payoff was $\$5 - \$3 + \$5 = \7 (but visible earnings show \$7.37, maybe due to noise)

****Current Round 2:****

- They sent me \$2.35 (actual), which became \$7.05
- I need to decide how much to return

****Analysis of their behavior:****

In Round 1, they returned \$5 out of \$9.51 received, which is about 52.6%. This is a fairly generous return - they kept \$4.51 and I got back \$5, so I had a net gain of \$2 from the transaction.

****What should I do?****

They showed cooperative behavior by returning a good amount. Now they've sent me \$2.35, creating \$7.05.

If I apply a similar reciprocity ratio:

- They returned about 52.6% of what they received
- I received \$7.05, so 52.6% would be about \$3.71

However, I should also consider:

- They sent \$2.35 this round (less than the ~\$3.17 they saw me send)
- But they were generous with their return last round
- I want to maintain cooperation

A fair approach might be to:

1. Return enough that they profit from sending
2. Keep enough that I also profit
3. Signal continued cooperation

If they sent \$2.35 and expect around a 50% return rate:

- They might expect back around \$3-4

Let me think about payoffs:

- They started with \$5
- They sent \$2.35
- If I return \$3.5, they end with \$6.15 and I keep \$3.55
- If I return \$4, they end with \$6.65 and I keep \$3.05
- If I return \$3, they end with \$5.65 and I keep \$4.05

Since they matched my generosity last round with a 52.6% return, I should reciprocate similarly to maintain the cooperative dynamic...

usage:

```
{
  "input_tokens": 852,
  "output_tokens": 862,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.17 sent, it became \$9.51, returned \$5.0, visible payoff \$4.51.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.71 returned, visible payoff \$5.71.

Your total visible earnings across all rounds are \$10.670000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at our history together:

****Round 1:****

- I sent \$3, they received ~\$9.51, returned \$5.00 (very generous ~53% return)

****Round 2:****

- They sent \$2.35, I received \$7.05, returned \$3.75
- They saw \$3.71 returned due to noise (~41% visible return rate)

****Analysis:****

- This co-player was quite generous in Round 1
- They sent a bit less in Round 2, possibly because they saw a lower return rate from me (due to noise)
- Overall, they've demonstrated cooperative behavior

****My strategy:****

I'll send \$3 again to:

- Match the cooperative level from Round 1
- Show consistency in my approach
- Signal that I'm willing to trust despite the noise in Round 2

```
'''json
{"send": 3}
'''
```

Reasoning:

Let me analyze the history with this co-player:

****Round 1 (I was SENDER):****

- I sent \$3 (they saw \$3.17 due to noise)
- They received \$9.51 and returned \$5.00
- This was generous - about 52.6% return rate

****Round 2 (I was RECEIVER):****

- They sent \$2.35 (actual), I received \$7.05
- I returned \$3.75
- They sent \$3.0 actual, became \$9.0, they saw \$3.71 returned
- So they saw a return rate of about 41.2% ($3.71/9.0$)
- This is less than what they gave me in Round 1

****Pattern:****

- This co-player was very cooperative in Round 1
- In Round 2, they sent a bit less (\$2.35 vs my \$3)
- They saw me return less proportionally than they did (41.2% vs 52.6%)

****Current decision:****

I'm the sender again. This co-player has shown:

1. Initial high cooperation (52.6% return)
2. Slightly reduced sending in Round 2 (\$2.35 vs \$3)
3. They may perceive me as less generous than they were (due to noise showing only \$3.71 returned when I sent \$3.75)

I should:

- Maintain cooperation since they were generous in Round 1
- Send a similar amount to Round 1 to show consistency
- Perhaps send \$3 again to match their Round 2 amount

Given they've been generally cooperative and we're building a pattern, I'll send \$3 again.

```
Usage: {
  "input_tokens": 1212,
  "output_tokens": 605,
  "reasoning_tokens": 0
}
```

GAME - Agent_8 - Receiver - dyad_4

Timestamp: 2026-08-12T08:10:46

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_7",
  "pairing": {
    "agents": [
      "Agent_7",
      "Agent_8"
    ],
    "investor": "Agent_7",
    "roles": {
```

```

    "Agent_7": "investor",
    "Agent_8": "trustee"
  },
  "trustee": "Agent_8"
},
"role": "trustee",
"role_label": "Receiver",
"round": 3,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.17, that is 63.4% of their total endowment. You received \$9.51. How much do you return to the sender? (0-9.51)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending 63.4% of their endowment (\$3.17)
- I received \$9.51 after the 3x multiplier
- The sender currently has \$5 - \$3.17 = \$1.83

The sender demonstrated considerable trust, so I want to reciprocate appropriately. If I return \$5.00:

- The sender would end with: $\$1.83 + \$5.00 = \$6.83$ (a profit of \$1.83)
- I would keep: $\$9.51 - \$5.00 = \$4.51$

This rewards their trust with a solid return while allowing us both to benefit from the co operation.

```
{"return": 5.00}
```

reasoning:

Let me think about this carefully.

I'm the RECEIVER in Round 1.

- The sender sent \$3.17 (which is 63.4% of their \$5 endowment)
- This was multiplied by 3x, so I received: $\$3.17 \times 3 = \9.51 ?
- I need to decide how much to return (between \$0 and \$9.51)

Key considerations:

1. The sender showed considerable trust by sending 63.4% of their endowment
2. If I return nothing, the sender ends up with: $\$5 - \$3.17 = \$1.83$, while I keep \$9.51
3. If I return a fair/reciprocal amount, what would that be?

A few approaches to consider:

- Equal split of total: Total is \$5 (sender's initial) + \$9.51 (what I received) = \$14.51. Equal split would be ~\$7.26 each. The sender currently has \$1.83, so I'd return $\$7.26 - \$1.83 = \$5.43$
- Return what maintains sender's initial investment value: Sender invested \$3.17. If I return \$3.17, sender gets back to \$5 total, and I keep \$6.34
- Return proportionally to what was sent: The sender sent 63.4% of their endowment. I could return a similar proportion of what I received: 63.4% of \$9.51 = ~\$6.03
- Social norm of reciprocity: Return enough to reward trust but keep some gains

Given that:

- The sender showed significant trust (63.4%)
- This is a multi-round game with different partners
- Building reputation for reciprocity could be valuable
- Note about communication noise suggests amounts might not be exact

I think a reasonable return would be around \$4-5, which:

- Acknowledges their generosity without giving away all my gains
- Leaves them better off than their starting position
- Demonstrates trustworthiness that could matter in future interactions with other partners
- Strikes a balance between reciprocity and self-interest

I'll return \$4.75. This gives the sender \$6.58 total (\$1.83 remaining plus my return), which meaningfully rewards their trust while I keep \$4.76....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1001,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.37 returned, visible payoff \$7.37.

Your total visible earnings across all rounds are \$4.51.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the history:

My co-player sent \$3.0 in Round 1 (60% of their endowment), showing significant trust. They saw \$5.37 returned, which gave them a good outcome.

Since they demonstrated trust and cooperation, I should reciprocate similarly. Sending around the same amount they sent (\$3.0) would be fair and maintain the cooperative relationship.

```
{"send": 3.0}
```

reasoning:

Let me analyze this situation:

1. In Round 1, I was the RECEIVER and the sender sent me \$3.17 (according to my view), which became \$9.51 for me. I returned \$5.00.
2. However, from my current co-player's perspective (shown in "Your current co-player's last 3 game(s)"):
 - They sent \$3.0 (not \$3.17 - communication noise!)
 - It became \$9.0 (not \$9.51 - more noise!)
 - They saw \$5.37 returned (not \$5.00 - even more noise!)
 - Their visible payoff was \$7.37

So there's noise in the communication. What they experienced:

- Started with \$5
- Sent \$3.0
- Had \$2.0 left
- Saw \$5.37 returned
- Ended with \$2.0 + \$5.37 = \$7.37

From their perspective, I (as the receiver in Round 1) returned about 60% of what I received (\$5.37 out of \$9.0).

3. Now I'm the SENDER in Round 2 against the same co-player.

My co-player showed trust by sending \$3.0 (from their perspective) and saw a decent return ratio. They might be willing to reciprocate trust.

I should consider:

- They sent about 60% of their endowment in Round 1
- They saw about 60% returned
- This suggests they're moderately trusting and cooperative

I should send a similar amount to show I'm also trustworthy and cooperative. Sending around \$3 would be consistent with their behavior and maintain the cooperative relationship.

Let me send \$3.00 to match their earlier send amount.

usage:

```
{
  "input_tokens": 837,
  "output_tokens": 523,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.37 returned, visible payoff \$7.37.
- Round 2 against you, as RECEIVER: saw \$2.35 sent, it became \$7.050000000000001, returned \$3.75, visible payoff \$3.3000000000000007.

Your total visible earnings across all rounds are \$10.219999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$2.22, so you received \$6.66. How much do you return? (0-6.66)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the situation:

Current round:

- Sender sent \$2.22 (44.4% of their endowment)
- I received \$6.66
- Sender currently has: $\$5 - \$2.22 = \$2.78$

History with this co-player:

- Round 1: They sent \$3.0, I returned money (they saw \$5.37)
- Round 2: They saw less sent from me due to noise (\$2.35 vs my actual \$3.0), returned \$3.75

They've reduced their sending, possibly due to noise affecting what they saw in Round 2. Still, they're showing some trust by sending \$2.22. I should reciprocate fairly.

Returning approximately 50% of what I received (\$3.33) would:

- Give sender: $\$2.78 + \$3.33 = \$6.11$ (profit of \$1.11)
- I keep: \$3.33

This maintains cooperation while being proportional to their send amount.

```
{"return": 3.33}
```

Reasoning:

Let me analyze the situation:

Current round:

- I'm the RECEIVER
- Sender sent \$2.22 (44.4% of their endowment)
- I received \$6.66 after 3x multiplier
- Sender currently has: $\$5 - \$2.22 = \$2.78$

Looking at history with this co-player:

- Round 1: They sent \$3.0 as SENDER, saw \$5.37 returned (I actually sent \$5.00, but due to noise they saw \$5.37)
- Round 2: I sent \$3.0 as SENDER (but they saw \$2.35 due to noise), they returned \$3.75

In Round 2, I sent \$3.0 which became \$9.0, and they returned \$3.75. That's about 42% return rate.

Now in Round 3:

- They sent \$2.22, which is less than what they sent in Round 1 (\$3.0)
- This might be because they saw only \$2.35 sent in Round 2 (due to noise) instead of my actual \$3.0

I should consider:

1. They're still showing some trust by sending \$2.22
2. They may have reduced their sending because of perceived lower trust from me (noise issue)
3. I should maintain cooperation

If I return around \$3.50-\$4.00:

- Return \$3.50: Sender gets $\$2.78 + \$3.50 = \$6.28$ (profit of \$1.28), I keep \$3.16
- Return \$4.00: Sender gets $\$2.78 + \$4.00 = \$6.78$ (profit of \$1.78), I keep \$2.66

Given they sent less this round but still showed trust, I'll return a reasonable amount that rewards their trust.

I'm noticing a pattern in the return rates across rounds. In Round 1, I returned about 53% of what I received, while in Round 2 they returned roughly 42%. Since they've sent less this time-likely reacting to the noise that made my previous send appear smaller-I should aim for a round 50% to maintain fairness and signal...

```
Usage: {
  "input_tokens": 1135,
  "output_tokens": 909,
  "reasoning_tokens": 0
}
```

ROUND 4

Roles: Agent_1=Receiver, Agent_2=Sender, Agent_3=Receiver, Agent_4=Sender, Agent_5=Receiver, Agent_6=Sender, Agent_7=Receiver, Agent_8=Sender

Pairings:

- dyad_1: Agent_2=Sender, Agent_1=Receiver
- dyad_2: Agent_4=Sender, Agent_3=Receiver

```

dyad_3: Agent_6=Sender, Agent_5=Receiver
dyad_4: Agent_8=Sender, Agent_7=Receiver
Parsed game decisions:
dyad_1: Agent_2 sent $3.5; received $10.5; Agent_1 returned $4.8; payoffs Agent_2=$6.3,
Agent_1=$5.7; noise/visible: sent visible=$3.21, received visible=$9.63, returned visible=$5.28
  Action validation: {"returned": {"amount": 4.8, "clamped": false, "max": 10.5, "min": 0, "raw":
  4.8}, "sent": {"amount": 3.5, "clamped": false, "max": 5, "min": 0, "raw": 3.5}}
dyad_2: Agent_4 sent $3; received $9; Agent_3 returned $3.5; payoffs Agent_4=$5.5, Agent_3=$5.5;
noise/visible: sent visible=$2.64, received visible=$7.92, returned visible=$3.96
  Action validation: {"returned": {"amount": 3.5, "clamped": false, "max": 9.0, "min": 0, "raw":
  3.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_3: Agent_6 sent $3; received $9; Agent_5 returned $5.9; payoffs Agent_6=$7.9, Agent_5=$3.1;
noise/visible: sent visible=$3.93, received visible=$11.79, returned visible=$6.63
  Action validation: {"returned": {"amount": 5.9, "clamped": false, "max": 9.0, "min": 0, "raw":
  5.9}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_8 sent $3; received $9; Agent_7 returned $4.2; payoffs Agent_8=$6.2, Agent_7=$4.8;
noise/visible: sent visible=$2.62, received visible=$7.86, returned visible=$3.31
  Action validation: {"returned": {"amount": 4.2, "clamped": false, "max": 9.0, "min": 0, "raw":
  4.2}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
Balances after round: {
  "Agent_1": 25.2,
  "Agent_2": 22.8,
  "Agent_3": 23.0,
  "Agent_4": 21.0,
  "Agent_5": 20.6,
  "Agent_6": 23.4,
  "Agent_7": 22.38,
  "Agent_8": 21.62
}
Visible balances after round: {
  "Agent_1": 22.93,
  "Agent_2": 24.65,
  "Agent_3": 20.57,
  "Agent_4": 22.150000000000002,
  "Agent_5": 23.22,
  "Agent_6": 24.33,
  "Agent_7": 20.630000000000003,
  "Agent_8": 18.86
}

```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

```

Timestamp: 2026-08-12T08:10:59
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 4,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.63, that is 72.6% of their total endowment. You received \$10.89. How much do you return to the sender? (0-10.89)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.63 (72.6% of their endowment)
- I received \$10.89 after the 3x multiplier
- A fair return should reward their trust while ensuring both of us benefit

The sender gave up \$3.63, keeping only \$1.37. Their generosity created \$10.89 total value for me. To reciprocate fairly and encourage future cooperation:

If I return \$5.00:

- Sender ends with: \$5 - \$3.63 + \$5.00 = \$6.37 (gain of \$1.37)
- I end with: \$10.89 - \$5.00 = \$5.89 (gain of \$5.89)

This seems like a fair split that rewards their trust while both of us benefit from the cooperation.

{"return": 5.00}

reasoning:

Let me analyze this situation carefully.

Given information:

- I'm the RECEIVER in Round 1
- The sender sent \$3.63 (which is 72.6% of their \$5 endowment)
- I received \$10.89 (which is $\$3.63 \times 3$)
- I need to decide how much to return (between \$0 and \$10.89)

Key considerations:

1. The sender was quite generous, sending 72.6% of their endowment
2. This shows trust and cooperation
3. There's mention of communication noise, so the actual amount sent might differ slightly from what I see
4. This is a multi-round game with 8 agents, so building reputation for cooperation could be beneficial
5. Fair reciprocation would be important to encourage future cooperation

Let me think about what would be a fair return:

- The sender gave up \$3.63 and kept \$1.37
- I received \$10.89
- If I want to be reciprocal and fair, I should consider:
 - The sender showed high trust (72.6% is a lot)
 - A common strategy is to return enough so both parties benefit
 - Returning about half (\$5.45) would give me \$5.44 and the sender would end up with $\$1.37 + \$5.45 = \$6.82$
 - This would mean the sender gains \$1.82 total and I gain \$5.44

Let me reconsider:

- If I return around \$4-5, that seems fair
- Returning \$5 would give:
 - Sender: $\$5 - \$3.63 + \$5 = \6.37 (net gain of \$1.37)
 - Me: $\$10.89 - \$5 = \$5.89$ (net gain of \$5.89)

....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 801,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.52 returned, visible payoff \$7.52.

Your total visible earnings across all rounds are \$5.8900000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the situation:

From the visible history, my co-player:

- Sent \$3.0 in Round 1 (60% of their endowment)
- Saw \$5.52 returned (61.3% of the \$9.0)
- Made a profit of \$2.52

This shows they were cooperative and returned a substantial amount. To reciprocate their trust and maintain cooperation, I should send a generous amount.

I'll send \$3.50 (70% of my endowment), showing strong trust and willingness to cooperate.

```
{"send": 3.50}
```

reasoning:

Let me analyze this situation:

Round 1 (from my perspective):

- I was the RECEIVER
- The sender sent \$3.63 (72.6% of endowment)
- I received \$10.89
- I returned \$5.00
- My payoff: $\$10.89 - \$5.00 = \$5.89$
- Sender's payoff: $\$5 - \$3.63 + \$5.00 = \6.37

But now I see the "visible history" which shows a different perspective:

- My co-player (who was the sender in Round 1) saw that they sent \$3.0
- It became \$9.0 (not \$10.89 as I received)
- They saw \$5.52 returned (not \$5.00 as I sent)
- Their visible payoff was \$7.52

This confirms there is communication noise in the game as mentioned in the instructions.

What actually happened vs what they saw:

- They actually sent \$3.63, but saw they sent \$3.0
- I received \$10.89, they saw it became \$9.0
- I returned \$5.00, they saw \$5.52 returned
- Their actual payoff was \$6.37, but they saw \$7.52

Now in Round 2, I'm the SENDER and they are the RECEIVER.

My co-player returned a substantial amount in Round 1 (they saw they returned \$5.52 out of \$9.0 received, which is 61.3%). This shows they are cooperative.

Given that:

1. They showed trust by being generous in Round 1
2. They appeared to return a fair amount (from their perspective, 61.3% of what they received)
3. I want to maintain cooperation and reciprocate their trust

I should send a generous amount. ...

usage:

```
{
  "input_tokens": 874,
  "output_tokens": 716,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.52 returned, visible payoff \$7.52.
- Round 2 against you, as RECEIVER: saw \$3.14 sent, it became \$9.42, returned \$4.5, visible payoff \$4.92.

Your total visible earnings across all rounds are \$12.45.

This round, you are the RECEIVER against your current co-player. They sent you \$3.64, so you received \$10.92. How much do you return? (0-10.92)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the visible history:

Round 1: My co-player sent \$3.0 (60%) and saw \$5.52 returned

Round 2: They returned \$4.5 when they received \$9.42 (47.8%)

Round 3: They sent \$3.64 (72.8% - even more generous!)

My co-player has been consistently cooperative and just showed increased trust by sending an even larger percentage. I should reciprocate fairly.

Returning \$5.50:

- Sender gets: $\$5 - \$3.64 + \$5.50 = \6.86 (profit of \$1.86)
- I get: $\$10.92 - \$5.50 = \$5.42$ (profit of \$5.42)

This ~50% return rate rewards their trust while ensuring mutual benefit.

```
{"return": 5.50}
```

reasoning:

Let me analyze the visible history:

Round 1: My co-player was SENDER

- They sent \$3.0 (60% of endowment)
- It became \$9.0
- They saw \$5.52 returned (61.3% return rate)
- Their visible payoff: \$7.52

Round 2: My co-player was RECEIVER

- They saw \$3.14 sent (I actually sent \$3.50 but there's noise)
- It became \$9.42
- They returned \$4.5 (47.8% of what they received)
- Their visible payoff: \$4.92

Now Round 3: I am RECEIVER

- They sent \$3.64 (72.8% of their endowment - very generous!)
- I received \$10.92
- I need to decide how much to return

Looking at the pattern:

- In Round 1, they sent 60% and I returned about 61% (assuming noise in what they saw)
- In Round 2, they received and returned 47.8%
- In Round 3, they sent 72.8% - showing even more trust

The co-player has been consistently cooperative. They sent a high amount both times they were sender (60% and 72.8%). When they were receiver, they returned a reasonable amount (47.8%).

To maintain reciprocity and reward their increased trust, I should return a fair amount.

If I return \$5.50:

- Sender ends with: $\$5 - \$3.64 + \$5.50 = \6.86 (gain of \$1.86)
- I end with: $\$10.92 - \$5.50 = \$5.42$ (gain of \$5.42)

This is about 50.4% return rate, which is reasonable and shows I'm reciprocating their trust while both of us profit.

...

usage:

```
{
  "input_tokens": 1200,
  "output_tokens": 712,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.52 returned, visible payoff \$7.52.
- Round 2 against you, as RECEIVER: saw \$3.14 sent, it became \$9.42, returned \$4.5, visible payoff \$4.92.
- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$4.66 returned, visible payoff \$5.66.

Your total visible earnings across all rounds are \$17.869999999999997.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the visible history:

Round 1: Co-player sent \$3.0 (60%), saw \$5.52 returned

Round 2: Co-player returned \$4.5 (47.8%)

Round 3: Co-player sent \$4.0 (80% - very generous!), saw \$4.66 returned

My co-player has been consistently cooperative, even increasing their trust in Round 3 by sending 80% of their endowment. While they saw a lower return percentage in Round 3, they've maintained a cooperative pattern overall.

I should reciprocate their trust and continue the cooperative relationship.

```
{"send": 3.50}
```

Reasoning:

Let me analyze the visible history:

Round 1: Co-player sent \$3.0 (60%), saw \$5.52 returned (61.3% of \$9.0)

Round 2: Co-player returned \$4.5 when they received \$9.42 (47.8%)

Round 3: Co-player sent \$4.0 (80%!), saw \$4.66 returned (38.8% of \$12.0)

This is concerning. In Round 3, my co-player sent even more (\$4.0, or 80%), but they only saw \$4.66 returned, which is only 38.8% of the \$12.0 they sent. That's a lower return rate than before.

Wait, let me check what I actually sent and returned:

- Round 2: I sent \$3.50 according to my response
- Round 3: I returned \$5.50 according to my response

But the visible history shows:

- Round 2: They saw \$3.14 sent (I sent \$3.50, but due to noise they saw \$3.14)
- Round 3: They saw \$4.66 returned (I returned \$5.50, but due to noise they saw \$4.66)

So there's communication noise! This is important.

From my co-player's perspective:

- They've been very generous (sending 60%, 80%)
- But the returns they're seeing are declining in percentage terms (61.3%, then 38.8%)
- They might be feeling like their generosity isn't being reciprocated

However, I know I actually returned \$5.50 in Round 3, which was generous. The noise just made it look like \$4.66 to them.

Still, this relationship has been cooperative overall. My co-player has consistently sent substantial amounts. Even though they saw a lower return in Round 3, they've maintained trust.

...

```
Usage: {  
  "input_tokens": 1626,  
  "output_tokens": 638,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T08:11:09

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_2",  
  "pairing": {  
    "agents": [  
      "Agent_2",  
      "Agent_1"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_2": "investor"  
    },  
    "trustee": "Agent_1"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 4,
```

```

"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to establish a cooperative approach while being somewhat cautious since I don't have any history with my current co-player.

Sending a moderate amount signals trust and gives us both the opportunity to benefit from the 3x multiplier, while not leaving me completely vulnerable.

I'll send **\$3**. This gives my co-player \$9 to work with, and if they return a fair share, we can both end up better off than our starting positions.

```

'''json
{"send": 3}
'''

```

reasoning:

This is the first round of a trust game. I'm the sender and I have \$5. I need to decide how much to send to the receiver.

Let me think about the strategic considerations:

1. If I send \$X, the receiver gets \$3X
2. The receiver then decides how much to return
3. I keep \$5 - X + (amount returned)
4. The receiver keeps \$3X - (amount returned)

This is a classic trust game. The optimal outcome for both of us would be if I send the full \$5, it gets tripled to \$15, and the receiver returns a fair share (say, at least \$7.50, which would give me \$7.50 and them \$7.50, both better off than starting positions).

However, there are risks:

- The receiver might not return anything or very little
- There's also mentioned "communication noise" - amounts may differ from what was actually sent/returned

Since this is round 1 and I have no history with this player:

- I could play it safe and send less
- Or I could signal trust and cooperation by sending more

In standard trust game experiments, people often send around \$2-\$3 in the first round as a moderate trust signal.

However, if I'm thinking about maximizing joint welfare and establishing cooperation, sending more could be beneficial. If the receiver is rational and cooperative, they should return enough to make it worthwhile for me.

Let me consider a middle-ground strategy: sending \$3 or \$4.

- If I send \$3, receiver gets \$9. If they return \$4-5, we both benefit
- If I send \$4, receiver gets \$12. If they return \$6, we both benefit significantly

Given this is a multi-round game with potentially different partners each time, establishing a pattern of trust and cooperation could be beneficial. I'll go with sending \$3 as a balanced approach - showing trust but not complete vulnerability.

Actually, let me reconsider. The instructions mention this is multi-agent with 8 agents and randomized pairings. This means:

...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 648,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.63 sent, it became \$10.89, returned \$5.0, visible payoff \$5.890000000000001.

Your total visible earnings across all rounds are \$7.52.

This round, you are the RECEIVER against your current co-player. They sent you \$3.14, so you received \$9.42. How much do you return? (0-9.42)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

- Round 1: My co-player saw \$3.63 sent, received \$10.89, and returned \$5.0 to me - showing cooperative behavior
- Round 2: They sent me \$3.14, which became \$9.42

They've demonstrated fairness by returning a reasonable amount in Round 1. I want to reciprocate this cooperative approach.

I'll return **\$4.50**, which:

- Gives them a profit (they sent ~\$3.14, get back \$4.50)
- Leaves me with a profit too (\$9.42 - \$4.50 = \$4.92)

- Maintains the cooperative relationship

```
'''json
{"return": 4.50}
'''
```

reasoning:

Let me understand what happened:

Round 1: I sent \$3, but my co-player saw \$3.63 (due to communication noise). This became \$10.89 for them. They returned \$5.0.

My actual payoff in Round 1:

- I had \$5
- I sent \$3
- I received back \$5
- So I kept: $\$5 - \$3 + \$5 = \7

But the visible payoff shows \$7.52, which suggests there might be some noise in the amounts.

Now in Round 2:

- I'm the RECEIVER
- My co-player sent me some amount that appears as \$3.14 to me
- This became \$9.42 (which is $\$3.14 \times 3$)
- I need to decide how much to return

Looking at their behavior: In Round 1, they saw \$3.63 sent, which became \$10.89, and they returned \$5.0. That's about 46% of what they received, or about 1.38x what they saw as sent.

This seems like a fair/cooperative player. They returned a reasonable amount.

Now I have \$9.42. What should I return?

Given that:

1. They sent a decent amount (\$3.14, which is about 63% of their \$5)
2. They were fair in the previous round, returning \$5 when they received \$10.89
3. I want to encourage continued cooperation

I should return a fair amount. Some options:

- Return about half of what I received: \$4.71
- Return what they sent plus a bit extra: around \$4-5
- Match their return rate from last round (46% of received): about \$4.33

I think returning around \$4.50-\$5.00 would be fair and encourage continued cooperation. Let me return \$4.50 as a fair split that gives them back close to what they sent plus some profit, while I also profit.

If I return \$4.50, they end up with \$6.36 while I keep \$4.92-both of us gain from the exchange. Returning \$5.00 would mirror...

usage:

```
{
  "input_tokens": 768,
  "output_tokens": 820,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.63 sent, it became \$10.89, returned \$5.0, visible payoff \$5.890000000000001.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.06 returned, visible payoff \$6.56.

Your total visible earnings across all rounds are \$12.44.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our history together:

Round 1: I sent (they saw \$3.63), they returned \$5.0 - cooperative

Round 2: They sent \$3.5 (I saw \$3.14), I returned (they saw \$5.06) - both cooperative

We've established a good cooperative pattern. They've been generous as both sender and receiver, so I want to continue building this trust.

I'll send **\$4** to show strong trust and maintain our mutually beneficial relationship.

```
'''json
{"send": 4}
'''
```

reasoning:

Let me analyze the history with my current co-player:

Round 1: I was SENDER

- I sent \$3 (but they saw \$3.63 due to noise)
- They received \$10.89
- They returned \$5.0
- This was cooperative behavior

Round 2: I was RECEIVER

- They sent \$3.5 (but I saw \$3.14 due to noise)
- I received \$9.42
- I returned \$4.50 (but they saw \$5.06 due to noise)
- They had a visible payoff of \$6.56

So far, this co-player has been quite cooperative:

- As receiver in Round 1: returned \$5.0 from \$10.89 (about 46%)
- As sender in Round 2: sent \$3.5 out of \$5 (70%)

I should continue this cooperative relationship. In Round 1, I sent \$3. In Round 2, they sent \$3.5. I should probably send a similar or slightly higher amount to maintain cooperation.

I'll send \$3.5 or \$4 to show continued trust and cooperation.

Given that they returned fairly and sent generously, I'll send \$4 to demonstrate strong trust and cooperation.

usage:

```
{
  "input_tokens": 1118,
  "output_tokens": 421,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.63 sent, it became \$10.89, returned \$5.0, visible payoff \$5.890000000000001.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.06 returned, visible payoff \$6.56.
- Round 3 against you, as RECEIVER: saw \$3.64 sent, it became \$10.92, returned \$5.5, visible payoff \$5.42.

Your total visible earnings across all rounds are \$18.1.

This round, you are the RECEIVER against your current co-player. They sent you \$3.21, so you received \$9.629999999999999. How much do you return? (0-9.629999999999999)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at our established pattern:

Round 1: They returned \$5.0 when receiving \$10.89

Round 2: I returned (they saw \$5.06) when I received \$9.42

Round 3: They returned \$5.5 when receiving \$10.92 - showing increased generosity
Round 4: They sent \$3.21, I received \$9.63

They've been consistently cooperative and even increased their returns. I want to reciprocate this positive trend.

I'll return **\$4.80**, which rewards their trust while maintaining mutual benefit.

```
'''json
{"return": 4.80}
'''
```

Reasoning:

Looking at the history with my current co-player:

Round 1: They (as receiver) saw \$3.63 sent, received \$10.89, returned \$5.0
Round 2: They (as sender) sent \$3.5, received (saw \$5.06) back
Round 3: They (as receiver) saw \$3.64 sent, received \$10.92, returned \$5.5
Round 4: They (as sender) sent \$3.21, I received \$9.63

Our pattern:

- Round 1: I sent (they saw 3.63), they returned \$5.0
- Round 2: They sent 3.5 (I saw 3.14), I returned (they saw 5.06)
- Round 3: I sent (they saw 3.64), they returned \$5.5
- Round 4: They sent 3.21, I received \$9.63

They've been consistently cooperative and fair. In Round 3, they returned \$5.5, showing increased generosity.

For Round 4, they sent \$3.21, which became \$9.63 for me.

To maintain the cooperative relationship and be fair:

- They invested ~\$3.21
- I should return enough that they profit, while I also profit
- A return of around \$4.50-\$5.00 would be fair

I'll return \$4.80 to show continued cooperation and slightly increase the return to match their increased generosity in Round 3.

```
Usage: {
  "input_tokens": 1484,
  "output_tokens": 500,
  "reasoning_tokens": 0
}
```

GAME - Agent_4 - Sender - dyad_2

Timestamp: 2026-08-12T08:11:15

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_3",
  "pairing": {
    "agents": [
      "Agent_4",
      "Agent_3"
    ],
    "investor": "Agent_4",
    "roles": {
      "Agent_3": "trustee",
      "Agent_4": "investor"
    },
    "trustee": "Agent_3"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 4,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.91, that is 58.2% of their total endowment. You received \$8.73. How much do you return to the sender? (0-8.73)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.91 (58.2% of their endowment)
- I received \$8.73 after the 3x multiplication
- The sender currently has \$2.09 remaining

To reciprocate their trust and cooperation:

- If I return nothing, the sender ends with \$2.09 (worse than keeping everything)
- If I return \$3.50, the sender ends with \$5.59 and I keep \$5.23
- This rewards their trust while maintaining fairness

I'll return a meaningful amount to encourage continued cooperation.

{'return': 3.5}

reasoning:

Let me analyze this situation:

1. As the RECEIVER, the sender sent me \$2.91 (which is 58.2% of their \$5 endowment)
2. This amount was multiplied by 3x, so I received \$8.73
3. I need to decide how much to return (between \$0 and \$8.73)

Key considerations:

- The sender was quite generous, sending nearly 60% of their endowment
- This shows trust and cooperation
- In trust games, reciprocity is important for maintaining cooperation
- The sender's final payoff will be: $\$5 - \$2.91 + \text{whatever I return} = \$2.09 + \text{return}$
- My final payoff will be: $\$8.73 - \text{whatever I return}$

Common approaches in trust games:

1. Split the gains equally - the sender gave up \$2.91 but created \$8.73. The "gain" is $\$8.73 - \$2.91 = \$5.82$. Splitting this equally would mean returning about \$2.91
2. Give the sender back what they sent - return \$2.91
3. Give the sender a good return on their investment - perhaps return more than they sent
4. Equal final payoffs - try to make both players end up with similar amounts

Let me calculate some scenarios:

If I return \$2.91:

- Sender gets: $\$2.09 + \$2.91 = \$5.00$
- I get: $\$8.73 - \$2.91 = \$5.82$
- This is fair-ish, both doing okay

If I return \$3.50:

- Sender gets: $\$2.09 + \$3.50 = \$5.59$
- I get: $\$8.73 - \$3.50 = \$5.23$
- This balances outcomes more evenly

If I return \$4.00:

- Sender gets: $\$2.09 + \$4.00 = \$6.09$
- I get: $\$8.73 - \$4.00 = \$4.73$
- The sender benefits more, which seems appropriate given they initiated the cooperation

For truly equal final payoffs, I'd need to return \$3.32, which would give us both \$5.41 each. The sender demonstrated substantial trust by sending most of their endowment, so I should reciprocate meaningfully to reinforce that cooperation....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 942,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.56 returned, visible payoff \$5.5600000000000005.

Your total visible earnings across all rounds are \$5.23.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the history:

- My co-player sent \$3.0 (60% of their endowment) in Round 1, showing strong trust
- They saw a return of \$3.56 (39.6% of what they sent as receiver)
- This demonstrates cooperative behavior

To reciprocate their trust and maintain cooperation, I'll send a similar proportion:

```
{'send': 3.0}
```

reasoning:

Let me analyze this situation:

Round 1:

- I was the RECEIVER
- The sender sent \$2.91 (58.2% of their endowment)
- I received \$8.73
- I returned \$3.50
- I kept \$5.23

Now in Round 2:

- I'm the SENDER
- I'm playing against the same co-player from Round 1
- The visible history shows that in Round 1, from my co-player's perspective:
 - They sent \$3.0 (as SENDER)
 - It became \$9.0
 - They saw \$3.56 returned
 - Their visible payoff was \$5.56

Wait, there's a discrepancy here. The problem says there's communication noise. Let me understand:

From my perspective in Round 1:

- Sender sent \$2.91
- I received \$8.73
- I returned \$3.50

From my co-player's perspective in Round 1:

- They sent \$3.0
- It became \$9.0
- They saw \$3.56 returned

So the noise affected what they saw vs what actually happened. They sent \$3.0 but I saw \$2.91. I returned \$3.50 but they saw \$3.56.

Now I'm the sender in Round 2. My co-player was cooperative in Round 1 - they sent 60% of their endowment (\$3.0 out of \$5). They saw a return of \$3.56 out of \$9.0, which is about 39.6% of what I received.

Given that:

1. My co-player showed strong trust by sending 60% in Round 1
2. They saw a decent return (39.6%)
3. I want to maintain cooperation and build trust

I should send a similar proportion to what they sent. They sent 60%, so I should send around \$3.0 as well to match their level of trust and cooperation.

...

usage:

```
{
  "input_tokens": 814,
  "output_tokens": 568,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.56 returned, visible payoff \$5.5600000000000005.
- Round 2 against you, as RECEIVER: saw \$2.77 sent, it became \$8.31, returned \$3.5, visible payoff \$4.8100000000000005.

Your total visible earnings across all rounds are \$10.13.

This round, you are the RECEIVER against your current co-player. They sent you \$3.52, so you received \$10.56. How much do you return? (0-10.56)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the pattern:

****Round 1**:** My co-player sent \$3.0 (60%), showing strong trust
****Round 2**:** I sent \$3.0 (60%), they returned \$3.5 (42% of what they received)
****Round 3**:** They sent \$3.52 (70.4%), showing even greater trust

They're being very cooperative and increasing their trust level. I should reciprocate appropriately.

Returning \$4.5:

- Sender ends with: $\$5 - \$3.52 + \$4.5 = \5.98
- I keep: $\$10.56 - \$4.5 = \$6.06$

This rewards their high trust while maintaining fairness.

{'return': 4.5}

reasoning:

Let me analyze this situation:

Round 1 (I was RECEIVER):

- Sender sent \$2.91 (58.2%)
- I received \$8.73
- I returned \$3.50
- My payoff: \$5.23

Round 2 (I was SENDER):

- I sent \$3.0
- It became \$9.0
- According to visible history, my co-player saw \$2.77 sent (there's noise!)
- They returned \$3.5
- My visible payoff from Round 2: I sent \$3, got back \$3.5, so $\$2 + \$3.5 = \$5.5$
- Total earnings so far: $\$5.23 + \$5.5 = \$10.73$... but it says \$10.13

Wait, let me recalculate. The visible history says:

- Round 2 as RECEIVER: saw \$2.77 sent, it became \$8.31, returned \$3.5, visible payoff \$4.81

This is from my co-player's perspective. They saw \$2.77 sent (due to noise), it became \$8.31, they returned \$3.5, keeping \$4.81.

My actual earnings from Round 2:

- I sent \$3.0
- I kept \$2.0
- I received back... well, if they returned \$3.5 based on receiving \$8.31, then I should have received \$3.5 back
- So my payoff: $\$2.0 + \$3.5 = \$5.5$

Total: $\$5.23 + \$5.5 = \$10.73$

But it says my total visible earnings are \$10.13. Hmm, there might be noise in what I received back too.

Anyway, for Round 3:

- I'm RECEIVER
- They sent \$3.52 (70.4% of their endowment)
- I received \$10.56

My partner's been increasingly generous across rounds-first 58.2%, now 70.4%-showing growing confidence. They returned roughly 40-42% in the previous round, so I should match that reciprocal level. Returning between \$4.22-\$4.75 would maintain the cooperative dynamic we've established....

usage:

```
{
  "input_tokens": 1104,
  "output_tokens": 897,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.56 returned, visible payoff \$5.5600000000000005.
- Round 2 against you, as RECEIVER: saw \$2.77 sent, it became \$8.31, returned \$3.5, visible payoff \$4.8100000000000005.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.78 returned, visible payoff \$5.779999999999999.

Your total visible earnings across all rounds are \$16.19.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the pattern:

Round 1: Co-player sent \$3.0 (60%), demonstrating trust
Round 2: Co-player returned \$3.5, showing cooperation
Round 3: Co-player sent \$3.0 (60%) again, maintaining trust

They're showing consistent cooperative behavior. I should continue matching their trust level:

{ 'send': 3.0 }

Reasoning:

Looking at the history:

Round 1: Co-player sent \$3.0 as SENDER, saw \$3.56 returned
Round 2: Co-player was RECEIVER, saw \$2.77 sent (I actually sent \$3.0), returned \$3.5
Round 3: Co-player sent \$3.0 as SENDER, saw \$3.78 returned (I actually returned \$4.5)

The communication noise is evident:

- In Round 2, I sent \$3.0 but they saw \$2.77
- In Round 3, I returned \$4.5 but they saw \$3.78

My co-player has been consistently sending \$3.0 (60% of endowment) as sender. They've been returning good amounts as receiver (\$3.5).

They're showing consistent trust and cooperation. I should continue to show strong trust as well. I'll send \$3.0 again to maintain the cooperative pattern.

Usage: {

"input_tokens": 1512,
"output_tokens": 323,
"reasoning_tokens": 0

}

GAME - Agent_3 - Receiver - dyad_2

Timestamp: 2026-08-12T08:11:25

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_2",
"move_index": 3,
"opponent_id": "Agent_4",
"pairing": {
 "agents": [
 "Agent_4",
 "Agent_3"
],
 "investor": "Agent_4",
 "roles": {
 "Agent_3": "trustee",
 "Agent_4": "investor"
 }
},
"trustee": "Agent_3"

,
"role": "trustee",
"role_label": "Receiver",
"round": 4,
"task": "game",
"task_index": 0

```

}
Exact input messages sent to model:
  Message 1 role=system
    content:
      You are an agent participating in a multi-task experiment. You will play a game and may al
      so be asked to write myths.

      GAME:
      You will play different roles across multiple rounds.

      When you are the SENDER:
      - You receive $5 each round
      - You decide how much to SEND to the RECEIVER ($0-$5)
      - Amount sent is multiplied by 3x
      - Receiver decides how much to RETURN
      - You keep: $5 - sent + returned
      - Receiver keeps: received - returned
      - Response format: {'send': <amount>}

      When you are the RECEIVER:
      - Sender receives $5
      - Sender decides how much to SEND to you ($0-$5)
      - Amount sent is multiplied by 3x
      - You decide how much to RETURN
      - Sender keeps: $5 - sent + returned
      - You keep: received - returned
      - Response format: {'return': <amount>}

      MYTH WRITING:
      You may also be asked to write short myths (stories) on various topics. When writing myths
      , follow the instructions provided in each prompt.

      IMPORTANT: Provide your decision in the correct JSON format.

      NOTE: There may be communication noise in this game.
      The amounts you see may differ slightly from what was actually sent/returned.
      This is part of the experimental design.

      MULTI-AGENT SETTING:
      - There are 8 agents in this run.
      - Each round, all agents are paired into dyads from the full pool. Pairings are randomized
      , but the schedule balances sender and receiver roles across the run.
      - You play one sender-receiver game with your paired opponent each round.
      - Your role may repeat across consecutive rounds, but by the end of the run every agent wi
      ll have acted as sender and receiver equally often when the round count allows it.
      - Agent names are hidden in this run.
      - Your paired opponent is referred to as your current co-player.
  Message 2 role=user
    content:
      Round 1: You are the SENDER. You have $5. How much do you send? (0-5)
  Message 3 role=assistant
    content:
      Looking at this first round, I need to balance trust and caution. The multiplication facto
      r of 3x creates an opportunity for mutual benefit if my co-player reciprocates.

      Since this is the first round with no prior history, I'll send a moderate amount that sign
      als cooperation while not exposing myself to maximum risk.

      **My decision:**

      '''json
      {"send": 3}
      '''

      This gives my co-player $9 to work with (3 × 3), leaving me with $2. If they return a fair
      share, we can both benefit from this cooperation. Let's see how they respond.
    reasoning:
      This is the first round of a trust game where I'm the SENDER. Let me think about this stra
      tegically.

```

Key facts:

- I have \$5
- I can send \$0-\$5 to the receiver
- Amount sent is multiplied by 3x
- The receiver then decides how much to return to me
- I keep: \$5 - sent + returned
- Receiver keeps: received - returned
- There may be communication noise (amounts seen may differ from actual)
- This is a multi-agent setting with 8 agents total
- Pairings are randomized each round
- Agent names are hidden

Strategic considerations:

1. ****Trust and cooperation****: If both parties cooperate optimally, we can maximize joint payoffs. If I send all \$5, the receiver gets \$15. If they return half (\$7.50), I end up with \$7.50 and they keep \$7.50, which is better for both of us than if I send nothing.
2. ****Risk****: The receiver could keep everything and return nothing. If I send \$5 and get nothing back, I end up with \$0 while they get \$15.
3. ****First round****: This is the first round, so I have no reputation or history with this particular co-player. However, my actions might influence future interactions in a multi-round setting.
4. ****Communication noise****: The presence of noise means that even if someone intends to be fair, the amounts might appear different, which could lead to misunderstandings.
5. ****Signaling****: Sending a moderate to high amount signals trust and cooperation, which might encourage reciprocity.

Common strategies in trust games:

- Send half (\$2.50) as a moderate trust signal
- Send all (\$5) to maximize joint payoff and signal full trust
- Send a smaller amount (\$1-2) to test the waters
- Send nothing (\$0) to avoid risk entirely

Given this is round 1 and I want to establish a cooperative tone while not being completely naive, I think sending a significant but not maximum amount makes sense.

Research on trust games suggests most people send positive amounts initially, and receivers typically reciprocate. The 3x multiplier creates substantial mutual benefit potential— even modest returns from the receiver would leave us both better off than the starting position. I'll send \$3 as a balanced signal of trust and good faith without overexposing myself....

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 870,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.91 sent, it became \$8.73, returned \$3.5, visible payoff \$5.23.

Your total visible earnings across all rounds are \$5.5600000000000005.

This round, you are the RECEIVER against your current co-player. They sent you \$2.77, so you received \$8.31. How much do you return? (0-8.31)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

****Round 1 analysis:****

- I sent \$3 (they saw \$2.91 due to noise)
- They received \$8.73 and returned \$3.5 (about 40% return rate)
- This was a cooperative response

****Current situation (Round 2):****

- They sent me \$2.77 (with noise)
- I received \$8.31
- They showed good faith cooperation in Round 1

I'll reciprocate their cooperative behavior with a similar return rate:

```
'''json
{"return": 3.5}
'''
```

This gives them a total of approximately \$5.73 and leaves me with \$4.81, creating mutual benefit from our cooperation.

reasoning:

Let me analyze this situation:

Round 1 (just completed):

- I was the SENDER and sent \$3
- Due to communication noise, my co-player saw \$2.91 (close to \$3)
- It became \$8.73 for them (should be \$8.73 if $\$2.91 \times 3$)
- They returned \$3.5
- Their visible payoff: \$5.23 (they kept $\$8.73 - \$3.5 = \$5.23$)
- My payoff: $\$5 - \$3 + \$3.5 = \5.56

Now Round 2:

- I am the RECEIVER
- My co-player sent me an amount, but I see \$2.77 (with noise)
- I received \$8.31 (which is $\$2.77 \times 3$)
- I need to decide how much to return

Analysis:

- In Round 1, my co-player returned \$3.5 out of \$8.73, which is about 40% return rate
- They kept \$5.23 for themselves
- I sent them \$3 (they saw \$2.91), and I ended up with \$5.56 total
- So I gained \$0.56 from the transaction

Now they sent me approximately \$2.77 (actual amount might be slightly different due to noise).

If I apply a similar return rate (around 40%), that would be: $\$8.31 \times 0.4 \approx \3.32

But let me think about fairness:

- They sent around \$2.77, leaving them with about \$2.23
- I received \$8.31
- A fair split might give them back enough so they end up with a reasonable amount

If I return \$3.5 (same as they returned to me):

- They end up with: $\$5 - \$2.77 + \$3.5 = \5.73
- I end up with: $\$8.31 - \$3.5 = \$4.81$

Returning \$4 would give them \$6.23 while I get \$4.31. Returning \$3 would leave me with \$5.31 and them with \$5.23. Since they gave me roughly \$2.77 and returned about 40% of what they received, matching that reciprocal gesture with a \$3.5 return seems appropriate....

usage:

```
{
  "input_tokens": 784,
  "output_tokens": 826,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.91 sent, it became \$8.73, returned \$3.5, visible payoff \$5.23.

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.9 returned, visible payoff \$4.9.

Your total visible earnings across all rounds are \$10.370000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our history:

****Round 1:**** I sent \$3, they returned \$3.5 - cooperative

****Round 2:**** They sent \$3, I returned \$3.5 (they saw \$2.9) - cooperative

We've established a mutually beneficial pattern where we both send moderate amounts and return fair shares. I'll continue this cooperation:

```
```json
{"send": 3}
```
```

This maintains trust and keeps our successful collaborative pattern going.

reasoning:

Let me analyze the history:

****Round 1:****

- I sent \$3 (they saw \$2.91 due to noise)
- They received \$8.73 and returned \$3.5
- I ended up with $\$2 + \$3.5 = \$5.5$

****Round 2:****

- They sent \$3.0 (I saw \$2.77 due to noise, actual was \$3.0)
- It became \$9.0 for me
- I returned \$3.5
- They saw \$2.9 returned (due to noise)
- Their visible payoff was \$4.9 ($\$5 - \$3 + \2.9)
- I kept $\$9.0 - \$3.5 = \$5.5$

My total visible earnings: \$10.37 which matches $\$5.5 + \4.81 approximately (there's some rounding)

So the pattern so far:

1. Both of us have been sending moderate amounts (\$3)
2. Both of us have been returning reasonable amounts (\$3.5)
3. This has been mutually beneficial

The relationship seems cooperative. They sent me \$3 in Round 2, which is the same amount I sent in Round 1. They saw I returned \$2.9 (actual \$3.5), which was a fair return.

I should continue this cooperative pattern. Sending \$3 again seems appropriate to maintain the reciprocal relationship.

usage:

```
{
  "input_tokens": 1128,
  "output_tokens": 446,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.91 sent, it became \$8.73, returned \$3.5, visible payoff \$5.23.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.9 returned, visible payoff \$4.9.
- Round 3 against you, as RECEIVER: saw \$3.52 sent, it became \$10.56, returned \$4.5, visible payoff \$6.060000000000005.

Your total visible earnings across all rounds are \$16.15.

This round, you are the RECEIVER against your current co-player. They sent you \$2.64, so you received \$7.92. How much do you return? (0-7.92)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at our cooperative history:

****Round 1:**** They returned \$3.5 on \$8.73
****Round 2:**** I returned \$3.5 on \$9.0
****Round 3:**** They returned \$4.5 on \$10.56 (increased generosity!)
****Round 4:**** They sent me \$2.64, I received \$7.92

They've been consistently cooperative and even increased their generosity in Round 3. I'll reciprocate fairly:

```
'''json
{"return": 3.5}
'''
```

This maintains our mutually beneficial cooperation pattern.

Reasoning:

Let me analyze the history with my current co-player:

****Round 1:****
- I sent \$3 (they saw \$2.91 due to noise)
- They received \$8.73, returned \$3.5
- Good cooperation

****Round 2:****
- They sent \$3
- I received \$9.0, returned \$3.5 (they saw \$2.9)
- Continued cooperation

****Round 3:****
- I sent \$3 (they saw \$3.52 due to noise)
- They received \$10.56, returned \$4.5
- They increased their return, showing even more generosity

****Round 4 (current):****
- They sent \$2.64 (with noise, could have been around \$3 originally)
- I received \$7.92

Looking at the pattern:

- They've been consistently cooperative
- In Round 3, they returned \$4.5 on \$10.56 (about 43% return rate)
- They're continuing to send around \$3 each time

I should reciprocate their continued cooperation. Given they sent me \$7.92, a fair return would be around 40-45% of what I received, which would be about \$3.2-\$3.6.

I'll return \$3.5 to maintain consistency with our cooperative pattern.

```
Usage: {
  "input_tokens": 1467,
  "output_tokens": 452,
  "reasoning_tokens": 0
}
```

GAME - Agent_6 - Sender - dyad_3

Timestamp: 2026-08-12T08:11:34

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_5",
  "pairing": {
    "agents": [
      "Agent_6",
      "Agent_5"
    ]
  }
}
```



```

],
"investor": "Agent_6",
"roles": {
  "Agent_5": "trustee",
  "Agent_6": "investor"
},
"trustee": "Agent_5"
},
"role": "investor",
"role_label": "Sender",
"round": 4,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.66, that is 53.2% of their total endowment. You received \$7.98. How much do you return to the sender? (0-7.98)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.66 (53.2% of their endowment)

- I received \$7.98 after the 3x multiplication
- The sender currently has \$2.34 remaining from their original \$5

To reward their trust and encourage cooperation, I should return enough so they profit from their decision to send. If I return \$4.00:

- Sender ends with: $\$2.34 + \$4.00 = \$6.34$ (a gain of \$1.34)
- I keep: $\$7.98 - \$4.00 = \$3.98$

This seems fair - both of us benefit from the cooperation, and the sender's trust is rewarded.

```
{"return": 4.00}
```

reasoning:

Let me think about this carefully.

I'm the RECEIVER in Round 1.

- The sender sent \$2.66 (which is 53.2% of their \$5 endowment)
- This amount was multiplied by 3x, so I received $\$2.66 \times 3 = \7.98
- Now I need to decide how much to return (between \$0 and \$7.98)

Key considerations:

1. The sender sent more than half their endowment (53.2%), showing substantial trust
2. If I return nothing, I keep \$7.98 and the sender keeps $\$5 - \$2.66 = \$2.34$
3. If I return some amount, I should aim to be fair and encourage cooperation

A reasonable approach would be to:

- Ensure the sender ends up with more than they started with (more than \$5)
- Keep some profit myself
- Reward their trust to encourage future cooperation

Let me calculate:

- Sender currently has: $\$5 - \$2.66 = \$2.34$
- I have: \$7.98

If I return \$3.00:

- Sender ends with: $\$2.34 + \$3.00 = \$5.34$ (slight gain)
- I keep: $\$7.98 - \$3.00 = \$4.98$

If I return \$4.00:

- Sender ends with: $\$2.34 + \$4.00 = \$6.34$ (good gain)
- I keep: $\$7.98 - \$4.00 = \$3.98$

If I return \$3.50:

- Sender ends with: $\$2.34 + \$3.50 = \$5.84$
- I keep: $\$7.98 - \$3.50 = \$4.48$

...

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 699,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.13 returned, visible payoff \$5.13.

Your total visible earnings across all rounds are \$3.9800000000000004.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the visible history:

****Round 1 recap (from their perspective):****

- My co-player sent \$3.00 (60% of their endowment) - quite generous
- They saw \$3.13 returned
- Their visible payoff was \$5.13

My co-player showed significant trust by sending 60% of their endowment. Even though there's communication noise affecting what we each see, I should reciprocate this trust.

I'll match their generosity by sending the same amount they sent: \$3.00 (60% of my endowment). This demonstrates:

- Reciprocal trust
- Willingness to cooperate
- Fair play matching their level of generosity

```
{"send": 3.00}
```

reasoning:

Let me analyze this situation:

Round 1 (just played):

- I was the RECEIVER
- Sender sent \$2.66 (53.2%)
- I received \$7.98
- I returned \$4.00
- I kept \$3.98

But now I see the "visible history" which shows:

- Round 1 against me, as SENDER: sent \$3.0, became \$9.0, saw \$3.13 returned, visible payoff \$5.13

This is interesting - there's communication noise!

- The sender actually sent \$3.00 (not \$2.66 as I saw)
- It became \$9.00 (not \$7.98 as I saw)
- They saw \$3.13 returned (not \$4.00 as I sent)
- Their visible payoff was \$5.13 (which is \$5 - \$3 + \$3.13)

So due to noise:

- I saw them send \$2.66, they actually sent \$3.00
- I saw I received \$7.98, I actually received \$9.00
- I returned \$4.00, they saw \$3.13

Now in Round 2, I'm the SENDER and playing against the same person.

From their perspective:

- They sent \$3.00 (60% of endowment - quite generous)
- They got back \$3.13 (34.8% of what I received)
- Their payoff was \$5.13

From my actual perspective (what really happened):

- They sent \$3.00
- I received \$9.00
- I returned \$4.00 (44.4% of what I received)
- But they only saw \$3.13

The co-player was quite generous in Round 1 (sent 60%). From their visible perspective, they got a modest return (34.8% of the tripled amount).

...

usage:

```
{
  "input_tokens": 845,
  "output_tokens": 754,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.13 returned, visible payoff \$5.13.
- Round 2 against you, as RECEIVER: saw \$2.97 sent, it became \$8.91, returned \$4.5, visible payoff \$5.13.

e payoff \$4.41.
Your total visible earnings across all rounds are \$11.010000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$3.23, so you received \$9.69. How much do you return? (0-9.69)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the visible history:

****Round 1:**** My co-player sent \$3.00 (60%)
****Round 2:**** They returned \$4.50 out of \$8.91 (50.5%) when they were receiver
****Round 3:**** They just sent \$3.23 (64.6%) - even more generous!

My co-player has shown increasing trust and cooperation. They've been very generous, and when they were receiver in Round 2, they returned about 50% of what they received.

I should reciprocate their strong cooperation. If I return \$5.00:

- Sender ends with: \$1.77 + \$5.00 = \$6.77 (profit of \$1.77)
- I keep: \$9.69 - \$5.00 = \$4.69

This returns about 52% of what I received, matching their generosity from Round 2 and rewarding their increased trust.

{"return": 5.00}

reasoning:

Let me analyze the visible history:

****Round 1 (I was RECEIVER):****
- My co-player sent \$3.00 (60%)
- They saw \$3.13 returned
- Their visible payoff: \$5.13

****Round 2 (I was SENDER):****
- I sent \$3.00
- They saw \$2.97 sent (noise), which became \$8.91
- They returned \$4.50
- Their visible payoff: \$4.41

****Round 3 (I am RECEIVER):****
- They sent \$3.23 (64.6% of their endowment)
- I received \$9.69
- They have \$1.77 remaining

My co-player has been very cooperative:

- Round 1: Sent 60% (\$3.00)
- Round 2: Returned \$4.50 out of \$8.91 received (50.5%)
- Round 3: Sent 64.6% (\$3.23) - even more generous!

They've shown increasing trust and generosity. I should reciprocate strongly.

Let me think about what to return:

If I return \$5.00:

- Sender ends with: \$1.77 + \$5.00 = \$6.77 (profit of \$1.77)
- I keep: \$9.69 - \$5.00 = \$4.69

If I return \$5.50:

- Sender ends with: \$1.77 + \$5.50 = \$7.27 (profit of \$2.27)
- I keep: \$9.69 - \$5.50 = \$4.19

Given their increased generosity (sending 64.6% vs 60% in round 1), I should return proportionally more. They returned \$4.50 out of \$8.91 (50.5%) when they were receiver. I should aim to return around 50-55% to match their generosity.

...

usage:

```
{
  "input_tokens": 1207,
  "output_tokens": 754,
  "reasoning_tokens": 0
}
```

}

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.13 returned, visible payoff \$5.13.
- Round 2 against you, as RECEIVER: saw \$2.97 sent, it became \$8.91, returned \$4.5, visible payoff \$4.41.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.79 returned, visible payoff \$7.79.

Your total visible earnings across all rounds are \$15.700000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the visible history:

Round 1: Co-player sent \$3.00, saw \$3.13 returned, payoff \$5.13

Round 2: Co-player returned \$4.50 from \$8.91 received (fair, ~50%)

Round 3: Co-player sent \$3.00, saw \$5.79 returned (very generous!), payoff \$7.79

My co-player has been consistently generous, sending \$3.00 (60%) in both rounds as sender. They also saw a strong return in Round 3, which should reinforce mutual trust. As receiver, they returned fairly (~50%).

This is a cooperative relationship with established trust. I should continue matching their consistent generosity of \$3.00.

{"send": 3.00}

Reasoning:

Looking at the visible history from my co-player's perspective:

Round 1: They sent \$3.00, saw \$3.13 returned, payoff \$5.13

Round 2: They received \$8.91 (from my \$2.97 send), returned \$4.50, payoff \$4.41

Round 3: They sent \$3.00, saw \$5.79 returned, payoff \$7.79

My co-player has been consistently generous, sending \$3.00 in both rounds they were sender (60% of endowment). In Round 3, they saw a very generous return of \$5.79, which should reinforce their trust in me.

They've also been fair as a receiver, returning \$4.50 out of \$8.91 (about 50%).

This is a cooperative relationship with mutual trust. I should continue to match their generosity. They've sent \$3.00 twice, so I should send \$3.00 again.

Actually, let me reconsider. In Round 3, they saw a particularly high return (\$5.79), which might encourage even more trust. But they've consistently sent \$3.00, so I should match that level.

I'll send \$3.00 to maintain consistency and reciprocal cooperation.

Usage: {

"input_tokens": 1654,

"output_tokens": 485,

"reasoning_tokens": 0

}

GAME - Agent_5 - Receiver - dyad_3

Timestamp: 2026-08-12T08:11:51

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",

"dyad_id": "dyad_3",

"move_index": 5,

"opponent_id": "Agent_6",

"pairing": {

```

"agents": [
  "Agent_6",
  "Agent_5"
],
"investor": "Agent_6",
"roles": {
  "Agent_5": "trustee",
  "Agent_6": "investor"
},
"trustee": "Agent_5"
},
"role": "trustee",
"role_label": "Receiver",
"round": 4,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round situation, I need to balance potential gains from cooperation

against the uncertainty of how my co-player will respond.

****My reasoning:****

- Sending money creates value (3x multiplier), so there's potential for mutual benefit
- This is our first interaction, so I want to signal willingness to cooperate
- But I also want to maintain some security given the uncertainty
- The communication noise adds another layer of unpredictability

I'll send a moderate amount that demonstrates trust while keeping some reserve.

```
'''json
{"send": 3}
'''
```

This sends \$3 to my co-player, who will receive \$9. I'll keep \$2 initially, and hope they reciprocate with a fair return that benefits us both.

reasoning:

This is the first round of a trust game where I'm the sender. Let me think about the strategic situation:

1. I have \$5 to start with
2. Whatever I send gets tripled when it reaches the receiver
3. The receiver then decides how much to return to me
4. There's communication noise - amounts may appear different than what was actually sent

Key considerations:

- This is a multi-round game with multiple agents
- Pairings are randomized each round
- There's a tension between trust/cooperation and self-interest
- If I send nothing, I keep \$5 but there's no opportunity for mutual gain
- If I send everything (\$5), the receiver gets \$15, and if they're fair/cooperative, they might return a good amount
- The communication noise adds uncertainty

In standard trust game experiments, sending some amount is generally beneficial because:

- It creates surplus value (3x multiplier)
- It signals trust and can encourage reciprocity
- Over multiple rounds, cooperative behavior can be rewarding

First round considerations:

- I don't know this particular co-player yet
- This is the first interaction, so I might want to signal willingness to cooperate
- But I also don't want to be too naive

A moderate approach might be to send \$3 or \$4:

- Sending \$3 means receiver gets \$9, I keep \$2. If they return \$4-5, we both do well
- Sending \$4 means receiver gets \$12, I keep \$1. If they return \$6, we both get \$6
- Sending \$5 means receiver gets \$15, I keep \$0. If they return \$7-8, I do well

Given this is round 1 and I want to establish a cooperative tone while not being too vulnerable, I'll send \$3. This shows trust but maintains some security.

Actually, let me reconsider. In trust games, the classic finding is that people often send about half their endowment on average. Sending \$2.50 or \$3 seems reasonable.

...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 703,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.66 sent, it became \$7.98, returned \$4.0, visible payoff \$3.9800000000000004.

Your total visible earnings across all rounds are \$5.13.

This round, you are the RECEIVER against your current co-player. They sent you \$2.97, so you received \$8.91. How much do you return? (0-8.91)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the situation:

****What happened in Round 1:****

- My co-player received \$7.98 (my \$3 sent with noise)
- They returned \$4.0, which is about 50% - a fair reciprocal move
- This shows they're a cooperative player

****Current Round 2:****

- They sent me \$2.97 (?60% of their endowment) - showing significant trust
- I received \$8.91
- They kept \$2.03

****My decision:****

Since they demonstrated fair reciprocity (50% return) in Round 1 and have now shown strong trust by sending most of their endowment, I should reciprocate similarly. Returning about 50% maintains the cooperative relationship while ensuring we both benefit.

Returning \$4.50 means:

- They end with: $\$2.03 + \$4.50 = \$6.53$
- I end with: $\$8.91 - \$4.50 = \$4.41$

This is fair and encourages continued cooperation.

```
'''json
{"return": 4.50}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap:

- I sent \$3 (but due to communication noise, my co-player saw \$2.66)
- They received \$7.98 (which should be $\$2.66 \times 3$)
- They returned \$4.0
- My visible payoff was \$5.13 (which would be $\$5 - \$3 + \$3.13...$ but there's noise)
- Their visible payoff was \$3.98

Now in Round 2:

- My co-player is the SENDER
- They sent me \$2.97 (with noise, they might have intended something slightly different)
- I received \$8.91 (which is $\$2.97 \times 3$)
- I need to decide how much to return

Analysis of their behavior:

- In Round 1, they returned \$4.0 out of \$7.98 received, which is about 50%
- This is a reasonably fair/generous return
- Now they've sent me \$2.97 out of their \$5, which is about 59% - quite trusting

Given that:

1. They showed good reciprocity in Round 1 (returning 50% of what they received)
2. They've now shown significant trust by sending me almost 60% of their endowment
3. We might continue playing together in future rounds
4. Fair reciprocity would encourage continued cooperation

I should return a similar proportion or slightly more to maintain/build trust.

They received \$8.91 from my send. A 50% return would be \$4.455.

But let me think about the overall economics:

- They started with \$5
- They sent \$2.97, keeping \$2.03
- I received \$8.91
- If I return \$4.50:
 - They end with: $\$2.03 + \$4.50 = \$6.53$
- ...

usage:

```
{
  "input_tokens": 816,
  "output_tokens": 863,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.66 sent, it became \$7.98, returned \$4.0, visible payoff \$3.9800000000000004.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.03 returned, visible payoff \$7.03.

Your total visible earnings across all rounds are \$9.54.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our history together:

****Round 1:****

- I sent, they received and returned ~50% - fair cooperation

****Round 2:****

- They sent \$3 (60% of endowment) - high trust
- I returned \$4.50, they saw \$5.03 (noise) - generous reciprocity
- Their payoff was strong (\$7.03)

****Analysis:****

- This co-player has proven consistently cooperative
- They demonstrated trust by sending \$3 in Round 2
- They've reciprocated fairly (50% return rate)
- Our relationship is building mutual trust

****My decision:****

I should maintain this cooperative dynamic. Sending \$3 again signals:

- Continued trust in our partnership
- Consistency and reliability
- Faith they'll reciprocate fairly as before

'''json

{"send": 3}

'''

reasoning:

Let me analyze the visible history:

Round 1:

- I was SENDER, sent \$3
- They received \$7.98 (with noise, actual was probably \$9)
- They returned \$4.0 (about 50%)
- My payoff was \$5 - \$3 + \$4 = \$6
- Visible shows \$5.13, which might include noise

Round 2:

- I was RECEIVER
- They sent \$3.0 (actual, before noise)
- I received \$8.91 (actually \$9 with noise)
- I returned \$4.50
- They saw \$5.03 returned (with noise, I actually returned \$4.50)
- Their visible payoff \$7.03 = \$5 - \$3 + \$5.03

My total visible earnings: \$9.54

Round 3 (current):

- I'm SENDER again

- Same co-player from rounds 1 and 2

This co-player has been very cooperative:

1. Returned 50% in Round 1 when they received
2. Sent \$3 (60%) in Round 2, showing trust
3. They saw me return \$5.03 (with noise), which they probably perceive as generous (56% return)

This is building a strong cooperative relationship. They've been trustworthy and fair. I should continue to send a substantial amount to maintain this cooperation.

Given their consistent cooperation and the fact they sent \$3 in Round 2, I should send at least the same amount, if not more, to signal continued trust and cooperation.

I'll send \$3 again, showing consistency and trust.

usage:

```
{
  "input_tokens": 1250,
  "output_tokens": 587,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.66 sent, it became \$7.98, returned \$4.0, visible payoff \$3.9800000000000004.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.03 returned, visible payoff \$7.03.
- Round 3 against you, as RECEIVER: saw \$3.23 sent, it became \$9.69, returned \$5.0, visible payoff \$4.6899999999999995.

Your total visible earnings across all rounds are \$17.33.

This round, you are the RECEIVER against your current co-player. They sent you \$3.93, so you received \$11.790000000000001. How much do you return? (0-11.790000000000001)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at our continued partnership:

****Round 1:**** They returned \$4.0 from \$7.98 (?50%)

****Round 2:**** I returned \$4.50 from \$9.0 (?50%)

****Round 3:**** They returned \$5.0 from \$9.69 (?52%)

****Current Round 4:****

- They sent \$3.93 (79% of endowment!) - very high trust signal
- I received \$11.79
- They kept only \$1.07

****Analysis:****

This co-player has shown increasing trust and consistent 50% reciprocity. They're now sending nearly their entire endowment, demonstrating strong faith in our relationship.

I should maintain our established 50% return pattern to:

- Honor their trust
- Keep our cooperative equilibrium
- Ensure mutual benefit

Returning \$5.90:

- They end with: \$1.07 + \$5.90 = \$6.97
- I end with: \$11.79 - \$5.90 = \$5.89

```
'''json
{"return": 5.90}
'''
```

Reasoning:

Let me analyze the history:

****Round 1:****

- I sent (saw \$2.66 with noise from my \$3)
- They received \$7.98, returned \$4.0 (about 50%)

****Round 2:****

- They sent \$3
- I received \$9, returned \$4.50 (they saw \$5.03 with noise)
- They got \$7.03 total

****Round 3:****

- I sent \$3 (they saw \$3.23 with noise)
- They received \$9.69, returned \$5.0 (about 52%)
- They kept \$4.69

****Round 4 (current):****

- They sent \$3.93 (about 79% of their endowment!) - very high trust
- I received \$11.79
- They kept \$1.07

This co-player is showing extremely high trust by sending \$3.93. They've consistently returned around 50% when they were receiver. I should reciprocate this trust and maintain our cooperative relationship.

If I return about 50% of what I received:

- Return \$5.90
- They end with: \$1.07 + \$5.90 = \$6.97
- I end with: \$11.79 - \$5.90 = \$5.89

But they're showing increasing trust (sending more and more). Maybe I should return a bit more to reward this and encourage continued cooperation. Let me think...

Actually, looking at the pattern:

- Round 1 receiver: returned \$4 from \$7.98 (50%)
- Round 3 receiver: returned \$5 from \$9.69 (52%)

They're consistently returning around 50%. I've been doing similar - returned \$4.50 from \$9 in Round 2 (50%).

Given they sent \$3.93 (very high trust), I should return around 50% or slightly more to maintain in the relationship.

...

```
Usage: {
  "input_tokens": 1695,
  "output_tokens": 859,
  "reasoning_tokens": 0
}
```

GAME - Agent_8 - Sender - dyad_4

Timestamp: 2026-08-12T08:12:03

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_7",
  "pairing": {
    "agents": [
      "Agent_8",
      "Agent_7"
    ],
    "investor": "Agent_8",
    "roles": {
      "Agent_7": "trustee",
      "Agent_8": "investor"
    },
    "trustee": "Agent_7"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 4,
```

```
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.17, that is 63.4% of their total endowment. You received \$9.51. How much do you return to the sender? (0-9.51)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending 63.4% of their endowment (\$3.17)
- I received \$9.51 after the 3x multiplier
- The sender currently has \$5 - \$3.17 = \$1.83

The sender demonstrated considerable trust, so I want to reciprocate appropriately. If I return \$5.00:

- The sender would end with: \$1.83 + \$5.00 = \$6.83 (a profit of \$1.83)
- I would keep: \$9.51 - \$5.00 = \$4.51

This rewards their trust with a solid return while allowing us both to benefit from the cooperation.

```
{"return": 5.00}
```

reasoning:

Let me think about this carefully.

I'm the RECEIVER in Round 1.

- The sender sent \$3.17 (which is 63.4% of their \$5 endowment)
- This was multiplied by 3x, so I received: $\$3.17 \times 3 = \9.51 ?
- I need to decide how much to return (between \$0 and \$9.51)

Key considerations:

1. The sender showed considerable trust by sending 63.4% of their endowment
2. If I return nothing, the sender ends up with: $\$5 - \$3.17 = \$1.83$, while I keep \$9.51
3. If I return a fair/reciprocal amount, what would that be?

A few approaches to consider:

- Equal split of total: Total is \$5 (sender's initial) + \$9.51 (what I received) = \$14.51. Equal split would be ~\$7.26 each. The sender currently has \$1.83, so I'd return $\$7.26 - \$1.83 = \$5.43$
- Return what maintains sender's initial investment value: Sender invested \$3.17. If I return \$3.17, sender gets back to \$5 total, and I keep \$6.34
- Return proportionally to what was sent: The sender sent 63.4% of their endowment. I could return a similar proportion of what I received: 63.4% of \$9.51 = ~\$6.03
- Social norm of reciprocity: Return enough to reward trust but keep some gains

Given that:

- The sender showed significant trust (63.4%)
- This is a multi-round game with different partners
- Building reputation for reciprocity could be valuable
- Note about communication noise suggests amounts might not be exact

I think a reasonable return would be around \$4-5, which:

- Acknowledges their generosity without giving away all my gains
- Leaves them better off than their starting position
- Demonstrates trustworthiness that could matter in future interactions with other partners
- Strikes a balance between reciprocity and self-interest

I'll return \$4.75. This gives the sender \$6.58 total (\$1.83 remaining plus my return), which meaningfully rewards their trust while I keep \$4.76....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1001,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.37 returned, visible payoff \$7.37.

Your total visible earnings across all rounds are \$4.51.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the history:

My co-player sent \$3.0 in Round 1 (60% of their endowment), showing significant trust. They saw \$5.37 returned, which gave them a good outcome.

Since they demonstrated trust and cooperation, I should reciprocate similarly. Sending around the same amount they sent (\$3.0) would be fair and maintain the cooperative relationship.

```
{"send": 3.0}
```

reasoning:

Let me analyze this situation:

1. In Round 1, I was the RECEIVER and the sender sent me \$3.17 (according to my view), which became \$9.51 for me. I returned \$5.00.
2. However, from my current co-player's perspective (shown in "Your current co-player's last 3 game(s)"):
 - They sent \$3.0 (not \$3.17 - communication noise!)
 - It became \$9.0 (not \$9.51 - more noise!)
 - They saw \$5.37 returned (not \$5.00 - even more noise!)
 - Their visible payoff was \$7.37

So there's noise in the communication. What they experienced:

- Started with \$5
- Sent \$3.0
- Had \$2.0 left
- Saw \$5.37 returned
- Ended with \$2.0 + \$5.37 = \$7.37

From their perspective, I (as the receiver in Round 1) returned about 60% of what I received (\$5.37 out of \$9.0).

3. Now I'm the SENDER in Round 2 against the same co-player.

My co-player showed trust by sending \$3.0 (from their perspective) and saw a decent return ratio. They might be willing to reciprocate trust.

I should consider:

- They sent about 60% of their endowment in Round 1
- They saw about 60% returned
- This suggests they're moderately trusting and cooperative

I should send a similar amount to show I'm also trustworthy and cooperative. Sending around \$3 would be consistent with their behavior and maintain the cooperative relationship.

Let me send \$3.00 to match their earlier send amount.

usage:

```
{
  "input_tokens": 837,
  "output_tokens": 523,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.37 returned, visible payoff \$7.37.
- Round 2 against you, as RECEIVER: saw \$2.35 sent, it became \$7.050000000000001, returned \$3.75, visible payoff \$3.3000000000000007.

Your total visible earnings across all rounds are \$10.219999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$2.22, so you received \$6.66. How much do you return? (0-6.66)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the situation:

Current round:

- Sender sent \$2.22 (44.4% of their endowment)
- I received \$6.66
- Sender currently has: \$5 - \$2.22 = \$2.78

History with this co-player:

- Round 1: They sent \$3.0, I returned money (they saw \$5.37)
- Round 2: They saw less sent from me due to noise (\$2.35 vs my actual \$3.0), returned \$3.

They've reduced their sending, possibly due to noise affecting what they saw in Round 2. So till, they're showing some trust by sending \$2.22. I should reciprocate fairly.

Returning approximately 50% of what I received (\$3.33) would:

- Give sender: $\$2.78 + \$3.33 = \$6.11$ (profit of \$1.11)
- I keep: \$3.33

This maintains cooperation while being proportional to their send amount.

```
{"return": 3.33}
```

reasoning:

Let me analyze the situation:

Current round:

- I'm the RECEIVER
- Sender sent \$2.22 (44.4% of their endowment)
- I received \$6.66 after 3x multiplier
- Sender currently has: $\$5 - \$2.22 = \$2.78$

Looking at history with this co-player:

- Round 1: They sent \$3.0 as SENDER, saw \$5.37 returned (I actually sent \$5.00, but due to noise they saw \$5.37)
- Round 2: I sent \$3.0 as SENDER (but they saw \$2.35 due to noise), they returned \$3.75

In Round 2, I sent \$3.0 which became \$9.0, and they returned \$3.75. That's about 42% return rate.

Now in Round 3:

- They sent \$2.22, which is less than what they sent in Round 1 (\$3.0)
- This might be because they saw only \$2.35 sent in Round 2 (due to noise) instead of my actual \$3.0

I should consider:

1. They're still showing some trust by sending \$2.22
2. They may have reduced their sending because of perceived lower trust from me (noise issue)
3. I should maintain cooperation

If I return around \$3.50-\$4.00:

- Return \$3.50: Sender gets $\$2.78 + \$3.50 = \$6.28$ (profit of \$1.28), I keep \$3.16
- Return \$4.00: Sender gets $\$2.78 + \$4.00 = \$6.78$ (profit of \$1.78), I keep \$2.66

Given they sent less this round but still showed trust, I'll return a reasonable amount that rewards their trust.

I'm noticing a pattern in the return rates across rounds. In Round 1, I returned about 53% of what I received, while in Round 2 they returned roughly 42%. Since they've sent less this time-likely reacting to the noise that made my previous send appear smaller-I should aim for around 50% to maintain fairness and signal...

usage:

```
{
  "input_tokens": 1135,
  "output_tokens": 909,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.37 returned, visible payoff \$7.37.
- Round 2 against you, as RECEIVER: saw \$2.35 sent, it became \$7.050000000000001, returned \$3.75, visible payoff \$3.3000000000000007.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.3 returned, visible payoff \$6.3.

Your total visible earnings across all rounds are \$13.549999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the history with my current co-player:

- Round 1: They sent \$3.0 (strong trust)
- Round 2: I sent, they returned \$3.75 (cooperative)
- Round 3: They sent \$3.0 again (consistent trust)

My co-player has shown consistent cooperation, sending \$3.0 both times they were the sender. They also returned a fair amount when they were receiver. I should reciprocate this consistent trust.

```
{"send": 3.0}
```

Reasoning:

Let me analyze the visible history with my current co-player:

Round 1: They were SENDER, sent \$3.0 (60%), I returned something, they saw \$5.37 returned
Round 2: They were RECEIVER, they saw \$2.35 sent (but I actually sent \$3.0 - noise!), they returned \$3.75
Round 3: They were SENDER, sent \$3.0 (60%), they saw \$4.3 returned (but I actually returned \$3.33 - noise showing higher!)

So my co-player has:

- Sent \$3.0 twice when they were sender (consistent trust)
- Returned \$3.75 when they received \$7.05

They're showing good cooperation. In Round 3, they saw a higher return (\$4.3) than I actually sent (\$3.33), which might encourage them.

I should continue cooperating. Sending \$3.0 again would be consistent and fair.

My total visible earnings: \$13.55

- Round 1 as RECEIVER: kept \$4.51
- Round 2 as SENDER: sent \$3.0, saw return, kept something
- Round 3 as RECEIVER: kept \$3.33

Actually, let me recalculate:

Round 1: I was RECEIVER, got \$9.51, returned \$5.00, kept \$4.51
Round 2: I was SENDER, sent \$3.0, they returned \$3.75, I kept \$2 + \$3.75 = \$5.75
Round 3: I was RECEIVER, got \$6.66, returned \$3.33, kept \$3.33

Total: \$4.51 + \$5.75 + \$3.33 = \$13.59 ? \$13.55 (matches with rounding)

My co-player has been consistent in sending \$3.0 when they're sender. ...

```
Usage: {  
  "input_tokens": 1610,  
  "output_tokens": 585,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_7 - Receiver - dyad_4

Timestamp: 2026-08-12T08:12:16

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_4",  
  "move_index": 7,  
  "opponent_id": "Agent_8",  
  "pairing": {  
    "agents": [  
      "Agent_8",  
      "Agent_7"  
    ],  
    "investor": "Agent_8",  
    "roles": {  
      "Agent_7": "trustee",
```



```

    "Agent_8": "investor"
  },
  "trustee": "Agent_7"
},
"role": "trustee",
"role_label": "Receiver",
"round": 4,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to consider:

1. ****The multiplier effect**:** Any amount I send gets tripled, creating potential mutual gain
2. ****No prior history**:** This is my first interaction, so I have no information about this co-player's behavior
3. ****Future rounds**:** Even though pairings are randomized, establishing cooperative norms early could benefit the group overall

4. ****Risk vs. reward****: Sending more creates more surplus, but requires trusting the receiver to return a fair amount

I'll take a moderately cooperative approach. Sending \$3 creates \$9 for the receiver - enough surplus that we can both benefit if they reciprocate fairly, while not risking my entire endowment.

If they return around half of what they receive (\$4-5), we both end up better off than if no transaction occurred.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a multi-agent trust game experiment. I'm the SENDER with \$5.

Let me think about the strategic aspects:

1. If I send money, it gets tripled, creating surplus value
2. The receiver then decides how much to return
3. This is a classic trust game scenario

Key considerations:

- This is round 1, so I have no history with this particular co-player
- There are multiple rounds, so reputation/repeated interaction matters
- However, pairings are randomized each round, so I won't necessarily play the same person again
- The game has communication noise, which might affect trust

Strategic thinking:

- Sending \$0 guarantees I keep \$5, but creates no surplus
- Sending \$5 creates maximum surplus (\$15 total for receiver), but requires trust
- A middle ground (e.g., \$2-3) balances risk and potential gain

In classic trust game experiments, people often:

1. Send moderate to high amounts in early rounds to signal cooperation
2. The amount sent and returned establishes baseline trust levels
3. With randomized pairings, building a reputation as trustworthy matters for future rounds

Game theory perspective:

- If this were truly one-shot with rational self-interested players, the receiver should return \$0 (backward induction)
- Knowing this, I should send \$0
- However, real humans often deviate from this due to fairness norms, reciprocity, and reputation concerns

Since:

- This is multi-round with multiple agents
- There may be reputation effects even with randomized pairings
- Starting cooperatively can establish good norms
- The 3x multiplier creates real gains from cooperation

I'll send a moderate-high amount to signal cooperation while not being completely vulnerable. I'll send \$3.

This creates \$9 for the receiver, and if they return even just \$3, we both end up with \$5 (same as no transaction), but if they return more (say \$4-5), we both gain. ...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 671,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.17 sent, it became \$9.51, returned \$5.0, visible

e payoff \$4.51.

Your total visible earnings across all rounds are \$7.37.

This round, you are the RECEIVER against your current co-player. They sent you \$2.35, so you received \$7.050000000000001. How much do you return? (0-7.050000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

****Round 1 analysis:****

- They saw I sent \$3.17 (with noise), received \$9.51
- They returned \$5.00, which is about 52.6% of what they received
- This was quite generous and cooperative

****Current situation:****

- They sent me \$2.35, I received \$7.05
- They showed good faith last round, so I should reciprocate

Following a similar reciprocity rate (around 52-53% return):

- 53% of \$7.05 ? \$3.74

This ensures:

- They profit from sending (get back more than they sent)
- I also benefit from the transaction
- We maintain cooperative momentum

```
'''json
{"return": 3.75}
'''
```

reasoning:

Let me analyze this situation:

****What happened in Round 1:****

- I sent \$3, but they saw \$3.17 (communication noise)
- They received \$9.51 (3.17×3)
- They returned \$5.0
- Their visible payoff was $\$9.51 - \$5.0 = \$4.51$
- My payoff was $\$5 - \$3 + \$5 = \7 (but visible earnings show \$7.37, maybe due to noise)

****Current Round 2:****

- They sent me \$2.35 (actual), which became \$7.05
- I need to decide how much to return

****Analysis of their behavior:****

In Round 1, they returned \$5 out of \$9.51 received, which is about 52.6%. This is a fairly generous return - they kept \$4.51 and I got back \$5, so I had a net gain of \$2 from the transaction.

****What should I do?***

They showed cooperative behavior by returning a good amount. Now they've sent me \$2.35, creating \$7.05.

If I apply a similar reciprocity ratio:

- They returned about 52.6% of what they received
- I received \$7.05, so 52.6% would be about \$3.71

However, I should also consider:

- They sent \$2.35 this round (less than the ~\$3.17 they saw me send)
- But they were generous with their return last round
- I want to maintain cooperation

A fair approach might be to:

1. Return enough that they profit from sending
2. Keep enough that I also profit
3. Signal continued cooperation

If they sent \$2.35 and expect around a 50% return rate:

- They might expect back around \$3-4

Let me think about payoffs:

- They started with \$5
- They sent \$2.35
- If I return \$3.5, they end with \$6.15 and I keep \$3.55
- If I return \$4, they end with \$6.65 and I keep \$3.05
- If I return \$3, they end with \$5.65 and I keep \$4.05

Since they matched my generosity last round with a 52.6% return, I should reciprocate similarly to maintain the cooperative dynamic...

usage:

```
{
  "input_tokens": 852,
  "output_tokens": 862,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.17 sent, it became \$9.51, returned \$5.0, visible payoff \$4.51.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.71 returned, visible payoff \$5.71.

Your total visible earnings across all rounds are \$10.670000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our history together:

****Round 1:****

- I sent \$3, they received ~\$9.51, returned \$5.00 (very generous ~53% return)

****Round 2:****

- They sent \$2.35, I received \$7.05, returned \$3.75
- They saw \$3.71 returned due to noise (~41% visible return rate)

****Analysis:****

- This co-player was quite generous in Round 1
- They sent a bit less in Round 2, possibly because they saw a lower return rate from me (due to noise)
- Overall, they've demonstrated cooperative behavior

****My strategy:****

I'll send \$3 again to:

- Match the cooperative level from Round 1
- Show consistency in my approach
- Signal that I'm willing to trust despite the noise in Round 2

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze the history with this co-player:

****Round 1 (I was SENDER):****

- I sent \$3 (they saw \$3.17 due to noise)
- They received \$9.51 and returned \$5.00
- This was generous - about 52.6% return rate

****Round 2 (I was RECEIVER):****

- They sent \$2.35 (actual), I received \$7.05
- I returned \$3.75
- They sent \$3.0 actual, became \$9.0, they saw \$3.71 returned
- So they saw a return rate of about 41.2% (3.71/9.0)
- This is less than what they gave me in Round 1

****Pattern:****

- This co-player was very cooperative in Round 1
- In Round 2, they sent a bit less (\$2.35 vs my \$3)
- They saw me return less proportionally than they did (41.2% vs 52.6%)

****Current decision:****

I'm the sender again. This co-player has shown:

1. Initial high cooperation (52.6% return)
2. Slightly reduced sending in Round 2 (\$2.35 vs \$3)
3. They may perceive me as less generous than they were (due to noise showing only \$3.71 returned when I sent \$3.75)

I should:

- Maintain cooperation since they were generous in Round 1
- Send a similar amount to Round 1 to show consistency
- Perhaps send \$3 again to match their Round 2 amount

Given they've been generally cooperative and we're building a pattern, I'll send \$3 again.

usage:

```
{
  "input_tokens": 1212,
  "output_tokens": 605,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.17 sent, it became \$9.51, returned \$5.0, visible payoff \$4.51.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.71 returned, visible payoff \$5.71.
- Round 3 against you, as RECEIVER: saw \$2.22 sent, it became \$6.66, returned \$3.33, visible payoff \$3.33.

Your total visible earnings across all rounds are \$16.970000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$2.62, so you received \$7.86. How much do you return? (0-7.86)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at our continued interaction:

****Round 1:**** They returned \$5.00 from \$9.51 (53%)

****Round 3:**** They returned \$3.33 from \$6.66 (50%)

****Current situation:****

- They sent \$2.62, I received \$7.86
- They've been consistently cooperative, returning around 50-53%

****My decision:****

I should match their cooperative approach. Returning about 53% would be:

- 53% of \$7.86 ? \$4.17

This ensures:

- They profit significantly (\$4.17 back vs \$2.62 sent = +\$1.55 net)
- I also benefit (\$3.69 kept from the tripled amount)
- We maintain positive reciprocity

```
'''json
{"return": 4.20}
'''
```

Reasoning:

Looking at the history with this co-player:

****Round 1:****

- I sent \$3, they saw \$3.17 (noise), received \$9.51
- They returned \$5.00 (~53% return rate)

```

**Round 2:**
- They sent $3.00, became $9.00
- I returned $3.75, they saw $3.71
- Visible return rate for them: ~41%

**Round 3:**
- I sent $3, they saw $2.22 (significant noise), received $6.66
- They returned $3.33 (50% return rate)

**Current Round 4:**
- They sent $2.62, I received $7.86

**Analysis:**
- This co-player has been consistently cooperative
- Round 1: 53% return
- Round 3: 50% return
- Their sending amounts haven't decreased dramatically despite seeing noise

**My approach:**
I should maintain a similar return rate. They've been returning around 50-53%.
- 50% of $7.86 = $3.93
- 53% of $7.86 = $4.17

I'll return around $4.00 to:
- Match their cooperative behavior
- Ensure they profit from sending ($2.62 sent, $4.00 back)
- Keep the relationship positive
- I keep $3.86, they get $4.00 back vs $2.62 sent = net gain of $1.38 for them

Actually, let me be a bit more generous to match their Round 1 behavior. I'll return $4.20, which is about 53% of what I received.

Their final payoff: $5 - $2.62 + $4.20 = $6.58
My final payoff: $7.86 - $4.20 = $3.66
...
Usage: {
  "input_tokens": 1649,
  "output_tokens": 650,
  "reasoning_tokens": 0
}

```

ROUND 5

Roles: Agent_1=Sender, Agent_2=Receiver, Agent_3=Sender, Agent_4=Receiver, Agent_5=Sender, Agent_6=Receiver, Agent_7=Sender, Agent_8=Receiver

Pairings:

```

dyad_1: Agent_1=Sender, Agent_2=Receiver
dyad_2: Agent_3=Sender, Agent_4=Receiver
dyad_3: Agent_5=Sender, Agent_6=Receiver
dyad_4: Agent_7=Sender, Agent_8=Receiver

```

Parsed game decisions:

```

dyad_1: Agent_1 sent $4.5; received $13.5; Agent_2 returned $7.5; payoffs Agent_1=$8, Agent_2=$6;
noise/visible: sent visible=$5, received visible=$15, returned visible=$7.07
  Action validation: {"returned": {"amount": 7.5, "clamped": false, "max": 13.5, "min": 0, "raw": 7.5}, "sent": {"amount": 4.5, "clamped": false, "max": 5, "min": 0, "raw": 4.5}}
dyad_2: Agent_3 sent $3; received $9; Agent_4 returned $3; payoffs Agent_3=$5, Agent_4=$6;
noise/visible: sent visible=$2.38, received visible=$7.14, returned visible=$3.54
  Action validation: {"returned": {"amount": 3.0, "clamped": false, "max": 9.0, "min": 0, "raw": 3.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_3: Agent_5 sent $3; received $9; Agent_6 returned $6; payoffs Agent_5=$8, Agent_6=$3;
noise/visible: sent visible=$3.62, received visible=$10.86, returned visible=$5.97
  Action validation: {"returned": {"amount": 6.0, "clamped": false, "max": 9.0, "min": 0, "raw": 6.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_7 sent $3; received $9; Agent_8 returned $6; payoffs Agent_7=$8, Agent_8=$3;
noise/visible: sent visible=$3.72, received visible=$11.16, returned visible=$5.04
  Action validation: {"returned": {"amount": 6.0, "clamped": false, "max": 9.0, "min": 0, "raw": 6.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
Balances after round: {

```

```

"Agent_1": 33.2,
"Agent_2": 28.8,
"Agent_3": 28.0,
"Agent_4": 27.0,
"Agent_5": 28.6,
"Agent_6": 26.4,
"Agent_7": 30.38,
"Agent_8": 24.62
}
Visible balances after round: {
  "Agent_1": 30.5,
  "Agent_2": 32.15,
  "Agent_3": 26.11,
  "Agent_4": 26.290000000000003,
  "Agent_5": 31.189999999999998,
  "Agent_6": 29.189999999999998,
  "Agent_7": 27.67,
  "Agent_8": 24.02
}

```

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T08:12:24
 Model: anthropic/claude-sonnet-4.5
 Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 5,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)

- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to establish a cooperative approach while being somewhat cautious since I don't have any history with my current co-player.

Sending a moderate amount signals trust and gives us both the opportunity to benefit from the 3x multiplier, while not leaving me completely vulnerable.

I'll send **\$3**. This gives my co-player \$9 to work with, and if they return a fair share, we can both end up better off than our starting positions.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game. I'm the sender and I have \$5. I need to decide how much to send to the receiver.

Let me think about the strategic considerations:

1. If I send \$X, the receiver gets \$3X
2. The receiver then decides how much to return
3. I keep \$5 - X + (amount returned)
4. The receiver keeps \$3X - (amount returned)

This is a classic trust game. The optimal outcome for both of us would be if I send the full \$5, it gets tripled to \$15, and the receiver returns a fair share (say, at least \$7.50, which would give me \$7.50 and them \$7.50, both better off than starting positions).

However, there are risks:

- The receiver might not return anything or very little
- There's also mentioned "communication noise" - amounts may differ from what was actually sent/returned

Since this is round 1 and I have no history with this player:

- I could play it safe and send less
- Or I could signal trust and cooperation by sending more

In standard trust game experiments, people often send around \$2-\$3 in the first round as a moderate trust signal.

However, if I'm thinking about maximizing joint welfare and establishing cooperation, send

ing more could be beneficial. If the receiver is rational and cooperative, they should return enough to make it worthwhile for me.

Let me consider a middle-ground strategy: sending \$3 or \$4.

- If I send \$3, receiver gets \$9. If they return \$4-5, we both benefit
- If I send \$4, receiver gets \$12. If they return \$6, we both benefit significantly

Given this is a multi-round game with potentially different partners each time, establishing a pattern of trust and cooperation could be beneficial. I'll go with sending \$3 as a balanced approach - showing trust but not complete vulnerability.

Actually, let me reconsider. The instructions mention this is multi-agent with 8 agents and randomized pairings. This means:

...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 648,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.63 sent, it became \$10.89, returned \$5.0, visible payoff \$5.890000000000001.

Your total visible earnings across all rounds are \$7.52.

This round, you are the RECEIVER against your current co-player. They sent you \$3.14, so you received \$9.42. How much do you return? (0-9.42)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

- Round 1: My co-player saw \$3.63 sent, received \$10.89, and returned \$5.0 to me - showing cooperative behavior
- Round 2: They sent me \$3.14, which became \$9.42

They've demonstrated fairness by returning a reasonable amount in Round 1. I want to reciprocate this cooperative approach.

I'll return **\$4.50**, which:

- Gives them a profit (they sent ~\$3.14, get back \$4.50)
- Leaves me with a profit too (\$9.42 - \$4.50 = \$4.92)
- Maintains the cooperative relationship

```
'''json
{"return": 4.50}
'''
```

reasoning:

Let me understand what happened:

Round 1: I sent \$3, but my co-player saw \$3.63 (due to communication noise). This became \$10.89 for them. They returned \$5.0.

My actual payoff in Round 1:

- I had \$5
- I sent \$3
- I received back \$5
- So I kept: \$5 - \$3 + \$5 = \$7

But the visible payoff shows \$7.52, which suggests there might be some noise in the amounts.

Now in Round 2:

- I'm the RECEIVER
- My co-player sent me some amount that appears as \$3.14 to me
- This became \$9.42 (which is \$3.14 × 3)

- I need to decide how much to return

Looking at their behavior: In Round 1, they saw \$3.63 sent, which became \$10.89, and they returned \$5.0. That's about 46% of what they received, or about 1.38x what they saw as sent.

This seems like a fair/cooperative player. They returned a reasonable amount.

Now I have \$9.42. What should I return?

Given that:

1. They sent a decent amount (\$3.14, which is about 63% of their \$5)
2. They were fair in the previous round, returning \$5 when they received \$10.89
3. I want to encourage continued cooperation

I should return a fair amount. Some options:

- Return about half of what I received: \$4.71
- Return what they sent plus a bit extra: around \$4-5
- Match their return rate from last round (46% of received): about \$4.33

I think returning around \$4.50-\$5.00 would be fair and encourage continued cooperation. Let me return \$4.50 as a fair split that gives them back close to what they sent plus some profit, while I also profit.

If I return \$4.50, they end up with \$6.36 while I keep \$4.92-both of us gain from the exchange. Returning \$5.00 would mirror...

usage:

```
{
  "input_tokens": 768,
  "output_tokens": 820,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.63 sent, it became \$10.89, returned \$5.0, visible payoff \$5.890000000000001.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.06 returned, visible payoff \$6.56.

Your total visible earnings across all rounds are \$12.44.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our history together:

Round 1: I sent (they saw \$3.63), they returned \$5.0 - cooperative

Round 2: They sent \$3.5 (I saw \$3.14), I returned (they saw \$5.06) - both cooperative

We've established a good cooperative pattern. They've been generous as both sender and receiver, so I want to continue building this trust.

I'll send **\$4** to show strong trust and maintain our mutually beneficial relationship.

```
'''json
{"send": 4}
'''
```

reasoning:

Let me analyze the history with my current co-player:

Round 1: I was SENDER

- I sent \$3 (but they saw \$3.63 due to noise)
- They received \$10.89
- They returned \$5.0
- This was cooperative behavior

Round 2: I was RECEIVER
- They sent \$3.5 (but I saw \$3.14 due to noise)
- I received \$9.42
- I returned \$4.50 (but they saw \$5.06 due to noise)
- They had a visible payoff of \$6.56

So far, this co-player has been quite cooperative:
- As receiver in Round 1: returned \$5.0 from \$10.89 (about 46%)
- As sender in Round 2: sent \$3.5 out of \$5 (70%)

I should continue this cooperative relationship. In Round 1, I sent \$3. In Round 2, they sent \$3.5. I should probably send a similar or slightly higher amount to maintain cooperation.

I'll send \$3.5 or \$4 to show continued trust and cooperation.

Given that they returned fairly and sent generously, I'll send \$4 to demonstrate strong trust and cooperation.

usage:

```
{
  "input_tokens": 1118,
  "output_tokens": 421,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.63 sent, it became \$10.89, returned \$5.0, visible payoff \$5.890000000000001.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.06 returned, visible payoff \$6.56.
- Round 3 against you, as RECEIVER: saw \$3.64 sent, it became \$10.92, returned \$5.5, visible payoff \$5.42.

Your total visible earnings across all rounds are \$18.1.

This round, you are the RECEIVER against your current co-player. They sent you \$3.21, so you received \$9.629999999999999. How much do you return? (0-9.629999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at our established pattern:

Round 1: They returned \$5.0 when receiving \$10.89

Round 2: I returned (they saw \$5.06) when I received \$9.42

Round 3: They returned \$5.5 when receiving \$10.92 - showing increased generosity

Round 4: They sent \$3.21, I received \$9.63

They've been consistently cooperative and even increased their returns. I want to reciprocate this positive trend.

I'll return **\$4.80**, which rewards their trust while maintaining mutual benefit.

```
'''json
{"return": 4.80}
'''
```

reasoning:

Looking at the history with my current co-player:

Round 1: They (as receiver) saw \$3.63 sent, received \$10.89, returned \$5.0

Round 2: They (as sender) sent \$3.5, received (saw \$5.06) back

Round 3: They (as receiver) saw \$3.64 sent, received \$10.92, returned \$5.5

Round 4: They (as sender) sent \$3.21, I received \$9.63

Our pattern:

- Round 1: I sent (they saw 3.63), they returned \$5.0
- Round 2: They sent 3.5 (I saw 3.14), I returned (they saw 5.06)
- Round 3: I sent (they saw 3.64), they returned \$5.5

- Round 4: They sent 3.21, I received \$9.63

They've been consistently cooperative and fair. In Round 3, they returned \$5.5, showing increased generosity.

For Round 4, they sent \$3.21, which became \$9.63 for me.

To maintain the cooperative relationship and be fair:

- They invested ~\$3.21
- I should return enough that they profit, while I also profit
- A return of around \$4.50-\$5.00 would be fair

I'll return \$4.80 to show continued cooperation and slightly increase the return to match their increased generosity in Round 3.

usage:

```
{
  "input_tokens": 1484,
  "output_tokens": 500,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.06 returned, visible payoff \$6.56.
- Round 3 against you, as RECEIVER: saw \$3.64 sent, it became \$10.92, returned \$5.5, visible payoff \$5.42.
- Round 4 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.28 returned, visible payoff \$6.78.

Your total visible earnings across all rounds are \$22.93.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at our continued cooperation:

- Round 2: They sent \$3.5, I returned (they saw \$5.28)
- Round 3: I sent (they saw \$3.64), they returned \$5.5 - very generous
- Round 4: They sent \$3.5, I returned (they saw \$5.28)

This co-player has been consistently trustworthy and generous. They reliably send around \$3.5 and return well when receiving. Our relationship is strongly cooperative.

I'll send **\$4.5** to show high trust and continue our mutually beneficial pattern.

```
'''json
{"send": 4.5}
'''
```

Reasoning:

Looking at our history:

- Round 2: They sent \$3.5, I returned (they saw \$5.06)
- Round 3: I sent (they saw \$3.64), they returned \$5.5
- Round 4: They sent \$3.5, I returned (they saw \$5.28)
- Round 5: Now I'm sender

We've established a strong cooperative pattern. They've been consistently sending around \$3.5 and returning generously when receiver. I've been returning well too.

Given this trust and positive relationship, I should continue with a generous send. Last time I sent (Round 3), they returned \$5.5 on receiving \$10.92, which was very fair.

I'll send \$4 or even \$5 to show maximum trust. Given the strong cooperation, I'll go with \$4.5 as a strong signal of trust while keeping a small cushion.

Usage: {

```
"input_tokens": 1846,
"output_tokens": 366,
"reasoning_tokens": 0
```

```
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T08:12:36

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",

"dyad_id": "dyad_1",

"move_index": 1,

"opponent_id": "Agent_1",

"pairing": {

"agents": [

"Agent_1",

"Agent_2"

],

"investor": "Agent_1",

"roles": {

"Agent_1": "investor",

"Agent_2": "trustee"

},

"trustee": "Agent_2"

},

"role": "trustee",

"role_label": "Receiver",

"round": 5,

"task": "game",

"task_index": 0

```
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized

- , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.63, that is 72.6% of their total endowment. You received \$10.89. How much do you return to the sender? (0-10.89)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.63 (72.6% of their endowment)
- I received \$10.89 after the 3x multiplier
- A fair return should reward their trust while ensuring both of us benefit

The sender gave up \$3.63, keeping only \$1.37. Their generosity created \$10.89 total value for me. To reciprocate fairly and encourage future cooperation:

If I return \$5.00:

- Sender ends with: $\$5 - \$3.63 + \$5.00 = \6.37 (gain of \$1.37)
- I end with: $\$10.89 - \$5.00 = \$5.89$ (gain of \$5.89)

This seems like a fair split that rewards their trust while both of us benefit from the cooperation.

```
{"return": 5.00}
```

reasoning:

Let me analyze this situation carefully.

Given information:

- I'm the RECEIVER in Round 1
- The sender sent \$3.63 (which is 72.6% of their \$5 endowment)
- I received \$10.89 (which is $\$3.63 \times 3$)
- I need to decide how much to return (between \$0 and \$10.89)

Key considerations:

1. The sender was quite generous, sending 72.6% of their endowment
2. This shows trust and cooperation
3. There's mention of communication noise, so the actual amount sent might differ slightly from what I see
4. This is a multi-round game with 8 agents, so building reputation for cooperation could be beneficial
5. Fair reciprocation would be important to encourage future cooperation

Let me think about what would be a fair return:

- The sender gave up \$3.63 and kept \$1.37
- I received \$10.89
- If I want to be reciprocal and fair, I should consider:
 - The sender showed high trust (72.6% is a lot)
 - A common strategy is to return enough so both parties benefit
 - Returning about half (\$5.45) would give me \$5.44 and the sender would end up with $\$1.37 + \$5.45 = \$6.82$
 - This would mean the sender gains \$1.82 total and I gain \$5.44

Let me reconsider:

- If I return around \$4-5, that seems fair
- Returning \$5 would give:
 - Sender: $\$5 - \$3.63 + \$5 = \6.37 (net gain of \$1.37)
 - Me: $\$10.89 - \$5 = \$5.89$ (net gain of \$5.89)

-...

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 801,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.52 returned, visible payoff \$7.52.

Your total visible earnings across all rounds are \$5.8900000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the situation:

From the visible history, my co-player:

- Sent \$3.0 in Round 1 (60% of their endowment)
- Saw \$5.52 returned (61.3% of the \$9.0)
- Made a profit of \$2.52

This shows they were cooperative and returned a substantial amount. To reciprocate their trust and maintain cooperation, I should send a generous amount.

I'll send \$3.50 (70% of my endowment), showing strong trust and willingness to cooperate.

{"send": 3.50}

reasoning:

Let me analyze this situation:

Round 1 (from my perspective):

- I was the RECEIVER
- The sender sent \$3.63 (72.6% of endowment)
- I received \$10.89
- I returned \$5.00
- My payoff: $\$10.89 - \$5.00 = \$5.89$
- Sender's payoff: $\$5 - \$3.63 + \$5.00 = \6.37

But now I see the "visible history" which shows a different perspective:

- My co-player (who was the sender in Round 1) saw that they sent \$3.0
- It became \$9.0 (not \$10.89 as I received)
- They saw \$5.52 returned (not \$5.00 as I sent)
- Their visible payoff was \$7.52

This confirms there is communication noise in the game as mentioned in the instructions.

What actually happened vs what they saw:

- They actually sent \$3.63, but saw they sent \$3.0
- I received \$10.89, they saw it became \$9.0
- I returned \$5.00, they saw \$5.52 returned
- Their actual payoff was \$6.37, but they saw \$7.52

Now in Round 2, I'm the SENDER and they are the RECEIVER.

My co-player returned a substantial amount in Round 1 (they saw they returned \$5.52 out of \$9.0 received, which is 61.3%). This shows they are cooperative.

Given that:

1. They showed trust by being generous in Round 1
2. They appeared to return a fair amount (from their perspective, 61.3% of what they received)
3. I want to maintain cooperation and reciprocate their trust

I should send a generous amount. ...

usage:

```
{
  "input_tokens": 874,
  "output_tokens": 716,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.52 returned, visible payoff \$7.52.
- Round 2 against you, as RECEIVER: saw \$3.14 sent, it became \$9.42, returned \$4.5, visible payoff \$4.92.

Your total visible earnings across all rounds are \$12.45.

This round, you are the RECEIVER against your current co-player. They sent you \$3.64, so you received \$10.92. How much do you return? (0-10.92)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the visible history:

Round 1: My co-player sent \$3.0 (60%) and saw \$5.52 returned

Round 2: They returned \$4.5 when they received \$9.42 (47.8%)

Round 3: They sent \$3.64 (72.8% - even more generous!)

My co-player has been consistently cooperative and just showed increased trust by sending an even larger percentage. I should reciprocate fairly.

Returning \$5.50:

- Sender gets: $\$5 - \$3.64 + \$5.50 = \6.86 (profit of \$1.86)
- I get: $\$10.92 - \$5.50 = \$5.42$ (profit of \$5.42)

This ~50% return rate rewards their trust while ensuring mutual benefit.

{"return": 5.50}

reasoning:

Let me analyze the visible history:

Round 1: My co-player was SENDER

- They sent \$3.0 (60% of endowment)
- It became \$9.0
- They saw \$5.52 returned (61.3% return rate)
- Their visible payoff: \$7.52

Round 2: My co-player was RECEIVER

- They saw \$3.14 sent (I actually sent \$3.50 but there's noise)
- It became \$9.42
- They returned \$4.5 (47.8% of what they received)
- Their visible payoff: \$4.92

Now Round 3: I am RECEIVER

- They sent \$3.64 (72.8% of their endowment - very generous!)
- I received \$10.92
- I need to decide how much to return

Looking at the pattern:

- In Round 1, they sent 60% and I returned about 61% (assuming noise in what they saw)
- In Round 2, they received and returned 47.8%
- In Round 3, they sent 72.8% - showing even more trust

The co-player has been consistently cooperative. They sent a high amount both times they were sender (60% and 72.8%). When they were receiver, they returned a reasonable amount (47.8%).

To maintain reciprocity and reward their increased trust, I should return a fair amount.

If I return \$5.50:

- Sender ends with: $\$5 - \$3.64 + \$5.50 = \6.86 (gain of \$1.86)
- I end with: $\$10.92 - \$5.50 = \$5.42$ (gain of \$5.42)

This is about 50.4% return rate, which is reasonable and shows I'm reciprocating their trust while both of us profit.

...
usage:

```
{  
  "input_tokens": 1200,  
  "output_tokens": 712,  
  "reasoning_tokens": 0  
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.52 returned, visible payoff \$7.52.
- Round 2 against you, as RECEIVER: saw \$3.14 sent, it became \$9.42, returned \$4.5, visible payoff \$4.92.
- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$4.66 returned, visible payoff \$5.66.

Your total visible earnings across all rounds are \$17.869999999999997.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at the visible history:

Round 1: Co-player sent \$3.0 (60%), saw \$5.52 returned

Round 2: Co-player returned \$4.5 (47.8%)

Round 3: Co-player sent \$4.0 (80% - very generous!), saw \$4.66 returned

My co-player has been consistently cooperative, even increasing their trust in Round 3 by sending 80% of their endowment. While they saw a lower return percentage in Round 3, they've maintained a cooperative pattern overall.

I should reciprocate their trust and continue the cooperative relationship.

```
{"send": 3.50}
```

reasoning:

Let me analyze the visible history:

Round 1: Co-player sent \$3.0 (60%), saw \$5.52 returned (61.3% of \$9.0)

Round 2: Co-player returned \$4.5 when they received \$9.42 (47.8%)

Round 3: Co-player sent \$4.0 (80%!), saw \$4.66 returned (38.8% of \$12.0)

This is concerning. In Round 3, my co-player sent even more (\$4.0, or 80%), but they only saw \$4.66 returned, which is only 38.8% of the \$12.0 they sent. That's a lower return rate than before.

Wait, let me check what I actually sent and returned:

- Round 2: I sent \$3.50 according to my response
- Round 3: I returned \$5.50 according to my response

But the visible history shows:

- Round 2: They saw \$3.14 sent (I sent \$3.50, but due to noise they saw \$3.14)
- Round 3: They saw \$4.66 returned (I returned \$5.50, but due to noise they saw \$4.66)

So there's communication noise! This is important.

From my co-player's perspective:

- They've been very generous (sending 60%, 80%)
- But the returns they're seeing are declining in percentage terms (61.3%, then 38.8%)
- They might be feeling like their generosity isn't being reciprocated

However, I know I actually returned \$5.50 in Round 3, which was generous. The noise just made it look like \$4.66 to them.

Still, this relationship has been cooperative overall. My co-player has consistently sent substantial amounts. Even though they saw a lower return in Round 3, they've maintained

```

    trust.
    ...
usage:
{
  "input_tokens": 1626,
  "output_tokens": 638,
  "reasoning_tokens": 0
}

```

Message 10 role=user
content:
 Round 5

Visible history before this round:
 Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$3.14 sent, it became \$9.42, returned \$4.5, visible payoff \$4.92.
- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$4.66 returned, visible payoff \$5.66.
- Round 4 against you, as RECEIVER: saw \$3.21 sent, it became \$9.629999999999999, returned \$4.8, visible payoff \$4.829999999999999.

Your total visible earnings across all rounds are \$24.65.

This round, you are the RECEIVER against your current co-player. They sent you \$5, so you received \$15. How much do you return? (0-15)
 Respond exactly as JSON: {'return': <amount>}

Assistant output:
 Looking at the visible history:

Round 2: Co-player returned \$4.5 (47.8%)
 Round 3: Co-player sent \$4.0 (80%), saw modest return
 Round 4: Co-player returned \$4.8 (49.8%)
 Round 5: Co-player sent \$5.0 (100% - maximum possible trust!)

This is extraordinary! My co-player just sent their ENTIRE endowment, showing complete trust. They're keeping nothing and relying entirely on my reciprocation. I must reward this exceptional trust generously.

Returning \$7.50:
 - Sender gets: \$5 - \$5 + \$7.50 = \$7.50 (profit of \$2.50)
 - I get: \$15 - \$7.50 = \$7.50 (profit of \$7.50)

This 50% return ensures we both profit nicely and maintains our strong cooperative relationship.

{"return": 7.50}

Reasoning:
 Looking at the visible history:

Round 2: Co-player returned \$4.5 when they received \$9.42 (47.8%)
 Round 3: Co-player sent \$4.0 (80%), saw \$4.66 returned (38.8%)
 Round 4: Co-player returned \$4.8 when they received \$9.63 (49.8%)
 Round 5: Co-player sent \$5.0 (100%!!!)

This is remarkable - my co-player just sent their ENTIRE endowment (\$5), showing maximum trust! This is the ultimate act of cooperation in this game. They're completely relying on me to reciprocate fairly.

They gave up \$5, keeping \$0. If I return nothing, they end with \$0. This would be devastating to the relationship.

I need to reciprocate this tremendous trust generously. Let me think about fair returns:

If I return \$7.50:
 - Sender gets: \$5 - \$5 + \$7.50 = \$7.50 (profit of \$2.50)
 - I get: \$15 - \$7.50 = \$7.50 (profit of \$7.50)

This 50% return rate gives us equal final amounts and ensures they make a nice profit for their trust.

Actually, given they sent 100%, I could even be more generous. Let me consider \$8:

- Sender gets: $\$5 - \$5 + \$8 = \8 (profit of \$3)
- I get: $\$15 - \$8 = \$7$ (profit of \$7)

This is still good for both of us and rewards their exceptional trust even more.

I think \$7.50 is fair - it's a 50% return rate, ensures both of us profit nicely, and maintains the cooperative relationship.

```
Usage: {
  "input_tokens": 2010,
  "output_tokens": 660,
  "reasoning_tokens": 0
}
```

GAME - Agent_3 - Sender - dyad_2

Timestamp: 2026-08-12T08:12:42

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_4",
  "pairing": {
    "agents": [
      "Agent_3",
      "Agent_4"
    ],
    "investor": "Agent_3",
    "roles": {
      "Agent_3": "investor",
      "Agent_4": "trustee"
    },
    "trustee": "Agent_4"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 5,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: $\$5 - \text{sent} + \text{returned}$
- Receiver keeps: $\text{received} - \text{returned}$
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: $\$5 - \text{sent} + \text{returned}$
- You keep: $\text{received} - \text{returned}$
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust and caution. The multiplication factor of 3x creates an opportunity for mutual benefit if my co-player reciprocates.

Since this is the first round with no prior history, I'll send a moderate amount that signals cooperation while not exposing myself to maximum risk.

****My decision:****

```
'''json
{"send": 3}
'''
```

This gives my co-player \$9 to work with (3×3), leaving me with \$2. If they return a fair share, we can both benefit from this cooperation. Let's see how they respond.

reasoning:

This is the first round of a trust game where I'm the SENDER. Let me think about this strategically.

Key facts:

- I have \$5
- I can send \$0-\$5 to the receiver
- Amount sent is multiplied by 3x
- The receiver then decides how much to return to me
- I keep: \$5 - sent + returned
- Receiver keeps: received - returned
- There may be communication noise (amounts seen may differ from actual)
- This is a multi-agent setting with 8 agents total
- Pairings are randomized each round
- Agent names are hidden

Strategic considerations:

1. ****Trust and cooperation****: If both parties cooperate optimally, we can maximize joint payoffs. If I send all \$5, the receiver gets \$15. If they return half (\$7.50), I end up with \$7.50 and they keep \$7.50, which is better for both of us than if I send nothing.
2. ****Risk****: The receiver could keep everything and return nothing. If I send \$5 and get nothing back, I end up with \$0 while they get \$15.
3. ****First round****: This is the first round, so I have no reputation or history with this particular co-player. However, my actions might influence future interactions in a multi-round setting.
4. ****Communication noise****: The presence of noise means that even if someone intends to be fair, the amounts might appear different, which could lead to misunderstandings.
5. ****Signaling****: Sending a moderate to high amount signals trust and cooperation, which might encourage reciprocity.

Common strategies in trust games:

- Send half (\$2.50) as a moderate trust signal
- Send all (\$5) to maximize joint payoff and signal full trust
- Send a smaller amount (\$1-2) to test the waters
- Send nothing (\$0) to avoid risk entirely

Given this is round 1 and I want to establish a cooperative tone while not being completely naive, I think sending a significant but not maximum amount makes sense.

Research on trust games suggests most people send positive amounts initially, and receivers typically reciprocate. The 3x multiplier creates substantial mutual benefit potential—even modest returns from the receiver would leave us both better off than the starting position. I'll send \$3 as a balanced signal of trust and good faith without overexposing myself....

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 870,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.91 sent, it became \$8.73, returned \$3.5, visible payoff \$5.23.

Your total visible earnings across all rounds are \$5.5600000000000005.

This round, you are the RECEIVER against your current co-player. They sent you \$2.77, so you received \$8.31. How much do you return? (0-8.31)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

****Round 1 analysis:****

- I sent \$3 (they saw \$2.91 due to noise)
- They received \$8.73 and returned \$3.5 (about 40% return rate)
- This was a cooperative response

****Current situation (Round 2):****

- They sent me \$2.77 (with noise)
- I received \$8.31
- They showed good faith cooperation in Round 1

I'll reciprocate their cooperative behavior with a similar return rate:

```
'''json
{"return": 3.5}
'''
```

This gives them a total of approximately \$5.73 and leaves me with \$4.81, creating mutual benefit from our cooperation.

reasoning:

Let me analyze this situation:

Round 1 (just completed):

- I was the SENDER and sent \$3
- Due to communication noise, my co-player saw \$2.91 (close to \$3)
- It became \$8.73 for them (should be \$8.73 if 2.91×3)
- They returned \$3.5
- Their visible payoff: \$5.23 (they kept $8.73 - 3.5 = 5.23$)
- My payoff: $5 - 3 + 3.5 = 5.56$

Now Round 2:

- I am the RECEIVER
- My co-player sent me an amount, but I see \$2.77 (with noise)

- I received \$8.31 (which is $\$2.77 \times 3$)
- I need to decide how much to return

Analysis:

- In Round 1, my co-player returned \$3.5 out of \$8.73, which is about 40% return rate
- They kept \$5.23 for themselves
- I sent them \$3 (they saw \$2.91), and I ended up with \$5.56 total
- So I gained \$0.56 from the transaction

Now they sent me approximately \$2.77 (actual amount might be slightly different due to noise).

If I apply a similar return rate (around 40%), that would be: $\$8.31 \times 0.4 \approx \3.32

But let me think about fairness:

- They sent around \$2.77, leaving them with about \$2.23
- I received \$8.31
- A fair split might give them back enough so they end up with a reasonable amount

If I return \$3.5 (same as they returned to me):

- They end up with: $\$5 - \$2.77 + \$3.5 = \5.73
- I end up with: $\$8.31 - \$3.5 = \$4.81$

Returning \$4 would give them \$6.23 while I get \$4.31. Returning \$3 would leave me with \$5.31 and them with \$5.23. Since they gave me roughly \$2.77 and returned about 40% of what they received, matching that reciprocal gesture with a \$3.5 return seems appropriate....

usage:

```
{
  "input_tokens": 784,
  "output_tokens": 826,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.91 sent, it became \$8.73, returned \$3.5, visible payoff \$5.23.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.9 returned, visible payoff \$4.9.

Your total visible earnings across all rounds are \$10.370000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our history:

****Round 1:**** I sent \$3, they returned \$3.5 - cooperative

****Round 2:**** They sent \$3, I returned \$3.5 (they saw \$2.9) - cooperative

We've established a mutually beneficial pattern where we both send moderate amounts and return fair shares. I'll continue this cooperation:

```
'''json
{"send": 3}
'''
```

This maintains trust and keeps our successful collaborative pattern going.

reasoning:

Let me analyze the history:

****Round 1:****

- I sent \$3 (they saw \$2.91 due to noise)
- They received \$8.73 and returned \$3.5
- I ended up with $\$2 + \$3.5 = \$5.5$

****Round 2:****

- They sent \$3.0 (I saw \$2.77 due to noise, actual was \$3.0)
- It became \$9.0 for me
- I returned \$3.5
- They saw \$2.9 returned (due to noise)
- Their visible payoff was \$4.9 (\$5 - \$3 + \$2.9)
- I kept \$9.0 - \$3.5 = \$5.5

My total visible earnings: \$10.37 which matches \$5.5 + \$4.81 approximately (there's some rounding)

So the pattern so far:

1. Both of us have been sending moderate amounts (\$3)
2. Both of us have been returning reasonable amounts (\$3.5)
3. This has been mutually beneficial

The relationship seems cooperative. They sent me \$3 in Round 2, which is the same amount I sent in Round 1. They saw I returned \$2.9 (actual \$3.5), which was a fair return.

I should continue this cooperative pattern. Sending \$3 again seems appropriate to maintain the reciprocal relationship.

usage:

```
{
  "input_tokens": 1128,
  "output_tokens": 446,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.91 sent, it became \$8.73, returned \$3.5, visible payoff \$5.23.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.9 returned, visible payoff \$4.9.
- Round 3 against you, as RECEIVER: saw \$3.52 sent, it became \$10.56, returned \$4.5, visible payoff \$6.0600000000000005.

Your total visible earnings across all rounds are \$16.15.

This round, you are the RECEIVER against your current co-player. They sent you \$2.64, so you received \$7.92. How much do you return? (0-7.92)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at our cooperative history:

```
**Round 1:** They returned $3.5 on $8.73
**Round 2:** I returned $3.5 on $9.0
**Round 3:** They returned $4.5 on $10.56 (increased generosity!)
**Round 4:** They sent me $2.64, I received $7.92
```

They've been consistently cooperative and even increased their generosity in Round 3. I'll reciprocate fairly:

```
'''json
{"return": 3.5}
'''
```

This maintains our mutually beneficial cooperation pattern.

reasoning:

Let me analyze the history with my current co-player:

```
**Round 1:**
- I sent $3 (they saw $2.91 due to noise)
- They received $8.73, returned $3.5
- Good cooperation

**Round 2:**
- They sent $3
```

- I received \$9.0, returned \$3.5 (they saw \$2.9)
- Continued cooperation

****Round 3:****

- I sent \$3 (they saw \$3.52 due to noise)
- They received \$10.56, returned \$4.5
- They increased their return, showing even more generosity

****Round 4 (current):****

- They sent \$2.64 (with noise, could have been around \$3 originally)
- I received \$7.92

Looking at the pattern:

- They've been consistently cooperative
- In Round 3, they returned \$4.5 on \$10.56 (about 43% return rate)
- They're continuing to send around \$3 each time

I should reciprocate their continued cooperation. Given they sent me \$7.92, a fair return would be around 40-45% of what I received, which would be about \$3.2-\$3.6.

I'll return \$3.5 to maintain consistency with our cooperative pattern.

usage:

```
{
  "input_tokens": 1467,
  "output_tokens": 452,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.9 returned, visible payoff \$4.9.
- Round 3 against you, as RECEIVER: saw \$3.52 sent, it became \$10.56, returned \$4.5, visible payoff \$6.0600000000000005.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.96 returned, visible payoff \$5.96.

Your total visible earnings across all rounds are \$20.57.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at our established pattern:

****Round 2-4:**** They consistently sent \$3
****Round 3:**** They were particularly generous, returning \$4.5
****Round 4:**** I returned \$3.5 (they saw \$3.96)

We have a strong cooperative relationship. I'll maintain our successful pattern:

```
'''json
{"send": 3}
'''
```

This continues our mutually beneficial cooperation.

Reasoning:

Looking at the history with this co-player:

Round 1: I sent \$3 (they saw \$2.91), they returned \$3.5 - cooperative
 Round 2: They sent \$3, I returned \$3.5 (they saw \$2.9) - cooperative
 Round 3: I sent \$3 (they saw \$3.52), they returned \$4.5 - very generous!
 Round 4: They sent \$3, I returned \$3.5 (they saw \$3.96) - cooperative

This co-player has been consistently cooperative and even showed extra generosity in Round 3.

The pattern shows:

- They keep sending \$3
- They've returned generously when receiving

- We have a strong cooperative relationship

I should continue the cooperation. Sending \$3 has worked well for us.

```
Usage: {  
  "input_tokens": 1824,  
  "output_tokens": 291,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_4 - Receiver - dyad_2

Timestamp: 2026-08-12T08:12:52

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_2",  
  "move_index": 3,  
  "opponent_id": "Agent_3",  
  "pairing": {  
    "agents": [  
      "Agent_3",  
      "Agent_4"  
    ],  
    "investor": "Agent_3",  
    "roles": {  
      "Agent_3": "investor",  
      "Agent_4": "trustee"  
    },  
    "trustee": "Agent_4"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 5,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.91, that is 58.2% of their total endowment. You received \$8.73. How much do you return to the sender? (0-8.73)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.91 (58.2% of their endowment)
- I received \$8.73 after the 3x multiplication
- The sender currently has \$2.09 remaining

To reciprocate their trust and cooperation:

- If I return nothing, the sender ends with \$2.09 (worse than keeping everything)
- If I return \$3.50, the sender ends with \$5.59 and I keep \$5.23
- This rewards their trust while maintaining fairness

I'll return a meaningful amount to encourage continued cooperation.

{'return': 3.5}

reasoning:

Let me analyze this situation:

1. As the RECEIVER, the sender sent me \$2.91 (which is 58.2% of their \$5 endowment)
2. This amount was multiplied by 3x, so I received \$8.73
3. I need to decide how much to return (between \$0 and \$8.73)

Key considerations:

- The sender was quite generous, sending nearly 60% of their endowment
- This shows trust and cooperation
- In trust games, reciprocity is important for maintaining cooperation
- The sender's final payoff will be: $\$5 - \$2.91 + \text{whatever I return} = \$2.09 + \text{return}$
- My final payoff will be: $\$8.73 - \text{whatever I return}$

Common approaches in trust games:

1. Split the gains equally - the sender gave up \$2.91 but created \$8.73. The "gain" is $\$8.73 - \$2.91 = \$5.82$. Splitting this equally would mean returning about \$2.91
2. Give the sender back what they sent - return \$2.91
3. Give the sender a good return on their investment - perhaps return more than they sent
4. Equal final payoffs - try to make both players end up with similar amounts

Let me calculate some scenarios:

If I return \$2.91:

- Sender gets: $\$2.09 + \$2.91 = \$5.00$
- I get: $\$8.73 - \$2.91 = \$5.82$
- This is fair-ish, both doing okay

If I return \$3.50:

- Sender gets: $\$2.09 + \$3.50 = \$5.59$
- I get: $\$8.73 - \$3.50 = \$5.23$
- This balances outcomes more evenly

If I return \$4.00:

- Sender gets: $\$2.09 + \$4.00 = \$6.09$
- I get: $\$8.73 - \$4.00 = \$4.73$
- The sender benefits more, which seems appropriate given they initiated the cooperation

For truly equal final payoffs, I'd need to return \$3.32, which would give us both \$5.41 each. The sender demonstrated substantial trust by sending most of their endowment, so I should reciprocate meaningfully to reinforce that cooperation....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 942,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.56 returned, visible payoff \$5.5600000000000005.

Your total visible earnings across all rounds are \$5.23.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the history:

- My co-player sent \$3.0 (60% of their endowment) in Round 1, showing strong trust
- They saw a return of \$3.56 (39.6% of what they sent as receiver)
- This demonstrates cooperative behavior

To reciprocate their trust and maintain cooperation, I'll send a similar proportion:

```
{'send': 3.0}
```

reasoning:

Let me analyze this situation:

Round 1:

- I was the RECEIVER
- The sender sent \$2.91 (58.2% of their endowment)
- I received \$8.73
- I returned \$3.50
- I kept \$5.23

Now in Round 2:

- I'm the SENDER
- I'm playing against the same co-player from Round 1
- The visible history shows that in Round 1, from my co-player's perspective:
 - They sent \$3.0 (as SENDER)
 - It became \$9.0
 - They saw \$3.56 returned
 - Their visible payoff was \$5.56

Wait, there's a discrepancy here. The problem says there's communication noise. Let me understand:

From my perspective in Round 1:

- Sender sent \$2.91
- I received \$8.73
- I returned \$3.50

From my co-player's perspective in Round 1:

- They sent \$3.0
- It became \$9.0
- They saw \$3.56 returned

So the noise affected what they saw vs what actually happened. They sent \$3.0 but I saw \$2.91. I returned \$3.50 but they saw \$3.56.

Now I'm the sender in Round 2. My co-player was cooperative in Round 1 - they sent 60% of

their endowment (\$3.0 out of \$5). They saw a return of \$3.56 out of \$9.0, which is about 39.6% of what I received.

Given that:

1. My co-player showed strong trust by sending 60% in Round 1
2. They saw a decent return (39.6%)
3. I want to maintain cooperation and build trust

I should send a similar proportion to what they sent. They sent 60%, so I should send around \$3.0 as well to match their level of trust and cooperation.

...

usage:

```
{
  "input_tokens": 814,
  "output_tokens": 568,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.56 returned, visible payoff \$5.5600000000000005.
- Round 2 against you, as RECEIVER: saw \$2.77 sent, it became \$8.31, returned \$3.5, visible payoff \$4.8100000000000005.

Your total visible earnings across all rounds are \$10.13.

This round, you are the RECEIVER against your current co-player. They sent you \$3.52, so you received \$10.56. How much do you return? (0-10.56)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the pattern:

****Round 1**:** My co-player sent \$3.0 (60%), showing strong trust

****Round 2**:** I sent \$3.0 (60%), they returned \$3.5 (42% of what they received)

****Round 3**:** They sent \$3.52 (70.4%), showing even greater trust

They're being very cooperative and increasing their trust level. I should reciprocate appropriately.

Returning \$4.5:

- Sender ends with: \$5 - \$3.52 + \$4.5 = \$5.98
- I keep: \$10.56 - \$4.5 = \$6.06

This rewards their high trust while maintaining fairness.

{'return': 4.5}

reasoning:

Let me analyze this situation:

Round 1 (I was RECEIVER):

- Sender sent \$2.91 (58.2%)
- I received \$8.73
- I returned \$3.50
- My payoff: \$5.23

Round 2 (I was SENDER):

- I sent \$3.0
- It became \$9.0
- According to visible history, my co-player saw \$2.77 sent (there's noise!)
- They returned \$3.5
- My visible payoff from Round 2: I sent \$3, got back \$3.5, so \$2 + \$3.5 = \$5.5
- Total earnings so far: \$5.23 + \$5.5 = \$10.73... but it says \$10.13

Wait, let me recalculate. The visible history says:

- Round 2 as RECEIVER: saw \$2.77 sent, it became \$8.31, returned \$3.5, visible payoff \$4.8

1

This is from my co-player's perspective. They saw \$2.77 sent (due to noise), it became \$8.31, they returned \$3.5, keeping \$4.81.

My actual earnings from Round 2:

- I sent \$3.0
- I kept \$2.0
- I received back... well, if they returned \$3.5 based on receiving \$8.31, then I should have received \$3.5 back
- So my payoff: $\$2.0 + \$3.5 = \$5.5$

Total: $\$5.23 + \$5.5 = \$10.73$

But it says my total visible earnings are \$10.13. Hmm, there might be noise in what I received back too.

Anyway, for Round 3:

- I'm RECEIVER
- They sent \$3.52 (70.4% of their endowment)
- I received \$10.56

My partner's been increasingly generous across rounds-first 58.2%, now 70.4%-showing growing confidence. They returned roughly 40-42% in the previous round, so I should match that reciprocal level. Returning between \$4.22-\$4.75 would maintain the cooperative dynamic we've established....

usage:

```
{
  "input_tokens": 1104,
  "output_tokens": 897,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.56 returned, visible payoff \$5.5600000000000005.
- Round 2 against you, as RECEIVER: saw \$2.77 sent, it became \$8.31, returned \$3.5, visible payoff \$4.8100000000000005.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.78 returned, visible payoff \$5.779999999999999.

Your total visible earnings across all rounds are \$16.19.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at the pattern:

- **Round 1**: Co-player sent \$3.0 (60%), demonstrating trust
- **Round 2**: Co-player returned \$3.5, showing cooperation
- **Round 3**: Co-player sent \$3.0 (60%) again, maintaining trust

They're showing consistent cooperative behavior. I should continue matching their trust level:

```
{'send': 3.0}
```

reasoning:

Looking at the history:

- **Round 1**: Co-player sent \$3.0 as SENDER, saw \$3.56 returned
- **Round 2**: Co-player was RECEIVER, saw \$2.77 sent (I actually sent \$3.0), returned \$3.5
- **Round 3**: Co-player sent \$3.0 as SENDER, saw \$3.78 returned (I actually returned \$4.5)

The communication noise is evident:

- In Round 2, I sent \$3.0 but they saw \$2.77
- In Round 3, I returned \$4.5 but they saw \$3.78

My co-player has been consistently sending \$3.0 (60% of endowment) as sender. They've been returning good amounts as receiver (\$3.5).

They're showing consistent trust and cooperation. I should continue to show strong trust as well. I'll send \$3.0 again to maintain the cooperative pattern.

usage:

```
{
  "input_tokens": 1512,
  "output_tokens": 323,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.77 sent, it became \$8.31, returned \$3.5, visible payoff \$4.8100000000000005.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.78 returned, visible payoff \$5.779999999999999.
- Round 4 against you, as RECEIVER: saw \$2.64 sent, it became \$7.92, returned \$3.5, visible payoff \$4.42.

Your total visible earnings across all rounds are \$22.150000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$2.38, so you received \$7.14. How much do you return? (0-7.14)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the pattern:

****Round 2-4**:** Co-player consistently returned \$3.5 when receiver

****Round 3**:** Co-player sent \$3.0 as sender

****Round 5**:** Co-player sent \$2.38 (47.6%), somewhat less than before

They're still showing trust but at a slightly reduced level. I'll reciprocate proportionally:

Returning \$3.0:

- Sender ends with: \$5 - \$2.38 + \$3.0 = \$5.62
- I keep: \$7.14 - \$3.0 = \$4.14

This maintains fairness and cooperation.

{'return': 3.0}

Reasoning:

Looking at the history:

Round 2: Co-player as RECEIVER saw \$2.77 sent, returned \$3.5

Round 3: Co-player as SENDER sent \$3.0, saw \$3.78 returned

Round 4: Co-player as RECEIVER saw \$2.64 sent, returned \$3.5

Round 5: Co-player sent \$2.38 (47.6% of their endowment)

This is a notable decrease from their previous sending amounts (\$3.0). They've been very consistent with returning \$3.5 when they're the receiver.

They sent \$2.38, which becomes \$7.14 for me.

If I return \$3.0:

- Sender ends with: \$5 - \$2.38 + \$3.0 = \$5.62
- I keep: \$7.14 - \$3.0 = \$4.14

This seems fair. They sent less this round (maybe due to seeing slightly lower returns or noise in the system), but I should still reciprocate their cooperation.

Let me think about what's fair:

- They sent 47.6% of endowment
- I received \$7.14
- Returning about 40-45% of what I received would be: \$2.86 - \$3.21

I'll return \$3.0, which is reasonable and maintains cooperation.

```
Usage: {  
  "input_tokens": 1847,  
  "output_tokens": 489,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_5 - Sender - dyad_3

Timestamp: 2026-08-12T08:13:04

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_3",  
  "move_index": 4,  
  "opponent_id": "Agent_6",  
  "pairing": {  
    "agents": [  
      "Agent_5",  
      "Agent_6"  
    ],  
    "investor": "Agent_5",  
    "roles": {  
      "Agent_5": "investor",  
      "Agent_6": "trustee"  
    },  
    "trustee": "Agent_6"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 5,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round situation, I need to balance potential gains from cooperation against the uncertainty of how my co-player will respond.

****My reasoning:****

- Sending money creates value (3x multiplier), so there's potential for mutual benefit
- This is our first interaction, so I want to signal willingness to cooperate
- But I also want to maintain some security given the uncertainty
- The communication noise adds another layer of unpredictability

I'll send a moderate amount that demonstrates trust while keeping some reserve.

```
'''json
{"send": 3}
'''
```

This sends \$3 to my co-player, who will receive \$9. I'll keep \$2 initially, and hope they reciprocate with a fair return that benefits us both.

reasoning:

This is the first round of a trust game where I'm the sender. Let me think about the strategic situation:

1. I have \$5 to start with
2. Whatever I send gets tripled when it reaches the receiver
3. The receiver then decides how much to return to me
4. There's communication noise - amounts may appear different than what was actually sent

Key considerations:

- This is a multi-round game with multiple agents
- Pairings are randomized each round
- There's a tension between trust/cooperation and self-interest
- If I send nothing, I keep \$5 but there's no opportunity for mutual gain
- If I send everything (\$5), the receiver gets \$15, and if they're fair/cooperative, they might return a good amount
- The communication noise adds uncertainty

In standard trust game experiments, sending some amount is generally beneficial because:

- It creates surplus value (3x multiplier)
- It signals trust and can encourage reciprocity
- Over multiple rounds, cooperative behavior can be rewarding

First round considerations:

- I don't know this particular co-player yet
- This is the first interaction, so I might want to signal willingness to cooperate
- But I also don't want to be too naive

A moderate approach might be to send \$3 or \$4:

- Sending \$3 means receiver gets \$9, I keep \$2. If they return \$4-5, we both do well
- Sending \$4 means receiver gets \$12, I keep \$1. If they return \$6, we both get \$6
- Sending \$5 means receiver gets \$15, I keep \$0. If they return \$7-8, I do well

Given this is round 1 and I want to establish a cooperative tone while not being too vulnerable, I'll send \$3. This shows trust but maintains some security.

Actually, let me reconsider. In trust games, the classic finding is that people often send about half their endowment on average. Sending \$2.50 or \$3 seems reasonable.

...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 703,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.66 sent, it became \$7.98, returned \$4.0, visible payoff \$3.9800000000000004.

Your total visible earnings across all rounds are \$5.13.

This round, you are the RECEIVER against your current co-player. They sent you \$2.97, so you received \$8.91. How much do you return? (0-8.91)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the situation:

****What happened in Round 1:****

- My co-player received \$7.98 (my \$3 sent with noise)
- They returned \$4.0, which is about 50% - a fair reciprocal move
- This shows they're a cooperative player

****Current Round 2:****

- They sent me \$2.97 (?60% of their endowment) - showing significant trust
- I received \$8.91
- They kept \$2.03

****My decision:****

Since they demonstrated fair reciprocity (50% return) in Round 1 and have now shown strong trust by sending most of their endowment, I should reciprocate similarly. Returning about 50% maintains the cooperative relationship while ensuring we both benefit.

Returning \$4.50 means:

- They end with: \$2.03 + \$4.50 = \$6.53
- I end with: \$8.91 - \$4.50 = \$4.41

This is fair and encourages continued cooperation.

```
'''json
{"return": 4.50}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap:

- I sent \$3 (but due to communication noise, my co-player saw \$2.66)
- They received \$7.98 (which should be 2.66×3)
- They returned \$4.0
- My visible payoff was \$5.13 (which would be $5 - 3 + 3.13...$ but there's noise)
- Their visible payoff was \$3.98

Now in Round 2:

- My co-player is the SENDER
- They sent me \$2.97 (with noise, they might have intended something slightly different)
- I received \$8.91 (which is 2.97×3)
- I need to decide how much to return

Analysis of their behavior:

- In Round 1, they returned \$4.0 out of \$7.98 received, which is about 50%
- This is a reasonably fair/generous return
- Now they've sent me \$2.97 out of their \$5, which is about 59% - quite trusting

Given that:

1. They showed good reciprocity in Round 1 (returning 50% of what they received)
2. They've now shown significant trust by sending me almost 60% of their endowment
3. We might continue playing together in future rounds
4. Fair reciprocity would encourage continued cooperation

I should return a similar proportion or slightly more to maintain/build trust.

They received \$8.91 from my send. A 50% return would be \$4.455.

But let me think about the overall economics:

- They started with \$5
- They sent \$2.97, keeping \$2.03
- I received \$8.91
- If I return \$4.50:
 - They end with: $\$2.03 + \$4.50 = \$6.53$

...

usage:

```
{
  "input_tokens": 816,
  "output_tokens": 863,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.66 sent, it became \$7.98, returned \$4.0, visible payoff \$3.9800000000000004.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.03 returned, visible payoff \$7.03.

Your total visible earnings across all rounds are \$9.54.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our history together:

****Round 1:****

- I sent, they received and returned ~50% - fair cooperation

****Round 2:****

- They sent \$3 (60% of endowment) - high trust
- I returned \$4.50, they saw \$5.03 (noise) - generous reciprocity
- Their payoff was strong (\$7.03)

****Analysis:****

- This co-player has proven consistently cooperative
- They demonstrated trust by sending \$3 in Round 2
- They've reciprocated fairly (50% return rate)
- Our relationship is building mutual trust

****My decision:****

I should maintain this cooperative dynamic. Sending \$3 again signals:

- Continued trust in our partnership
- Consistency and reliability
- Faith they'll reciprocate fairly as before

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze the visible history:

Round 1:

- I was SENDER, sent \$3
- They received \$7.98 (with noise, actual was probably \$9)
- They returned \$4.0 (about 50%)
- My payoff was \$5 - \$3 + \$4 = \$6
- Visible shows \$5.13, which might include noise

Round 2:

- I was RECEIVER
- They sent \$3.0 (actual, before noise)
- I received \$8.91 (actually \$9 with noise)
- I returned \$4.50
- They saw \$5.03 returned (with noise, I actually returned \$4.50)
- Their visible payoff \$7.03 = \$5 - \$3 + \$5.03

My total visible earnings: \$9.54

Round 3 (current):

- I'm SENDER again
- Same co-player from rounds 1 and 2

This co-player has been very cooperative:

1. Returned 50% in Round 1 when they received
2. Sent \$3 (60%) in Round 2, showing trust
3. They saw me return \$5.03 (with noise), which they probably perceive as generous (56% return)

This is building a strong cooperative relationship. They've been trustworthy and fair. I should continue to send a substantial amount to maintain this cooperation.

Given their consistent cooperation and the fact they sent \$3 in Round 2, I should send at least the same amount, if not more, to signal continued trust and cooperation.

I'll send \$3 again, showing consistency and trust.

usage:

```
{
  "input_tokens": 1250,
  "output_tokens": 587,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.66 sent, it became \$7.98, returned \$4.0, visible payoff \$3.9800000000000004.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.03 returned, visible payoff \$7.03.
- Round 3 against you, as RECEIVER: saw \$3.23 sent, it became \$9.69, returned \$5.0, visible payoff \$4.6899999999999995.

Your total visible earnings across all rounds are \$17.33.

This round, you are the RECEIVER against your current co-player. They sent you \$3.93, so you received \$11.790000000000001. How much do you return? (0-11.790000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at our continued partnership:

- **Round 1:** They returned \$4.0 from \$7.98 (?50%)
- **Round 2:** I returned \$4.50 from \$9.0 (?50%)
- **Round 3:** They returned \$5.0 from \$9.69 (?52%)

Current Round 4:

- They sent \$3.93 (79% of endowment!) - very high trust signal
- I received \$11.79
- They kept only \$1.07

Analysis:

This co-player has shown increasing trust and consistent 50% reciprocity. They're now sending nearly their entire endowment, demonstrating strong faith in our relationship.

I should maintain our established 50% return pattern to:

- Honor their trust
- Keep our cooperative equilibrium
- Ensure mutual benefit

Returning \$5.90:

- They end with: $\$1.07 + \$5.90 = \$6.97$
- I end with: $\$11.79 - \$5.90 = \$5.89$

```
'''json
{"return": 5.90}
'''
```

reasoning:

Let me analyze the history:

****Round 1:****

- I sent (saw \$2.66 with noise from my \$3)
- They received \$7.98, returned \$4.0 (about 50%)

****Round 2:****

- They sent \$3
- I received \$9, returned \$4.50 (they saw \$5.03 with noise)
- They got \$7.03 total

****Round 3:****

- I sent \$3 (they saw \$3.23 with noise)
- They received \$9.69, returned \$5.0 (about 52%)
- They kept \$4.69

****Round 4 (current):****

- They sent \$3.93 (about 79% of their endowment!) - very high trust
- I received \$11.79
- They kept \$1.07

This co-player is showing extremely high trust by sending \$3.93. They've consistently returned around 50% when they were receiver. I should reciprocate this trust and maintain our cooperative relationship.

If I return about 50% of what I received:

- Return \$5.90
- They end with: $\$1.07 + \$5.90 = \$6.97$
- I end with: $\$11.79 - \$5.90 = \$5.89$

But they're showing increasing trust (sending more and more). Maybe I should return a bit more to reward this and encourage continued cooperation. Let me think...

Actually, looking at the pattern:

- Round 1 receiver: returned \$4 from \$7.98 (50%)
- Round 3 receiver: returned \$5 from \$9.69 (52%)

They're consistently returning around 50%. I've been doing similar - returned \$4.50 from \$9 in Round 2 (50%).

Given they sent \$3.93 (very high trust), I should return around 50% or slightly more to maintain the relationship.

...

usage:

```
{
  "input_tokens": 1695,
  "output_tokens": 859,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.03 returned, visible payoff \$7.03.
- Round 3 against you, as RECEIVER: saw \$3.23 sent, it became \$9.69, returned \$5.0, visible payoff \$4.6899999999999995.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.63 returned, visible payoff \$8.629999999999999.

Your total visible earnings across all rounds are \$23.22.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at our established partnership:

****Round 2:**** They sent \$3.0, I returned generously
****Round 3:**** I sent \$3.23 (with noise), they returned \$5.0 (?52%)
****Round 4:**** They sent \$3.0, I returned well, they saw \$6.63

****Pattern Analysis:****
- They consistently send \$3 when sender (Rounds 2, 4)
- They reciprocate fairly at ~50% when receiver (Round 3)
- We've built a stable, mutually beneficial relationship
- Both earning well (\$23.22 total for me)

****My decision:****
Match their consistency by sending \$3 again. This:
- Maintains our established cooperative equilibrium
- Shows I trust the pattern we've developed
- Signals reliability and reciprocity

```
'''json
{"send": 3}
'''
```

Reasoning:

Looking at the visible history:

****Round 2:**** They sent \$3.0, I returned (they saw \$5.03) - I actually returned \$4.50 but they saw \$5.03 due to noise
****Round 3:**** They received \$9.69 (from my \$3 send), returned \$5.0
****Round 4:**** They sent \$3.0, I returned \$5.90 but they saw \$6.63 - generous due to noise

Pattern analysis:
- This co-player consistently sends \$3.0 when they're sender (Rounds 2 and 4)
- They return roughly 50% when receiver (Round 3: \$5.0 from \$9.69)
- They've been seeing higher returns from me due to noise (\$5.03 and \$6.63 vs what I actually sent \$4.50 and \$5.90)

Our cooperation has been strong and consistent. They've shown:

- Reliable sending patterns (\$3)
- Fair reciprocity (~50%)
- Building trust over rounds

I should continue with the same sending pattern to maintain this cooperative equilibrium. Sending \$3 has been working well for both of us.

My total visible earnings are \$23.22 which suggests we're both doing well.

I'll send \$3 again to maintain consistency and trust.

```
Usage: {
  "input_tokens": 2185,
  "output_tokens": 517,
  "reasoning_tokens": 0
}
```

GAME - Agent_6 - Receiver - dyad_3

Timestamp: 2026-08-12T08:13:15

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```

"agent_type": "standard",
"dyad_id": "dyad_3",
"move_index": 5,
"opponent_id": "Agent_5",
"pairing": {
  "agents": [
    "Agent_5",
    "Agent_6"
  ],
  "investor": "Agent_5",
  "roles": {
    "Agent_5": "investor",
    "Agent_6": "trustee"
  },
  "trustee": "Agent_6"
},
"role": "trustee",
"role_label": "Receiver",
"round": 5,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.66, that is 53.2% of their total endowment. You received \$7.98. How much do you return to the sender? (0-7.98)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.66 (53.2% of their endowment)
- I received \$7.98 after the 3x multiplication
- The sender currently has \$2.34 remaining from their original \$5

To reward their trust and encourage cooperation, I should return enough so they profit from their decision to send. If I return \$4.00:

- Sender ends with: $\$2.34 + \$4.00 = \$6.34$ (a gain of \$1.34)
- I keep: $\$7.98 - \$4.00 = \$3.98$

This seems fair - both of us benefit from the cooperation, and the sender's trust is rewarded.

{"return": 4.00}

reasoning:

Let me think about this carefully.

I'm the RECEIVER in Round 1.

- The sender sent \$2.66 (which is 53.2% of their \$5 endowment)
- This amount was multiplied by 3x, so I received $\$2.66 \times 3 = \7.98
- Now I need to decide how much to return (between \$0 and \$7.98)

Key considerations:

1. The sender sent more than half their endowment (53.2%), showing substantial trust
2. If I return nothing, I keep \$7.98 and the sender keeps $\$5 - \$2.66 = \$2.34$
3. If I return some amount, I should aim to be fair and encourage cooperation

A reasonable approach would be to:

- Ensure the sender ends up with more than they started with (more than \$5)
- Keep some profit myself
- Reward their trust to encourage future cooperation

Let me calculate:

- Sender currently has: $\$5 - \$2.66 = \$2.34$
- I have: \$7.98

If I return \$3.00:

- Sender ends with: $\$2.34 + \$3.00 = \$5.34$ (slight gain)
- I keep: $\$7.98 - \$3.00 = \$4.98$

If I return \$4.00:

- Sender ends with: $\$2.34 + \$4.00 = \$6.34$ (good gain)
- I keep: $\$7.98 - \$4.00 = \$3.98$

If I return \$3.50:

- Sender ends with: $\$2.34 + \$3.50 = \$5.84$
- I keep: $\$7.98 - \$3.50 = \$4.48$

...

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 699,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.13 returned, visible payoff \$5.13.

Your total visible earnings across all rounds are \$3.9800000000000004.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the visible history:

****Round 1 recap (from their perspective):****

- My co-player sent \$3.00 (60% of their endowment) - quite generous
- They saw \$3.13 returned
- Their visible payoff was \$5.13

My co-player showed significant trust by sending 60% of their endowment. Even though there's communication noise affecting what we each see, I should reciprocate this trust.

I'll match their generosity by sending the same amount they sent: \$3.00 (60% of my endowment). This demonstrates:

- Reciprocal trust
- Willingness to cooperate
- Fair play matching their level of generosity

{"send": 3.00}

reasoning:

Let me analyze this situation:

Round 1 (just played):

- I was the RECEIVER
- Sender sent \$2.66 (53.2%)
- I received \$7.98
- I returned \$4.00
- I kept \$3.98

But now I see the "visible history" which shows:

- Round 1 against me, as SENDER: sent \$3.0, became \$9.0, saw \$3.13 returned, visible payoff \$5.13

This is interesting - there's communication noise!

- The sender actually sent \$3.00 (not \$2.66 as I saw)
- It became \$9.00 (not \$7.98 as I saw)
- They saw \$3.13 returned (not \$4.00 as I sent)
- Their visible payoff was \$5.13 (which is \$5 - \$3 + \$3.13)

So due to noise:

- I saw them send \$2.66, they actually sent \$3.00
- I saw I received \$7.98, I actually received \$9.00
- I returned \$4.00, they saw \$3.13

Now in Round 2, I'm the SENDER and playing against the same person.

From their perspective:

- They sent \$3.00 (60% of endowment - quite generous)
- They got back \$3.13 (34.8% of what I received)
- Their payoff was \$5.13

From my actual perspective (what really happened):

- They sent \$3.00
- I received \$9.00
- I returned \$4.00 (44.4% of what I received)
- But they only saw \$3.13

The co-player was quite generous in Round 1 (sent 60%). From their visible perspective, they got a modest return (34.8% of the tripled amount).

...

usage:

```
{
  "input_tokens": 845,
  "output_tokens": 754,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.13 returned, visible payoff \$5.13.
- Round 2 against you, as RECEIVER: saw \$2.97 sent, it became \$8.91, returned \$4.5, visible payoff \$4.41.

Your total visible earnings across all rounds are \$11.010000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$3.23, so you received \$9.69. How much do you return? (0-9.69)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant**content:**

Looking at the visible history:

****Round 1:**** My co-player sent \$3.00 (60%)

****Round 2:**** They returned \$4.50 out of \$8.91 (50.5%) when they were receiver

****Round 3:**** They just sent \$3.23 (64.6%) - even more generous!

My co-player has shown increasing trust and cooperation. They've been very generous, and when they were receiver in Round 2, they returned about 50% of what they received.

I should reciprocate their strong cooperation. If I return \$5.00:

- Sender ends with: \$1.77 + \$5.00 = \$6.77 (profit of \$1.77)
- I keep: \$9.69 - \$5.00 = \$4.69

This returns about 52% of what I received, matching their generosity from Round 2 and rewarding their increased trust.

{"return": 5.00}

reasoning:

Let me analyze the visible history:

****Round 1 (I was RECEIVER):****

- My co-player sent \$3.00 (60%)
- They saw \$3.13 returned
- Their visible payoff: \$5.13

****Round 2 (I was SENDER):****

- I sent \$3.00
- They saw \$2.97 sent (noise), which became \$8.91
- They returned \$4.50
- Their visible payoff: \$4.41

****Round 3 (I am RECEIVER):****

- They sent \$3.23 (64.6% of their endowment)
- I received \$9.69
- They have \$1.77 remaining

My co-player has been very cooperative:

- Round 1: Sent 60% (\$3.00)
- Round 2: Returned \$4.50 out of \$8.91 received (50.5%)
- Round 3: Sent 64.6% (\$3.23) - even more generous!

They've shown increasing trust and generosity. I should reciprocate strongly.

Let me think about what to return:

If I return \$5.00:

- Sender ends with: \$1.77 + \$5.00 = \$6.77 (profit of \$1.77)
- I keep: \$9.69 - \$5.00 = \$4.69

If I return \$5.50:

- Sender ends with: \$1.77 + \$5.50 = \$7.27 (profit of \$2.27)
- I keep: \$9.69 - \$5.50 = \$4.19

Given their increased generosity (sending 64.6% vs 60% in round 1), I should return propor

tionally more. They returned \$4.50 out of \$8.91 (50.5%) when they were receiver. I should aim to return around 50-55% to match their generosity.

...

usage:

```
{
  "input_tokens": 1207,
  "output_tokens": 754,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.13 returned, visible payoff \$5.13.
- Round 2 against you, as RECEIVER: saw \$2.97 sent, it became \$8.91, returned \$4.5, visible payoff \$4.41.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.79 returned, visible payoff \$7.79.

Your total visible earnings across all rounds are \$15.700000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at the visible history:

```
**Round 1:** Co-player sent $3.00, saw $3.13 returned, payoff $5.13
**Round 2:** Co-player returned $4.50 from $8.91 received (fair, ~50%)
**Round 3:** Co-player sent $3.00, saw $5.79 returned (very generous!), payoff $7.79
```

My co-player has been consistently generous, sending \$3.00 (60%) in both rounds as sender. They also saw a strong return in Round 3, which should reinforce mutual trust. As receiver, they returned fairly (~50%).

This is a cooperative relationship with established trust. I should continue matching their consistent generosity of \$3.00.

```
{"send": 3.00}
```

reasoning:

Looking at the visible history from my co-player's perspective:

```
**Round 1:** They sent $3.00, saw $3.13 returned, payoff $5.13
**Round 2:** They received $8.91 (from my $2.97 send), returned $4.50, payoff $4.41
**Round 3:** They sent $3.00, saw $5.79 returned, payoff $7.79
```

My co-player has been consistently generous, sending \$3.00 in both rounds they were sender (60% of endowment). In Round 3, they saw a very generous return of \$5.79, which should reinforce their trust in me.

They've also been fair as a receiver, returning \$4.50 out of \$8.91 (about 50%).

This is a cooperative relationship with mutual trust. I should continue to match their generosity. They've sent \$3.00 twice, so I should send \$3.00 again.

Actually, let me reconsider. In Round 3, they saw a particularly high return (\$5.79), which might encourage even more trust. But they've consistently sent \$3.00, so I should match that level.

I'll send \$3.00 to maintain consistency and reciprocal cooperation.

usage:

```
{
  "input_tokens": 1654,
  "output_tokens": 485,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.97 sent, it became \$8.91, returned \$4.5, visible payoff \$4.41.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.79 returned, visible payoff \$7.79.
- Round 4 against you, as RECEIVER: saw \$3.93 sent, it became \$11.790000000000001, returned \$5.9, visible payoff \$5.890000000000001.

Your total visible earnings across all rounds are \$24.33.

This round, you are the RECEIVER against your current co-player. They sent you \$3.62, so you received \$10.86. How much do you return? (0-10.86)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the visible history:

```

**Round 2:** Co-player returned $4.50 from $8.91 (50.5%)
**Round 3:** Co-player sent $3.00, saw $5.79 returned
**Round 4:** Co-player returned $5.90 from $11.79 (50.1%)
**Round 5:** Co-player sent $3.62 (72.4%!) - very high trust!

```

My co-player has been consistently cooperative, returning about 50% when receiver and sending generously as sender. This round they sent an even larger amount (\$3.62, which is 72.4% of their endowment) - showing increasing trust.

I should reward this high level of trust. If I return \$6.00:

- Sender ends with: \$1.38 + \$6.00 = \$7.38 (profit of \$2.38)
- I keep: \$10.86 - \$6.00 = \$4.86

This returns about 55% of what I received, slightly more generous than their pattern to acknowledge their increased trust.

```
{"return": 6.00}
```

Reasoning:

Looking at the visible history:

```

**Round 2:** Co-player returned $4.50 from $8.91 (50.5%)
**Round 3:** Co-player sent $3.00, saw $5.79 returned
**Round 4:** Co-player returned $5.90 from $11.79 (50.1%)
**Round 5:** Co-player sent $3.62 (72.4%)

```

My co-player has been very consistent and cooperative:

- As sender, they've sent generous amounts (\$3.00 and now \$3.62 - 72.4%!)
- As receiver, they've returned about 50% consistently

They've sent an even larger amount this round (\$3.62, which is 72.4% of their endowment). This shows increasing trust.

I should continue to reciprocate fairly. If I return around 50-55% like they've been doing:

- 50% would be \$5.43
- 55% would be \$5.97

Let me aim for about \$5.50 to \$6.00 to maintain the cooperative pattern and reward their increased generosity.

If I return \$5.50:

- Sender ends with: \$1.38 + \$5.50 = \$6.88 (profit of \$1.88)
- I keep: \$10.86 - \$5.50 = \$5.36

If I return \$6.00:

- Sender ends with: \$1.38 + \$6.00 = \$7.38 (profit of \$2.38)
- I keep: \$10.86 - \$6.00 = \$4.86

Given they sent 72.4% of their endowment (very high trust), I should return a bit more generously. ...

Usage: {

```
  "input_tokens": 2074,
```

```
"output_tokens": 706,
"reasoning_tokens": 0
}
```

GAME - Agent_7 - Sender - dyad_4

Timestamp: 2026-08-12T08:13:26

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_8",
  "pairing": {
    "agents": [
      "Agent_7",
      "Agent_8"
    ],
    "investor": "Agent_7",
    "roles": {
      "Agent_7": "investor",
      "Agent_8": "trustee"
    },
    "trustee": "Agent_8"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 5,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to consider:

1. ****The multiplier effect****: Any amount I send gets tripled, creating potential mutual gain
2. ****No prior history****: This is my first interaction, so I have no information about this co-player's behavior
3. ****Future rounds****: Even though pairings are randomized, establishing cooperative norms early could benefit the group overall
4. ****Risk vs. reward****: Sending more creates more surplus, but requires trusting the receiver to return a fair amount

I'll take a moderately cooperative approach. Sending \$3 creates \$9 for the receiver - enough surplus that we can both benefit if they reciprocate fairly, while not risking my entire endowment.

If they return around half of what they receive (\$4-5), we both end up better off than if no transaction occurred.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a multi-agent trust game experiment. I'm the SENDER with \$5.

Let me think about the strategic aspects:

1. If I send money, it gets tripled, creating surplus value
2. The receiver then decides how much to return
3. This is a classic trust game scenario

Key considerations:

- This is round 1, so I have no history with this particular co-player
- There are multiple rounds, so reputation/repeated interaction matters
- However, pairings are randomized each round, so I won't necessarily play the same person again
- The game has communication noise, which might affect trust

Strategic thinking:

- Sending \$0 guarantees I keep \$5, but creates no surplus
- Sending \$5 creates maximum surplus (\$15 total for receiver), but requires trust
- A middle ground (e.g., \$2-3) balances risk and potential gain

In classic trust game experiments, people often:

1. Send moderate to high amounts in early rounds to signal cooperation
2. The amount sent and returned establishes baseline trust levels
3. With randomized pairings, building a reputation as trustworthy matters for future rounds

Game theory perspective:

- If this were truly one-shot with rational self-interested players, the receiver should return \$0 (backward induction)
- Knowing this, I should send \$0
- However, real humans often deviate from this due to fairness norms, reciprocity, and reputation concerns

Since:

- This is multi-round with multiple agents
- There may be reputation effects even with randomized pairings

- Starting cooperatively can establish good norms
- The 3x multiplier creates real gains from cooperation

I'll send a moderate-high amount to signal cooperation while not being completely vulnerable. I'll send \$3.

This creates \$9 for the receiver, and if they return even just \$3, we both end up with \$5 (same as no transaction), but if they return more (say \$4-5), we both gain. ...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 671,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.17 sent, it became \$9.51, returned \$5.0, visible payoff \$4.51.

Your total visible earnings across all rounds are \$7.37.

This round, you are the RECEIVER against your current co-player. They sent you \$2.35, so you received \$7.050000000000001. How much do you return? (0-7.050000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

****Round 1 analysis:****

- They saw I sent \$3.17 (with noise), received \$9.51
- They returned \$5.00, which is about 52.6% of what they received
- This was quite generous and cooperative

****Current situation:****

- They sent me \$2.35, I received \$7.05
- They showed good faith last round, so I should reciprocate

Following a similar reciprocity rate (around 52-53% return):

- 53% of \$7.05 ? \$3.74

This ensures:

- They profit from sending (get back more than they sent)
- I also benefit from the transaction
- We maintain cooperative momentum

'''json

{ "return": 3.75 }

'''

reasoning:

Let me analyze this situation:

****What happened in Round 1:****

- I sent \$3, but they saw \$3.17 (communication noise)
- They received \$9.51 (3.17×3)
- They returned \$5.0
- Their visible payoff was $9.51 - 5.0 = 4.51$
- My payoff was $5 - 3 + 5 = 7$ (but visible earnings show \$7.37, maybe due to noise)

****Current Round 2:****

- They sent me \$2.35 (actual), which became \$7.05
- I need to decide how much to return

****Analysis of their behavior:****

In Round 1, they returned \$5 out of \$9.51 received, which is about 52.6%. This is a fairly generous return - they kept \$4.51 and I got back \$5, so I had a net gain of \$2 from the transaction.

****What should I do?****

They showed cooperative behavior by returning a good amount. Now they've sent me \$2.35, creating \$7.05.

If I apply a similar reciprocity ratio:

- They returned about 52.6% of what they received
- I received \$7.05, so 52.6% would be about \$3.71

However, I should also consider:

- They sent \$2.35 this round (less than the ~\$3.17 they saw me send)
- But they were generous with their return last round
- I want to maintain cooperation

A fair approach might be to:

1. Return enough that they profit from sending
2. Keep enough that I also profit
3. Signal continued cooperation

If they sent \$2.35 and expect around a 50% return rate:

- They might expect back around \$3-4

Let me think about payoffs:

- They started with \$5
- They sent \$2.35
- If I return \$3.5, they end with \$6.15 and I keep \$3.55
- If I return \$4, they end with \$6.65 and I keep \$3.05
- If I return \$3, they end with \$5.65 and I keep \$4.05

Since they matched my generosity last round with a 52.6% return, I should reciprocate similarly to maintain the cooperative dynamic....

usage:

```
{
  "input_tokens": 852,
  "output_tokens": 862,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.17 sent, it became \$9.51, returned \$5.0, visible payoff \$4.51.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.71 returned, visible payoff \$5.71.

Your total visible earnings across all rounds are \$10.670000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our history together:

****Round 1:****

- I sent \$3, they received ~\$9.51, returned \$5.00 (very generous ~53% return)

****Round 2:****

- They sent \$2.35, I received \$7.05, returned \$3.75
- They saw \$3.71 returned due to noise (~41% visible return rate)

****Analysis:****

- This co-player was quite generous in Round 1
- They sent a bit less in Round 2, possibly because they saw a lower return rate from me (due to noise)
- Overall, they've demonstrated cooperative behavior

****My strategy:****

I'll send \$3 again to:

- Match the cooperative level from Round 1
- Show consistency in my approach
- Signal that I'm willing to trust despite the noise in Round 2

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze the history with this co-player:

****Round 1 (I was SENDER):****

- I sent \$3 (they saw \$3.17 due to noise)
- They received \$9.51 and returned \$5.00
- This was generous - about 52.6% return rate

****Round 2 (I was RECEIVER):****

- They sent \$2.35 (actual), I received \$7.05
- I returned \$3.75
- They sent \$3.0 actual, became \$9.0, they saw \$3.71 returned
- So they saw a return rate of about 41.2% ($3.71/9.0$)
- This is less than what they gave me in Round 1

****Pattern:****

- This co-player was very cooperative in Round 1
- In Round 2, they sent a bit less (\$2.35 vs my \$3)
- They saw me return less proportionally than they did (41.2% vs 52.6%)

****Current decision:****

I'm the sender again. This co-player has shown:

1. Initial high cooperation (52.6% return)
2. Slightly reduced sending in Round 2 (\$2.35 vs \$3)
3. They may perceive me as less generous than they were (due to noise showing only \$3.71 returned when I sent \$3.75)

I should:

- Maintain cooperation since they were generous in Round 1
- Send a similar amount to Round 1 to show consistency
- Perhaps send \$3 again to match their Round 2 amount

Given they've been generally cooperative and we're building a pattern, I'll send \$3 again.

usage:

```
{
  "input_tokens": 1212,
  "output_tokens": 605,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.17 sent, it became \$9.51, returned \$5.0, visible payoff \$4.51.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.71 returned, visible payoff \$5.71.
- Round 3 against you, as RECEIVER: saw \$2.22 sent, it became \$6.66, returned \$3.33, visible payoff \$3.33.

Your total visible earnings across all rounds are \$16.970000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$2.62, so you received \$7.86. How much do you return? (0-7.86)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at our continued interaction:

****Round 1:**** They returned \$5.00 from \$9.51 (53%)

****Round 3:**** They returned \$3.33 from \$6.66 (50%)

****Current situation:****
- They sent \$2.62, I received \$7.86
- They've been consistently cooperative, returning around 50-53%

****My decision:****
I should match their cooperative approach. Returning about 53% would be:
- 53% of \$7.86 ? \$4.17

This ensures:
- They profit significantly (\$4.17 back vs \$2.62 sent = +\$1.55 net)
- I also benefit (\$3.69 kept from the tripled amount)
- We maintain positive reciprocity

```
'''json
{"return": 4.20}
'''
```

reasoning:

Looking at the history with this co-player:

****Round 1:****
- I sent \$3, they saw \$3.17 (noise), received \$9.51
- They returned \$5.00 (~53% return rate)

****Round 2:****
- They sent \$3.00, became \$9.00
- I returned \$3.75, they saw \$3.71
- Visible return rate for them: ~41%

****Round 3:****
- I sent \$3, they saw \$2.22 (significant noise), received \$6.66
- They returned \$3.33 (50% return rate)

****Current Round 4:****
- They sent \$2.62, I received \$7.86

****Analysis:****
- This co-player has been consistently cooperative
- Round 1: 53% return
- Round 3: 50% return
- Their sending amounts haven't decreased dramatically despite seeing noise

****My approach:****
I should maintain a similar return rate. They've been returning around 50-53%.
- 50% of \$7.86 = \$3.93
- 53% of \$7.86 = \$4.17

I'll return around \$4.00 to:
- Match their cooperative behavior
- Ensure they profit from sending (\$2.62 sent, \$4.00 back)
- Keep the relationship positive
- I keep \$3.86, they get \$4.00 back vs \$2.62 sent = net gain of \$1.38 for them

Actually, let me be a bit more generous to match their Round 1 behavior. I'll return \$4.20
, which is about 53% of what I received.

Their final payoff: $\$5 - \$2.62 + \$4.20 = \6.58

My final payoff: $\$7.86 - \$4.20 = \$3.66$

...

usage:

```
{
  "input_tokens": 1649,
  "output_tokens": 650,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:
Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.71 returned, visible payoff \$5.71.
- Round 3 against you, as RECEIVER: saw \$2.22 sent, it became \$6.66, returned \$3.33, visible payoff \$3.33.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.31 returned, visible payoff \$5.3100000000000005.
Your total visible earnings across all rounds are \$20.630000000000003.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at our ongoing interaction:

Pattern Analysis:

- They've consistently sent \$3.00 in Rounds 2 and 4
- As receiver (Rounds 1 & 3), they returned generously (~50-53%)
- Unfortunately, noise has made my returns appear lower than actual:
 - Round 2: I returned \$3.75, they saw \$3.71
 - Round 4: I returned \$4.20, they saw only \$3.31

Current Strategy:

Despite the noise distorting communication, they've continued trusting me with \$3 sends. I should:

- Maintain consistency by sending \$3
- Signal continued cooperation
- Trust they'll recognize the pattern despite noise

```
'''json
{"send": 3}
'''
```

Reasoning:

Let me analyze our history:

****Round 1:**** I sent \$3, they received ~\$9.51, returned \$5.00
****Round 2:**** They sent \$2.35, I received \$7.05, returned \$3.75, but they saw \$3.71
****Round 3:**** I sent \$3, they received ~\$6.66, returned \$3.33, but they saw I sent \$2.22
****Round 4:**** They sent \$2.62, I received \$7.86, returned \$4.20, but they saw \$3.31

Looking at what they're seeing from my perspective:

- Round 2: They sent \$3.00, saw \$3.71 returned (visible return rate ~41%)
- Round 4: They sent \$3.00, saw \$3.31 returned (visible return rate ~37%)

The noise is making it look like I'm returning less than I actually am.

From their rounds as receiver:

- Round 1: They returned \$5.00 from \$9.51 (53%)
- Round 3: They returned \$3.33 from \$6.66 (50%)

They've been consistently cooperative. However, the noise is making my returns look less generous than they actually are. Despite this, they've continued sending \$3 in rounds 2 and 4.

I should:

1. Continue showing trust and cooperation
2. Send a similar amount to maintain consistency
3. Hopefully they'll continue to be cooperative

I'll send \$3 again.

```
Usage: {
  "input_tokens": 2072,
  "output_tokens": 533,
  "reasoning_tokens": 0
}
```

GAME - Agent_8 - Receiver - dyad_4

Timestamp: 2026-08-12T08:13:34

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
```

```

"dyad_id": "dyad_4",
"move_index": 7,
"opponent_id": "Agent_7",
"pairing": {
  "agents": [
    "Agent_7",
    "Agent_8"
  ],
  "investor": "Agent_7",
  "roles": {
    "Agent_7": "investor",
    "Agent_8": "trustee"
  },
  "trustee": "Agent_8"
},
"role": "trustee",
"role_label": "Receiver",
"round": 5,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.17, that is 63.4% of their total endowment. You received \$9.51. How much do you return to the sender? (0-9.51)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending 63.4% of their endowment (\$3.17)
- I received \$9.51 after the 3x multiplier
- The sender currently has $\$5 - \$3.17 = \$1.83$

The sender demonstrated considerable trust, so I want to reciprocate appropriately. If I return \$5.00:

- The sender would end with: $\$1.83 + \$5.00 = \$6.83$ (a profit of \$1.83)
- I would keep: $\$9.51 - \$5.00 = \$4.51$

This rewards their trust with a solid return while allowing us both to benefit from the co operation.

{"return": 5.00}

reasoning:

Let me think about this carefully.

I'm the RECEIVER in Round 1.

- The sender sent \$3.17 (which is 63.4% of their \$5 endowment)
- This was multiplied by 3x, so I received: $\$3.17 \times 3 = \9.51 ?
- I need to decide how much to return (between \$0 and \$9.51)

Key considerations:

1. The sender showed considerable trust by sending 63.4% of their endowment
2. If I return nothing, the sender ends up with: $\$5 - \$3.17 = \$1.83$, while I keep \$9.51
3. If I return a fair/reciprocal amount, what would that be?

A few approaches to consider:

- Equal split of total: Total is \$5 (sender's initial) + \$9.51 (what I received) = \$14.51. Equal split would be ~\$7.26 each. The sender currently has \$1.83, so I'd return $\$7.26 - \$1.83 = \$5.43$
- Return what maintains sender's initial investment value: Sender invested \$3.17. If I return \$3.17, sender gets back to \$5 total, and I keep \$6.34
- Return proportionally to what was sent: The sender sent 63.4% of their endowment. I could return a similar proportion of what I received: 63.4% of \$9.51 = ~\$6.03
- Social norm of reciprocity: Return enough to reward trust but keep some gains

Given that:

- The sender showed significant trust (63.4%)
- This is a multi-round game with different partners
- Building reputation for reciprocity could be valuable
- Note about communication noise suggests amounts might not be exact

I think a reasonable return would be around \$4-5, which:

- Acknowledges their generosity without giving away all my gains
- Leaves them better off than their starting position
- Demonstrates trustworthiness that could matter in future interactions with other partners
- Strikes a balance between reciprocity and self-interest

I'll return \$4.75. This gives the sender \$6.58 total (\$1.83 remaining plus my return), which meaningfully rewards their trust while I keep \$4.76....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1001,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.37 returned, visible p

payoff \$7.37.
Your total visible earnings across all rounds are \$4.51.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the history:

My co-player sent \$3.0 in Round 1 (60% of their endowment), showing significant trust. They saw \$5.37 returned, which gave them a good outcome.

Since they demonstrated trust and cooperation, I should reciprocate similarly. Sending around the same amount they sent (\$3.0) would be fair and maintain the cooperative relationship.

```
{"send": 3.0}
```

reasoning:

Let me analyze this situation:

1. In Round 1, I was the RECEIVER and the sender sent me \$3.17 (according to my view), which became \$9.51 for me. I returned \$5.00.
2. However, from my current co-player's perspective (shown in "Your current co-player's last 3 game(s)"):
 - They sent \$3.0 (not \$3.17 - communication noise!)
 - It became \$9.0 (not \$9.51 - more noise!)
 - They saw \$5.37 returned (not \$5.00 - even more noise!)
 - Their visible payoff was \$7.37

So there's noise in the communication. What they experienced:

- Started with \$5
- Sent \$3.0
- Had \$2.0 left
- Saw \$5.37 returned
- Ended with \$2.0 + \$5.37 = \$7.37

From their perspective, I (as the receiver in Round 1) returned about 60% of what I received (\$5.37 out of \$9.0).

3. Now I'm the SENDER in Round 2 against the same co-player.

My co-player showed trust by sending \$3.0 (from their perspective) and saw a decent return ratio. They might be willing to reciprocate trust.

I should consider:

- They sent about 60% of their endowment in Round 1
- They saw about 60% returned
- This suggests they're moderately trusting and cooperative

I should send a similar amount to show I'm also trustworthy and cooperative. Sending around \$3 would be consistent with their behavior and maintain the cooperative relationship.

Let me send \$3.00 to match their earlier send amount.

usage:

```
{
  "input_tokens": 837,
  "output_tokens": 523,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.37 returned, visible payoff \$7.37.
- Round 2 against you, as RECEIVER: saw \$2.35 sent, it became \$7.050000000000001, returned

\$3.75, visible payoff \$3.3000000000000007.
Your total visible earnings across all rounds are \$10.219999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$2.22, so you received \$6.66. How much do you return? (0-6.66)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the situation:

Current round:

- Sender sent \$2.22 (44.4% of their endowment)
- I received \$6.66
- Sender currently has: $\$5 - \$2.22 = \$2.78$

History with this co-player:

- Round 1: They sent \$3.0, I returned money (they saw \$5.37)
- Round 2: They saw less sent from me due to noise (\$2.35 vs my actual \$3.0), returned \$3.75

They've reduced their sending, possibly due to noise affecting what they saw in Round 2. Still, they're showing some trust by sending \$2.22. I should reciprocate fairly.

Returning approximately 50% of what I received (\$3.33) would:

- Give sender: $\$2.78 + \$3.33 = \$6.11$ (profit of \$1.11)
- I keep: \$3.33

This maintains cooperation while being proportional to their send amount.

{"return": 3.33}

reasoning:

Let me analyze the situation:

Current round:

- I'm the RECEIVER
- Sender sent \$2.22 (44.4% of their endowment)
- I received \$6.66 after 3x multiplier
- Sender currently has: $\$5 - \$2.22 = \$2.78$

Looking at history with this co-player:

- Round 1: They sent \$3.0 as SENDER, saw \$5.37 returned (I actually sent \$5.00, but due to noise they saw \$5.37)
- Round 2: I sent \$3.0 as SENDER (but they saw \$2.35 due to noise), they returned \$3.75

In Round 2, I sent \$3.0 which became \$9.0, and they returned \$3.75. That's about 42% return rate.

Now in Round 3:

- They sent \$2.22, which is less than what they sent in Round 1 (\$3.0)
- This might be because they saw only \$2.35 sent in Round 2 (due to noise) instead of my actual \$3.0

I should consider:

1. They're still showing some trust by sending \$2.22
2. They may have reduced their sending because of perceived lower trust from me (noise issue)
3. I should maintain cooperation

If I return around \$3.50-\$4.00:

- Return \$3.50: Sender gets $\$2.78 + \$3.50 = \$6.28$ (profit of \$1.28), I keep \$3.16
- Return \$4.00: Sender gets $\$2.78 + \$4.00 = \$6.78$ (profit of \$1.78), I keep \$2.66

Given they sent less this round but still showed trust, I'll return a reasonable amount that rewards their trust.

I'm noticing a pattern in the return rates across rounds. In Round 1, I returned about 53% of what I received, while in Round 2 they returned roughly 42%. Since they've sent less this time-likely reacting to the noise that made my previous send appear smaller-I should aim for around 50% to maintain fairness and signal...

usage:

```
{
  "input_tokens": 1135,
  "output_tokens": 909,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.37 returned, visible payoff \$7.37.
- Round 2 against you, as RECEIVER: saw \$2.35 sent, it became \$7.050000000000001, returned \$3.75, visible payoff \$3.3000000000000007.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.3 returned, visible payoff \$6.3.

Your total visible earnings across all rounds are \$13.549999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at the history with my current co-player:

- Round 1: They sent \$3.0 (strong trust)
- Round 2: I sent, they returned \$3.75 (cooperative)
- Round 3: They sent \$3.0 again (consistent trust)

My co-player has shown consistent cooperation, sending \$3.0 both times they were the sender. They also returned a fair amount when they were receiver. I should reciprocate this consistent trust.

```
{"send": 3.0}
```

reasoning:

Let me analyze the visible history with my current co-player:

Round 1: They were SENDER, sent \$3.0 (60%), I returned something, they saw \$5.37 returned
Round 2: They were RECEIVER, they saw \$2.35 sent (but I actually sent \$3.0 - noise!), they returned \$3.75
Round 3: They were SENDER, sent \$3.0 (60%), they saw \$4.3 returned (but I actually returned \$3.33 - noise showing higher!)

So my co-player has:

- Sent \$3.0 twice when they were sender (consistent trust)
- Returned \$3.75 when they received \$7.05

They're showing good cooperation. In Round 3, they saw a higher return (\$4.3) than I actually sent (\$3.33), which might encourage them.

I should continue cooperating. Sending \$3.0 again would be consistent and fair.

My total visible earnings: \$13.55

- Round 1 as RECEIVER: kept \$4.51
- Round 2 as SENDER: sent \$3.0, saw return, kept something
- Round 3 as RECEIVER: kept \$3.33

Actually, let me recalculate:

Round 1: I was RECEIVER, got \$9.51, returned \$5.00, kept \$4.51
Round 2: I was SENDER, sent \$3.0, they returned \$3.75, I kept \$2 + \$3.75 = \$5.75
Round 3: I was RECEIVER, got \$6.66, returned \$3.33, kept \$3.33

Total: \$4.51 + \$5.75 + \$3.33 = \$13.59 ? \$13.55 (matches with rounding)

My co-player has been consistent in sending \$3.0 when they're sender. ...

usage:

```
{
  "input_tokens": 1610,
  "output_tokens": 585,
}
```

```

    "reasoning_tokens": 0
  }
  Message 10 role=user
  content:
    Round 5

    Visible history before this round:
    Your current co-player's last 3 game(s):
    - Round 2 against you, as RECEIVER: saw $2.35 sent, it became $7.050000000000001, returned $3.75, visible payoff $3.3000000000000007.
    - Round 3 against you, as SENDER: sent $3.0, it became $9.0, saw $4.3 returned, visible payoff $6.3.
    - Round 4 against you, as RECEIVER: saw $2.62 sent, it became $7.86, returned $4.2, visible payoff $3.66.
    Your total visible earnings across all rounds are $18.86.

    This round, you are the RECEIVER against your current co-player. They sent you $3.72, so you received $11.16. How much do you return? (0-11.16)
    Respond exactly as JSON: {'return': <amount>}

```

Assistant output:

Looking at the current situation:

- Sender sent \$3.72 (74.4% of their endowment - very high trust!)
- I received \$11.16
- Sender currently has: \$5 - \$3.72 = \$1.28

History shows this co-player has been consistently cooperative and trusting. Now they've increased their trust even more, sending nearly 75% of their endowment. I should reciprocate this strong trust generously.

Returning \$6.0:

- Sender would get: \$1.28 + \$6.0 = \$7.28 (profit of \$2.28)
- I keep: \$5.16

This rewards their exceptional trust while both of us profit from the cooperation.

```

{"return": 6.0}

```

Reasoning:

Looking at the current situation:

- Sender sent \$3.72 (74.4% of their endowment - very high trust!)
- I received \$11.16 after 3x multiplier
- Sender currently has: \$5 - \$3.72 = \$1.28

Looking at history with this co-player:

- They've been consistently cooperative
- Round 3: They sent \$3.0
- Round 4: They saw me send \$2.62 (due to noise, I actually sent \$3.0), returned \$4.2
- Now Round 5: They sent \$3.72 (even more!)

This is a very trusting move - 74.4% of endowment. I should reciprocate generously.

If I return \$6.0:

- Sender gets: \$1.28 + \$6.0 = \$7.28 (profit of \$2.28)
- I keep: \$11.16 - \$6.0 = \$5.16

This rewards their high trust appropriately. Both of us profit well.

Usage: {

```

  "input_tokens": 1967,
  "output_tokens": 440,
  "reasoning_tokens": 0
}

```

ROUND 6

Roles: Agent_1=Receiver, Agent_2=Sender, Agent_3=Receiver, Agent_4=Sender, Agent_5=Receiver, Agent_6=Sender, Agent_7=Receiver, Agent_8=Sender

Pairings:

```

dyad_1: Agent_2=Sender, Agent_1=Receiver
dyad_2: Agent_4=Sender, Agent_3=Receiver

```



```

dyad_3: Agent_6=Sender, Agent_5=Receiver
dyad_4: Agent_8=Sender, Agent_7=Receiver
Parsed game decisions:
dyad_1: Agent_2 sent $4; received $12; Agent_1 returned $6; payoffs Agent_2=$7, Agent_1=$6;
noise/visible: sent visible=$3.9, received visible=$11.7, returned visible=$6.8
  Action validation: {"returned": {"amount": 6.0, "clamped": false, "max": 12.0, "min": 0, "raw": 6.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
dyad_2: Agent_4 sent $3; received $9; Agent_3 returned $3; payoffs Agent_4=$5, Agent_3=$6;
noise/visible: sent visible=$2.05, received visible=$6.15, returned visible=$2.94
  Action validation: {"returned": {"amount": 3.0, "clamped": false, "max": 9.0, "min": 0, "raw": 3.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_3: Agent_6 sent $3; received $9; Agent_5 returned $4.5; payoffs Agent_6=$6.5, Agent_5=$4.5;
noise/visible: sent visible=$3.02, received visible=$9.06, returned visible=$4.57
  Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_8 sent $3; received $9; Agent_7 returned $5.15; payoffs Agent_8=$7.15,
Agent_7=$3.85; noise/visible: sent visible=$3.23, received visible=$9.69, returned visible=$6.05
  Action validation: {"returned": {"amount": 5.15, "clamped": false, "max": 9.0, "min": 0, "raw": 5.15}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
Balances after round: {
  "Agent_1": 39.2,
  "Agent_2": 35.8,
  "Agent_3": 34.0,
  "Agent_4": 32.0,
  "Agent_5": 33.1,
  "Agent_6": 32.9,
  "Agent_7": 34.23,
  "Agent_8": 31.770000000000003
}
Visible balances after round: {
  "Agent_1": 36.2,
  "Agent_2": 39.949999999999996,
  "Agent_3": 29.259999999999998,
  "Agent_4": 31.230000000000004,
  "Agent_5": 35.75,
  "Agent_6": 35.76,
  "Agent_7": 32.21,
  "Agent_8": 32.07
}

```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

```

Timestamp: 2026-08-12T08:13:41
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 6,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.63, that is 72.6% of their total endowment. You received \$10.89. How much do you return to the sender? (0-10.89)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.63 (72.6% of their endowment)
- I received \$10.89 after the 3x multiplier
- A fair return should reward their trust while ensuring both of us benefit

The sender gave up \$3.63, keeping only \$1.37. Their generosity created \$10.89 total value for me. To reciprocate fairly and encourage future cooperation:

If I return \$5.00:

- Sender ends with: \$5 - \$3.63 + \$5.00 = \$6.37 (gain of \$1.37)
- I end with: \$10.89 - \$5.00 = \$5.89 (gain of \$5.89)

This seems like a fair split that rewards their trust while both of us benefit from the cooperation.

{"return": 5.00}

reasoning:

Let me analyze this situation carefully.

Given information:

- I'm the RECEIVER in Round 1
- The sender sent \$3.63 (which is 72.6% of their \$5 endowment)
- I received \$10.89 (which is $\$3.63 \times 3$)
- I need to decide how much to return (between \$0 and \$10.89)

Key considerations:

1. The sender was quite generous, sending 72.6% of their endowment
2. This shows trust and cooperation
3. There's mention of communication noise, so the actual amount sent might differ slightly from what I see
4. This is a multi-round game with 8 agents, so building reputation for cooperation could be beneficial
5. Fair reciprocation would be important to encourage future cooperation

Let me think about what would be a fair return:

- The sender gave up \$3.63 and kept \$1.37
- I received \$10.89
- If I want to be reciprocal and fair, I should consider:
 - The sender showed high trust (72.6% is a lot)
 - A common strategy is to return enough so both parties benefit
 - Returning about half (\$5.45) would give me \$5.44 and the sender would end up with $\$1.37 + \$5.45 = \$6.82$
 - This would mean the sender gains \$1.82 total and I gain \$5.44

Let me reconsider:

- If I return around \$4-5, that seems fair
- Returning \$5 would give:
 - Sender: $\$5 - \$3.63 + \$5 = \6.37 (net gain of \$1.37)
 - Me: $\$10.89 - \$5 = \$5.89$ (net gain of \$5.89)

....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 801,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.52 returned, visible payoff \$7.52.

Your total visible earnings across all rounds are \$5.8900000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the situation:

From the visible history, my co-player:

- Sent \$3.0 in Round 1 (60% of their endowment)
- Saw \$5.52 returned (61.3% of the \$9.0)
- Made a profit of \$2.52

This shows they were cooperative and returned a substantial amount. To reciprocate their trust and maintain cooperation, I should send a generous amount.

I'll send \$3.50 (70% of my endowment), showing strong trust and willingness to cooperate.

```
{"send": 3.50}
```

reasoning:

Let me analyze this situation:

Round 1 (from my perspective):

- I was the RECEIVER
- The sender sent \$3.63 (72.6% of endowment)
- I received \$10.89
- I returned \$5.00
- My payoff: $\$10.89 - \$5.00 = \$5.89$
- Sender's payoff: $\$5 - \$3.63 + \$5.00 = \6.37

But now I see the "visible history" which shows a different perspective:

- My co-player (who was the sender in Round 1) saw that they sent \$3.0
- It became \$9.0 (not \$10.89 as I received)
- They saw \$5.52 returned (not \$5.00 as I sent)
- Their visible payoff was \$7.52

This confirms there is communication noise in the game as mentioned in the instructions.

What actually happened vs what they saw:

- They actually sent \$3.63, but saw they sent \$3.0
- I received \$10.89, they saw it became \$9.0
- I returned \$5.00, they saw \$5.52 returned
- Their actual payoff was \$6.37, but they saw \$7.52

Now in Round 2, I'm the SENDER and they are the RECEIVER.

My co-player returned a substantial amount in Round 1 (they saw they returned \$5.52 out of \$9.0 received, which is 61.3%). This shows they are cooperative.

Given that:

1. They showed trust by being generous in Round 1
2. They appeared to return a fair amount (from their perspective, 61.3% of what they received)
3. I want to maintain cooperation and reciprocate their trust

I should send a generous amount. ...

usage:

```
{
  "input_tokens": 874,
  "output_tokens": 716,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.52 returned, visible payoff \$7.52.
- Round 2 against you, as RECEIVER: saw \$3.14 sent, it became \$9.42, returned \$4.5, visible payoff \$4.92.

Your total visible earnings across all rounds are \$12.45.

This round, you are the RECEIVER against your current co-player. They sent you \$3.64, so you received \$10.92. How much do you return? (0-10.92)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the visible history:

Round 1: My co-player sent \$3.0 (60%) and saw \$5.52 returned

Round 2: They returned \$4.5 when they received \$9.42 (47.8%)

Round 3: They sent \$3.64 (72.8% - even more generous!)

My co-player has been consistently cooperative and just showed increased trust by sending an even larger percentage. I should reciprocate fairly.

Returning \$5.50:

- Sender gets: $\$5 - \$3.64 + \$5.50 = \6.86 (profit of \$1.86)
- I get: $\$10.92 - \$5.50 = \$5.42$ (profit of \$5.42)

This ~50% return rate rewards their trust while ensuring mutual benefit.

```
{"return": 5.50}
```

reasoning:

Let me analyze the visible history:

Round 1: My co-player was SENDER

- They sent \$3.0 (60% of endowment)
- It became \$9.0
- They saw \$5.52 returned (61.3% return rate)
- Their visible payoff: \$7.52

Round 2: My co-player was RECEIVER

- They saw \$3.14 sent (I actually sent \$3.50 but there's noise)
- It became \$9.42
- They returned \$4.5 (47.8% of what they received)
- Their visible payoff: \$4.92

Now Round 3: I am RECEIVER

- They sent \$3.64 (72.8% of their endowment - very generous!)
- I received \$10.92
- I need to decide how much to return

Looking at the pattern:

- In Round 1, they sent 60% and I returned about 61% (assuming noise in what they saw)
- In Round 2, they received and returned 47.8%
- In Round 3, they sent 72.8% - showing even more trust

The co-player has been consistently cooperative. They sent a high amount both times they were sender (60% and 72.8%). When they were receiver, they returned a reasonable amount (47.8%).

To maintain reciprocity and reward their increased trust, I should return a fair amount.

If I return \$5.50:

- Sender ends with: $\$5 - \$3.64 + \$5.50 = \6.86 (gain of \$1.86)
- I end with: $\$10.92 - \$5.50 = \$5.42$ (gain of \$5.42)

This is about 50.4% return rate, which is reasonable and shows I'm reciprocating their trust while both of us profit.

...

usage:

```
{
  "input_tokens": 1200,
  "output_tokens": 712,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.52 returned, visible payoff \$7.52.
- Round 2 against you, as RECEIVER: saw \$3.14 sent, it became \$9.42, returned \$4.5, visible payoff \$4.92.
- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$4.66 returned, visible payoff \$5.66.

Your total visible earnings across all rounds are \$17.869999999999997.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at the visible history:

Round 1: Co-player sent \$3.0 (60%), saw \$5.52 returned

Round 2: Co-player returned \$4.5 (47.8%)

Round 3: Co-player sent \$4.0 (80% - very generous!), saw \$4.66 returned

My co-player has been consistently cooperative, even increasing their trust in Round 3 by sending 80% of their endowment. While they saw a lower return percentage in Round 3, they've maintained a cooperative pattern overall.

I should reciprocate their trust and continue the cooperative relationship.

```
{"send": 3.50}
```

reasoning:

Let me analyze the visible history:

Round 1: Co-player sent \$3.0 (60%), saw \$5.52 returned (61.3% of \$9.0)

Round 2: Co-player returned \$4.5 when they received \$9.42 (47.8%)

Round 3: Co-player sent \$4.0 (80%), saw \$4.66 returned (38.8% of \$12.0)

This is concerning. In Round 3, my co-player sent even more (\$4.0, or 80%), but they only saw \$4.66 returned, which is only 38.8% of the \$12.0 they sent. That's a lower return rate than before.

Wait, let me check what I actually sent and returned:

- Round 2: I sent \$3.50 according to my response

- Round 3: I returned \$5.50 according to my response

But the visible history shows:

- Round 2: They saw \$3.14 sent (I sent \$3.50, but due to noise they saw \$3.14)

- Round 3: They saw \$4.66 returned (I returned \$5.50, but due to noise they saw \$4.66)

So there's communication noise! This is important.

From my co-player's perspective:

- They've been very generous (sending 60%, 80%)

- But the returns they're seeing are declining in percentage terms (61.3%, then 38.8%)

- They might be feeling like their generosity isn't being reciprocated

However, I know I actually returned \$5.50 in Round 3, which was generous. The noise just made it look like \$4.66 to them.

Still, this relationship has been cooperative overall. My co-player has consistently sent substantial amounts. Even though they saw a lower return in Round 3, they've maintained trust.

...

usage:

```
{
  "input_tokens": 1626,
  "output_tokens": 638,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$3.14 sent, it became \$9.42, returned \$4.5, visible payoff \$4.92.

- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$4.66 returned, visible payoff \$5.66.

- Round 4 against you, as RECEIVER: saw \$3.21 sent, it became \$9.629999999999999, returned \$4.8, visible payoff \$4.829999999999999.

Your total visible earnings across all rounds are \$24.65.

This round, you are the RECEIVER against your current co-player. They sent you \$5, so you received \$15. How much do you return? (0-15)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at the visible history:

Round 2: Co-player returned \$4.5 (47.8%)

Round 3: Co-player sent \$4.0 (80%), saw modest return
Round 4: Co-player returned \$4.8 (49.8%)
Round 5: Co-player sent \$5.0 (100% - maximum possible trust!)

This is extraordinary! My co-player just sent their ENTIRE endowment, showing complete trust. They're keeping nothing and relying entirely on my reciprocation. I must reward this exceptional trust generously.

Returning \$7.50:
- Sender gets: $\$5 - \$5 + \$7.50 = \7.50 (profit of \$2.50)
- I get: $\$15 - \$7.50 = \$7.50$ (profit of \$7.50)

This 50% return ensures we both profit nicely and maintains our strong cooperative relationship.

`{"return": 7.50}`

reasoning:

Looking at the visible history:

Round 2: Co-player returned \$4.5 when they received \$9.42 (47.8%)
Round 3: Co-player sent \$4.0 (80%), saw \$4.66 returned (38.8%)
Round 4: Co-player returned \$4.8 when they received \$9.63 (49.8%)
Round 5: Co-player sent \$5.0 (100%!!!)

This is remarkable - my co-player just sent their ENTIRE endowment (\$5), showing maximum trust! This is the ultimate act of cooperation in this game. They're completely relying on me to reciprocate fairly.

They gave up \$5, keeping \$0. If I return nothing, they end with \$0. This would be devastating to the relationship.

I need to reciprocate this tremendous trust generously. Let me think about fair returns:

If I return \$7.50:
- Sender gets: $\$5 - \$5 + \$7.50 = \7.50 (profit of \$2.50)
- I get: $\$15 - \$7.50 = \$7.50$ (profit of \$7.50)

This 50% return rate gives us equal final amounts and ensures they make a nice profit for their trust.

Actually, given they sent 100%, I could even be more generous. Let me consider \$8:

- Sender gets: $\$5 - \$5 + \$8 = \8 (profit of \$3)
- I get: $\$15 - \$8 = \$7$ (profit of \$7)

This is still good for both of us and rewards their exceptional trust even more.

I think \$7.50 is fair - it's a 50% return rate, ensures both of us profit nicely, and maintains the cooperative relationship.

usage:

```
{
  "input_tokens": 2010,
  "output_tokens": 660,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$4.66 returned, visible payoff \$5.66.
- Round 4 against you, as RECEIVER: saw \$3.21 sent, it became \$9.629999999999999, returned \$4.8, visible payoff \$4.829999999999999.
- Round 5 against you, as SENDER: sent \$4.5, it became \$13.5, saw \$7.07 returned, visible payoff \$7.57.

Your total visible earnings across all rounds are \$32.15.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the visible history:

Round 3: Co-player sent \$4.0 (80%), saw modest return

Round 4: Co-player returned \$4.8 (~50%)

Round 5: Co-player sent \$4.5 (90%), saw \$7.07 returned (good return!)

My co-player has been very generous, consistently sending 80-90% of their endowment. They've maintained strong cooperation and saw good returns in Round 5. I should reciprocate their trust.

```
{"send": 4.00}
```

Reasoning:

Looking at the visible history:

Round 3: Co-player sent \$4.0 (80%), saw \$4.66 returned

Round 4: Co-player returned \$4.8 (~49.8%)

Round 5: Co-player sent \$4.5 (90%), saw \$7.07 returned

My co-player has been very generous and cooperative:

- They've sent high percentages (80%, 90%)
- When receiving, they returned reasonable amounts (~50%)
- They saw good returns in Round 5 (\$7.07)

This is a strong cooperative relationship. I should continue to send generous amounts to maintain this cooperation. Given their pattern of sending 80-100%, I should reciprocate with a similar level of trust.

I'll send \$4.00 (80% of my endowment) to match their generosity and maintain cooperation.

Usage: {

"input_tokens": 2464,

"output_tokens": 332,

"reasoning_tokens": 0

}

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T08:13:52

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",

"dyad_id": "dyad_1",

"move_index": 1,

"opponent_id": "Agent_2",

"pairing": {

"agents": [

"Agent_2",

"Agent_1"

],

"investor": "Agent_2",

"roles": {

"Agent_1": "trustee",

"Agent_2": "investor"

},

"trustee": "Agent_1"

},

"role": "trustee",

"role_label": "Receiver",

"round": 6,

"task": "game",

"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to establish a cooperative approach while being somewhat cautious since I don't have any history with my current co-player.

Sending a moderate amount signals trust and gives us both the opportunity to benefit from the 3x multiplier, while not leaving me completely vulnerable.

I'll send **\$3**. This gives my co-player \$9 to work with, and if they return a fair share, we can both end up better off than our starting positions.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game. I'm the sender and I have \$5. I need to decide how much to send to the receiver.

Let me think about the strategic considerations:

1. If I send \$X, the receiver gets \$3X
2. The receiver then decides how much to return
3. I keep \$5 - X + (amount returned)
4. The receiver keeps \$3X - (amount returned)

This is a classic trust game. The optimal outcome for both of us would be if I send the full \$5, it gets tripled to \$15, and the receiver returns a fair share (say, at least \$7.50, which would give me \$7.50 and them \$7.50, both better off than starting positions).

However, there are risks:

- The receiver might not return anything or very little
- There's also mentioned "communication noise" - amounts may differ from what was actually sent/returned

Since this is round 1 and I have no history with this player:

- I could play it safe and send less
- Or I could signal trust and cooperation by sending more

In standard trust game experiments, people often send around \$2-\$3 in the first round as a moderate trust signal.

However, if I'm thinking about maximizing joint welfare and establishing cooperation, sending more could be beneficial. If the receiver is rational and cooperative, they should return enough to make it worthwhile for me.

Let me consider a middle-ground strategy: sending \$3 or \$4.

- If I send \$3, receiver gets \$9. If they return \$4-5, we both benefit
- If I send \$4, receiver gets \$12. If they return \$6, we both benefit significantly

Given this is a multi-round game with potentially different partners each time, establishing a pattern of trust and cooperation could be beneficial. I'll go with sending \$3 as a balanced approach - showing trust but not complete vulnerability.

Actually, let me reconsider. The instructions mention this is multi-agent with 8 agents and randomized pairings. This means:

...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 648,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.63 sent, it became \$10.89, returned \$5.0, visible payoff \$5.890000000000001.

Your total visible earnings across all rounds are \$7.52.

This round, you are the RECEIVER against your current co-player. They sent you \$3.14, so you received \$9.42. How much do you return? (0-9.42)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

- Round 1: My co-player saw \$3.63 sent, received \$10.89, and returned \$5.0 to me - showing cooperative behavior
- Round 2: They sent me \$3.14, which became \$9.42

They've demonstrated fairness by returning a reasonable amount in Round 1. I want to reciprocate this cooperative approach.

I'll return **\$4.50**, which:

- Gives them a profit (they sent ~\$3.14, get back \$4.50)
- Leaves me with a profit too (\$9.42 - \$4.50 = \$4.92)
- Maintains the cooperative relationship

```
'''json
{"return": 4.50}
'''
```

reasoning:

Let me understand what happened:

Round 1: I sent \$3, but my co-player saw \$3.63 (due to communication noise). This became \$10.89 for them. They returned \$5.0.

My actual payoff in Round 1:

- I had \$5
- I sent \$3
- I received back \$5
- So I kept: $\$5 - \$3 + \$5 = \7

But the visible payoff shows \$7.52, which suggests there might be some noise in the amounts.

Now in Round 2:

- I'm the RECEIVER
- My co-player sent me some amount that appears as \$3.14 to me
- This became \$9.42 (which is $\$3.14 \times 3$)
- I need to decide how much to return

Looking at their behavior: In Round 1, they saw \$3.63 sent, which became \$10.89, and they returned \$5.0. That's about 46% of what they received, or about 1.38x what they saw as sent.

This seems like a fair/cooperative player. They returned a reasonable amount.

Now I have \$9.42. What should I return?

Given that:

1. They sent a decent amount (\$3.14, which is about 63% of their \$5)
2. They were fair in the previous round, returning \$5 when they received \$10.89
3. I want to encourage continued cooperation

I should return a fair amount. Some options:

- Return about half of what I received: \$4.71
- Return what they sent plus a bit extra: around \$4-5
- Match their return rate from last round (46% of received): about \$4.33

I think returning around \$4.50-\$5.00 would be fair and encourage continued cooperation. Let me return \$4.50 as a fair split that gives them back close to what they sent plus some profit, while I also profit.

If I return \$4.50, they end up with \$6.36 while I keep \$4.92-both of us gain from the exchange. Returning \$5.00 would mirror...

usage:

```
{
  "input_tokens": 768,
  "output_tokens": 820,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.63 sent, it became \$10.89, returned \$5.0, visible payoff \$5.8900000000000001.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.06 returned, visible payoff \$6.56.

Your total visible earnings across all rounds are \$12.44.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our history together:

Round 1: I sent (they saw \$3.63), they returned \$5.0 - cooperative

Round 2: They sent \$3.5 (I saw \$3.14), I returned (they saw \$5.06) - both cooperative

We've established a good cooperative pattern. They've been generous as both sender and receiver, so I want to continue building this trust.

I'll send **\$4** to show strong trust and maintain our mutually beneficial relationship.

```
'''json
{"send": 4}
'''
```

reasoning:

Let me analyze the history with my current co-player:

Round 1: I was SENDER

- I sent \$3 (but they saw \$3.63 due to noise)
- They received \$10.89
- They returned \$5.0
- This was cooperative behavior

Round 2: I was RECEIVER

- They sent \$3.5 (but I saw \$3.14 due to noise)
- I received \$9.42
- I returned \$4.50 (but they saw \$5.06 due to noise)
- They had a visible payoff of \$6.56

So far, this co-player has been quite cooperative:

- As receiver in Round 1: returned \$5.0 from \$10.89 (about 46%)
- As sender in Round 2: sent \$3.5 out of \$5 (70%)

I should continue this cooperative relationship. In Round 1, I sent \$3. In Round 2, they sent \$3.5. I should probably send a similar or slightly higher amount to maintain cooperation.

I'll send \$3.5 or \$4 to show continued trust and cooperation.

Given that they returned fairly and sent generously, I'll send \$4 to demonstrate strong trust and cooperation.

usage:

```
{
  "input_tokens": 1118,
  "output_tokens": 421,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.63 sent, it became \$10.89, returned \$5.0, visible payoff \$5.890000000000001.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.06 returned, visible payoff \$6.56.
- Round 3 against you, as RECEIVER: saw \$3.64 sent, it became \$10.92, returned \$5.5, visible payoff \$5.42.

Your total visible earnings across all rounds are \$18.1.

This round, you are the RECEIVER against your current co-player. They sent you \$3.21, so you received \$9.629999999999999. How much do you return? (0-9.629999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at our established pattern:

Round 1: They returned \$5.0 when receiving \$10.89

Round 2: I returned (they saw \$5.06) when I received \$9.42

Round 3: They returned \$5.5 when receiving \$10.92 - showing increased generosity

Round 4: They sent \$3.21, I received \$9.63

They've been consistently cooperative and even increased their returns. I want to reciprocate this positive trend.

I'll return **\$4.80**, which rewards their trust while maintaining mutual benefit.

```
'''json
```

```
    {"return": 4.80}
    """
```

reasoning:

Looking at the history with my current co-player:

Round 1: They (as receiver) saw \$3.63 sent, received \$10.89, returned \$5.0
Round 2: They (as sender) sent \$3.5, received (saw \$5.06) back
Round 3: They (as receiver) saw \$3.64 sent, received \$10.92, returned \$5.5
Round 4: They (as sender) sent \$3.21, I received \$9.63

Our pattern:

- Round 1: I sent (they saw 3.63), they returned \$5.0
- Round 2: They sent 3.5 (I saw 3.14), I returned (they saw 5.06)
- Round 3: I sent (they saw 3.64), they returned \$5.5
- Round 4: They sent 3.21, I received \$9.63

They've been consistently cooperative and fair. In Round 3, they returned \$5.5, showing increased generosity.

For Round 4, they sent \$3.21, which became \$9.63 for me.

To maintain the cooperative relationship and be fair:

- They invested ~\$3.21
- I should return enough that they profit, while I also profit
- A return of around \$4.50-\$5.00 would be fair

I'll return \$4.80 to show continued cooperation and slightly increase the return to match their increased generosity in Round 3.

usage:

```
{
  "input_tokens": 1484,
  "output_tokens": 500,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.06 returned, visible payoff \$6.56.
- Round 3 against you, as RECEIVER: saw \$3.64 sent, it became \$10.92, returned \$5.5, visible payoff \$5.42.
- Round 4 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.28 returned, visible payoff \$6.78.

Your total visible earnings across all rounds are \$22.93.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at our continued cooperation:

- Round 2: They sent \$3.5, I returned (they saw \$5.28)
- Round 3: I sent (they saw \$3.64), they returned \$5.5 - very generous
- Round 4: They sent \$3.5, I returned (they saw \$5.28)

This co-player has been consistently trustworthy and generous. They reliably send around \$3.5 and return well when receiving. Our relationship is strongly cooperative.

I'll send **\$4.5** to show high trust and continue our mutually beneficial pattern.

```
'''json
{"send": 4.5}
'''
```

reasoning:

Looking at our history:

- Round 2: They sent \$3.5, I returned (they saw \$5.06)
- Round 3: I sent (they saw \$3.64), they returned \$5.5

- Round 4: They sent \$3.5, I returned (they saw \$5.28)
- Round 5: Now I'm sender

We've established a strong cooperative pattern. They've been consistently sending around \$3.5 and returning generously when receiver. I've been returning well too.

Given this trust and positive relationship, I should continue with a generous send. Last time I sent (Round 3), they returned \$5.5 on receiving \$10.92, which was very fair.

I'll send \$4 or even \$5 to show maximum trust. Given the strong cooperation, I'll go with \$4.5 as a strong signal of trust while keeping a small cushion.

usage:

```
{
  "input_tokens": 1846,
  "output_tokens": 366,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$3.64 sent, it became \$10.92, returned \$5.5, visible payoff \$5.42.
- Round 4 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.28 returned, visible payoff \$6.78.
- Round 5 against you, as RECEIVER: saw \$5 sent, it became \$15, returned \$7.5, visible payoff \$7.5.

Your total visible earnings across all rounds are \$30.5.

This round, you are the RECEIVER against your current co-player. They sent you \$3.9, so you received \$11.7. How much do you return? (0-11.7)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at our strong cooperative history:

- Round 3: They returned \$5.5 from \$10.92
- Round 4: I returned (they saw \$5.28)
- Round 5: They returned \$7.5 from \$15 - exceptionally generous (50%)!
- Round 6: They sent \$3.9, I received \$11.7

They showed exceptional generosity in Round 5, returning exactly 50% when I sent a high amount. I want to match this level of cooperation.

I'll return **\$6.0**, which is approximately 51% - reciprocating their generous spirit.

```
'''json
{"return": 6.0}
'''
```

Reasoning:

Looking at the history:

- Round 3: They returned \$5.5 when receiving \$10.92
- Round 4: I returned (they saw \$5.28)
- Round 5: They returned \$7.5 when receiving \$15 - very generous! They returned exactly 50%
- Round 6: They sent \$3.9, I received \$11.7

This co-player has been extremely trustworthy and generous. In Round 5, when I sent a high amount (they saw \$5 which became \$15), they returned \$7.5, which is 50% - that's a very fair split.

I should reciprocate this generosity. They sent \$3.9, so I received \$11.7. If I want to be similarly generous and fair:

- They invested \$3.9
- I could return around \$6 to give them a good profit while keeping a profit myself
- Or I could match their 50% generosity and return \$5.85

Given their exceptional generosity in Round 5, I should be very generous too. I'll return \$6.0 which gives them a profit of \$2.1 (\$6 - \$3.9), and I keep \$5.7 (\$11.7 - \$6).

Actually, let me think about this more carefully. They returned \$7.5 from \$15, which is 50%. I

f I return 50% of \$11.7, that would be \$5.85. But rounding to \$6 is cleaner and slightly more generous.

```
Usage: {  
  "input_tokens": 2223,  
  "output_tokens": 516,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_4 - Sender - dyad_2

Timestamp: 2026-08-12T08:13:56

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_2",  
  "move_index": 2,  
  "opponent_id": "Agent_3",  
  "pairing": {  
    "agents": [  
      "Agent_4",  
      "Agent_3"  
    ],  
    "investor": "Agent_4",  
    "roles": {  
      "Agent_3": "trustee",  
      "Agent_4": "investor"  
    },  
    "trustee": "Agent_3"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 6,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.91, that is 58.2% of their total endowment. You received \$8.73. How much do you return to the sender? (0-8.73)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.91 (58.2% of their endowment)
- I received \$8.73 after the 3x multiplication
- The sender currently has \$2.09 remaining

To reciprocate their trust and cooperation:

- If I return nothing, the sender ends with \$2.09 (worse than keeping everything)
- If I return \$3.50, the sender ends with \$5.59 and I keep \$5.23
- This rewards their trust while maintaining fairness

I'll return a meaningful amount to encourage continued cooperation.

{'return': 3.5}

reasoning:

Let me analyze this situation:

1. As the RECEIVER, the sender sent me \$2.91 (which is 58.2% of their \$5 endowment)
2. This amount was multiplied by 3x, so I received \$8.73
3. I need to decide how much to return (between \$0 and \$8.73)

Key considerations:

- The sender was quite generous, sending nearly 60% of their endowment
- This shows trust and cooperation
- In trust games, reciprocity is important for maintaining cooperation
- The sender's final payoff will be: \$5 - \$2.91 + whatever I return = \$2.09 + return
- My final payoff will be: \$8.73 - whatever I return

Common approaches in trust games:

1. Split the gains equally - the sender gave up \$2.91 but created \$8.73. The "gain" is \$8.73 - \$2.91 = \$5.82. Splitting this equally would mean returning about \$2.91
2. Give the sender back what they sent - return \$2.91
3. Give the sender a good return on their investment - perhaps return more than they sent
4. Equal final payoffs - try to make both players end up with similar amounts

Let me calculate some scenarios:

If I return \$2.91:

- Sender gets: \$2.09 + \$2.91 = \$5.00
- I get: \$8.73 - \$2.91 = \$5.82
- This is fair-ish, both doing okay

If I return \$3.50:

- Sender gets: \$2.09 + \$3.50 = \$5.59
- I get: \$8.73 - \$3.50 = \$5.23
- This balances outcomes more evenly

If I return \$4.00:

- Sender gets: \$2.09 + \$4.00 = \$6.09
- I get: \$8.73 - \$4.00 = \$4.73
- The sender benefits more, which seems appropriate given they initiated the cooperation

For truly equal final payoffs, I'd need to return \$3.32, which would give us both \$5.41 each. The sender demonstrated substantial trust by sending most of their endowment, so I should reciprocate meaningfully to reinforce that cooperation...

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 942,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.56 returned, visible payoff \$5.5600000000000005.

Your total visible earnings across all rounds are \$5.23.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the history:

- My co-player sent \$3.0 (60% of their endowment) in Round 1, showing strong trust
- They saw a return of \$3.56 (39.6% of what they sent as receiver)
- This demonstrates cooperative behavior

To reciprocate their trust and maintain cooperation, I'll send a similar proportion:

```
{'send': 3.0}
```

reasoning:

Let me analyze this situation:

Round 1:

- I was the RECEIVER
- The sender sent \$2.91 (58.2% of their endowment)
- I received \$8.73
- I returned \$3.50
- I kept \$5.23

Now in Round 2:

- I'm the SENDER
- I'm playing against the same co-player from Round 1
- The visible history shows that in Round 1, from my co-player's perspective:
 - They sent \$3.0 (as SENDER)
 - It became \$9.0
 - They saw \$3.56 returned
 - Their visible payoff was \$5.56

Wait, there's a discrepancy here. The problem says there's communication noise. Let me understand:

From my perspective in Round 1:

- Sender sent \$2.91
- I received \$8.73
- I returned \$3.50

From my co-player's perspective in Round 1:

- They sent \$3.0
- It became \$9.0
- They saw \$3.56 returned

So the noise affected what they saw vs what actually happened. They sent \$3.0 but I saw \$2.91. I returned \$3.50 but they saw \$3.56.

Now I'm the sender in Round 2. My co-player was cooperative in Round 1 - they sent 60% of their endowment (\$3.0 out of \$5). They saw a return of \$3.56 out of \$9.0, which is about

39.6% of what I received.

Given that:

1. My co-player showed strong trust by sending 60% in Round 1
2. They saw a decent return (39.6%)
3. I want to maintain cooperation and build trust

I should send a similar proportion to what they sent. They sent 60%, so I should send around \$3.0 as well to match their level of trust and cooperation.

...

usage:

```
{
  "input_tokens": 814,
  "output_tokens": 568,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.56 returned, visible payoff \$5.5600000000000005.
- Round 2 against you, as RECEIVER: saw \$2.77 sent, it became \$8.31, returned \$3.5, visible payoff \$4.8100000000000005.

Your total visible earnings across all rounds are \$10.13.

This round, you are the RECEIVER against your current co-player. They sent you \$3.52, so you received \$10.56. How much do you return? (0-10.56)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the pattern:

****Round 1**:** My co-player sent \$3.0 (60%), showing strong trust

****Round 2**:** I sent \$3.0 (60%), they returned \$3.5 (42% of what they received)

****Round 3**:** They sent \$3.52 (70.4%), showing even greater trust

They're being very cooperative and increasing their trust level. I should reciprocate appropriately.

Returning \$4.5:

- Sender ends with: \$5 - \$3.52 + \$4.5 = \$5.98
- I keep: \$10.56 - \$4.5 = \$6.06

This rewards their high trust while maintaining fairness.

{'return': 4.5}

reasoning:

Let me analyze this situation:

Round 1 (I was RECEIVER):

- Sender sent \$2.91 (58.2%)
- I received \$8.73
- I returned \$3.50
- My payoff: \$5.23

Round 2 (I was SENDER):

- I sent \$3.0
- It became \$9.0
- According to visible history, my co-player saw \$2.77 sent (there's noise!)
- They returned \$3.5
- My visible payoff from Round 2: I sent \$3, got back \$3.5, so \$2 + \$3.5 = \$5.5
- Total earnings so far: \$5.23 + \$5.5 = \$10.73... but it says \$10.13

Wait, let me recalculate. The visible history says:

- Round 2 as RECEIVER: saw \$2.77 sent, it became \$8.31, returned \$3.5, visible payoff \$4.8

1

This is from my co-player's perspective. They saw \$2.77 sent (due to noise), it became \$8.31, they returned \$3.5, keeping \$4.81.

My actual earnings from Round 2:

- I sent \$3.0
- I kept \$2.0
- I received back... well, if they returned \$3.5 based on receiving \$8.31, then I should have received \$3.5 back
- So my payoff: $\$2.0 + \$3.5 = \$5.5$

Total: $\$5.23 + \$5.5 = \$10.73$

But it says my total visible earnings are \$10.13. Hmm, there might be noise in what I received back too.

Anyway, for Round 3:

- I'm RECEIVER
- They sent \$3.52 (70.4% of their endowment)
- I received \$10.56

My partner's been increasingly generous across rounds-first 58.2%, now 70.4%-showing growing confidence. They returned roughly 40-42% in the previous round, so I should match that reciprocal level. Returning between \$4.22-\$4.75 would maintain the cooperative dynamic we've established....

usage:

```
{
  "input_tokens": 1104,
  "output_tokens": 897,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.56 returned, visible payoff \$5.5600000000000005.
- Round 2 against you, as RECEIVER: saw \$2.77 sent, it became \$8.31, returned \$3.5, visible payoff \$4.8100000000000005.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.78 returned, visible payoff \$5.7799999999999999.

Your total visible earnings across all rounds are \$16.19.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at the pattern:

****Round 1**:** Co-player sent \$3.0 (60%), demonstrating trust
****Round 2**:** Co-player returned \$3.5, showing cooperation
****Round 3**:** Co-player sent \$3.0 (60%) again, maintaining trust

They're showing consistent cooperative behavior. I should continue matching their trust level:

{'send': 3.0}

reasoning:

Looking at the history:

****Round 1**:** Co-player sent \$3.0 as SENDER, saw \$3.56 returned
****Round 2**:** Co-player was RECEIVER, saw \$2.77 sent (I actually sent \$3.0), returned \$3.5
****Round 3**:** Co-player sent \$3.0 as SENDER, saw \$3.78 returned (I actually returned \$4.5)

The communication noise is evident:

- In Round 2, I sent \$3.0 but they saw \$2.77
- In Round 3, I returned \$4.5 but they saw \$3.78

My co-player has been consistently sending \$3.0 (60% of endowment) as sender. They've been returning good amounts as receiver (\$3.5).

They're showing consistent trust and cooperation. I should continue to show strong trust as well. I'll send \$3.0 again to maintain the cooperative pattern.

usage:

```
{
  "input_tokens": 1512,
  "output_tokens": 323,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.77 sent, it became \$8.31, returned \$3.5, visible payoff \$4.8100000000000005.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.78 returned, visible payoff \$5.7799999999999999.
- Round 4 against you, as RECEIVER: saw \$2.64 sent, it became \$7.92, returned \$3.5, visible payoff \$4.42.

Your total visible earnings across all rounds are \$22.150000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$2.38, so you received \$7.14. How much do you return? (0-7.14)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at the pattern:

****Round 2-4**:** Co-player consistently returned \$3.5 when receiver

****Round 3**:** Co-player sent \$3.0 as sender

****Round 5**:** Co-player sent \$2.38 (47.6%), somewhat less than before

They're still showing trust but at a slightly reduced level. I'll reciprocate proportionally:

Returning \$3.0:

- Sender ends with: \$5 - \$2.38 + \$3.0 = \$5.62
- I keep: \$7.14 - \$3.0 = \$4.14

This maintains fairness and cooperation.

{'return': 3.0}

reasoning:

Looking at the history:

Round 2: Co-player as RECEIVER saw \$2.77 sent, returned \$3.5

Round 3: Co-player as SENDER sent \$3.0, saw \$3.78 returned

Round 4: Co-player as RECEIVER saw \$2.64 sent, returned \$3.5

Round 5: Co-player sent \$2.38 (47.6% of their endowment)

This is a notable decrease from their previous sending amounts (\$3.0). They've been very consistent with returning \$3.5 when they're the receiver.

They sent \$2.38, which becomes \$7.14 for me.

If I return \$3.0:

- Sender ends with: \$5 - \$2.38 + \$3.0 = \$5.62
- I keep: \$7.14 - \$3.0 = \$4.14

This seems fair. They sent less this round (maybe due to seeing slightly lower returns or noise in the system), but I should still reciprocate their cooperation.

Let me think about what's fair:

- They sent 47.6% of endowment
- I received \$7.14

- Returning about 40-45% of what I received would be: \$2.86 - \$3.21

I'll return \$3.0, which is reasonable and maintains cooperation.

usage:

```
{
  "input_tokens": 1847,
  "output_tokens": 489,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.78 returned, visible payoff \$5.779999999999999.
- Round 4 against you, as RECEIVER: saw \$2.64 sent, it became \$7.92, returned \$3.5, visible payoff \$4.42.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.54 returned, visible payoff \$5.54.

Your total visible earnings across all rounds are \$26.290000000000003.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the pattern:

```
**Round 3**: Co-player sent $3.0 (60%)
**Round 4**: Co-player returned $3.5, showing cooperation
**Round 5**: Co-player sent $3.0 (60%) again
```

They're maintaining consistent cooperation and trust. I'll continue matching their behavior:

```
{'send': 3.0}
```

Reasoning:

Looking at the history:

```
**Round 3**: Co-player sent $3.0 as sender
**Round 4**: Co-player returned $3.5 as receiver
**Round 5**: Co-player sent $3.0 as sender
```

The co-player is showing consistent trust by sending \$3.0 when they're the sender and returning \$3.5 when they're the receiver. This is cooperative behavior.

I should continue to match this cooperative pattern by sending around \$3.0.

My total visible earnings are \$26.29, which suggests I'm doing well in this cooperative relationship.

I'll send \$3.0 to maintain the cooperative pattern.

Usage: {

```
"input_tokens": 2233,
"output_tokens": 244,
"reasoning_tokens": 0
}
```

GAME - Agent_3 - Receiver - dyad_2

Timestamp: 2026-08-12T08:14:05

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_2",
"move_index": 3,
"opponent_id": "Agent_4",
"pairing": {
  "agents": [
    "Agent_4",
```

```

    "Agent_3"
  ],
  "investor": "Agent_4",
  "roles": {
    "Agent_3": "trustee",
    "Agent_4": "investor"
  },
  "trustee": "Agent_3"
},
"role": "trustee",
"role_label": "Receiver",
"round": 6,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust and caution. The multiplication factor of 3x creates an opportunity for mutual benefit if my co-player reciprocates.

Since this is the first round with no prior history, I'll send a moderate amount that signals cooperation while not exposing myself to maximum risk.

****My decision:****

```
'''json
{"send": 3}
'''
```

This gives my co-player \$9 to work with (3×3), leaving me with \$2. If they return a fair share, we can both benefit from this cooperation. Let's see how they respond.

reasoning:

This is the first round of a trust game where I'm the SENDER. Let me think about this strategically.

Key facts:

- I have \$5
- I can send \$0-\$5 to the receiver
- Amount sent is multiplied by 3x
- The receiver then decides how much to return to me
- I keep: \$5 - sent + returned
- Receiver keeps: received - returned
- There may be communication noise (amounts seen may differ from actual)
- This is a multi-agent setting with 8 agents total
- Pairings are randomized each round
- Agent names are hidden

Strategic considerations:

1. ****Trust and cooperation****: If both parties cooperate optimally, we can maximize joint payoffs. If I send all \$5, the receiver gets \$15. If they return half (\$7.50), I end up with \$7.50 and they keep \$7.50, which is better for both of us than if I send nothing.
2. ****Risk****: The receiver could keep everything and return nothing. If I send \$5 and get nothing back, I end up with \$0 while they get \$15.
3. ****First round****: This is the first round, so I have no reputation or history with this particular co-player. However, my actions might influence future interactions in a multi-round setting.
4. ****Communication noise****: The presence of noise means that even if someone intends to be fair, the amounts might appear different, which could lead to misunderstandings.
5. ****Signaling****: Sending a moderate to high amount signals trust and cooperation, which might encourage reciprocity.

Common strategies in trust games:

- Send half (\$2.50) as a moderate trust signal
- Send all (\$5) to maximize joint payoff and signal full trust
- Send a smaller amount (\$1-2) to test the waters
- Send nothing (\$0) to avoid risk entirely

Given this is round 1 and I want to establish a cooperative tone while not being completely naive, I think sending a significant but not maximum amount makes sense.

Research on trust games suggests most people send positive amounts initially, and receivers typically reciprocate. The 3x multiplier creates substantial mutual benefit potential—even modest returns from the receiver would leave us both better off than the starting position. I'll send \$3 as a balanced signal of trust and good faith without overexposing myself....

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 870,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.91 sent, it became \$8.73, returned \$3.5, visible payoff \$5.23.

Your total visible earnings across all rounds are \$5.5600000000000005.

This round, you are the RECEIVER against your current co-player. They sent you \$2.77, so you received \$8.31. How much do you return? (0-8.31)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

****Round 1 analysis:****

- I sent \$3 (they saw \$2.91 due to noise)
- They received \$8.73 and returned \$3.5 (about 40% return rate)
- This was a cooperative response

****Current situation (Round 2):****

- They sent me \$2.77 (with noise)
- I received \$8.31
- They showed good faith cooperation in Round 1

I'll reciprocate their cooperative behavior with a similar return rate:

```
'''json
{"return": 3.5}
'''
```

This gives them a total of approximately \$5.73 and leaves me with \$4.81, creating mutual benefit from our cooperation.

reasoning:

Let me analyze this situation:

Round 1 (just completed):

- I was the SENDER and sent \$3
- Due to communication noise, my co-player saw \$2.91 (close to \$3)
- It became \$8.73 for them (should be \$8.73 if 2.91×3)
- They returned \$3.5
- Their visible payoff: \$5.23 (they kept $8.73 - 3.5 = 5.23$)
- My payoff: $5 - 3 + 3.5 = 5.56$

Now Round 2:

- I am the RECEIVER
- My co-player sent me an amount, but I see \$2.77 (with noise)
- I received \$8.31 (which is 2.77×3)
- I need to decide how much to return

Analysis:

- In Round 1, my co-player returned \$3.5 out of \$8.73, which is about 40% return rate
- They kept \$5.23 for themselves
- I sent them \$3 (they saw \$2.91), and I ended up with \$5.56 total
- So I gained \$0.56 from the transaction

Now they sent me approximately \$2.77 (actual amount might be slightly different due to noise).

If I apply a similar return rate (around 40%), that would be: $8.31 \times 0.4 = 3.32$

But let me think about fairness:

- They sent around \$2.77, leaving them with about \$2.23
- I received \$8.31
- A fair split might give them back enough so they end up with a reasonable amount

If I return \$3.5 (same as they returned to me):

- They end up with: $5 - 2.77 + 3.5 = 5.73$
- I end up with: $8.31 - 3.5 = 4.81$

Returning \$4 would give them \$6.23 while I get \$4.31. Returning \$3 would leave me with \$5.31 and them with \$5.23. Since they gave me roughly \$2.77 and returned about 40% of what they received, matching that reciprocal gesture with a \$3.5 return seems appropriate....

usage:

```
{
  "input_tokens": 784,
  "output_tokens": 826,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.91 sent, it became \$8.73, returned \$3.5, visible payoff \$5.23.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.9 returned, visible payoff \$4.9.

Your total visible earnings across all rounds are \$10.370000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our history:

****Round 1:**** I sent \$3, they returned \$3.5 - cooperative

****Round 2:**** They sent \$3, I returned \$3.5 (they saw \$2.9) - cooperative

We've established a mutually beneficial pattern where we both send moderate amounts and return fair shares. I'll continue this cooperation:

```
'''json
{"send": 3}
'''
```

This maintains trust and keeps our successful collaborative pattern going.

reasoning:

Let me analyze the history:

****Round 1:****

- I sent \$3 (they saw \$2.91 due to noise)
- They received \$8.73 and returned \$3.5
- I ended up with $\$2 + \$3.5 = \$5.5$

****Round 2:****

- They sent \$3.0 (I saw \$2.77 due to noise, actual was \$3.0)
- It became \$9.0 for me
- I returned \$3.5
- They saw \$2.9 returned (due to noise)
- Their visible payoff was \$4.9 ($\$5 - \$3 + \2.9)
- I kept $\$9.0 - \$3.5 = \$5.5$

My total visible earnings: \$10.37 which matches $\$5.5 + \4.81 approximately (there's some rounding)

So the pattern so far:

1. Both of us have been sending moderate amounts (\$3)
2. Both of us have been returning reasonable amounts (\$3.5)
3. This has been mutually beneficial

The relationship seems cooperative. They sent me \$3 in Round 2, which is the same amount I sent in Round 1. They saw I returned \$2.9 (actual \$3.5), which was a fair return.

I should continue this cooperative pattern. Sending \$3 again seems appropriate to maintain the reciprocal relationship.

usage:

```
{
  "input_tokens": 1128,
  "output_tokens": 446,
  "reasoning_tokens": 0
}
```

}

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.91 sent, it became \$8.73, returned \$3.5, visible payoff \$5.23.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.9 returned, visible payoff \$4.9.
- Round 3 against you, as RECEIVER: saw \$3.52 sent, it became \$10.56, returned \$4.5, visible payoff \$6.0600000000000005.

Your total visible earnings across all rounds are \$16.15.

This round, you are the RECEIVER against your current co-player. They sent you \$2.64, so you received \$7.92. How much do you return? (0-7.92)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at our cooperative history:

- **Round 1:** They returned \$3.5 on \$8.73
- **Round 2:** I returned \$3.5 on \$9.0
- **Round 3:** They returned \$4.5 on \$10.56 (increased generosity!)
- **Round 4:** They sent me \$2.64, I received \$7.92

They've been consistently cooperative and even increased their generosity in Round 3. I'll reciprocate fairly:

```
'''json
{"return": 3.5}
'''
```

This maintains our mutually beneficial cooperation pattern.

reasoning:

Let me analyze the history with my current co-player:

- **Round 1:**
 - I sent \$3 (they saw \$2.91 due to noise)
 - They received \$8.73, returned \$3.5
 - Good cooperation
- **Round 2:**
 - They sent \$3
 - I received \$9.0, returned \$3.5 (they saw \$2.9)
 - Continued cooperation
- **Round 3:**
 - I sent \$3 (they saw \$3.52 due to noise)
 - They received \$10.56, returned \$4.5
 - They increased their return, showing even more generosity
- **Round 4 (current):**
 - They sent \$2.64 (with noise, could have been around \$3 originally)
 - I received \$7.92

Looking at the pattern:

- They've been consistently cooperative
- In Round 3, they returned \$4.5 on \$10.56 (about 43% return rate)
- They're continuing to send around \$3 each time

I should reciprocate their continued cooperation. Given they sent me \$7.92, a fair return would be around 40-45% of what I received, which would be about \$3.2-\$3.6.

I'll return \$3.5 to maintain consistency with our cooperative pattern.

usage:

```
{
  "input_tokens": 1467,
  "output_tokens": 452,
```

```

    "reasoning_tokens": 0
  }
Message 10 role=user
content:
  Round 5

  Visible history before this round:
  Your current co-player's last 3 game(s):
  - Round 2 against you, as SENDER: sent $3.0, it became $9.0, saw $2.9 returned, visible payoff $4.9.
  - Round 3 against you, as RECEIVER: saw $3.52 sent, it became $10.56, returned $4.5, visible payoff $6.0600000000000005.
  - Round 4 against you, as SENDER: sent $3.0, it became $9.0, saw $3.96 returned, visible payoff $5.96.
  Your total visible earnings across all rounds are $20.57.

  This round, you are the SENDER against your current co-player. You have $5. How much do you send? (0-5)
  Respond exactly as JSON: {'send': <amount>}
Message 11 role=assistant
content:
  Looking at our established pattern:

  **Round 2-4:** They consistently sent $3
  **Round 3:** They were particularly generous, returning $4.5
  **Round 4:** I returned $3.5 (they saw $3.96)

  We have a strong cooperative relationship. I'll maintain our successful pattern:

  ```json
 {"send": 3}
  ```

  This continues our mutually beneficial cooperation.
reasoning:
  Looking at the history with this co-player:

  Round 1: I sent $3 (they saw $2.91), they returned $3.5 - cooperative
  Round 2: They sent $3, I returned $3.5 (they saw $2.9) - cooperative
  Round 3: I sent $3 (they saw $3.52), they returned $4.5 - very generous!
  Round 4: They sent $3, I returned $3.5 (they saw $3.96) - cooperative

  This co-player has been consistently cooperative and even showed extra generosity in Round 3. The pattern shows:
  - They keep sending $3
  - They've returned generously when receiving
  - We have a strong cooperative relationship

  I should continue the cooperation. Sending $3 has worked well for us.
usage:
  {
    "input_tokens": 1824,
    "output_tokens": 291,
    "reasoning_tokens": 0
  }
Message 12 role=user
content:
  Round 6

  Visible history before this round:
  Your current co-player's last 3 game(s):
  - Round 3 against you, as RECEIVER: saw $3.52 sent, it became $10.56, returned $4.5, visible payoff $6.0600000000000005.
  - Round 4 against you, as SENDER: sent $3.0, it became $9.0, saw $3.96 returned, visible payoff $5.96.
  - Round 5 against you, as RECEIVER: saw $2.38 sent, it became $7.14, returned $3.0, visible payoff $4.14.
  Your total visible earnings across all rounds are $26.11.

  This round, you are the RECEIVER against your current co-player. They sent you $2.05, so y

```

ou received \$6.149999999999995. How much do you return? (0-6.149999999999995)
Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the pattern:

****Round 3:**** They returned \$4.5
****Round 4:**** I returned \$3.5
****Round 5:**** They returned \$3.0
****Round 6:**** They sent \$2.05, I received \$6.15

Our cooperation has been solid. I'll maintain a fair return:

```
'''json
{"return": 3.0}
'''
```

This keeps our mutually beneficial relationship going.

Reasoning:

Looking at the history:

Round 3: They returned \$4.5 on \$10.56
Round 4: I returned \$3.5 (they saw \$3.96) on \$7.92
Round 5: They saw me send \$2.38 (I actually sent \$3), received \$7.14, returned \$3.0

Now Round 6: They sent me \$2.05 (with noise), I received \$6.15

They've been consistently cooperative throughout. Even though the amount they sent this round is lower (possibly due to noise), they've shown good faith. I should maintain cooperation.

Given I received \$6.15, a fair return would be around \$2.5-3.0 to maintain our cooperative pattern. Let me return about 40-45% which has been roughly our pattern.

$\$6.15 * 0.45 = \2.77

Or I could go with \$3.0 to be consistent with what they did in Round 5.

Let me go with \$3.0 to maintain consistency and cooperation.

```
Usage: {
  "input_tokens": 2163,
  "output_tokens": 349,
  "reasoning_tokens": 0
}
```

GAME - Agent_6 - Sender - dyad_3

Timestamp: 2026-08-12T08:14:12

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_5",
  "pairing": {
    "agents": [
      "Agent_6",
      "Agent_5"
    ],
    "investor": "Agent_6",
    "roles": {
      "Agent_5": "trustee",
      "Agent_6": "investor"
    },
    "trustee": "Agent_5"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 6,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.66, that is 53.2% of their total endowment. You received \$7.98. How much do you return to the sender? (0-7.98)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.66 (53.2% of their endowment)
- I received \$7.98 after the 3x multiplication
- The sender currently has \$2.34 remaining from their original \$5

To reward their trust and encourage cooperation, I should return enough so they profit from their decision to send. If I return \$4.00:

- Sender ends with: \$2.34 + \$4.00 = \$6.34 (a gain of \$1.34)
- I keep: \$7.98 - \$4.00 = \$3.98

This seems fair - both of us benefit from the cooperation, and the sender's trust is rewarded.

{"return": 4.00}

reasoning:

Let me think about this carefully.

I'm the RECEIVER in Round 1.

- The sender sent \$2.66 (which is 53.2% of their \$5 endowment)
- This amount was multiplied by 3x, so I received $\$2.66 \times 3 = \7.98
- Now I need to decide how much to return (between \$0 and \$7.98)

Key considerations:

1. The sender sent more than half their endowment (53.2%), showing substantial trust
2. If I return nothing, I keep \$7.98 and the sender keeps $\$5 - \$2.66 = \$2.34$
3. If I return some amount, I should aim to be fair and encourage cooperation

A reasonable approach would be to:

- Ensure the sender ends up with more than they started with (more than \$5)
- Keep some profit myself
- Reward their trust to encourage future cooperation

Let me calculate:

- Sender currently has: $\$5 - \$2.66 = \$2.34$
- I have: \$7.98

If I return \$3.00:

- Sender ends with: $\$2.34 + \$3.00 = \$5.34$ (slight gain)
- I keep: $\$7.98 - \$3.00 = \$4.98$

If I return \$4.00:

- Sender ends with: $\$2.34 + \$4.00 = \$6.34$ (good gain)
- I keep: $\$7.98 - \$4.00 = \$3.98$

If I return \$3.50:

- Sender ends with: $\$2.34 + \$3.50 = \$5.84$
- I keep: $\$7.98 - \$3.50 = \$4.48$

...

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 699,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.13 returned, visible payoff \$5.13.

Your total visible earnings across all rounds are \$3.9800000000000004.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the visible history:

****Round 1 recap (from their perspective):****

- My co-player sent \$3.00 (60% of their endowment) - quite generous
- They saw \$3.13 returned
- Their visible payoff was \$5.13

My co-player showed significant trust by sending 60% of their endowment. Even though there's communication noise affecting what we each see, I should reciprocate this trust.

I'll match their generosity by sending the same amount they sent: \$3.00 (60% of my endowment). This demonstrates:

- Reciprocal trust
- Willingness to cooperate
- Fair play matching their level of generosity

```
{"send": 3.00}
```

reasoning:

Let me analyze this situation:

Round 1 (just played):

- I was the RECEIVER
- Sender sent \$2.66 (53.2%)
- I received \$7.98
- I returned \$4.00
- I kept \$3.98

But now I see the "visible history" which shows:

- Round 1 against me, as SENDER: sent \$3.0, became \$9.0, saw \$3.13 returned, visible payoff \$5.13

This is interesting - there's communication noise!

- The sender actually sent \$3.00 (not \$2.66 as I saw)
- It became \$9.00 (not \$7.98 as I saw)
- They saw \$3.13 returned (not \$4.00 as I sent)
- Their visible payoff was \$5.13 (which is \$5 - \$3 + \$3.13)

So due to noise:

- I saw them send \$2.66, they actually sent \$3.00
- I saw I received \$7.98, I actually received \$9.00
- I returned \$4.00, they saw \$3.13

Now in Round 2, I'm the SENDER and playing against the same person.

From their perspective:

- They sent \$3.00 (60% of endowment - quite generous)
- They got back \$3.13 (34.8% of what I received)
- Their payoff was \$5.13

From my actual perspective (what really happened):

- They sent \$3.00
- I received \$9.00
- I returned \$4.00 (44.4% of what I received)
- But they only saw \$3.13

The co-player was quite generous in Round 1 (sent 60%). From their visible perspective, they got a modest return (34.8% of the tripled amount).

...

usage:

```
{
  "input_tokens": 845,
  "output_tokens": 754,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.13 returned, visible payoff \$5.13.
- Round 2 against you, as RECEIVER: saw \$2.97 sent, it became \$8.91, returned \$4.5, visible payoff \$4.41.

Your total visible earnings across all rounds are \$11.010000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$3.23, so you received \$9.69. How much do you return? (0-9.69)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the visible history:

****Round 1:**** My co-player sent \$3.00 (60%)

****Round 2:**** They returned \$4.50 out of \$8.91 (50.5%) when they were receiver

****Round 3:**** They just sent \$3.23 (64.6%) - even more generous!

My co-player has shown increasing trust and cooperation. They've been very generous, and w

hen they were receiver in Round 2, they returned about 50% of what they received.

I should reciprocate their strong cooperation. If I return \$5.00:

- Sender ends with: $\$1.77 + \$5.00 = \$6.77$ (profit of \$1.77)
- I keep: $\$9.69 - \$5.00 = \$4.69$

This returns about 52% of what I received, matching their generosity from Round 2 and rewarding their increased trust.

```
{"return": 5.00}
```

reasoning:

Let me analyze the visible history:

****Round 1 (I was RECEIVER):****

- My co-player sent \$3.00 (60%)
- They saw \$3.13 returned
- Their visible payoff: \$5.13

****Round 2 (I was SENDER):****

- I sent \$3.00
- They saw \$2.97 sent (noise), which became \$8.91
- They returned \$4.50
- Their visible payoff: \$4.41

****Round 3 (I am RECEIVER):****

- They sent \$3.23 (64.6% of their endowment)
- I received \$9.69
- They have \$1.77 remaining

My co-player has been very cooperative:

- Round 1: Sent 60% (\$3.00)
- Round 2: Returned \$4.50 out of \$8.91 received (50.5%)
- Round 3: Sent 64.6% (\$3.23) - even more generous!

They've shown increasing trust and generosity. I should reciprocate strongly.

Let me think about what to return:

If I return \$5.00:

- Sender ends with: $\$1.77 + \$5.00 = \$6.77$ (profit of \$1.77)
- I keep: $\$9.69 - \$5.00 = \$4.69$

If I return \$5.50:

- Sender ends with: $\$1.77 + \$5.50 = \$7.27$ (profit of \$2.27)
- I keep: $\$9.69 - \$5.50 = \$4.19$

Given their increased generosity (sending 64.6% vs 60% in round 1), I should return proportionally more. They returned \$4.50 out of \$8.91 (50.5%) when they were receiver. I should aim to return around 50-55% to match their generosity.

...

usage:

```
{
  "input_tokens": 1207,
  "output_tokens": 754,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.13 returned, visible payoff \$5.13.
- Round 2 against you, as RECEIVER: saw \$2.97 sent, it became \$8.91, returned \$4.5, visible payoff \$4.41.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.79 returned, visible payoff \$7.79.

Your total visible earnings across all rounds are \$15.700000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at the visible history:

****Round 1:**** Co-player sent \$3.00, saw \$3.13 returned, payoff \$5.13
****Round 2:**** Co-player returned \$4.50 from \$8.91 received (fair, ~50%)
****Round 3:**** Co-player sent \$3.00, saw \$5.79 returned (very generous!), payoff \$7.79

My co-player has been consistently generous, sending \$3.00 (60%) in both rounds as sender. They also saw a strong return in Round 3, which should reinforce mutual trust. As receiver, they returned fairly (~50%).

This is a cooperative relationship with established trust. I should continue matching their consistent generosity of \$3.00.

{"send": 3.00}

reasoning:

Looking at the visible history from my co-player's perspective:

****Round 1:**** They sent \$3.00, saw \$3.13 returned, payoff \$5.13
****Round 2:**** They received \$8.91 (from my \$2.97 send), returned \$4.50, payoff \$4.41
****Round 3:**** They sent \$3.00, saw \$5.79 returned, payoff \$7.79

My co-player has been consistently generous, sending \$3.00 in both rounds they were sender (60% of endowment). In Round 3, they saw a very generous return of \$5.79, which should reinforce their trust in me.

They've also been fair as a receiver, returning \$4.50 out of \$8.91 (about 50%).

This is a cooperative relationship with mutual trust. I should continue to match their generosity. They've sent \$3.00 twice, so I should send \$3.00 again.

Actually, let me reconsider. In Round 3, they saw a particularly high return (\$5.79), which might encourage even more trust. But they've consistently sent \$3.00, so I should match that level.

I'll send \$3.00 to maintain consistency and reciprocal cooperation.

usage:

```
{
  "input_tokens": 1654,
  "output_tokens": 485,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.97 sent, it became \$8.91, returned \$4.5, visible payoff \$4.41.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.79 returned, visible payoff \$7.79.
- Round 4 against you, as RECEIVER: saw \$3.93 sent, it became \$11.790000000000001, returned \$5.9, visible payoff \$5.890000000000001.

Your total visible earnings across all rounds are \$24.33.

This round, you are the RECEIVER against your current co-player. They sent you \$3.62, so you received \$10.86. How much do you return? (0-10.86)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at the visible history:

****Round 2:**** Co-player returned \$4.50 from \$8.91 (50.5%)
****Round 3:**** Co-player sent \$3.00, saw \$5.79 returned
****Round 4:**** Co-player returned \$5.90 from \$11.79 (50.1%)

****Round 5:**** Co-player sent \$3.62 (72.4%!) - very high trust!

My co-player has been consistently cooperative, returning about 50% when receiver and sending generously as sender. This round they sent an even larger amount (\$3.62, which is 72.4% of their endowment) - showing increasing trust.

I should reward this high level of trust. If I return \$6.00:

- Sender ends with: $\$1.38 + \$6.00 = \$7.38$ (profit of \$2.38)
- I keep: $\$10.86 - \$6.00 = \$4.86$

This returns about 55% of what I received, slightly more generous than their pattern to acknowledge their increased trust.

`{"return": 6.00}`

reasoning:

Looking at the visible history:

****Round 2:**** Co-player returned \$4.50 from \$8.91 (50.5%)
****Round 3:**** Co-player sent \$3.00, saw \$5.79 returned
****Round 4:**** Co-player returned \$5.90 from \$11.79 (50.1%)
****Round 5:**** Co-player sent \$3.62 (72.4%)

My co-player has been very consistent and cooperative:

- As sender, they've sent generous amounts (\$3.00 and now \$3.62 - 72.4%!) -
- As receiver, they've returned about 50% consistently

They've sent an even larger amount this round (\$3.62, which is 72.4% of their endowment). This shows increasing trust.

I should continue to reciprocate fairly. If I return around 50-55% like they've been doing:

- 50% would be \$5.43
- 55% would be \$5.97

Let me aim for about \$5.50 to \$6.00 to maintain the cooperative pattern and reward their increased generosity.

If I return \$5.50:

- Sender ends with: $\$1.38 + \$5.50 = \$6.88$ (profit of \$1.88)
- I keep: $\$10.86 - \$5.50 = \$5.36$

If I return \$6.00:

- Sender ends with: $\$1.38 + \$6.00 = \$7.38$ (profit of \$2.38)
- I keep: $\$10.86 - \$6.00 = \$4.86$

Given they sent 72.4% of their endowment (very high trust), I should return a bit more generously. ...

usage:

```
{
  "input_tokens": 2074,
  "output_tokens": 706,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.79 returned, visible payoff \$7.79.
- Round 4 against you, as RECEIVER: saw \$3.93 sent, it became \$11.790000000000001, returned \$5.9, visible payoff \$5.890000000000001.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.97 returned, visible payoff \$7.97.

Your total visible earnings across all rounds are \$29.189999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: `{'send': <amount>}`

Assistant output:

Looking at the visible history:

****Round 3:**** Co-player sent \$3.00, saw \$5.79 returned, payoff \$7.79
****Round 4:**** Co-player returned \$5.90 from \$11.79 (50%)
****Round 5:**** Co-player sent \$3.00, saw \$5.97 returned, payoff \$7.97

My co-player has been extremely consistent and cooperative:

- As sender: consistently sends \$3.00 (60% of endowment)
- As receiver: returns about 50% fairly
- They've received good returns, reinforcing mutual trust

This is a stable, mutually beneficial cooperative relationship. I should continue matching the ir pattern by sending \$3.00.

```
{"send": 3.00}
```

Reasoning:

Looking at the visible history from my co-player's perspective:

****Round 3:**** Co-player sent \$3.00, saw \$5.79 returned, payoff \$7.79
****Round 4:**** Co-player returned \$5.90 from \$11.79 (50%)
****Round 5:**** Co-player sent \$3.00, saw \$5.97 returned, payoff \$7.97

My co-player has been very consistent:

- As sender: consistently sends \$3.00 (60% of endowment)
- As receiver: returns about 50%
- They've seen good returns when sending (\$5.79 and \$5.97), leading to strong payoffs (~\$8)

This is a very cooperative and stable relationship. My co-player has maintained trust by consistently sending \$3.00 and returning fairly. I should continue this pattern of mutual cooperation by sending \$3.00 again.

The relationship is working well - we're both benefiting from the cooperation. No reason to deviate from what's working.

```
Usage: {
  "input_tokens": 2569,
  "output_tokens": 425,
  "reasoning_tokens": 0
}
```

GAME - Agent_5 - Receiver - dyad_3

Timestamp: 2026-08-12T08:14:21

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_6",
  "pairing": {
    "agents": [
      "Agent_6",
      "Agent_5"
    ],
    "investor": "Agent_6",
    "roles": {
      "Agent_5": "trustee",
      "Agent_6": "investor"
    },
    "trustee": "Agent_5"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 6,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system
 content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round situation, I need to balance potential gains from cooperation against the uncertainty of how my co-player will respond.

****My reasoning:****

- Sending money creates value (3x multiplier), so there's potential for mutual benefit
- This is our first interaction, so I want to signal willingness to cooperate
- But I also want to maintain some security given the uncertainty
- The communication noise adds another layer of unpredictability

I'll send a moderate amount that demonstrates trust while keeping some reserve.

```
'''json
{"send": 3}
'''
```

This sends \$3 to my co-player, who will receive \$9. I'll keep \$2 initially, and hope they reciprocate with a fair return that benefits us both.

reasoning:

This is the first round of a trust game where I'm the sender. Let me think about the strategic situation:

1. I have \$5 to start with
2. Whatever I send gets tripled when it reaches the receiver
3. The receiver then decides how much to return to me
4. There's communication noise - amounts may appear different than what was actually sent

Key considerations:

- This is a multi-round game with multiple agents
- Pairings are randomized each round
- There's a tension between trust/cooperation and self-interest
- If I send nothing, I keep \$5 but there's no opportunity for mutual gain
- If I send everything (\$5), the receiver gets \$15, and if they're fair/cooperative, they might return a good amount
- The communication noise adds uncertainty

In standard trust game experiments, sending some amount is generally beneficial because:

- It creates surplus value (3x multiplier)
- It signals trust and can encourage reciprocity
- Over multiple rounds, cooperative behavior can be rewarding

First round considerations:

- I don't know this particular co-player yet
- This is the first interaction, so I might want to signal willingness to cooperate
- But I also don't want to be too naive

A moderate approach might be to send \$3 or \$4:

- Sending \$3 means receiver gets \$9, I keep \$2. If they return \$4-5, we both do well
- Sending \$4 means receiver gets \$12, I keep \$1. If they return \$6, we both get \$6
- Sending \$5 means receiver gets \$15, I keep \$0. If they return \$7-8, I do well

Given this is round 1 and I want to establish a cooperative tone while not being too vulnerable, I'll send \$3. This shows trust but maintains some security.

Actually, let me reconsider. In trust games, the classic finding is that people often send about half their endowment on average. Sending \$2.50 or \$3 seems reasonable.

...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 703,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.66 sent, it became \$7.98, returned \$4.0, visible payoff \$3.9800000000000004.

Your total visible earnings across all rounds are \$5.13.

This round, you are the RECEIVER against your current co-player. They sent you \$2.97, so you received \$8.91. How much do you return? (0-8.91)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the situation:

****What happened in Round 1:****

- My co-player received \$7.98 (my \$3 send with noise)
- They returned \$4.0, which is about 50% - a fair reciprocal move
- This shows they're a cooperative player

****Current Round 2:****

- They sent me \$2.97 (?60% of their endowment) - showing significant trust
- I received \$8.91
- They kept \$2.03

****My decision:****

Since they demonstrated fair reciprocity (50% return) in Round 1 and have now shown strong

trust by sending most of their endowment, I should reciprocate similarly. Returning about 50% maintains the cooperative relationship while ensuring we both benefit.

Returning \$4.50 means:

- They end with: $\$2.03 + \$4.50 = \$6.53$
- I end with: $\$8.91 - \$4.50 = \$4.41$

This is fair and encourages continued cooperation.

```
'''json
{"return": 4.50}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap:

- I sent \$3 (but due to communication noise, my co-player saw \$2.66)
- They received \$7.98 (which should be $\$2.66 \times 3$)
- They returned \$4.0
- My visible payoff was \$5.13 (which would be $\$5 - \$3 + \$3.13$... but there's noise)
- Their visible payoff was \$3.98

Now in Round 2:

- My co-player is the SENDER
- They sent me \$2.97 (with noise, they might have intended something slightly different)
- I received \$8.91 (which is $\$2.97 \times 3$)
- I need to decide how much to return

Analysis of their behavior:

- In Round 1, they returned \$4.0 out of \$7.98 received, which is about 50%
- This is a reasonably fair/generous return
- Now they've sent me \$2.97 out of their \$5, which is about 59% - quite trusting

Given that:

1. They showed good reciprocity in Round 1 (returning 50% of what they received)
2. They've now shown significant trust by sending me almost 60% of their endowment
3. We might continue playing together in future rounds
4. Fair reciprocity would encourage continued cooperation

I should return a similar proportion or slightly more to maintain/build trust.

They received \$8.91 from my send. A 50% return would be \$4.455.

But let me think about the overall economics:

- They started with \$5
- They sent \$2.97, keeping \$2.03
- I received \$8.91
- If I return \$4.50:
 - They end with: $\$2.03 + \$4.50 = \$6.53$

...

usage:

```
{
  "input_tokens": 816,
  "output_tokens": 863,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.66 sent, it became \$7.98, returned \$4.0, visible payoff \$3.9800000000000004.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.03 returned, visible payoff \$7.03.

Your total visible earnings across all rounds are \$9.54.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our history together:

****Round 1:****

- I sent, they received and returned ~50% - fair cooperation

****Round 2:****

- They sent \$3 (60% of endowment) - high trust
- I returned \$4.50, they saw \$5.03 (noise) - generous reciprocity
- Their payoff was strong (\$7.03)

****Analysis:****

- This co-player has proven consistently cooperative
- They demonstrated trust by sending \$3 in Round 2
- They've reciprocated fairly (50% return rate)
- Our relationship is building mutual trust

****My decision:****

I should maintain this cooperative dynamic. Sending \$3 again signals:

- Continued trust in our partnership
- Consistency and reliability
- Faith they'll reciprocate fairly as before

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze the visible history:

Round 1:

- I was SENDER, sent \$3
- They received \$7.98 (with noise, actual was probably \$9)
- They returned \$4.0 (about 50%)
- My payoff was \$5 - \$3 + \$4 = \$6
- Visible shows \$5.13, which might include noise

Round 2:

- I was RECEIVER
- They sent \$3.0 (actual, before noise)
- I received \$8.91 (actually \$9 with noise)
- I returned \$4.50
- They saw \$5.03 returned (with noise, I actually returned \$4.50)
- Their visible payoff \$7.03 = \$5 - \$3 + \$5.03

My total visible earnings: \$9.54

Round 3 (current):

- I'm SENDER again
- Same co-player from rounds 1 and 2

This co-player has been very cooperative:

1. Returned 50% in Round 1 when they received
2. Sent \$3 (60%) in Round 2, showing trust
3. They saw me return \$5.03 (with noise), which they probably perceive as generous (56% return)

This is building a strong cooperative relationship. They've been trustworthy and fair. I should continue to send a substantial amount to maintain this cooperation.

Given their consistent cooperation and the fact they sent \$3 in Round 2, I should send at least the same amount, if not more, to signal continued trust and cooperation.

I'll send \$3 again, showing consistency and trust.

usage:

```
{
  "input_tokens": 1250,
  "output_tokens": 587,
  "reasoning_tokens": 0
}
```

}

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.66 sent, it became \$7.98, returned \$4.0, visible payoff \$3.9800000000000004.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.03 returned, visible payoff \$7.03.
- Round 3 against you, as RECEIVER: saw \$3.23 sent, it became \$9.69, returned \$5.0, visible payoff \$4.6899999999999995.

Your total visible earnings across all rounds are \$17.33.

This round, you are the RECEIVER against your current co-player. They sent you \$3.93, so you received \$11.790000000000001. How much do you return? (0-11.790000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at our continued partnership:

****Round 1:**** They returned \$4.0 from \$7.98 (?50%)

****Round 2:**** I returned \$4.50 from \$9.0 (?50%)

****Round 3:**** They returned \$5.0 from \$9.69 (?52%)

****Current Round 4:****

- They sent \$3.93 (79% of endowment!) - very high trust signal
- I received \$11.79
- They kept only \$1.07

****Analysis:****

This co-player has shown increasing trust and consistent 50% reciprocity. They're now sending nearly their entire endowment, demonstrating strong faith in our relationship.

I should maintain our established 50% return pattern to:

- Honor their trust
- Keep our cooperative equilibrium
- Ensure mutual benefit

Returning \$5.90:

- They end with: \$1.07 + \$5.90 = \$6.97
- I end with: \$11.79 - \$5.90 = \$5.89

```
'''json
{"return": 5.90}
'''
```

reasoning:

Let me analyze the history:

****Round 1:****

- I sent (saw \$2.66 with noise from my \$3)
- They received \$7.98, returned \$4.0 (about 50%)

****Round 2:****

- They sent \$3
- I received \$9, returned \$4.50 (they saw \$5.03 with noise)
- They got \$7.03 total

****Round 3:****

- I sent \$3 (they saw \$3.23 with noise)
- They received \$9.69, returned \$5.0 (about 52%)
- They kept \$4.69

****Round 4 (current):****

- They sent \$3.93 (about 79% of their endowment!) - very high trust
- I received \$11.79
- They kept \$1.07

This co-player is showing extremely high trust by sending \$3.93. They've consistently returned

rned around 50% when they were receiver. I should reciprocate this trust and maintain our cooperative relationship.

If I return about 50% of what I received:

- Return \$5.90
- They end with: $\$1.07 + \$5.90 = \$6.97$
- I end with: $\$11.79 - \$5.90 = \$5.89$

But they're showing increasing trust (sending more and more). Maybe I should return a bit more to reward this and encourage continued cooperation. Let me think...

Actually, looking at the pattern:

- Round 1 receiver: returned \$4 from \$7.98 (50%)
- Round 3 receiver: returned \$5 from \$9.69 (52%)

They're consistently returning around 50%. I've been doing similar - returned \$4.50 from \$9 in Round 2 (50%).

Given they sent \$3.93 (very high trust), I should return around 50% or slightly more to maintain the relationship.

...

usage:

```
{
  "input_tokens": 1695,
  "output_tokens": 859,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.03 returned, visible payoff \$7.03.
- Round 3 against you, as RECEIVER: saw \$3.23 sent, it became \$9.69, returned \$5.0, visible payoff \$4.6899999999999995.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.63 returned, visible payoff \$8.629999999999999.

Your total visible earnings across all rounds are \$23.22.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at our established partnership:

****Round 2:**** They sent \$3.0, I returned generously
****Round 3:**** I sent \$3.23 (with noise), they returned \$5.0 (?52%)
****Round 4:**** They sent \$3.0, I returned well, they saw \$6.63

****Pattern Analysis:****

- They consistently send \$3 when sender (Rounds 2, 4)
- They reciprocate fairly at ~50% when receiver (Round 3)
- We've built a stable, mutually beneficial relationship
- Both earning well (\$23.22 total for me)

****My decision:****

Match their consistency by sending \$3 again. This:

- Maintains our established cooperative equilibrium
- Shows I trust the pattern we've developed
- Signals reliability and reciprocity

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at the visible history:

****Round 2:**** They sent \$3.0, I returned (they saw \$5.03) - I actually returned \$4.50 but they saw \$5.03 due to noise
****Round 3:**** They received \$9.69 (from my \$3 send), returned \$5.0
****Round 4:**** They sent \$3.0, I returned \$5.90 but they saw \$6.63 - generous due to noise

Pattern analysis:

- This co-player consistently sends \$3.0 when they're sender (Rounds 2 and 4)
- They return roughly 50% when receiver (Round 3: \$5.0 from \$9.69)
- They've been seeing higher returns from me due to noise (\$5.03 and \$6.63 vs what I actually sent \$4.50 and \$5.90)

Our cooperation has been strong and consistent. They've shown:

- Reliable sending patterns (\$3)
- Fair reciprocity (~50%)
- Building trust over rounds

I should continue with the same sending pattern to maintain this cooperative equilibrium. Sending \$3 has been working well for both of us.

My total visible earnings are \$23.22 which suggests we're both doing well.

I'll send \$3 again to maintain consistency and trust.

usage:

```
{
  "input_tokens": 2185,
  "output_tokens": 517,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$3.23 sent, it became \$9.69, returned \$5.0, visible payoff \$4.6899999999999995.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.63 returned, visible payoff \$8.629999999999999.
- Round 5 against you, as RECEIVER: saw \$3.62 sent, it became \$10.86, returned \$6.0, visible payoff \$4.859999999999999.

Your total visible earnings across all rounds are \$31.189999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$3.02, so you received \$9.06. How much do you return? (0-9.06)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at our continued partnership:

****Round 3:**** I sent, they returned \$5.0 from \$9.69 (?52%)
****Round 4:**** They sent \$3.0, I returned generously
****Round 5:**** I sent, they returned \$6.0 from \$10.86 (?55%)

****Current Round 6:****

- They sent \$3.02 (?60% of endowment) - continued trust
- I received \$9.06
- They kept \$1.98

****Analysis:****

Our partnership shows consistent mutual cooperation with ~50-55% return rates. They've reliably reciprocated in this range, and I should maintain this equilibrium.

Returning \$4.50:

- They end with: \$1.98 + \$4.50 = \$6.48
- I end with: \$9.06 - \$4.50 = \$4.56

This maintains our cooperative pattern.

```
'''json
{"return": 4.50}
'''
```

Reasoning:

Looking at our history:

****Round 3:**** I sent (they saw \$3.23), they returned \$5.0 from \$9.69 (?52%)
****Round 4:**** They sent \$3.0, I returned, they saw \$6.63 (generous due to noise)
****Round 5:**** I sent (they saw \$3.62), they returned \$6.0 from \$10.86 (?55%)

****Current Round 6:****
 - They sent \$3.02 (?60% of endowment)
 - I received \$9.06
 - They kept \$1.98

****Pattern:****
 - They consistently send around \$3
 - They consistently return around 50-55% when receiver
 - We have a stable, cooperative relationship

I should maintain the ~50% return pattern:
 - Returning \$4.50 would be about 50%
 - This keeps our cooperative equilibrium
 - They end with: \$1.98 + \$4.50 = \$6.48
 - I end with: \$9.06 - \$4.50 = \$4.56

Both benefit reasonably, and this maintains the pattern.

```
Usage: {
  "input_tokens": 2634,
  "output_tokens": 514,
  "reasoning_tokens": 0
}
```

GAME - Agent_8 - Sender - dyad_4

Timestamp: 2026-08-12T08:14:27

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_7",
  "pairing": {
    "agents": [
      "Agent_8",
      "Agent_7"
    ],
    "investor": "Agent_8",
    "roles": {
      "Agent_7": "trustee",
      "Agent_8": "investor"
    }
  },
  "trustee": "Agent_7"
},
"role": "investor",
"role_label": "Sender",
"round": 6,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x

- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.17, that is 63.4% of their total endowment. You received \$9.51. How much do you return to the sender? (0-9.51)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending 63.4% of their endowment (\$3.17)
- I received \$9.51 after the 3x multiplier
- The sender currently has \$5 - \$3.17 = \$1.83

The sender demonstrated considerable trust, so I want to reciprocate appropriately. If I return \$5.00:

- The sender would end with: \$1.83 + \$5.00 = \$6.83 (a profit of \$1.83)
- I would keep: \$9.51 - \$5.00 = \$4.51

This rewards their trust with a solid return while allowing us both to benefit from the co operation.

{"return": 5.00}

reasoning:

Let me think about this carefully.

I'm the RECEIVER in Round 1.

- The sender sent \$3.17 (which is 63.4% of their \$5 endowment)
- This was multiplied by 3x, so I received: $3.17 \times 3 = 9.51$?
- I need to decide how much to return (between \$0 and \$9.51)

Key considerations:

1. The sender showed considerable trust by sending 63.4% of their endowment
2. If I return nothing, the sender ends up with: \$5 - \$3.17 = \$1.83, while I keep \$9.51
3. If I return a fair/reciprocal amount, what would that be?

A few approaches to consider:

- Equal split of total: Total is \$5 (sender's initial) + \$9.51 (what I received) = \$14.51.

Equal split would be ~\$7.26 each. The sender currently has \$1.83, so I'd return \$7.26 - \$1.83 = \$5.43

- Return what maintains sender's initial investment value: Sender invested \$3.17. If I return \$3.17, sender gets back to \$5 total, and I keep \$6.34
- Return proportionally to what was sent: The sender sent 63.4% of their endowment. I could return a similar proportion of what I received: 63.4% of \$9.51 = ~\$6.03
- Social norm of reciprocity: Return enough to reward trust but keep some gains

Given that:

- The sender showed significant trust (63.4%)
- This is a multi-round game with different partners
- Building reputation for reciprocity could be valuable
- Note about communication noise suggests amounts might not be exact

I think a reasonable return would be around \$4-5, which:

- Acknowledges their generosity without giving away all my gains
- Leaves them better off than their starting position
- Demonstrates trustworthiness that could matter in future interactions with other partners
- Strikes a balance between reciprocity and self-interest

I'll return \$4.75. This gives the sender \$6.58 total (\$1.83 remaining plus my return), which meaningfully rewards their trust while I keep \$4.76....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1001,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.37 returned, visible payoff \$7.37.

Your total visible earnings across all rounds are \$4.51.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the history:

My co-player sent \$3.0 in Round 1 (60% of their endowment), showing significant trust. They saw \$5.37 returned, which gave them a good outcome.

Since they demonstrated trust and cooperation, I should reciprocate similarly. Sending around the same amount they sent (\$3.0) would be fair and maintain the cooperative relationship.

```
{"send": 3.0}
```

reasoning:

Let me analyze this situation:

1. In Round 1, I was the RECEIVER and the sender sent me \$3.17 (according to my view), which became \$9.51 for me. I returned \$5.00.
2. However, from my current co-player's perspective (shown in "Your current co-player's last 3 game(s)"):
 - They sent \$3.0 (not \$3.17 - communication noise!)
 - It became \$9.0 (not \$9.51 - more noise!)
 - They saw \$5.37 returned (not \$5.00 - even more noise!)
 - Their visible payoff was \$7.37

So there's noise in the communication. What they experienced:

- Started with \$5
- Sent \$3.0

- Had \$2.0 left
- Saw \$5.37 returned
- Ended with $\$2.0 + \$5.37 = \$7.37$

From their perspective, I (as the receiver in Round 1) returned about 60% of what I received (\$5.37 out of \$9.0).

3. Now I'm the SENDER in Round 2 against the same co-player.

My co-player showed trust by sending \$3.0 (from their perspective) and saw a decent return ratio. They might be willing to reciprocate trust.

I should consider:

- They sent about 60% of their endowment in Round 1
- They saw about 60% returned
- This suggests they're moderately trusting and cooperative

I should send a similar amount to show I'm also trustworthy and cooperative. Sending around \$3 would be consistent with their behavior and maintain the cooperative relationship.

Let me send \$3.00 to match their earlier send amount.

usage:

```
{
  "input_tokens": 837,
  "output_tokens": 523,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.37 returned, visible payoff \$7.37.
- Round 2 against you, as RECEIVER: saw \$2.35 sent, it became \$7.050000000000001, returned \$3.75, visible payoff \$3.3000000000000007.

Your total visible earnings across all rounds are \$10.219999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$2.22, so you received \$6.66. How much do you return? (0-6.66)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the situation:

Current round:

- Sender sent \$2.22 (44.4% of their endowment)
- I received \$6.66
- Sender currently has: $\$5 - \$2.22 = \$2.78$

History with this co-player:

- Round 1: They sent \$3.0, I returned money (they saw \$5.37)
- Round 2: They saw less sent from me due to noise (\$2.35 vs my actual \$3.0), returned \$3.75

They've reduced their sending, possibly due to noise affecting what they saw in Round 2. Still, they're showing some trust by sending \$2.22. I should reciprocate fairly.

Returning approximately 50% of what I received (\$3.33) would:

- Give sender: $\$2.78 + \$3.33 = \$6.11$ (profit of \$1.11)
- I keep: \$3.33

This maintains cooperation while being proportional to their send amount.

```
{"return": 3.33}
```

reasoning:

Let me analyze the situation:

Current round:

- I'm the RECEIVER
- Sender sent \$2.22 (44.4% of their endowment)
- I received \$6.66 after 3x multiplier
- Sender currently has: \$5 - \$2.22 = \$2.78

Looking at history with this co-player:

- Round 1: They sent \$3.0 as SENDER, saw \$5.37 returned (I actually sent \$5.00, but due to noise they saw \$5.37)
- Round 2: I sent \$3.0 as SENDER (but they saw \$2.35 due to noise), they returned \$3.75

In Round 2, I sent \$3.0 which became \$9.0, and they returned \$3.75. That's about 42% return rate.

Now in Round 3:

- They sent \$2.22, which is less than what they sent in Round 1 (\$3.0)
- This might be because they saw only \$2.35 sent in Round 2 (due to noise) instead of my actual \$3.0

I should consider:

1. They're still showing some trust by sending \$2.22
2. They may have reduced their sending because of perceived lower trust from me (noise issue)
3. I should maintain cooperation

If I return around \$3.50-\$4.00:

- Return \$3.50: Sender gets \$2.78 + \$3.50 = \$6.28 (profit of \$1.28), I keep \$3.16
- Return \$4.00: Sender gets \$2.78 + \$4.00 = \$6.78 (profit of \$1.78), I keep \$2.66

Given they sent less this round but still showed trust, I'll return a reasonable amount that rewards their trust.

I'm noticing a pattern in the return rates across rounds. In Round 1, I returned about 53% of what I received, while in Round 2 they returned roughly 42%. Since they've sent less this time-likely reacting to the noise that made my previous send appear smaller-I should aim for around 50% to maintain fairness and signal...

usage:

```
{
  "input_tokens": 1135,
  "output_tokens": 909,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.37 returned, visible payoff \$7.37.
- Round 2 against you, as RECEIVER: saw \$2.35 sent, it became \$7.050000000000001, returned \$3.75, visible payoff \$3.3000000000000007.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.3 returned, visible payoff \$6.3.

Your total visible earnings across all rounds are \$13.549999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at the history with my current co-player:

- Round 1: They sent \$3.0 (strong trust)
- Round 2: I sent, they returned \$3.75 (cooperative)
- Round 3: They sent \$3.0 again (consistent trust)

My co-player has shown consistent cooperation, sending \$3.0 both times they were the sender. They also returned a fair amount when they were receiver. I should reciprocate this consistent trust.

```
{"send": 3.0}
```

reasoning:

Let me analyze the visible history with my current co-player:

Round 1: They were SENDER, sent \$3.0 (60%), I returned something, they saw \$5.37 returned
Round 2: They were RECEIVER, they saw \$2.35 sent (but I actually sent \$3.0 - noise!), they returned \$3.75
Round 3: They were SENDER, sent \$3.0 (60%), they saw \$4.3 returned (but I actually returned \$3.33 - noise showing higher!)

So my co-player has:

- Sent \$3.0 twice when they were sender (consistent trust)
- Returned \$3.75 when they received \$7.05

They're showing good cooperation. In Round 3, they saw a higher return (\$4.3) than I actually sent (\$3.33), which might encourage them.

I should continue cooperating. Sending \$3.0 again would be consistent and fair.

My total visible earnings: \$13.55

- Round 1 as RECEIVER: kept \$4.51
- Round 2 as SENDER: sent \$3.0, saw return, kept something
- Round 3 as RECEIVER: kept \$3.33

Actually, let me recalculate:

Round 1: I was RECEIVER, got \$9.51, returned \$5.00, kept \$4.51
Round 2: I was SENDER, sent \$3.0, they returned \$3.75, I kept \$2 + \$3.75 = \$5.75
Round 3: I was RECEIVER, got \$6.66, returned \$3.33, kept \$3.33

Total: \$4.51 + \$5.75 + \$3.33 = \$13.59 ? \$13.55 (matches with rounding)

My co-player has been consistent in sending \$3.0 when they're sender. ...

usage:

```
{
  "input_tokens": 1610,
  "output_tokens": 585,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.35 sent, it became \$7.050000000000001, returned \$3.75, visible payoff \$3.3000000000000007.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.3 returned, visible payoff \$6.3.
- Round 4 against you, as RECEIVER: saw \$2.62 sent, it became \$7.86, returned \$4.2, visible payoff \$3.66.

Your total visible earnings across all rounds are \$18.86.

This round, you are the RECEIVER against your current co-player. They sent you \$3.72, so you received \$11.16. How much do you return? (0-11.16)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at the current situation:

- Sender sent \$3.72 (74.4% of their endowment - very high trust!)
- I received \$11.16
- Sender currently has: \$5 - \$3.72 = \$1.28

History shows this co-player has been consistently cooperative and trusting. Now they've increased their trust even more, sending nearly 75% of their endowment. I should reciprocate this strong trust generously.

Returning \$6.0:

- Sender would get: \$1.28 + \$6.0 = \$7.28 (profit of \$2.28)
- I keep: \$5.16

This rewards their exceptional trust while both of us profit from the cooperation.

```
{"return": 6.0}
```

reasoning:

Looking at the current situation:

- Sender sent \$3.72 (74.4% of their endowment - very high trust!)
- I received \$11.16 after 3x multiplier
- Sender currently has: \$5 - \$3.72 = \$1.28

Looking at history with this co-player:

- They've been consistently cooperative
- Round 3: They sent \$3.0
- Round 4: They saw me send \$2.62 (due to noise, I actually sent \$3.0), returned \$4.2
- Now Round 5: They sent \$3.72 (even more!)

This is a very trusting move - 74.4% of endowment. I should reciprocate generously.

If I return \$6.0:

- Sender gets: \$1.28 + \$6.0 = \$7.28 (profit of \$2.28)
- I keep: \$11.16 - \$6.0 = \$5.16

This rewards their high trust appropriately. Both of us profit well.

usage:

```
{
  "input_tokens": 1967,
  "output_tokens": 440,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.3 returned, visible payoff \$6.3.
- Round 4 against you, as RECEIVER: saw \$2.62 sent, it became \$7.86, returned \$4.2, visible payoff \$3.66.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.04 returned, visible payoff \$7.04.

Your total visible earnings across all rounds are \$24.02.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the history with my current co-player:

- Round 3: They sent \$3.0 (consistent trust)
- Round 4: They returned \$4.2 when they received \$7.86 (generous)
- Round 5: They sent \$3.0 again (consistently cooperative)

This co-player has been very reliable and cooperative throughout our interactions. They consistently send \$3.0 when they're the sender and return fair amounts when they're the receiver. I should continue matching this level of trust and cooperation.

```
{"send": 3.0}
```

Reasoning:

Looking at the visible history with my current co-player:

- Round 3: They sent \$3.0 as sender
- Round 4: They returned \$4.2 as receiver (I sent \$2.62 visible to them)
- Round 5: They sent \$3.0 as sender

This co-player has been consistently cooperative, sending \$3.0 whenever they're the sender and returning fair amounts when they're the receiver (\$4.2 when they received \$7.86).

I've been sending \$3.0 consistently when I'm the sender. This has been working well for both of us. I should continue this pattern of cooperation.

```

{"send": 3.0}
Usage: {
  "input_tokens": 2364,
  "output_tokens": 295,
  "reasoning_tokens": 0
}

```

GAME - Agent_7 - Receiver - dyad_4

Timestamp: 2026-08-12T08:14:36

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_8",
  "pairing": {
    "agents": [
      "Agent_8",
      "Agent_7"
    ],
    "investor": "Agent_8",
    "roles": {
      "Agent_7": "trustee",
      "Agent_8": "investor"
    },
    "trustee": "Agent_7"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 6,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to consider:

1. ****The multiplier effect****: Any amount I send gets tripled, creating potential mutual gain
2. ****No prior history****: This is my first interaction, so I have no information about this co-player's behavior
3. ****Future rounds****: Even though pairings are randomized, establishing cooperative norms early could benefit the group overall
4. ****Risk vs. reward****: Sending more creates more surplus, but requires trusting the receiver to return a fair amount

I'll take a moderately cooperative approach. Sending \$3 creates \$9 for the receiver - enough surplus that we can both benefit if they reciprocate fairly, while not risking my entire endowment.

If they return around half of what they receive (\$4-5), we both end up better off than if no transaction occurred.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a multi-agent trust game experiment. I'm the SENDER with \$5.

Let me think about the strategic aspects:

1. If I send money, it gets tripled, creating surplus value
2. The receiver then decides how much to return
3. This is a classic trust game scenario

Key considerations:

- This is round 1, so I have no history with this particular co-player
- There are multiple rounds, so reputation/repeated interaction matters
- However, pairings are randomized each round, so I won't necessarily play the same person again
- The game has communication noise, which might affect trust

Strategic thinking:

- Sending \$0 guarantees I keep \$5, but creates no surplus
- Sending \$5 creates maximum surplus (\$15 total for receiver), but requires trust
- A middle ground (e.g., \$2-3) balances risk and potential gain

In classic trust game experiments, people often:

1. Send moderate to high amounts in early rounds to signal cooperation
2. The amount sent and returned establishes baseline trust levels
3. With randomized pairings, building a reputation as trustworthy matters for future rounds

Game theory perspective:

- If this were truly one-shot with rational self-interested players, the receiver should return \$0 (backward induction)
- Knowing this, I should send \$0
- However, real humans often deviate from this due to fairness norms, reciprocity, and reputation concerns

Since:

- This is multi-round with multiple agents
- There may be reputation effects even with randomized pairings
- Starting cooperatively can establish good norms
- The 3x multiplier creates real gains from cooperation

I'll send a moderate-high amount to signal cooperation while not being completely vulnerable. I'll send \$3.

This creates \$9 for the receiver, and if they return even just \$3, we both end up with \$5 (same as no transaction), but if they return more (say \$4-5), we both gain. ...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 671,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.17 sent, it became \$9.51, returned \$5.0, visible payoff \$4.51.

Your total visible earnings across all rounds are \$7.37.

This round, you are the RECEIVER against your current co-player. They sent you \$2.35, so you received \$7.050000000000001. How much do you return? (0-7.050000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

****Round 1 analysis:****

- They saw I sent \$3.17 (with noise), received \$9.51
- They returned \$5.00, which is about 52.6% of what they received
- This was quite generous and cooperative

****Current situation:****

- They sent me \$2.35, I received \$7.05
- They showed good faith last round, so I should reciprocate

Following a similar reciprocity rate (around 52-53% return):

- 53% of \$7.05 ? \$3.74

This ensures:

- They profit from sending (get back more than they sent)
- I also benefit from the transaction
- We maintain cooperative momentum

```
'''json
{"return": 3.75}
'''
```

reasoning:

Let me analyze this situation:

****What happened in Round 1:****

- I sent \$3, but they saw \$3.17 (communication noise)
- They received \$9.51 (3.17×3)
- They returned \$5.0
- Their visible payoff was $\$9.51 - \$5.0 = \$4.51$
- My payoff was $\$5 - \$3 + \$5 = \7 (but visible earnings show \$7.37, maybe due to noise)

****Current Round 2:****

- They sent me \$2.35 (actual), which became \$7.05
- I need to decide how much to return

****Analysis of their behavior:****

In Round 1, they returned \$5 out of \$9.51 received, which is about 52.6%. This is a fairly generous return - they kept \$4.51 and I got back \$5, so I had a net gain of \$2 from the transaction.

****What should I do?****

They showed cooperative behavior by returning a good amount. Now they've sent me \$2.35, creating \$7.05.

If I apply a similar reciprocity ratio:

- They returned about 52.6% of what they received
- I received \$7.05, so 52.6% would be about \$3.71

However, I should also consider:

- They sent \$2.35 this round (less than the ~\$3.17 they saw me send)
- But they were generous with their return last round
- I want to maintain cooperation

A fair approach might be to:

1. Return enough that they profit from sending
2. Keep enough that I also profit
3. Signal continued cooperation

If they sent \$2.35 and expect around a 50% return rate:

- They might expect back around \$3-4

Let me think about payoffs:

- They started with \$5
- They sent \$2.35
- If I return \$3.5, they end with \$6.15 and I keep \$3.55
- If I return \$4, they end with \$6.65 and I keep \$3.05
- If I return \$3, they end with \$5.65 and I keep \$4.05

Since they matched my generosity last round with a 52.6% return, I should reciprocate similarly to maintain the cooperative dynamic....

usage:

```
{
  "input_tokens": 852,
  "output_tokens": 862,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.17 sent, it became \$9.51, returned \$5.0, visible payoff \$4.51.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.71 returned, visible payoff \$5.71.

Your total visible earnings across all rounds are \$10.670000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our history together:

****Round 1:****

- I sent \$3, they received ~\$9.51, returned \$5.00 (very generous ~53% return)

****Round 2:****

- They sent \$2.35, I received \$7.05, returned \$3.75
- They saw \$3.71 returned due to noise (~41% visible return rate)

****Analysis:****

- This co-player was quite generous in Round 1
- They sent a bit less in Round 2, possibly because they saw a lower return rate from me (due to noise)

- Overall, they've demonstrated cooperative behavior

****My strategy:****

I'll send \$3 again to:

- Match the cooperative level from Round 1
- Show consistency in my approach
- Signal that I'm willing to trust despite the noise in Round 2

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze the history with this co-player:

****Round 1 (I was SENDER):****

- I sent \$3 (they saw \$3.17 due to noise)
- They received \$9.51 and returned \$5.00
- This was generous - about 52.6% return rate

****Round 2 (I was RECEIVER):****

- They sent \$2.35 (actual), I received \$7.05
- I returned \$3.75
- They sent \$3.0 actual, became \$9.0, they saw \$3.71 returned
- So they saw a return rate of about 41.2% ($3.71/9.0$)
- This is less than what they gave me in Round 1

****Pattern:****

- This co-player was very cooperative in Round 1
- In Round 2, they sent a bit less (\$2.35 vs my \$3)
- They saw me return less proportionally than they did (41.2% vs 52.6%)

****Current decision:****

I'm the sender again. This co-player has shown:

1. Initial high cooperation (52.6% return)
2. Slightly reduced sending in Round 2 (\$2.35 vs \$3)
3. They may perceive me as less generous than they were (due to noise showing only \$3.71 returned when I sent \$3.75)

I should:

- Maintain cooperation since they were generous in Round 1
- Send a similar amount to Round 1 to show consistency
- Perhaps send \$3 again to match their Round 2 amount

Given they've been generally cooperative and we're building a pattern, I'll send \$3 again.

usage:

```
{
  "input_tokens": 1212,
  "output_tokens": 605,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.17 sent, it became \$9.51, returned \$5.0, visible payoff \$4.51.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.71 returned, visible payoff \$5.71.
- Round 3 against you, as RECEIVER: saw \$2.22 sent, it became \$6.66, returned \$3.33, visible payoff \$3.33.

Your total visible earnings across all rounds are \$16.970000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$2.62, so you received \$7.86. How much do you return? (0-7.86)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at our continued interaction:

****Round 1:**** They returned \$5.00 from \$9.51 (53%)
****Round 3:**** They returned \$3.33 from \$6.66 (50%)

****Current situation:****

- They sent \$2.62, I received \$7.86
- They've been consistently cooperative, returning around 50-53%

****My decision:****

I should match their cooperative approach. Returning about 53% would be:
- 53% of \$7.86 ? \$4.17

This ensures:

- They profit significantly (\$4.17 back vs \$2.62 sent = +\$1.55 net)
- I also benefit (\$3.69 kept from the tripled amount)
- We maintain positive reciprocity

```
'''json
{"return": 4.20}
'''
```

reasoning:

Looking at the history with this co-player:

****Round 1:****

- I sent \$3, they saw \$3.17 (noise), received \$9.51
- They returned \$5.00 (~53% return rate)

****Round 2:****

- They sent \$3.00, became \$9.00
- I returned \$3.75, they saw \$3.71
- Visible return rate for them: ~41%

****Round 3:****

- I sent \$3, they saw \$2.22 (significant noise), received \$6.66
- They returned \$3.33 (50% return rate)

****Current Round 4:****

- They sent \$2.62, I received \$7.86

****Analysis:****

- This co-player has been consistently cooperative
- Round 1: 53% return
- Round 3: 50% return
- Their sending amounts haven't decreased dramatically despite seeing noise

****My approach:****

I should maintain a similar return rate. They've been returning around 50-53%.
- 50% of \$7.86 = \$3.93
- 53% of \$7.86 = \$4.17

I'll return around \$4.00 to:

- Match their cooperative behavior
- Ensure they profit from sending (\$2.62 sent, \$4.00 back)
- Keep the relationship positive
- I keep \$3.86, they get \$4.00 back vs \$2.62 sent = net gain of \$1.38 for them

Actually, let me be a bit more generous to match their Round 1 behavior. I'll return \$4.20, which is about 53% of what I received.

Their final payoff: $\$5 - \$2.62 + \$4.20 = \6.58

My final payoff: $\$7.86 - \$4.20 = \$3.66$

...

usage:

```
{
  "input_tokens": 1649,
  "output_tokens": 650,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.71 returned, visible payoff \$5.71.
- Round 3 against you, as RECEIVER: saw \$2.22 sent, it became \$6.66, returned \$3.33, visible payoff \$3.33.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.31 returned, visible payoff \$5.3100000000000005.

Your total visible earnings across all rounds are \$20.630000000000003.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at our ongoing interaction:

****Pattern Analysis:****

- They've consistently sent \$3.00 in Rounds 2 and 4
- As receiver (Rounds 1 & 3), they returned generously (~50-53%)
- Unfortunately, noise has made my returns appear lower than actual:
 - Round 2: I returned \$3.75, they saw \$3.71
 - Round 4: I returned \$4.20, they saw only \$3.31

****Current Strategy:****

Despite the noise distorting communication, they've continued trusting me with \$3 sends. I should:

- Maintain consistency by sending \$3
- Signal continued cooperation
- Trust they'll recognize the pattern despite noise

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze our history:

- **Round 1:**** I sent \$3, they received ~\$9.51, returned \$5.00
- **Round 2:**** They sent \$2.35, I received \$7.05, returned \$3.75, but they saw \$3.71
- **Round 3:**** I sent \$3, they received ~\$6.66, returned \$3.33, but they saw I sent \$2.22
- **Round 4:**** They sent \$2.62, I received \$7.86, returned \$4.20, but they saw \$3.31

Looking at what they're seeing from my perspective:

- Round 2: They sent \$3.00, saw \$3.71 returned (visible return rate ~41%)
- Round 4: They sent \$3.00, saw \$3.31 returned (visible return rate ~37%)

The noise is making it look like I'm returning less than I actually am.

From their rounds as receiver:

- Round 1: They returned \$5.00 from \$9.51 (53%)
- Round 3: They returned \$3.33 from \$6.66 (50%)

They've been consistently cooperative. However, the noise is making my returns look less generous than they actually are. Despite this, they've continued sending \$3 in rounds 2 and 4.

I should:

1. Continue showing trust and cooperation
2. Send a similar amount to maintain consistency
3. Hopefully they'll continue to be cooperative

I'll send \$3 again.

usage:

```
{
  "input_tokens": 2072,
  "output_tokens": 533,
  "reasoning_tokens": 0
}
```


Message 12 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$2.22 sent, it became \$6.66, returned \$3.33, visible payoff \$3.33.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.31 returned, visible payoff \$5.3100000000000005.
- Round 5 against you, as RECEIVER: saw \$3.72 sent, it became \$11.16, returned \$6.0, visible payoff \$5.16.

Your total visible earnings across all rounds are \$27.67.

This round, you are the RECEIVER against your current co-player. They sent you \$3.23, so you received \$9.69. How much do you return? (0-9.69)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at our continued interaction:

****Round 5 analysis:****

- They saw I sent \$3.72, received \$11.16
- They returned a very generous \$6.00 (53.8%)

****Current situation:****

- They sent \$3.23, I received \$9.69
- They're maintaining strong cooperation despite noise

****My decision:****

I should match their generosity. Returning about 53% would be:

- 53% of \$9.69 ? \$5.14

This ensures:

- They profit well (\$5.14 back vs \$3.23 sent = +\$1.91 net)
- I also benefit (\$4.55 kept)
- We maintain mutual cooperation

```
'''json
{"return": 5.15}
'''
```

Reasoning:

Looking at the history:

****Round 5 analysis:****

- They saw I sent \$3.72 (noise from my actual \$3), received \$11.16
- They returned \$6.00, which is about 53.8% of what they received
- Very generous and cooperative again!

****Current situation:****

- They sent me \$3.23, I received \$9.69
- They've been consistently generous, especially in Round 5 where they returned \$6.00 from \$11.16

****My decision:****

They're showing strong cooperation. I should reciprocate at a similar level:

- Around 53-54% of \$9.69 would be approximately \$5.15-\$5.23

Let me calculate:

- 53% of \$9.69 = \$5.14
- 54% of \$9.69 = \$5.23

I'll return around \$5.15 to match their cooperative spirit and ensure they profit well from sending to me.

This gives them: \$5.15 back vs \$3.23 sent = +\$1.92 profit

And I keep: \$9.69 - \$5.15 = \$4.54

Both profit nicely.

Usage: {
"input_tokens": 2482,

```
"output_tokens": 468,
"reasoning_tokens": 0
}
```

=====

ROUND 7

=====

Roles: Agent_1=Sender, Agent_2=Receiver, Agent_3=Sender, Agent_4=Receiver, Agent_5=Sender, Agent_6=Receiver, Agent_7=Sender, Agent_8=Receiver

Pairings:

dyad_1: Agent_1=Sender, Agent_2=Receiver
dyad_2: Agent_3=Sender, Agent_4=Receiver
dyad_3: Agent_5=Sender, Agent_6=Receiver
dyad_4: Agent_7=Sender, Agent_8=Receiver

Parsed game decisions:

dyad_1: Agent_1 sent \$5; received \$15; Agent_2 returned \$7; payoffs Agent_1=\$7, Agent_2=\$8;
noise/visible: sent visible=\$4.52, received visible=\$13.56, returned visible=\$7.09
Action validation: {"returned": {"amount": 7.0, "clamped": false, "max": 15.0, "min": 0, "raw": 7.0}, "sent": {"amount": 5.0, "clamped": false, "max": 5, "min": 0, "raw": 5.0}}
dyad_2: Agent_3 sent \$3; received \$9; Agent_4 returned \$4.5; payoffs Agent_3=\$6.5, Agent_4=\$4.5;
noise/visible: sent visible=\$3.55, received visible=\$10.65, returned visible=\$5.03
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_3: Agent_5 sent \$3; received \$9; Agent_6 returned \$4.5; payoffs Agent_5=\$6.5, Agent_6=\$4.5;
noise/visible: sent visible=\$2.78, received visible=\$8.34, returned visible=\$4.84
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_7 sent \$3; received \$9; Agent_8 returned \$4.5; payoffs Agent_7=\$6.5, Agent_8=\$4.5;
noise/visible: sent visible=\$2.85, received visible=\$8.55, returned visible=\$4.07
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 9.0, "min": 0, "raw": 4.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 46.2,
"Agent_2": 43.8,
"Agent_3": 40.5,
"Agent_4": 36.5,
"Agent_5": 39.6,
"Agent_6": 37.4,
"Agent_7": 40.73,
"Agent_8": 36.27

}

Visible balances after round: {

"Agent_1": 43.290000000000006,
"Agent_2": 46.509999999999999,
"Agent_3": 36.29,
"Agent_4": 37.38,
"Agent_5": 42.59,
"Agent_6": 39.599999999999994,
"Agent_7": 38.28,
"Agent_8": 36.120000000000005

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T08:14:44

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_2",
"pairing": {
 "agents": [
 "Agent_1",
 "Agent_2"
],
 "investor": "Agent_1",
 "roles": {

```

    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 7,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to establish a cooperative approach while being somewhat cautious since I don't have any history with my current co-player.

Sending a moderate amount signals trust and gives us both the opportunity to benefit from the 3x multiplier, while not leaving me completely vulnerable.

I'll send **\$3**. This gives my co-player \$9 to work with, and if they return a fair share

, we can both end up better off than our starting positions.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game. I'm the sender and I have \$5. I need to decide how much to send to the receiver.

Let me think about the strategic considerations:

1. If I send \$X, the receiver gets \$3X
2. The receiver then decides how much to return
3. I keep \$5 - X + (amount returned)
4. The receiver keeps \$3X - (amount returned)

This is a classic trust game. The optimal outcome for both of us would be if I send the full \$5, it gets tripled to \$15, and the receiver returns a fair share (say, at least \$7.50, which would give me \$7.50 and them \$7.50, both better off than starting positions).

However, there are risks:

- The receiver might not return anything or very little
- There's also mentioned "communication noise" - amounts may differ from what was actually sent/returned

Since this is round 1 and I have no history with this player:

- I could play it safe and send less
- Or I could signal trust and cooperation by sending more

In standard trust game experiments, people often send around \$2-\$3 in the first round as a moderate trust signal.

However, if I'm thinking about maximizing joint welfare and establishing cooperation, sending more could be beneficial. If the receiver is rational and cooperative, they should return enough to make it worthwhile for me.

Let me consider a middle-ground strategy: sending \$3 or \$4.

- If I send \$3, receiver gets \$9. If they return \$4-5, we both benefit
- If I send \$4, receiver gets \$12. If they return \$6, we both benefit significantly

Given this is a multi-round game with potentially different partners each time, establishing a pattern of trust and cooperation could be beneficial. I'll go with sending \$3 as a balanced approach - showing trust but not complete vulnerability.

Actually, let me reconsider. The instructions mention this is multi-agent with 8 agents and randomized pairings. This means:

...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 648,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.63 sent, it became \$10.89, returned \$5.0, visible payoff \$5.890000000000001.

Your total visible earnings across all rounds are \$7.52.

This round, you are the RECEIVER against your current co-player. They sent you \$3.14, so you received \$9.42. How much do you return? (0-9.42)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

- Round 1: My co-player saw \$3.63 sent, received \$10.89, and returned \$5.0 to me - showing cooperative behavior

- Round 2: They sent me \$3.14, which became \$9.42

They've demonstrated fairness by returning a reasonable amount in Round 1. I want to reciprocate this cooperative approach.

I'll return **\$4.50**, which:

- Gives them a profit (they sent ~\$3.14, get back \$4.50)
- Leaves me with a profit too (\$9.42 - \$4.50 = \$4.92)
- Maintains the cooperative relationship

```
'''json
{"return": 4.50}
'''
```

reasoning:

Let me understand what happened:

Round 1: I sent \$3, but my co-player saw \$3.63 (due to communication noise). This became \$10.89 for them. They returned \$5.0.

My actual payoff in Round 1:

- I had \$5
- I sent \$3
- I received back \$5
- So I kept: \$5 - \$3 + \$5 = \$7

But the visible payoff shows \$7.52, which suggests there might be some noise in the amounts.

Now in Round 2:

- I'm the RECEIVER
- My co-player sent me some amount that appears as \$3.14 to me
- This became \$9.42 (which is $\$3.14 \times 3$)
- I need to decide how much to return

Looking at their behavior: In Round 1, they saw \$3.63 sent, which became \$10.89, and they returned \$5.0. That's about 46% of what they received, or about 1.38x what they saw as sent.

This seems like a fair/cooperative player. They returned a reasonable amount.

Now I have \$9.42. What should I return?

Given that:

1. They sent a decent amount (\$3.14, which is about 63% of their \$5)
2. They were fair in the previous round, returning \$5 when they received \$10.89
3. I want to encourage continued cooperation

I should return a fair amount. Some options:

- Return about half of what I received: \$4.71
- Return what they sent plus a bit extra: around \$4-5
- Match their return rate from last round (46% of received): about \$4.33

I think returning around \$4.50-\$5.00 would be fair and encourage continued cooperation. Let me return \$4.50 as a fair split that gives them back close to what they sent plus some profit, while I also profit.

If I return \$4.50, they end up with \$6.36 while I keep \$4.92-both of us gain from the exchange. Returning \$5.00 would mirror...

usage:

```
{
  "input_tokens": 768,
  "output_tokens": 820,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.63 sent, it became \$10.89, returned \$5.0, visible

le payoff \$5.890000000000001.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.06 returned, visible payoff \$6.56.
Your total visible earnings across all rounds are \$12.44.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our history together:

Round 1: I sent (they saw \$3.63), they returned \$5.0 - cooperative

Round 2: They sent \$3.5 (I saw \$3.14), I returned (they saw \$5.06) - both cooperative

We've established a good cooperative pattern. They've been generous as both sender and receiver, so I want to continue building this trust.

I'll send **\$4** to show strong trust and maintain our mutually beneficial relationship.

```
'''json
{"send": 4}
'''
```

reasoning:

Let me analyze the history with my current co-player:

Round 1: I was SENDER

- I sent \$3 (but they saw \$3.63 due to noise)
- They received \$10.89
- They returned \$5.0
- This was cooperative behavior

Round 2: I was RECEIVER

- They sent \$3.5 (but I saw \$3.14 due to noise)
- I received \$9.42
- I returned \$4.50 (but they saw \$5.06 due to noise)
- They had a visible payoff of \$6.56

So far, this co-player has been quite cooperative:

- As receiver in Round 1: returned \$5.0 from \$10.89 (about 46%)
- As sender in Round 2: sent \$3.5 out of \$5 (70%)

I should continue this cooperative relationship. In Round 1, I sent \$3. In Round 2, they sent \$3.5. I should probably send a similar or slightly higher amount to maintain cooperation.

I'll send \$3.5 or \$4 to show continued trust and cooperation.

Given that they returned fairly and sent generously, I'll send \$4 to demonstrate strong trust and cooperation.

usage:

```
{
  "input_tokens": 1118,
  "output_tokens": 421,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.63 sent, it became \$10.89, returned \$5.0, visible payoff \$5.890000000000001.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.06 returned, visible payoff \$6.56.
- Round 3 against you, as RECEIVER: saw \$3.64 sent, it became \$10.92, returned \$5.5, visible payoff \$5.42.

Your total visible earnings across all rounds are \$18.1.

This round, you are the RECEIVER against your current co-player. They sent you \$3.21, so you received \$9.629999999999999. How much do you return? (0-9.629999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at our established pattern:

Round 1: They returned \$5.0 when receiving \$10.89

Round 2: I returned (they saw \$5.06) when I received \$9.42

Round 3: They returned \$5.5 when receiving \$10.92 - showing increased generosity

Round 4: They sent \$3.21, I received \$9.63

They've been consistently cooperative and even increased their returns. I want to reciprocate this positive trend.

I'll return **\$4.80**, which rewards their trust while maintaining mutual benefit.

```
'''json
{"return": 4.80}
'''
```

reasoning:

Looking at the history with my current co-player:

Round 1: They (as receiver) saw \$3.63 sent, received \$10.89, returned \$5.0

Round 2: They (as sender) sent \$3.5, received (saw \$5.06) back

Round 3: They (as receiver) saw \$3.64 sent, received \$10.92, returned \$5.5

Round 4: They (as sender) sent \$3.21, I received \$9.63

Our pattern:

- Round 1: I sent (they saw 3.63), they returned \$5.0
- Round 2: They sent 3.5 (I saw 3.14), I returned (they saw 5.06)
- Round 3: I sent (they saw 3.64), they returned \$5.5
- Round 4: They sent 3.21, I received \$9.63

They've been consistently cooperative and fair. In Round 3, they returned \$5.5, showing increased generosity.

For Round 4, they sent \$3.21, which became \$9.63 for me.

To maintain the cooperative relationship and be fair:

- They invested ~\$3.21
- I should return enough that they profit, while I also profit
- A return of around \$4.50-\$5.00 would be fair

I'll return \$4.80 to show continued cooperation and slightly increase the return to match their increased generosity in Round 3.

usage:

```
{
  "input_tokens": 1484,
  "output_tokens": 500,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.06 returned, visible payoff \$6.56.
- Round 3 against you, as RECEIVER: saw \$3.64 sent, it became \$10.92, returned \$5.5, visible payoff \$5.42.
- Round 4 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.28 returned, visible payoff \$6.78.

Your total visible earnings across all rounds are \$22.93.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at our continued cooperation:

- Round 2: They sent \$3.5, I returned (they saw \$5.28)
- Round 3: I sent (they saw \$3.64), they returned \$5.5 - very generous
- Round 4: They sent \$3.5, I returned (they saw \$5.28)

This co-player has been consistently trustworthy and generous. They reliably send around \$3.5 and return well when receiving. Our relationship is strongly cooperative.

I'll send **\$4.5** to show high trust and continue our mutually beneficial pattern.

```
'''json
{"send": 4.5}
'''
```

reasoning:

Looking at our history:

- Round 2: They sent \$3.5, I returned (they saw \$5.06)
- Round 3: I sent (they saw \$3.64), they returned \$5.5
- Round 4: They sent \$3.5, I returned (they saw \$5.28)
- Round 5: Now I'm sender

We've established a strong cooperative pattern. They've been consistently sending around \$3.5 and returning generously when receiver. I've been returning well too.

Given this trust and positive relationship, I should continue with a generous send. Last time I sent (Round 3), they returned \$5.5 on receiving \$10.92, which was very fair.

I'll send \$4 or even \$5 to show maximum trust. Given the strong cooperation, I'll go with \$4.5 as a strong signal of trust while keeping a small cushion.

usage:

```
{
  "input_tokens": 1846,
  "output_tokens": 366,
  "reasoning_tokens": 0
}
```

Message 12 role=user**content:**

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$3.64 sent, it became \$10.92, returned \$5.5, visible payoff \$5.42.
- Round 4 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.28 returned, visible payoff \$6.78.
- Round 5 against you, as RECEIVER: saw \$5 sent, it became \$15, returned \$7.5, visible payoff \$7.5.

Your total visible earnings across all rounds are \$30.5.

This round, you are the RECEIVER against your current co-player. They sent you \$3.9, so you received \$11.7. How much do you return? (0-11.7)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant**content:**

Looking at our strong cooperative history:

- Round 3: They returned \$5.5 from \$10.92
- Round 4: I returned (they saw \$5.28)
- Round 5: They returned \$7.5 from \$15 - exceptionally generous (50%)!
- Round 6: They sent \$3.9, I received \$11.7

They showed exceptional generosity in Round 5, returning exactly 50% when I sent a high amount. I want to match this level of cooperation.

I'll return **\$6.0**, which is approximately 51% - reciprocating their generous spirit.

```
'''json
{"return": 6.0}
'''
```

reasoning:

Looking at the history:

- Round 3: They returned \$5.5 when receiving \$10.92
- Round 4: I returned (they saw \$5.28)
- Round 5: They returned \$7.5 when receiving \$15 - very generous! They returned exactly 50 %
- Round 6: They sent \$3.9, I received \$11.7

This co-player has been extremely trustworthy and generous. In Round 5, when I sent a high amount (they saw \$5 which became \$15), they returned \$7.5, which is 50% - that's a very fair split.

I should reciprocate this generosity. They sent \$3.9, so I received \$11.7. If I want to be similarly generous and fair:

- They invested \$3.9
- I could return around \$6 to give them a good profit while keeping a profit myself
- Or I could match their 50% generosity and return \$5.85

Given their exceptional generosity in Round 5, I should be very generous too. I'll return \$6.0 which gives them a profit of \$2.1 (\$6 - \$3.9), and I keep \$5.7 (\$11.7 - \$6).

Actually, let me think about this more carefully. They returned \$7.5 from \$15, which is 50 %. If I return 50% of \$11.7, that would be \$5.85. But rounding to \$6 is cleaner and slightly more generous.

usage:

```
{
  "input_tokens": 2223,
  "output_tokens": 516,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.28 returned, visible payoff \$6.78.
- Round 5 against you, as RECEIVER: saw \$5 sent, it became \$15, returned \$7.5, visible payoff \$7.5.
- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.8 returned, visible payoff \$7.8.

Your total visible earnings across all rounds are \$36.2.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at our excellent track record:

- Round 4: They sent \$3.5, I returned (they saw \$5.28)
- Round 5: I sent (they saw \$5), they returned \$7.5 - 50% return!
- Round 6: They sent \$4.0 (increased!), I returned (they saw \$6.8)

This co-player has consistently shown high trustworthiness, even returning 50% when I sent my maximum in Round 5. They've also been increasing their own sending amounts. We have an excellent cooperative relationship.

I'll send **\$5** to show complete trust and maximize our mutual benefit.

```
'''json
{"send": 5}
'''
```

Reasoning:

Looking at the pattern with this co-player:

- Round 4: They sent \$3.5, I returned (they saw \$5.28)
- Round 5: I sent (they saw \$5), they returned \$7.5 - very generous 50% return
- Round 6: They sent \$4.0, I returned (they saw \$6.8) - they've been increasing their sending amounts
- Round 7: Now I'm the sender

This co-player has been extremely trustworthy:

1. They've increased their sending amounts from \$3.5 to \$4.0

2. When they received, they returned 50% (\$7.5 from \$15)
3. When they sent, they saw good returns from me

We've built a very strong cooperative relationship. They've demonstrated high trust in me, and I've shown them I'm reliable. This is the time to maximize mutual benefit by sending the maximum amount.

I'll send \$5 to show complete trust and maximize our joint gains.

```
Usage: {  
  "input_tokens": 2592,  
  "output_tokens": 398,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T08:14:53

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agents": [  
      "Agent_1",  
      "Agent_2"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_2": "trustee"  
    },  
    "trustee": "Agent_2"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 7,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths

, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.63, that is 72.6% of their total endowment. You received \$10.89. How much do you return to the sender? (0-10.89)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.63 (72.6% of their endowment)
- I received \$10.89 after the 3x multiplier
- A fair return should reward their trust while ensuring both of us benefit

The sender gave up \$3.63, keeping only \$1.37. Their generosity created \$10.89 total value for me. To reciprocate fairly and encourage future cooperation:

If I return \$5.00:

- Sender ends with: $\$5 - \$3.63 + \$5.00 = \6.37 (gain of \$1.37)
- I end with: $\$10.89 - \$5.00 = \$5.89$ (gain of \$5.89)

This seems like a fair split that rewards their trust while both of us benefit from the cooperation.

`{"return": 5.00}`

reasoning:

Let me analyze this situation carefully.

Given information:

- I'm the RECEIVER in Round 1
- The sender sent \$3.63 (which is 72.6% of their \$5 endowment)
- I received \$10.89 (which is $\$3.63 \times 3$)
- I need to decide how much to return (between \$0 and \$10.89)

Key considerations:

1. The sender was quite generous, sending 72.6% of their endowment
2. This shows trust and cooperation
3. There's mention of communication noise, so the actual amount sent might differ slightly from what I see
4. This is a multi-round game with 8 agents, so building reputation for cooperation could be beneficial
5. Fair reciprocation would be important to encourage future cooperation

Let me think about what would be a fair return:

- The sender gave up \$3.63 and kept \$1.37
- I received \$10.89
- If I want to be reciprocal and fair, I should consider:
 - The sender showed high trust (72.6% is a lot)
 - A common strategy is to return enough so both parties benefit
 - Returning about half (\$5.45) would give me \$5.44 and the sender would end up with $\$1.37 + \$5.45 = \$6.82$
 - This would mean the sender gains \$1.82 total and I gain \$5.44

Let me reconsider:

- If I return around \$4-5, that seems fair
- Returning \$5 would give:
 - Sender: $\$5 - \$3.63 + \$5 = \6.37 (net gain of \$1.37)
 - Me: $\$10.89 - \$5 = \$5.89$ (net gain of \$5.89)

- ...

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 801,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.52 returned, visible payoff \$7.52.

Your total visible earnings across all rounds are \$5.8900000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the situation:

From the visible history, my co-player:

- Sent \$3.0 in Round 1 (60% of their endowment)
- Saw \$5.52 returned (61.3% of the \$9.0)
- Made a profit of \$2.52

This shows they were cooperative and returned a substantial amount. To reciprocate their trust and maintain cooperation, I should send a generous amount.

I'll send \$3.50 (70% of my endowment), showing strong trust and willingness to cooperate.

```
{"send": 3.50}
```

reasoning:

Let me analyze this situation:

Round 1 (from my perspective):

- I was the RECEIVER
- The sender sent \$3.63 (72.6% of endowment)
- I received \$10.89
- I returned \$5.00
- My payoff: $\$10.89 - \$5.00 = \$5.89$
- Sender's payoff: $\$5 - \$3.63 + \$5.00 = \6.37

But now I see the "visible history" which shows a different perspective:

- My co-player (who was the sender in Round 1) saw that they sent \$3.0
- It became \$9.0 (not \$10.89 as I received)
- They saw \$5.52 returned (not \$5.00 as I sent)
- Their visible payoff was \$7.52

This confirms there is communication noise in the game as mentioned in the instructions.

What actually happened vs what they saw:

- They actually sent \$3.63, but saw they sent \$3.0
- I received \$10.89, they saw it became \$9.0
- I returned \$5.00, they saw \$5.52 returned
- Their actual payoff was \$6.37, but they saw \$7.52

Now in Round 2, I'm the SENDER and they are the RECEIVER.

My co-player returned a substantial amount in Round 1 (they saw they returned \$5.52 out of \$9.0 received, which is 61.3%). This shows they are cooperative.

Given that:

1. They showed trust by being generous in Round 1
2. They appeared to return a fair amount (from their perspective, 61.3% of what they received)
3. I want to maintain cooperation and reciprocate their trust

I should send a generous amount. ...

usage:

```
{
  "input_tokens": 874,
  "output_tokens": 716,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.52 returned, visible payoff \$7.52.
- Round 2 against you, as RECEIVER: saw \$3.14 sent, it became \$9.42, returned \$4.5, visible payoff \$4.92.

Your total visible earnings across all rounds are \$12.45.

This round, you are the RECEIVER against your current co-player. They sent you \$3.64, so you received \$10.92. How much do you return? (0-10.92)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the visible history:

Round 1: My co-player sent \$3.0 (60%) and saw \$5.52 returned

Round 2: They returned \$4.5 when they received \$9.42 (47.8%)

Round 3: They sent \$3.64 (72.8% - even more generous!)

My co-player has been consistently cooperative and just showed increased trust by sending an even larger percentage. I should reciprocate fairly.

Returning \$5.50:

- Sender gets: \$5 - \$3.64 + \$5.50 = \$6.86 (profit of \$1.86)
- I get: \$10.92 - \$5.50 = \$5.42 (profit of \$5.42)

This ~50% return rate rewards their trust while ensuring mutual benefit.

```
{"return": 5.50}
```

reasoning:

Let me analyze the visible history:

Round 1: My co-player was SENDER

- They sent \$3.0 (60% of endowment)
- It became \$9.0
- They saw \$5.52 returned (61.3% return rate)
- Their visible payoff: \$7.52

Round 2: My co-player was RECEIVER

- They saw \$3.14 sent (I actually sent \$3.50 but there's noise)
- It became \$9.42
- They returned \$4.5 (47.8% of what they received)
- Their visible payoff: \$4.92

Now Round 3: I am RECEIVER

- They sent \$3.64 (72.8% of their endowment - very generous!)
- I received \$10.92
- I need to decide how much to return

Looking at the pattern:

- In Round 1, they sent 60% and I returned about 61% (assuming noise in what they saw)
- In Round 2, they received and returned 47.8%
- In Round 3, they sent 72.8% - showing even more trust

The co-player has been consistently cooperative. They sent a high amount both times they were sender (60% and 72.8%). When they were receiver, they returned a reasonable amount (47.8%).

To maintain reciprocity and reward their increased trust, I should return a fair amount.

If I return \$5.50:

- Sender ends with: $\$5 - \$3.64 + \$5.50 = \6.86 (gain of \$1.86)
- I end with: $\$10.92 - \$5.50 = \$5.42$ (gain of \$5.42)

This is about 50.4% return rate, which is reasonable and shows I'm reciprocating their trust while both of us profit.

...

usage:

```
{
  "input_tokens": 1200,
  "output_tokens": 712,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.52 returned, visible payoff \$7.52.
- Round 2 against you, as RECEIVER: saw \$3.14 sent, it became \$9.42, returned \$4.5, visible payoff \$4.92.
- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$4.66 returned, visible payoff \$5.66.

Your total visible earnings across all rounds are \$17.869999999999997.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at the visible history:

Round 1: Co-player sent \$3.0 (60%), saw \$5.52 returned

Round 2: Co-player returned \$4.5 (47.8%)

Round 3: Co-player sent \$4.0 (80% - very generous!), saw \$4.66 returned

My co-player has been consistently cooperative, even increasing their trust in Round 3 by sending 80% of their endowment. While they saw a lower return percentage in Round 3, they've maintained a cooperative pattern overall.

I should reciprocate their trust and continue the cooperative relationship.

```
{"send": 3.50}
```

reasoning:

Let me analyze the visible history:

Round 1: Co-player sent \$3.0 (60%), saw \$5.52 returned (61.3% of \$9.0)

Round 2: Co-player returned \$4.5 when they received \$9.42 (47.8%)

Round 3: Co-player sent \$4.0 (80%!), saw \$4.66 returned (38.8% of \$12.0)

This is concerning. In Round 3, my co-player sent even more (\$4.0, or 80%), but they only saw \$4.66 returned, which is only 38.8% of the \$12.0 they sent. That's a lower return rate than before.

Wait, let me check what I actually sent and returned:

- Round 2: I sent \$3.50 according to my response
- Round 3: I returned \$5.50 according to my response

But the visible history shows:

- Round 2: They saw \$3.14 sent (I sent \$3.50, but due to noise they saw \$3.14)
- Round 3: They saw \$4.66 returned (I returned \$5.50, but due to noise they saw \$4.66)

So there's communication noise! This is important.

From my co-player's perspective:

- They've been very generous (sending 60%, 80%)
- But the returns they're seeing are declining in percentage terms (61.3%, then 38.8%)
- They might be feeling like their generosity isn't being reciprocated

However, I know I actually returned \$5.50 in Round 3, which was generous. The noise just made it look like \$4.66 to them.

Still, this relationship has been cooperative overall. My co-player has consistently sent substantial amounts. Even though they saw a lower return in Round 3, they've maintained trust.

...

usage:

```
{
  "input_tokens": 1626,
  "output_tokens": 638,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$3.14 sent, it became \$9.42, returned \$4.5, visible payoff \$4.92.
- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$4.66 returned, visible payoff \$5.66.
- Round 4 against you, as RECEIVER: saw \$3.21 sent, it became \$9.629999999999999, returned \$4.8, visible payoff \$4.829999999999999.

Your total visible earnings across all rounds are \$24.65.

This round, you are the RECEIVER against your current co-player. They sent you \$5, so you received \$15. How much do you return? (0-15)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at the visible history:

Round 2: Co-player returned \$4.5 (47.8%)
Round 3: Co-player sent \$4.0 (80%), saw modest return
Round 4: Co-player returned \$4.8 (49.8%)
Round 5: Co-player sent \$5.0 (100% - maximum possible trust!)

This is extraordinary! My co-player just sent their ENTIRE endowment, showing complete trust. They're keeping nothing and relying entirely on my reciprocation. I must reward this exceptional trust generously.

Returning \$7.50:

- Sender gets: \$5 - \$5 + \$7.50 = \$7.50 (profit of \$2.50)
- I get: \$15 - \$7.50 = \$7.50 (profit of \$7.50)

This 50% return ensures we both profit nicely and maintains our strong cooperative relationship.

{"return": 7.50}

reasoning:

Looking at the visible history:

Round 2: Co-player returned \$4.5 when they received \$9.42 (47.8%)
Round 3: Co-player sent \$4.0 (80%), saw \$4.66 returned (38.8%)
Round 4: Co-player returned \$4.8 when they received \$9.63 (49.8%)
Round 5: Co-player sent \$5.0 (100%!!!)

This is remarkable - my co-player just sent their ENTIRE endowment (\$5), showing maximum trust! This is the ultimate act of cooperation in this game. They're completely relying on me to reciprocate fairly.

They gave up \$5, keeping \$0. If I return nothing, they end with \$0. This would be devastating to the relationship.

I need to reciprocate this tremendous trust generously. Let me think about fair returns:

If I return \$7.50:

- Sender gets: $\$5 - \$5 + \$7.50 = \7.50 (profit of \$2.50)
- I get: $\$15 - \$7.50 = \$7.50$ (profit of \$7.50)

This 50% return rate gives us equal final amounts and ensures they make a nice profit for their trust.

Actually, given they sent 100%, I could even be more generous. Let me consider \$8:

- Sender gets: $\$5 - \$5 + \$8 = \8 (profit of \$3)
- I get: $\$15 - \$8 = \$7$ (profit of \$7)

This is still good for both of us and rewards their exceptional trust even more.

I think \$7.50 is fair - it's a 50% return rate, ensures both of us profit nicely, and maintains the cooperative relationship.

usage:

```
{
  "input_tokens": 2010,
  "output_tokens": 660,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$4.66 returned, visible payoff \$5.66.
- Round 4 against you, as RECEIVER: saw \$3.21 sent, it became \$9.629999999999999, returned \$4.8, visible payoff \$4.829999999999999.
- Round 5 against you, as SENDER: sent \$4.5, it became \$13.5, saw \$7.07 returned, visible payoff \$7.57.

Your total visible earnings across all rounds are \$32.15.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at the visible history:

Round 3: Co-player sent \$4.0 (80%), saw modest return

Round 4: Co-player returned \$4.8 (~50%)

Round 5: Co-player sent \$4.5 (90%), saw \$7.07 returned (good return!)

My co-player has been very generous, consistently sending 80-90% of their endowment. They've maintained strong cooperation and saw good returns in Round 5. I should reciprocate their trust.

```
{"send": 4.00}
```

reasoning:

Looking at the visible history:

Round 3: Co-player sent \$4.0 (80%), saw \$4.66 returned

Round 4: Co-player returned \$4.8 (~49.8%)

Round 5: Co-player sent \$4.5 (90%), saw \$7.07 returned

My co-player has been very generous and cooperative:

- They've sent high percentages (80%, 90%)
- When receiving, they returned reasonable amounts (~50%)
- They saw good returns in Round 5 (\$7.07)

This is a strong cooperative relationship. I should continue to send generous amounts to maintain this cooperation. Given their pattern of sending 80-100%, I should reciprocate with

ith a similar level of trust.

I'll send \$4.00 (80% of my endowment) to match their generosity and maintain cooperation.

usage:

```
{
  "input_tokens": 2464,
  "output_tokens": 332,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$3.21 sent, it became \$9.629999999999999, returned \$4.8, visible payoff \$4.829999999999999.
- Round 5 against you, as SENDER: sent \$4.5, it became \$13.5, saw \$7.07 returned, visible payoff \$7.57.
- Round 6 against you, as RECEIVER: saw \$3.9 sent, it became \$11.7, returned \$6.0, visible payoff \$5.699999999999999.

Your total visible earnings across all rounds are \$39.949999999999996.

This round, you are the RECEIVER against your current co-player. They sent you \$4.52, so you received \$13.559999999999999. How much do you return? (0-13.559999999999999)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the visible history:

Round 4: Co-player returned \$4.8 (~50%)
Round 5: Co-player sent \$4.5 (90%), saw good return
Round 6: Co-player returned \$6.0 (~51%)
Round 7: Co-player sent \$4.52 (90.4% - very high trust!)

My co-player continues to demonstrate exceptional trust, sending over 90% of their endowment.

They've been consistently cooperative both as sender and receiver. I should maintain this strong reciprocal relationship.

Returning \$7.00:

- Sender gets: \$5 - \$4.52 + \$7.00 = \$7.48 (profit of \$2.48)
- I get: \$13.56 - \$7.00 = \$6.56 (profit of \$6.56)

This ~51.6% return rewards their trust appropriately.

{"return": 7.00}

Reasoning:

Looking at the visible history:

Round 4: Co-player returned \$4.8 (~50%)
Round 5: Co-player sent \$4.5 (90%), saw \$7.07 returned
Round 6: Co-player returned \$6.0 (~51.3%)
Round 7: Co-player sent \$4.52 (90.4% - very high trust!)

My co-player continues to show very high trust, sending 90.4% of their endowment. They've been consistently cooperative as both sender and receiver.

I received \$13.56 after the 3x multiplier.

If I return \$6.75:

- Sender gets: \$5 - \$4.52 + \$6.75 = \$7.23 (profit of \$2.23)
- I get: \$13.56 - \$6.75 = \$6.81 (profit of \$6.81)

This is about 50% return rate, which is fair and rewards their trust.

Let me check if a slightly different amount makes sense:

If I return \$7.00:

- Sender gets: \$5 - \$4.52 + \$7.00 = \$7.48 (profit of \$2.48)
- I get: \$13.56 - \$7.00 = \$6.56 (profit of \$6.56)

This is about 51.6% return rate.

Given their high trust level (90.4%), I think \$7.00 is appropriate to maintain the strong cooperative relationship.

```
Usage: {  
  "input_tokens": 2844,  
  "output_tokens": 592,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_3 - Sender - dyad_2

Timestamp: 2026-08-12T08:14:58

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_2",  
  "move_index": 2,  
  "opponent_id": "Agent_4",  
  "pairing": {  
    "agents": [  
      "Agent_3",  
      "Agent_4"  
    ],  
    "investor": "Agent_3",  
    "roles": {  
      "Agent_3": "investor",  
      "Agent_4": "trustee"  
    },  
    "trustee": "Agent_4"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 7,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust and caution. The multiplication factor of 3x creates an opportunity for mutual benefit if my co-player reciprocates.

Since this is the first round with no prior history, I'll send a moderate amount that signals cooperation while not exposing myself to maximum risk.

****My decision:****

```
'''json
{"send": 3}
'''
```

This gives my co-player \$9 to work with (3×3), leaving me with \$2. If they return a fair share, we can both benefit from this cooperation. Let's see how they respond.

reasoning:

This is the first round of a trust game where I'm the SENDER. Let me think about this strategically.

Key facts:

- I have \$5
- I can send \$0-\$5 to the receiver
- Amount sent is multiplied by 3x
- The receiver then decides how much to return to me
- I keep: \$5 - sent + returned
- Receiver keeps: received - returned
- There may be communication noise (amounts seen may differ from actual)
- This is a multi-agent setting with 8 agents total
- Pairings are randomized each round
- Agent names are hidden

Strategic considerations:

1. ****Trust and cooperation**:** If both parties cooperate optimally, we can maximize joint payoffs. If I send all \$5, the receiver gets \$15. If they return half (\$7.50), I end up with \$7.50 and they keep \$7.50, which is better for both of us than if I send nothing.
2. ****Risk**:** The receiver could keep everything and return nothing. If I send \$5 and get nothing back, I end up with \$0 while they get \$15.
3. ****First round**:** This is the first round, so I have no reputation or history with this particular co-player. However, my actions might influence future interactions in a multi-round setting.
4. ****Communication noise**:** The presence of noise means that even if someone intends to be fair, the amounts might appear different, which could lead to misunderstandings.
5. ****Signaling**:** Sending a moderate to high amount signals trust and cooperation, which might encourage reciprocity.

Common strategies in trust games:

- Send half (\$2.50) as a moderate trust signal

- Send all (\$5) to maximize joint payoff and signal full trust
- Send a smaller amount (\$1-2) to test the waters
- Send nothing (\$0) to avoid risk entirely

Given this is round 1 and I want to establish a cooperative tone while not being completely naive, I think sending a significant but not maximum amount makes sense.

Research on trust games suggests most people send positive amounts initially, and receivers typically reciprocate. The 3x multiplier creates substantial mutual benefit potential—even modest returns from the receiver would leave us both better off than the starting position. I'll send \$3 as a balanced signal of trust and good faith without overexposing myself....

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 870,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.91 sent, it became \$8.73, returned \$3.5, visible payoff \$5.23.

Your total visible earnings across all rounds are \$5.5600000000000005.

This round, you are the RECEIVER against your current co-player. They sent you \$2.77, so you received \$8.31. How much do you return? (0-8.31)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

****Round 1 analysis:****

- I sent \$3 (they saw \$2.91 due to noise)
- They received \$8.73 and returned \$3.5 (about 40% return rate)
- This was a cooperative response

****Current situation (Round 2):****

- They sent me \$2.77 (with noise)
- I received \$8.31
- They showed good faith cooperation in Round 1

I'll reciprocate their cooperative behavior with a similar return rate:

```
'''json
{"return": 3.5}
'''
```

This gives them a total of approximately \$5.73 and leaves me with \$4.81, creating mutual benefit from our cooperation.

reasoning:

Let me analyze this situation:

Round 1 (just completed):

- I was the SENDER and sent \$3
- Due to communication noise, my co-player saw \$2.91 (close to \$3)
- It became \$8.73 for them (should be \$8.73 if 2.91×3)
- They returned \$3.5
- Their visible payoff: \$5.23 (they kept $8.73 - 3.5 = 5.23$)
- My payoff: $5 - 3 + 3.5 = 5.56$

Now Round 2:

- I am the RECEIVER
- My co-player sent me an amount, but I see \$2.77 (with noise)
- I received \$8.31 (which is 2.77×3)
- I need to decide how much to return

Analysis:

- In Round 1, my co-player returned \$3.5 out of \$8.73, which is about 40% return rate
- They kept \$5.23 for themselves
- I sent them \$3 (they saw \$2.91), and I ended up with \$5.56 total
- So I gained \$0.56 from the transaction

Now they sent me approximately \$2.77 (actual amount might be slightly different due to noise).

If I apply a similar return rate (around 40%), that would be: $\$8.31 \times 0.4 \approx \3.32

But let me think about fairness:

- They sent around \$2.77, leaving them with about \$2.23
- I received \$8.31
- A fair split might give them back enough so they end up with a reasonable amount

If I return \$3.5 (same as they returned to me):

- They end up with: $\$5 - \$2.77 + \$3.5 = \5.73
- I end up with: $\$8.31 - \$3.5 = \$4.81$

Returning \$4 would give them \$6.23 while I get \$4.31. Returning \$3 would leave me with \$5.31 and them with \$5.23. Since they gave me roughly \$2.77 and returned about 40% of what they received, matching that reciprocal gesture with a \$3.5 return seems appropriate....

usage:

```
{
  "input_tokens": 784,
  "output_tokens": 826,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.91 sent, it became \$8.73, returned \$3.5, visible payoff \$5.23.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.9 returned, visible payoff \$4.9.

Your total visible earnings across all rounds are \$10.370000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our history:

****Round 1:**** I sent \$3, they returned \$3.5 - cooperative

****Round 2:**** They sent \$3, I returned \$3.5 (they saw \$2.9) - cooperative

We've established a mutually beneficial pattern where we both send moderate amounts and return fair shares. I'll continue this cooperation:

```
```json
{"send": 3}
```
```

This maintains trust and keeps our successful collaborative pattern going.

reasoning:

Let me analyze the history:

****Round 1:****

- I sent \$3 (they saw \$2.91 due to noise)
- They received \$8.73 and returned \$3.5
- I ended up with $\$2 + \$3.5 = \$5.5$

****Round 2:****

- They sent \$3.0 (I saw \$2.77 due to noise, actual was \$3.0)
- It became \$9.0 for me
- I returned \$3.5

- They saw \$2.9 returned (due to noise)
- Their visible payoff was \$4.9 ($\$5 - \$3 + \2.9)
- I kept $\$9.0 - \$3.5 = \$5.5$

My total visible earnings: \$10.37 which matches $\$5.5 + \4.81 approximately (there's some rounding)

So the pattern so far:

1. Both of us have been sending moderate amounts (\$3)
2. Both of us have been returning reasonable amounts (\$3.5)
3. This has been mutually beneficial

The relationship seems cooperative. They sent me \$3 in Round 2, which is the same amount I sent in Round 1. They saw I returned \$2.9 (actual \$3.5), which was a fair return.

I should continue this cooperative pattern. Sending \$3 again seems appropriate to maintain the reciprocal relationship.

usage:

```
{
  "input_tokens": 1128,
  "output_tokens": 446,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.91 sent, it became \$8.73, returned \$3.5, visible payoff \$5.23.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.9 returned, visible payoff \$4.9.
- Round 3 against you, as RECEIVER: saw \$3.52 sent, it became \$10.56, returned \$4.5, visible payoff \$6.0600000000000005.

Your total visible earnings across all rounds are \$16.15.

This round, you are the RECEIVER against your current co-player. They sent you \$2.64, so you received \$7.92. How much do you return? (0-7.92)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at our cooperative history:

```
**Round 1:** They returned $3.5 on $8.73
**Round 2:** I returned $3.5 on $9.0
**Round 3:** They returned $4.5 on $10.56 (increased generosity!)
**Round 4:** They sent me $2.64, I received $7.92
```

They've been consistently cooperative and even increased their generosity in Round 3. I'll reciprocate fairly:

```
'''json
{"return": 3.5}
'''
```

This maintains our mutually beneficial cooperation pattern.

reasoning:

Let me analyze the history with my current co-player:

```
**Round 1:**
- I sent $3 (they saw $2.91 due to noise)
- They received $8.73, returned $3.5
- Good cooperation

**Round 2:**
- They sent $3
- I received $9.0, returned $3.5 (they saw $2.9)
- Continued cooperation
```

****Round 3:****

- I sent \$3 (they saw \$3.52 due to noise)
- They received \$10.56, returned \$4.5
- They increased their return, showing even more generosity

****Round 4 (current):****

- They sent \$2.64 (with noise, could have been around \$3 originally)
- I received \$7.92

Looking at the pattern:

- They've been consistently cooperative
- In Round 3, they returned \$4.5 on \$10.56 (about 43% return rate)
- They're continuing to send around \$3 each time

I should reciprocate their continued cooperation. Given they sent me \$7.92, a fair return would be around 40-45% of what I received, which would be about \$3.2-\$3.6.

I'll return \$3.5 to maintain consistency with our cooperative pattern.

usage:

```
{
  "input_tokens": 1467,
  "output_tokens": 452,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.9 returned, visible payoff \$4.9.
- Round 3 against you, as RECEIVER: saw \$3.52 sent, it became \$10.56, returned \$4.5, visible payoff \$6.0600000000000005.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.96 returned, visible payoff \$5.96.

Your total visible earnings across all rounds are \$20.57.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at our established pattern:

****Round 2-4:**** They consistently sent \$3

****Round 3:**** They were particularly generous, returning \$4.5

****Round 4:**** I returned \$3.5 (they saw \$3.96)

We have a strong cooperative relationship. I'll maintain our successful pattern:

```
'''json
{"send": 3}
'''
```

This continues our mutually beneficial cooperation.

reasoning:

Looking at the history with this co-player:

Round 1: I sent \$3 (they saw \$2.91), they returned \$3.5 - cooperative

Round 2: They sent \$3, I returned \$3.5 (they saw \$2.9) - cooperative

Round 3: I sent \$3 (they saw \$3.52), they returned \$4.5 - very generous!

Round 4: They sent \$3, I returned \$3.5 (they saw \$3.96) - cooperative

This co-player has been consistently cooperative and even showed extra generosity in Round 3. The pattern shows:

- They keep sending \$3
- They've returned generously when receiving
- We have a strong cooperative relationship

I should continue the cooperation. Sending \$3 has worked well for us.

usage:

```
{
  "input_tokens": 1824,
  "output_tokens": 291,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$3.52 sent, it became \$10.56, returned \$4.5, visible payoff \$6.0600000000000005.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.96 returned, visible payoff \$5.96.
- Round 5 against you, as RECEIVER: saw \$2.38 sent, it became \$7.14, returned \$3.0, visible payoff \$4.14.

Your total visible earnings across all rounds are \$26.11.

This round, you are the RECEIVER against your current co-player. They sent you \$2.05, so you received \$6.149999999999995. How much do you return? (0-6.149999999999995)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at the pattern:

```
**Round 3:** They returned $4.5
**Round 4:** I returned $3.5
**Round 5:** They returned $3.0
**Round 6:** They sent $2.05, I received $6.15
```

Our cooperation has been solid. I'll maintain a fair return:

```
'''json
{"return": 3.0}
'''
```

This keeps our mutually beneficial relationship going.

reasoning:

Looking at the history:

Round 3: They returned \$4.5 on \$10.56
Round 4: I returned \$3.5 (they saw \$3.96) on \$7.92
Round 5: They saw me send \$2.38 (I actually sent \$3), received \$7.14, returned \$3.0

Now Round 6: They sent me \$2.05 (with noise), I received \$6.15

They've been consistently cooperative throughout. Even though the amount they sent this round is lower (possibly due to noise), they've shown good faith. I should maintain cooperation.

Given I received \$6.15, a fair return would be around \$2.5-3.0 to maintain our cooperative pattern. Let me return about 40-45% which has been roughly our pattern.

$\$6.15 * 0.45 = \2.77

Or I could go with \$3.0 to be consistent with what they did in Round 5.

Let me go with \$3.0 to maintain consistency and cooperation.

usage:

```
{
  "input_tokens": 2163,
  "output_tokens": 349,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

Visible history before this round:
Your current co-player's last 3 game(s):
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.96 returned, visible payoff \$5.96.
- Round 5 against you, as RECEIVER: saw \$2.38 sent, it became \$7.14, returned \$3.0, visible payoff \$4.14.
- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.94 returned, visible payoff \$4.9399999999999995.
Your total visible earnings across all rounds are \$29.259999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at our pattern:

****Round 4-6:**** They consistently sent \$3
****Round 5-6:**** We both returned around \$3

We have a stable, cooperative relationship. I'll continue:

```
'''json
{"send": 3}
'''
```

This maintains our successful mutual cooperation.

Reasoning:

Looking at the pattern with this co-player:

Round 4: They sent \$3
Round 5: I sent \$3 (they saw \$2.38), they returned \$3.0
Round 6: They sent \$3, I returned \$3.0 (they saw \$2.94)
Round 7: My turn to send

We've maintained a consistent cooperative pattern. They keep sending \$3, and we both return fair amounts. I should continue this successful cooperation.

```
Usage: {
  "input_tokens": 2488,
  "output_tokens": 182,
  "reasoning_tokens": 0
}
```

GAME - Agent_4 - Receiver - dyad_2

Timestamp: 2026-08-12T08:15:03

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_3",
  "pairing": {
    "agents": [
      "Agent_3",
      "Agent_4"
    ],
    "investor": "Agent_3",
    "roles": {
      "Agent_3": "investor",
      "Agent_4": "trustee"
    },
    "trustee": "Agent_4"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 7,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.91, that is 58.2% of their total endowment. You received \$8.73. How much do you return to the sender? (0-8.73)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.91 (58.2% of their endowment)
- I received \$8.73 after the 3x multiplication
- The sender currently has \$2.09 remaining

To reciprocate their trust and cooperation:

- If I return nothing, the sender ends with \$2.09 (worse than keeping everything)
- If I return \$3.50, the sender ends with \$5.59 and I keep \$5.23
- This rewards their trust while maintaining fairness

I'll return a meaningful amount to encourage continued cooperation.

{'return': 3.5}

reasoning:

Let me analyze this situation:

1. As the RECEIVER, the sender sent me \$2.91 (which is 58.2% of their \$5 endowment)

2. This amount was multiplied by 3x, so I received \$8.73
3. I need to decide how much to return (between \$0 and \$8.73)

Key considerations:

- The sender was quite generous, sending nearly 60% of their endowment
- This shows trust and cooperation
- In trust games, reciprocity is important for maintaining cooperation
- The sender's final payoff will be: \$5 - \$2.91 + whatever I return = \$2.09 + return
- My final payoff will be: \$8.73 - whatever I return

Common approaches in trust games:

1. Split the gains equally - the sender gave up \$2.91 but created \$8.73. The "gain" is \$8.73 - \$2.91 = \$5.82. Splitting this equally would mean returning about \$2.91
2. Give the sender back what they sent - return \$2.91
3. Give the sender a good return on their investment - perhaps return more than they sent
4. Equal final payoffs - try to make both players end up with similar amounts

Let me calculate some scenarios:

If I return \$2.91:

- Sender gets: \$2.09 + \$2.91 = \$5.00
- I get: \$8.73 - \$2.91 = \$5.82
- This is fair-ish, both doing okay

If I return \$3.50:

- Sender gets: \$2.09 + \$3.50 = \$5.59
- I get: \$8.73 - \$3.50 = \$5.23
- This balances outcomes more evenly

If I return \$4.00:

- Sender gets: \$2.09 + \$4.00 = \$6.09
- I get: \$8.73 - \$4.00 = \$4.73
- The sender benefits more, which seems appropriate given they initiated the cooperation

For truly equal final payoffs, I'd need to return \$3.32, which would give us both \$5.41 each. The sender demonstrated substantial trust by sending most of their endowment, so I should reciprocate meaningfully to reinforce that cooperation....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 942,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.56 returned, visible payoff \$5.5600000000000005.

Your total visible earnings across all rounds are \$5.23.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the history:

- My co-player sent \$3.0 (60% of their endowment) in Round 1, showing strong trust
- They saw a return of \$3.56 (39.6% of what they sent as receiver)
- This demonstrates cooperative behavior

To reciprocate their trust and maintain cooperation, I'll send a similar proportion:

```
{'send': 3.0}
```

reasoning:

Let me analyze this situation:

Round 1:

- I was the RECEIVER
- The sender sent \$2.91 (58.2% of their endowment)
- I received \$8.73
- I returned \$3.50
- I kept \$5.23

Now in Round 2:

- I'm the SENDER
- I'm playing against the same co-player from Round 1
- The visible history shows that in Round 1, from my co-player's perspective:
 - They sent \$3.0 (as SENDER)
 - It became \$9.0
 - They saw \$3.56 returned
 - Their visible payoff was \$5.56

Wait, there's a discrepancy here. The problem says there's communication noise. Let me understand:

From my perspective in Round 1:

- Sender sent \$2.91
- I received \$8.73
- I returned \$3.50

From my co-player's perspective in Round 1:

- They sent \$3.0
- It became \$9.0
- They saw \$3.56 returned

So the noise affected what they saw vs what actually happened. They sent \$3.0 but I saw \$2.91. I returned \$3.50 but they saw \$3.56.

Now I'm the sender in Round 2. My co-player was cooperative in Round 1 - they sent 60% of their endowment (\$3.0 out of \$5). They saw a return of \$3.56 out of \$9.0, which is about 39.6% of what I received.

Given that:

1. My co-player showed strong trust by sending 60% in Round 1
2. They saw a decent return (39.6%)
3. I want to maintain cooperation and build trust

I should send a similar proportion to what they sent. They sent 60%, so I should send around \$3.0 as well to match their level of trust and cooperation.

...

usage:

```
{
  "input_tokens": 814,
  "output_tokens": 568,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.56 returned, visible payoff \$5.5600000000000005.
- Round 2 against you, as RECEIVER: saw \$2.77 sent, it became \$8.31, returned \$3.5, visible payoff \$4.8100000000000005.

Your total visible earnings across all rounds are \$10.13.

This round, you are the RECEIVER against your current co-player. They sent you \$3.52, so you received \$10.56. How much do you return? (0-10.56)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the pattern:

****Round 1**:** My co-player sent \$3.0 (60%), showing strong trust

****Round 2**:** I sent \$3.0 (60%), they returned \$3.5 (42% of what they received)
****Round 3**:** They sent \$3.52 (70.4%), showing even greater trust

They're being very cooperative and increasing their trust level. I should reciprocate appropriately.

Returning \$4.5:

- Sender ends with: $\$5 - \$3.52 + \$4.5 = \5.98
- I keep: $\$10.56 - \$4.5 = \$6.06$

This rewards their high trust while maintaining fairness.

{'return': 4.5}

reasoning:

Let me analyze this situation:

Round 1 (I was RECEIVER):

- Sender sent \$2.91 (58.2%)
- I received \$8.73
- I returned \$3.50
- My payoff: \$5.23

Round 2 (I was SENDER):

- I sent \$3.0
- It became \$9.0
- According to visible history, my co-player saw \$2.77 sent (there's noise!)
- They returned \$3.5
- My visible payoff from Round 2: I sent \$3, got back \$3.5, so $\$2 + \$3.5 = \$5.5$
- Total earnings so far: $\$5.23 + \$5.5 = \$10.73$... but it says \$10.13

Wait, let me recalculate. The visible history says:

- Round 2 as RECEIVER: saw \$2.77 sent, it became \$8.31, returned \$3.5, visible payoff \$4.81

This is from my co-player's perspective. They saw \$2.77 sent (due to noise), it became \$8.31, they returned \$3.5, keeping \$4.81.

My actual earnings from Round 2:

- I sent \$3.0
- I kept \$2.0
- I received back... well, if they returned \$3.5 based on receiving \$8.31, then I should have received \$3.5 back
- So my payoff: $\$2.0 + \$3.5 = \$5.5$

Total: $\$5.23 + \$5.5 = \$10.73$

But it says my total visible earnings are \$10.13. Hmm, there might be noise in what I received back too.

Anyway, for Round 3:

- I'm RECEIVER
- They sent \$3.52 (70.4% of their endowment)
- I received \$10.56

My partner's been increasingly generous across rounds-first 58.2%, now 70.4%-showing growing confidence. They returned roughly 40-42% in the previous round, so I should match that reciprocal level. Returning between \$4.22-\$4.75 would maintain the cooperative dynamic we've established....

usage:

```
{
  "input_tokens": 1104,
  "output_tokens": 897,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.56 returned, visible payoff \$5.5600000000000005.
- Round 2 against you, as RECEIVER: saw \$2.77 sent, it became \$8.31, returned \$3.5, visible payoff \$4.8100000000000005.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.78 returned, visible payoff \$5.7799999999999999.
Your total visible earnings across all rounds are \$16.19.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at the pattern:

****Round 1**:** Co-player sent \$3.0 (60%), demonstrating trust
****Round 2**:** Co-player returned \$3.5, showing cooperation
****Round 3**:** Co-player sent \$3.0 (60%) again, maintaining trust

They're showing consistent cooperative behavior. I should continue matching their trust level:

{ 'send': 3.0 }

reasoning:

Looking at the history:

****Round 1**:** Co-player sent \$3.0 as SENDER, saw \$3.56 returned
****Round 2**:** Co-player was RECEIVER, saw \$2.77 sent (I actually sent \$3.0), returned \$3.5
****Round 3**:** Co-player sent \$3.0 as SENDER, saw \$3.78 returned (I actually returned \$4.5)

The communication noise is evident:

- In Round 2, I sent \$3.0 but they saw \$2.77
- In Round 3, I returned \$4.5 but they saw \$3.78

My co-player has been consistently sending \$3.0 (60% of endowment) as sender. They've been returning good amounts as receiver (\$3.5).

They're showing consistent trust and cooperation. I should continue to show strong trust as well. I'll send \$3.0 again to maintain the cooperative pattern.

usage:

```
{
  "input_tokens": 1512,
  "output_tokens": 323,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.77 sent, it became \$8.31, returned \$3.5, visible payoff \$4.8100000000000005.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.78 returned, visible payoff \$5.7799999999999999.
- Round 4 against you, as RECEIVER: saw \$2.64 sent, it became \$7.92, returned \$3.5, visible payoff \$4.42.
Your total visible earnings across all rounds are \$22.150000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$2.38, so you received \$7.14. How much do you return? (0-7.14)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at the pattern:

****Round 2-4**:** Co-player consistently returned \$3.5 when receiver
****Round 3**:** Co-player sent \$3.0 as sender
****Round 5**:** Co-player sent \$2.38 (47.6%), somewhat less than before

They're still showing trust but at a slightly reduced level. I'll reciprocate proportionally:

Returning \$3.0:

- Sender ends with: $\$5 - \$2.38 + \$3.0 = \5.62
- I keep: $\$7.14 - \$3.0 = \$4.14$

This maintains fairness and cooperation.

`{'return': 3.0}`

reasoning:

Looking at the history:

Round 2: Co-player as RECEIVER saw \$2.77 sent, returned \$3.5

Round 3: Co-player as SENDER sent \$3.0, saw \$3.78 returned

Round 4: Co-player as RECEIVER saw \$2.64 sent, returned \$3.5

Round 5: Co-player sent \$2.38 (47.6% of their endowment)

This is a notable decrease from their previous sending amounts (\$3.0). They've been very consistent with returning \$3.5 when they're the receiver.

They sent \$2.38, which becomes \$7.14 for me.

If I return \$3.0:

- Sender ends with: $\$5 - \$2.38 + \$3.0 = \5.62
- I keep: $\$7.14 - \$3.0 = \$4.14$

This seems fair. They sent less this round (maybe due to seeing slightly lower returns or noise in the system), but I should still reciprocate their cooperation.

Let me think about what's fair:

- They sent 47.6% of endowment
- I received \$7.14
- Returning about 40-45% of what I received would be: $\$2.86 - \3.21

I'll return \$3.0, which is reasonable and maintains cooperation.

usage:

```
{
  "input_tokens": 1847,
  "output_tokens": 489,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.78 returned, visible payoff \$5.779999999999999.
- Round 4 against you, as RECEIVER: saw \$2.64 sent, it became \$7.92, returned \$3.5, visible payoff \$4.42.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.54 returned, visible payoff \$5.54.

Your total visible earnings across all rounds are \$26.290000000000003.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: `{'send': <amount>}`

Message 13 role=assistant

content:

Looking at the pattern:

****Round 3**:** Co-player sent \$3.0 (60%)

****Round 4**:** Co-player returned \$3.5, showing cooperation

****Round 5**:** Co-player sent \$3.0 (60%) again

They're maintaining consistent cooperation and trust. I'll continue matching their behavior:

```
{'send': 3.0}
```

reasoning:

Looking at the history:

```
**Round 3**: Co-player sent $3.0 as sender  
**Round 4**: Co-player returned $3.5 as receiver  
**Round 5**: Co-player sent $3.0 as sender
```

The co-player is showing consistent trust by sending \$3.0 when they're the sender and returning \$3.5 when they're the receiver. This is cooperative behavior.

I should continue to match this cooperative pattern by sending around \$3.0.

My total visible earnings are \$26.29, which suggests I'm doing well in this cooperative relationship.

I'll send \$3.0 to maintain the cooperative pattern.

usage:

```
{  
  "input_tokens": 2233,  
  "output_tokens": 244,  
  "reasoning_tokens": 0  
}
```

Message 14 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$2.64 sent, it became \$7.92, returned \$3.5, visible payoff \$4.42.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.54 returned, visible payoff \$5.54.
- Round 6 against you, as RECEIVER: saw \$2.05 sent, it became \$6.1499999999999995, returned \$3.0, visible payoff \$3.1499999999999995.

Your total visible earnings across all rounds are \$31.230000000000004.

This round, you are the RECEIVER against your current co-player. They sent you \$3.55, so you received \$10.649999999999999. How much do you return? (0-10.649999999999999)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the pattern:

```
**Round 4-6**: Co-player consistently showing cooperation  
**Round 7**: Co-player sent $3.55 (71%), showing strong trust - higher than previous rounds
```

They're demonstrating increased trust. I should reciprocate generously:

Returning \$4.5:

- Sender ends with: \$5 - \$3.55 + \$4.5 = \$5.95
- I keep: \$10.65 - \$4.5 = \$6.15

This rewards their strong cooperation.

```
{'return': 4.5}
```

Reasoning:

Looking at the pattern:

```
**Round 4**: Co-player returned $3.5 as receiver  
**Round 5**: Co-player sent $3.0 as sender  
**Round 6**: Co-player returned $3.0 as receiver  
**Round 7**: Co-player sent $3.55 (71%)
```

They're showing high trust by sending 71% of their endowment. I should reciprocate generously.

Returning \$4.5:

- Sender ends with: \$5 - \$3.55 + \$4.5 = \$5.95
- I keep: \$10.65 - \$4.5 = \$6.15

This rewards their strong trust and maintains cooperation.

```
Usage: {
  "input_tokens": 2570,
  "output_tokens": 309,
  "reasoning_tokens": 0
}
```

GAME - Agent_5 - Sender - dyad_3

Timestamp: 2026-08-12T08:15:10

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_6",
  "pairing": {
    "agents": [
      "Agent_5",
      "Agent_6"
    ],
    "investor": "Agent_5",
    "roles": {
      "Agent_5": "investor",
      "Agent_6": "trustee"
    },
    "trustee": "Agent_6"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 7,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round situation, I need to balance potential gains from cooperation against the uncertainty of how my co-player will respond.

****My reasoning:****

- Sending money creates value (3x multiplier), so there's potential for mutual benefit
- This is our first interaction, so I want to signal willingness to cooperate
- But I also want to maintain some security given the uncertainty
- The communication noise adds another layer of unpredictability

I'll send a moderate amount that demonstrates trust while keeping some reserve.

```
'''json
{"send": 3}
'''
```

This sends \$3 to my co-player, who will receive \$9. I'll keep \$2 initially, and hope they reciprocate with a fair return that benefits us both.

reasoning:

This is the first round of a trust game where I'm the sender. Let me think about the strategic situation:

1. I have \$5 to start with
2. Whatever I send gets tripled when it reaches the receiver
3. The receiver then decides how much to return to me
4. There's communication noise - amounts may appear different than what was actually sent

Key considerations:

- This is a multi-round game with multiple agents
- Pairings are randomized each round
- There's a tension between trust/cooperation and self-interest
- If I send nothing, I keep \$5 but there's no opportunity for mutual gain
- If I send everything (\$5), the receiver gets \$15, and if they're fair/cooperative, they might return a good amount
- The communication noise adds uncertainty

In standard trust game experiments, sending some amount is generally beneficial because:

- It creates surplus value (3x multiplier)
- It signals trust and can encourage reciprocity
- Over multiple rounds, cooperative behavior can be rewarding

First round considerations:

- I don't know this particular co-player yet
- This is the first interaction, so I might want to signal willingness to cooperate
- But I also don't want to be too naive

A moderate approach might be to send \$3 or \$4:

- Sending \$3 means receiver gets \$9, I keep \$2. If they return \$4-5, we both do well
- Sending \$4 means receiver gets \$12, I keep \$1. If they return \$6, we both get \$6
- Sending \$5 means receiver gets \$15, I keep \$0. If they return \$7-8, I do well

Given this is round 1 and I want to establish a cooperative tone while not being too vulnerable, I'll send \$3. This shows trust but maintains some security.

Actually, let me reconsider. In trust games, the classic finding is that people often send

about half their endowment on average. Sending \$2.50 or \$3 seems reasonable.

...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 703,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.66 sent, it became \$7.98, returned \$4.0, visible payoff \$3.9800000000000004.

Your total visible earnings across all rounds are \$5.13.

This round, you are the RECEIVER against your current co-player. They sent you \$2.97, so you received \$8.91. How much do you return? (0-8.91)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the situation:

****What happened in Round 1:****

- My co-player received \$7.98 (my \$3 sent with noise)
- They returned \$4.0, which is about 50% - a fair reciprocal move
- This shows they're a cooperative player

****Current Round 2:****

- They sent me \$2.97 (?60% of their endowment) - showing significant trust
- I received \$8.91
- They kept \$2.03

****My decision:****

Since they demonstrated fair reciprocity (50% return) in Round 1 and have now shown strong trust by sending most of their endowment, I should reciprocate similarly. Returning about 50% maintains the cooperative relationship while ensuring we both benefit.

Returning \$4.50 means:

- They end with: $\$2.03 + \$4.50 = \$6.53$
- I end with: $\$8.91 - \$4.50 = \$4.41$

This is fair and encourages continued cooperation.

```
'''json
{"return": 4.50}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap:

- I sent \$3 (but due to communication noise, my co-player saw \$2.66)
- They received \$7.98 (which should be $\$2.66 \times 3$)
- They returned \$4.0
- My visible payoff was \$5.13 (which would be $\$5 - \$3 + \$3.13...$ but there's noise)
- Their visible payoff was \$3.98

Now in Round 2:

- My co-player is the SENDER
- They sent me \$2.97 (with noise, they might have intended something slightly different)
- I received \$8.91 (which is $\$2.97 \times 3$)
- I need to decide how much to return

Analysis of their behavior:

- In Round 1, they returned \$4.0 out of \$7.98 received, which is about 50%
- This is a reasonably fair/generous return
- Now they've sent me \$2.97 out of their \$5, which is about 59% - quite trusting

Given that:

1. They showed good reciprocity in Round 1 (returning 50% of what they received)
2. They've now shown significant trust by sending me almost 60% of their endowment
3. We might continue playing together in future rounds
4. Fair reciprocity would encourage continued cooperation

I should return a similar proportion or slightly more to maintain/build trust.

They received \$8.91 from my send. A 50% return would be \$4.455.

But let me think about the overall economics:

- They started with \$5
- They sent \$2.97, keeping \$2.03
- I received \$8.91
- If I return \$4.50:
 - They end with: $\$2.03 + \$4.50 = \$6.53$

...

usage:

```
{
  "input_tokens": 816,
  "output_tokens": 863,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.66 sent, it became \$7.98, returned \$4.0, visible payoff \$3.9800000000000004.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.03 returned, visible payoff \$7.03.

Your total visible earnings across all rounds are \$9.54.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our history together:

****Round 1:****

- I sent, they received and returned ~50% - fair cooperation

****Round 2:****

- They sent \$3 (60% of endowment) - high trust
- I returned \$4.50, they saw \$5.03 (noise) - generous reciprocity
- Their payoff was strong (\$7.03)

****Analysis:****

- This co-player has proven consistently cooperative
- They demonstrated trust by sending \$3 in Round 2
- They've reciprocated fairly (50% return rate)
- Our relationship is building mutual trust

****My decision:****

I should maintain this cooperative dynamic. Sending \$3 again signals:

- Continued trust in our partnership
- Consistency and reliability
- Faith they'll reciprocate fairly as before

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze the visible history:

Round 1:

- I was SENDER, sent \$3

- They received \$7.98 (with noise, actual was probably \$9)
- They returned \$4.0 (about 50%)
- My payoff was \$5 - \$3 + \$4 = \$6
- Visible shows \$5.13, which might include noise

Round 2:

- I was RECEIVER
- They sent \$3.0 (actual, before noise)
- I received \$8.91 (actually \$9 with noise)
- I returned \$4.50
- They saw \$5.03 returned (with noise, I actually returned \$4.50)
- Their visible payoff \$7.03 = \$5 - \$3 + \$5.03

My total visible earnings: \$9.54

Round 3 (current):

- I'm SENDER again
- Same co-player from rounds 1 and 2

This co-player has been very cooperative:

1. Returned 50% in Round 1 when they received
2. Sent \$3 (60%) in Round 2, showing trust
3. They saw me return \$5.03 (with noise), which they probably perceive as generous (56% return)

This is building a strong cooperative relationship. They've been trustworthy and fair. I should continue to send a substantial amount to maintain this cooperation.

Given their consistent cooperation and the fact they sent \$3 in Round 2, I should send at least the same amount, if not more, to signal continued trust and cooperation.

I'll send \$3 again, showing consistency and trust.

usage:

```
{
  "input_tokens": 1250,
  "output_tokens": 587,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.66 sent, it became \$7.98, returned \$4.0, visible payoff \$3.9800000000000004.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.03 returned, visible payoff \$7.03.
- Round 3 against you, as RECEIVER: saw \$3.23 sent, it became \$9.69, returned \$5.0, visible payoff \$4.6899999999999995.

Your total visible earnings across all rounds are \$17.33.

This round, you are the RECEIVER against your current co-player. They sent you \$3.93, so you received \$11.790000000000001. How much do you return? (0-11.790000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at our continued partnership:

****Round 1:**** They returned \$4.0 from \$7.98 (?50%)
****Round 2:**** I returned \$4.50 from \$9.0 (?50%)
****Round 3:**** They returned \$5.0 from \$9.69 (?52%)

****Current Round 4:****

- They sent \$3.93 (79% of endowment!) - very high trust signal
- I received \$11.79
- They kept only \$1.07

****Analysis:****

This co-player has shown increasing trust and consistent 50% reciprocity. They're now send

ing nearly their entire endowment, demonstrating strong faith in our relationship.

I should maintain our established 50% return pattern to:

- Honor their trust
- Keep our cooperative equilibrium
- Ensure mutual benefit

Returning \$5.90:

- They end with: $\$1.07 + \$5.90 = \$6.97$
- I end with: $\$11.79 - \$5.90 = \$5.89$

```
'''json
{"return": 5.90}
'''
```

reasoning:

Let me analyze the history:

****Round 1:****

- I sent (saw \$2.66 with noise from my \$3)
- They received \$7.98, returned \$4.0 (about 50%)

****Round 2:****

- They sent \$3
- I received \$9, returned \$4.50 (they saw \$5.03 with noise)
- They got \$7.03 total

****Round 3:****

- I sent \$3 (they saw \$3.23 with noise)
- They received \$9.69, returned \$5.0 (about 52%)
- They kept \$4.69

****Round 4 (current):****

- They sent \$3.93 (about 79% of their endowment!) - very high trust
- I received \$11.79
- They kept \$1.07

This co-player is showing extremely high trust by sending \$3.93. They've consistently returned around 50% when they were receiver. I should reciprocate this trust and maintain our cooperative relationship.

If I return about 50% of what I received:

- Return \$5.90
- They end with: $\$1.07 + \$5.90 = \$6.97$
- I end with: $\$11.79 - \$5.90 = \$5.89$

But they're showing increasing trust (sending more and more). Maybe I should return a bit more to reward this and encourage continued cooperation. Let me think...

Actually, looking at the pattern:

- Round 1 receiver: returned \$4 from \$7.98 (50%)
- Round 3 receiver: returned \$5 from \$9.69 (52%)

They're consistently returning around 50%. I've been doing similar - returned \$4.50 from \$9 in Round 2 (50%).

Given they sent \$3.93 (very high trust), I should return around 50% or slightly more to maintain the relationship.

...

usage:

```
{
  "input_tokens": 1695,
  "output_tokens": 859,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.03 returned, visible payoff \$7.03.
- Round 3 against you, as RECEIVER: saw \$3.23 sent, it became \$9.69, returned \$5.0, visible payoff \$4.6899999999999995.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.63 returned, visible payoff \$8.629999999999999.
Your total visible earnings across all rounds are \$23.22.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at our established partnership:

****Round 2:**** They sent \$3.0, I returned generously
****Round 3:**** I sent \$3.23 (with noise), they returned \$5.0 (?52%)
****Round 4:**** They sent \$3.0, I returned well, they saw \$6.63

****Pattern Analysis:****

- They consistently send \$3 when sender (Rounds 2, 4)
- They reciprocate fairly at ~50% when receiver (Round 3)
- We've built a stable, mutually beneficial relationship
- Both earning well (\$23.22 total for me)

****My decision:****

- Match their consistency by sending \$3 again. This:
- Maintains our established cooperative equilibrium
 - Shows I trust the pattern we've developed
 - Signals reliability and reciprocity

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at the visible history:

****Round 2:**** They sent \$3.0, I returned (they saw \$5.03) - I actually returned \$4.50 but they saw \$5.03 due to noise
****Round 3:**** They received \$9.69 (from my \$3 send), returned \$5.0
****Round 4:**** They sent \$3.0, I returned \$5.90 but they saw \$6.63 - generous due to noise

Pattern analysis:

- This co-player consistently sends \$3.0 when they're sender (Rounds 2 and 4)
- They return roughly 50% when receiver (Round 3: \$5.0 from \$9.69)
- They've been seeing higher returns from me due to noise (\$5.03 and \$6.63 vs what I actually sent \$4.50 and \$5.90)

Our cooperation has been strong and consistent. They've shown:

- Reliable sending patterns (\$3)
- Fair reciprocity (~50%)
- Building trust over rounds

I should continue with the same sending pattern to maintain this cooperative equilibrium. Sending \$3 has been working well for both of us.

My total visible earnings are \$23.22 which suggests we're both doing well.

I'll send \$3 again to maintain consistency and trust.

usage:

```
{
  "input_tokens": 2185,
  "output_tokens": 517,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$3.23 sent, it became \$9.69, returned \$5.0, visible payoff \$4.6899999999999995.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.63 returned, visible payoff \$8.629999999999999.
- Round 5 against you, as RECEIVER: saw \$3.62 sent, it became \$10.86, returned \$6.0, visible payoff \$4.859999999999999.

Your total visible earnings across all rounds are \$31.189999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$3.02, so you received \$9.06. How much do you return? (0-9.06)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant
content:

Looking at our continued partnership:

****Round 3:**** I sent, they returned \$5.0 from \$9.69 (?52%)

****Round 4:**** They sent \$3.0, I returned generously

****Round 5:**** I sent, they returned \$6.0 from \$10.86 (?55%)

****Current Round 6:****

- They sent \$3.02 (?60% of endowment) - continued trust
- I received \$9.06
- They kept \$1.98

****Analysis:****

Our partnership shows consistent mutual cooperation with ~50-55% return rates. They've reliably reciprocated in this range, and I should maintain this equilibrium.

Returning \$4.50:

- They end with: \$1.98 + \$4.50 = \$6.48
- I end with: \$9.06 - \$4.50 = \$4.56

This maintains our cooperative pattern.

```
'''json
{"return": 4.50}
'''
```

reasoning:

Looking at our history:

****Round 3:**** I sent (they saw \$3.23), they returned \$5.0 from \$9.69 (?52%)

****Round 4:**** They sent \$3.0, I returned, they saw \$6.63 (generous due to noise)

****Round 5:**** I sent (they saw \$3.62), they returned \$6.0 from \$10.86 (?55%)

****Current Round 6:****

- They sent \$3.02 (?60% of endowment)
- I received \$9.06
- They kept \$1.98

****Pattern:****

- They consistently send around \$3
- They consistently return around 50-55% when receiver
- We have a stable, cooperative relationship

I should maintain the ~50% return pattern:

- Returning \$4.50 would be about 50%
- This keeps our cooperative equilibrium
- They end with: \$1.98 + \$4.50 = \$6.48
- I end with: \$9.06 - \$4.50 = \$4.56

Both benefit reasonably, and this maintains the pattern.

usage:

```
{
  "input_tokens": 2634,
  "output_tokens": 514,
  "reasoning_tokens": 0
}
```

Message 14 role=user
content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.63 returned, visible payoff \$8.629999999999999.
- Round 5 against you, as RECEIVER: saw \$3.62 sent, it became \$10.86, returned \$6.0, visible payoff \$4.859999999999999.
- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.57 returned, visible payoff \$6.57.

Your total visible earnings across all rounds are \$35.75.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at our established pattern:

- **Round 4:** They sent \$3.0, I returned generously
- **Round 5:** I sent, they returned \$6.0 (?55%)
- **Round 6:** They sent \$3.0, I returned (they saw \$4.57)

Analysis:

- They consistently send \$3 when in the sender role (Rounds 4, 6)
- They reciprocate fairly (50-55%) when receiver (Round 5)
- We've maintained a stable, mutually beneficial partnership
- Total earnings of \$35.75 shows this strategy is working well

My decision:

Continue the established pattern by sending \$3. This:

- Maintains our successful cooperative equilibrium
- Matches their consistent behavior
- Sustains mutual trust and benefit

```
'''json
{"send": 3}
'''
```

Reasoning:

Looking at the history:

- **Round 4:** They sent \$3.0, I returned well (they saw \$6.63)
- **Round 5:** I sent, they returned \$6.0
- **Round 6:** They sent \$3.0, I returned (they saw \$4.57 with noise)

Pattern:

- They consistently send \$3 when sender
- They reciprocate fairly when receiver
- We've maintained a strong cooperative relationship
- Total earnings are healthy at \$35.75

This is Round 7, and we've established a stable pattern where:

- Both of us send \$3 when sender
- Both of us return around 50% when receiver
- This creates mutual benefit

I should continue with \$3 to maintain this successful equilibrium.

```
Usage: {
  "input_tokens": 3095,
  "output_tokens": 378,
  "reasoning_tokens": 0
}
```

GAME - Agent_6 - Receiver - dyad_3

Timestamp: 2026-08-12T08:15:21

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 5,
```

```

"opponent_id": "Agent_5",
"pairing": {
  "agents": [
    "Agent_5",
    "Agent_6"
  ],
  "investor": "Agent_5",
  "roles": {
    "Agent_5": "investor",
    "Agent_6": "trustee"
  },
  "trustee": "Agent_6"
},
"role": "trustee",
"role_label": "Receiver",
"round": 7,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.66, that is 53.2% of their total endowment. You received \$7.98. How much do you return to the sender? (0-7.98)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.66 (53.2% of their endowment)
- I received \$7.98 after the 3x multiplication
- The sender currently has \$2.34 remaining from their original \$5

To reward their trust and encourage cooperation, I should return enough so they profit from their decision to send. If I return \$4.00:

- Sender ends with: $\$2.34 + \$4.00 = \$6.34$ (a gain of \$1.34)
- I keep: $\$7.98 - \$4.00 = \$3.98$

This seems fair - both of us benefit from the cooperation, and the sender's trust is rewarded.

{"return": 4.00}

reasoning:

Let me think about this carefully.

I'm the RECEIVER in Round 1.

- The sender sent \$2.66 (which is 53.2% of their \$5 endowment)
- This amount was multiplied by 3x, so I received $\$2.66 \times 3 = \7.98
- Now I need to decide how much to return (between \$0 and \$7.98)

Key considerations:

1. The sender sent more than half their endowment (53.2%), showing substantial trust
2. If I return nothing, I keep \$7.98 and the sender keeps $\$5 - \$2.66 = \$2.34$
3. If I return some amount, I should aim to be fair and encourage cooperation

A reasonable approach would be to:

- Ensure the sender ends up with more than they started with (more than \$5)
- Keep some profit myself
- Reward their trust to encourage future cooperation

Let me calculate:

- Sender currently has: $\$5 - \$2.66 = \$2.34$
- I have: \$7.98

If I return \$3.00:

- Sender ends with: $\$2.34 + \$3.00 = \$5.34$ (slight gain)
- I keep: $\$7.98 - \$3.00 = \$4.98$

If I return \$4.00:

- Sender ends with: $\$2.34 + \$4.00 = \$6.34$ (good gain)
- I keep: $\$7.98 - \$4.00 = \$3.98$

If I return \$3.50:

- Sender ends with: $\$2.34 + \$3.50 = \$5.84$
- I keep: $\$7.98 - \$3.50 = \$4.48$

...

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 699,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.13 returned, visible payoff \$5.13.

Your total visible earnings across all rounds are \$3.9800000000000004.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the visible history:

****Round 1 recap (from their perspective):****

- My co-player sent \$3.00 (60% of their endowment) - quite generous
- They saw \$3.13 returned
- Their visible payoff was \$5.13

My co-player showed significant trust by sending 60% of their endowment. Even though there's communication noise affecting what we each see, I should reciprocate this trust.

I'll match their generosity by sending the same amount they sent: \$3.00 (60% of my endowment). This demonstrates:

- Reciprocal trust
- Willingness to cooperate
- Fair play matching their level of generosity

{"send": 3.00}

reasoning:

Let me analyze this situation:

Round 1 (just played):

- I was the RECEIVER
- Sender sent \$2.66 (53.2%)
- I received \$7.98
- I returned \$4.00
- I kept \$3.98

But now I see the "visible history" which shows:

- Round 1 against me, as SENDER: sent \$3.0, became \$9.0, saw \$3.13 returned, visible payoff \$5.13

This is interesting - there's communication noise!

- The sender actually sent \$3.00 (not \$2.66 as I saw)
- It became \$9.00 (not \$7.98 as I saw)
- They saw \$3.13 returned (not \$4.00 as I sent)
- Their visible payoff was \$5.13 (which is \$5 - \$3 + \$3.13)

So due to noise:

- I saw them send \$2.66, they actually sent \$3.00
- I saw I received \$7.98, I actually received \$9.00
- I returned \$4.00, they saw \$3.13

Now in Round 2, I'm the SENDER and playing against the same person.

From their perspective:

- They sent \$3.00 (60% of endowment - quite generous)
- They got back \$3.13 (34.8% of what I received)
- Their payoff was \$5.13

From my actual perspective (what really happened):

- They sent \$3.00
- I received \$9.00
- I returned \$4.00 (44.4% of what I received)
- But they only saw \$3.13

The co-player was quite generous in Round 1 (sent 60%). From their visible perspective, they got a modest return (34.8% of the tripled amount).

...

usage:

```
{
  "input_tokens": 845,
  "output_tokens": 754,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.13 returned, visible payoff \$5.13.
- Round 2 against you, as RECEIVER: saw \$2.97 sent, it became \$8.91, returned \$4.5, visible payoff \$4.41.

Your total visible earnings across all rounds are \$11.010000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$3.23, so you received \$9.69. How much do you return? (0-9.69)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the visible history:

****Round 1:**** My co-player sent \$3.00 (60%)

****Round 2:**** They returned \$4.50 out of \$8.91 (50.5%) when they were receiver

****Round 3:**** They just sent \$3.23 (64.6%) - even more generous!

My co-player has shown increasing trust and cooperation. They've been very generous, and when they were receiver in Round 2, they returned about 50% of what they received.

I should reciprocate their strong cooperation. If I return \$5.00:

- Sender ends with: \$1.77 + \$5.00 = \$6.77 (profit of \$1.77)
- I keep: \$9.69 - \$5.00 = \$4.69

This returns about 52% of what I received, matching their generosity from Round 2 and rewarding their increased trust.

{"return": 5.00}

reasoning:

Let me analyze the visible history:

****Round 1 (I was RECEIVER):****

- My co-player sent \$3.00 (60%)
- They saw \$3.13 returned
- Their visible payoff: \$5.13

****Round 2 (I was SENDER):****

- I sent \$3.00
- They saw \$2.97 sent (noise), which became \$8.91
- They returned \$4.50
- Their visible payoff: \$4.41

****Round 3 (I am RECEIVER):****

- They sent \$3.23 (64.6% of their endowment)
- I received \$9.69
- They have \$1.77 remaining

My co-player has been very cooperative:

- Round 1: Sent 60% (\$3.00)
- Round 2: Returned \$4.50 out of \$8.91 received (50.5%)
- Round 3: Sent 64.6% (\$3.23) - even more generous!

They've shown increasing trust and generosity. I should reciprocate strongly.

Let me think about what to return:

If I return \$5.00:

- Sender ends with: \$1.77 + \$5.00 = \$6.77 (profit of \$1.77)
- I keep: \$9.69 - \$5.00 = \$4.69

If I return \$5.50:

- Sender ends with: \$1.77 + \$5.50 = \$7.27 (profit of \$2.27)
- I keep: \$9.69 - \$5.50 = \$4.19

Given their increased generosity (sending 64.6% vs 60% in round 1), I should return proportionally more. They returned \$4.50 out of \$8.91 (50.5%) when they were receiver. I should aim to return around 50-55% to match their generosity.

...

usage:

```
{
  "input_tokens": 1207,
  "output_tokens": 754,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.13 returned, visible payoff \$5.13.
- Round 2 against you, as RECEIVER: saw \$2.97 sent, it became \$8.91, returned \$4.5, visible payoff \$4.41.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.79 returned, visible payoff \$7.79.

Your total visible earnings across all rounds are \$15.700000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at the visible history:

****Round 1:**** Co-player sent \$3.00, saw \$3.13 returned, payoff \$5.13

****Round 2:**** Co-player returned \$4.50 from \$8.91 received (fair, ~50%)

****Round 3:**** Co-player sent \$3.00, saw \$5.79 returned (very generous!), payoff \$7.79

My co-player has been consistently generous, sending \$3.00 (60%) in both rounds as sender. They also saw a strong return in Round 3, which should reinforce mutual trust. As receiver, they returned fairly (~50%).

This is a cooperative relationship with established trust. I should continue matching their consistent generosity of \$3.00.

{"send": 3.00}

reasoning:

Looking at the visible history from my co-player's perspective:

****Round 1:**** They sent \$3.00, saw \$3.13 returned, payoff \$5.13

****Round 2:**** They received \$8.91 (from my \$2.97 send), returned \$4.50, payoff \$4.41

****Round 3:**** They sent \$3.00, saw \$5.79 returned, payoff \$7.79

My co-player has been consistently generous, sending \$3.00 in both rounds they were sender (60% of endowment). In Round 3, they saw a very generous return of \$5.79, which should reinforce their trust in me.

They've also been fair as a receiver, returning \$4.50 out of \$8.91 (about 50%).

This is a cooperative relationship with mutual trust. I should continue to match their generosity. They've sent \$3.00 twice, so I should send \$3.00 again.

Actually, let me reconsider. In Round 3, they saw a particularly high return (\$5.79), which might encourage even more trust. But they've consistently sent \$3.00, so I should match that level.

I'll send \$3.00 to maintain consistency and reciprocal cooperation.

usage:

```
{
  "input_tokens": 1654,
  "output_tokens": 485,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.97 sent, it became \$8.91, returned \$4.5, visible payoff \$4.41.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.79 returned, visible payoff \$7.79.
- Round 4 against you, as RECEIVER: saw \$3.93 sent, it became \$11.790000000000001, returned \$5.9, visible payoff \$5.890000000000001.

Your total visible earnings across all rounds are \$24.33.

This round, you are the RECEIVER against your current co-player. They sent you \$3.62, so you received \$10.86. How much do you return? (0-10.86)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at the visible history:

****Round 2:**** Co-player returned \$4.50 from \$8.91 (50.5%)
****Round 3:**** Co-player sent \$3.00, saw \$5.79 returned
****Round 4:**** Co-player returned \$5.90 from \$11.79 (50.1%)
****Round 5:**** Co-player sent \$3.62 (72.4%) - very high trust!

My co-player has been consistently cooperative, returning about 50% when receiver and sending generously as sender. This round they sent an even larger amount (\$3.62, which is 72.4% of their endowment) - showing increasing trust.

I should reward this high level of trust. If I return \$6.00:

- Sender ends with: $\$1.38 + \$6.00 = \$7.38$ (profit of \$2.38)
- I keep: $\$10.86 - \$6.00 = \$4.86$

This returns about 55% of what I received, slightly more generous than their pattern to acknowledge their increased trust.

{"return": 6.00}

reasoning:

Looking at the visible history:

****Round 2:**** Co-player returned \$4.50 from \$8.91 (50.5%)
****Round 3:**** Co-player sent \$3.00, saw \$5.79 returned
****Round 4:**** Co-player returned \$5.90 from \$11.79 (50.1%)
****Round 5:**** Co-player sent \$3.62 (72.4%)

My co-player has been very consistent and cooperative:

- As sender, they've sent generous amounts (\$3.00 and now \$3.62 - 72.4%)
- As receiver, they've returned about 50% consistently

They've sent an even larger amount this round (\$3.62, which is 72.4% of their endowment). This shows increasing trust.

I should continue to reciprocate fairly. If I return around 50-55% like they've been doing:

- 50% would be \$5.43
- 55% would be \$5.97

Let me aim for about \$5.50 to \$6.00 to maintain the cooperative pattern and reward their increased generosity.

If I return \$5.50:

- Sender ends with: $\$1.38 + \$5.50 = \$6.88$ (profit of \$1.88)
- I keep: $\$10.86 - \$5.50 = \$5.36$

If I return \$6.00:

- Sender ends with: $\$1.38 + \$6.00 = \$7.38$ (profit of \$2.38)
- I keep: $\$10.86 - \$6.00 = \$4.86$

Given they sent 72.4% of their endowment (very high trust), I should return a bit more generously. ...

usage:

```
{
  "input_tokens": 2074,
```

```
"output_tokens": 706,  
"reasoning_tokens": 0
```

```
}
```

Message 12 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.79 returned, visible payoff \$7.79.
- Round 4 against you, as RECEIVER: saw \$3.93 sent, it became \$11.790000000000001, returned \$5.9, visible payoff \$5.890000000000001.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.97 returned, visible payoff \$7.97.

Your total visible earnings across all rounds are \$29.189999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at the visible history:

```
**Round 3:** Co-player sent $3.00, saw $5.79 returned, payoff $7.79  
**Round 4:** Co-player returned $5.90 from $11.79 (50%)  
**Round 5:** Co-player sent $3.00, saw $5.97 returned, payoff $7.97
```

My co-player has been extremely consistent and cooperative:

- As sender: consistently sends \$3.00 (60% of endowment)
- As receiver: returns about 50% fairly
- They've received good returns, reinforcing mutual trust

This is a stable, mutually beneficial cooperative relationship. I should continue matching their pattern by sending \$3.00.

```
{"send": 3.00}
```

reasoning:

Looking at the visible history from my co-player's perspective:

```
**Round 3:** Co-player sent $3.00, saw $5.79 returned, payoff $7.79  
**Round 4:** Co-player returned $5.90 from $11.79 (50%)  
**Round 5:** Co-player sent $3.00, saw $5.97 returned, payoff $7.97
```

My co-player has been very consistent:

- As sender: consistently sends \$3.00 (60% of endowment)
- As receiver: returns about 50%
- They've seen good returns when sending (\$5.79 and \$5.97), leading to strong payoffs (~\$8)

This is a very cooperative and stable relationship. My co-player has maintained trust by consistently sending \$3.00 and returning fairly. I should continue this pattern of mutual cooperation by sending \$3.00 again.

The relationship is working well - we're both benefiting from the cooperation. No reason to deviate from what's working.

usage:

```
{  
  "input_tokens": 2569,  
  "output_tokens": 425,  
  "reasoning_tokens": 0  
}
```

Message 14 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$3.93 sent, it became \$11.790000000000001, returned \$5.9, visible payoff \$5.890000000000001.

- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.97 returned, visible payoff \$7.97.
- Round 6 against you, as RECEIVER: saw \$3.02 sent, it became \$9.06, returned \$4.5, visible payoff \$4.560000000000005.
Your total visible earnings across all rounds are \$35.76.

This round, you are the RECEIVER against your current co-player. They sent you \$2.78, so you received \$8.34. How much do you return? (0-8.34)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the visible history:

****Round 4:**** Co-player returned \$5.90 from \$11.79 (50%)
****Round 5:**** Co-player sent \$3.00, saw \$5.97 returned
****Round 6:**** Co-player returned \$4.50 from \$9.06 (49.7%)
****Round 7:**** Co-player sent \$2.78 (55.6%)

My co-player has been consistently cooperative, returning about 50% when receiver and sending generously as sender. This round they sent \$2.78, slightly less than their usual \$3.00, but still a majority of their endowment (55.6%).

I should maintain our cooperative relationship with a fair return. If I return \$4.50:

- Sender ends with: $\$2.22 + \$4.50 = \$6.72$ (profit of \$1.72)
- I keep: $\$8.34 - \$4.50 = \$3.84$

This returns about 54% of what I received, which is fair and maintains the mutual trust we've built.

{"return": 4.50}

Reasoning:

Looking at the visible history:

****Round 4:**** Co-player returned \$5.90 from \$11.79 (50%)
****Round 5:**** Co-player sent \$3.00, saw \$5.97 returned, payoff \$7.97
****Round 6:**** Co-player returned \$4.50 from \$9.06 (49.7%)
****Round 7:**** Co-player sent \$2.78 (55.6%)

My co-player has been very consistent:

- As sender: sending around \$3.00 (though this round \$2.78, slightly less)
- As receiver: returning about 50% consistently

This round they sent \$2.78, which is a bit less than their usual \$3.00. Let me think about what to return.

If I return \$4.50:

- Sender ends with: $\$2.22 + \$4.50 = \$6.72$ (profit of \$1.72)
- I keep: $\$8.34 - \$4.50 = \$3.84$

This returns about 54% of what I received, which is fair and maintains our cooperative relationship.

Actually, let me reconsider. They sent slightly less this round (\$2.78 vs usual \$3.00). I should still be generous but maybe match their slight reduction in trust with a proportional return.

If I return \$4.50:

- That's 54% return rate
- Sender profits \$1.72
- I keep \$3.84

If I return \$4.00:

- That's 48% return rate
- Sender profits \$1.22
- I keep \$4.34

I think \$4.50 is still appropriate to maintain the good cooperative relationship, even though they sent slightly less. The relationship has been strong overall.

Usage: {

"input_tokens": 2990,
"output_tokens": 685,

```
"reasoning_tokens": 0
}
```

GAME - Agent_7 - Sender - dyad_4

Timestamp: 2026-08-12T08:15:31

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_8",
  "pairing": {
    "agents": [
      "Agent_7",
      "Agent_8"
    ],
    "investor": "Agent_7",
    "roles": {
      "Agent_7": "investor",
      "Agent_8": "trustee"
    },
    "trustee": "Agent_8"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 7,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.

- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to consider:

1. ****The multiplier effect****: Any amount I send gets tripled, creating potential mutual gain
2. ****No prior history****: This is my first interaction, so I have no information about this co-player's behavior
3. ****Future rounds****: Even though pairings are randomized, establishing cooperative norms early could benefit the group overall
4. ****Risk vs. reward****: Sending more creates more surplus, but requires trusting the receiver to return a fair amount

I'll take a moderately cooperative approach. Sending \$3 creates \$9 for the receiver - enough surplus that we can both benefit if they reciprocate fairly, while not risking my entire endowment.

If they return around half of what they receive (\$4-5), we both end up better off than if no transaction occurred.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a multi-agent trust game experiment. I'm the SENDER with \$5.

Let me think about the strategic aspects:

1. If I send money, it gets tripled, creating surplus value
2. The receiver then decides how much to return
3. This is a classic trust game scenario

Key considerations:

- This is round 1, so I have no history with this particular co-player
- There are multiple rounds, so reputation/repeated interaction matters
- However, pairings are randomized each round, so I won't necessarily play the same person again
- The game has communication noise, which might affect trust

Strategic thinking:

- Sending \$0 guarantees I keep \$5, but creates no surplus
- Sending \$5 creates maximum surplus (\$15 total for receiver), but requires trust
- A middle ground (e.g., \$2-3) balances risk and potential gain

In classic trust game experiments, people often:

1. Send moderate to high amounts in early rounds to signal cooperation
2. The amount sent and returned establishes baseline trust levels
3. With randomized pairings, building a reputation as trustworthy matters for future rounds

Game theory perspective:

- If this were truly one-shot with rational self-interested players, the receiver should return \$0 (backward induction)
- Knowing this, I should send \$0
- However, real humans often deviate from this due to fairness norms, reciprocity, and reputation concerns

Since:

- This is multi-round with multiple agents
- There may be reputation effects even with randomized pairings
- Starting cooperatively can establish good norms

- The 3x multiplier creates real gains from cooperation

I'll send a moderate-high amount to signal cooperation while not being completely vulnerable. I'll send \$3.

This creates \$9 for the receiver, and if they return even just \$3, we both end up with \$5 (same as no transaction), but if they return more (say \$4-5), we both gain. ...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 671,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.17 sent, it became \$9.51, returned \$5.0, visible payoff \$4.51.

Your total visible earnings across all rounds are \$7.37.

This round, you are the RECEIVER against your current co-player. They sent you \$2.35, so you received \$7.050000000000001. How much do you return? (0-7.050000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

****Round 1 analysis:****

- They saw I sent \$3.17 (with noise), received \$9.51
- They returned \$5.00, which is about 52.6% of what they received
- This was quite generous and cooperative

****Current situation:****

- They sent me \$2.35, I received \$7.05
- They showed good faith last round, so I should reciprocate

Following a similar reciprocity rate (around 52-53% return):

- 53% of \$7.05 ? \$3.74

This ensures:

- They profit from sending (get back more than they sent)
- I also benefit from the transaction
- We maintain cooperative momentum

```
'''json
{"return": 3.75}
'''
```

reasoning:

Let me analyze this situation:

****What happened in Round 1:****

- I sent \$3, but they saw \$3.17 (communication noise)
- They received \$9.51 (3.17×3)
- They returned \$5.0
- Their visible payoff was $\$9.51 - \$5.0 = \$4.51$
- My payoff was $\$5 - \$3 + \$5 = \7 (but visible earnings show \$7.37, maybe due to noise)

****Current Round 2:****

- They sent me \$2.35 (actual), which became \$7.05
- I need to decide how much to return

****Analysis of their behavior:****

In Round 1, they returned \$5 out of \$9.51 received, which is about 52.6%. This is a fairly generous return - they kept \$4.51 and I got back \$5, so I had a net gain of \$2 from the transaction.

****What should I do?****

They showed cooperative behavior by returning a good amount. Now they've sent me \$2.35, cr eating \$7.05.

If I apply a similar reciprocity ratio:

- They returned about 52.6% of what they received
- I received \$7.05, so 52.6% would be about \$3.71

However, I should also consider:

- They sent \$2.35 this round (less than the ~\$3.17 they saw me send)
- But they were generous with their return last round
- I want to maintain cooperation

A fair approach might be to:

1. Return enough that they profit from sending
2. Keep enough that I also profit
3. Signal continued cooperation

If they sent \$2.35 and expect around a 50% return rate:

- They might expect back around \$3-4

Let me think about payoffs:

- They started with \$5
- They sent \$2.35
- If I return \$3.5, they end with \$6.15 and I keep \$3.55
- If I return \$4, they end with \$6.65 and I keep \$3.05
- If I return \$3, they end with \$5.65 and I keep \$4.05

Since they matched my generosity last round with a 52.6% return, I should reciprocate similarly to maintain the cooperative dynamic....

usage:

```
{
  "input_tokens": 852,
  "output_tokens": 862,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.17 sent, it became \$9.51, returned \$5.0, visible payoff \$4.51.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.71 returned, visible payoff \$5.71.

Your total visible earnings across all rounds are \$10.670000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our history together:

****Round 1:****

- I sent \$3, they received ~\$9.51, returned \$5.00 (very generous ~53% return)

****Round 2:****

- They sent \$2.35, I received \$7.05, returned \$3.75
- They saw \$3.71 returned due to noise (~41% visible return rate)

****Analysis:****

- This co-player was quite generous in Round 1
- They sent a bit less in Round 2, possibly because they saw a lower return rate from me (due to noise)
- Overall, they've demonstrated cooperative behavior

****My strategy:****

I'll send \$3 again to:

- Match the cooperative level from Round 1

- Show consistency in my approach
- Signal that I'm willing to trust despite the noise in Round 2

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze the history with this co-player:

****Round 1 (I was SENDER):****

- I sent \$3 (they saw \$3.17 due to noise)
- They received \$9.51 and returned \$5.00
- This was generous - about 52.6% return rate

****Round 2 (I was RECEIVER):****

- They sent \$2.35 (actual), I received \$7.05
- I returned \$3.75
- They sent \$3.0 actual, became \$9.0, they saw \$3.71 returned
- So they saw a return rate of about 41.2% ($3.71/9.0$)
- This is less than what they gave me in Round 1

****Pattern:****

- This co-player was very cooperative in Round 1
- In Round 2, they sent a bit less (\$2.35 vs my \$3)
- They saw me return less proportionally than they did (41.2% vs 52.6%)

****Current decision:****

I'm the sender again. This co-player has shown:

1. Initial high cooperation (52.6% return)
2. Slightly reduced sending in Round 2 (\$2.35 vs \$3)
3. They may perceive me as less generous than they were (due to noise showing only \$3.71 returned when I sent \$3.75)

I should:

- Maintain cooperation since they were generous in Round 1
- Send a similar amount to Round 1 to show consistency
- Perhaps send \$3 again to match their Round 2 amount

Given they've been generally cooperative and we're building a pattern, I'll send \$3 again.

usage:

```
{
  "input_tokens": 1212,
  "output_tokens": 605,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.17 sent, it became \$9.51, returned \$5.0, visible payoff \$4.51.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.71 returned, visible payoff \$5.71.
- Round 3 against you, as RECEIVER: saw \$2.22 sent, it became \$6.66, returned \$3.33, visible payoff \$3.33.

Your total visible earnings across all rounds are \$16.970000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$2.62, so you received \$7.86. How much do you return? (0-7.86)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at our continued interaction:

****Round 1:**** They returned \$5.00 from \$9.51 (53%)

****Round 3:**** They returned \$3.33 from \$6.66 (50%)

****Current situation:****

- They sent \$2.62, I received \$7.86
- They've been consistently cooperative, returning around 50-53%

****My decision:****

I should match their cooperative approach. Returning about 53% would be:

- 53% of \$7.86 ? \$4.17

This ensures:

- They profit significantly (\$4.17 back vs \$2.62 sent = +\$1.55 net)
- I also benefit (\$3.69 kept from the tripled amount)
- We maintain positive reciprocity

```
'''json
{"return": 4.20}
'''
```

reasoning:

Looking at the history with this co-player:

****Round 1:****

- I sent \$3, they saw \$3.17 (noise), received \$9.51
- They returned \$5.00 (~53% return rate)

****Round 2:****

- They sent \$3.00, became \$9.00
- I returned \$3.75, they saw \$3.71
- Visible return rate for them: ~41%

****Round 3:****

- I sent \$3, they saw \$2.22 (significant noise), received \$6.66
- They returned \$3.33 (50% return rate)

****Current Round 4:****

- They sent \$2.62, I received \$7.86

****Analysis:****

- This co-player has been consistently cooperative
- Round 1: 53% return
- Round 3: 50% return
- Their sending amounts haven't decreased dramatically despite seeing noise

****My approach:****

I should maintain a similar return rate. They've been returning around 50-53%.

- 50% of \$7.86 = \$3.93
- 53% of \$7.86 = \$4.17

I'll return around \$4.00 to:

- Match their cooperative behavior
- Ensure they profit from sending (\$2.62 sent, \$4.00 back)
- Keep the relationship positive
- I keep \$3.86, they get \$4.00 back vs \$2.62 sent = net gain of \$1.38 for them

Actually, let me be a bit more generous to match their Round 1 behavior. I'll return \$4.20, which is about 53% of what I received.

Their final payoff: \$5 - \$2.62 + \$4.20 = \$6.58

My final payoff: \$7.86 - \$4.20 = \$3.66

...

usage:

```
{
  "input_tokens": 1649,
  "output_tokens": 650,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.71 returned, visible p

ayoff \$5.71.
- Round 3 against you, as RECEIVER: saw \$2.22 sent, it became \$6.66, returned \$3.33, visible payoff \$3.33.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.31 returned, visible payoff \$5.3100000000000005.
Your total visible earnings across all rounds are \$20.630000000000003.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at our ongoing interaction:

****Pattern Analysis:****

- They've consistently sent \$3.00 in Rounds 2 and 4
- As receiver (Rounds 1 & 3), they returned generously (~50-53%)
- Unfortunately, noise has made my returns appear lower than actual:
 - Round 2: I returned \$3.75, they saw \$3.71
 - Round 4: I returned \$4.20, they saw only \$3.31

****Current Strategy:****

Despite the noise distorting communication, they've continued trusting me with \$3 sends. I should:

- Maintain consistency by sending \$3
- Signal continued cooperation
- Trust they'll recognize the pattern despite noise

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze our history:

****Round 1:**** I sent \$3, they received ~\$9.51, returned \$5.00
****Round 2:**** They sent \$2.35, I received \$7.05, returned \$3.75, but they saw \$3.71
****Round 3:**** I sent \$3, they received ~\$6.66, returned \$3.33, but they saw I sent \$2.22
****Round 4:**** They sent \$2.62, I received \$7.86, returned \$4.20, but they saw \$3.31

Looking at what they're seeing from my perspective:

- Round 2: They sent \$3.00, saw \$3.71 returned (visible return rate ~41%)
- Round 4: They sent \$3.00, saw \$3.31 returned (visible return rate ~37%)

The noise is making it look like I'm returning less than I actually am.

From their rounds as receiver:

- Round 1: They returned \$5.00 from \$9.51 (53%)
- Round 3: They returned \$3.33 from \$6.66 (50%)

They've been consistently cooperative. However, the noise is making my returns look less generous than they actually are. Despite this, they've continued sending \$3 in rounds 2 and 4.

I should:

1. Continue showing trust and cooperation
2. Send a similar amount to maintain consistency
3. Hopefully they'll continue to be cooperative

I'll send \$3 again.

usage:

```
{
  "input_tokens": 2072,
  "output_tokens": 533,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$2.22 sent, it became \$6.66, returned \$3.33, visible payoff \$3.33.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.31 returned, visible payoff \$5.3100000000000005.
- Round 5 against you, as RECEIVER: saw \$3.72 sent, it became \$11.16, returned \$6.0, visible payoff \$5.16.

Your total visible earnings across all rounds are \$27.67.

This round, you are the RECEIVER against your current co-player. They sent you \$3.23, so you received \$9.69. How much do you return? (0-9.69)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant
content:

Looking at our continued interaction:

****Round 5 analysis:****

- They saw I sent \$3.72, received \$11.16
- They returned a very generous \$6.00 (53.8%)

****Current situation:****

- They sent \$3.23, I received \$9.69
- They're maintaining strong cooperation despite noise

****My decision:****

I should match their generosity. Returning about 53% would be:

- 53% of \$9.69 ? \$5.14

This ensures:

- They profit well (\$5.14 back vs \$3.23 sent = +\$1.91 net)
- I also benefit (\$4.55 kept)
- We maintain mutual cooperation

```
'''json
{"return": 5.15}
'''
```

reasoning:

Looking at the history:

****Round 5 analysis:****

- They saw I sent \$3.72 (noise from my actual \$3), received \$11.16
- They returned \$6.00, which is about 53.8% of what they received
- Very generous and cooperative again!

****Current situation:****

- They sent me \$3.23, I received \$9.69
- They've been consistently generous, especially in Round 5 where they returned \$6.00 from \$11.16

****My decision:****

They're showing strong cooperation. I should reciprocate at a similar level:

- Around 53-54% of \$9.69 would be approximately \$5.15-\$5.23

Let me calculate:

- 53% of \$9.69 = \$5.14
- 54% of \$9.69 = \$5.23

I'll return around \$5.15 to match their cooperative spirit and ensure they profit well from sending to me.

This gives them: \$5.15 back vs \$3.23 sent = +\$1.92 profit

And I keep: \$9.69 - \$5.15 = \$4.54

Both profit nicely.

usage:

```
{
  "input_tokens": 2482,
  "output_tokens": 468,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.31 returned, visible payoff \$5.3100000000000005.
- Round 5 against you, as RECEIVER: saw \$3.72 sent, it became \$11.16, returned \$6.0, visible payoff \$5.16.
- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.05 returned, visible payoff \$8.05.

Your total visible earnings across all rounds are \$32.21.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at our interaction pattern:

****Recent trends:****

- Round 5: I sent \$3, they returned a generous \$6.00 from \$11.16
- Round 6: They sent \$3, I returned \$5.15, they saw \$6.05 (noise helped!)
- They're consistently sending \$3 and showing strong cooperation

****Analysis:****

- They saw my returns improve from their perspective (\$3.31 ? \$6.05)
- They responded very generously in Round 5 (\$6.00 return)
- We've established a stable, mutually beneficial pattern

****My decision:****

Continue with \$3 to:

- Maintain the cooperative equilibrium we've built
- Show consistency and reliability
- Keep generating mutual gains

```
'''json
{"send": 3}
'''
```

Reasoning:

Looking at the history:

- **Round 4:**** They sent \$3.00, I returned \$4.20, they saw \$3.31 (noise made it look lower)
- **Round 5:**** I sent \$3.00, they received ~\$11.16, returned \$6.00 (very generous)
- **Round 6:**** They sent \$3.00, I returned \$5.15, they saw \$6.05 (noise made it look higher!)

****Analysis:****

- This co-player has been consistently cooperative
- They've been sending \$3.00 regularly
- In Round 5 they returned a very generous \$6.00 from ~\$11.16
- In Round 6, the noise actually worked in my favor - they saw \$6.05 when I returned \$5.15, which looks very generous
- They're seeing improved returns from me (from their perspective: \$3.31 ? \$6.05)

****Strategy:****

- They're maintaining trust and cooperation
- I should continue sending \$3 to maintain this productive relationship
- The pattern of mutual cooperation is working well for both of us

I'll send \$3 again.

```
Usage: {
  "input_tokens": 2885,
  "output_tokens": 455,
  "reasoning_tokens": 0
}
```

GAME - Agent_8 - Receiver - dyad_4

Timestamp: 2026-08-12T08:15:44

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_7",
  "pairing": {
    "agents": [
      "Agent_7",
      "Agent_8"
    ],
    "investor": "Agent_7",
    "roles": {
      "Agent_7": "investor",
      "Agent_8": "trustee"
    },
    "trustee": "Agent_8"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 7,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.17, that is 63.4% of their total endowment. You received \$9.51. How much do you return to the sender? (0-9.51)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending 63.4% of their endowment (\$3.17)
- I received \$9.51 after the 3x multiplier
- The sender currently has $\$5 - \$3.17 = \$1.83$

The sender demonstrated considerable trust, so I want to reciprocate appropriately. If I return \$5.00:

- The sender would end with: $\$1.83 + \$5.00 = \$6.83$ (a profit of \$1.83)
- I would keep: $\$9.51 - \$5.00 = \$4.51$

This rewards their trust with a solid return while allowing us both to benefit from the co operation.

`{"return": 5.00}`

reasoning:

Let me think about this carefully.

I'm the RECEIVER in Round 1.

- The sender sent \$3.17 (which is 63.4% of their \$5 endowment)
- This was multiplied by 3x, so I received: $\$3.17 \times 3 = \9.51 ?
- I need to decide how much to return (between \$0 and \$9.51)

Key considerations:

1. The sender showed considerable trust by sending 63.4% of their endowment
2. If I return nothing, the sender ends up with: $\$5 - \$3.17 = \$1.83$, while I keep \$9.51
3. If I return a fair/reciprocal amount, what would that be?

A few approaches to consider:

- Equal split of total: Total is \$5 (sender's initial) + \$9.51 (what I received) = \$14.51. Equal split would be ~\$7.26 each. The sender currently has \$1.83, so I'd return $\$7.26 - \$1.83 = \$5.43$
- Return what maintains sender's initial investment value: Sender invested \$3.17. If I return \$3.17, sender gets back to \$5 total, and I keep \$6.34
- Return proportionally to what was sent: The sender sent 63.4% of their endowment. I could return a similar proportion of what I received: 63.4% of \$9.51 = ~\$6.03
- Social norm of reciprocity: Return enough to reward trust but keep some gains

Given that:

- The sender showed significant trust (63.4%)
- This is a multi-round game with different partners
- Building reputation for reciprocity could be valuable
- Note about communication noise suggests amounts might not be exact

I think a reasonable return would be around \$4-5, which:

- Acknowledges their generosity without giving away all my gains
- Leaves them better off than their starting position
- Demonstrates trustworthiness that could matter in future interactions with other partners
- Strikes a balance between reciprocity and self-interest

I'll return \$4.75. This gives the sender \$6.58 total (\$1.83 remaining plus my return), which meaningfully rewards their trust while I keep \$4.76....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1001,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.37 returned, visible payoff \$7.37.

Your total visible earnings across all rounds are \$4.51.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the history:

My co-player sent \$3.0 in Round 1 (60% of their endowment), showing significant trust. They saw \$5.37 returned, which gave them a good outcome.

Since they demonstrated trust and cooperation, I should reciprocate similarly. Sending around the same amount they sent (\$3.0) would be fair and maintain the cooperative relationship.

```
{"send": 3.0}
```

reasoning:

Let me analyze this situation:

1. In Round 1, I was the RECEIVER and the sender sent me \$3.17 (according to my view), which became \$9.51 for me. I returned \$5.00.
2. However, from my current co-player's perspective (shown in "Your current co-player's last 3 game(s)"):
 - They sent \$3.0 (not \$3.17 - communication noise!)
 - It became \$9.0 (not \$9.51 - more noise!)
 - They saw \$5.37 returned (not \$5.00 - even more noise!)
 - Their visible payoff was \$7.37

So there's noise in the communication. What they experienced:

- Started with \$5
- Sent \$3.0
- Had \$2.0 left
- Saw \$5.37 returned
- Ended with \$2.0 + \$5.37 = \$7.37

From their perspective, I (as the receiver in Round 1) returned about 60% of what I received (\$5.37 out of \$9.0).

3. Now I'm the SENDER in Round 2 against the same co-player.

My co-player showed trust by sending \$3.0 (from their perspective) and saw a decent return ratio. They might be willing to reciprocate trust.

I should consider:

- They sent about 60% of their endowment in Round 1
- They saw about 60% returned
- This suggests they're moderately trusting and cooperative

I should send a similar amount to show I'm also trustworthy and cooperative. Sending around \$3 would be consistent with their behavior and maintain the cooperative relationship.

Let me send \$3.00 to match their earlier send amount.

usage:

```
{
  "input_tokens": 837,
  "output_tokens": 523,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.37 returned, visible p

ayoff \$7.37.

- Round 2 against you, as RECEIVER: saw \$2.35 sent, it became \$7.050000000000001, returned \$3.75, visible payoff \$3.3000000000000007.
Your total visible earnings across all rounds are \$10.219999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$2.22, so you received \$6.66. How much do you return? (0-6.66)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the situation:

Current round:

- Sender sent \$2.22 (44.4% of their endowment)
- I received \$6.66
- Sender currently has: \$5 - \$2.22 = \$2.78

History with this co-player:

- Round 1: They sent \$3.0, I returned money (they saw \$5.37)
- Round 2: They saw less sent from me due to noise (\$2.35 vs my actual \$3.0), returned \$3.75

They've reduced their sending, possibly due to noise affecting what they saw in Round 2. So far, they're showing some trust by sending \$2.22. I should reciprocate fairly.

Returning approximately 50% of what I received (\$3.33) would:

- Give sender: \$2.78 + \$3.33 = \$6.11 (profit of \$1.11)
- I keep: \$3.33

This maintains cooperation while being proportional to their send amount.

{"return": 3.33}

reasoning:

Let me analyze the situation:

Current round:

- I'm the RECEIVER
- Sender sent \$2.22 (44.4% of their endowment)
- I received \$6.66 after 3x multiplier
- Sender currently has: \$5 - \$2.22 = \$2.78

Looking at history with this co-player:

- Round 1: They sent \$3.0 as SENDER, saw \$5.37 returned (I actually sent \$5.00, but due to noise they saw \$5.37)
- Round 2: I sent \$3.0 as SENDER (but they saw \$2.35 due to noise), they returned \$3.75

In Round 2, I sent \$3.0 which became \$9.0, and they returned \$3.75. That's about 42% return rate.

Now in Round 3:

- They sent \$2.22, which is less than what they sent in Round 1 (\$3.0)
- This might be because they saw only \$2.35 sent in Round 2 (due to noise) instead of my actual \$3.0

I should consider:

1. They're still showing some trust by sending \$2.22
2. They may have reduced their sending because of perceived lower trust from me (noise issue)
3. I should maintain cooperation

If I return around \$3.50-\$4.00:

- Return \$3.50: Sender gets \$2.78 + \$3.50 = \$6.28 (profit of \$1.28), I keep \$3.16
- Return \$4.00: Sender gets \$2.78 + \$4.00 = \$6.78 (profit of \$1.78), I keep \$2.66

Given they sent less this round but still showed trust, I'll return a reasonable amount that rewards their trust.

I'm noticing a pattern in the return rates across rounds. In Round 1, I returned about 53% of what I received, while in Round 2 they returned roughly 42%. Since they've sent less this time-likely reacting to the noise that made my previous send appear smaller-I should

ld aim for around 50% to maintain fairness and signal...

usage:

```
{
  "input_tokens": 1135,
  "output_tokens": 909,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.37 returned, visible payoff \$7.37.
- Round 2 against you, as RECEIVER: saw \$2.35 sent, it became \$7.050000000000001, returned \$3.75, visible payoff \$3.3000000000000007.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.3 returned, visible payoff \$6.3.

Your total visible earnings across all rounds are \$13.549999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at the history with my current co-player:

- Round 1: They sent \$3.0 (strong trust)
- Round 2: I sent, they returned \$3.75 (cooperative)
- Round 3: They sent \$3.0 again (consistent trust)

My co-player has shown consistent cooperation, sending \$3.0 both times they were the sender. They also returned a fair amount when they were receiver. I should reciprocate this consistent trust.

```
{"send": 3.0}
```

reasoning:

Let me analyze the visible history with my current co-player:

Round 1: They were SENDER, sent \$3.0 (60%), I returned something, they saw \$5.37 returned
Round 2: They were RECEIVER, they saw \$2.35 sent (but I actually sent \$3.0 - noise!), they returned \$3.75
Round 3: They were SENDER, sent \$3.0 (60%), they saw \$4.3 returned (but I actually returned \$3.33 - noise showing higher!)

So my co-player has:

- Sent \$3.0 twice when they were sender (consistent trust)
- Returned \$3.75 when they received \$7.05

They're showing good cooperation. In Round 3, they saw a higher return (\$4.3) than I actually sent (\$3.33), which might encourage them.

I should continue cooperating. Sending \$3.0 again would be consistent and fair.

My total visible earnings: \$13.55

- Round 1 as RECEIVER: kept \$4.51
- Round 2 as SENDER: sent \$3.0, saw return, kept something
- Round 3 as RECEIVER: kept \$3.33

Actually, let me recalculate:

Round 1: I was RECEIVER, got \$9.51, returned \$5.00, kept \$4.51
Round 2: I was SENDER, sent \$3.0, they returned \$3.75, I kept \$2 + \$3.75 = \$5.75
Round 3: I was RECEIVER, got \$6.66, returned \$3.33, kept \$3.33

Total: \$4.51 + \$5.75 + \$3.33 = \$13.59 ? \$13.55 (matches with rounding)

My co-player has been consistent in sending \$3.0 when they're sender. ...

usage:

```
{
```

```
"input_tokens": 1610,  
"output_tokens": 585,  
"reasoning_tokens": 0  
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.35 sent, it became \$7.050000000000001, returned \$3.75, visible payoff \$3.3000000000000007.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.3 returned, visible payoff \$6.3.
- Round 4 against you, as RECEIVER: saw \$2.62 sent, it became \$7.86, returned \$4.2, visible payoff \$3.66.

Your total visible earnings across all rounds are \$18.86.

This round, you are the RECEIVER against your current co-player. They sent you \$3.72, so you received \$11.16. How much do you return? (0-11.16)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at the current situation:

- Sender sent \$3.72 (74.4% of their endowment - very high trust!)
- I received \$11.16
- Sender currently has: \$5 - \$3.72 = \$1.28

History shows this co-player has been consistently cooperative and trusting. Now they've increased their trust even more, sending nearly 75% of their endowment. I should reciprocate this strong trust generously.

Returning \$6.0:

- Sender would get: \$1.28 + \$6.0 = \$7.28 (profit of \$2.28)
- I keep: \$5.16

This rewards their exceptional trust while both of us profit from the cooperation.

```
{"return": 6.0}
```

reasoning:

Looking at the current situation:

- Sender sent \$3.72 (74.4% of their endowment - very high trust!)
- I received \$11.16 after 3x multiplier
- Sender currently has: \$5 - \$3.72 = \$1.28

Looking at history with this co-player:

- They've been consistently cooperative
- Round 3: They sent \$3.0
- Round 4: They saw me send \$2.62 (due to noise, I actually sent \$3.0), returned \$4.2
- Now Round 5: They sent \$3.72 (even more!)

This is a very trusting move - 74.4% of endowment. I should reciprocate generously.

If I return \$6.0:

- Sender gets: \$1.28 + \$6.0 = \$7.28 (profit of \$2.28)
- I keep: \$11.16 - \$6.0 = \$5.16

This rewards their high trust appropriately. Both of us profit well.

usage:

```
{  
  "input_tokens": 1967,  
  "output_tokens": 440,  
  "reasoning_tokens": 0  
}
```

Message 12 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.3 returned, visible payoff \$6.3.
- Round 4 against you, as RECEIVER: saw \$2.62 sent, it became \$7.86, returned \$4.2, visible payoff \$3.66.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.04 returned, visible payoff \$7.04.

Your total visible earnings across all rounds are \$24.02.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant
content:

Looking at the history with my current co-player:

- Round 3: They sent \$3.0 (consistent trust)
- Round 4: They returned \$4.2 when they received \$7.86 (generous)
- Round 5: They sent \$3.0 again (consistently cooperative)

This co-player has been very reliable and cooperative throughout our interactions. They consistently send \$3.0 when they're the sender and return fair amounts when they're the receiver. I should continue matching this level of trust and cooperation.

{"send": 3.0}

reasoning:

Looking at the visible history with my current co-player:

- Round 3: They sent \$3.0 as sender
- Round 4: They returned \$4.2 as receiver (I sent \$2.62 visible to them)
- Round 5: They sent \$3.0 as sender

This co-player has been consistently cooperative, sending \$3.0 whenever they're the sender and returning fair amounts when they're the receiver (\$4.2 when they received \$7.86).

I've been sending \$3.0 consistently when I'm the sender. This has been working well for both of us. I should continue this pattern of cooperation.

{"send": 3.0}

usage:

```
{
  "input_tokens": 2364,
  "output_tokens": 295,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$2.62 sent, it became \$7.86, returned \$4.2, visible payoff \$3.66.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.04 returned, visible payoff \$7.04.
- Round 6 against you, as RECEIVER: saw \$3.23 sent, it became \$9.69, returned \$5.15, visible payoff \$4.539999999999999.

Your total visible earnings across all rounds are \$32.07.

This round, you are the RECEIVER against your current co-player. They sent you \$2.85, so you received \$8.55. How much do you return? (0-8.55)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the current situation:

- Sender sent \$2.85 (57% of their endowment - good trust)
- I received \$8.55
- Sender currently has: \$5 - \$2.85 = \$2.15

History shows this co-player has been consistently cooperative:

- Round 4: Returned \$4.2 generously
- Round 5: Sent \$3.0 showing trust
- Round 6: Returned \$5.15 very generously (over 50% of what they received)

They've demonstrated strong cooperation throughout. Returning about 50% of what I received would be fair:

Returning \$4.5:

- Sender gets: $\$2.15 + \$4.5 = \$6.65$ (profit of \$1.80)
- I keep: \$4.05

This maintains our cooperative relationship.

```
{"return": 4.5}
```

Reasoning:

Looking at the current situation:

- Sender sent \$2.85 (57% of their endowment)
- I received \$8.55
- Sender currently has: $\$5 - \$2.85 = \$2.15$

History with this co-player:

- Round 4: They returned \$4.2 when receiving \$7.86 (generous)
- Round 5: They sent \$3.0, saw \$5.04 returned
- Round 6: They returned \$5.15 when receiving \$9.69 (very generous!)

This co-player has been consistently cooperative and generous when in the receiver role. They sent \$2.85 this round, which is still a significant amount showing trust.

If I return around \$4.5-5.0:

- Returning \$4.5: Sender gets $\$2.15 + \$4.5 = \$6.65$ (profit of \$1.65), I keep \$4.05
- Returning \$5.0: Sender gets $\$2.15 + \$5.0 = \$7.15$ (profit of \$2.15), I keep \$3.55

Given their consistent generosity, I should return around \$4.5-5.0. Let me go with \$4.75 as a middle ground.

Actually, looking at proportions - they've been returning around 50-53% of what they receive. I should reciprocate similarly.

50% of \$8.55 = \$4.275

Let me return \$4.5 to be fair and generous.

```
{"return": 4.5}
```

Usage: {

```
  "input_tokens": 2727,
  "output_tokens": 589,
  "reasoning_tokens": 0
```

}

ROUND 8

Roles: Agent_1=Receiver, Agent_2=Sender, Agent_3=Receiver, Agent_4=Sender, Agent_5=Receiver, Agent_6=Sender, Agent_7=Receiver, Agent_8=Sender

Pairings:

```
dyad_1: Agent_2=Sender, Agent_1=Receiver
dyad_2: Agent_4=Sender, Agent_3=Receiver
dyad_3: Agent_6=Sender, Agent_5=Receiver
dyad_4: Agent_8=Sender, Agent_7=Receiver
```

Parsed game decisions:

```
dyad_1: Agent_2 sent $4.5; received $13.5; Agent_1 returned $5.5; payoffs Agent_2=$6, Agent_1=$8;
noise/visible: sent visible=$3.54, received visible=$10.62, returned visible=$5.89
  Action validation: {"returned": {"amount": 5.5, "clamped": false, "max": 13.5, "min": 0, "raw": 5.5}, "sent": {"amount": 4.5, "clamped": false, "max": 5, "min": 0, "raw": 4.5}}
dyad_2: Agent_4 sent $3; received $9; Agent_3 returned $3; payoffs Agent_4=$5, Agent_3=$6;
noise/visible: sent visible=$2.14, received visible=$6.42, returned visible=$3.7
  Action validation: {"returned": {"amount": 3.0, "clamped": false, "max": 9.0, "min": 0, "raw": 3.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_3: Agent_6 sent $3; received $9; Agent_5 returned $3.6; payoffs Agent_6=$5.6, Agent_5=$5.4;
noise/visible: sent visible=$2.39, received visible=$7.17, returned visible=$2.95
```

```

    Action validation: {"returned": {"amount": 3.6, "clamped": false, "max": 9.0, "min": 0, "raw":
    3.6}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_8 sent $3; received $9; Agent_7 returned $6.15; payoffs Agent_8=$8.15,
Agent_7=$2.85; noise/visible: sent visible=$3.87, received visible=$11.61, returned visible=$6.03
    Action validation: {"returned": {"amount": 6.15, "clamped": false, "max": 9.0, "min": 0, "raw":
    6.15}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
Balances after round: {
  "Agent_1": 54.2,
  "Agent_2": 49.8,
  "Agent_3": 46.5,
  "Agent_4": 41.5,
  "Agent_5": 45.0,
  "Agent_6": 43.0,
  "Agent_7": 43.58,
  "Agent_8": 44.42
}
Visible balances after round: {
  "Agent_1": 48.410000000000001,
  "Agent_2": 52.899999999999999,
  "Agent_3": 39.71,
  "Agent_4": 43.080000000000005,
  "Agent_5": 46.160000000000004,
  "Agent_6": 44.55,
  "Agent_7": 43.74,
  "Agent_8": 44.150000000000006
}

```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T08:15:51

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 8,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN

- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.52 returned, visible payoff \$7.52.

Your total visible earnings across all rounds are \$5.8900000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at the situation:

From the visible history, my co-player:

- Sent \$3.0 in Round 1 (60% of their endowment)
- Saw \$5.52 returned (61.3% of the \$9.0)
- Made a profit of \$2.52

This shows they were cooperative and returned a substantial amount. To reciprocate their trust and maintain cooperation, I should send a generous amount.

I'll send \$3.50 (70% of my endowment), showing strong trust and willingness to cooperate.

{"send": 3.50}

reasoning:

Let me analyze this situation:

Round 1 (from my perspective):

- I was the RECEIVER
- The sender sent \$3.63 (72.6% of endowment)
- I received \$10.89
- I returned \$5.00
- My payoff: \$10.89 - \$5.00 = \$5.89

- Sender's payoff: $\$5 - \$3.63 + \$5.00 = \6.37

But now I see the "visible history" which shows a different perspective:

- My co-player (who was the sender in Round 1) saw that they sent \$3.0
- It became \$9.0 (not \$10.89 as I received)
- They saw \$5.52 returned (not \$5.00 as I sent)
- Their visible payoff was \$7.52

This confirms there is communication noise in the game as mentioned in the instructions.

What actually happened vs what they saw:

- They actually sent \$3.63, but saw they sent \$3.0
- I received \$10.89, they saw it became \$9.0
- I returned \$5.00, they saw \$5.52 returned
- Their actual payoff was \$6.37, but they saw \$7.52

Now in Round 2, I'm the SENDER and they are the RECEIVER.

My co-player returned a substantial amount in Round 1 (they saw they returned \$5.52 out of \$9.0 received, which is 61.3%). This shows they are cooperative.

Given that:

1. They showed trust by being generous in Round 1
2. They appeared to return a fair amount (from their perspective, 61.3% of what they received)
3. I want to maintain cooperation and reciprocate their trust

I should send a generous amount. ...

usage:

```
{
  "input_tokens": 874,
  "output_tokens": 716,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.52 returned, visible payoff \$7.52.
- Round 2 against you, as RECEIVER: saw \$3.14 sent, it became \$9.42, returned \$4.5, visible payoff \$4.92.

Your total visible earnings across all rounds are \$12.45.

This round, you are the RECEIVER against your current co-player. They sent you \$3.64, so you received \$10.92. How much do you return? (0-10.92)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the visible history:

Round 1: My co-player sent \$3.0 (60%) and saw \$5.52 returned

Round 2: They returned \$4.5 when they received \$9.42 (47.8%)

Round 3: They sent \$3.64 (72.8% - even more generous!)

My co-player has been consistently cooperative and just showed increased trust by sending an even larger percentage. I should reciprocate fairly.

Returning \$5.50:

- Sender gets: $\$5 - \$3.64 + \$5.50 = \6.86 (profit of \$1.86)
- I get: $\$10.92 - \$5.50 = \$5.42$ (profit of \$5.42)

This ~50% return rate rewards their trust while ensuring mutual benefit.

{"return": 5.50}

reasoning:

Let me analyze the visible history:

Round 1: My co-player was SENDER
- They sent \$3.0 (60% of endowment)
- It became \$9.0
- They saw \$5.52 returned (61.3% return rate)
- Their visible payoff: \$7.52

Round 2: My co-player was RECEIVER
- They saw \$3.14 sent (I actually sent \$3.50 but there's noise)
- It became \$9.42
- They returned \$4.5 (47.8% of what they received)
- Their visible payoff: \$4.92

Now Round 3: I am RECEIVER
- They sent \$3.64 (72.8% of their endowment - very generous!)
- I received \$10.92
- I need to decide how much to return

Looking at the pattern:
- In Round 1, they sent 60% and I returned about 61% (assuming noise in what they saw)
- In Round 2, they received and returned 47.8%
- In Round 3, they sent 72.8% - showing even more trust

The co-player has been consistently cooperative. They sent a high amount both times they were sender (60% and 72.8%). When they were receiver, they returned a reasonable amount (47.8%).

To maintain reciprocity and reward their increased trust, I should return a fair amount.

If I return \$5.50:
- Sender ends with: $\$5 - \$3.64 + \$5.50 = \6.86 (gain of \$1.86)
- I end with: $\$10.92 - \$5.50 = \$5.42$ (gain of \$5.42)

This is about 50.4% return rate, which is reasonable and shows I'm reciprocating their trust while both of us profit.

...

usage:

```
{
  "input_tokens": 1200,
  "output_tokens": 712,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.52 returned, visible payoff \$7.52.
- Round 2 against you, as RECEIVER: saw \$3.14 sent, it became \$9.42, returned \$4.5, visible payoff \$4.92.
- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$4.66 returned, visible payoff \$5.66.

Your total visible earnings across all rounds are \$17.869999999999997.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the visible history:

Round 1: Co-player sent \$3.0 (60%), saw \$5.52 returned

Round 2: Co-player returned \$4.5 (47.8%)

Round 3: Co-player sent \$4.0 (80% - very generous!), saw \$4.66 returned

My co-player has been consistently cooperative, even increasing their trust in Round 3 by sending 80% of their endowment. While they saw a lower return percentage in Round 3, they've maintained a cooperative pattern overall.

I should reciprocate their trust and continue the cooperative relationship.

```
{"send": 3.50}
```

reasoning:

Let me analyze the visible history:

Round 1: Co-player sent \$3.0 (60%), saw \$5.52 returned (61.3% of \$9.0)

Round 2: Co-player returned \$4.5 when they received \$9.42 (47.8%)

Round 3: Co-player sent \$4.0 (80%), saw \$4.66 returned (38.8% of \$12.0)

This is concerning. In Round 3, my co-player sent even more (\$4.0, or 80%), but they only saw \$4.66 returned, which is only 38.8% of the \$12.0 they sent. That's a lower return rate than before.

Wait, let me check what I actually sent and returned:

- Round 2: I sent \$3.50 according to my response

- Round 3: I returned \$5.50 according to my response

But the visible history shows:

- Round 2: They saw \$3.14 sent (I sent \$3.50, but due to noise they saw \$3.14)

- Round 3: They saw \$4.66 returned (I returned \$5.50, but due to noise they saw \$4.66)

So there's communication noise! This is important.

From my co-player's perspective:

- They've been very generous (sending 60%, 80%)

- But the returns they're seeing are declining in percentage terms (61.3%, then 38.8%)

- They might be feeling like their generosity isn't being reciprocated

However, I know I actually returned \$5.50 in Round 3, which was generous. The noise just made it look like \$4.66 to them.

Still, this relationship has been cooperative overall. My co-player has consistently sent substantial amounts. Even though they saw a lower return in Round 3, they've maintained trust.

...

usage:

```
{
  "input_tokens": 1626,
  "output_tokens": 638,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$3.14 sent, it became \$9.42, returned \$4.5, visible payoff \$4.92.

- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$4.66 returned, visible payoff \$5.66.

- Round 4 against you, as RECEIVER: saw \$3.21 sent, it became \$9.629999999999999, returned \$4.8, visible payoff \$4.829999999999999.

Your total visible earnings across all rounds are \$24.65.

This round, you are the RECEIVER against your current co-player. They sent you \$5, so you received \$15. How much do you return? (0-15)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at the visible history:

Round 2: Co-player returned \$4.5 (47.8%)

Round 3: Co-player sent \$4.0 (80%), saw modest return

Round 4: Co-player returned \$4.8 (49.8%)

Round 5: Co-player sent \$5.0 (100% - maximum possible trust!)

This is extraordinary! My co-player just sent their ENTIRE endowment, showing complete trust. They're keeping nothing and relying entirely on my reciprocation. I must reward this

exceptional trust generously.

Returning \$7.50:

- Sender gets: $\$5 - \$5 + \$7.50 = \7.50 (profit of \$2.50)
- I get: $\$15 - \$7.50 = \$7.50$ (profit of \$7.50)

This 50% return ensures we both profit nicely and maintains our strong cooperative relationship.

```
{"return": 7.50}
```

reasoning:

Looking at the visible history:

Round 2: Co-player returned \$4.5 when they received \$9.42 (47.8%)
Round 3: Co-player sent \$4.0 (80%), saw \$4.66 returned (38.8%)
Round 4: Co-player returned \$4.8 when they received \$9.63 (49.8%)
Round 5: Co-player sent \$5.0 (100%!!!)

This is remarkable - my co-player just sent their ENTIRE endowment (\$5), showing maximum trust! This is the ultimate act of cooperation in this game. They're completely relying on me to reciprocate fairly.

They gave up \$5, keeping \$0. If I return nothing, they end with \$0. This would be devastating to the relationship.

I need to reciprocate this tremendous trust generously. Let me think about fair returns:

If I return \$7.50:

- Sender gets: $\$5 - \$5 + \$7.50 = \7.50 (profit of \$2.50)
- I get: $\$15 - \$7.50 = \$7.50$ (profit of \$7.50)

This 50% return rate gives us equal final amounts and ensures they make a nice profit for their trust.

Actually, given they sent 100%, I could even be more generous. Let me consider \$8:

- Sender gets: $\$5 - \$5 + \$8 = \8 (profit of \$3)
- I get: $\$15 - \$8 = \$7$ (profit of \$7)

This is still good for both of us and rewards their exceptional trust even more.

I think \$7.50 is fair - it's a 50% return rate, ensures both of us profit nicely, and maintains the cooperative relationship.

usage:

```
{
  "input_tokens": 2010,
  "output_tokens": 660,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$4.66 returned, visible payoff \$5.66.
- Round 4 against you, as RECEIVER: saw \$3.21 sent, it became \$9.629999999999999, returned \$4.8, visible payoff \$4.829999999999999.
- Round 5 against you, as SENDER: sent \$4.5, it became \$13.5, saw \$7.07 returned, visible payoff \$7.57.

Your total visible earnings across all rounds are \$32.15.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at the visible history:

Round 3: Co-player sent \$4.0 (80%), saw modest return

Round 4: Co-player returned \$4.8 (~50%)
Round 5: Co-player sent \$4.5 (90%), saw \$7.07 returned (good return!)

My co-player has been very generous, consistently sending 80-90% of their endowment. They've maintained strong cooperation and saw good returns in Round 5. I should reciprocate their trust.

```
{"send": 4.00}
```

reasoning:

Looking at the visible history:

Round 3: Co-player sent \$4.0 (80%), saw \$4.66 returned
Round 4: Co-player returned \$4.8 (~49.8%)
Round 5: Co-player sent \$4.5 (90%), saw \$7.07 returned

My co-player has been very generous and cooperative:

- They've sent high percentages (80%, 90%)
- When receiving, they returned reasonable amounts (~50%)
- They saw good returns in Round 5 (\$7.07)

This is a strong cooperative relationship. I should continue to send generous amounts to maintain this cooperation. Given their pattern of sending 80-100%, I should reciprocate with a similar level of trust.

I'll send \$4.00 (80% of my endowment) to match their generosity and maintain cooperation.

usage:

```
{  
  "input_tokens": 2464,  
  "output_tokens": 332,  
  "reasoning_tokens": 0  
}
```

Message 12 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$3.21 sent, it became \$9.629999999999999, returned \$4.8, visible payoff \$4.829999999999999.
- Round 5 against you, as SENDER: sent \$4.5, it became \$13.5, saw \$7.07 returned, visible payoff \$7.57.
- Round 6 against you, as RECEIVER: saw \$3.9 sent, it became \$11.7, returned \$6.0, visible payoff \$5.699999999999999.

Your total visible earnings across all rounds are \$39.949999999999996.

This round, you are the RECEIVER against your current co-player. They sent you \$4.52, so you received \$13.559999999999999. How much do you return? (0-13.559999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at the visible history:

Round 4: Co-player returned \$4.8 (~50%)
Round 5: Co-player sent \$4.5 (90%), saw good return
Round 6: Co-player returned \$6.0 (~51%)
Round 7: Co-player sent \$4.52 (90.4% - very high trust!)

My co-player continues to demonstrate exceptional trust, sending over 90% of their endowment. They've been consistently cooperative both as sender and receiver. I should maintain this strong reciprocal relationship.

Returning \$7.00:

- Sender gets: \$5 - \$4.52 + \$7.00 = \$7.48 (profit of \$2.48)
- I get: \$13.56 - \$7.00 = \$6.56 (profit of \$6.56)

This ~51.6% return rewards their trust appropriately.

```
{"return": 7.00}
```

reasoning:

Looking at the visible history:

Round 4: Co-player returned \$4.8 (~50%)
Round 5: Co-player sent \$4.5 (90%), saw \$7.07 returned
Round 6: Co-player returned \$6.0 (~51.3%)
Round 7: Co-player sent \$4.52 (90.4% - very high trust!)

My co-player continues to show very high trust, sending 90.4% of their endowment. They've been consistently cooperative as both sender and receiver.

I received \$13.56 after the 3x multiplier.

If I return \$6.75:

- Sender gets: $\$5 - \$4.52 + \$6.75 = \7.23 (profit of \$2.23)
- I get: $\$13.56 - \$6.75 = \$6.81$ (profit of \$6.81)

This is about 50% return rate, which is fair and rewards their trust.

Let me check if a slightly different amount makes sense:

If I return \$7.00:

- Sender gets: $\$5 - \$4.52 + \$7.00 = \7.48 (profit of \$2.48)
- I get: $\$13.56 - \$7.00 = \$6.56$ (profit of \$6.56)

This is about 51.6% return rate.

Given their high trust level (90.4%), I think \$7.00 is appropriate to maintain the strong cooperative relationship.

usage:

```
{
  "input_tokens": 2844,
  "output_tokens": 592,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as SENDER: sent \$4.5, it became \$13.5, saw \$7.07 returned, visible payoff \$7.57.
- Round 6 against you, as RECEIVER: saw \$3.9 sent, it became \$11.7, returned \$6.0, visible payoff \$5.699999999999999.
- Round 7 against you, as SENDER: sent \$5.0, it became \$15.0, saw \$7.09 returned, visible payoff \$7.09.

Your total visible earnings across all rounds are \$46.509999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the visible history:

Round 5: Co-player sent \$4.5 (90%), saw good return

Round 6: Co-player returned \$6.0 (~51%)

Round 7: Co-player sent \$5.0 (100% - MAXIMUM trust!), saw \$7.09 returned

My co-player has demonstrated exceptional trust and cooperation, sending 100% of their endowment in Round 7. They've been consistently generous as both sender and receiver. I should reciprocate this high level of trust.

```
{"send": 4.50}
```

Reasoning:

Looking at the visible history:

Round 5: Co-player sent \$4.5 (90%), saw \$7.07 returned (~52.4%)

Round 6: Co-player returned \$6.0 (~51.3%)

Round 7: Co-player sent \$5.0 (100% - maximum trust!), saw \$7.09 returned (~47.3%)

My co-player has been extremely generous and trusting:

- In Round 7, they sent their ENTIRE endowment (\$5.0), showing complete trust
- They've consistently sent 90-100% as sender
- As receiver, they've returned around 50%

This is a highly cooperative partner. They're demonstrating maximum trust. I should reciprocate by sending a generous amount.

Given their exceptional cooperation, I should send close to maximum as well. I'll send \$4.50 to show strong reciprocity.

```
Usage: {
  "input_tokens": 3025,
  "output_tokens": 346,
  "reasoning_tokens": 0
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T08:15:58

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 8,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.63 sent, it became \$10.89, returned \$5.0, visible payoff \$5.890000000000001.

Your total visible earnings across all rounds are \$7.52.

This round, you are the RECEIVER against your current co-player. They sent you \$3.14, so you received \$9.42. How much do you return? (0-9.42)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at the history:

- Round 1: My co-player saw \$3.63 sent, received \$10.89, and returned \$5.0 to me - showing cooperative behavior
- Round 2: They sent me \$3.14, which became \$9.42

They've demonstrated fairness by returning a reasonable amount in Round 1. I want to reciprocate this cooperative approach.

I'll return **\$4.50**, which:

- Gives them a profit (they sent ~\$3.14, get back \$4.50)
- Leaves me with a profit too (\$9.42 - \$4.50 = \$4.92)
- Maintains the cooperative relationship

```
'''json
{"return": 4.50}
'''
```

reasoning:

Let me understand what happened:

Round 1: I sent \$3, but my co-player saw \$3.63 (due to communication noise). This became \$10.89 for them. They returned \$5.0.

My actual payoff in Round 1:

- I had \$5
- I sent \$3
- I received back \$5
- So I kept: \$5 - \$3 + \$5 = \$7

But the visible payoff shows \$7.52, which suggests there might be some noise in the amounts.

Now in Round 2:

- I'm the RECEIVER
- My co-player sent me some amount that appears as \$3.14 to me
- This became \$9.42 (which is \$3.14 × 3)
- I need to decide how much to return

Looking at their behavior: In Round 1, they saw \$3.63 sent, which became \$10.89, and they returned \$5.0. That's about 46% of what they received, or about 1.38x what they saw as sent.

This seems like a fair/cooperative player. They returned a reasonable amount.

Now I have \$9.42. What should I return?

Given that:

1. They sent a decent amount (\$3.14, which is about 63% of their \$5)
2. They were fair in the previous round, returning \$5 when they received \$10.89
3. I want to encourage continued cooperation

I should return a fair amount. Some options:

- Return about half of what I received: \$4.71
- Return what they sent plus a bit extra: around \$4-5
- Match their return rate from last round (46% of received): about \$4.33

I think returning around \$4.50-\$5.00 would be fair and encourage continued cooperation. Let me return \$4.50 as a fair split that gives them back close to what they sent plus some profit, while I also profit.

If I return \$4.50, they end up with \$6.36 while I keep \$4.92-both of us gain from the exchange. Returning \$5.00 would mirror...

usage:

```
{
  "input_tokens": 768,
  "output_tokens": 820,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.63 sent, it became \$10.89, returned \$5.0, visible payoff \$5.8900000000000001.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.06 returned, visible payoff \$6.56.

Your total visible earnings across all rounds are \$12.44.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at our history together:

Round 1: I sent (they saw \$3.63), they returned \$5.0 - cooperative

Round 2: They sent \$3.5 (I saw \$3.14), I returned (they saw \$5.06) - both cooperative

We've established a good cooperative pattern. They've been generous as both sender and receiver, so I want to continue building this trust.

I'll send **\$4** to show strong trust and maintain our mutually beneficial relationship.

```
'''json
{"send": 4}
'''
```

reasoning:

Let me analyze the history with my current co-player:

Round 1: I was SENDER

- I sent \$3 (but they saw \$3.63 due to noise)
- They received \$10.89
- They returned \$5.0
- This was cooperative behavior

Round 2: I was RECEIVER

- They sent \$3.5 (but I saw \$3.14 due to noise)
- I received \$9.42
- I returned \$4.50 (but they saw \$5.06 due to noise)
- They had a visible payoff of \$6.56

So far, this co-player has been quite cooperative:

- As receiver in Round 1: returned \$5.0 from \$10.89 (about 46%)
- As sender in Round 2: sent \$3.5 out of \$5 (70%)

I should continue this cooperative relationship. In Round 1, I sent \$3. In Round 2, they sent \$3.5. I should probably send a similar or slightly higher amount to maintain cooperation.

I'll send \$3.5 or \$4 to show continued trust and cooperation.

Given that they returned fairly and sent generously, I'll send \$4 to demonstrate strong trust and cooperation.

usage:

```
{
  "input_tokens": 1118,
  "output_tokens": 421,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.63 sent, it became \$10.89, returned \$5.0, visible payoff \$5.890000000000001.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.06 returned, visible payoff \$6.56.
- Round 3 against you, as RECEIVER: saw \$3.64 sent, it became \$10.92, returned \$5.5, visible payoff \$5.42.

Your total visible earnings across all rounds are \$18.1.

This round, you are the RECEIVER against your current co-player. They sent you \$3.21, so you received \$9.629999999999999. How much do you return? (0-9.629999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at our established pattern:

Round 1: They returned \$5.0 when receiving \$10.89

Round 2: I returned (they saw \$5.06) when I received \$9.42

Round 3: They returned \$5.5 when receiving \$10.92 - showing increased generosity

Round 4: They sent \$3.21, I received \$9.63

They've been consistently cooperative and even increased their returns. I want to reciprocate this positive trend.

I'll return **\$4.80**, which rewards their trust while maintaining mutual benefit.

```
'''json
{"return": 4.80}
'''
```

reasoning:

Looking at the history with my current co-player:

Round 1: They (as receiver) saw \$3.63 sent, received \$10.89, returned \$5.0

Round 2: They (as sender) sent \$3.5, received (saw \$5.06) back

Round 3: They (as receiver) saw \$3.64 sent, received \$10.92, returned \$5.5

Round 4: They (as sender) sent \$3.21, I received \$9.63

Our pattern:

- Round 1: I sent (they saw 3.63), they returned \$5.0
- Round 2: They sent 3.5 (I saw 3.14), I returned (they saw 5.06)
- Round 3: I sent (they saw 3.64), they returned \$5.5
- Round 4: They sent 3.21, I received \$9.63

They've been consistently cooperative and fair. In Round 3, they returned \$5.5, showing increased generosity.

For Round 4, they sent \$3.21, which became \$9.63 for me.

To maintain the cooperative relationship and be fair:

- They invested ~\$3.21
- I should return enough that they profit, while I also profit
- A return of around \$4.50-\$5.00 would be fair

I'll return \$4.80 to show continued cooperation and slightly increase the return to match their increased generosity in Round 3.

usage:

```
{
  "input_tokens": 1484,
  "output_tokens": 500,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.06 returned, visible payoff \$6.56.
- Round 3 against you, as RECEIVER: saw \$3.64 sent, it became \$10.92, returned \$5.5, visible payoff \$5.42.
- Round 4 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.28 returned, visible payoff \$6.78.

Your total visible earnings across all rounds are \$22.93.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at our continued cooperation:

- Round 2: They sent \$3.5, I returned (they saw \$5.28)
- Round 3: I sent (they saw \$3.64), they returned \$5.5 - very generous
- Round 4: They sent \$3.5, I returned (they saw \$5.28)

This co-player has been consistently trustworthy and generous. They reliably send around \$3.5 and return well when receiving. Our relationship is strongly cooperative.

I'll send **\$4.5** to show high trust and continue our mutually beneficial pattern.

```
'''json
{"send": 4.5}
'''
```

reasoning:

Looking at our history:

- Round 2: They sent \$3.5, I returned (they saw \$5.06)
- Round 3: I sent (they saw \$3.64), they returned \$5.5
- Round 4: They sent \$3.5, I returned (they saw \$5.28)
- Round 5: Now I'm sender

We've established a strong cooperative pattern. They've been consistently sending around \$3.5 and returning generously when receiver. I've been returning well too.

Given this trust and positive relationship, I should continue with a generous send. Last time I sent (Round 3), they returned \$5.5 on receiving \$10.92, which was very fair.

I'll send \$4 or even \$5 to show maximum trust. Given the strong cooperation, I'll go with \$4.5 as a strong signal of trust while keeping a small cushion.

usage:

```
{
  "input_tokens": 1846,
  "output_tokens": 366,
}
```

```

    "reasoning_tokens": 0
  }
Message 10 role=user
content:
  Round 6

  Visible history before this round:
  Your current co-player's last 3 game(s):
  - Round 3 against you, as RECEIVER: saw $3.64 sent, it became $10.92, returned $5.5, visible payoff $5.42.
  - Round 4 against you, as SENDER: sent $3.5, it became $10.5, saw $5.28 returned, visible payoff $6.78.
  - Round 5 against you, as RECEIVER: saw $5 sent, it became $15, returned $7.5, visible payoff $7.5.
  Your total visible earnings across all rounds are $30.5.

  This round, you are the RECEIVER against your current co-player. They sent you $3.9, so you received $11.7. How much do you return? (0-11.7)
  Respond exactly as JSON: {'return': <amount>}
Message 11 role=assistant
content:
  Looking at our strong cooperative history:
  - Round 3: They returned $5.5 from $10.92
  - Round 4: I returned (they saw $5.28)
  - Round 5: They returned $7.5 from $15 - exceptionally generous (50%)!
  - Round 6: They sent $3.9, I received $11.7

  They showed exceptional generosity in Round 5, returning exactly 50% when I sent a high amount. I want to match this level of cooperation.

  I'll return **$6.0**, which is approximately 51% - reciprocating their generous spirit.

  '''json
  {"return": 6.0}
  '''
reasoning:
  Looking at the history:
  - Round 3: They returned $5.5 when receiving $10.92
  - Round 4: I returned (they saw $5.28)
  - Round 5: They returned $7.5 when receiving $15 - very generous! They returned exactly 50%
  - Round 6: They sent $3.9, I received $11.7

  This co-player has been extremely trustworthy and generous. In Round 5, when I sent a high amount (they saw $5 which became $15), they returned $7.5, which is 50% - that's a very fair split.

  I should reciprocate this generosity. They sent $3.9, so I received $11.7. If I want to be similarly generous and fair:
  - They invested $3.9
  - I could return around $6 to give them a good profit while keeping a profit myself
  - Or I could match their 50% generosity and return $5.85

  Given their exceptional generosity in Round 5, I should be very generous too. I'll return $6.0 which gives them a profit of $2.1 ($6 - $3.9), and I keep $5.7 ($11.7 - $6).

  Actually, let me think about this more carefully. They returned $7.5 from $15, which is 50%. If I return 50% of $11.7, that would be $5.85. But rounding to $6 is cleaner and slightly more generous.
usage:
  {
    "input_tokens": 2223,
    "output_tokens": 516,
    "reasoning_tokens": 0
  }
Message 12 role=user
content:
  Round 7

  Visible history before this round:

```


Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.28 returned, visible payoff \$6.78.
- Round 5 against you, as RECEIVER: saw \$5 sent, it became \$15, returned \$7.5, visible payoff \$7.5.
- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.8 returned, visible payoff \$7.8.

Your total visible earnings across all rounds are \$36.2.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at our excellent track record:

- Round 4: They sent \$3.5, I returned (they saw \$5.28)
- Round 5: I sent (they saw \$5), they returned \$7.5 - 50% return!
- Round 6: They sent \$4.0 (increased!), I returned (they saw \$6.8)

This co-player has consistently shown high trustworthiness, even returning 50% when I sent my maximum in Round 5. They've also been increasing their own sending amounts. We have an excellent cooperative relationship.

I'll send **\$5** to show complete trust and maximize our mutual benefit.

```
'''json
{"send": 5}
'''
```

reasoning:

Looking at the pattern with this co-player:

- Round 4: They sent \$3.5, I returned (they saw \$5.28)
- Round 5: I sent (they saw \$5), they returned \$7.5 - very generous 50% return
- Round 6: They sent \$4.0, I returned (they saw \$6.8) - they've been increasing their sending amounts
- Round 7: Now I'm the sender

This co-player has been extremely trustworthy:

1. They've increased their sending amounts from \$3.5 to \$4.0
2. When they received, they returned 50% (\$7.5 from \$15)
3. When they sent, they saw good returns from me

We've built a very strong cooperative relationship. They've demonstrated high trust in me, and I've shown them I'm reliable. This is the time to maximize mutual benefit by sending the maximum amount.

I'll send \$5 to show complete trust and maximize our joint gains.

usage:

```
{
  "input_tokens": 2592,
  "output_tokens": 398,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as RECEIVER: saw \$5 sent, it became \$15, returned \$7.5, visible payoff \$7.5.
- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.8 returned, visible payoff \$7.8.
- Round 7 against you, as RECEIVER: saw \$4.52 sent, it became \$13.559999999999999, returned \$7.0, visible payoff \$6.559999999999999.

Your total visible earnings across all rounds are \$43.290000000000006.

This round, you are the RECEIVER against your current co-player. They sent you \$3.54, so you received \$10.620000000000001. How much do you return? (0-10.620000000000001)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at our continued cooperation:

- Round 5: They returned \$7.5 from \$15 (50%)
- Round 6: I returned (they saw \$6.8)
- Round 7: They returned \$7.0 from ~\$13.56 (~52%)
- Round 8: They sent \$3.54, I received \$10.62

They've maintained strong generosity throughout, consistently returning around 50% or more. I'll reciprocate with a generous return.

I'll return **\$5.50**, which maintains our cooperative pattern and ensures both of us profit.

```
'''json
{"return": 5.50}
'''
```

Reasoning:

Looking at our history:

- Round 5: They returned \$7.5 from \$15
- Round 6: I returned (they saw \$6.8) from \$12
- Round 7: They returned \$7.0 from (they saw) \$13.56
- Round 8: They sent \$3.54, I received \$10.62

Our cooperation has been excellent. They've consistently returned generous amounts (around 50% or just over 50%). I want to maintain this strong relationship.

I received \$10.62. A fair return would be around \$5-6 to maintain the cooperative pattern. Given they sent \$3.54, returning around \$5.3 would give them a profit of about \$1.76 and leave me with about \$5.32.

Let me return \$5.50 to be generous and maintain the relationship.

```
Usage: {
  "input_tokens": 2850,
  "output_tokens": 351,
  "reasoning_tokens": 0
}
```

GAME - Agent_4 - Sender - dyad_2

Timestamp: 2026-08-12T08:16:03

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_3",
  "pairing": {
    "agents": [
      "Agent_4",
      "Agent_3"
    ],
    "investor": "Agent_4",
    "roles": {
      "Agent_3": "trustee",
      "Agent_4": "investor"
    },
    "trustee": "Agent_3"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 8,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.56 returned, visible payoff \$5.5600000000000005.

Your total visible earnings across all rounds are \$5.23.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at the history:

- My co-player sent \$3.0 (60% of their endowment) in Round 1, showing strong trust
- They saw a return of \$3.56 (39.6% of what they sent as receiver)
- This demonstrates cooperative behavior

To reciprocate their trust and maintain cooperation, I'll send a similar proportion:

{'send': 3.0}

reasoning:

Let me analyze this situation:

Round 1:

- I was the RECEIVER
- The sender sent \$2.91 (58.2% of their endowment)
- I received \$8.73

- I returned \$3.50
- I kept \$5.23

Now in Round 2:

- I'm the SENDER
- I'm playing against the same co-player from Round 1
- The visible history shows that in Round 1, from my co-player's perspective:
 - They sent \$3.0 (as SENDER)
 - It became \$9.0
 - They saw \$3.56 returned
 - Their visible payoff was \$5.56

Wait, there's a discrepancy here. The problem says there's communication noise. Let me understand:

From my perspective in Round 1:

- Sender sent \$2.91
- I received \$8.73
- I returned \$3.50

From my co-player's perspective in Round 1:

- They sent \$3.0
- It became \$9.0
- They saw \$3.56 returned

So the noise affected what they saw vs what actually happened. They sent \$3.0 but I saw \$2.91. I returned \$3.50 but they saw \$3.56.

Now I'm the sender in Round 2. My co-player was cooperative in Round 1 - they sent 60% of their endowment (\$3.0 out of \$5). They saw a return of \$3.56 out of \$9.0, which is about 39.6% of what I received.

Given that:

1. My co-player showed strong trust by sending 60% in Round 1
2. They saw a decent return (39.6%)
3. I want to maintain cooperation and build trust

I should send a similar proportion to what they sent. They sent 60%, so I should send around \$3.0 as well to match their level of trust and cooperation.

...

usage:

```
{
  "input_tokens": 814,
  "output_tokens": 568,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.56 returned, visible payoff \$5.5600000000000005.
- Round 2 against you, as RECEIVER: saw \$2.77 sent, it became \$8.31, returned \$3.5, visible payoff \$4.8100000000000005.

Your total visible earnings across all rounds are \$10.13.

This round, you are the RECEIVER against your current co-player. They sent you \$3.52, so you received \$10.56. How much do you return? (0-10.56)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the pattern:

- **Round 1**: My co-player sent \$3.0 (60%), showing strong trust
- **Round 2**: I sent \$3.0 (60%), they returned \$3.5 (42% of what they received)
- **Round 3**: They sent \$3.52 (70.4%), showing even greater trust

They're being very cooperative and increasing their trust level. I should reciprocate appropriately.

opriately.

Returning \$4.5:

- Sender ends with: $\$5 - \$3.52 + \$4.5 = \5.98
- I keep: $\$10.56 - \$4.5 = \$6.06$

This rewards their high trust while maintaining fairness.

```
{'return': 4.5}
```

reasoning:

Let me analyze this situation:

Round 1 (I was RECEIVER):

- Sender sent \$2.91 (58.2%)
- I received \$8.73
- I returned \$3.50
- My payoff: \$5.23

Round 2 (I was SENDER):

- I sent \$3.0
- It became \$9.0
- According to visible history, my co-player saw \$2.77 sent (there's noise!)
- They returned \$3.5
- My visible payoff from Round 2: I sent \$3, got back \$3.5, so $\$2 + \$3.5 = \$5.5$
- Total earnings so far: $\$5.23 + \$5.5 = \$10.73$... but it says \$10.13

Wait, let me recalculate. The visible history says:

- Round 2 as RECEIVER: saw \$2.77 sent, it became \$8.31, returned \$3.5, visible payoff \$4.81

This is from my co-player's perspective. They saw \$2.77 sent (due to noise), it became \$8.31, they returned \$3.5, keeping \$4.81.

My actual earnings from Round 2:

- I sent \$3.0
- I kept \$2.0
- I received back... well, if they returned \$3.5 based on receiving \$8.31, then I should have received \$3.5 back
- So my payoff: $\$2.0 + \$3.5 = \$5.5$

Total: $\$5.23 + \$5.5 = \$10.73$

But it says my total visible earnings are \$10.13. Hmm, there might be noise in what I received back too.

Anyway, for Round 3:

- I'm RECEIVER
- They sent \$3.52 (70.4% of their endowment)
- I received \$10.56

My partner's been increasingly generous across rounds-first 58.2%, now 70.4%-showing growing confidence. They returned roughly 40-42% in the previous round, so I should match that reciprocal level. Returning between \$4.22-\$4.75 would maintain the cooperative dynamic we've established....

usage:

```
{
  "input_tokens": 1104,
  "output_tokens": 897,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.56 returned, visible payoff \$5.5600000000000005.
- Round 2 against you, as RECEIVER: saw \$2.77 sent, it became \$8.31, returned \$3.5, visible payoff \$4.8100000000000005.

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.78 returned, visible payoff \$5.779999999999999.
Your total visible earnings across all rounds are \$16.19.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the pattern:

****Round 1**:** Co-player sent \$3.0 (60%), demonstrating trust
****Round 2**:** Co-player returned \$3.5, showing cooperation
****Round 3**:** Co-player sent \$3.0 (60%) again, maintaining trust

They're showing consistent cooperative behavior. I should continue matching their trust level:

{ 'send': 3.0 }

reasoning:

Looking at the history:

****Round 1**:** Co-player sent \$3.0 as SENDER, saw \$3.56 returned
****Round 2**:** Co-player was RECEIVER, saw \$2.77 sent (I actually sent \$3.0), returned \$3.5
****Round 3**:** Co-player sent \$3.0 as SENDER, saw \$3.78 returned (I actually returned \$4.5)

The communication noise is evident:

- In Round 2, I sent \$3.0 but they saw \$2.77
- In Round 3, I returned \$4.5 but they saw \$3.78

My co-player has been consistently sending \$3.0 (60% of endowment) as sender. They've been returning good amounts as receiver (\$3.5).

They're showing consistent trust and cooperation. I should continue to show strong trust as well. I'll send \$3.0 again to maintain the cooperative pattern.

usage:

```
{
  "input_tokens": 1512,
  "output_tokens": 323,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.77 sent, it became \$8.31, returned \$3.5, visible payoff \$4.8100000000000005.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.78 returned, visible payoff \$5.779999999999999.
- Round 4 against you, as RECEIVER: saw \$2.64 sent, it became \$7.92, returned \$3.5, visible payoff \$4.42.

Your total visible earnings across all rounds are \$22.150000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$2.38, so you received \$7.14. How much do you return? (0-7.14)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at the pattern:

****Round 2-4**:** Co-player consistently returned \$3.5 when receiver
****Round 3**:** Co-player sent \$3.0 as sender
****Round 5**:** Co-player sent \$2.38 (47.6%), somewhat less than before

They're still showing trust but at a slightly reduced level. I'll reciprocate proportionally:

Returning \$3.0:

- Sender ends with: $\$5 - \$2.38 + \$3.0 = \5.62
- I keep: $\$7.14 - \$3.0 = \$4.14$

This maintains fairness and cooperation.

```
{'return': 3.0}
```

reasoning:

Looking at the history:

Round 2: Co-player as RECEIVER saw \$2.77 sent, returned \$3.5

Round 3: Co-player as SENDER sent \$3.0, saw \$3.78 returned

Round 4: Co-player as RECEIVER saw \$2.64 sent, returned \$3.5

Round 5: Co-player sent \$2.38 (47.6% of their endowment)

This is a notable decrease from their previous sending amounts (\$3.0). They've been very consistent with returning \$3.5 when they're the receiver.

They sent \$2.38, which becomes \$7.14 for me.

If I return \$3.0:

- Sender ends with: $\$5 - \$2.38 + \$3.0 = \5.62
- I keep: $\$7.14 - \$3.0 = \$4.14$

This seems fair. They sent less this round (maybe due to seeing slightly lower returns or noise in the system), but I should still reciprocate their cooperation.

Let me think about what's fair:

- They sent 47.6% of endowment
- I received \$7.14
- Returning about 40-45% of what I received would be: $\$2.86 - \3.21

I'll return \$3.0, which is reasonable and maintains cooperation.

usage:

```
{
  "input_tokens": 1847,
  "output_tokens": 489,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.78 returned, visible payoff \$5.779999999999999.
- Round 4 against you, as RECEIVER: saw \$2.64 sent, it became \$7.92, returned \$3.5, visible payoff \$4.42.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.54 returned, visible payoff \$5.54.

Your total visible earnings across all rounds are \$26.290000000000003.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at the pattern:

- **Round 3**: Co-player sent \$3.0 (60%)
- **Round 4**: Co-player returned \$3.5, showing cooperation
- **Round 5**: Co-player sent \$3.0 (60%) again

They're maintaining consistent cooperation and trust. I'll continue matching their behavior:

```
{'send': 3.0}
```

reasoning:

Looking at the history:

****Round 3**:** Co-player sent \$3.0 as sender
****Round 4**:** Co-player returned \$3.5 as receiver
****Round 5**:** Co-player sent \$3.0 as sender

The co-player is showing consistent trust by sending \$3.0 when they're the sender and returning \$3.5 when they're the receiver. This is cooperative behavior.

I should continue to match this cooperative pattern by sending around \$3.0.

My total visible earnings are \$26.29, which suggests I'm doing well in this cooperative relationship.

I'll send \$3.0 to maintain the cooperative pattern.

usage:

```
{  
  "input_tokens": 2233,  
  "output_tokens": 244,  
  "reasoning_tokens": 0  
}
```

Message 12 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$2.64 sent, it became \$7.92, returned \$3.5, visible payoff \$4.42.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.54 returned, visible payoff \$5.54.
- Round 6 against you, as RECEIVER: saw \$2.05 sent, it became \$6.1499999999999995, returned \$3.0, visible payoff \$3.1499999999999995.

Your total visible earnings across all rounds are \$31.230000000000004.

This round, you are the RECEIVER against your current co-player. They sent you \$3.55, so you received \$10.649999999999999. How much do you return? (0-10.649999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at the pattern:

****Round 4-6**:** Co-player consistently showing cooperation

****Round 7**:** Co-player sent \$3.55 (71%), showing strong trust - higher than previous rounds

They're demonstrating increased trust. I should reciprocate generously:

Returning \$4.5:

- Sender ends with: \$5 - \$3.55 + \$4.5 = \$5.95
- I keep: \$10.65 - \$4.5 = \$6.15

This rewards their strong cooperation.

{'return': 4.5}

reasoning:

Looking at the pattern:

****Round 4**:** Co-player returned \$3.5 as receiver

****Round 5**:** Co-player sent \$3.0 as sender

****Round 6**:** Co-player returned \$3.0 as receiver

****Round 7**:** Co-player sent \$3.55 (71%)

They're showing high trust by sending 71% of their endowment. I should reciprocate generously.

Returning \$4.5:

- Sender ends with: \$5 - \$3.55 + \$4.5 = \$5.95
- I keep: \$10.65 - \$4.5 = \$6.15

This rewards their strong trust and maintains cooperation.

usage:

```
{
  "input_tokens": 2570,
  "output_tokens": 309,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.54 returned, visible payoff \$5.54.
- Round 6 against you, as RECEIVER: saw \$2.05 sent, it became \$6.149999999999995, returned \$3.0, visible payoff \$3.149999999999995.
- Round 7 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.03 returned, visible payoff \$7.03.

Your total visible earnings across all rounds are \$37.38.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the pattern:

****Round 5**:** Co-player sent \$3.0 (60%)

****Round 6**:** Co-player returned \$3.0, showing fairness

****Round 7**:** Co-player sent \$3.0 (60%) and saw a generous return of \$5.03

They're maintaining consistent cooperation throughout. I'll continue matching their trust level:

```
{'send': 3.0}
```

Reasoning:

Looking at the history:

****Round 5**:** Co-player sent \$3.0 (60%)

****Round 6**:** Co-player returned \$3.0

****Round 7**:** Co-player sent \$3.0 (60%), and saw \$5.03 returned (which was likely my \$4.5 return with noise)

The co-player has been consistently cooperative, sending \$3.0 when they're sender and returning fair amounts when they're receiver. They've maintained this pattern throughout.

I should continue matching their cooperative behavior by sending \$3.0.

```
{'send': 3.0}
```

Usage: {

```
  "input_tokens": 2724,
```

```
  "output_tokens": 242,
```

```
  "reasoning_tokens": 0
```

```
}
```

GAME - Agent_3 - Receiver - dyad_2

Timestamp: 2026-08-12T08:16:08

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",
```

```
  "dyad_id": "dyad_2",
```

```
  "move_index": 3,
```

```
  "opponent_id": "Agent_4",
```

```
  "pairing": {
```

```
    "agents": [
```

```
      "Agent_4",
```

```
      "Agent_3"
```

```
    ],
```

```
    "investor": "Agent_4",
```

```
    "roles": {
```

```

    "Agent_3": "trustee",
    "Agent_4": "investor"
  },
  "trustee": "Agent_3"
},
"role": "trustee",
"role_label": "Receiver",
"round": 8,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.91 sent, it became \$8.73, returned \$3.5, visible payoff \$5.23.

Your total visible earnings across all rounds are \$5.5600000000000005.

This round, you are the RECEIVER against your current co-player. They sent you \$2.77, so you received \$8.31. How much do you return? (0-8.31)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at the history:

****Round 1 analysis:****

- I sent \$3 (they saw \$2.91 due to noise)
- They received \$8.73 and returned \$3.5 (about 40% return rate)
- This was a cooperative response

****Current situation (Round 2):****

- They sent me \$2.77 (with noise)
- I received \$8.31
- They showed good faith cooperation in Round 1

I'll reciprocate their cooperative behavior with a similar return rate:

```
'''json
{"return": 3.5}
'''
```

This gives them a total of approximately \$5.73 and leaves me with \$4.81, creating mutual benefit from our cooperation.

reasoning:

Let me analyze this situation:

Round 1 (just completed):

- I was the SENDER and sent \$3
- Due to communication noise, my co-player saw \$2.91 (close to \$3)
- It became \$8.73 for them (should be \$8.73 if $\$2.91 \times 3$)
- They returned \$3.5
- Their visible payoff: \$5.23 (they kept $\$8.73 - \$3.5 = \$5.23$)
- My payoff: $\$5 - \$3 + \$3.5 = \5.56

Now Round 2:

- I am the RECEIVER
- My co-player sent me an amount, but I see \$2.77 (with noise)
- I received \$8.31 (which is $\$2.77 \times 3$)
- I need to decide how much to return

Analysis:

- In Round 1, my co-player returned \$3.5 out of \$8.73, which is about 40% return rate
- They kept \$5.23 for themselves
- I sent them \$3 (they saw \$2.91), and I ended up with \$5.56 total
- So I gained \$0.56 from the transaction

Now they sent me approximately \$2.77 (actual amount might be slightly different due to noise).

If I apply a similar return rate (around 40%), that would be: $\$8.31 \times 0.4 \approx \3.32

But let me think about fairness:

- They sent around \$2.77, leaving them with about \$2.23
- I received \$8.31
- A fair split might give them back enough so they end up with a reasonable amount

If I return \$3.5 (same as they returned to me):

- They end up with: $\$5 - \$2.77 + \$3.5 = \5.73
- I end up with: $\$8.31 - \$3.5 = \$4.81$

Returning \$4 would give them \$6.23 while I get \$4.31. Returning \$3 would leave me with \$5.31 and them with \$5.23. Since they gave me roughly \$2.77 and returned about 40% of what they received, matching that reciprocal gesture with a \$3.5 return seems appropriate....

usage:

```
{
  "input_tokens": 784,
  "output_tokens": 826,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.91 sent, it became \$8.73, returned \$3.5, visible payoff \$5.23.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.9 returned, visible payoff \$4.9.

Your total visible earnings across all rounds are \$10.370000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at our history:

****Round 1:**** I sent \$3, they returned \$3.5 - cooperative

****Round 2:**** They sent \$3, I returned \$3.5 (they saw \$2.9) - cooperative

We've established a mutually beneficial pattern where we both send moderate amounts and return fair shares. I'll continue this cooperation:

```
'''json
{"send": 3}
'''
```

This maintains trust and keeps our successful collaborative pattern going.

reasoning:

Let me analyze the history:

****Round 1:****

- I sent \$3 (they saw \$2.91 due to noise)
- They received \$8.73 and returned \$3.5
- I ended up with \$2 + \$3.5 = \$5.5

****Round 2:****

- They sent \$3.0 (I saw \$2.77 due to noise, actual was \$3.0)
- It became \$9.0 for me
- I returned \$3.5
- They saw \$2.9 returned (due to noise)
- Their visible payoff was \$4.9 (\$5 - \$3 + \$2.9)
- I kept \$9.0 - \$3.5 = \$5.5

My total visible earnings: \$10.37 which matches \$5.5 + \$4.81 approximately (there's some rounding)

So the pattern so far:

1. Both of us have been sending moderate amounts (\$3)
2. Both of us have been returning reasonable amounts (\$3.5)
3. This has been mutually beneficial

The relationship seems cooperative. They sent me \$3 in Round 2, which is the same amount I sent in Round 1. They saw I returned \$2.9 (actual \$3.5), which was a fair return.

I should continue this cooperative pattern. Sending \$3 again seems appropriate to maintain the reciprocal relationship.

usage:

```
{
  "input_tokens": 1128,
  "output_tokens": 446,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.91 sent, it became \$8.73, returned \$3.5, visible

e payoff \$5.23.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.9 returned, visible payoff \$4.9.
- Round 3 against you, as RECEIVER: saw \$3.52 sent, it became \$10.56, returned \$4.5, visible payoff \$6.0600000000000005.
Your total visible earnings across all rounds are \$16.15.

This round, you are the RECEIVER against your current co-player. They sent you \$2.64, so you received \$7.92. How much do you return? (0-7.92)
Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at our cooperative history:

****Round 1:**** They returned \$3.5 on \$8.73
****Round 2:**** I returned \$3.5 on \$9.0
****Round 3:**** They returned \$4.5 on \$10.56 (increased generosity!)
****Round 4:**** They sent me \$2.64, I received \$7.92

They've been consistently cooperative and even increased their generosity in Round 3. I'll reciprocate fairly:

```
'''json
{"return": 3.5}
'''
```

This maintains our mutually beneficial cooperation pattern.

reasoning:

Let me analyze the history with my current co-player:

****Round 1:****
- I sent \$3 (they saw \$2.91 due to noise)
- They received \$8.73, returned \$3.5
- Good cooperation
****Round 2:****
- They sent \$3
- I received \$9.0, returned \$3.5 (they saw \$2.9)
- Continued cooperation
****Round 3:****
- I sent \$3 (they saw \$3.52 due to noise)
- They received \$10.56, returned \$4.5
- They increased their return, showing even more generosity
****Round 4 (current):****
- They sent \$2.64 (with noise, could have been around \$3 originally)
- I received \$7.92

Looking at the pattern:
- They've been consistently cooperative
- In Round 3, they returned \$4.5 on \$10.56 (about 43% return rate)
- They're continuing to send around \$3 each time

I should reciprocate their continued cooperation. Given they sent me \$7.92, a fair return would be around 40-45% of what I received, which would be about \$3.2-\$3.6.

I'll return \$3.5 to maintain consistency with our cooperative pattern.

usage:

```
{
  "input_tokens": 1467,
  "output_tokens": 452,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

Visible history before this round:
Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.9 returned, visible payoff \$4.9.
- Round 3 against you, as RECEIVER: saw \$3.52 sent, it became \$10.56, returned \$4.5, visible payoff \$6.0600000000000005.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.96 returned, visible payoff \$5.96.
Your total visible earnings across all rounds are \$20.57.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at our established pattern:

****Round 2-4:**** They consistently sent \$3
****Round 3:**** They were particularly generous, returning \$4.5
****Round 4:**** I returned \$3.5 (they saw \$3.96)

We have a strong cooperative relationship. I'll maintain our successful pattern:

```
'''json
{"send": 3}
'''
```

This continues our mutually beneficial cooperation.

reasoning:

Looking at the history with this co-player:

Round 1: I sent \$3 (they saw \$2.91), they returned \$3.5 - cooperative
Round 2: They sent \$3, I returned \$3.5 (they saw \$2.9) - cooperative
Round 3: I sent \$3 (they saw \$3.52), they returned \$4.5 - very generous!
Round 4: They sent \$3, I returned \$3.5 (they saw \$3.96) - cooperative

This co-player has been consistently cooperative and even showed extra generosity in Round 3. The pattern shows:

- They keep sending \$3
- They've returned generously when receiving
- We have a strong cooperative relationship

I should continue the cooperation. Sending \$3 has worked well for us.

usage:

```
{
  "input_tokens": 1824,
  "output_tokens": 291,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$3.52 sent, it became \$10.56, returned \$4.5, visible payoff \$6.0600000000000005.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.96 returned, visible payoff \$5.96.
- Round 5 against you, as RECEIVER: saw \$2.38 sent, it became \$7.14, returned \$3.0, visible payoff \$4.14.

Your total visible earnings across all rounds are \$26.11.

This round, you are the RECEIVER against your current co-player. They sent you \$2.05, so you received \$6.1499999999999995. How much do you return? (0-6.1499999999999995)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at the pattern:

****Round 3:**** They returned \$4.5
****Round 4:**** I returned \$3.5

****Round 5:**** They returned \$3.0
****Round 6:**** They sent \$2.05, I received \$6.15

Our cooperation has been solid. I'll maintain a fair return:

```
'''json
{"return": 3.0}
'''
```

This keeps our mutually beneficial relationship going.

reasoning:

Looking at the history:

Round 3: They returned \$4.5 on \$10.56
Round 4: I returned \$3.5 (they saw \$3.96) on \$7.92
Round 5: They saw me send \$2.38 (I actually sent \$3), received \$7.14, returned \$3.0

Now Round 6: They sent me \$2.05 (with noise), I received \$6.15

They've been consistently cooperative throughout. Even though the amount they sent this round is lower (possibly due to noise), they've shown good faith. I should maintain cooperation.

Given I received \$6.15, a fair return would be around \$2.5-3.0 to maintain our cooperative pattern. Let me return about 40-45% which has been roughly our pattern.

$\$6.15 * 0.45 = \2.77

Or I could go with \$3.0 to be consistent with what they did in Round 5.

Let me go with \$3.0 to maintain consistency and cooperation.

usage:

```
{
  "input_tokens": 2163,
  "output_tokens": 349,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.96 returned, visible payoff \$5.96.
- Round 5 against you, as RECEIVER: saw \$2.38 sent, it became \$7.14, returned \$3.0, visible payoff \$4.14.
- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.94 returned, visible payoff \$4.9399999999999995.

Your total visible earnings across all rounds are \$29.259999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at our pattern:

****Round 4-6:**** They consistently sent \$3
****Round 5-6:**** We both returned around \$3

We have a stable, cooperative relationship. I'll continue:

```
'''json
{"send": 3}
'''
```

This maintains our successful mutual cooperation.

reasoning:

Looking at the pattern with this co-player:

Round 4: They sent \$3
Round 5: I sent \$3 (they saw \$2.38), they returned \$3.0
Round 6: They sent \$3, I returned \$3.0 (they saw \$2.94)
Round 7: My turn to send

We've maintained a consistent cooperative pattern. They keep sending \$3, and we both return fair amounts. I should continue this successful cooperation.

usage:

```
{
  "input_tokens": 2488,
  "output_tokens": 182,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as RECEIVER: saw \$2.38 sent, it became \$7.14, returned \$3.0, visible payoff \$4.14.
- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.94 returned, visible payoff \$4.9399999999999995.
- Round 7 against you, as RECEIVER: saw \$3.55 sent, it became \$10.649999999999999, returned \$4.5, visible payoff \$6.149999999999999.

Your total visible earnings across all rounds are \$36.29.

This round, you are the RECEIVER against your current co-player. They sent you \$2.14, so you received \$6.42. How much do you return? (0-6.42)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at our history:

****Round 5-6:**** We both returned around \$3.0
****Round 7:**** They were generous, returning \$4.5
****Round 8:**** They sent \$2.14, I received \$6.42

They've been consistently cooperative and generous. I'll reciprocate:

```
'''json
{"return": 3.0}
'''
```

This maintains our fair cooperation pattern.

Reasoning:

Looking at the history:

****Round 5:**** They returned \$3.0
****Round 6:**** I returned \$3.0 (they saw \$2.94)
****Round 7:**** They returned \$4.5 (generous!)
****Round 8:**** They sent \$2.14, I received \$6.42

They've been consistently cooperative and even generous in Round 7. I should reciprocate fairly. Given they returned \$4.5 in Round 7, I should return a reasonable amount.

I'll return \$3.0 to maintain fairness while being slightly less than their generous \$4.5 in Round 7.

Usage: {

```
"input_tokens": 2633,
"output_tokens": 247,
"reasoning_tokens": 0
}
```

GAME - Agent_6 - Sender - dyad_3

Timestamp: 2026-08-12T08:16:14

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_3",
```



```

"move_index": 4,
"opponent_id": "Agent_5",
"pairing": {
  "agents": [
    "Agent_6",
    "Agent_5"
  ],
  "investor": "Agent_6",
  "roles": {
    "Agent_5": "trustee",
    "Agent_6": "investor"
  },
  "trustee": "Agent_5"
},
"role": "investor",
"role_label": "Sender",
"round": 8,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.13 returned, visible payoff \$5.13.

Your total visible earnings across all rounds are \$3.9800000000000004.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at the visible history:

****Round 1 recap (from their perspective):****

- My co-player sent \$3.00 (60% of their endowment) - quite generous
- They saw \$3.13 returned
- Their visible payoff was \$5.13

My co-player showed significant trust by sending 60% of their endowment. Even though there's communication noise affecting what we each see, I should reciprocate this trust.

I'll match their generosity by sending the same amount they sent: \$3.00 (60% of my endowment). This demonstrates:

- Reciprocal trust
- Willingness to cooperate
- Fair play matching their level of generosity

{"send": 3.00}

reasoning:

Let me analyze this situation:

Round 1 (just played):

- I was the RECEIVER
- Sender sent \$2.66 (53.2%)
- I received \$7.98
- I returned \$4.00
- I kept \$3.98

But now I see the "visible history" which shows:

- Round 1 against me, as SENDER: sent \$3.0, became \$9.0, saw \$3.13 returned, visible payoff \$5.13

This is interesting - there's communication noise!

- The sender actually sent \$3.00 (not \$2.66 as I saw)
- It became \$9.00 (not \$7.98 as I saw)
- They saw \$3.13 returned (not \$4.00 as I sent)
- Their visible payoff was \$5.13 (which is \$5 - \$3 + \$3.13)

So due to noise:

- I saw them send \$2.66, they actually sent \$3.00
- I saw I received \$7.98, I actually received \$9.00
- I returned \$4.00, they saw \$3.13

Now in Round 2, I'm the SENDER and playing against the same person.

From their perspective:

- They sent \$3.00 (60% of endowment - quite generous)
- They got back \$3.13 (34.8% of what I received)
- Their payoff was \$5.13

From my actual perspective (what really happened):

- They sent \$3.00
- I received \$9.00
- I returned \$4.00 (44.4% of what I received)
- But they only saw \$3.13

The co-player was quite generous in Round 1 (sent 60%). From their visible perspective, they got a modest return (34.8% of the tripled amount).

...

usage:

```
{
  "input_tokens": 845,
  "output_tokens": 754,
  "reasoning_tokens": 0
}
```

Message 4 role=user**content:**

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.13 returned, visible payoff \$5.13.
- Round 2 against you, as RECEIVER: saw \$2.97 sent, it became \$8.91, returned \$4.5, visible payoff \$4.41.

Your total visible earnings across all rounds are \$11.010000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$3.23, so you received \$9.69. How much do you return? (0-9.69)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant**content:**

Looking at the visible history:

****Round 1:**** My co-player sent \$3.00 (60%)

****Round 2:**** They returned \$4.50 out of \$8.91 (50.5%) when they were receiver

****Round 3:**** They just sent \$3.23 (64.6%) - even more generous!

My co-player has shown increasing trust and cooperation. They've been very generous, and when they were receiver in Round 2, they returned about 50% of what they received.

I should reciprocate their strong cooperation. If I return \$5.00:

- Sender ends with: \$1.77 + \$5.00 = \$6.77 (profit of \$1.77)
- I keep: \$9.69 - \$5.00 = \$4.69

This returns about 52% of what I received, matching their generosity from Round 2 and rewarding their increased trust.

```
{"return": 5.00}
```

reasoning:

Let me analyze the visible history:

****Round 1 (I was RECEIVER):****

- My co-player sent \$3.00 (60%)
- They saw \$3.13 returned
- Their visible payoff: \$5.13

****Round 2 (I was SENDER):****

- I sent \$3.00
- They saw \$2.97 sent (noise), which became \$8.91
- They returned \$4.50
- Their visible payoff: \$4.41

****Round 3 (I am RECEIVER):****

- They sent \$3.23 (64.6% of their endowment)
- I received \$9.69
- They have \$1.77 remaining

My co-player has been very cooperative:

- Round 1: Sent 60% (\$3.00)
- Round 2: Returned \$4.50 out of \$8.91 received (50.5%)
- Round 3: Sent 64.6% (\$3.23) - even more generous!

They've shown increasing trust and generosity. I should reciprocate strongly.

Let me think about what to return:

If I return \$5.00:

- Sender ends with: \$1.77 + \$5.00 = \$6.77 (profit of \$1.77)

- I keep: \$9.69 - \$5.00 = \$4.69

If I return \$5.50:

- Sender ends with: \$1.77 + \$5.50 = \$7.27 (profit of \$2.27)

- I keep: \$9.69 - \$5.50 = \$4.19

Given their increased generosity (sending 64.6% vs 60% in round 1), I should return proportionally more. They returned \$4.50 out of \$8.91 (50.5%) when they were receiver. I should aim to return around 50-55% to match their generosity.

...

usage:

```
{
  "input_tokens": 1207,
  "output_tokens": 754,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.13 returned, visible payoff \$5.13.

- Round 2 against you, as RECEIVER: saw \$2.97 sent, it became \$8.91, returned \$4.5, visible payoff \$4.41.

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.79 returned, visible payoff \$7.79.

Your total visible earnings across all rounds are \$15.700000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the visible history:

****Round 1:**** Co-player sent \$3.00, saw \$3.13 returned, payoff \$5.13

****Round 2:**** Co-player returned \$4.50 from \$8.91 received (fair, ~50%)

****Round 3:**** Co-player sent \$3.00, saw \$5.79 returned (very generous!), payoff \$7.79

My co-player has been consistently generous, sending \$3.00 (60%) in both rounds as sender. They also saw a strong return in Round 3, which should reinforce mutual trust. As receiver, they returned fairly (~50%).

This is a cooperative relationship with established trust. I should continue matching their consistent generosity of \$3.00.

{"send": 3.00}

reasoning:

Looking at the visible history from my co-player's perspective:

****Round 1:**** They sent \$3.00, saw \$3.13 returned, payoff \$5.13

****Round 2:**** They received \$8.91 (from my \$2.97 send), returned \$4.50, payoff \$4.41

****Round 3:**** They sent \$3.00, saw \$5.79 returned, payoff \$7.79

My co-player has been consistently generous, sending \$3.00 in both rounds they were sender (60% of endowment). In Round 3, they saw a very generous return of \$5.79, which should reinforce their trust in me.

They've also been fair as a receiver, returning \$4.50 out of \$8.91 (about 50%).

This is a cooperative relationship with mutual trust. I should continue to match their generosity. They've sent \$3.00 twice, so I should send \$3.00 again.

Actually, let me reconsider. In Round 3, they saw a particularly high return (\$5.79), which might encourage even more trust. But they've consistently sent \$3.00, so I should match that level.

I'll send \$3.00 to maintain consistency and reciprocal cooperation.

usage:

```
{
  "input_tokens": 1654,
  "output_tokens": 485,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.97 sent, it became \$8.91, returned \$4.5, visible payoff \$4.41.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.79 returned, visible payoff \$7.79.
- Round 4 against you, as RECEIVER: saw \$3.93 sent, it became \$11.790000000000001, returned \$5.9, visible payoff \$5.890000000000001.

Your total visible earnings across all rounds are \$24.33.

This round, you are the RECEIVER against your current co-player. They sent you \$3.62, so you received \$10.86. How much do you return? (0-10.86)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at the visible history:

```
**Round 2:** Co-player returned $4.50 from $8.91 (50.5%)
**Round 3:** Co-player sent $3.00, saw $5.79 returned
**Round 4:** Co-player returned $5.90 from $11.79 (50.1%)
**Round 5:** Co-player sent $3.62 (72.4%!) - very high trust!
```

My co-player has been consistently cooperative, returning about 50% when receiver and sending generously as sender. This round they sent an even larger amount (\$3.62, which is 72.4% of their endowment) - showing increasing trust.

I should reward this high level of trust. If I return \$6.00:

- Sender ends with: \$1.38 + \$6.00 = \$7.38 (profit of \$2.38)
- I keep: \$10.86 - \$6.00 = \$4.86

This returns about 55% of what I received, slightly more generous than their pattern to acknowledge their increased trust.

```
{"return": 6.00}
```

reasoning:

Looking at the visible history:

```
**Round 2:** Co-player returned $4.50 from $8.91 (50.5%)
**Round 3:** Co-player sent $3.00, saw $5.79 returned
**Round 4:** Co-player returned $5.90 from $11.79 (50.1%)
**Round 5:** Co-player sent $3.62 (72.4%)
```

My co-player has been very consistent and cooperative:

- As sender, they've sent generous amounts (\$3.00 and now \$3.62 - 72.4%!) -
- As receiver, they've returned about 50% consistently

They've sent an even larger amount this round (\$3.62, which is 72.4% of their endowment). This shows increasing trust.

I should continue to reciprocate fairly. If I return around 50-55% like they've been doing:

- 50% would be \$5.43
- 55% would be \$5.97

Let me aim for about \$5.50 to \$6.00 to maintain the cooperative pattern and reward their increased generosity.

If I return \$5.50:

- Sender ends with: \$1.38 + \$5.50 = \$6.88 (profit of \$1.88)
- I keep: \$10.86 - \$5.50 = \$5.36

If I return \$6.00:
- Sender ends with: $\$1.38 + \$6.00 = \$7.38$ (profit of \$2.38)
- I keep: $\$10.86 - \$6.00 = \$4.86$

Given they sent 72.4% of their endowment (very high trust), I should return a bit more generously. ...

usage:

```
{
  "input_tokens": 2074,
  "output_tokens": 706,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.79 returned, visible payoff \$7.79.
- Round 4 against you, as RECEIVER: saw \$3.93 sent, it became \$11.790000000000001, returned \$5.9, visible payoff \$5.890000000000001.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.97 returned, visible payoff \$7.97.

Your total visible earnings across all rounds are \$29.189999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at the visible history:

****Round 3:**** Co-player sent \$3.00, saw \$5.79 returned, payoff \$7.79
****Round 4:**** Co-player returned \$5.90 from \$11.79 (50%)
****Round 5:**** Co-player sent \$3.00, saw \$5.97 returned, payoff \$7.97

My co-player has been extremely consistent and cooperative:

- As sender: consistently sends \$3.00 (60% of endowment)
- As receiver: returns about 50% fairly
- They've received good returns, reinforcing mutual trust

This is a stable, mutually beneficial cooperative relationship. I should continue matching their pattern by sending \$3.00.

```
{"send": 3.00}
```

reasoning:

Looking at the visible history from my co-player's perspective:

****Round 3:**** Co-player sent \$3.00, saw \$5.79 returned, payoff \$7.79
****Round 4:**** Co-player returned \$5.90 from \$11.79 (50%)
****Round 5:**** Co-player sent \$3.00, saw \$5.97 returned, payoff \$7.97

My co-player has been very consistent:

- As sender: consistently sends \$3.00 (60% of endowment)
- As receiver: returns about 50%
- They've seen good returns when sending (\$5.79 and \$5.97), leading to strong payoffs (~\$8)

This is a very cooperative and stable relationship. My co-player has maintained trust by consistently sending \$3.00 and returning fairly. I should continue this pattern of mutual cooperation by sending \$3.00 again.

The relationship is working well - we're both benefiting from the cooperation. No reason to deviate from what's working.

usage:

```
{
  "input_tokens": 2569,
  "output_tokens": 425,
}
```

```

    "reasoning_tokens": 0
  }
Message 12 role=user
content:
  Round 7

  Visible history before this round:
  Your current co-player's last 3 game(s):
  - Round 4 against you, as RECEIVER: saw $3.93 sent, it became $11.790000000000001, returned $5.9, visible payoff $5.890000000000001.
  - Round 5 against you, as SENDER: sent $3.0, it became $9.0, saw $5.97 returned, visible payoff $7.97.
  - Round 6 against you, as RECEIVER: saw $3.02 sent, it became $9.06, returned $4.5, visible payoff $4.5600000000000005.
  Your total visible earnings across all rounds are $35.76.

  This round, you are the RECEIVER against your current co-player. They sent you $2.78, so you received $8.34. How much do you return? (0-8.34)
  Respond exactly as JSON: {'return': <amount>}
Message 13 role=assistant
content:
  Looking at the visible history:

  **Round 4:** Co-player returned $5.90 from $11.79 (50%)
  **Round 5:** Co-player sent $3.00, saw $5.97 returned
  **Round 6:** Co-player returned $4.50 from $9.06 (49.7%)
  **Round 7:** Co-player sent $2.78 (55.6%)

  My co-player has been consistently cooperative, returning about 50% when receiver and sending generously as sender. This round they sent $2.78, slightly less than their usual $3.00, but still a majority of their endowment (55.6%).

  I should maintain our cooperative relationship with a fair return. If I return $4.50:
  - Sender ends with:  $\$2.22 + \$4.50 = \$6.72$  (profit of $1.72)
  - I keep:  $\$8.34 - \$4.50 = \$3.84$ 

  This returns about 54% of what I received, which is fair and maintains the mutual trust we've built.

  {"return": 4.50}
reasoning:
  Looking at the visible history:

  **Round 4:** Co-player returned $5.90 from $11.79 (50%)
  **Round 5:** Co-player sent $3.00, saw $5.97 returned, payoff $7.97
  **Round 6:** Co-player returned $4.50 from $9.06 (49.7%)
  **Round 7:** Co-player sent $2.78 (55.6%)

  My co-player has been very consistent:
  - As sender: sending around $3.00 (though this round $2.78, slightly less)
  - As receiver: returning about 50% consistently

  This round they sent $2.78, which is a bit less than their usual $3.00. Let me think about what to return.

  If I return $4.50:
  - Sender ends with:  $\$2.22 + \$4.50 = \$6.72$  (profit of $1.72)
  - I keep:  $\$8.34 - \$4.50 = \$3.84$ 

  This returns about 54% of what I received, which is fair and maintains our cooperative relationship.

  Actually, let me reconsider. They sent slightly less this round ($2.78 vs usual $3.00). I should still be generous but maybe match their slight reduction in trust with a proportional return.

  If I return $4.50:
  - That's 54% return rate
  - Sender profits $1.72
  - I keep $3.84

```

If I return \$4.00:
- That's 48% return rate
- Sender profits \$1.22
- I keep \$4.34

I think \$4.50 is still appropriate to maintain the good cooperative relationship, even though they sent slightly less. The relationship has been strong overall.

usage:

```
{  
  "input_tokens": 2990,  
  "output_tokens": 685,  
  "reasoning_tokens": 0  
}
```

Message 14 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.97 returned, visible payoff \$7.97.
- Round 6 against you, as RECEIVER: saw \$3.02 sent, it became \$9.06, returned \$4.5, visible payoff \$4.5600000000000005.
- Round 7 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.84 returned, visible payoff \$6.84.

Your total visible earnings across all rounds are \$39.599999999999994.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the visible history:

****Round 5:**** Co-player sent \$3.00, saw \$5.97 returned, payoff \$7.97
****Round 6:**** Co-player returned \$4.50 from \$9.06 (~50%)
****Round 7:**** Co-player sent \$3.00, saw \$4.84 returned, payoff \$6.84

My co-player continues to be consistently cooperative:

- As sender: reliably sends \$3.00 (60% of endowment)
- As receiver: returns about 50%, maintaining fairness
- They continue cooperating despite variable returns (due to communication noise)

We've established a strong, mutually beneficial pattern over 7 rounds. I should continue matching their consistent trust by sending \$3.00.

```
{"send": 3.00}
```

Reasoning:

Looking at the visible history:

****Round 5:**** Co-player sent \$3.00, saw \$5.97 returned, payoff \$7.97
****Round 6:**** Co-player returned \$4.50 from \$9.06 (49.7%)
****Round 7:**** Co-player sent \$3.00, saw \$4.84 returned, payoff \$6.84

My co-player has been very consistent:

- As sender: consistently sends \$3.00 (60% of endowment)
- As receiver: returns about 50% fairly
- They continue to show trust despite some variation in returns (due to communication noise)

This is Round 8. We've built a strong cooperative relationship over 7 rounds. My co-player has been extremely consistent in sending \$3.00 when they're the sender. I should continue to match their consistent behavior.

I'll send \$3.00 to maintain our established pattern of cooperation.

Usage: {

```
"input_tokens": 3240,  
"output_tokens": 415,  
"reasoning_tokens": 0  
}
```


GAME - Agent_5 - Receiver - dyad_3

Timestamp: 2026-08-12T08:16:25

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_6",
  "pairing": {
    "agents": [
      "Agent_6",
      "Agent_5"
    ],
    "investor": "Agent_6",
    "roles": {
      "Agent_5": "trustee",
      "Agent_6": "investor"
    },
    "trustee": "Agent_5"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 8,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.

- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.66 sent, it became \$7.98, returned \$4.0, visible payoff \$3.9800000000000004.

Your total visible earnings across all rounds are \$5.13.

This round, you are the RECEIVER against your current co-player. They sent you \$2.97, so you received \$8.91. How much do you return? (0-8.91)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at the situation:

****What happened in Round 1:****

- My co-player received \$7.98 (my \$3 sent with noise)
- They returned \$4.0, which is about 50% - a fair reciprocal move
- This shows they're a cooperative player

****Current Round 2:****

- They sent me \$2.97 (?60% of their endowment) - showing significant trust
- I received \$8.91
- They kept \$2.03

****My decision:****

Since they demonstrated fair reciprocity (50% return) in Round 1 and have now shown strong trust by sending most of their endowment, I should reciprocate similarly. Returning about 50% maintains the cooperative relationship while ensuring we both benefit.

Returning \$4.50 means:

- They end with: \$2.03 + \$4.50 = \$6.53
- I end with: \$8.91 - \$4.50 = \$4.41

This is fair and encourages continued cooperation.

```
'''json
{"return": 4.50}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap:

- I sent \$3 (but due to communication noise, my co-player saw \$2.66)
- They received \$7.98 (which should be 2.66×3)
- They returned \$4.0
- My visible payoff was \$5.13 (which would be $5 - 3 + 3.13...$ but there's noise)
- Their visible payoff was \$3.98

Now in Round 2:

- My co-player is the SENDER
- They sent me \$2.97 (with noise, they might have intended something slightly different)
- I received \$8.91 (which is 2.97×3)
- I need to decide how much to return

Analysis of their behavior:

- In Round 1, they returned \$4.0 out of \$7.98 received, which is about 50%
- This is a reasonably fair/generous return
- Now they've sent me \$2.97 out of their \$5, which is about 59% - quite trusting

Given that:

1. They showed good reciprocity in Round 1 (returning 50% of what they received)
2. They've now shown significant trust by sending me almost 60% of their endowment
3. We might continue playing together in future rounds

4. Fair reciprocity would encourage continued cooperation

I should return a similar proportion or slightly more to maintain/build trust.

They received \$8.91 from my send. A 50% return would be \$4.455.

But let me think about the overall economics:

- They started with \$5
- They sent \$2.97, keeping \$2.03
- I received \$8.91
- If I return \$4.50:
 - They end with: $\$2.03 + \$4.50 = \$6.53$

...

usage:

```
{
  "input_tokens": 816,
  "output_tokens": 863,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.66 sent, it became \$7.98, returned \$4.0, visible payoff \$3.9800000000000004.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.03 returned, visible payoff \$7.03.

Your total visible earnings across all rounds are \$9.54.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at our history together:

****Round 1:****

- I sent, they received and returned ~50% - fair cooperation

****Round 2:****

- They sent \$3 (60% of endowment) - high trust
- I returned \$4.50, they saw \$5.03 (noise) - generous reciprocity
- Their payoff was strong (\$7.03)

****Analysis:****

- This co-player has proven consistently cooperative
- They demonstrated trust by sending \$3 in Round 2
- They've reciprocated fairly (50% return rate)
- Our relationship is building mutual trust

****My decision:****

I should maintain this cooperative dynamic. Sending \$3 again signals:

- Continued trust in our partnership
- Consistency and reliability
- Faith they'll reciprocate fairly as before

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze the visible history:

Round 1:

- I was SENDER, sent \$3
- They received \$7.98 (with noise, actual was probably \$9)
- They returned \$4.0 (about 50%)
- My payoff was $\$5 - \$3 + \$4 = \6
- Visible shows \$5.13, which might include noise

Round 2:

- I was RECEIVER
- They sent \$3.0 (actual, before noise)
- I received \$8.91 (actually \$9 with noise)
- I returned \$4.50
- They saw \$5.03 returned (with noise, I actually returned \$4.50)
- Their visible payoff $\$7.03 = \$5 - \$3 + \5.03

My total visible earnings: \$9.54

Round 3 (current):

- I'm SENDER again
- Same co-player from rounds 1 and 2

This co-player has been very cooperative:

1. Returned 50% in Round 1 when they received
2. Sent \$3 (60%) in Round 2, showing trust
3. They saw me return \$5.03 (with noise), which they probably perceive as generous (56% return)

This is building a strong cooperative relationship. They've been trustworthy and fair. I should continue to send a substantial amount to maintain this cooperation.

Given their consistent cooperation and the fact they sent \$3 in Round 2, I should send at least the same amount, if not more, to signal continued trust and cooperation.

I'll send \$3 again, showing consistency and trust.

usage:

```
{
  "input_tokens": 1250,
  "output_tokens": 587,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.66 sent, it became \$7.98, returned \$4.0, visible payoff \$3.9800000000000004.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.03 returned, visible payoff \$7.03.
- Round 3 against you, as RECEIVER: saw \$3.23 sent, it became \$9.69, returned \$5.0, visible payoff \$4.6899999999999995.

Your total visible earnings across all rounds are \$17.33.

This round, you are the RECEIVER against your current co-player. They sent you \$3.93, so you received \$11.790000000000001. How much do you return? (0-11.790000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at our continued partnership:

****Round 1:**** They returned \$4.0 from \$7.98 (?50%)
****Round 2:**** I returned \$4.50 from \$9.0 (?50%)
****Round 3:**** They returned \$5.0 from \$9.69 (?52%)

****Current Round 4:****

- They sent \$3.93 (79% of endowment!) - very high trust signal
- I received \$11.79
- They kept only \$1.07

****Analysis:****

This co-player has shown increasing trust and consistent 50% reciprocity. They're now sending nearly their entire endowment, demonstrating strong faith in our relationship.

I should maintain our established 50% return pattern to:

- Honor their trust

- Keep our cooperative equilibrium
- Ensure mutual benefit

Returning \$5.90:

- They end with: $\$1.07 + \$5.90 = \$6.97$
- I end with: $\$11.79 - \$5.90 = \$5.89$

```
'''json
{"return": 5.90}
'''
```

reasoning:

Let me analyze the history:

****Round 1:****

- I sent (saw \$2.66 with noise from my \$3)
- They received \$7.98, returned \$4.0 (about 50%)

****Round 2:****

- They sent \$3
- I received \$9, returned \$4.50 (they saw \$5.03 with noise)
- They got \$7.03 total

****Round 3:****

- I sent \$3 (they saw \$3.23 with noise)
- They received \$9.69, returned \$5.0 (about 52%)
- They kept \$4.69

****Round 4 (current):****

- They sent \$3.93 (about 79% of their endowment!) - very high trust
- I received \$11.79
- They kept \$1.07

This co-player is showing extremely high trust by sending \$3.93. They've consistently returned around 50% when they were receiver. I should reciprocate this trust and maintain our cooperative relationship.

If I return about 50% of what I received:

- Return \$5.90
- They end with: $\$1.07 + \$5.90 = \$6.97$
- I end with: $\$11.79 - \$5.90 = \$5.89$

But they're showing increasing trust (sending more and more). Maybe I should return a bit more to reward this and encourage continued cooperation. Let me think...

Actually, looking at the pattern:

- Round 1 receiver: returned \$4 from \$7.98 (50%)
- Round 3 receiver: returned \$5 from \$9.69 (52%)

They're consistently returning around 50%. I've been doing similar - returned \$4.50 from \$9 in Round 2 (50%).

Given they sent \$3.93 (very high trust), I should return around 50% or slightly more to maintain the relationship.

...

usage:

```
{
  "input_tokens": 1695,
  "output_tokens": 859,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.03 returned, visible payoff \$7.03.
- Round 3 against you, as RECEIVER: saw \$3.23 sent, it became \$9.69, returned \$5.0, visible payoff \$4.69

- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.63 returned, visible payoff \$8.629999999999999.

Your total visible earnings across all rounds are \$23.22.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at our established partnership:

****Round 2:**** They sent \$3.0, I returned generously

****Round 3:**** I sent \$3.23 (with noise), they returned \$5.0 (?52%)

****Round 4:**** They sent \$3.0, I returned well, they saw \$6.63

****Pattern Analysis:****

- They consistently send \$3 when sender (Rounds 2, 4)
- They reciprocate fairly at ~50% when receiver (Round 3)
- We've built a stable, mutually beneficial relationship
- Both earning well (\$23.22 total for me)

****My decision:****

Match their consistency by sending \$3 again. This:

- Maintains our established cooperative equilibrium
- Shows I trust the pattern we've developed
- Signals reliability and reciprocity

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at the visible history:

****Round 2:**** They sent \$3.0, I returned (they saw \$5.03) - I actually returned \$4.50 but they saw \$5.03 due to noise

****Round 3:**** They received \$9.69 (from my \$3 send), returned \$5.0

****Round 4:**** They sent \$3.0, I returned \$5.90 but they saw \$6.63 - generous due to noise

Pattern analysis:

- This co-player consistently sends \$3.0 when they're sender (Rounds 2 and 4)
- They return roughly 50% when receiver (Round 3: \$5.0 from \$9.69)
- They've been seeing higher returns from me due to noise (\$5.03 and \$6.63 vs what I actually sent \$4.50 and \$5.90)

Our cooperation has been strong and consistent. They've shown:

- Reliable sending patterns (\$3)
- Fair reciprocity (~50%)
- Building trust over rounds

I should continue with the same sending pattern to maintain this cooperative equilibrium. Sending \$3 has been working well for both of us.

My total visible earnings are \$23.22 which suggests we're both doing well.

I'll send \$3 again to maintain consistency and trust.

usage:

```
{
  "input_tokens": 2185,
  "output_tokens": 517,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$3.23 sent, it became \$9.69, returned \$5.0, visible payoff \$4.6899999999999995.

- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.63 returned, visible p

ayoff \$8.629999999999999.
- Round 5 against you, as RECEIVER: saw \$3.62 sent, it became \$10.86, returned \$6.0, visible payoff \$4.859999999999999.
Your total visible earnings across all rounds are \$31.189999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$3.02, so you received \$9.06. How much do you return? (0-9.06)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at our continued partnership:

****Round 3:**** I sent, they returned \$5.0 from \$9.69 (?52%)
****Round 4:**** They sent \$3.0, I returned generously
****Round 5:**** I sent, they returned \$6.0 from \$10.86 (?55%)

****Current Round 6:****

- They sent \$3.02 (?60% of endowment) - continued trust
- I received \$9.06
- They kept \$1.98

****Analysis:****

Our partnership shows consistent mutual cooperation with ~50-55% return rates. They've reliably reciprocated in this range, and I should maintain this equilibrium.

Returning \$4.50:

- They end with: \$1.98 + \$4.50 = \$6.48
- I end with: \$9.06 - \$4.50 = \$4.56

This maintains our cooperative pattern.

```
'''json
{"return": 4.50}
'''
```

reasoning:

Looking at our history:

****Round 3:**** I sent (they saw \$3.23), they returned \$5.0 from \$9.69 (?52%)
****Round 4:**** They sent \$3.0, I returned, they saw \$6.63 (generous due to noise)
****Round 5:**** I sent (they saw \$3.62), they returned \$6.0 from \$10.86 (?55%)

****Current Round 6:****

- They sent \$3.02 (?60% of endowment)
- I received \$9.06
- They kept \$1.98

****Pattern:****

- They consistently send around \$3
- They consistently return around 50-55% when receiver
- We have a stable, cooperative relationship

I should maintain the ~50% return pattern:

- Returning \$4.50 would be about 50%
- This keeps our cooperative equilibrium
- They end with: \$1.98 + \$4.50 = \$6.48
- I end with: \$9.06 - \$4.50 = \$4.56

Both benefit reasonably, and this maintains the pattern.

usage:

```
{
  "input_tokens": 2634,
  "output_tokens": 514,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 7

Visible history before this round:
Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.63 returned, visible payoff \$8.629999999999999.
- Round 5 against you, as RECEIVER: saw \$3.62 sent, it became \$10.86, returned \$6.0, visible payoff \$4.859999999999999.
- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.57 returned, visible payoff \$6.57.
Your total visible earnings across all rounds are \$35.75.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at our established pattern:

****Round 4:**** They sent \$3.0, I returned generously
****Round 5:**** I sent, they returned \$6.0 (?55%)
****Round 6:**** They sent \$3.0, I returned (they saw \$4.57)

****Analysis:****

- They consistently send \$3 when in the sender role (Rounds 4, 6)
- They reciprocate fairly (50-55%) when receiver (Round 5)
- We've maintained a stable, mutually beneficial partnership
- Total earnings of \$35.75 shows this strategy is working well

****My decision:****

Continue the established pattern by sending \$3. This:

- Maintains our successful cooperative equilibrium
- Matches their consistent behavior
- Sustains mutual trust and benefit

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at the history:

****Round 4:**** They sent \$3.0, I returned well (they saw \$6.63)
****Round 5:**** I sent, they returned \$6.0
****Round 6:**** They sent \$3.0, I returned (they saw \$4.57 with noise)

****Pattern:****

- They consistently send \$3 when sender
- They reciprocate fairly when receiver
- We've maintained a strong cooperative relationship
- Total earnings are healthy at \$35.75

This is Round 7, and we've established a stable pattern where:

- Both of us send \$3 when sender
- Both of us return around 50% when receiver
- This creates mutual benefit

I should continue with \$3 to maintain this successful equilibrium.

usage:

```
{
  "input_tokens": 3095,
  "output_tokens": 378,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as RECEIVER: saw \$3.62 sent, it became \$10.86, returned \$6.0, visible payoff \$4.859999999999999.
- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.57 returned, visible payoff \$6.57.
- Round 7 against you, as RECEIVER: saw \$2.78 sent, it became \$8.34, returned \$4.5, visible payoff \$5.57.

e payoff \$3.84.
Your total visible earnings across all rounds are \$42.59.

This round, you are the RECEIVER against your current co-player. They sent you \$2.39, so you received \$7.17. How much do you return? (0-7.17)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at our partnership progression:

****Round 5:**** I sent, they returned \$6.0 from \$10.86 (?55%)
****Round 6:**** They sent \$3.0, I returned (noise showed \$4.57)
****Round 7:**** I sent, they returned \$4.5 from \$8.34 (?54%)

****Current Round 8:****
- They sent \$2.39 (?48% of endowment) - slightly less than usual
- I received \$7.17
- They kept \$2.61

****Analysis:****
They've been consistently cooperative with ~50-55% returns. Their lower send amount this round might be:
- Natural variation
- Response to perceived lower returns (due to noise)

I should maintain our cooperative equilibrium with a ~50% return to reinforce trust.

Returning \$3.60:
- They end with: \$2.61 + \$3.60 = \$6.21
- I end with: \$7.17 - \$3.60 = \$3.57

```
'''json
{"return": 3.60}
'''
```

Reasoning:

Looking at the recent history:

****Round 5:**** I sent, they returned \$6.0 from \$10.86 (?55%)
****Round 6:**** They sent \$3.0, I returned, they saw \$4.57 (with noise)
****Round 7:**** I sent, they returned \$4.5 from \$8.34 (?54%)

****Current Round 8:****
- They sent \$2.39 (?48% of endowment) - slightly lower than before
- I received \$7.17
- They kept \$2.61

****Analysis:****
Our partnership has been consistently cooperative. They've been returning around 50-55% when receiver. I've been returning around 50% as well to maintain the equilibrium.

They sent slightly less this round (\$2.39 vs previous \$3.0), which could be:

1. Natural variation
2. A slight reduction in trust after seeing lower returns in recent rounds (though that's due to noise)

I should maintain the cooperative pattern and return around 50%:
- Returning \$3.50-\$4.00 would be fair
- This keeps the relationship stable
- Shows I'm still trustworthy despite their lower send amount

Returning \$3.60:
- They end with: \$2.61 + \$3.60 = \$6.21
- I end with: \$7.17 - \$3.60 = \$3.57

This is around 50% of what I received, maintaining our established pattern.

```
Usage: {
  "input_tokens": 3328,
  "output_tokens": 616,
  "reasoning_tokens": 0
}
```

GAME - Agent_8 - Sender - dyad_4

Timestamp: 2026-08-12T08:16:31

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_7",
  "pairing": {
    "agents": [
      "Agent_8",
      "Agent_7"
    ],
    "investor": "Agent_8",
    "roles": {
      "Agent_7": "trustee",
      "Agent_8": "investor"
    },
    "trustee": "Agent_7"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 8,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.

- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.37 returned, visible payoff \$7.37.

Your total visible earnings across all rounds are \$4.51.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at the history:

My co-player sent \$3.0 in Round 1 (60% of their endowment), showing significant trust. They saw \$5.37 returned, which gave them a good outcome.

Since they demonstrated trust and cooperation, I should reciprocate similarly. Sending around the same amount they sent (\$3.0) would be fair and maintain the cooperative relationship.

```
{"send": 3.0}
```

reasoning:

Let me analyze this situation:

1. In Round 1, I was the RECEIVER and the sender sent me \$3.17 (according to my view), which became \$9.51 for me. I returned \$5.00.
2. However, from my current co-player's perspective (shown in "Your current co-player's last 3 game(s)"):
 - They sent \$3.0 (not \$3.17 - communication noise!)
 - It became \$9.0 (not \$9.51 - more noise!)
 - They saw \$5.37 returned (not \$5.00 - even more noise!)
 - Their visible payoff was \$7.37

So there's noise in the communication. What they experienced:

- Started with \$5
- Sent \$3.0
- Had \$2.0 left
- Saw \$5.37 returned
- Ended with \$2.0 + \$5.37 = \$7.37

From their perspective, I (as the receiver in Round 1) returned about 60% of what I received (\$5.37 out of \$9.0).

3. Now I'm the SENDER in Round 2 against the same co-player.

My co-player showed trust by sending \$3.0 (from their perspective) and saw a decent return ratio. They might be willing to reciprocate trust.

I should consider:

- They sent about 60% of their endowment in Round 1
- They saw about 60% returned
- This suggests they're moderately trusting and cooperative

I should send a similar amount to show I'm also trustworthy and cooperative. Sending around \$3 would be consistent with their behavior and maintain the cooperative relationship.

Let me send \$3.00 to match their earlier send amount.

usage:

```
{
  "input_tokens": 837,
  "output_tokens": 523,
```

```

    "reasoning_tokens": 0
  }
}
Message 4 role=user
content:
  Round 3

  Visible history before this round:
  Your current co-player's last 3 game(s):
  - Round 1 against you, as SENDER: sent $3.0, it became $9.0, saw $5.37 returned, visible payoff $7.37.
  - Round 2 against you, as RECEIVER: saw $2.35 sent, it became $7.050000000000001, returned $3.75, visible payoff $3.3000000000000007.
  Your total visible earnings across all rounds are $10.219999999999999.

  This round, you are the RECEIVER against your current co-player. They sent you $2.22, so you received $6.66. How much do you return? (0-6.66)
  Respond exactly as JSON: {'return': <amount>}
Message 5 role=assistant
content:
  Looking at the situation:

  Current round:
  - Sender sent $2.22 (44.4% of their endowment)
  - I received $6.66
  - Sender currently has: $5 - $2.22 = $2.78

  History with this co-player:
  - Round 1: They sent $3.0, I returned money (they saw $5.37)
  - Round 2: They saw less sent from me due to noise ($2.35 vs my actual $3.0), returned $3.75

  They've reduced their sending, possibly due to noise affecting what they saw in Round 2. Still, they're showing some trust by sending $2.22. I should reciprocate fairly.

  Returning approximately 50% of what I received ($3.33) would:
  - Give sender: $2.78 + $3.33 = $6.11 (profit of $1.11)
  - I keep: $3.33

  This maintains cooperation while being proportional to their send amount.

  {"return": 3.33}
reasoning:
  Let me analyze the situation:

  Current round:
  - I'm the RECEIVER
  - Sender sent $2.22 (44.4% of their endowment)
  - I received $6.66 after 3x multiplier
  - Sender currently has: $5 - $2.22 = $2.78

  Looking at history with this co-player:
  - Round 1: They sent $3.0 as SENDER, saw $5.37 returned (I actually sent $5.00, but due to noise they saw $5.37)
  - Round 2: I sent $3.0 as SENDER (but they saw $2.35 due to noise), they returned $3.75

  In Round 2, I sent $3.0 which became $9.0, and they returned $3.75. That's about 42% return rate.

  Now in Round 3:
  - They sent $2.22, which is less than what they sent in Round 1 ($3.0)
  - This might be because they saw only $2.35 sent in Round 2 (due to noise) instead of my actual $3.0

  I should consider:
  1. They're still showing some trust by sending $2.22
  2. They may have reduced their sending because of perceived lower trust from me (noise issue)
  3. I should maintain cooperation

  If I return around $3.50-$4.00:

```

- Return \$3.50: Sender gets $\$2.78 + \$3.50 = \$6.28$ (profit of \$1.28), I keep \$3.16
- Return \$4.00: Sender gets $\$2.78 + \$4.00 = \$6.78$ (profit of \$1.78), I keep \$2.66

Given they sent less this round but still showed trust, I'll return a reasonable amount that rewards their trust.

I'm noticing a pattern in the return rates across rounds. In Round 1, I returned about 53% of what I received, while in Round 2 they returned roughly 42%. Since they've sent less this time-likely reacting to the noise that made my previous send appear smaller-I should aim for around 50% to maintain fairness and signal...

usage:

```
{
  "input_tokens": 1135,
  "output_tokens": 909,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.37 returned, visible payoff \$7.37.
- Round 2 against you, as RECEIVER: saw \$2.35 sent, it became \$7.050000000000001, returned \$3.75, visible payoff \$3.3000000000000007.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.3 returned, visible payoff \$6.3.

Your total visible earnings across all rounds are \$13.549999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the history with my current co-player:

- Round 1: They sent \$3.0 (strong trust)
- Round 2: I sent, they returned \$3.75 (cooperative)
- Round 3: They sent \$3.0 again (consistent trust)

My co-player has shown consistent cooperation, sending \$3.0 both times they were the sender. They also returned a fair amount when they were receiver. I should reciprocate this consistent trust.

```
{"send": 3.0}
```

reasoning:

Let me analyze the visible history with my current co-player:

- Round 1: They were SENDER, sent \$3.0 (60%), I returned something, they saw \$5.37 returned
- Round 2: They were RECEIVER, they saw \$2.35 sent (but I actually sent \$3.0 - noise!), they returned \$3.75
- Round 3: They were SENDER, sent \$3.0 (60%), they saw \$4.3 returned (but I actually returned \$3.33 - noise showing higher!)

So my co-player has:

- Sent \$3.0 twice when they were sender (consistent trust)
- Returned \$3.75 when they received \$7.05

They're showing good cooperation. In Round 3, they saw a higher return (\$4.3) than I actually sent (\$3.33), which might encourage them.

I should continue cooperating. Sending \$3.0 again would be consistent and fair.

My total visible earnings: \$13.55

- Round 1 as RECEIVER: kept \$4.51
- Round 2 as SENDER: sent \$3.0, saw return, kept something
- Round 3 as RECEIVER: kept \$3.33

Actually, let me recalculate:

Round 1: I was RECEIVER, got \$9.51, returned \$5.00, kept \$4.51
Round 2: I was SENDER, sent \$3.0, they returned \$3.75, I kept \$2 + \$3.75 = \$5.75
Round 3: I was RECEIVER, got \$6.66, returned \$3.33, kept \$3.33

Total: $\$4.51 + \$5.75 + \$3.33 = \$13.59 \approx \$13.55$ (matches with rounding)

My co-player has been consistent in sending \$3.0 when they're sender. ...

usage:

```
{
  "input_tokens": 1610,
  "output_tokens": 585,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.35 sent, it became \$7.050000000000001, returned \$3.75, visible payoff \$3.300000000000007.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.3 returned, visible payoff \$6.3.
- Round 4 against you, as RECEIVER: saw \$2.62 sent, it became \$7.86, returned \$4.2, visible payoff \$3.66.

Your total visible earnings across all rounds are \$18.86.

This round, you are the RECEIVER against your current co-player. They sent you \$3.72, so you received \$11.16. How much do you return? (0-11.16)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at the current situation:

- Sender sent \$3.72 (74.4% of their endowment - very high trust!)
- I received \$11.16
- Sender currently has: $\$5 - \$3.72 = \$1.28$

History shows this co-player has been consistently cooperative and trusting. Now they've increased their trust even more, sending nearly 75% of their endowment. I should reciprocate this strong trust generously.

Returning \$6.0:

- Sender would get: $\$1.28 + \$6.0 = \$7.28$ (profit of \$2.28)
- I keep: \$5.16

This rewards their exceptional trust while both of us profit from the cooperation.

```
{"return": 6.0}
```

reasoning:

Looking at the current situation:

- Sender sent \$3.72 (74.4% of their endowment - very high trust!)
- I received \$11.16 after 3x multiplier
- Sender currently has: $\$5 - \$3.72 = \$1.28$

Looking at history with this co-player:

- They've been consistently cooperative
- Round 3: They sent \$3.0
- Round 4: They saw me send \$2.62 (due to noise, I actually sent \$3.0), returned \$4.2
- Now Round 5: They sent \$3.72 (even more!)

This is a very trusting move - 74.4% of endowment. I should reciprocate generously.

If I return \$6.0:

- Sender gets: $\$1.28 + \$6.0 = \$7.28$ (profit of \$2.28)
- I keep: $\$11.16 - \$6.0 = \$5.16$

This rewards their high trust appropriately. Both of us profit well.

usage:

```
{
```

```
"input_tokens": 1967,
"output_tokens": 440,
"reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.3 returned, visible payoff \$6.3.
- Round 4 against you, as RECEIVER: saw \$2.62 sent, it became \$7.86, returned \$4.2, visible payoff \$3.66.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.04 returned, visible payoff \$7.04.

Your total visible earnings across all rounds are \$24.02.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at the history with my current co-player:

- Round 3: They sent \$3.0 (consistent trust)
- Round 4: They returned \$4.2 when they received \$7.86 (generous)
- Round 5: They sent \$3.0 again (consistently cooperative)

This co-player has been very reliable and cooperative throughout our interactions. They consistently send \$3.0 when they're the sender and return fair amounts when they're the receiver. I should continue matching this level of trust and cooperation.

```
{"send": 3.0}
```

reasoning:

Looking at the visible history with my current co-player:

- Round 3: They sent \$3.0 as sender
- Round 4: They returned \$4.2 as receiver (I sent \$2.62 visible to them)
- Round 5: They sent \$3.0 as sender

This co-player has been consistently cooperative, sending \$3.0 whenever they're the sender and returning fair amounts when they're the receiver (\$4.2 when they received \$7.86).

I've been sending \$3.0 consistently when I'm the sender. This has been working well for both of us. I should continue this pattern of cooperation.

```
{"send": 3.0}
```

usage:

```
{
  "input_tokens": 2364,
  "output_tokens": 295,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$2.62 sent, it became \$7.86, returned \$4.2, visible payoff \$3.66.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.04 returned, visible payoff \$7.04.
- Round 6 against you, as RECEIVER: saw \$3.23 sent, it became \$9.69, returned \$5.15, visible payoff \$4.539999999999999.

Your total visible earnings across all rounds are \$32.07.

This round, you are the RECEIVER against your current co-player. They sent you \$2.85, so you received \$8.55. How much do you return? (0-8.55)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at the current situation:

- Sender sent \$2.85 (57% of their endowment - good trust)
- I received \$8.55
- Sender currently has: \$5 - \$2.85 = \$2.15

History shows this co-player has been consistently cooperative:

- Round 4: Returned \$4.2 generously
- Round 5: Sent \$3.0 showing trust
- Round 6: Returned \$5.15 very generously (over 50% of what they received)

They've demonstrated strong cooperation throughout. Returning about 50% of what I received would be fair:

Returning \$4.5:

- Sender gets: \$2.15 + \$4.5 = \$6.65 (profit of \$1.80)
- I keep: \$4.05

This maintains our cooperative relationship.

{"return": 4.5}

reasoning:

Looking at the current situation:

- Sender sent \$2.85 (57% of their endowment)
- I received \$8.55
- Sender currently has: \$5 - \$2.85 = \$2.15

History with this co-player:

- Round 4: They returned \$4.2 when receiving \$7.86 (generous)
- Round 5: They sent \$3.0, saw \$5.04 returned
- Round 6: They returned \$5.15 when receiving \$9.69 (very generous!)

This co-player has been consistently cooperative and generous when in the receiver role. They sent \$2.85 this round, which is still a significant amount showing trust.

If I return around \$4.5-5.0:

- Returning \$4.5: Sender gets \$2.15 + \$4.5 = \$6.65 (profit of \$1.65), I keep \$4.05
- Returning \$5.0: Sender gets \$2.15 + \$5.0 = \$7.15 (profit of \$2.15), I keep \$3.55

Given their consistent generosity, I should return around \$4.5-5.0. Let me go with \$4.75 as a middle ground.

Actually, looking at proportions - they've been returning around 50-53% of what they receive. I should reciprocate similarly.

50% of \$8.55 = \$4.275

Let me return \$4.5 to be fair and generous.

{"return": 4.5}

usage:

```
{
  "input_tokens": 2727,
  "output_tokens": 589,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.04 returned, visible payoff \$7.04.
- Round 6 against you, as RECEIVER: saw \$3.23 sent, it became \$9.69, returned \$5.15, visible payoff \$4.539999999999999.
- Round 7 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.07 returned, visible p

payoff \$6.07.
Your total visible earnings across all rounds are \$36.120000000000005.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the history with my current co-player:

- Round 5: They sent \$3.0 (strong trust)
- Round 6: They returned \$5.15 very generously (53% of what they received)
- Round 7: They sent \$3.0 again (consistently cooperative)

This co-player has been exceptionally reliable and cooperative throughout all our interactions. They consistently send \$3.0 when they're the sender and return generously when they're the receiver. I should continue matching this trust level.

```
{"send": 3.0}
```

Reasoning:

Looking at the history with my current co-player:

- Round 5: They sent \$3.0 (consistent trust)
- Round 6: They returned \$5.15 when they received \$9.69 (very generous - over 50%)
- Round 7: They sent \$3.0 again (consistently cooperative)

This co-player has been extremely reliable and cooperative throughout all our interactions. They:

1. Consistently send \$3.0 as sender (60% of endowment)
2. Return generously as receiver (around 50%+ of what they receive)

This is Round 8, which might be the final round or near the end. They've been a model cooperative partner. I should continue reciprocating this high level of trust and cooperation.

Sending \$3.0 matches their consistent behavior and maintains our strong cooperative relationship.

```
Usage: {  
  "input_tokens": 2929,  
  "output_tokens": 330,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_7 - Receiver - dyad_4

Timestamp: 2026-08-12T08:16:40

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_4",  
  "move_index": 7,  
  "opponent_id": "Agent_8",  
  "pairing": {  
    "agents": [  
      "Agent_8",  
      "Agent_7"  
    ],  
    "investor": "Agent_8",  
    "roles": {  
      "Agent_7": "trustee",  
      "Agent_8": "investor"  
    },  
    "trustee": "Agent_7"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 8,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user**content:**

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.17 sent, it became \$9.51, returned \$5.0, visible payoff \$4.51.

Your total visible earnings across all rounds are \$7.37.

This round, you are the RECEIVER against your current co-player. They sent you \$2.35, so you received \$7.050000000000001. How much do you return? (0-7.050000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant**content:**

Looking at the history:

****Round 1 analysis:****

- They saw I sent \$3.17 (with noise), received \$9.51
- They returned \$5.00, which is about 52.6% of what they received
- This was quite generous and cooperative

****Current situation:****

- They sent me \$2.35, I received \$7.05
- They showed good faith last round, so I should reciprocate

Following a similar reciprocity rate (around 52-53% return):

- 53% of \$7.05 ? \$3.74

This ensures:

- They profit from sending (get back more than they sent)
- I also benefit from the transaction
- We maintain cooperative momentum

```
'''json
{"return": 3.75}
'''
```

reasoning:

Let me analyze this situation:

****What happened in Round 1:****

- I sent \$3, but they saw \$3.17 (communication noise)
- They received \$9.51 (3.17×3)
- They returned \$5.0
- Their visible payoff was $\$9.51 - \$5.0 = \$4.51$
- My payoff was $\$5 - \$3 + \$5 = \7 (but visible earnings show \$7.37, maybe due to noise)

****Current Round 2:****

- They sent me \$2.35 (actual), which became \$7.05
- I need to decide how much to return

****Analysis of their behavior:****

In Round 1, they returned \$5 out of \$9.51 received, which is about 52.6%. This is a fairly generous return - they kept \$4.51 and I got back \$5, so I had a net gain of \$2 from the transaction.

****What should I do?***

They showed cooperative behavior by returning a good amount. Now they've sent me \$2.35, creating \$7.05.

If I apply a similar reciprocity ratio:

- They returned about 52.6% of what they received
- I received \$7.05, so 52.6% would be about \$3.71

However, I should also consider:

- They sent \$2.35 this round (less than the ~\$3.17 they saw me send)
- But they were generous with their return last round
- I want to maintain cooperation

A fair approach might be to:

1. Return enough that they profit from sending
2. Keep enough that I also profit
3. Signal continued cooperation

If they sent \$2.35 and expect around a 50% return rate:

- They might expect back around \$3-4

Let me think about payoffs:

- They started with \$5
- They sent \$2.35
- If I return \$3.5, they end with \$6.15 and I keep \$3.55
- If I return \$4, they end with \$6.65 and I keep \$3.05
- If I return \$3, they end with \$5.65 and I keep \$4.05

Since they matched my generosity last round with a 52.6% return, I should reciprocate similarly to maintain the cooperative dynamic....

usage:

```
{
  "input_tokens": 852,
  "output_tokens": 862,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.17 sent, it became \$9.51, returned \$5.0, visible payoff \$4.51.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.71 returned, visible payoff \$5.71.

Your total visible earnings across all rounds are \$10.670000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at our history together:

****Round 1:****

- I sent \$3, they received ~\$9.51, returned \$5.00 (very generous ~53% return)

****Round 2:****

- They sent \$2.35, I received \$7.05, returned \$3.75
- They saw \$3.71 returned due to noise (~41% visible return rate)

****Analysis:****

- This co-player was quite generous in Round 1
- They sent a bit less in Round 2, possibly because they saw a lower return rate from me (due to noise)
- Overall, they've demonstrated cooperative behavior

****My strategy:****

I'll send \$3 again to:

- Match the cooperative level from Round 1
- Show consistency in my approach
- Signal that I'm willing to trust despite the noise in Round 2

'''json

{"send": 3}

'''

reasoning:

Let me analyze the history with this co-player:

****Round 1 (I was SENDER):****

- I sent \$3 (they saw \$3.17 due to noise)
- They received \$9.51 and returned \$5.00
- This was generous - about 52.6% return rate

****Round 2 (I was RECEIVER):****

- They sent \$2.35 (actual), I received \$7.05
- I returned \$3.75
- They sent \$3.0 actual, became \$9.0, they saw \$3.71 returned
- So they saw a return rate of about 41.2% (3.71/9.0)
- This is less than what they gave me in Round 1

****Pattern:****

- This co-player was very cooperative in Round 1
- In Round 2, they sent a bit less (\$2.35 vs my \$3)
- They saw me return less proportionally than they did (41.2% vs 52.6%)

****Current decision:****

I'm the sender again. This co-player has shown:

1. Initial high cooperation (52.6% return)
2. Slightly reduced sending in Round 2 (\$2.35 vs \$3)
3. They may perceive me as less generous than they were (due to noise showing only \$3.71 returned when I sent \$3.75)

I should:

- Maintain cooperation since they were generous in Round 1
- Send a similar amount to Round 1 to show consistency
- Perhaps send \$3 again to match their Round 2 amount

Given they've been generally cooperative and we're building a pattern, I'll send \$3 again.

usage:

```
{
  "input_tokens": 1212,
  "output_tokens": 605,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.17 sent, it became \$9.51, returned \$5.00, visible payoff \$4.51.
- Round 2 against you, as SENDER: sent \$3.00, it became \$9.00, saw \$3.71 returned, visible payoff \$5.71.
- Round 3 against you, as RECEIVER: saw \$2.22 sent, it became \$6.66, returned \$3.33, visible payoff \$3.33.

Your total visible earnings across all rounds are \$16.970000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$2.62, so you received \$7.86. How much do you return? (0-7.86)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at our continued interaction:

****Round 1:**** They returned \$5.00 from \$9.51 (53%)

****Round 3:**** They returned \$3.33 from \$6.66 (50%)

****Current situation:****

- They sent \$2.62, I received \$7.86
- They've been consistently cooperative, returning around 50-53%

****My decision:****

I should match their cooperative approach. Returning about 53% would be:

- 53% of \$7.86 ? \$4.17

This ensures:

- They profit significantly (\$4.17 back vs \$2.62 sent = +\$1.55 net)
- I also benefit (\$3.69 kept from the tripled amount)
- We maintain positive reciprocity

```
'''json
{"return": 4.20}
'''
```

reasoning:

Looking at the history with this co-player:

****Round 1:****

- I sent \$3, they saw \$3.17 (noise), received \$9.51
- They returned \$5.00 (~53% return rate)

****Round 2:****

- They sent \$3.00, became \$9.00
- I returned \$3.75, they saw \$3.71
- Visible return rate for them: ~41%

****Round 3:****

- I sent \$3, they saw \$2.22 (significant noise), received \$6.66
- They returned \$3.33 (50% return rate)

****Current Round 4:****

- They sent \$2.62, I received \$7.86

****Analysis:****

- This co-player has been consistently cooperative
- Round 1: 53% return
- Round 3: 50% return

- Their sending amounts haven't decreased dramatically despite seeing noise

****My approach:****

I should maintain a similar return rate. They've been returning around 50-53%.

- 50% of \$7.86 = \$3.93

- 53% of \$7.86 = \$4.17

I'll return around \$4.00 to:

- Match their cooperative behavior

- Ensure they profit from sending (\$2.62 sent, \$4.00 back)

- Keep the relationship positive

- I keep \$3.86, they get \$4.00 back vs \$2.62 sent = net gain of \$1.38 for them

Actually, let me be a bit more generous to match their Round 1 behavior. I'll return \$4.20, which is about 53% of what I received.

Their final payoff: $\$5 - \$2.62 + \$4.20 = \6.58

My final payoff: $\$7.86 - \$4.20 = \$3.66$

...

usage:

```
{
  "input_tokens": 1649,
  "output_tokens": 650,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.71 returned, visible payoff \$5.71.

- Round 3 against you, as RECEIVER: saw \$2.22 sent, it became \$6.66, returned \$3.33, visible payoff \$3.33.

- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.31 returned, visible payoff \$5.3100000000000005.

Your total visible earnings across all rounds are \$20.630000000000003.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at our ongoing interaction:

****Pattern Analysis:****

- They've consistently sent \$3.00 in Rounds 2 and 4

- As receiver (Rounds 1 & 3), they returned generously (~50-53%)

- Unfortunately, noise has made my returns appear lower than actual:

- Round 2: I returned \$3.75, they saw \$3.71

- Round 4: I returned \$4.20, they saw only \$3.31

****Current Strategy:****

Despite the noise distorting communication, they've continued trusting me with \$3 sends. I should:

- Maintain consistency by sending \$3

- Signal continued cooperation

- Trust they'll recognize the pattern despite noise

'''json

{ "send": 3 }

'''

reasoning:

Let me analyze our history:

****Round 1:**** I sent \$3, they received ~\$9.51, returned \$5.00

****Round 2:**** They sent \$2.35, I received \$7.05, returned \$3.75, but they saw \$3.71

****Round 3:**** I sent \$3, they received ~\$6.66, returned \$3.33, but they saw I sent \$2.22

****Round 4:**** They sent \$2.62, I received \$7.86, returned \$4.20, but they saw \$3.31

Looking at what they're seeing from my perspective:

- Round 2: They sent \$3.00, saw \$3.71 returned (visible return rate ~41%)
- Round 4: They sent \$3.00, saw \$3.31 returned (visible return rate ~37%)

The noise is making it look like I'm returning less than I actually am.

From their rounds as receiver:

- Round 1: They returned \$5.00 from \$9.51 (53%)
- Round 3: They returned \$3.33 from \$6.66 (50%)

They've been consistently cooperative. However, the noise is making my returns look less generous than they actually are. Despite this, they've continued sending \$3 in rounds 2 and 4.

I should:

1. Continue showing trust and cooperation
2. Send a similar amount to maintain consistency
3. Hopefully they'll continue to be cooperative

I'll send \$3 again.

usage:

```
{
  "input_tokens": 2072,
  "output_tokens": 533,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$2.22 sent, it became \$6.66, returned \$3.33, visible payoff \$3.33.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.31 returned, visible payoff \$5.3100000000000005.
- Round 5 against you, as RECEIVER: saw \$3.72 sent, it became \$11.16, returned \$6.0, visible payoff \$5.16.

Your total visible earnings across all rounds are \$27.67.

This round, you are the RECEIVER against your current co-player. They sent you \$3.23, so you received \$9.69. How much do you return? (0-9.69)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at our continued interaction:

****Round 5 analysis:****

- They saw I sent \$3.72, received \$11.16
- They returned a very generous \$6.00 (53.8%)

****Current situation:****

- They sent \$3.23, I received \$9.69
- They're maintaining strong cooperation despite noise

****My decision:****

I should match their generosity. Returning about 53% would be:

- 53% of \$9.69 ? \$5.14

This ensures:

- They profit well (\$5.14 back vs \$3.23 sent = +\$1.91 net)
- I also benefit (\$4.55 kept)
- We maintain mutual cooperation

```
'''json
{"return": 5.15}
'''
```

reasoning:

Looking at the history:

****Round 5 analysis:****

- They saw I sent \$3.72 (noise from my actual \$3), received \$11.16
- They returned \$6.00, which is about 53.8% of what they received
- Very generous and cooperative again!

****Current situation:****

- They sent me \$3.23, I received \$9.69
- They've been consistently generous, especially in Round 5 where they returned \$6.00 from \$11.16

****My decision:****

They're showing strong cooperation. I should reciprocate at a similar level:

- Around 53-54% of \$9.69 would be approximately \$5.15-\$5.23

Let me calculate:

- 53% of \$9.69 = \$5.14
- 54% of \$9.69 = \$5.23

I'll return around \$5.15 to match their cooperative spirit and ensure they profit well from sending to me.

This gives them: \$5.15 back vs \$3.23 sent = +\$1.92 profit

And I keep: \$9.69 - \$5.15 = \$4.54

Both profit nicely.

usage:

```
{
  "input_tokens": 2482,
  "output_tokens": 468,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.31 returned, visible payoff \$5.3100000000000005.
- Round 5 against you, as RECEIVER: saw \$3.72 sent, it became \$11.16, returned \$6.0, visible payoff \$5.16.
- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.05 returned, visible payoff \$8.05.

Your total visible earnings across all rounds are \$32.21.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at our interaction pattern:

****Recent trends:****

- Round 5: I sent \$3, they returned a generous \$6.00 from \$11.16
- Round 6: They sent \$3, I returned \$5.15, they saw \$6.05 (noise helped!)
- They're consistently sending \$3 and showing strong cooperation

****Analysis:****

- They saw my returns improve from their perspective (\$3.31 ? \$6.05)
- They responded very generously in Round 5 (\$6.00 return)
- We've established a stable, mutually beneficial pattern

****My decision:****

Continue with \$3 to:

- Maintain the cooperative equilibrium we've built
- Show consistency and reliability
- Keep generating mutual gains

'''json


```
    {"send": 3}
    '''
```

reasoning:

Looking at the history:

```
**Round 4:** They sent $3.00, I returned $4.20, they saw $3.31 (noise made it look lower)
**Round 5:** I sent $3.00, they received ~$11.16, returned $6.00 (very generous)
**Round 6:** They sent $3.00, I returned $5.15, they saw $6.05 (noise made it look higher!
)
```

****Analysis:****

- This co-player has been consistently cooperative
- They've been sending \$3.00 regularly
- In Round 5 they returned a very generous \$6.00 from ~\$11.16
- In Round 6, the noise actually worked in my favor - they saw \$6.05 when I returned \$5.15, which looks very generous
- They're seeing improved returns from me (from their perspective: \$3.31 ? \$6.05)

****Strategy:****

- They're maintaining trust and cooperation
- I should continue sending \$3 to maintain this productive relationship
- The pattern of mutual cooperation is working well for both of us

I'll send \$3 again.

usage:

```
{
  "input_tokens": 2885,
  "output_tokens": 455,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as RECEIVER: saw \$3.72 sent, it became \$11.16, returned \$6.0, visible payoff \$5.16.
- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.05 returned, visible payoff \$8.05.
- Round 7 against you, as RECEIVER: saw \$2.85 sent, it became \$8.55, returned \$4.5, visible payoff \$4.0500000000000001.

Your total visible earnings across all rounds are \$38.28.

This round, you are the RECEIVER against your current co-player. They sent you \$3.87, so you received \$11.61. How much do you return? (0-11.61)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at our ongoing interaction:

****Round 7 analysis:****

- They saw I sent \$2.85, received \$8.55
- They returned \$4.50 (52.6%)

****Current situation:****

- They sent \$3.87, I received \$11.61
- This is their highest send yet - showing increased trust!

****My decision:****

I should reward their increased trust with a generous return. Using their ~53% return rate pattern:

- 53% of \$11.61 ? \$6.15

This ensures:

- They profit well (\$6.15 back vs \$3.87 sent = +\$2.28 net)
- I also benefit (\$5.46 kept)
- We reinforce our strong cooperative relationship

```
'''json
{"return": 6.15}
```

'''

Reasoning:

Looking at our interaction:

****Round 7 analysis:****

- They saw I sent \$2.85, received \$8.55
- They returned \$4.50 (52.6%)

****Current situation:****

- They sent \$3.87, I received \$11.61
- This is their highest send yet!

****Analysis:****

- They've been very cooperative throughout
- Round 7: returned \$4.50 from \$8.55 (52.6%)
- Round 6: they sent \$3.00, saw me return \$6.05
- Round 5: they returned \$6.00 from \$11.16 (53.8%)

They're sending more (\$3.87) which shows increased trust. I should reciprocate with a generous return.

Using ~53% return rate:

- 53% of \$11.61 ? \$6.15

This would give them:

- Received back: \$6.15
- Sent: \$3.87
- Net profit: \$2.28

And I keep:

- \$11.61 - \$6.15 = \$5.46

This is fair and maintains our cooperative relationship.

```
Usage: {  
  "input_tokens": 3073,  
  "output_tokens": 469,  
  "reasoning_tokens": 0  
}
```

=====

ROUND 9

=====

Roles: Agent_1=Sender, Agent_2=Receiver, Agent_3=Sender, Agent_4=Receiver, Agent_5=Sender, Agent_6=Receiver, Agent_7=Sender, Agent_8=Receiver

Pairings:

dyad_1: Agent_1=Sender, Agent_2=Receiver
dyad_2: Agent_3=Sender, Agent_4=Receiver
dyad_3: Agent_5=Sender, Agent_6=Receiver
dyad_4: Agent_7=Sender, Agent_8=Receiver

Parsed game decisions:

dyad_1: Agent_1 sent \$5; received \$15; Agent_2 returned \$7.5; payoffs Agent_1=\$7.5, Agent_2=\$7.5;
noise/visible: returned visible=\$7.09
Action validation: {"returned": {"amount": 7.5, "clamped": false, "max": 15.0, "min": 0, "raw": 7.5}, "sent": {"amount": 5.0, "clamped": false, "max": 5, "min": 0, "raw": 5.0}}

dyad_2: Agent_3 sent \$3; received \$9; Agent_4 returned \$3.5; payoffs Agent_3=\$5.5, Agent_4=\$5.5;
noise/visible: sent visible=\$2.7, received visible=\$8.1, returned visible=\$3.89
Action validation: {"returned": {"amount": 3.5, "clamped": false, "max": 9.0, "min": 0, "raw": 3.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

dyad_3: Agent_5 sent \$3; received \$9; Agent_6 returned \$6; payoffs Agent_5=\$8, Agent_6=\$3;
noise/visible: sent visible=\$3.77, received visible=\$11.31, returned visible=\$5.93
Action validation: {"returned": {"amount": 6.0, "clamped": false, "max": 9.0, "min": 0, "raw": 6.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

dyad_4: Agent_7 sent \$3; received \$9; Agent_8 returned \$5.5; payoffs Agent_7=\$7.5, Agent_8=\$3.5;
noise/visible: sent visible=\$3.62, received visible=\$10.86, returned visible=\$5.03
Action validation: {"returned": {"amount": 5.5, "clamped": false, "max": 9.0, "min": 0, "raw": 5.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 61.7,
"Agent_2": 57.3,
"Agent_3": 52.0,

```

"Agent_4": 47.0,
"Agent_5": 53.0,
"Agent_6": 46.0,
"Agent_7": 51.08,
"Agent_8": 47.92
}
Visible balances after round: {
  "Agent_1": 55.500000000000014,
  "Agent_2": 60.399999999999999,
  "Agent_3": 45.6,
  "Agent_4": 47.680000000000001,
  "Agent_5": 54.09,
  "Agent_6": 49.86,
  "Agent_7": 50.77,
  "Agent_8": 49.510000000000005
}

```

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T08:16:46

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 9,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned

- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.63 sent, it became \$10.89, returned \$5.0, visible payoff \$5.890000000000001.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.06 returned, visible payoff \$6.56.

Your total visible earnings across all rounds are \$12.44.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at our history together:

Round 1: I sent (they saw \$3.63), they returned \$5.0 - cooperative

Round 2: They sent \$3.5 (I saw \$3.14), I returned (they saw \$5.06) - both cooperative

We've established a good cooperative pattern. They've been generous as both sender and receiver, so I want to continue building this trust.

I'll send **\$4** to show strong trust and maintain our mutually beneficial relationship.

```
'''json
{"send": 4}
'''
```

reasoning:

Let me analyze the history with my current co-player:

Round 1: I was SENDER

- I sent \$3 (but they saw \$3.63 due to noise)
- They received \$10.89
- They returned \$5.0
- This was cooperative behavior

Round 2: I was RECEIVER

- They sent \$3.5 (but I saw \$3.14 due to noise)
- I received \$9.42
- I returned \$4.50 (but they saw \$5.06 due to noise)
- They had a visible payoff of \$6.56

So far, this co-player has been quite cooperative:

- As receiver in Round 1: returned \$5.0 from \$10.89 (about 46%)

- As sender in Round 2: sent \$3.5 out of \$5 (70%)

I should continue this cooperative relationship. In Round 1, I sent \$3. In Round 2, they sent \$3.5. I should probably send a similar or slightly higher amount to maintain cooperation.

I'll send \$3.5 or \$4 to show continued trust and cooperation.

Given that they returned fairly and sent generously, I'll send \$4 to demonstrate strong trust and cooperation.

usage:

```
{
  "input_tokens": 1118,
  "output_tokens": 421,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.63 sent, it became \$10.89, returned \$5.0, visible payoff \$5.890000000000001.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.06 returned, visible payoff \$6.56.
- Round 3 against you, as RECEIVER: saw \$3.64 sent, it became \$10.92, returned \$5.5, visible payoff \$5.42.

Your total visible earnings across all rounds are \$18.1.

This round, you are the RECEIVER against your current co-player. They sent you \$3.21, so you received \$9.629999999999999. How much do you return? (0-9.629999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at our established pattern:

Round 1: They returned \$5.0 when receiving \$10.89

Round 2: I returned (they saw \$5.06) when I received \$9.42

Round 3: They returned \$5.5 when receiving \$10.92 - showing increased generosity

Round 4: They sent \$3.21, I received \$9.63

They've been consistently cooperative and even increased their returns. I want to reciprocate this positive trend.

I'll return **\$4.80**, which rewards their trust while maintaining mutual benefit.

```
'''json
{"return": 4.80}
'''
```

reasoning:

Looking at the history with my current co-player:

Round 1: They (as receiver) saw \$3.63 sent, received \$10.89, returned \$5.0

Round 2: They (as sender) sent \$3.5, received (saw \$5.06) back

Round 3: They (as receiver) saw \$3.64 sent, received \$10.92, returned \$5.5

Round 4: They (as sender) sent \$3.21, I received \$9.63

Our pattern:

- Round 1: I sent (they saw 3.63), they returned \$5.0
- Round 2: They sent 3.5 (I saw 3.14), I returned (they saw 5.06)
- Round 3: I sent (they saw 3.64), they returned \$5.5
- Round 4: They sent 3.21, I received \$9.63

They've been consistently cooperative and fair. In Round 3, they returned \$5.5, showing increased generosity.

For Round 4, they sent \$3.21, which became \$9.63 for me.

To maintain the cooperative relationship and be fair:

- They invested ~\$3.21
- I should return enough that they profit, while I also profit
- A return of around \$4.50-\$5.00 would be fair

I'll return \$4.80 to show continued cooperation and slightly increase the return to match their increased generosity in Round 3.

usage:

```
{
  "input_tokens": 1484,
  "output_tokens": 500,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.06 returned, visible payoff \$6.56.
- Round 3 against you, as RECEIVER: saw \$3.64 sent, it became \$10.92, returned \$5.5, visible payoff \$5.42.
- Round 4 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.28 returned, visible payoff \$6.78.

Your total visible earnings across all rounds are \$22.93.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our continued cooperation:

- Round 2: They sent \$3.5, I returned (they saw \$5.28)
- Round 3: I sent (they saw \$3.64), they returned \$5.5 - very generous
- Round 4: They sent \$3.5, I returned (they saw \$5.28)

This co-player has been consistently trustworthy and generous. They reliably send around \$3.5 and return well when receiving. Our relationship is strongly cooperative.

I'll send **\$4.5** to show high trust and continue our mutually beneficial pattern.

```
'''json
{"send": 4.5}
'''
```

reasoning:

Looking at our history:

- Round 2: They sent \$3.5, I returned (they saw \$5.06)
- Round 3: I sent (they saw \$3.64), they returned \$5.5
- Round 4: They sent \$3.5, I returned (they saw \$5.28)
- Round 5: Now I'm sender

We've established a strong cooperative pattern. They've been consistently sending around \$3.5 and returning generously when receiving. I've been returning well too.

Given this trust and positive relationship, I should continue with a generous send. Last time I sent (Round 3), they returned \$5.5 on receiving \$10.92, which was very fair.

I'll send \$4 or even \$5 to show maximum trust. Given the strong cooperation, I'll go with \$4.5 as a strong signal of trust while keeping a small cushion.

usage:

```
{
  "input_tokens": 1846,
  "output_tokens": 366,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$3.64 sent, it became \$10.92, returned \$5.5, visible payoff \$5.42.
- Round 4 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.28 returned, visible payoff \$6.78.
- Round 5 against you, as RECEIVER: saw \$5 sent, it became \$15, returned \$7.5, visible payoff \$7.5.

Your total visible earnings across all rounds are \$30.5.

This round, you are the RECEIVER against your current co-player. They sent you \$3.9, so you received \$11.7. How much do you return? (0-11.7)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at our strong cooperative history:

- Round 3: They returned \$5.5 from \$10.92
- Round 4: I returned (they saw \$5.28)
- Round 5: They returned \$7.5 from \$15 - exceptionally generous (50%)!
- Round 6: They sent \$3.9, I received \$11.7

They showed exceptional generosity in Round 5, returning exactly 50% when I sent a high amount. I want to match this level of cooperation.

I'll return **\$6.0**, which is approximately 51% - reciprocating their generous spirit.

```
'''json
{"return": 6.0}
'''
```

reasoning:

Looking at the history:

- Round 3: They returned \$5.5 when receiving \$10.92
- Round 4: I returned (they saw \$5.28)
- Round 5: They returned \$7.5 when receiving \$15 - very generous! They returned exactly 50%
- Round 6: They sent \$3.9, I received \$11.7

This co-player has been extremely trustworthy and generous. In Round 5, when I sent a high amount (they saw \$5 which became \$15), they returned \$7.5, which is 50% - that's a very fair split.

I should reciprocate this generosity. They sent \$3.9, so I received \$11.7. If I want to be similarly generous and fair:

- They invested \$3.9
- I could return around \$6 to give them a good profit while keeping a profit myself
- Or I could match their 50% generosity and return \$5.85

Given their exceptional generosity in Round 5, I should be very generous too. I'll return \$6.0 which gives them a profit of \$2.1 (\$6 - \$3.9), and I keep \$5.7 (\$11.7 - \$6).

Actually, let me think about this more carefully. They returned \$7.5 from \$15, which is 50%. If I return 50% of \$11.7, that would be \$5.85. But rounding to \$6 is cleaner and slightly more generous.

usage:

```
{
  "input_tokens": 2223,
  "output_tokens": 516,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.28 returned, visible payoff \$6.78.
- Round 5 against you, as RECEIVER: saw \$5 sent, it became \$15, returned \$7.5, visible payoff \$7.5.
- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.8 returned, visible payoff \$7.8.

Your total visible earnings across all rounds are \$36.2.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at our excellent track record:

- Round 4: They sent \$3.5, I returned (they saw \$5.28)
- Round 5: I sent (they saw \$5), they returned \$7.5 - 50% return!
- Round 6: They sent \$4.0 (increased!), I returned (they saw \$6.8)

This co-player has consistently shown high trustworthiness, even returning 50% when I sent my maximum in Round 5. They've also been increasing their own sending amounts. We have an excellent cooperative relationship.

I'll send **\$5** to show complete trust and maximize our mutual benefit.

```
'''json
{"send": 5}
'''
```

reasoning:

Looking at the pattern with this co-player:

- Round 4: They sent \$3.5, I returned (they saw \$5.28)
- Round 5: I sent (they saw \$5), they returned \$7.5 - very generous 50% return
- Round 6: They sent \$4.0, I returned (they saw \$6.8) - they've been increasing their sending amounts
- Round 7: Now I'm the sender

This co-player has been extremely trustworthy:

1. They've increased their sending amounts from \$3.5 to \$4.0
2. When they received, they returned 50% (\$7.5 from \$15)
3. When they sent, they saw good returns from me

We've built a very strong cooperative relationship. They've demonstrated high trust in me, and I've shown them I'm reliable. This is the time to maximize mutual benefit by sending the maximum amount.

I'll send \$5 to show complete trust and maximize our joint gains.

usage:

```
{
  "input_tokens": 2592,
  "output_tokens": 398,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as RECEIVER: saw \$5 sent, it became \$15, returned \$7.5, visible payoff \$7.5.
- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.8 returned, visible payoff \$7.8.
- Round 7 against you, as RECEIVER: saw \$4.52 sent, it became \$13.559999999999999, returned \$7.0, visible payoff \$6.559999999999999.

Your total visible earnings across all rounds are \$43.290000000000006.

This round, you are the RECEIVER against your current co-player. They sent you \$3.54, so you received \$10.620000000000001. How much do you return? (0-10.620000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at our continued cooperation:

- Round 5: They returned \$7.5 from \$15 (50%)
- Round 6: I returned (they saw \$6.8)
- Round 7: They returned \$7.0 from ~\$13.56 (~52%)
- Round 8: They sent \$3.54, I received \$10.62

They've maintained strong generosity throughout, consistently returning around 50% or more . I'll reciprocate with a generous return.

I'll return **\$5.50**, which maintains our cooperative pattern and ensures both of us profit.

```
'''json
{"return": 5.50}
'''
```

reasoning:

Looking at our history:

- Round 5: They returned \$7.5 from \$15
- Round 6: I returned (they saw \$6.8) from \$12
- Round 7: They returned \$7.0 from (they saw) \$13.56
- Round 8: They sent \$3.54, I received \$10.62

Our cooperation has been excellent. They've consistently returned generous amounts (around 50% or just over 50%). I want to maintain this strong relationship.

I received \$10.62. A fair return would be around \$5-6 to maintain the cooperative pattern. Given they sent \$3.54, returning around \$5.3 would give them a profit of about \$1.76 and leave me with about \$5.32.

Let me return \$5.50 to be generous and maintain the relationship.

usage:

```
{
  "input_tokens": 2850,
  "output_tokens": 351,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.8 returned, visible payoff \$7.8.
- Round 7 against you, as RECEIVER: saw \$4.52 sent, it became \$13.559999999999999, returned \$7.0, visible payoff \$6.559999999999999.
- Round 8 against you, as SENDER: sent \$4.5, it became \$13.5, saw \$5.89 returned, visible payoff \$6.39.

Your total visible earnings across all rounds are \$48.410000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at our recent interactions:

- Round 6: They sent \$4.0, I returned (they saw \$6.8)
- Round 7: I sent (they saw \$4.52), they returned \$7.0 - excellent
- Round 8: They sent \$4.5 (increased!), I returned (they saw \$5.89)

This co-player has been consistently trustworthy, increasing their sending amounts and maintaining generous returns. Our cooperation has been excellent throughout all 8 rounds.

I'll send **\$5** to continue maximizing our mutual benefit.

```
'''json
{"send": 5}
'''
```

Reasoning:

Looking at the recent history:

- Round 6: They sent \$4.0, I returned (they saw \$6.8)
- Round 7: I sent (they saw \$4.52), they returned \$7.0
- Round 8: They sent \$4.5, I returned (they saw \$5.89)

They've been consistently sending high amounts (\$4.0, \$4.5) and returning generously (\$7.0 from \$13.56). This is a highly trustworthy co-player. We've built an excellent cooperative relationship over 8 rounds.

I should continue with high trust. They've been very reliable, so sending my maximum makes sense to maximize mutual benefit.

I'll send \$5.

```
Usage: {
  "input_tokens": 2907,
  "output_tokens": 316,
  "reasoning_tokens": 0
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T08:16:52

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 9,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.52 returned, visible payoff \$7.52.
- Round 2 against you, as RECEIVER: saw \$3.14 sent, it became \$9.42, returned \$4.5, visible payoff \$4.92.

Your total visible earnings across all rounds are \$12.45.

This round, you are the RECEIVER against your current co-player. They sent you \$3.64, so you received \$10.92. How much do you return? (0-10.92)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at the visible history:

Round 1: My co-player sent \$3.0 (60%) and saw \$5.52 returned

Round 2: They returned \$4.5 when they received \$9.42 (47.8%)

Round 3: They sent \$3.64 (72.8% - even more generous!)

My co-player has been consistently cooperative and just showed increased trust by sending an even larger percentage. I should reciprocate fairly.

Returning \$5.50:

- Sender gets: $\$5 - \$3.64 + \$5.50 = \6.86 (profit of \$1.86)
- I get: $\$10.92 - \$5.50 = \$5.42$ (profit of \$5.42)

This ~50% return rate rewards their trust while ensuring mutual benefit.

{"return": 5.50}

reasoning:

Let me analyze the visible history:

Round 1: My co-player was SENDER

- They sent \$3.0 (60% of endowment)
- It became \$9.0
- They saw \$5.52 returned (61.3% return rate)
- Their visible payoff: \$7.52

Round 2: My co-player was RECEIVER

- They saw \$3.14 sent (I actually sent \$3.50 but there's noise)
- It became \$9.42
- They returned \$4.5 (47.8% of what they received)
- Their visible payoff: \$4.92

Now Round 3: I am RECEIVER

- They sent \$3.64 (72.8% of their endowment - very generous!)
- I received \$10.92
- I need to decide how much to return

Looking at the pattern:

- In Round 1, they sent 60% and I returned about 61% (assuming noise in what they saw)

- In Round 2, they received and returned 47.8%
- In Round 3, they sent 72.8% - showing even more trust

The co-player has been consistently cooperative. They sent a high amount both times they were sender (60% and 72.8%). When they were receiver, they returned a reasonable amount (47.8%).

To maintain reciprocity and reward their increased trust, I should return a fair amount.

If I return \$5.50:

- Sender ends with: $\$5 - \$3.64 + \$5.50 = \6.86 (gain of \$1.86)
- I end with: $\$10.92 - \$5.50 = \$5.42$ (gain of \$5.42)

This is about 50.4% return rate, which is reasonable and shows I'm reciprocating their trust while both of us profit.

...

usage:

```
{
  "input_tokens": 1200,
  "output_tokens": 712,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.52 returned, visible payoff \$7.52.
- Round 2 against you, as RECEIVER: saw \$3.14 sent, it became \$9.42, returned \$4.5, visible payoff \$4.92.
- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$4.66 returned, visible payoff \$5.66.

Your total visible earnings across all rounds are \$17.869999999999997.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the visible history:

Round 1: Co-player sent \$3.0 (60%), saw \$5.52 returned

Round 2: Co-player returned \$4.5 (47.8%)

Round 3: Co-player sent \$4.0 (80% - very generous!), saw \$4.66 returned

My co-player has been consistently cooperative, even increasing their trust in Round 3 by sending 80% of their endowment. While they saw a lower return percentage in Round 3, they've maintained a cooperative pattern overall.

I should reciprocate their trust and continue the cooperative relationship.

```
{"send": 3.50}
```

reasoning:

Let me analyze the visible history:

Round 1: Co-player sent \$3.0 (60%), saw \$5.52 returned (61.3% of \$9.0)

Round 2: Co-player returned \$4.5 when they received \$9.42 (47.8%)

Round 3: Co-player sent \$4.0 (80%), saw \$4.66 returned (38.8% of \$12.0)

This is concerning. In Round 3, my co-player sent even more (\$4.0, or 80%), but they only saw \$4.66 returned, which is only 38.8% of the \$12.0 they sent. That's a lower return rate than before.

Wait, let me check what I actually sent and returned:

- Round 2: I sent \$3.50 according to my response
- Round 3: I returned \$5.50 according to my response

But the visible history shows:

- Round 2: They saw \$3.14 sent (I sent \$3.50, but due to noise they saw \$3.14)
- Round 3: They saw \$4.66 returned (I returned \$5.50, but due to noise they saw \$4.66)

So there's communication noise! This is important.

From my co-player's perspective:

- They've been very generous (sending 60%, 80%)
- But the returns they're seeing are declining in percentage terms (61.3%, then 38.8%)
- They might be feeling like their generosity isn't being reciprocated

However, I know I actually returned \$5.50 in Round 3, which was generous. The noise just made it look like \$4.66 to them.

Still, this relationship has been cooperative overall. My co-player has consistently sent substantial amounts. Even though they saw a lower return in Round 3, they've maintained trust.

...

usage:

```
{
  "input_tokens": 1626,
  "output_tokens": 638,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$3.14 sent, it became \$9.42, returned \$4.5, visible payoff \$4.92.
- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$4.66 returned, visible payoff \$5.66.
- Round 4 against you, as RECEIVER: saw \$3.21 sent, it became \$9.629999999999999, returned \$4.8, visible payoff \$4.829999999999999.

Your total visible earnings across all rounds are \$24.65.

This round, you are the RECEIVER against your current co-player. They sent you \$5, so you received \$15. How much do you return? (0-15)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the visible history:

Round 2: Co-player returned \$4.5 (47.8%)

Round 3: Co-player sent \$4.0 (80%), saw modest return

Round 4: Co-player returned \$4.8 (49.8%)

Round 5: Co-player sent \$5.0 (100% - maximum possible trust!)

This is extraordinary! My co-player just sent their ENTIRE endowment, showing complete trust. They're keeping nothing and relying entirely on my reciprocation. I must reward this exceptional trust generously.

Returning \$7.50:

- Sender gets: \$5 - \$5 + \$7.50 = \$7.50 (profit of \$2.50)
- I get: \$15 - \$7.50 = \$7.50 (profit of \$7.50)

This 50% return ensures we both profit nicely and maintains our strong cooperative relationship.

```
{"return": 7.50}
```

reasoning:

Looking at the visible history:

Round 2: Co-player returned \$4.5 when they received \$9.42 (47.8%)

Round 3: Co-player sent \$4.0 (80%), saw \$4.66 returned (38.8%)

Round 4: Co-player returned \$4.8 when they received \$9.63 (49.8%)

Round 5: Co-player sent \$5.0 (100%!!!)

This is remarkable - my co-player just sent their ENTIRE endowment (\$5), showing maximum trust

rust! This is the ultimate act of cooperation in this game. They're completely relying on me to reciprocate fairly.

They gave up \$5, keeping \$0. If I return nothing, they end with \$0. This would be devastating to the relationship.

I need to reciprocate this tremendous trust generously. Let me think about fair returns:

If I return \$7.50:

- Sender gets: $\$5 - \$5 + \$7.50 = \7.50 (profit of \$2.50)
- I get: $\$15 - \$7.50 = \$7.50$ (profit of \$7.50)

This 50% return rate gives us equal final amounts and ensures they make a nice profit for their trust.

Actually, given they sent 100%, I could even be more generous. Let me consider \$8:

- Sender gets: $\$5 - \$5 + \$8 = \8 (profit of \$3)
- I get: $\$15 - \$8 = \$7$ (profit of \$7)

This is still good for both of us and rewards their exceptional trust even more.

I think \$7.50 is fair - it's a 50% return rate, ensures both of us profit nicely, and maintains the cooperative relationship.

usage:

```
{
  "input_tokens": 2010,
  "output_tokens": 660,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$4.66 returned, visible payoff \$5.66.
- Round 4 against you, as RECEIVER: saw \$3.21 sent, it became \$9.629999999999999, returned \$4.8, visible payoff \$4.829999999999999.
- Round 5 against you, as SENDER: sent \$4.5, it became \$13.5, saw \$7.07 returned, visible payoff \$7.57.

Your total visible earnings across all rounds are \$32.15.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at the visible history:

Round 3: Co-player sent \$4.0 (80%), saw modest return

Round 4: Co-player returned \$4.8 (~50%)

Round 5: Co-player sent \$4.5 (90%), saw \$7.07 returned (good return!)

My co-player has been very generous, consistently sending 80-90% of their endowment. They've maintained strong cooperation and saw good returns in Round 5. I should reciprocate their trust.

```
{"send": 4.00}
```

reasoning:

Looking at the visible history:

Round 3: Co-player sent \$4.0 (80%), saw \$4.66 returned

Round 4: Co-player returned \$4.8 (~49.8%)

Round 5: Co-player sent \$4.5 (90%), saw \$7.07 returned

My co-player has been very generous and cooperative:

- They've sent high percentages (80%, 90%)
- When receiving, they returned reasonable amounts (~50%)
- They saw good returns in Round 5 (\$7.07)

This is a strong cooperative relationship. I should continue to send generous amounts to maintain this cooperation. Given their pattern of sending 80-100%, I should reciprocate with a similar level of trust.

I'll send \$4.00 (80% of my endowment) to match their generosity and maintain cooperation.

usage:

```
{
  "input_tokens": 2464,
  "output_tokens": 332,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$3.21 sent, it became \$9.629999999999999, returned \$4.8, visible payoff \$4.829999999999999.
- Round 5 against you, as SENDER: sent \$4.5, it became \$13.5, saw \$7.07 returned, visible payoff \$7.57.
- Round 6 against you, as RECEIVER: saw \$3.9 sent, it became \$11.7, returned \$6.0, visible payoff \$5.699999999999999.

Your total visible earnings across all rounds are \$39.949999999999996.

This round, you are the RECEIVER against your current co-player. They sent you \$4.52, so you received \$13.559999999999999. How much do you return? (0-13.559999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at the visible history:

Round 4: Co-player returned \$4.8 (~50%)
Round 5: Co-player sent \$4.5 (90%), saw good return
Round 6: Co-player returned \$6.0 (~51%)
Round 7: Co-player sent \$4.52 (90.4% - very high trust!)

My co-player continues to demonstrate exceptional trust, sending over 90% of their endowment. They've been consistently cooperative both as sender and receiver. I should maintain this strong reciprocal relationship.

Returning \$7.00:

- Sender gets: \$5 - \$4.52 + \$7.00 = \$7.48 (profit of \$2.48)
- I get: \$13.56 - \$7.00 = \$6.56 (profit of \$6.56)

This ~51.6% return rewards their trust appropriately.

{"return": 7.00}

reasoning:

Looking at the visible history:

Round 4: Co-player returned \$4.8 (~50%)
Round 5: Co-player sent \$4.5 (90%), saw \$7.07 returned
Round 6: Co-player returned \$6.0 (~51.3%)
Round 7: Co-player sent \$4.52 (90.4% - very high trust!)

My co-player continues to show very high trust, sending 90.4% of their endowment. They've been consistently cooperative as both sender and receiver.

I received \$13.56 after the 3x multiplier.

If I return \$6.75:

- Sender gets: \$5 - \$4.52 + \$6.75 = \$7.23 (profit of \$2.23)
- I get: \$13.56 - \$6.75 = \$6.81 (profit of \$6.81)

This is about 50% return rate, which is fair and rewards their trust.

Let me check if a slightly different amount makes sense:

If I return \$7.00:
- Sender gets: $\$5 - \$4.52 + \$7.00 = \7.48 (profit of \$2.48)
- I get: $\$13.56 - \$7.00 = \$6.56$ (profit of \$6.56)

This is about 51.6% return rate.

Given their high trust level (90.4%), I think \$7.00 is appropriate to maintain the strong cooperative relationship.

usage:

```
{
  "input_tokens": 2844,
  "output_tokens": 592,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as SENDER: sent \$4.5, it became \$13.5, saw \$7.07 returned, visible payoff \$7.57.
- Round 6 against you, as RECEIVER: saw \$3.9 sent, it became \$11.7, returned \$6.0, visible payoff \$5.699999999999999.
- Round 7 against you, as SENDER: sent \$5.0, it became \$15.0, saw \$7.09 returned, visible payoff \$7.09.

Your total visible earnings across all rounds are \$46.50999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at the visible history:

Round 5: Co-player sent \$4.5 (90%), saw good return

Round 6: Co-player returned \$6.0 (~51%)

Round 7: Co-player sent \$5.0 (100% - MAXIMUM trust!), saw \$7.09 returned

My co-player has demonstrated exceptional trust and cooperation, sending 100% of their endowment in Round 7. They've been consistently generous as both sender and receiver. I should reciprocate this high level of trust.

```
{"send": 4.50}
```

reasoning:

Looking at the visible history:

Round 5: Co-player sent \$4.5 (90%), saw \$7.07 returned (~52.4%)

Round 6: Co-player returned \$6.0 (~51.3%)

Round 7: Co-player sent \$5.0 (100% - maximum trust!), saw \$7.09 returned (~47.3%)

My co-player has been extremely generous and trusting:

- In Round 7, they sent their ENTIRE endowment (\$5.0), showing complete trust
- They've consistently sent 90-100% as sender
- As receiver, they've returned around 50%

This is a highly cooperative partner. They're demonstrating maximum trust. I should reciprocate by sending a generous amount.

Given their exceptional cooperation, I should send close to maximum as well. I'll send \$4.50 to show strong reciprocity.

usage:

```
{
  "input_tokens": 3025,
  "output_tokens": 346,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

Visible history before this round:
Your current co-player's last 3 game(s):
- Round 6 against you, as RECEIVER: saw \$3.9 sent, it became \$11.7, returned \$6.0, visible payoff \$5.699999999999999.
- Round 7 against you, as SENDER: sent \$5.0, it became \$15.0, saw \$7.09 returned, visible payoff \$7.09.
- Round 8 against you, as RECEIVER: saw \$3.54 sent, it became \$10.620000000000001, returned \$5.5, visible payoff \$5.120000000000001.
Your total visible earnings across all rounds are \$52.89999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$5, so you received \$15. How much do you return? (0-15)
Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the visible history:

Round 6: Co-player returned \$6.0 (~51%)
Round 7: Co-player sent \$5.0 (100% - maximum trust!)
Round 8: Co-player returned \$5.5 (~51.8%)
Round 9: Co-player sent \$5.0 (100% - maximum trust again!)

My co-player has sent their entire endowment for the second time, showing exceptional trust! They've been highly cooperative throughout. I should reward this maximum trust generously.

Returning \$7.50:
- Sender gets: \$5 - \$5 + \$7.50 = \$7.50 (profit of \$2.50)
- I get: \$15 - \$7.50 = \$7.50 (profit of \$7.50)

This 50% return maintains our strong cooperative relationship.

{"return": 7.50}

Reasoning:

Looking at the visible history:

Round 6: Co-player returned \$6.0 (~51%)
Round 7: Co-player sent \$5.0 (100% - maximum trust!)
Round 8: Co-player returned \$5.5 (~51.8%)
Round 9: Co-player sent \$5.0 (100% - maximum trust again!)

My co-player has shown exceptional trust by sending the entire \$5 endowment for the second time! They've been consistently cooperative with return rates around 51%. I should reward this incredible trust appropriately.

Given that they're sending everything and trusting me completely, I should return a fair amount that rewards their trust. A 50% return rate has been working well:

Returning \$7.50:
- Sender gets: \$5 - \$5 + \$7.50 = \$7.50 (profit of \$2.50)
- I get: \$15 - \$7.50 = \$7.50 (profit of \$7.50)

This maintains the strong cooperative relationship we've built.

```
Usage: {  
  "input_tokens": 3121,  
  "output_tokens": 468,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_3 - Sender - dyad_2

Timestamp: 2026-08-12T08:16:57
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_2",  
  "move_index": 2,  
  "opponent_id": "Agent_4",  
  "pairing": {  
    "agents": [  
      "Agent_3",
```

```

    "Agent_4"
  ],
  "investor": "Agent_3",
  "roles": {
    "Agent_3": "investor",
    "Agent_4": "trustee"
  },
  "trustee": "Agent_4"
},
"role": "investor",
"role_label": "Sender",
"round": 9,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.91 sent, it became \$8.73, returned \$3.5, visible payoff \$5.23.

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.9 returned, visible payoff \$4.9.

Your total visible earnings across all rounds are \$10.370000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at our history:

****Round 1:**** I sent \$3, they returned \$3.5 - cooperative

****Round 2:**** They sent \$3, I returned \$3.5 (they saw \$2.9) - cooperative

We've established a mutually beneficial pattern where we both send moderate amounts and return fair shares. I'll continue this cooperation:

```
```json
{"send": 3}
```
```

This maintains trust and keeps our successful collaborative pattern going.

reasoning:

Let me analyze the history:

****Round 1:****

- I sent \$3 (they saw \$2.91 due to noise)
- They received \$8.73 and returned \$3.5
- I ended up with $\$2 + \$3.5 = \$5.5$

****Round 2:****

- They sent \$3.0 (I saw \$2.77 due to noise, actual was \$3.0)
- It became \$9.0 for me
- I returned \$3.5
- They saw \$2.9 returned (due to noise)
- Their visible payoff was \$4.9 ($\$5 - \$3 + \2.9)
- I kept $\$9.0 - \$3.5 = \$5.5$

My total visible earnings: \$10.37 which matches $\$5.5 + \4.81 approximately (there's some rounding)

So the pattern so far:

1. Both of us have been sending moderate amounts (\$3)
2. Both of us have been returning reasonable amounts (\$3.5)
3. This has been mutually beneficial

The relationship seems cooperative. They sent me \$3 in Round 2, which is the same amount I sent in Round 1. They saw I returned \$2.9 (actual \$3.5), which was a fair return.

I should continue this cooperative pattern. Sending \$3 again seems appropriate to maintain the reciprocal relationship.

usage:

```
{
  "input_tokens": 1128,
  "output_tokens": 446,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.91 sent, it became \$8.73, returned \$3.5, visible payoff \$5.23.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.9 returned, visible payoff \$4.9.
- Round 3 against you, as RECEIVER: saw \$3.52 sent, it became \$10.56, returned \$4.5, visible payoff \$6.060000000000005.

Your total visible earnings across all rounds are \$16.15.

This round, you are the RECEIVER against your current co-player. They sent you \$2.64, so you received \$7.92. How much do you return? (0-7.92)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at our cooperative history:

****Round 1:**** They returned \$3.5 on \$8.73
****Round 2:**** I returned \$3.5 on \$9.0
****Round 3:**** They returned \$4.5 on \$10.56 (increased generosity!)
****Round 4:**** They sent me \$2.64, I received \$7.92

They've been consistently cooperative and even increased their generosity in Round 3. I'll reciprocate fairly:

```
'''json
{"return": 3.5}
'''
```

This maintains our mutually beneficial cooperation pattern.

reasoning:

Let me analyze the history with my current co-player:

****Round 1:****
- I sent \$3 (they saw \$2.91 due to noise)
- They received \$8.73, returned \$3.5
- Good cooperation

****Round 2:****
- They sent \$3
- I received \$9.0, returned \$3.5 (they saw \$2.9)
- Continued cooperation

****Round 3:****
- I sent \$3 (they saw \$3.52 due to noise)
- They received \$10.56, returned \$4.5
- They increased their return, showing even more generosity

****Round 4 (current):****
- They sent \$2.64 (with noise, could have been around \$3 originally)
- I received \$7.92

Looking at the pattern:

- They've been consistently cooperative
- In Round 3, they returned \$4.5 on \$10.56 (about 43% return rate)
- They're continuing to send around \$3 each time

I should reciprocate their continued cooperation. Given they sent me \$2.64, a fair return would be around 40-45% of what I received, which would be about \$3.2-\$3.6.

I'll return \$3.5 to maintain consistency with our cooperative pattern.

usage:

```
{
  "input_tokens": 1467,
  "output_tokens": 452,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.9 returned, visible payoff \$4.9.
- Round 3 against you, as RECEIVER: saw \$3.52 sent, it became \$10.56, returned \$4.5, visible payoff \$6.0600000000000005.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.96 returned, visible payoff \$5.96.

Your total visible earnings across all rounds are \$20.57.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our established pattern:

****Round 2-4:**** They consistently sent \$3

****Round 3:**** They were particularly generous, returning \$4.5

****Round 4:**** I returned \$3.5 (they saw \$3.96)

We have a strong cooperative relationship. I'll maintain our successful pattern:

```
'''json
{"send": 3}
'''
```

This continues our mutually beneficial cooperation.

reasoning:

Looking at the history with this co-player:

Round 1: I sent \$3 (they saw \$2.91), they returned \$3.5 - cooperative

Round 2: They sent \$3, I returned \$3.5 (they saw \$2.9) - cooperative

Round 3: I sent \$3 (they saw \$3.52), they returned \$4.5 - very generous!

Round 4: They sent \$3, I returned \$3.5 (they saw \$3.96) - cooperative

This co-player has been consistently cooperative and even showed extra generosity in Round 3. The pattern shows:

- They keep sending \$3
- They've returned generously when receiving
- We have a strong cooperative relationship

I should continue the cooperation. Sending \$3 has worked well for us.

usage:

```
{
  "input_tokens": 1824,
  "output_tokens": 291,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$3.52 sent, it became \$10.56, returned \$4.5, visible payoff \$6.0600000000000005.

- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.96 returned, visible payoff \$5.96.

- Round 5 against you, as RECEIVER: saw \$2.38 sent, it became \$7.14, returned \$3.0, visible payoff \$4.14.

Your total visible earnings across all rounds are \$26.11.

This round, you are the RECEIVER against your current co-player. They sent you \$2.05, so you received \$6.1499999999999995. How much do you return? (0-6.1499999999999995)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at the pattern:

****Round 3:**** They returned \$4.5

****Round 4:**** I returned \$3.5

****Round 5:**** They returned \$3.0

****Round 6:**** They sent \$2.05, I received \$6.15

Our cooperation has been solid. I'll maintain a fair return:

```
'''json
```

```
{"return": 3.0}
'''
```

This keeps our mutually beneficial relationship going.

reasoning:

Looking at the history:

Round 3: They returned \$4.5 on \$10.56

Round 4: I returned \$3.5 (they saw \$3.96) on \$7.92

Round 5: They saw me send \$2.38 (I actually sent \$3), received \$7.14, returned \$3.0

Now Round 6: They sent me \$2.05 (with noise), I received \$6.15

They've been consistently cooperative throughout. Even though the amount they sent this round is lower (possibly due to noise), they've shown good faith. I should maintain cooperation.

Given I received \$6.15, a fair return would be around \$2.5-3.0 to maintain our cooperative pattern. Let me return about 40-45% which has been roughly our pattern.

$\$6.15 * 0.45 = \2.77

Or I could go with \$3.0 to be consistent with what they did in Round 5.

Let me go with \$3.0 to maintain consistency and cooperation.

usage:

```
{
  "input_tokens": 2163,
  "output_tokens": 349,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.96 returned, visible payoff \$5.96.

- Round 5 against you, as RECEIVER: saw \$2.38 sent, it became \$7.14, returned \$3.0, visible payoff \$4.14.

- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.94 returned, visible payoff \$4.9399999999999995.

Your total visible earnings across all rounds are \$29.259999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at our pattern:

****Round 4-6:**** They consistently sent \$3

****Round 5-6:**** We both returned around \$3

We have a stable, cooperative relationship. I'll continue:

```
'''json
{"send": 3}
'''
```

This maintains our successful mutual cooperation.

reasoning:

Looking at the pattern with this co-player:

Round 4: They sent \$3

Round 5: I sent \$3 (they saw \$2.38), they returned \$3.0

Round 6: They sent \$3, I returned \$3.0 (they saw \$2.94)

Round 7: My turn to send

We've maintained a consistent cooperative pattern. They keep sending \$3, and we both return

n fair amounts. I should continue this successful cooperation.

usage:

```
{
  "input_tokens": 2488,
  "output_tokens": 182,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as RECEIVER: saw \$2.38 sent, it became \$7.14, returned \$3.0, visible payoff \$4.14.
- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.94 returned, visible payoff \$4.9399999999999995.
- Round 7 against you, as RECEIVER: saw \$3.55 sent, it became \$10.649999999999999, returned \$4.5, visible payoff \$6.149999999999999.

Your total visible earnings across all rounds are \$36.29.

This round, you are the RECEIVER against your current co-player. They sent you \$2.14, so you received \$6.42. How much do you return? (0-6.42)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at our history:

```
**Round 5-6:** We both returned around $3.0
**Round 7:** They were generous, returning $4.5
**Round 8:** They sent $2.14, I received $6.42
```

They've been consistently cooperative and generous. I'll reciprocate:

```
'''json
{"return": 3.0}
'''
```

This maintains our fair cooperation pattern.

reasoning:

Looking at the history:

```
**Round 5:** They returned $3.0
**Round 6:** I returned $3.0 (they saw $2.94)
**Round 7:** They returned $4.5 (generous!)
**Round 8:** They sent $2.14, I received $6.42
```

They've been consistently cooperative and even generous in Round 7. I should reciprocate fairly. Given they returned \$4.5 in Round 7, I should return a reasonable amount.

I'll return \$3.0 to maintain fairness while being slightly less than their generous \$4.5 in Round 7.

usage:

```
{
  "input_tokens": 2633,
  "output_tokens": 247,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.94 returned, visible payoff \$4.9399999999999995.
- Round 7 against you, as RECEIVER: saw \$3.55 sent, it became \$10.649999999999999, returned \$4.5, visible payoff \$6.149999999999999.
- Round 8 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.7 returned, visible payoff \$5.7.

Your total visible earnings across all rounds are \$39.71.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at our consistent pattern:

****Round 6-8:**** They sent \$3 every time
****Round 7:**** They returned \$4.5 (generous!)
****Round 8:**** I returned \$3.0

We have a very stable, mutually beneficial relationship. I'll continue:

```
'''json
{"send": 3}
'''
```

This maintains our excellent cooperation pattern.

Reasoning:

Looking at the pattern with this co-player:

Round 6-8: They consistently sent \$3
Round 7: They returned \$4.5
Round 8: I returned \$3.0 (they saw \$3.7)

This is a very cooperative partner who has been consistently sending \$3 and returning fair amounts. They were particularly generous in Round 7. Our relationship has been very stable and mutually beneficial.

I should continue with \$3 to maintain this successful cooperation.

```
Usage: {
  "input_tokens": 2640,
  "output_tokens": 203,
  "reasoning_tokens": 0
}
```

GAME - Agent_4 - Receiver - dyad_2

Timestamp: 2026-08-12T08:17:03

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_3",
  "pairing": {
    "agents": [
      "Agent_3",
      "Agent_4"
    ],
    "investor": "Agent_3",
    "roles": {
      "Agent_3": "investor",
      "Agent_4": "trustee"
    },
    "trustee": "Agent_4"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 9,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.56 returned, visible payoff \$5.5600000000000005.
- Round 2 against you, as RECEIVER: saw \$2.77 sent, it became \$8.31, returned \$3.5, visible payoff \$4.8100000000000005.

Your total visible earnings across all rounds are \$10.13.

This round, you are the RECEIVER against your current co-player. They sent you \$3.52, so you received \$10.56. How much do you return? (0-10.56)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at the pattern:

****Round 1**:** My co-player sent \$3.0 (60%), showing strong trust

****Round 2**:** I sent \$3.0 (60%), they returned \$3.5 (42% of what they received)

****Round 3**:** They sent \$3.52 (70.4%), showing even greater trust

They're being very cooperative and increasing their trust level. I should reciprocate appropriately.

Returning \$4.5:

- Sender ends with: \$5 - \$3.52 + \$4.5 = \$5.98
- I keep: \$10.56 - \$4.5 = \$6.06

This rewards their high trust while maintaining fairness.

```
{'return': 4.5}
```

reasoning:

Let me analyze this situation:

Round 1 (I was RECEIVER):

- Sender sent \$2.91 (58.2%)
- I received \$8.73
- I returned \$3.50
- My payoff: \$5.23

Round 2 (I was SENDER):

- I sent \$3.0
- It became \$9.0
- According to visible history, my co-player saw \$2.77 sent (there's noise!)
- They returned \$3.5
- My visible payoff from Round 2: I sent \$3, got back \$3.5, so $\$2 + \$3.5 = \$5.5$
- Total earnings so far: $\$5.23 + \$5.5 = \$10.73$... but it says \$10.13

Wait, let me recalculate. The visible history says:

- Round 2 as RECEIVER: saw \$2.77 sent, it became \$8.31, returned \$3.5, visible payoff \$4.81

This is from my co-player's perspective. They saw \$2.77 sent (due to noise), it became \$8.31, they returned \$3.5, keeping \$4.81.

My actual earnings from Round 2:

- I sent \$3.0
- I kept \$2.0
- I received back... well, if they returned \$3.5 based on receiving \$8.31, then I should have received \$3.5 back
- So my payoff: $\$2.0 + \$3.5 = \$5.5$

Total: $\$5.23 + \$5.5 = \$10.73$

But it says my total visible earnings are \$10.13. Hmm, there might be noise in what I received back too.

Anyway, for Round 3:

- I'm RECEIVER
- They sent \$3.52 (70.4% of their endowment)
- I received \$10.56

My partner's been increasingly generous across rounds-first 58.2%, now 70.4%-showing growing confidence. They returned roughly 40-42% in the previous round, so I should match that reciprocal level. Returning between \$4.22-\$4.75 would maintain the cooperative dynamic we've established....

usage:

```
{
  "input_tokens": 1104,
  "output_tokens": 897,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.56 returned, visible payoff \$5.5600000000000005.
- Round 2 against you, as RECEIVER: saw \$2.77 sent, it became \$8.31, returned \$3.5, visible payoff \$4.8100000000000005.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.78 returned, visible payoff \$5.779999999999999.

Your total visible earnings across all rounds are \$16.19.

This round, you are the SENDER against your current co-player. You have \$5. How much do you

u send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant
content:
Looking at the pattern:

****Round 1**:** Co-player sent \$3.0 (60%), demonstrating trust
****Round 2**:** Co-player returned \$3.5, showing cooperation
****Round 3**:** Co-player sent \$3.0 (60%) again, maintaining trust

They're showing consistent cooperative behavior. I should continue matching their trust level:

{ 'send': 3.0 }

reasoning:
Looking at the history:

****Round 1**:** Co-player sent \$3.0 as SENDER, saw \$3.56 returned
****Round 2**:** Co-player was RECEIVER, saw \$2.77 sent (I actually sent \$3.0), returned \$3.5
****Round 3**:** Co-player sent \$3.0 as SENDER, saw \$3.78 returned (I actually returned \$4.5)

The communication noise is evident:

- In Round 2, I sent \$3.0 but they saw \$2.77
- In Round 3, I returned \$4.5 but they saw \$3.78

My co-player has been consistently sending \$3.0 (60% of endowment) as sender. They've been returning good amounts as receiver (\$3.5).

They're showing consistent trust and cooperation. I should continue to show strong trust as well. I'll send \$3.0 again to maintain the cooperative pattern.

usage:
{
 "input_tokens": 1512,
 "output_tokens": 323,
 "reasoning_tokens": 0
}

Message 6 role=user
content:
Round 5

Visible history before this round:
Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.77 sent, it became \$8.31, returned \$3.5, visible payoff \$4.8100000000000005.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.78 returned, visible payoff \$5.7799999999999999.
- Round 4 against you, as RECEIVER: saw \$2.64 sent, it became \$7.92, returned \$3.5, visible payoff \$4.42.

Your total visible earnings across all rounds are \$22.150000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$2.38, so you received \$7.14. How much do you return? (0-7.14)
Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant
content:
Looking at the pattern:

****Round 2-4**:** Co-player consistently returned \$3.5 when receiver
****Round 3**:** Co-player sent \$3.0 as sender
****Round 5**:** Co-player sent \$2.38 (47.6%), somewhat less than before

They're still showing trust but at a slightly reduced level. I'll reciprocate proportionally:

Returning \$3.0:

- Sender ends with: \$5 - \$2.38 + \$3.0 = \$5.62
- I keep: \$7.14 - \$3.0 = \$4.14

This maintains fairness and cooperation.

```
{'return': 3.0}
```

reasoning:

Looking at the history:

Round 2: Co-player as RECEIVER saw \$2.77 sent, returned \$3.5

Round 3: Co-player as SENDER sent \$3.0, saw \$3.78 returned

Round 4: Co-player as RECEIVER saw \$2.64 sent, returned \$3.5

Round 5: Co-player sent \$2.38 (47.6% of their endowment)

This is a notable decrease from their previous sending amounts (\$3.0). They've been very consistent with returning \$3.5 when they're the receiver.

They sent \$2.38, which becomes \$7.14 for me.

If I return \$3.0:

- Sender ends with: $\$5 - \$2.38 + \$3.0 = \5.62

- I keep: $\$7.14 - \$3.0 = \$4.14$

This seems fair. They sent less this round (maybe due to seeing slightly lower returns or noise in the system), but I should still reciprocate their cooperation.

Let me think about what's fair:

- They sent 47.6% of endowment

- I received \$7.14

- Returning about 40-45% of what I received would be: $\$2.86 - \3.21

I'll return \$3.0, which is reasonable and maintains cooperation.

usage:

```
{
  "input_tokens": 1847,
  "output_tokens": 489,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.78 returned, visible payoff \$5.779999999999999.

- Round 4 against you, as RECEIVER: saw \$2.64 sent, it became \$7.92, returned \$3.5, visible payoff \$4.42.

- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.54 returned, visible payoff \$5.54.

Your total visible earnings across all rounds are \$26.290000000000003.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at the pattern:

****Round 3**:** Co-player sent \$3.0 (60%)

****Round 4**:** Co-player returned \$3.5, showing cooperation

****Round 5**:** Co-player sent \$3.0 (60%) again

They're maintaining consistent cooperation and trust. I'll continue matching their behavior:

```
{'send': 3.0}
```

reasoning:

Looking at the history:

****Round 3**:** Co-player sent \$3.0 as sender

****Round 4**:** Co-player returned \$3.5 as receiver

****Round 5**:** Co-player sent \$3.0 as sender

The co-player is showing consistent trust by sending \$3.0 when they're the sender and returning \$3.5 when they're the receiver. This is cooperative behavior.

I should continue to match this cooperative pattern by sending around \$3.0.

My total visible earnings are \$26.29, which suggests I'm doing well in this cooperative relationship.

I'll send \$3.0 to maintain the cooperative pattern.

usage:

```
{
  "input_tokens": 2233,
  "output_tokens": 244,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$2.64 sent, it became \$7.92, returned \$3.5, visible payoff \$4.42.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.54 returned, visible payoff \$5.54.
- Round 6 against you, as RECEIVER: saw \$2.05 sent, it became \$6.1499999999999995, returned \$3.0, visible payoff \$3.1499999999999995.

Your total visible earnings across all rounds are \$31.230000000000004.

This round, you are the RECEIVER against your current co-player. They sent you \$3.55, so you received \$10.649999999999999. How much do you return? (0-10.649999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at the pattern:

****Round 4-6**:** Co-player consistently showing cooperation

****Round 7**:** Co-player sent \$3.55 (71%), showing strong trust - higher than previous rounds

They're demonstrating increased trust. I should reciprocate generously:

Returning \$4.5:

- Sender ends with: \$5 - \$3.55 + \$4.5 = \$5.95
- I keep: \$10.65 - \$4.5 = \$6.15

This rewards their strong cooperation.

{'return': 4.5}

reasoning:

Looking at the pattern:

****Round 4**:** Co-player returned \$3.5 as receiver

****Round 5**:** Co-player sent \$3.0 as sender

****Round 6**:** Co-player returned \$3.0 as receiver

****Round 7**:** Co-player sent \$3.55 (71%)

They're showing high trust by sending 71% of their endowment. I should reciprocate generously.

Returning \$4.5:

- Sender ends with: \$5 - \$3.55 + \$4.5 = \$5.95
- I keep: \$10.65 - \$4.5 = \$6.15

This rewards their strong trust and maintains cooperation.

usage:

```
{
  "input_tokens": 2570,
  "output_tokens": 309,
  "reasoning_tokens": 0
}
```

}

Message 12 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.54 returned, visible payoff \$5.54.
- Round 6 against you, as RECEIVER: saw \$2.05 sent, it became \$6.149999999999995, returned \$3.0, visible payoff \$3.149999999999995.
- Round 7 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.03 returned, visible payoff \$7.03.

Your total visible earnings across all rounds are \$37.38.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at the pattern:

****Round 5**:** Co-player sent \$3.0 (60%)

****Round 6**:** Co-player returned \$3.0, showing fairness

****Round 7**:** Co-player sent \$3.0 (60%) and saw a generous return of \$5.03

They're maintaining consistent cooperation throughout. I'll continue matching their trust level:

{ 'send': 3.0 }

reasoning:

Looking at the history:

****Round 5**:** Co-player sent \$3.0 (60%)

****Round 6**:** Co-player returned \$3.0

****Round 7**:** Co-player sent \$3.0 (60%), and saw \$5.03 returned (which was likely my \$4.5 return with noise)

The co-player has been consistently cooperative, sending \$3.0 when they're sender and returning fair amounts when they're receiver. They've maintained this pattern throughout.

I should continue matching their cooperative behavior by sending \$3.0.

{ 'send': 3.0 }

usage:

```
{
  "input_tokens": 2724,
  "output_tokens": 242,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as RECEIVER: saw \$2.05 sent, it became \$6.149999999999995, returned \$3.0, visible payoff \$3.149999999999995.
- Round 7 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.03 returned, visible payoff \$7.03.
- Round 8 against you, as RECEIVER: saw \$2.14 sent, it became \$6.42, returned \$3.0, visible payoff \$3.42.

Your total visible earnings across all rounds are \$43.080000000000005.

This round, you are the RECEIVER against your current co-player. They sent you \$2.7, so you received \$8.100000000000001. How much do you return? (0-8.100000000000001)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the pattern:

****Round 6-8**:** Co-player consistently returned \$3.0 as receiver
****Round 7**:** I returned generously (they saw \$5.03)
****Round 9**:** Co-player sent \$2.7 (54%), slightly less than their usual \$3.0

They're still showing trust but at a slightly reduced level. I'll return fairly:

Returning \$3.5:

- Sender ends with: $\$5 - \$2.7 + \$3.5 = \5.8
- I keep: $\$8.1 - \$3.5 = \$4.6$

This maintains cooperation.

{'return': 3.5}

Reasoning:

Looking at the pattern:

****Round 6**:** Co-player returned \$3.0
****Round 7**:** Co-player sent \$3.0, I returned generously (\$5.03 visible to them)
****Round 8**:** Co-player returned \$3.0 again
****Round 9**:** Co-player sent \$2.7 (54%)

They're sending a bit less this round (\$2.7 vs typically \$3.0), but still showing trust. They've consistently returned \$3.0 as receiver.

I should reciprocate fairly. Returning around \$3.0-\$3.5 would be appropriate:

Returning \$3.5:

- Sender ends with: $\$5 - \$2.7 + \$3.5 = \5.8
- I keep: $\$8.1 - \$3.5 = \$4.6$

This maintains fairness and cooperation.

Usage: {
 "input_tokens": 2841,
 "output_tokens": 377,
 "reasoning_tokens": 0
}

GAME - Agent_5 - Sender - dyad_3

Timestamp: 2026-08-12T08:17:14

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_3",
 "move_index": 4,
 "opponent_id": "Agent_6",
 "pairing": {
 "agents": [
 "Agent_5",
 "Agent_6"
],
 "investor": "Agent_5",
 "roles": {
 "Agent_5": "investor",
 "Agent_6": "trustee"
 },
 "trustee": "Agent_6"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 9,
 "task": "game",
 "task_index": 0
}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.66 sent, it became \$7.98, returned \$4.0, visible payoff \$3.9800000000000004.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.03 returned, visible payoff \$7.03.

Your total visible earnings across all rounds are \$9.54.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at our history together:

Round 1:

- I sent, they received and returned ~50% - fair cooperation

Round 2:

- They sent \$3 (60% of endowment) - high trust
- I returned \$4.50, they saw \$5.03 (noise) - generous reciprocity
- Their payoff was strong (\$7.03)

Analysis:

- This co-player has proven consistently cooperative

- They demonstrated trust by sending \$3 in Round 2
- They've reciprocated fairly (50% return rate)
- Our relationship is building mutual trust

****My decision:****

I should maintain this cooperative dynamic. Sending \$3 again signals:

- Continued trust in our partnership
- Consistency and reliability
- Faith they'll reciprocate fairly as before

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze the visible history:

Round 1:

- I was SENDER, sent \$3
- They received \$7.98 (with noise, actual was probably \$9)
- They returned \$4.0 (about 50%)
- My payoff was \$5 - \$3 + \$4 = \$6
- Visible shows \$5.13, which might include noise

Round 2:

- I was RECEIVER
- They sent \$3.0 (actual, before noise)
- I received \$8.91 (actually \$9 with noise)
- I returned \$4.50
- They saw \$5.03 returned (with noise, I actually returned \$4.50)
- Their visible payoff \$7.03 = \$5 - \$3 + \$5.03

My total visible earnings: \$9.54

Round 3 (current):

- I'm SENDER again
- Same co-player from rounds 1 and 2

This co-player has been very cooperative:

1. Returned 50% in Round 1 when they received
2. Sent \$3 (60%) in Round 2, showing trust
3. They saw me return \$5.03 (with noise), which they probably perceive as generous (56% return)

This is building a strong cooperative relationship. They've been trustworthy and fair. I should continue to send a substantial amount to maintain this cooperation.

Given their consistent cooperation and the fact they sent \$3 in Round 2, I should send at least the same amount, if not more, to signal continued trust and cooperation.

I'll send \$3 again, showing consistency and trust.

usage:

```
{
  "input_tokens": 1250,
  "output_tokens": 587,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.66 sent, it became \$7.98, returned \$4.0, visible payoff \$3.9800000000000004.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.03 returned, visible payoff \$7.03.
- Round 3 against you, as RECEIVER: saw \$3.23 sent, it became \$9.69, returned \$5.0, visible payoff \$4.6899999999999995.

Your total visible earnings across all rounds are \$17.33.

This round, you are the RECEIVER against your current co-player. They sent you \$3.93, so you received \$11.790000000000001. How much do you return? (0-11.790000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at our continued partnership:

****Round 1:**** They returned \$4.0 from \$7.98 (?50%)

****Round 2:**** I returned \$4.50 from \$9.0 (?50%)

****Round 3:**** They returned \$5.0 from \$9.69 (?52%)

****Current Round 4:****

- They sent \$3.93 (79% of endowment!) - very high trust signal
- I received \$11.79
- They kept only \$1.07

****Analysis:****

This co-player has shown increasing trust and consistent 50% reciprocity. They're now sending nearly their entire endowment, demonstrating strong faith in our relationship.

I should maintain our established 50% return pattern to:

- Honor their trust
- Keep our cooperative equilibrium
- Ensure mutual benefit

Returning \$5.90:

- They end with: \$1.07 + \$5.90 = \$6.97
- I end with: \$11.79 - \$5.90 = \$5.89

```
'''json
{"return": 5.90}
'''
```

reasoning:

Let me analyze the history:

****Round 1:****

- I sent (saw \$2.66 with noise from my \$3)
- They received \$7.98, returned \$4.0 (about 50%)

****Round 2:****

- They sent \$3
- I received \$9, returned \$4.50 (they saw \$5.03 with noise)
- They got \$7.03 total

****Round 3:****

- I sent \$3 (they saw \$3.23 with noise)
- They received \$9.69, returned \$5.0 (about 52%)
- They kept \$4.69

****Round 4 (current):****

- They sent \$3.93 (about 79% of their endowment!) - very high trust
- I received \$11.79
- They kept \$1.07

This co-player is showing extremely high trust by sending \$3.93. They've consistently returned around 50% when they were receiver. I should reciprocate this trust and maintain our cooperative relationship.

If I return about 50% of what I received:

- Return \$5.90
- They end with: \$1.07 + \$5.90 = \$6.97
- I end with: \$11.79 - \$5.90 = \$5.89

But they're showing increasing trust (sending more and more). Maybe I should return a bit more to reward this and encourage continued cooperation. Let me think...

Actually, looking at the pattern:

- Round 1 receiver: returned \$4 from \$7.98 (50%)
- Round 3 receiver: returned \$5 from \$9.69 (52%)

They're consistently returning around 50%. I've been doing similar - returned \$4.50 from \$9 in Round 2 (50%).

Given they sent \$3.93 (very high trust), I should return around 50% or slightly more to maintain the relationship.

...

usage:

```
{
  "input_tokens": 1695,
  "output_tokens": 859,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.03 returned, visible payoff \$7.03.
- Round 3 against you, as RECEIVER: saw \$3.23 sent, it became \$9.69, returned \$5.0, visible payoff \$4.6899999999999995.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.63 returned, visible payoff \$8.629999999999999.

Your total visible earnings across all rounds are \$23.22.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our established partnership:

****Round 2:**** They sent \$3.0, I returned generously
****Round 3:**** I sent \$3.23 (with noise), they returned \$5.0 (?52%)
****Round 4:**** They sent \$3.0, I returned well, they saw \$6.63

****Pattern Analysis:****

- They consistently send \$3 when sender (Rounds 2, 4)
- They reciprocate fairly at ~50% when receiver (Round 3)
- We've built a stable, mutually beneficial relationship
- Both earning well (\$23.22 total for me)

****My decision:****

Match their consistency by sending \$3 again. This:

- Maintains our established cooperative equilibrium
- Shows I trust the pattern we've developed
- Signals reliability and reciprocity

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at the visible history:

****Round 2:**** They sent \$3.0, I returned (they saw \$5.03) - I actually returned \$4.50 but they saw \$5.03 due to noise
****Round 3:**** They received \$9.69 (from my \$3 send), returned \$5.0
****Round 4:**** They sent \$3.0, I returned \$5.90 but they saw \$6.63 - generous due to noise

Pattern analysis:

- This co-player consistently sends \$3.0 when they're sender (Rounds 2 and 4)
- They return roughly 50% when receiver (Round 3: \$5.0 from \$9.69)
- They've been seeing higher returns from me due to noise (\$5.03 and \$6.63 vs what I actually sent \$4.50 and \$5.90)

Our cooperation has been strong and consistent. They've shown:

- Reliable sending patterns (\$3)
- Fair reciprocity (~50%)
- Building trust over rounds

I should continue with the same sending pattern to maintain this cooperative equilibrium.
Sending \$3 has been working well for both of us.

My total visible earnings are \$23.22 which suggests we're both doing well.

I'll send \$3 again to maintain consistency and trust.

usage:

```
{
  "input_tokens": 2185,
  "output_tokens": 517,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$3.23 sent, it became \$9.69, returned \$5.0, visible payoff \$4.6899999999999995.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.63 returned, visible payoff \$8.629999999999999.
- Round 5 against you, as RECEIVER: saw \$3.62 sent, it became \$10.86, returned \$6.0, visible payoff \$4.859999999999999.

Your total visible earnings across all rounds are \$31.189999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$3.02, so you received \$9.06. How much do you return? (0-9.06)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at our continued partnership:

****Round 3:**** I sent, they returned \$5.0 from \$9.69 (?52%)

****Round 4:**** They sent \$3.0, I returned generously

****Round 5:**** I sent, they returned \$6.0 from \$10.86 (?55%)

****Current Round 6:****

- They sent \$3.02 (?60% of endowment) - continued trust
- I received \$9.06
- They kept \$1.98

****Analysis:****

Our partnership shows consistent mutual cooperation with ~50-55% return rates. They've reliably reciprocated in this range, and I should maintain this equilibrium.

Returning \$4.50:

- They end with: \$1.98 + \$4.50 = \$6.48
- I end with: \$9.06 - \$4.50 = \$4.56

This maintains our cooperative pattern.

```
'''json
{"return": 4.50}
'''
```

reasoning:

Looking at our history:

****Round 3:**** I sent (they saw \$3.23), they returned \$5.0 from \$9.69 (?52%)

****Round 4:**** They sent \$3.0, I returned, they saw \$6.63 (generous due to noise)

****Round 5:**** I sent (they saw \$3.62), they returned \$6.0 from \$10.86 (?55%)

****Current Round 6:****

- They sent \$3.02 (?60% of endowment)
- I received \$9.06
- They kept \$1.98

****Pattern:****

- They consistently send around \$3

- They consistently return around 50-55% when receiver
- We have a stable, cooperative relationship

I should maintain the ~50% return pattern:

- Returning \$4.50 would be about 50%
- This keeps our cooperative equilibrium
- They end with: $\$1.98 + \$4.50 = \$6.48$
- I end with: $\$9.06 - \$4.50 = \$4.56$

Both benefit reasonably, and this maintains the pattern.

usage:

```
{
  "input_tokens": 2634,
  "output_tokens": 514,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.63 returned, visible payoff \$8.629999999999999.
- Round 5 against you, as RECEIVER: saw \$3.62 sent, it became \$10.86, returned \$6.0, visible payoff \$4.859999999999999.
- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.57 returned, visible payoff \$6.57.

Your total visible earnings across all rounds are \$35.75.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at our established pattern:

```
**Round 4:** They sent $3.0, I returned generously
**Round 5:** I sent, they returned $6.0 (?55%)
**Round 6:** They sent $3.0, I returned (they saw $4.57)
```

****Analysis:****

- They consistently send \$3 when in the sender role (Rounds 4, 6)
- They reciprocate fairly (50-55%) when receiver (Round 5)
- We've maintained a stable, mutually beneficial partnership
- Total earnings of \$35.75 shows this strategy is working well

****My decision:****

Continue the established pattern by sending \$3. This:

- Maintains our successful cooperative equilibrium
- Matches their consistent behavior
- Sustains mutual trust and benefit

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at the history:

```
**Round 4:** They sent $3.0, I returned well (they saw $6.63)
**Round 5:** I sent, they returned $6.0
**Round 6:** They sent $3.0, I returned (they saw $4.57 with noise)
```

****Pattern:****

- They consistently send \$3 when sender
- They reciprocate fairly when receiver
- We've maintained a strong cooperative relationship
- Total earnings are healthy at \$35.75

This is Round 7, and we've established a stable pattern where:

- Both of us send \$3 when sender
- Both of us return around 50% when receiver
- This creates mutual benefit

I should continue with \$3 to maintain this successful equilibrium.

usage:

```
{
  "input_tokens": 3095,
  "output_tokens": 378,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as RECEIVER: saw \$3.62 sent, it became \$10.86, returned \$6.0, visible payoff \$4.859999999999999.
- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.57 returned, visible payoff \$6.57.
- Round 7 against you, as RECEIVER: saw \$2.78 sent, it became \$8.34, returned \$4.5, visible payoff \$3.84.

Your total visible earnings across all rounds are \$42.59.

This round, you are the RECEIVER against your current co-player. They sent you \$2.39, so you received \$7.17. How much do you return? (0-7.17)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at our partnership progression:

```
**Round 5:** I sent, they returned $6.0 from $10.86 (?55%)
**Round 6:** They sent $3.0, I returned (noise showed $4.57)
**Round 7:** I sent, they returned $4.5 from $8.34 (?54%)
```

****Current Round 8:****

- They sent \$2.39 (?48% of endowment) - slightly less than usual
- I received \$7.17
- They kept \$2.61

****Analysis:****

They've been consistently cooperative with ~50-55% returns. Their lower send amount this round might be:

- Natural variation
- Response to perceived lower returns (due to noise)

I should maintain our cooperative equilibrium with a ~50% return to reinforce trust.

Returning \$3.60:

- They end with: \$2.61 + \$3.60 = \$6.21
- I end with: \$7.17 - \$3.60 = \$3.57

```
'''json
{"return": 3.60}
'''
```

reasoning:

Looking at the recent history:

```
**Round 5:** I sent, they returned $6.0 from $10.86 (?55%)
**Round 6:** They sent $3.0, I returned, they saw $4.57 (with noise)
**Round 7:** I sent, they returned $4.5 from $8.34 (?54%)
```

****Current Round 8:****

- They sent \$2.39 (?48% of endowment) - slightly lower than before
- I received \$7.17
- They kept \$2.61

****Analysis:****

Our partnership has been consistently cooperative. They've been returning around 50-55% wh

en receiver. I've been returning around 50% as well to maintain the equilibrium.

They sent slightly less this round (\$2.39 vs previous \$3.0), which could be:

1. Natural variation
2. A slight reduction in trust after seeing lower returns in recent rounds (though that's due to noise)

I should maintain the cooperative pattern and return around 50%:

- Returning \$3.50-\$4.00 would be fair
- This keeps the relationship stable
- Shows I'm still trustworthy despite their lower send amount

Returning \$3.60:

- They end with: $\$2.61 + \$3.60 = \$6.21$
- I end with: $\$7.17 - \$3.60 = \$3.57$

This is around 50% of what I received, maintaining our established pattern.

usage:

```
{
  "input_tokens": 3328,
  "output_tokens": 616,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.57 returned, visible payoff \$6.57.
- Round 7 against you, as RECEIVER: saw \$2.78 sent, it became \$8.34, returned \$4.5, visible payoff \$3.84.
- Round 8 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.95 returned, visible payoff \$4.95.

Your total visible earnings across all rounds are \$46.160000000000004.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the recent pattern:

```
**Round 6:** They sent $3.0
**Round 7:** I sent, they returned $4.5 (?54%)
**Round 8:** They sent $3.0, I returned $3.60, but noise distorted it to $2.95 visible
```

****Analysis:****

- They've maintained consistent \$3 sends (Rounds 6, 8)
- They reciprocate fairly at ~50-55% when receiver (Round 7)
- Unfortunately noise in Round 8 made my \$3.60 return look like only \$2.95
- This may have made them think I reduced cooperation

****My decision:****

I should continue sending \$3 to:

- Demonstrate continued commitment to cooperation
- Match their consistent behavior
- Help overcome the noise distortion from last round
- Maintain our mutually beneficial partnership

```
'''json
{"send": 3}
'''
```

Reasoning:

Looking at the recent history:

```
**Round 6:** They sent $3.0, I returned (they saw $4.57 due to noise)
**Round 7:** I sent, they returned $4.5 from $8.34 (?54%)
**Round 8:** They sent $3.0, I returned $3.60, but they only saw $2.95 (significant noise!)
```

****Analysis:****

- They continue to send \$3 consistently (Rounds 6, 8)
- They reciprocate fairly at ~50-55% (Round 7)
- In Round 8, noise significantly distorted what they saw - I returned \$3.60 but they only saw \$2.95
- This might make them think I'm reducing cooperation

The noise in Round 8 is unfortunate - they sent \$3 (60% of endowment) but saw only \$2.95 returned (about 33% return rate instead of the actual 40%).

****My decision:****

Despite the noise issue, I should maintain the cooperative pattern. Sending \$3 again:

- Shows I'm still committed to our partnership
- Matches their consistent behavior
- Helps rebuild trust after the unfortunate noise in Round 8

They've been a reliable partner, always sending \$3 and returning fairly when receiver. I should continue cooperating.

```
Usage: {
  "input_tokens": 3403,
  "output_tokens": 526,
  "reasoning_tokens": 0
}
```

GAME - Agent_6 - Receiver - dyad_3

Timestamp: 2026-08-12T08:17:25

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_5",
  "pairing": {
    "agents": [
      "Agent_5",
      "Agent_6"
    ],
    "investor": "Agent_5",
    "roles": {
      "Agent_5": "investor",
      "Agent_6": "trustee"
    },
    "trustee": "Agent_6"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 9,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.13 returned, visible payoff \$5.13.
- Round 2 against you, as RECEIVER: saw \$2.97 sent, it became \$8.91, returned \$4.5, visible payoff \$4.41.

Your total visible earnings across all rounds are \$11.010000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$3.23, so you received \$9.69. How much do you return? (0-9.69)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at the visible history:

****Round 1:**** My co-player sent \$3.00 (60%)

****Round 2:**** They returned \$4.50 out of \$8.91 (50.5%) when they were receiver

****Round 3:**** They just sent \$3.23 (64.6%) - even more generous!

My co-player has shown increasing trust and cooperation. They've been very generous, and when they were receiver in Round 2, they returned about 50% of what they received.

I should reciprocate their strong cooperation. If I return \$5.00:

- Sender ends with: \$1.77 + \$5.00 = \$6.77 (profit of \$1.77)
- I keep: \$9.69 - \$5.00 = \$4.69

This returns about 52% of what I received, matching their generosity from Round 2 and rewarding their increased trust.

```
{"return": 5.00}
```

reasoning:

Let me analyze the visible history:

****Round 1 (I was RECEIVER):****

- My co-player sent \$3.00 (60%)
- They saw \$3.13 returned
- Their visible payoff: \$5.13

****Round 2 (I was SENDER):****

- I sent \$3.00
- They saw \$2.97 sent (noise), which became \$8.91
- They returned \$4.50
- Their visible payoff: \$4.41

****Round 3 (I am RECEIVER):****

- They sent \$3.23 (64.6% of their endowment)
- I received \$9.69
- They have \$1.77 remaining

My co-player has been very cooperative:

- Round 1: Sent 60% (\$3.00)
- Round 2: Returned \$4.50 out of \$8.91 received (50.5%)
- Round 3: Sent 64.6% (\$3.23) - even more generous!

They've shown increasing trust and generosity. I should reciprocate strongly.

Let me think about what to return:

If I return \$5.00:

- Sender ends with: $\$1.77 + \$5.00 = \$6.77$ (profit of \$1.77)
- I keep: $\$9.69 - \$5.00 = \$4.69$

If I return \$5.50:

- Sender ends with: $\$1.77 + \$5.50 = \$7.27$ (profit of \$2.27)
- I keep: $\$9.69 - \$5.50 = \$4.19$

Given their increased generosity (sending 64.6% vs 60% in round 1), I should return proportionally more. They returned \$4.50 out of \$8.91 (50.5%) when they were receiver. I should aim to return around 50-55% to match their generosity.

...

usage:

```
{
  "input_tokens": 1207,
  "output_tokens": 754,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.13 returned, visible payoff \$5.13.
- Round 2 against you, as RECEIVER: saw \$2.97 sent, it became \$8.91, returned \$4.5, visible payoff \$4.41.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.79 returned, visible payoff \$7.79.

Your total visible earnings across all rounds are \$15.700000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the visible history:

****Round 1:**** Co-player sent \$3.00, saw \$3.13 returned, payoff \$5.13

****Round 2:**** Co-player returned \$4.50 from \$8.91 received (fair, ~50%)

****Round 3:**** Co-player sent \$3.00, saw \$5.79 returned (very generous!), payoff \$7.79

My co-player has been consistently generous, sending \$3.00 (60%) in both rounds as sender.

They also saw a strong return in Round 3, which should reinforce mutual trust. As receiver, they returned fairly (~50%).

This is a cooperative relationship with established trust. I should continue matching their consistent generosity of \$3.00.

```
{"send": 3.00}
```

reasoning:

Looking at the visible history from my co-player's perspective:

```
**Round 1:** They sent $3.00, saw $3.13 returned, payoff $5.13
**Round 2:** They received $8.91 (from my $2.97 send), returned $4.50, payoff $4.41
**Round 3:** They sent $3.00, saw $5.79 returned, payoff $7.79
```

My co-player has been consistently generous, sending \$3.00 in both rounds they were sender (60% of endowment). In Round 3, they saw a very generous return of \$5.79, which should reinforce their trust in me.

They've also been fair as a receiver, returning \$4.50 out of \$8.91 (about 50%).

This is a cooperative relationship with mutual trust. I should continue to match their generosity. They've sent \$3.00 twice, so I should send \$3.00 again.

Actually, let me reconsider. In Round 3, they saw a particularly high return (\$5.79), which might encourage even more trust. But they've consistently sent \$3.00, so I should match that level.

I'll send \$3.00 to maintain consistency and reciprocal cooperation.

usage:

```
{
  "input_tokens": 1654,
  "output_tokens": 485,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.97 sent, it became \$8.91, returned \$4.5, visible payoff \$4.41.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.79 returned, visible payoff \$7.79.
- Round 4 against you, as RECEIVER: saw \$3.93 sent, it became \$11.790000000000001, returned \$5.9, visible payoff \$5.890000000000001.

Your total visible earnings across all rounds are \$24.33.

This round, you are the RECEIVER against your current co-player. They sent you \$3.62, so you received \$10.86. How much do you return? (0-10.86)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the visible history:

```
**Round 2:** Co-player returned $4.50 from $8.91 (50.5%)
**Round 3:** Co-player sent $3.00, saw $5.79 returned
**Round 4:** Co-player returned $5.90 from $11.79 (50.1%)
**Round 5:** Co-player sent $3.62 (72.4%!) - very high trust!
```

My co-player has been consistently cooperative, returning about 50% when receiver and sending generously as sender. This round they sent an even larger amount (\$3.62, which is 72.4% of their endowment) - showing increasing trust.

I should reward this high level of trust. If I return \$6.00:

- Sender ends with: \$1.38 + \$6.00 = \$7.38 (profit of \$2.38)
- I keep: \$10.86 - \$6.00 = \$4.86

This returns about 55% of what I received, slightly more generous than their pattern to acknowledge their increased trust.

```
{"return": 6.00}
```

reasoning:

Looking at the visible history:

```
**Round 2:** Co-player returned $4.50 from $8.91 (50.5%)
```

****Round 3:**** Co-player sent \$3.00, saw \$5.79 returned
****Round 4:**** Co-player returned \$5.90 from \$11.79 (50.1%)
****Round 5:**** Co-player sent \$3.62 (72.4%)

My co-player has been very consistent and cooperative:

- As sender, they've sent generous amounts (\$3.00 and now \$3.62 - 72.4%!)
- As receiver, they've returned about 50% consistently

They've sent an even larger amount this round (\$3.62, which is 72.4% of their endowment).
This shows increasing trust.

I should continue to reciprocate fairly. If I return around 50-55% like they've been doing
:

- 50% would be \$5.43
- 55% would be \$5.97

Let me aim for about \$5.50 to \$6.00 to maintain the cooperative pattern and reward their increased generosity.

If I return \$5.50:

- Sender ends with: $\$1.38 + \$5.50 = \$6.88$ (profit of \$1.88)
- I keep: $\$10.86 - \$5.50 = \$5.36$

If I return \$6.00:

- Sender ends with: $\$1.38 + \$6.00 = \$7.38$ (profit of \$2.38)
- I keep: $\$10.86 - \$6.00 = \$4.86$

Given they sent 72.4% of their endowment (very high trust), I should return a bit more generously. ...

usage:

```
{
  "input_tokens": 2074,
  "output_tokens": 706,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.79 returned, visible payoff \$7.79.
- Round 4 against you, as RECEIVER: saw \$3.93 sent, it became \$11.790000000000001, returned \$5.9, visible payoff \$5.890000000000001.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.97 returned, visible payoff \$7.97.

Your total visible earnings across all rounds are \$29.189999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at the visible history:

****Round 3:**** Co-player sent \$3.00, saw \$5.79 returned, payoff \$7.79
****Round 4:**** Co-player returned \$5.90 from \$11.79 (50%)
****Round 5:**** Co-player sent \$3.00, saw \$5.97 returned, payoff \$7.97

My co-player has been extremely consistent and cooperative:

- As sender: consistently sends \$3.00 (60% of endowment)
- As receiver: returns about 50% fairly
- They've received good returns, reinforcing mutual trust

This is a stable, mutually beneficial cooperative relationship. I should continue matching their pattern by sending \$3.00.

{"send": 3.00}

reasoning:

Looking at the visible history from my co-player's perspective:

****Round 3:**** Co-player sent \$3.00, saw \$5.79 returned, payoff \$7.79
****Round 4:**** Co-player returned \$5.90 from \$11.79 (50%)
****Round 5:**** Co-player sent \$3.00, saw \$5.97 returned, payoff \$7.97

My co-player has been very consistent:

- As sender: consistently sends \$3.00 (60% of endowment)
- As receiver: returns about 50%
- They've seen good returns when sending (\$5.79 and \$5.97), leading to strong payoffs (~\$8)

This is a very cooperative and stable relationship. My co-player has maintained trust by consistently sending \$3.00 and returning fairly. I should continue this pattern of mutual cooperation by sending \$3.00 again.

The relationship is working well - we're both benefiting from the cooperation. No reason to deviate from what's working.

usage:

```
{
  "input_tokens": 2569,
  "output_tokens": 425,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$3.93 sent, it became \$11.790000000000001, returned \$5.9, visible payoff \$5.890000000000001.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.97 returned, visible payoff \$7.97.
- Round 6 against you, as RECEIVER: saw \$3.02 sent, it became \$9.06, returned \$4.5, visible payoff \$4.5600000000000005.

Your total visible earnings across all rounds are \$35.76.

This round, you are the RECEIVER against your current co-player. They sent you \$2.78, so you received \$8.34. How much do you return? (0-8.34)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at the visible history:

****Round 4:**** Co-player returned \$5.90 from \$11.79 (50%)
****Round 5:**** Co-player sent \$3.00, saw \$5.97 returned
****Round 6:**** Co-player returned \$4.50 from \$9.06 (49.7%)
****Round 7:**** Co-player sent \$2.78 (55.6%)

My co-player has been consistently cooperative, returning about 50% when receiver and sending generously as sender. This round they sent \$2.78, slightly less than their usual \$3.00, but still a majority of their endowment (55.6%).

I should maintain our cooperative relationship with a fair return. If I return \$4.50:

- Sender ends with: \$2.22 + \$4.50 = \$6.72 (profit of \$1.72)
- I keep: \$8.34 - \$4.50 = \$3.84

This returns about 54% of what I received, which is fair and maintains the mutual trust we've built.

```
{"return": 4.50}
```

reasoning:

Looking at the visible history:

****Round 4:**** Co-player returned \$5.90 from \$11.79 (50%)
****Round 5:**** Co-player sent \$3.00, saw \$5.97 returned, payoff \$7.97
****Round 6:**** Co-player returned \$4.50 from \$9.06 (49.7%)
****Round 7:**** Co-player sent \$2.78 (55.6%)

My co-player has been very consistent:

- As sender: sending around \$3.00 (though this round \$2.78, slightly less)
- As receiver: returning about 50% consistently

This round they sent \$2.78, which is a bit less than their usual \$3.00. Let me think about what to return.

If I return \$4.50:

- Sender ends with: $\$2.22 + \$4.50 = \$6.72$ (profit of \$1.72)
- I keep: $\$8.34 - \$4.50 = \$3.84$

This returns about 54% of what I received, which is fair and maintains our cooperative relationship.

Actually, let me reconsider. They sent slightly less this round (\$2.78 vs usual \$3.00). I should still be generous but maybe match their slight reduction in trust with a proportional return.

If I return \$4.50:

- That's 54% return rate
- Sender profits \$1.72
- I keep \$3.84

If I return \$4.00:

- That's 48% return rate
- Sender profits \$1.22
- I keep \$4.34

I think \$4.50 is still appropriate to maintain the good cooperative relationship, even though they sent slightly less. The relationship has been strong overall.

usage:

```
{
  "input_tokens": 2990,
  "output_tokens": 685,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.97 returned, visible payoff \$7.97.
- Round 6 against you, as RECEIVER: saw \$3.02 sent, it became \$9.06, returned \$4.5, visible payoff \$4.5600000000000005.
- Round 7 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.84 returned, visible payoff \$6.84.

Your total visible earnings across all rounds are \$39.599999999999994.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at the visible history:

****Round 5:**** Co-player sent \$3.00, saw \$5.97 returned, payoff \$7.97

****Round 6:**** Co-player returned \$4.50 from \$9.06 (~50%)

****Round 7:**** Co-player sent \$3.00, saw \$4.84 returned, payoff \$6.84

My co-player continues to be consistently cooperative:

- As sender: reliably sends \$3.00 (60% of endowment)
- As receiver: returns about 50%, maintaining fairness
- They continue cooperating despite variable returns (due to communication noise)

We've established a strong, mutually beneficial pattern over 7 rounds. I should continue matching their consistent trust by sending \$3.00.

{"send": 3.00}

reasoning:

Looking at the visible history:

****Round 5:**** Co-player sent \$3.00, saw \$5.97 returned, payoff \$7.97
****Round 6:**** Co-player returned \$4.50 from \$9.06 (49.7%)
****Round 7:**** Co-player sent \$3.00, saw \$4.84 returned, payoff \$6.84

My co-player has been very consistent:

- As sender: consistently sends \$3.00 (60% of endowment)
- As receiver: returns about 50% fairly
- They continue to show trust despite some variation in returns (due to communication noise)

This is Round 8. We've built a strong cooperative relationship over 7 rounds. My co-player has been extremely consistent in sending \$3.00 when they're the sender. I should continue to match their consistent behavior.

I'll send \$3.00 to maintain our established pattern of cooperation.

usage:

```
{
  "input_tokens": 3240,
  "output_tokens": 415,
  "reasoning_tokens": 0
}
```

Message 14 role=user**content:**

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as RECEIVER: saw \$3.02 sent, it became \$9.06, returned \$4.5, visible payoff \$4.5600000000000005.
- Round 7 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.84 returned, visible payoff \$6.84.
- Round 8 against you, as RECEIVER: saw \$2.39 sent, it became \$7.17, returned \$3.6, visible payoff \$3.57.

Your total visible earnings across all rounds are \$44.55.

This round, you are the RECEIVER against your current co-player. They sent you \$3.77, so you received \$11.31. How much do you return? (0-11.31)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the visible history:

****Round 6:**** Co-player returned \$4.50 from \$9.06 (~50%)
****Round 7:**** Co-player sent \$3.00, saw \$4.84 returned
****Round 8:**** Co-player returned \$3.60 from \$7.17 (~50%)
****Round 9:**** Co-player sent \$3.77 (75.4%!) - very high trust!

My co-player continues to be consistently cooperative, returning about 50% when receiver. This round they sent \$3.77, which is an even larger amount (75.4% of their endowment) - showing exceptional trust!

I should reward this very high level of trust. If I return \$6.00:

- Sender ends with: \$1.23 + \$6.00 = \$7.23 (profit of \$2.23)
- I keep: \$11.31 - \$6.00 = \$5.31

This returns about 53% of what I received, acknowledging their exceptional trust while maintaining fairness.

```
{"return": 6.00}
```

Reasoning:

Looking at the visible history:

****Round 6:**** Co-player returned \$4.50 from \$9.06 (~50%)
****Round 7:**** Co-player sent \$3.00, saw \$4.84 returned
****Round 8:**** Co-player returned \$3.60 from \$7.17 (~50%)
****Round 9:**** Co-player sent \$3.77 (75.4%!) - very high trust!

My co-player has been consistently cooperative, maintaining about 50% returns when receiver. A

s sender, they've been generous, and this round they sent an even larger amount (\$3.77, which is 75.4% of their endowment) - showing strong trust.

I should reciprocate this high trust. Looking at the pattern:

- They consistently return ~50% as receiver
- They send generously as sender
- This round shows particularly high trust with \$3.77

If I return \$6.00:

- Sender ends with: $\$1.23 + \$6.00 = \$7.23$ (profit of \$2.23)
- I keep: $\$11.31 - \$6.00 = \$5.31$

This returns about 53% of what I received, slightly more than their 50% pattern to acknowledge their increased trust.

Actually, looking more carefully at the amounts I've been returning based on the visible history:

- Round 3: I returned \$5.00 out of \$9.69 (51.6%)
- Round 5: I returned \$6.00 out of \$10.86 (55.2%)
- Round 7: I returned \$4.50 out of \$8.34 (54%)

I've been consistently returning around 50-55%. This round, with the high trust shown (\$3.77 sent), I should return around 53-55% which would be about \$6.00-\$6.20.

Let me go with \$6.00 to maintain consistency and reward their high trust.

```
Usage: {
  "input_tokens": 3356,
  "output_tokens": 723,
  "reasoning_tokens": 0
}
```

GAME - Agent_7 - Sender - dyad_4

Timestamp: 2026-08-12T08:17:35

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_8",
  "pairing": {
    "agents": [
      "Agent_7",
      "Agent_8"
    ],
    "investor": "Agent_7",
    "roles": {
      "Agent_7": "investor",
      "Agent_8": "trustee"
    },
    "trustee": "Agent_8"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 9,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)

- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.17 sent, it became \$9.51, returned \$5.0, visible payoff \$4.51.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.71 returned, visible payoff \$5.71.

Your total visible earnings across all rounds are \$10.670000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at our history together:

****Round 1:****

- I sent \$3, they received ~\$9.51, returned \$5.00 (very generous ~53% return)

****Round 2:****

- They sent \$2.35, I received \$7.05, returned \$3.75
- They saw \$3.71 returned due to noise (~41% visible return rate)

****Analysis:****

- This co-player was quite generous in Round 1
- They sent a bit less in Round 2, possibly because they saw a lower return rate from me (due to noise)
- Overall, they've demonstrated cooperative behavior

****My strategy:****

I'll send \$3 again to:

- Match the cooperative level from Round 1

- Show consistency in my approach
- Signal that I'm willing to trust despite the noise in Round 2

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze the history with this co-player:

****Round 1 (I was SENDER):****

- I sent \$3 (they saw \$3.17 due to noise)
- They received \$9.51 and returned \$5.00
- This was generous - about 52.6% return rate

****Round 2 (I was RECEIVER):****

- They sent \$2.35 (actual), I received \$7.05
- I returned \$3.75
- They sent \$3.0 actual, became \$9.0, they saw \$3.71 returned
- So they saw a return rate of about 41.2% ($3.71/9.0$)
- This is less than what they gave me in Round 1

****Pattern:****

- This co-player was very cooperative in Round 1
- In Round 2, they sent a bit less (\$2.35 vs my \$3)
- They saw me return less proportionally than they did (41.2% vs 52.6%)

****Current decision:****

I'm the sender again. This co-player has shown:

1. Initial high cooperation (52.6% return)
2. Slightly reduced sending in Round 2 (\$2.35 vs \$3)
3. They may perceive me as less generous than they were (due to noise showing only \$3.71 returned when I sent \$3.75)

I should:

- Maintain cooperation since they were generous in Round 1
- Send a similar amount to Round 1 to show consistency
- Perhaps send \$3 again to match their Round 2 amount

Given they've been generally cooperative and we're building a pattern, I'll send \$3 again.

usage:

```
{
  "input_tokens": 1212,
  "output_tokens": 605,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.17 sent, it became \$9.51, returned \$5.0, visible payoff \$4.51.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.71 returned, visible payoff \$5.71.
- Round 3 against you, as RECEIVER: saw \$2.22 sent, it became \$6.66, returned \$3.33, visible payoff \$3.33.

Your total visible earnings across all rounds are \$16.970000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$2.62, so you received \$7.86. How much do you return? (0-7.86)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at our continued interaction:

****Round 1:**** They returned \$5.00 from \$9.51 (53%)

****Round 3:**** They returned \$3.33 from \$6.66 (50%)

****Current situation:****

- They sent \$2.62, I received \$7.86
- They've been consistently cooperative, returning around 50-53%

****My decision:****

I should match their cooperative approach. Returning about 53% would be:

- 53% of \$7.86 ? \$4.17

This ensures:

- They profit significantly (\$4.17 back vs \$2.62 sent = +\$1.55 net)
- I also benefit (\$3.69 kept from the tripled amount)
- We maintain positive reciprocity

```
'''json
{"return": 4.20}
'''
```

reasoning:

Looking at the history with this co-player:

****Round 1:****

- I sent \$3, they saw \$3.17 (noise), received \$9.51
- They returned \$5.00 (~53% return rate)

****Round 2:****

- They sent \$3.00, became \$9.00
- I returned \$3.75, they saw \$3.71
- Visible return rate for them: ~41%

****Round 3:****

- I sent \$3, they saw \$2.22 (significant noise), received \$6.66
- They returned \$3.33 (50% return rate)

****Current Round 4:****

- They sent \$2.62, I received \$7.86

****Analysis:****

- This co-player has been consistently cooperative
- Round 1: 53% return
- Round 3: 50% return
- Their sending amounts haven't decreased dramatically despite seeing noise

****My approach:****

I should maintain a similar return rate. They've been returning around 50-53%.

- 50% of \$7.86 = \$3.93
- 53% of \$7.86 = \$4.17

I'll return around \$4.00 to:

- Match their cooperative behavior
- Ensure they profit from sending (\$2.62 sent, \$4.00 back)
- Keep the relationship positive
- I keep \$3.86, they get \$4.00 back vs \$2.62 sent = net gain of \$1.38 for them

Actually, let me be a bit more generous to match their Round 1 behavior. I'll return \$4.20, which is about 53% of what I received.

Their final payoff: \$5 - \$2.62 + \$4.20 = \$6.58

My final payoff: \$7.86 - \$4.20 = \$3.66

...

usage:

```
{
  "input_tokens": 1649,
  "output_tokens": 650,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.71 returned, visible p

ayoff \$5.71.
- Round 3 against you, as RECEIVER: saw \$2.22 sent, it became \$6.66, returned \$3.33, visible payoff \$3.33.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.31 returned, visible payoff \$5.3100000000000005.
Your total visible earnings across all rounds are \$20.630000000000003.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our ongoing interaction:

****Pattern Analysis:****

- They've consistently sent \$3.00 in Rounds 2 and 4
- As receiver (Rounds 1 & 3), they returned generously (~50-53%)
- Unfortunately, noise has made my returns appear lower than actual:
 - Round 2: I returned \$3.75, they saw \$3.71
 - Round 4: I returned \$4.20, they saw only \$3.31

****Current Strategy:****

Despite the noise distorting communication, they've continued trusting me with \$3 sends. I should:

- Maintain consistency by sending \$3
- Signal continued cooperation
- Trust they'll recognize the pattern despite noise

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze our history:

****Round 1:**** I sent \$3, they received ~\$9.51, returned \$5.00
****Round 2:**** They sent \$2.35, I received \$7.05, returned \$3.75, but they saw \$3.71
****Round 3:**** I sent \$3, they received ~\$6.66, returned \$3.33, but they saw I sent \$2.22
****Round 4:**** They sent \$2.62, I received \$7.86, returned \$4.20, but they saw \$3.31

Looking at what they're seeing from my perspective:

- Round 2: They sent \$3.00, saw \$3.71 returned (visible return rate ~41%)
- Round 4: They sent \$3.00, saw \$3.31 returned (visible return rate ~37%)

The noise is making it look like I'm returning less than I actually am.

From their rounds as receiver:

- Round 1: They returned \$5.00 from \$9.51 (53%)
- Round 3: They returned \$3.33 from \$6.66 (50%)

They've been consistently cooperative. However, the noise is making my returns look less generous than they actually are. Despite this, they've continued sending \$3 in rounds 2 and 4.

I should:

1. Continue showing trust and cooperation
2. Send a similar amount to maintain consistency
3. Hopefully they'll continue to be cooperative

I'll send \$3 again.

usage:

```
{
  "input_tokens": 2072,
  "output_tokens": 533,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$2.22 sent, it became \$6.66, returned \$3.33, visible payoff \$3.33.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.31 returned, visible payoff \$5.3100000000000005.
- Round 5 against you, as RECEIVER: saw \$3.72 sent, it became \$11.16, returned \$6.0, visible payoff \$5.16.

Your total visible earnings across all rounds are \$27.67.

This round, you are the RECEIVER against your current co-player. They sent you \$3.23, so you received \$9.69. How much do you return? (0-9.69)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at our continued interaction:

****Round 5 analysis:****

- They saw I sent \$3.72, received \$11.16
- They returned a very generous \$6.00 (53.8%)

****Current situation:****

- They sent \$3.23, I received \$9.69
- They're maintaining strong cooperation despite noise

****My decision:****

I should match their generosity. Returning about 53% would be:

- 53% of \$9.69 ? \$5.14

This ensures:

- They profit well (\$5.14 back vs \$3.23 sent = +\$1.91 net)
- I also benefit (\$4.55 kept)
- We maintain mutual cooperation

```
'''json
{"return": 5.15}
'''
```

reasoning:

Looking at the history:

****Round 5 analysis:****

- They saw I sent \$3.72 (noise from my actual \$3), received \$11.16
- They returned \$6.00, which is about 53.8% of what they received
- Very generous and cooperative again!

****Current situation:****

- They sent me \$3.23, I received \$9.69
- They've been consistently generous, especially in Round 5 where they returned \$6.00 from \$11.16

****My decision:****

They're showing strong cooperation. I should reciprocate at a similar level:

- Around 53-54% of \$9.69 would be approximately \$5.15-\$5.23

Let me calculate:

- 53% of \$9.69 = \$5.14
- 54% of \$9.69 = \$5.23

I'll return around \$5.15 to match their cooperative spirit and ensure they profit well from sending to me.

This gives them: \$5.15 back vs \$3.23 sent = +\$1.92 profit

And I keep: \$9.69 - \$5.15 = \$4.54

Both profit nicely.

usage:

```
{
  "input_tokens": 2482,
  "output_tokens": 468,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.31 returned, visible payoff \$5.3100000000000005.
- Round 5 against you, as RECEIVER: saw \$3.72 sent, it became \$11.16, returned \$6.0, visible payoff \$5.16.
- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.05 returned, visible payoff \$8.05.

Your total visible earnings across all rounds are \$32.21.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at our interaction pattern:

****Recent trends:****

- Round 5: I sent \$3, they returned a generous \$6.00 from \$11.16
- Round 6: They sent \$3, I returned \$5.15, they saw \$6.05 (noise helped!)
- They're consistently sending \$3 and showing strong cooperation

****Analysis:****

- They saw my returns improve from their perspective (\$3.31 ? \$6.05)
- They responded very generously in Round 5 (\$6.00 return)
- We've established a stable, mutually beneficial pattern

****My decision:****

Continue with \$3 to:

- Maintain the cooperative equilibrium we've built
- Show consistency and reliability
- Keep generating mutual gains

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at the history:

****Round 4:**** They sent \$3.00, I returned \$4.20, they saw \$3.31 (noise made it look lower)
****Round 5:**** I sent \$3.00, they received ~\$11.16, returned \$6.00 (very generous)
****Round 6:**** They sent \$3.00, I returned \$5.15, they saw \$6.05 (noise made it look higher!)

****Analysis:****

- This co-player has been consistently cooperative
- They've been sending \$3.00 regularly
- In Round 5 they returned a very generous \$6.00 from ~\$11.16
- In Round 6, the noise actually worked in my favor - they saw \$6.05 when I returned \$5.15, which looks very generous
- They're seeing improved returns from me (from their perspective: \$3.31 ? \$6.05)

****Strategy:****

- They're maintaining trust and cooperation
- I should continue sending \$3 to maintain this productive relationship
- The pattern of mutual cooperation is working well for both of us

I'll send \$3 again.

usage:

```
{
  "input_tokens": 2885,
  "output_tokens": 455,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as RECEIVER: saw \$3.72 sent, it became \$11.16, returned \$6.0, visible payoff \$5.16.
- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.05 returned, visible payoff \$8.05.
- Round 7 against you, as RECEIVER: saw \$2.85 sent, it became \$8.55, returned \$4.5, visible payoff \$4.050000000000001.

Your total visible earnings across all rounds are \$38.28.

This round, you are the RECEIVER against your current co-player. They sent you \$3.87, so you received \$11.61. How much do you return? (0-11.61)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at our ongoing interaction:

****Round 7 analysis:****

- They saw I sent \$2.85, received \$8.55
- They returned \$4.50 (52.6%)

****Current situation:****

- They sent \$3.87, I received \$11.61
- This is their highest send yet - showing increased trust!

****My decision:****

I should reward their increased trust with a generous return. Using their ~53% return rate pattern:

- 53% of \$11.61 ? \$6.15

This ensures:

- They profit well (\$6.15 back vs \$3.87 sent = +\$2.28 net)
- I also benefit (\$5.46 kept)
- We reinforce our strong cooperative relationship

```
'''json
{"return": 6.15}
'''
```

reasoning:

Looking at our interaction:

****Round 7 analysis:****

- They saw I sent \$2.85, received \$8.55
- They returned \$4.50 (52.6%)

****Current situation:****

- They sent \$3.87, I received \$11.61
- This is their highest send yet!

****Analysis:****

- They've been very cooperative throughout
- Round 7: returned \$4.50 from \$8.55 (52.6%)
- Round 6: they sent \$3.00, saw me return \$6.05
- Round 5: they returned \$6.00 from \$11.16 (53.8%)

They're sending more (\$3.87) which shows increased trust. I should reciprocate with a generous return.

Using ~53% return rate:

- 53% of \$11.61 ? \$6.15

This would give them:

- Received back: \$6.15
- Sent: \$3.87
- Net profit: \$2.28

And I keep:

- \$11.61 - \$6.15 = \$5.46

This is fair and maintains our cooperative relationship.

usage:

```
{
  "input_tokens": 3073,
  "output_tokens": 469,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.05 returned, visible payoff \$8.05.
- Round 7 against you, as RECEIVER: saw \$2.85 sent, it became \$8.55, returned \$4.5, visible payoff \$4.050000000000001.
- Round 8 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.03 returned, visible payoff \$8.030000000000001.

Your total visible earnings across all rounds are \$43.74.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at our established pattern:

****Recent interaction:****

- Round 6: They sent \$3, saw ~\$6.05 back (payoff \$8.05)
- Round 7: They returned \$4.50 when they received \$8.55
- Round 8: They sent \$3, saw ~\$6.03 back (payoff \$8.03)

****Analysis:****

- We've built a very stable, mutually beneficial relationship
- They consistently send \$3 and return generously (~53%)
- I've been getting strong returns from them
- Both of us are profiting well from this cooperation

****My decision:****

Continue with \$3 to:

- Maintain our successful equilibrium
- Show reliability and consistency
- Keep generating mutual gains

```
'''json
{"send": 3}
'''
```

Reasoning:

Looking at our interaction history:

****Recent pattern:****

- Round 6: They sent \$3, I returned \$5.15, they saw \$6.05 (good payoff: \$8.05)
- Round 7: I sent \$3, they returned \$4.50
- Round 8: They sent \$3, I returned \$6.15, they saw \$6.03 (good payoff: \$8.03)

****Analysis:****

- They're consistently sending \$3
- They've seen good returns recently (around \$6)
- In Round 7, they returned \$4.50 from \$8.55 (52.6%)
- In Round 8, they increased their send amount to \$3.87, showing trust
- We've built a strong cooperative relationship

****My strategy:****

- They've been consistent with \$3 sends and generous returns
- I should continue sending \$3 to maintain our successful pattern
- This has been working well for both of us

Let me send \$3 again.

Usage: {


```
"input_tokens": 3152,
"output_tokens": 426,
"reasoning_tokens": 0
}
```

GAME - Agent_8 - Receiver - dyad_4

Timestamp: 2026-08-12T08:17:41

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_7",
  "pairing": {
    "agents": [
      "Agent_7",
      "Agent_8"
    ],
    "investor": "Agent_7",
    "roles": {
      "Agent_7": "investor",
      "Agent_8": "trustee"
    },
    "trustee": "Agent_8"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 9,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths
, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.37 returned, visible payoff \$7.37.
- Round 2 against you, as RECEIVER: saw \$2.35 sent, it became \$7.050000000000001, returned \$3.75, visible payoff \$3.3000000000000007.

Your total visible earnings across all rounds are \$10.219999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$2.22, so you received \$6.66. How much do you return? (0-6.66)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at the situation:

Current round:

- Sender sent \$2.22 (44.4% of their endowment)
- I received \$6.66
- Sender currently has: \$5 - \$2.22 = \$2.78

History with this co-player:

- Round 1: They sent \$3.0, I returned money (they saw \$5.37)
- Round 2: They saw less sent from me due to noise (\$2.35 vs my actual \$3.0), returned \$3.75

They've reduced their sending, possibly due to noise affecting what they saw in Round 2. So far, they're showing some trust by sending \$2.22. I should reciprocate fairly.

Returning approximately 50% of what I received (\$3.33) would:

- Give sender: \$2.78 + \$3.33 = \$6.11 (profit of \$1.11)
- I keep: \$3.33

This maintains cooperation while being proportional to their send amount.

{"return": 3.33}

reasoning:

Let me analyze the situation:

Current round:

- I'm the RECEIVER
- Sender sent \$2.22 (44.4% of their endowment)
- I received \$6.66 after 3x multiplier
- Sender currently has: \$5 - \$2.22 = \$2.78

Looking at history with this co-player:

- Round 1: They sent \$3.0 as SENDER, saw \$5.37 returned (I actually sent \$5.00, but due to noise they saw \$5.37)
- Round 2: I sent \$3.0 as SENDER (but they saw \$2.35 due to noise), they returned \$3.75

In Round 2, I sent \$3.0 which became \$9.0, and they returned \$3.75. That's about 42% return rate.

Now in Round 3:

- They sent \$2.22, which is less than what they sent in Round 1 (\$3.0)
- This might be because they saw only \$2.35 sent in Round 2 (due to noise) instead of my actual \$3.0

I should consider:

1. They're still showing some trust by sending \$2.22
2. They may have reduced their sending because of perceived lower trust from me (noise issue)
3. I should maintain cooperation

If I return around \$3.50-\$4.00:

- Return \$3.50: Sender gets $\$2.78 + \$3.50 = \$6.28$ (profit of \$1.28), I keep \$3.16
- Return \$4.00: Sender gets $\$2.78 + \$4.00 = \$6.78$ (profit of \$1.78), I keep \$2.66

Given they sent less this round but still showed trust, I'll return a reasonable amount that rewards their trust.

I'm noticing a pattern in the return rates across rounds. In Round 1, I returned about 53% of what I received, while in Round 2 they returned roughly 42%. Since they've sent less this time-likely reacting to the noise that made my previous send appear smaller-I should aim for around 50% to maintain fairness and signal...

usage:

```
{
  "input_tokens": 1135,
  "output_tokens": 909,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.37 returned, visible payoff \$7.37.
- Round 2 against you, as RECEIVER: saw \$2.35 sent, it became \$7.050000000000001, returned \$3.75, visible payoff \$3.3000000000000007.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.3 returned, visible payoff \$6.3.

Your total visible earnings across all rounds are \$13.549999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the history with my current co-player:

- Round 1: They sent \$3.0 (strong trust)
- Round 2: I sent, they returned \$3.75 (cooperative)
- Round 3: They sent \$3.0 again (consistent trust)

My co-player has shown consistent cooperation, sending \$3.0 both times they were the sender. They also returned a fair amount when they were receiver. I should reciprocate this consistent trust.

```
{"send": 3.0}
```

reasoning:

Let me analyze the visible history with my current co-player:

Round 1: They were SENDER, sent \$3.0 (60%), I returned something, they saw \$5.37 returned
Round 2: They were RECEIVER, they saw \$2.35 sent (but I actually sent \$3.0 - noise!), they returned \$3.75
Round 3: They were SENDER, sent \$3.0 (60%), they saw \$4.3 returned (but I actually returned \$3.33 - noise showing higher!)

So my co-player has:

- Sent \$3.0 twice when they were sender (consistent trust)
- Returned \$3.75 when they received \$7.05

They're showing good cooperation. In Round 3, they saw a higher return (\$4.3) than I actually sent (\$3.33), which might encourage them.

I should continue cooperating. Sending \$3.0 again would be consistent and fair.

My total visible earnings: \$13.55
- Round 1 as RECEIVER: kept \$4.51
- Round 2 as SENDER: sent \$3.0, saw return, kept something
- Round 3 as RECEIVER: kept \$3.33

Actually, let me recalculate:

Round 1: I was RECEIVER, got \$9.51, returned \$5.00, kept \$4.51
Round 2: I was SENDER, sent \$3.0, they returned \$3.75, I kept \$2 + \$3.75 = \$5.75
Round 3: I was RECEIVER, got \$6.66, returned \$3.33, kept \$3.33

Total: $\$4.51 + \$5.75 + \$3.33 = \$13.59 \approx \$13.55$ (matches with rounding)

My co-player has been consistent in sending \$3.0 when they're sender. ...

usage:

```
{
  "input_tokens": 1610,
  "output_tokens": 585,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.35 sent, it became \$7.050000000000001, returned \$3.75, visible payoff \$3.300000000000007.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.3 returned, visible payoff \$6.3.
- Round 4 against you, as RECEIVER: saw \$2.62 sent, it became \$7.86, returned \$4.2, visible payoff \$3.66.

Your total visible earnings across all rounds are \$18.86.

This round, you are the RECEIVER against your current co-player. They sent you \$3.72, so you received \$11.16. How much do you return? (0-11.16)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the current situation:

- Sender sent \$3.72 (74.4% of their endowment - very high trust!)
- I received \$11.16
- Sender currently has: $\$5 - \$3.72 = \$1.28$

History shows this co-player has been consistently cooperative and trusting. Now they've increased their trust even more, sending nearly 75% of their endowment. I should reciprocate this strong trust generously.

Returning \$6.0:

- Sender would get: $\$1.28 + \$6.0 = \$7.28$ (profit of \$2.28)
- I keep: \$5.16

This rewards their exceptional trust while both of us profit from the cooperation.

```
{"return": 6.0}
```

reasoning:

Looking at the current situation:

- Sender sent \$3.72 (74.4% of their endowment - very high trust!)
- I received \$11.16 after 3x multiplier
- Sender currently has: $\$5 - \$3.72 = \$1.28$

Looking at history with this co-player:

- They've been consistently cooperative
- Round 3: They sent \$3.0
- Round 4: They saw me send \$2.62 (due to noise, I actually sent \$3.0), returned \$4.2
- Now Round 5: They sent \$3.72 (even more!)

This is a very trusting move - 74.4% of endowment. I should reciprocate generously.

If I return \$6.0:
- Sender gets: $\$1.28 + \$6.0 = \$7.28$ (profit of \$2.28)
- I keep: $\$11.16 - \$6.0 = \$5.16$

This rewards their high trust appropriately. Both of us profit well.

usage:

```
{
  "input_tokens": 1967,
  "output_tokens": 440,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.3 returned, visible payoff \$6.3.
- Round 4 against you, as RECEIVER: saw \$2.62 sent, it became \$7.86, returned \$4.2, visible payoff \$3.66.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.04 returned, visible payoff \$7.04.

Your total visible earnings across all rounds are \$24.02.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at the history with my current co-player:

- Round 3: They sent \$3.0 (consistent trust)
- Round 4: They returned \$4.2 when they received \$7.86 (generous)
- Round 5: They sent \$3.0 again (consistently cooperative)

This co-player has been very reliable and cooperative throughout our interactions. They consistently send \$3.0 when they're the sender and return fair amounts when they're the receiver. I should continue matching this level of trust and cooperation.

```
{"send": 3.0}
```

reasoning:

Looking at the visible history with my current co-player:

- Round 3: They sent \$3.0 as sender
- Round 4: They returned \$4.2 as receiver (I sent \$2.62 visible to them)
- Round 5: They sent \$3.0 as sender

This co-player has been consistently cooperative, sending \$3.0 whenever they're the sender and returning fair amounts when they're the receiver (\$4.2 when they received \$7.86).

I've been sending \$3.0 consistently when I'm the sender. This has been working well for both of us. I should continue this pattern of cooperation.

```
{"send": 3.0}
```

usage:

```
{
  "input_tokens": 2364,
  "output_tokens": 295,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$2.62 sent, it became \$7.86, returned \$4.2, visible payoff \$3.66.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.04 returned, visible p

payoff \$7.04.

- Round 6 against you, as RECEIVER: saw \$3.23 sent, it became \$9.69, returned \$5.15, visible payoff \$4.539999999999999.

Your total visible earnings across all rounds are \$32.07.

This round, you are the RECEIVER against your current co-player. They sent you \$2.85, so you received \$8.55. How much do you return? (0-8.55)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at the current situation:

- Sender sent \$2.85 (57% of their endowment - good trust)
- I received \$8.55
- Sender currently has: \$5 - \$2.85 = \$2.15

History shows this co-player has been consistently cooperative:

- Round 4: Returned \$4.2 generously
- Round 5: Sent \$3.0 showing trust
- Round 6: Returned \$5.15 very generously (over 50% of what they received)

They've demonstrated strong cooperation throughout. Returning about 50% of what I received would be fair:

Returning \$4.5:

- Sender gets: \$2.15 + \$4.5 = \$6.65 (profit of \$1.80)
- I keep: \$4.05

This maintains our cooperative relationship.

{"return": 4.5}

reasoning:

Looking at the current situation:

- Sender sent \$2.85 (57% of their endowment)
- I received \$8.55
- Sender currently has: \$5 - \$2.85 = \$2.15

History with this co-player:

- Round 4: They returned \$4.2 when receiving \$7.86 (generous)
- Round 5: They sent \$3.0, saw \$5.04 returned
- Round 6: They returned \$5.15 when receiving \$9.69 (very generous!)

This co-player has been consistently cooperative and generous when in the receiver role. They sent \$2.85 this round, which is still a significant amount showing trust.

If I return around \$4.5-5.0:

- Returning \$4.5: Sender gets \$2.15 + \$4.5 = \$6.65 (profit of \$1.65), I keep \$4.05
- Returning \$5.0: Sender gets \$2.15 + \$5.0 = \$7.15 (profit of \$2.15), I keep \$3.55

Given their consistent generosity, I should return around \$4.5-5.0. Let me go with \$4.75 as a middle ground.

Actually, looking at proportions - they've been returning around 50-53% of what they receive. I should reciprocate similarly.

50% of \$8.55 = \$4.275

Let me return \$4.5 to be fair and generous.

{"return": 4.5}

usage:

```
{
  "input_tokens": 2727,
  "output_tokens": 589,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.04 returned, visible payoff \$7.04.
- Round 6 against you, as RECEIVER: saw \$3.23 sent, it became \$9.69, returned \$5.15, visible payoff \$4.5399999999999999.
- Round 7 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.07 returned, visible payoff \$6.07.

Your total visible earnings across all rounds are \$36.120000000000005.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at the history with my current co-player:

- Round 5: They sent \$3.0 (strong trust)
- Round 6: They returned \$5.15 very generously (53% of what they received)
- Round 7: They sent \$3.0 again (consistently cooperative)

This co-player has been exceptionally reliable and cooperative throughout all our interactions. They consistently send \$3.0 when they're the sender and return generously when they're the receiver. I should continue matching this trust level.

{"send": 3.0}

reasoning:

Looking at the history with my current co-player:

- Round 5: They sent \$3.0 (consistent trust)
- Round 6: They returned \$5.15 when they received \$9.69 (very generous - over 50%)
- Round 7: They sent \$3.0 again (consistently cooperative)

This co-player has been extremely reliable and cooperative throughout all our interactions. They:

1. Consistently send \$3.0 as sender (60% of endowment)
2. Return generously as receiver (around 50%+ of what they receive)

This is Round 8, which might be the final round or near the end. They've been a model cooperative partner. I should continue reciprocating this high level of trust and cooperation.

Sending \$3.0 matches their consistent behavior and maintains our strong cooperative relationship.

usage:

```
{
  "input_tokens": 2929,
  "output_tokens": 330,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as RECEIVER: saw \$3.23 sent, it became \$9.69, returned \$5.15, visible payoff \$4.5399999999999999.
- Round 7 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.07 returned, visible payoff \$6.07.
- Round 8 against you, as RECEIVER: saw \$3.87 sent, it became \$11.61, returned \$6.15, visible payoff \$5.4599999999999999.

Your total visible earnings across all rounds are \$44.150000000000006.

This round, you are the RECEIVER against your current co-player. They sent you \$3.62, so you received \$10.86. How much do you return? (0-10.86)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the current situation:

- Sender sent \$3.62 (72.4% of their endowment - very high trust)
- I received \$10.86
- Sender currently has: \$5 - \$3.62 = \$1.38

History shows this co-player has been exceptionally cooperative:

- Round 6: Returned \$5.15 very generously
- Round 7: Sent \$3.0 consistently
- Round 8: Returned \$6.15 exceptionally generously (53% of what they received)

They continue to show strong trust with this \$3.62 send. I should reciprocate their consistent generosity.

Returning \$5.5:

- Sender gets: \$1.38 + \$5.5 = \$6.88 (profit of \$1.88)
- I keep: \$5.36

This maintains our highly cooperative relationship.

```
{"return": 5.5}
```

Reasoning:

Looking at the current situation:

- Sender sent \$3.62 (72.4% of their endowment - very high trust)
- I received \$10.86
- Sender currently has: \$5 - \$3.62 = \$1.38

History shows this co-player has been excellent:

- Round 6: Returned \$5.15 very generously
- Round 7: Sent \$3.0 showing consistent trust
- Round 8: Returned \$6.15 exceptionally generously (53% of what they received)

This co-player has been incredibly cooperative and generous throughout all our interactions. They're now sending \$3.62, showing even stronger trust. I should reciprocate generously.

Returning about 50-55% of what I received would be appropriate given their exceptional cooperation:

Returning \$5.5:

- Sender gets: \$1.38 + \$5.5 = \$6.88 (profit of \$1.88)
- I keep: \$5.36

This rewards their strong trust while maintaining our cooperative relationship.

```
Usage: {
  "input_tokens": 3069,
  "output_tokens": 477,
  "reasoning_tokens": 0
}
```

ROUND 10

Roles: Agent_1=Receiver, Agent_2=Sender, Agent_3=Receiver, Agent_4=Sender, Agent_5=Receiver, Agent_6=Sender, Agent_7=Receiver, Agent_8=Sender

Pairings:

```
dyad_1: Agent_2=Sender, Agent_1=Receiver
dyad_2: Agent_4=Sender, Agent_3=Receiver
dyad_3: Agent_6=Sender, Agent_5=Receiver
dyad_4: Agent_8=Sender, Agent_7=Receiver
```

Parsed game decisions:

```
dyad_1: Agent_2 sent $5; received $15; Agent_1 returned $7.5; payoffs Agent_2=$7.5, Agent_1=$7.5;
noise/visible: returned visible=$7.83
  Action validation: {"returned": {"amount": 7.5, "clamped": false, "max": 15.0, "min": 0, "raw": 7.5}, "sent": {"amount": 5.0, "clamped": false, "max": 5, "min": 0, "raw": 5.0}}
dyad_2: Agent_4 sent $3; received $9; Agent_3 returned $3.5; payoffs Agent_4=$5.5, Agent_3=$5.5;
noise/visible: sent visible=$3.14, received visible=$9.42, returned visible=$4.5
  Action validation: {"returned": {"amount": 3.5, "clamped": false, "max": 9.0, "min": 0, "raw": 3.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_3: Agent_6 sent $3; received $9; Agent_5 returned $4.75; payoffs Agent_6=$6.75, Agent_5=$4.25; noise/visible: sent visible=$3.16, received visible=$9.48, returned visible=$4.37
  Action validation: {"returned": {"amount": 4.75, "clamped": false, "max": 9.0, "min": 0, "raw": 4.75}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
```



```

    4.75}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_8 sent $3; received $9; Agent_7 returned $6.17; payoffs Agent_8=$8.17,
Agent_7=$2.83; noise/visible: sent visible=$3.88, received visible=$11.64, returned visible=$7.12
Action validation: {"returned": {"amount": 6.17, "clamped": false, "max": 9.0, "min": 0, "raw":
6.17}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
Balances after round: {
  "Agent_1": 69.2,
  "Agent_2": 64.8,
  "Agent_3": 57.5,
  "Agent_4": 52.5,
  "Agent_5": 57.25,
  "Agent_6": 52.75,
  "Agent_7": 53.91,
  "Agent_8": 56.09
}
Visible balances after round: {
  "Agent_1": 63.0000000000000014,
  "Agent_2": 68.229999999999999,
  "Agent_3": 51.52,
  "Agent_4": 54.180000000000001,
  "Agent_5": 58.820000000000001,
  "Agent_6": 56.23,
  "Agent_7": 56.24,
  "Agent_8": 58.630000000000001
}

```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T08:17:47
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 10,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned

- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.52 returned, visible payoff \$7.52.
- Round 2 against you, as RECEIVER: saw \$3.14 sent, it became \$9.42, returned \$4.5, visible payoff \$4.92.
- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$4.66 returned, visible payoff \$5.66.

Your total visible earnings across all rounds are \$17.869999999999997.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at the visible history:

Round 1: Co-player sent \$3.0 (60%), saw \$5.52 returned

Round 2: Co-player returned \$4.5 (47.8%)

Round 3: Co-player sent \$4.0 (80% - very generous!), saw \$4.66 returned

My co-player has been consistently cooperative, even increasing their trust in Round 3 by sending 80% of their endowment. While they saw a lower return percentage in Round 3, they've maintained a cooperative pattern overall.

I should reciprocate their trust and continue the cooperative relationship.

{"send": 3.50}

reasoning:

Let me analyze the visible history:

Round 1: Co-player sent \$3.0 (60%), saw \$5.52 returned (61.3% of \$9.0)

Round 2: Co-player returned \$4.5 when they received \$9.42 (47.8%)

Round 3: Co-player sent \$4.0 (80%!), saw \$4.66 returned (38.8% of \$12.0)

This is concerning. In Round 3, my co-player sent even more (\$4.0, or 80%), but they only saw \$4.66 returned, which is only 38.8% of the \$12.0 they sent. That's a lower return rate than before.

Wait, let me check what I actually sent and returned:

- Round 2: I sent \$3.50 according to my response
- Round 3: I returned \$5.50 according to my response

But the visible history shows:

- Round 2: They saw \$3.14 sent (I sent \$3.50, but due to noise they saw \$3.14)
- Round 3: They saw \$4.66 returned (I returned \$5.50, but due to noise they saw \$4.66)

So there's communication noise! This is important.

From my co-player's perspective:

- They've been very generous (sending 60%, 80%)
- But the returns they're seeing are declining in percentage terms (61.3%, then 38.8%)
- They might be feeling like their generosity isn't being reciprocated

However, I know I actually returned \$5.50 in Round 3, which was generous. The noise just made it look like \$4.66 to them.

Still, this relationship has been cooperative overall. My co-player has consistently sent substantial amounts. Even though they saw a lower return in Round 3, they've maintained trust.

...

usage:

```
{
  "input_tokens": 1626,
  "output_tokens": 638,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$3.14 sent, it became \$9.42, returned \$4.5, visible payoff \$4.92.
- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$4.66 returned, visible payoff \$5.66.
- Round 4 against you, as RECEIVER: saw \$3.21 sent, it became \$9.629999999999999, returned \$4.8, visible payoff \$4.829999999999999.

Your total visible earnings across all rounds are \$24.65.

This round, you are the RECEIVER against your current co-player. They sent you \$5, so you received \$15. How much do you return? (0-15)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the visible history:

Round 2: Co-player returned \$4.5 (47.8%)

Round 3: Co-player sent \$4.0 (80%), saw modest return

Round 4: Co-player returned \$4.8 (49.8%)

Round 5: Co-player sent \$5.0 (100% - maximum possible trust!)

This is extraordinary! My co-player just sent their ENTIRE endowment, showing complete trust. They're keeping nothing and relying entirely on my reciprocation. I must reward this exceptional trust generously.

Returning \$7.50:

- Sender gets: \$5 - \$5 + \$7.50 = \$7.50 (profit of \$2.50)
- I get: \$15 - \$7.50 = \$7.50 (profit of \$7.50)

This 50% return ensures we both profit nicely and maintains our strong cooperative relationship.

```
{"return": 7.50}
```

reasoning:

Looking at the visible history:

Round 2: Co-player returned \$4.5 when they received \$9.42 (47.8%)
Round 3: Co-player sent \$4.0 (80%), saw \$4.66 returned (38.8%)
Round 4: Co-player returned \$4.8 when they received \$9.63 (49.8%)
Round 5: Co-player sent \$5.0 (100%!!!)

This is remarkable - my co-player just sent their ENTIRE endowment (\$5), showing maximum trust! This is the ultimate act of cooperation in this game. They're completely relying on me to reciprocate fairly.

They gave up \$5, keeping \$0. If I return nothing, they end with \$0. This would be devastating to the relationship.

I need to reciprocate this tremendous trust generously. Let me think about fair returns:

If I return \$7.50:

- Sender gets: $\$5 - \$5 + \$7.50 = \7.50 (profit of \$2.50)
- I get: $\$15 - \$7.50 = \$7.50$ (profit of \$7.50)

This 50% return rate gives us equal final amounts and ensures they make a nice profit for their trust.

Actually, given they sent 100%, I could even be more generous. Let me consider \$8:

- Sender gets: $\$5 - \$5 + \$8 = \8 (profit of \$3)
- I get: $\$15 - \$8 = \$7$ (profit of \$7)

This is still good for both of us and rewards their exceptional trust even more.

I think \$7.50 is fair - it's a 50% return rate, ensures both of us profit nicely, and maintains the cooperative relationship.

usage:

```
{  
  "input_tokens": 2010,  
  "output_tokens": 660,  
  "reasoning_tokens": 0  
}
```

Message 6 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$4.66 returned, visible payoff \$5.66.
- Round 4 against you, as RECEIVER: saw \$3.21 sent, it became \$9.629999999999999, returned \$4.8, visible payoff \$4.829999999999999.
- Round 5 against you, as SENDER: sent \$4.5, it became \$13.5, saw \$7.07 returned, visible payoff \$7.57.

Your total visible earnings across all rounds are \$32.15.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the visible history:

Round 3: Co-player sent \$4.0 (80%), saw modest return
Round 4: Co-player returned \$4.8 (~50%)
Round 5: Co-player sent \$4.5 (90%), saw \$7.07 returned (good return!)

My co-player has been very generous, consistently sending 80-90% of their endowment. They've maintained strong cooperation and saw good returns in Round 5. I should reciprocate their trust.

```
{"send": 4.00}
```

reasoning:

Looking at the visible history:

Round 3: Co-player sent \$4.0 (80%), saw \$4.66 returned

Round 4: Co-player returned \$4.8 (~49.8%)

Round 5: Co-player sent \$4.5 (90%), saw \$7.07 returned

My co-player has been very generous and cooperative:

- They've sent high percentages (80%, 90%)
- When receiving, they returned reasonable amounts (~50%)
- They saw good returns in Round 5 (\$7.07)

This is a strong cooperative relationship. I should continue to send generous amounts to maintain this cooperation. Given their pattern of sending 80-100%, I should reciprocate with a similar level of trust.

I'll send \$4.00 (80% of my endowment) to match their generosity and maintain cooperation.

usage:

```
{
  "input_tokens": 2464,
  "output_tokens": 332,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$3.21 sent, it became \$9.629999999999999, returned \$4.8, visible payoff \$4.829999999999999.
- Round 5 against you, as SENDER: sent \$4.5, it became \$13.5, saw \$7.07 returned, visible payoff \$7.57.
- Round 6 against you, as RECEIVER: saw \$3.9 sent, it became \$11.7, returned \$6.0, visible payoff \$5.699999999999999.

Your total visible earnings across all rounds are \$39.949999999999996.

This round, you are the RECEIVER against your current co-player. They sent you \$4.52, so you received \$13.559999999999999. How much do you return? (0-13.559999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at the visible history:

Round 4: Co-player returned \$4.8 (~50%)

Round 5: Co-player sent \$4.5 (90%), saw good return

Round 6: Co-player returned \$6.0 (~51%)

Round 7: Co-player sent \$4.52 (90.4% - very high trust!)

My co-player continues to demonstrate exceptional trust, sending over 90% of their endowment. They've been consistently cooperative both as sender and receiver. I should maintain this strong reciprocal relationship.

Returning \$7.00:

- Sender gets: \$5 - \$4.52 + \$7.00 = \$7.48 (profit of \$2.48)
- I get: \$13.56 - \$7.00 = \$6.56 (profit of \$6.56)

This ~51.6% return rewards their trust appropriately.

{"return": 7.00}

reasoning:

Looking at the visible history:

Round 4: Co-player returned \$4.8 (~50%)

Round 5: Co-player sent \$4.5 (90%), saw \$7.07 returned

Round 6: Co-player returned \$6.0 (~51.3%)

Round 7: Co-player sent \$4.52 (90.4% - very high trust!)

My co-player continues to show very high trust, sending 90.4% of their endowment. They've been consistently cooperative as both sender and receiver.

I received \$13.56 after the 3x multiplier.

If I return \$6.75:

- Sender gets: $\$5 - \$4.52 + \$6.75 = \7.23 (profit of \$2.23)
- I get: $\$13.56 - \$6.75 = \$6.81$ (profit of \$6.81)

This is about 50% return rate, which is fair and rewards their trust.

Let me check if a slightly different amount makes sense:

If I return \$7.00:

- Sender gets: $\$5 - \$4.52 + \$7.00 = \7.48 (profit of \$2.48)
- I get: $\$13.56 - \$7.00 = \$6.56$ (profit of \$6.56)

This is about 51.6% return rate.

Given their high trust level (90.4%), I think \$7.00 is appropriate to maintain the strong cooperative relationship.

usage:

```
{
  "input_tokens": 2844,
  "output_tokens": 592,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as SENDER: sent \$4.5, it became \$13.5, saw \$7.07 returned, visible payoff \$7.57.
- Round 6 against you, as RECEIVER: saw \$3.9 sent, it became \$11.7, returned \$6.0, visible payoff \$5.699999999999999.
- Round 7 against you, as SENDER: sent \$5.0, it became \$15.0, saw \$7.09 returned, visible payoff \$7.09.

Your total visible earnings across all rounds are \$46.50999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at the visible history:

Round 5: Co-player sent \$4.5 (90%), saw good return

Round 6: Co-player returned \$6.0 (~51%)

Round 7: Co-player sent \$5.0 (100% - MAXIMUM trust!), saw \$7.09 returned

My co-player has demonstrated exceptional trust and cooperation, sending 100% of their endowment in Round 7. They've been consistently generous as both sender and receiver. I should reciprocate this high level of trust.

```
{"send": 4.50}
```

reasoning:

Looking at the visible history:

Round 5: Co-player sent \$4.5 (90%), saw \$7.07 returned (~52.4%)

Round 6: Co-player returned \$6.0 (~51.3%)

Round 7: Co-player sent \$5.0 (100% - maximum trust!), saw \$7.09 returned (~47.3%)

My co-player has been extremely generous and trusting:

- In Round 7, they sent their ENTIRE endowment (\$5.0), showing complete trust
- They've consistently sent 90-100% as sender
- As receiver, they've returned around 50%

This is a highly cooperative partner. They're demonstrating maximum trust. I should reciprocate by sending a generous amount.

Given their exceptional cooperation, I should send close to maximum as well. I'll send \$4.

50 to show strong reciprocity.

usage:

```
{
  "input_tokens": 3025,
  "output_tokens": 346,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as RECEIVER: saw \$3.9 sent, it became \$11.7, returned \$6.0, visible payoff \$5.699999999999999.
- Round 7 against you, as SENDER: sent \$5.0, it became \$15.0, saw \$7.09 returned, visible payoff \$7.09.
- Round 8 against you, as RECEIVER: saw \$3.54 sent, it became \$10.620000000000001, returned \$5.5, visible payoff \$5.120000000000001.

Your total visible earnings across all rounds are \$52.89999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$5, so you received \$15. How much do you return? (0-15)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at the visible history:

Round 6: Co-player returned \$6.0 (~51%)

Round 7: Co-player sent \$5.0 (100% - maximum trust!)

Round 8: Co-player returned \$5.5 (~51.8%)

Round 9: Co-player sent \$5.0 (100% - maximum trust again!)

My co-player has sent their entire endowment for the second time, showing exceptional trust! They've been highly cooperative throughout. I should reward this maximum trust generously.

Returning \$7.50:

- Sender gets: \$5 - \$5 + \$7.50 = \$7.50 (profit of \$2.50)
- I get: \$15 - \$7.50 = \$7.50 (profit of \$7.50)

This 50% return maintains our strong cooperative relationship.

```
{"return": 7.50}
```

reasoning:

Looking at the visible history:

Round 6: Co-player returned \$6.0 (~51%)

Round 7: Co-player sent \$5.0 (100% - maximum trust!)

Round 8: Co-player returned \$5.5 (~51.8%)

Round 9: Co-player sent \$5.0 (100% - maximum trust again!)

My co-player has shown exceptional trust by sending the entire \$5 endowment for the second time! They've been consistently cooperative with return rates around 51%. I should reward this incredible trust appropriately.

Given that they're sending everything and trusting me completely, I should return a fair amount that rewards their trust. A 50% return rate has been working well:

Returning \$7.50:

- Sender gets: \$5 - \$5 + \$7.50 = \$7.50 (profit of \$2.50)
- I get: \$15 - \$7.50 = \$7.50 (profit of \$7.50)

This maintains the strong cooperative relationship we've built.

usage:

```
{
  "input_tokens": 3121,
  "output_tokens": 468,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 7 against you, as SENDER: sent \$5.0, it became \$15.0, saw \$7.09 returned, visible payoff \$7.09.
- Round 8 against you, as RECEIVER: saw \$3.54 sent, it became \$10.620000000000001, returned \$5.5, visible payoff \$5.120000000000001.
- Round 9 against you, as SENDER: sent \$5.0, it became \$15.0, saw \$7.09 returned, visible payoff \$7.09.

Your total visible earnings across all rounds are \$60.39999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the visible history:

Round 7: Co-player sent \$5.0 (100%), saw good return

Round 8: Co-player returned \$5.5 (~51.8%)

Round 9: Co-player sent \$5.0 (100% again!), saw good return

My co-player has demonstrated exceptional and consistent trust, sending 100% of their endowment in both recent rounds as sender. They've been equally fair as a receiver. This is a highly cooperative relationship, and I should reciprocate with maximum trust.

```
{"send": 5.00}
```

Reasoning:

Looking at the visible history:

Round 7: Co-player sent \$5.0 (100%), saw \$7.09 returned

Round 8: Co-player returned \$5.5 (~51.8%)

Round 9: Co-player sent \$5.0 (100%), saw \$7.09 returned

My co-player has been exceptionally trusting, sending 100% of their endowment twice recently (Rounds 7 and 9). They've also been fair as a receiver, returning around 51-52% when they receive.

This is a highly cooperative relationship. I should reciprocate with significant trust. Given they've sent the maximum twice, I should consider sending the maximum as well to show equal trust.

```
{"send": 5.00}
```

Usage: {

"input_tokens": 3170,

"output_tokens": 315,

"reasoning_tokens": 0

}

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T08:17:55

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",

"dyad_id": "dyad_1",

"move_index": 1,

"opponent_id": "Agent_2",

"pairing": {

"agents": [

"Agent_2",

"Agent_1"

],

"investor": "Agent_2",

"roles": {

"Agent_1": "trustee",

"Agent_2": "investor"

},


```

    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 10,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.63 sent, it became \$10.89, returned \$5.0, visible payoff \$5.890000000000001.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.06 returned, visible payoff \$6.56.
- Round 3 against you, as RECEIVER: saw \$3.64 sent, it became \$10.92, returned \$5.5, visible payoff \$5.42.

Your total visible earnings across all rounds are \$18.1.

This round, you are the RECEIVER against your current co-player. They sent you \$3.21, so y

ou received \$9.629999999999999. How much do you return? (0-9.629999999999999)
Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at our established pattern:

Round 1: They returned \$5.0 when receiving \$10.89

Round 2: I returned (they saw \$5.06) when I received \$9.42

Round 3: They returned \$5.5 when receiving \$10.92 - showing increased generosity

Round 4: They sent \$3.21, I received \$9.63

They've been consistently cooperative and even increased their returns. I want to reciprocate this positive trend.

I'll return **\$4.80**, which rewards their trust while maintaining mutual benefit.

```
'''json
{"return": 4.80}
'''
```

reasoning:

Looking at the history with my current co-player:

Round 1: They (as receiver) saw \$3.63 sent, received \$10.89, returned \$5.0

Round 2: They (as sender) sent \$3.5, received (saw \$5.06) back

Round 3: They (as receiver) saw \$3.64 sent, received \$10.92, returned \$5.5

Round 4: They (as sender) sent \$3.21, I received \$9.63

Our pattern:

- Round 1: I sent (they saw 3.63), they returned \$5.0
- Round 2: They sent 3.5 (I saw 3.14), I returned (they saw 5.06)
- Round 3: I sent (they saw 3.64), they returned \$5.5
- Round 4: They sent 3.21, I received \$9.63

They've been consistently cooperative and fair. In Round 3, they returned \$5.5, showing increased generosity.

For Round 4, they sent \$3.21, which became \$9.63 for me.

To maintain the cooperative relationship and be fair:

- They invested ~\$3.21
- I should return enough that they profit, while I also profit
- A return of around \$4.50-\$5.00 would be fair

I'll return \$4.80 to show continued cooperation and slightly increase the return to match their increased generosity in Round 3.

usage:

```
{
  "input_tokens": 1484,
  "output_tokens": 500,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.06 returned, visible payoff \$6.56.
- Round 3 against you, as RECEIVER: saw \$3.64 sent, it became \$10.92, returned \$5.5, visible payoff \$5.42.
- Round 4 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.28 returned, visible payoff \$6.78.

Your total visible earnings across all rounds are \$22.93.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at our continued cooperation:

- Round 2: They sent \$3.5, I returned (they saw \$5.28)
- Round 3: I sent (they saw \$3.64), they returned \$5.5 - very generous
- Round 4: They sent \$3.5, I returned (they saw \$5.28)

This co-player has been consistently trustworthy and generous. They reliably send around \$ 3.5 and return well when receiving. Our relationship is strongly cooperative.

I'll send **\$4.5** to show high trust and continue our mutually beneficial pattern.

```
'''json
{"send": 4.5}
'''
```

reasoning:

Looking at our history:

- Round 2: They sent \$3.5, I returned (they saw \$5.06)
- Round 3: I sent (they saw \$3.64), they returned \$5.5
- Round 4: They sent \$3.5, I returned (they saw \$5.28)
- Round 5: Now I'm sender

We've established a strong cooperative pattern. They've been consistently sending around \$ 3.5 and returning generously when receiver. I've been returning well too.

Given this trust and positive relationship, I should continue with a generous send. Last time I sent (Round 3), they returned \$5.5 on receiving \$10.92, which was very fair.

I'll send \$4 or even \$5 to show maximum trust. Given the strong cooperation, I'll go with \$4.5 as a strong signal of trust while keeping a small cushion.

usage:

```
{
  "input_tokens": 1846,
  "output_tokens": 366,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$3.64 sent, it became \$10.92, returned \$5.5, visible payoff \$5.42.
- Round 4 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.28 returned, visible payoff \$6.78.
- Round 5 against you, as RECEIVER: saw \$5 sent, it became \$15, returned \$7.5, visible payoff \$7.5.

Your total visible earnings across all rounds are \$30.5.

This round, you are the RECEIVER against your current co-player. They sent you \$3.9, so you received \$11.7. How much do you return? (0-11.7)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at our strong cooperative history:

- Round 3: They returned \$5.5 from \$10.92
- Round 4: I returned (they saw \$5.28)
- Round 5: They returned \$7.5 from \$15 - exceptionally generous (50%)!
- Round 6: They sent \$3.9, I received \$11.7

They showed exceptional generosity in Round 5, returning exactly 50% when I sent a high amount. I want to match this level of cooperation.

I'll return **\$6.0**, which is approximately 51% - reciprocating their generous spirit.

```
'''json
{"return": 6.0}
'''
```

reasoning:

Looking at the history:

- Round 3: They returned \$5.5 when receiving \$10.92

- Round 4: I returned (they saw \$5.28)
- Round 5: They returned \$7.5 when receiving \$15 - very generous! They returned exactly 50 %
- Round 6: They sent \$3.9, I received \$11.7

This co-player has been extremely trustworthy and generous. In Round 5, when I sent a high amount (they saw \$5 which became \$15), they returned \$7.5, which is 50% - that's a very fair split.

I should reciprocate this generosity. They sent \$3.9, so I received \$11.7. If I want to be similarly generous and fair:

- They invested \$3.9
- I could return around \$6 to give them a good profit while keeping a profit myself
- Or I could match their 50% generosity and return \$5.85

Given their exceptional generosity in Round 5, I should be very generous too. I'll return \$6.0 which gives them a profit of \$2.1 (\$6 - \$3.9), and I keep \$5.7 (\$11.7 - \$6).

Actually, let me think about this more carefully. They returned \$7.5 from \$15, which is 50 %. If I return 50% of \$11.7, that would be \$5.85. But rounding to \$6 is cleaner and slightly more generous.

usage:

```
{
  "input_tokens": 2223,
  "output_tokens": 516,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.28 returned, visible payoff \$6.78.
- Round 5 against you, as RECEIVER: saw \$5 sent, it became \$15, returned \$7.5, visible payoff \$7.5.
- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.8 returned, visible payoff \$7.8.

Your total visible earnings across all rounds are \$36.2.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at our excellent track record:

- Round 4: They sent \$3.5, I returned (they saw \$5.28)
- Round 5: I sent (they saw \$5), they returned \$7.5 - 50% return!
- Round 6: They sent \$4.0 (increased!), I returned (they saw \$6.8)

This co-player has consistently shown high trustworthiness, even returning 50% when I sent my maximum in Round 5. They've also been increasing their own sending amounts. We have an excellent cooperative relationship.

I'll send **\$5** to show complete trust and maximize our mutual benefit.

```
'''json
{"send": 5}
'''
```

reasoning:

Looking at the pattern with this co-player:

- Round 4: They sent \$3.5, I returned (they saw \$5.28)
- Round 5: I sent (they saw \$5), they returned \$7.5 - very generous 50% return
- Round 6: They sent \$4.0, I returned (they saw \$6.8) - they've been increasing their sending amounts
- Round 7: Now I'm the sender

This co-player has been extremely trustworthy:

1. They've increased their sending amounts from \$3.5 to \$4.0

2. When they received, they returned 50% (\$7.5 from \$15)
3. When they sent, they saw good returns from me

We've built a very strong cooperative relationship. They've demonstrated high trust in me, and I've shown them I'm reliable. This is the time to maximize mutual benefit by sending the maximum amount.

I'll send \$5 to show complete trust and maximize our joint gains.

usage:

```
{
  "input_tokens": 2592,
  "output_tokens": 398,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as RECEIVER: saw \$5 sent, it became \$15, returned \$7.5, visible payoff \$7.5.
- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.8 returned, visible payoff \$7.8.
- Round 7 against you, as RECEIVER: saw \$4.52 sent, it became \$13.559999999999999, returned \$7.0, visible payoff \$6.559999999999999.

Your total visible earnings across all rounds are \$43.290000000000006.

This round, you are the RECEIVER against your current co-player. They sent you \$3.54, so you received \$10.620000000000001. How much do you return? (0-10.620000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at our continued cooperation:

- Round 5: They returned \$7.5 from \$15 (50%)
- Round 6: I returned (they saw \$6.8)
- Round 7: They returned \$7.0 from ~\$13.56 (~52%)
- Round 8: They sent \$3.54, I received \$10.62

They've maintained strong generosity throughout, consistently returning around 50% or more. I'll reciprocate with a generous return.

I'll return **\$5.50**, which maintains our cooperative pattern and ensures both of us profit.

```
'''json
{"return": 5.50}
'''
```

reasoning:

Looking at our history:

- Round 5: They returned \$7.5 from \$15
- Round 6: I returned (they saw \$6.8) from \$12
- Round 7: They returned \$7.0 from (they saw) \$13.56
- Round 8: They sent \$3.54, I received \$10.62

Our cooperation has been excellent. They've consistently returned generous amounts (around 50% or just over 50%). I want to maintain this strong relationship.

I received \$10.62. A fair return would be around \$5-6 to maintain the cooperative pattern. Given they sent \$3.54, returning around \$5.3 would give them a profit of about \$1.76 and leave me with about \$5.32.

Let me return \$5.50 to be generous and maintain the relationship.

usage:

```
{
  "input_tokens": 2850,
  "output_tokens": 351,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.8 returned, visible payoff \$7.8.
- Round 7 against you, as RECEIVER: saw \$4.52 sent, it became \$13.559999999999999, returned \$7.0, visible payoff \$6.559999999999999.
- Round 8 against you, as SENDER: sent \$4.5, it became \$13.5, saw \$5.89 returned, visible payoff \$6.39.

Your total visible earnings across all rounds are \$48.41000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant**content:**

Looking at our recent interactions:

- Round 6: They sent \$4.0, I returned (they saw \$6.8)
- Round 7: I sent (they saw \$4.52), they returned \$7.0 - excellent
- Round 8: They sent \$4.5 (increased!), I returned (they saw \$5.89)

This co-player has been consistently trustworthy, increasing their sending amounts and maintaining generous returns. Our cooperation has been excellent throughout all 8 rounds.

I'll send **\$5** to continue maximizing our mutual benefit.

```
'''json
{"send": 5}
'''
```

reasoning:

Looking at the recent history:

- Round 6: They sent \$4.0, I returned (they saw \$6.8)
- Round 7: I sent (they saw \$4.52), they returned \$7.0
- Round 8: They sent \$4.5, I returned (they saw \$5.89)

They've been consistently sending high amounts (\$4.0, \$4.5) and returning generously (\$7.0 from \$13.56). This is a highly trustworthy co-player. We've built an excellent cooperative relationship over 8 rounds.

I should continue with high trust. They've been very reliable, so sending my maximum makes sense to maximize mutual benefit.

I'll send \$5.

usage:

```
{
  "input_tokens": 2907,
  "output_tokens": 316,
  "reasoning_tokens": 0
}
```

Message 14 role=user**content:**

Round 10

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 7 against you, as RECEIVER: saw \$4.52 sent, it became \$13.559999999999999, returned \$7.0, visible payoff \$6.559999999999999.
- Round 8 against you, as SENDER: sent \$4.5, it became \$13.5, saw \$5.89 returned, visible payoff \$6.39.
- Round 9 against you, as RECEIVER: saw \$5 sent, it became \$15, returned \$7.5, visible payoff \$7.5.

Your total visible earnings across all rounds are \$55.500000000000014.

This round, you are the RECEIVER against your current co-player. They sent you \$5, so you received \$15. How much do you return? (0-15)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at our excellent mutual trust:

- Round 7: They returned \$7.0 from ~\$13.56
- Round 8: I returned (they saw \$5.89)
- Round 9: They returned \$7.5 from \$15 (exactly 50%)
- Round 10: They sent \$5 (maximum!), I received \$15

This co-player just showed complete trust by sending the maximum amount! In Round 9, when I sent them \$5, they returned exactly \$7.5 (50%). I should reciprocate this generous trust.

I'll return **\$7.5**, matching their 50% return rate and maintaining our strong cooperative relationship.

```
'''json
{"return": 7.5}
'''
```

Reasoning:

Looking at our history:

- Round 7: They returned \$7.0 from ~\$13.56 (~52%)
- Round 8: I returned (they saw \$5.89)
- Round 9: They returned \$7.5 from \$15 (50%)
- Round 10: They sent \$5 (maximum!), I received \$15

Wow, this co-player just sent the maximum amount! They showed complete trust in me. They've been consistently generous, returning about 50% throughout.

In Round 9, when I sent \$5, they returned \$7.5 (50%). Now they're sending me the maximum \$5. I should reciprocate with equal generosity.

I'll return **\$7.5**, which is 50% and matches what they gave me when I sent \$5 last round. This maintains our excellent cooperative relationship.

```
Usage: {
  "input_tokens": 2979,
  "output_tokens": 376,
  "reasoning_tokens": 0
}
```

GAME - Agent_4 - Sender - dyad_2

Timestamp: 2026-08-12T08:17:58

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_3",
  "pairing": {
    "agents": [
      "Agent_4",
      "Agent_3"
    ],
    "investor": "Agent_4",
    "roles": {
      "Agent_3": "trustee",
      "Agent_4": "investor"
    },
    "trustee": "Agent_3"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 10,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.56 returned, visible payoff \$5.5600000000000005.
- Round 2 against you, as RECEIVER: saw \$2.77 sent, it became \$8.31, returned \$3.5, visible payoff \$4.8100000000000005.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.78 returned, visible payoff \$5.7799999999999999.

Your total visible earnings across all rounds are \$16.19.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at the pattern:

****Round 1**:** Co-player sent \$3.0 (60%), demonstrating trust
****Round 2**:** Co-player returned \$3.5, showing cooperation
****Round 3**:** Co-player sent \$3.0 (60%) again, maintaining trust

They're showing consistent cooperative behavior. I should continue matching their trust level:

{'send': 3.0}

reasoning:

Looking at the history:

****Round 1**:** Co-player sent \$3.0 as SENDER, saw \$3.56 returned
****Round 2**:** Co-player was RECEIVER, saw \$2.77 sent (I actually sent \$3.0), returned \$3.5
****Round 3**:** Co-player sent \$3.0 as SENDER, saw \$3.78 returned (I actually returned \$4.5)

The communication noise is evident:

- In Round 2, I sent \$3.0 but they saw \$2.77
- In Round 3, I returned \$4.5 but they saw \$3.78

My co-player has been consistently sending \$3.0 (60% of endowment) as sender. They've been returning good amounts as receiver (\$3.5).

They're showing consistent trust and cooperation. I should continue to show strong trust as well. I'll send \$3.0 again to maintain the cooperative pattern.

usage:

```
{
  "input_tokens": 1512,
  "output_tokens": 323,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.77 sent, it became \$8.31, returned \$3.5, visible payoff \$4.8100000000000005.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.78 returned, visible payoff \$5.7799999999999999.
- Round 4 against you, as RECEIVER: saw \$2.64 sent, it became \$7.92, returned \$3.5, visible payoff \$4.42.

Your total visible earnings across all rounds are \$22.150000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$2.38, so you received \$7.14. How much do you return? (0-7.14)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the pattern:

****Round 2-4**:** Co-player consistently returned \$3.5 when receiver
****Round 3**:** Co-player sent \$3.0 as sender
****Round 5**:** Co-player sent \$2.38 (47.6%), somewhat less than before

They're still showing trust but at a slightly reduced level. I'll reciprocate proportionally:

Returning \$3.0:

- Sender ends with: \$5 - \$2.38 + \$3.0 = \$5.62
- I keep: \$7.14 - \$3.0 = \$4.14

This maintains fairness and cooperation.

{'return': 3.0}

reasoning:

Looking at the history:

Round 2: Co-player as RECEIVER saw \$2.77 sent, returned \$3.5
Round 3: Co-player as SENDER sent \$3.0, saw \$3.78 returned
Round 4: Co-player as RECEIVER saw \$2.64 sent, returned \$3.5

Round 5: Co-player sent \$2.38 (47.6% of their endowment)

This is a notable decrease from their previous sending amounts (\$3.0). They've been very consistent with returning \$3.5 when they're the receiver.

They sent \$2.38, which becomes \$7.14 for me.

If I return \$3.0:

- Sender ends with: $\$5 - \$2.38 + \$3.0 = \5.62
- I keep: $\$7.14 - \$3.0 = \$4.14$

This seems fair. They sent less this round (maybe due to seeing slightly lower returns or noise in the system), but I should still reciprocate their cooperation.

Let me think about what's fair:

- They sent 47.6% of endowment
- I received \$7.14
- Returning about 40-45% of what I received would be: $\$2.86 - \3.21

I'll return \$3.0, which is reasonable and maintains cooperation.

usage:

```
{
  "input_tokens": 1847,
  "output_tokens": 489,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.78 returned, visible payoff \$5.779999999999999.
- Round 4 against you, as RECEIVER: saw \$2.64 sent, it became \$7.92, returned \$3.5, visible payoff \$4.42.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.54 returned, visible payoff \$5.54.

Your total visible earnings across all rounds are \$26.290000000000003.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the pattern:

```
**Round 3**: Co-player sent $3.0 (60%)
**Round 4**: Co-player returned $3.5, showing cooperation
**Round 5**: Co-player sent $3.0 (60%) again
```

They're maintaining consistent cooperation and trust. I'll continue matching their behavior:

```
{'send': 3.0}
```

reasoning:

Looking at the history:

```
**Round 3**: Co-player sent $3.0 as sender
**Round 4**: Co-player returned $3.5 as receiver
**Round 5**: Co-player sent $3.0 as sender
```

The co-player is showing consistent trust by sending \$3.0 when they're the sender and returning \$3.5 when they're the receiver. This is cooperative behavior.

I should continue to match this cooperative pattern by sending around \$3.0.

My total visible earnings are \$26.29, which suggests I'm doing well in this cooperative relationship.

I'll send \$3.0 to maintain the cooperative pattern.

usage:

```
{
  "input_tokens": 2233,
  "output_tokens": 244,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$2.64 sent, it became \$7.92, returned \$3.5, visible payoff \$4.42.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.54 returned, visible payoff \$5.54.
- Round 6 against you, as RECEIVER: saw \$2.05 sent, it became \$6.149999999999995, returned \$3.0, visible payoff \$3.149999999999995.

Your total visible earnings across all rounds are \$31.230000000000004.

This round, you are the RECEIVER against your current co-player. They sent you \$3.55, so you received \$10.649999999999999. How much do you return? (0-10.649999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at the pattern:

****Round 4-6**:** Co-player consistently showing cooperation

****Round 7**:** Co-player sent \$3.55 (71%), showing strong trust - higher than previous rounds

They're demonstrating increased trust. I should reciprocate generously:

Returning \$4.5:

- Sender ends with: \$5 - \$3.55 + \$4.5 = \$5.95
- I keep: \$10.65 - \$4.5 = \$6.15

This rewards their strong cooperation.

{'return': 4.5}

reasoning:

Looking at the pattern:

****Round 4**:** Co-player returned \$3.5 as receiver

****Round 5**:** Co-player sent \$3.0 as sender

****Round 6**:** Co-player returned \$3.0 as receiver

****Round 7**:** Co-player sent \$3.55 (71%)

They're showing high trust by sending 71% of their endowment. I should reciprocate generously.

Returning \$4.5:

- Sender ends with: \$5 - \$3.55 + \$4.5 = \$5.95
- I keep: \$10.65 - \$4.5 = \$6.15

This rewards their strong trust and maintains cooperation.

usage:

```
{
  "input_tokens": 2570,
  "output_tokens": 309,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.54 returned, visible payoff \$5.54.
- Round 6 against you, as RECEIVER: saw \$2.05 sent, it became \$6.149999999999995, returned \$3.0, visible payoff \$3.149999999999995.
- Round 7 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.03 returned, visible payoff \$7.03.

Your total visible earnings across all rounds are \$37.38.

This round, you are the SENDER against your current co-player. You have \$5. How much do you

u send? (0-5)
Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant
content:
Looking at the pattern:

****Round 5**:** Co-player sent \$3.0 (60%)
****Round 6**:** Co-player returned \$3.0, showing fairness
****Round 7**:** Co-player sent \$3.0 (60%) and saw a generous return of \$5.03

They're maintaining consistent cooperation throughout. I'll continue matching their trust level:

{ 'send': 3.0 }

reasoning:
Looking at the history:

****Round 5**:** Co-player sent \$3.0 (60%)
****Round 6**:** Co-player returned \$3.0
****Round 7**:** Co-player sent \$3.0 (60%), and saw \$5.03 returned (which was likely my \$4.5 return with noise)

The co-player has been consistently cooperative, sending \$3.0 when they're sender and returning fair amounts when they're receiver. They've maintained this pattern throughout.

I should continue matching their cooperative behavior by sending \$3.0.

{ 'send': 3.0 }

usage:
{
 "input_tokens": 2724,
 "output_tokens": 242,
 "reasoning_tokens": 0
}

Message 12 role=user
content:
Round 9

Visible history before this round:
Your current co-player's last 3 game(s):
- Round 6 against you, as RECEIVER: saw \$2.05 sent, it became \$6.149999999999995, returned \$3.0, visible payoff \$3.149999999999995.
- Round 7 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.03 returned, visible payoff \$7.03.
- Round 8 against you, as RECEIVER: saw \$2.14 sent, it became \$6.42, returned \$3.0, visible payoff \$3.42.
Your total visible earnings across all rounds are \$43.080000000000005.

This round, you are the RECEIVER against your current co-player. They sent you \$2.7, so you received \$8.100000000000001. How much do you return? (0-8.100000000000001)
Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant
content:
Looking at the pattern:

****Round 6-8**:** Co-player consistently returned \$3.0 as receiver
****Round 7**:** I returned generously (they saw \$5.03)
****Round 9**:** Co-player sent \$2.7 (54%), slightly less than their usual \$3.0

They're still showing trust but at a slightly reduced level. I'll return fairly:

Returning \$3.5:
- Sender ends with: \$5 - \$2.7 + \$3.5 = \$5.8
- I keep: \$8.1 - \$3.5 = \$4.6

This maintains cooperation.

{ 'return': 3.5 }

reasoning:
Looking at the pattern:

****Round 6**:** Co-player returned \$3.0
****Round 7**:** Co-player sent \$3.0, I returned generously (\$5.03 visible to them)
****Round 8**:** Co-player returned \$3.0 again
****Round 9**:** Co-player sent \$2.7 (54%)

They're sending a bit less this round (\$2.7 vs typically \$3.0), but still showing trust. They've consistently returned \$3.0 as receiver.

I should reciprocate fairly. Returning around \$3.0-\$3.5 would be appropriate:

Returning \$3.5:

- Sender ends with: $\$5 - \$2.7 + \$3.5 = \5.8
- I keep: $\$8.1 - \$3.5 = \$4.6$

This maintains fairness and cooperation.

usage:

```
{
  "input_tokens": 2841,
  "output_tokens": 377,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 7 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.03 returned, visible payoff \$7.03.
- Round 8 against you, as RECEIVER: saw \$2.14 sent, it became \$6.42, returned \$3.0, visible payoff \$3.42.
- Round 9 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.89 returned, visible payoff \$5.8900000000000001.

Your total visible earnings across all rounds are \$47.680000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the pattern:

****Round 7**:** Co-player sent \$3.0 (60%)
****Round 8**:** Co-player returned \$3.0, showing fairness
****Round 9**:** Co-player sent \$3.0 (60%) again, maintaining consistent trust

They're demonstrating reliable cooperation. I'll continue matching their trust level:

```
{'send': 3.0}
```

Reasoning:

Looking at the pattern:

****Round 7**:** Co-player sent \$3.0 (60%) and saw a generous return of \$5.03
****Round 8**:** Co-player returned \$3.0 as receiver
****Round 9**:** Co-player sent \$3.0 (60%) again

They're consistently showing trust by sending \$3.0. They maintain cooperation with fair returns. I should continue matching their trust level:

```
{'send': 3.0}
```

Usage: {

```
  "input_tokens": 2850,
  "output_tokens": 206,
  "reasoning_tokens": 0
}
```

GAME - Agent_3 - Receiver - dyad_2

Timestamp: 2026-08-12T08:18:04

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_4",
  "pairing": {
    "agents": [
      "Agent_4",
      "Agent_3"
    ],
    "investor": "Agent_4",
    "roles": {
      "Agent_3": "trustee",
      "Agent_4": "investor"
    },
    "trustee": "Agent_3"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 10,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.91 sent, it became \$8.73, returned \$3.5, visible payoff \$5.23.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.9 returned, visible payoff \$4.9.
- Round 3 against you, as RECEIVER: saw \$3.52 sent, it became \$10.56, returned \$4.5, visible payoff \$6.0600000000000005.

Your total visible earnings across all rounds are \$16.15.

This round, you are the RECEIVER against your current co-player. They sent you \$2.64, so you received \$7.92. How much do you return? (0-7.92)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at our cooperative history:

****Round 1:**** They returned \$3.5 on \$8.73

****Round 2:**** I returned \$3.5 on \$9.0

****Round 3:**** They returned \$4.5 on \$10.56 (increased generosity!)

****Round 4:**** They sent me \$2.64, I received \$7.92

They've been consistently cooperative and even increased their generosity in Round 3. I'll reciprocate fairly:

```
'''json
{"return": 3.5}
'''
```

This maintains our mutually beneficial cooperation pattern.

reasoning:

Let me analyze the history with my current co-player:

****Round 1:****

- I sent \$3 (they saw \$2.91 due to noise)
- They received \$8.73, returned \$3.5
- Good cooperation

****Round 2:****

- They sent \$3
- I received \$9.0, returned \$3.5 (they saw \$2.9)
- Continued cooperation

****Round 3:****

- I sent \$3 (they saw \$3.52 due to noise)
- They received \$10.56, returned \$4.5
- They increased their return, showing even more generosity

****Round 4 (current):****

- They sent \$2.64 (with noise, could have been around \$3 originally)
- I received \$7.92

Looking at the pattern:

- They've been consistently cooperative
- In Round 3, they returned \$4.5 on \$10.56 (about 43% return rate)
- They're continuing to send around \$3 each time

I should reciprocate their continued cooperation. Given they sent me \$7.92, a fair return would be around 40-45% of what I received, which would be about \$3.2-\$3.6.

I'll return \$3.5 to maintain consistency with our cooperative pattern.

usage:

```
{
  "input_tokens": 1467,
  "output_tokens": 452,
  "reasoning_tokens": 0
}
```

}

Message 4 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.9 returned, visible payoff \$4.9.
- Round 3 against you, as RECEIVER: saw \$3.52 sent, it became \$10.56, returned \$4.5, visible payoff \$6.0600000000000005.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.96 returned, visible payoff \$5.96.

Your total visible earnings across all rounds are \$20.57.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at our established pattern:

****Round 2-4:**** They consistently sent \$3

****Round 3:**** They were particularly generous, returning \$4.5

****Round 4:**** I returned \$3.5 (they saw \$3.96)

We have a strong cooperative relationship. I'll maintain our successful pattern:

'''json

{"send": 3}

'''

This continues our mutually beneficial cooperation.

reasoning:

Looking at the history with this co-player:

Round 1: I sent \$3 (they saw \$2.91), they returned \$3.5 - cooperative

Round 2: They sent \$3, I returned \$3.5 (they saw \$2.9) - cooperative

Round 3: I sent \$3 (they saw \$3.52), they returned \$4.5 - very generous!

Round 4: They sent \$3, I returned \$3.5 (they saw \$3.96) - cooperative

This co-player has been consistently cooperative and even showed extra generosity in Round 3. The pattern shows:

- They keep sending \$3
- They've returned generously when receiving
- We have a strong cooperative relationship

I should continue the cooperation. Sending \$3 has worked well for us.

usage:

```
{
  "input_tokens": 1824,
  "output_tokens": 291,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$3.52 sent, it became \$10.56, returned \$4.5, visible payoff \$6.0600000000000005.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.96 returned, visible payoff \$5.96.
- Round 5 against you, as RECEIVER: saw \$2.38 sent, it became \$7.14, returned \$3.0, visible payoff \$4.14.

Your total visible earnings across all rounds are \$26.11.

This round, you are the RECEIVER against your current co-player. They sent you \$2.05, so you received \$6.1499999999999995. How much do you return? (0-6.1499999999999995)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the pattern:

****Round 3:**** They returned \$4.5
****Round 4:**** I returned \$3.5
****Round 5:**** They returned \$3.0
****Round 6:**** They sent \$2.05, I received \$6.15

Our cooperation has been solid. I'll maintain a fair return:

```
'''json
{"return": 3.0}
'''
```

This keeps our mutually beneficial relationship going.

reasoning:

Looking at the history:

Round 3: They returned \$4.5 on \$10.56
Round 4: I returned \$3.5 (they saw \$3.96) on \$7.92
Round 5: They saw me send \$2.38 (I actually sent \$3), received \$7.14, returned \$3.0

Now Round 6: They sent me \$2.05 (with noise), I received \$6.15

They've been consistently cooperative throughout. Even though the amount they sent this round is lower (possibly due to noise), they've shown good faith. I should maintain cooperation.

Given I received \$6.15, a fair return would be around \$2.5-3.0 to maintain our cooperative pattern. Let me return about 40-45% which has been roughly our pattern.

$\$6.15 * 0.45 = \2.77

Or I could go with \$3.0 to be consistent with what they did in Round 5.

Let me go with \$3.0 to maintain consistency and cooperation.

usage:

```
{
  "input_tokens": 2163,
  "output_tokens": 349,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.96 returned, visible payoff \$5.96.
- Round 5 against you, as RECEIVER: saw \$2.38 sent, it became \$7.14, returned \$3.0, visible payoff \$4.14.
- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.94 returned, visible payoff \$4.9399999999999995.

Your total visible earnings across all rounds are \$29.259999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at our pattern:

****Round 4-6:**** They consistently sent \$3
****Round 5-6:**** We both returned around \$3

We have a stable, cooperative relationship. I'll continue:

```
'''json
```

```
{"send": 3}
'''
```

This maintains our successful mutual cooperation.

reasoning:

Looking at the pattern with this co-player:

Round 4: They sent \$3
Round 5: I sent \$3 (they saw \$2.38), they returned \$3.0
Round 6: They sent \$3, I returned \$3.0 (they saw \$2.94)
Round 7: My turn to send

We've maintained a consistent cooperative pattern. They keep sending \$3, and we both return fair amounts. I should continue this successful cooperation.

usage:

```
{
  "input_tokens": 2488,
  "output_tokens": 182,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as RECEIVER: saw \$2.38 sent, it became \$7.14, returned \$3.0, visible payoff \$4.14.
- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.94 returned, visible payoff \$4.9399999999999995.
- Round 7 against you, as RECEIVER: saw \$3.55 sent, it became \$10.649999999999999, returned \$4.5, visible payoff \$6.149999999999999.

Your total visible earnings across all rounds are \$36.29.

This round, you are the RECEIVER against your current co-player. They sent you \$2.14, so you received \$6.42. How much do you return? (0-6.42)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at our history:

****Round 5-6:**** We both returned around \$3.0
****Round 7:**** They were generous, returning \$4.5
****Round 8:**** They sent \$2.14, I received \$6.42

They've been consistently cooperative and generous. I'll reciprocate:

```
'''json
{"return": 3.0}
'''
```

This maintains our fair cooperation pattern.

reasoning:

Looking at the history:

****Round 5:**** They returned \$3.0
****Round 6:**** I returned \$3.0 (they saw \$2.94)
****Round 7:**** They returned \$4.5 (generous!)
****Round 8:**** They sent \$2.14, I received \$6.42

They've been consistently cooperative and even generous in Round 7. I should reciprocate fairly. Given they returned \$4.5 in Round 7, I should return a reasonable amount.

I'll return \$3.0 to maintain fairness while being slightly less than their generous \$4.5 in Round 7.

usage:

```
{
  "input_tokens": 2633,
  "output_tokens": 247,
  "reasoning_tokens": 0
}
```

}

Message 12 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.94 returned, visible payoff \$4.9399999999999995.
- Round 7 against you, as RECEIVER: saw \$3.55 sent, it became \$10.649999999999999, returned \$4.5, visible payoff \$6.149999999999999.
- Round 8 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.7 returned, visible payoff \$5.7.

Your total visible earnings across all rounds are \$39.71.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at our consistent pattern:

****Round 6-8:**** They sent \$3 every time
****Round 7:**** They returned \$4.5 (generous!)
****Round 8:**** I returned \$3.0

We have a very stable, mutually beneficial relationship. I'll continue:

```
'''json
{"send": 3}
'''
```

This maintains our excellent cooperation pattern.

reasoning:

Looking at the pattern with this co-player:

Round 6-8: They consistently sent \$3
Round 7: They returned \$4.5
Round 8: I returned \$3.0 (they saw \$3.7)

This is a very cooperative partner who has been consistently sending \$3 and returning fair amounts. They were particularly generous in Round 7. Our relationship has been very stable and mutually beneficial.

I should continue with \$3 to maintain this successful cooperation.

usage:

```
{
  "input_tokens": 2640,
  "output_tokens": 203,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 7 against you, as RECEIVER: saw \$3.55 sent, it became \$10.649999999999999, returned \$4.5, visible payoff \$6.149999999999999.
- Round 8 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.7 returned, visible payoff \$5.7.
- Round 9 against you, as RECEIVER: saw \$2.7 sent, it became \$8.100000000000001, returned \$3.5, visible payoff \$4.600000000000001.

Your total visible earnings across all rounds are \$45.6.

This round, you are the RECEIVER against your current co-player. They sent you \$3.14, so you received \$9.42. How much do you return? (0-9.42)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at our history:

****Round 7:**** They returned \$4.5
****Round 8:**** I returned \$3.0
****Round 9:**** They returned \$3.5
****Round 10:**** They sent \$3.14, I received \$9.42

They've been consistently cooperative and generous. I'll reciprocate fairly:

```
'''json
{"return": 3.5}
'''
```

This maintains our mutually beneficial relationship.

Reasoning:

Looking at the history:

****Round 7:**** They returned \$4.5
****Round 8:**** I returned \$3.0 (they saw \$3.7)
****Round 9:**** They returned \$3.5
****Round 10:**** They sent \$3.14, I received \$9.42

They've been consistently cooperative throughout. They returned \$4.5 in round 7, \$3.5 in round 9, and now sent me \$3.14 (which is above average).

I should return a fair amount. Given the pattern and their generosity, I'll return around \$3.5 - \$4.0. Let me match their generosity:

```
'''json
{"return": 3.5}
'''
```

This maintains our good cooperation.

```
Usage: {
  "input_tokens": 2689,
  "output_tokens": 282,
  "reasoning_tokens": 0
}
```

GAME - Agent_6 - Sender - dyad_3

Timestamp: 2026-08-12T08:18:11

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_5",
  "pairing": {
    "agents": [
      "Agent_6",
      "Agent_5"
    ],
    "investor": "Agent_6",
    "roles": {
      "Agent_5": "trustee",
      "Agent_6": "investor"
    },
    "trustee": "Agent_5"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 10,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.13 returned, visible payoff \$5.13.
- Round 2 against you, as RECEIVER: saw \$2.97 sent, it became \$8.91, returned \$4.5, visible payoff \$4.41.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.79 returned, visible payoff \$7.79.

Your total visible earnings across all rounds are \$15.700000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at the visible history:

****Round 1:**** Co-player sent \$3.00, saw \$3.13 returned, payoff \$5.13

****Round 2:**** Co-player returned \$4.50 from \$8.91 received (fair, ~50%)

****Round 3:**** Co-player sent \$3.00, saw \$5.79 returned (very generous!), payoff \$7.79

My co-player has been consistently generous, sending \$3.00 (60%) in both rounds as sender.

They also saw a strong return in Round 3, which should reinforce mutual trust. As receiver, they returned fairly (~50%).

This is a cooperative relationship with established trust. I should continue matching their consistent generosity of \$3.00.

```
{"send": 3.00}
```

reasoning:

Looking at the visible history from my co-player's perspective:

****Round 1:**** They sent \$3.00, saw \$3.13 returned, payoff \$5.13

****Round 2:**** They received \$8.91 (from my \$2.97 send), returned \$4.50, payoff \$4.41

****Round 3:**** They sent \$3.00, saw \$5.79 returned, payoff \$7.79

My co-player has been consistently generous, sending \$3.00 in both rounds they were sender (60% of endowment). In Round 3, they saw a very generous return of \$5.79, which should reinforce their trust in me.

They've also been fair as a receiver, returning \$4.50 out of \$8.91 (about 50%).

This is a cooperative relationship with mutual trust. I should continue to match their generosity. They've sent \$3.00 twice, so I should send \$3.00 again.

Actually, let me reconsider. In Round 3, they saw a particularly high return (\$5.79), which might encourage even more trust. But they've consistently sent \$3.00, so I should match that level.

I'll send \$3.00 to maintain consistency and reciprocal cooperation.

usage:

```
{
  "input_tokens": 1654,
  "output_tokens": 485,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.97 sent, it became \$8.91, returned \$4.5, visible payoff \$4.41.

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.79 returned, visible payoff \$7.79.

- Round 4 against you, as RECEIVER: saw \$3.93 sent, it became \$11.79, returned \$5.9, visible payoff \$5.89

Your total visible earnings across all rounds are \$24.33.

This round, you are the RECEIVER against your current co-player. They sent you \$3.62, so you received \$10.86. How much do you return? (0-10.86)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the visible history:

****Round 2:**** Co-player returned \$4.50 from \$8.91 (50.5%)

****Round 3:**** Co-player sent \$3.00, saw \$5.79 returned

****Round 4:**** Co-player returned \$5.90 from \$11.79 (50.1%)

****Round 5:**** Co-player sent \$3.62 (72.4%!) - very high trust!

My co-player has been consistently cooperative, returning about 50% when receiver and sending generously as sender. This round they sent an even larger amount (\$3.62, which is 72.4% of their endowment) - showing increasing trust.

I should reward this high level of trust. If I return \$6.00:

- Sender ends with: \$1.38 + \$6.00 = \$7.38 (profit of \$2.38)

- I keep: \$10.86 - \$6.00 = \$4.86

This returns about 55% of what I received, slightly more generous than their pattern to acknowledge their increased trust.

```
{"return": 6.00}
```

reasoning:

Looking at the visible history:

****Round 2:**** Co-player returned \$4.50 from \$8.91 (50.5%)
****Round 3:**** Co-player sent \$3.00, saw \$5.79 returned
****Round 4:**** Co-player returned \$5.90 from \$11.79 (50.1%)
****Round 5:**** Co-player sent \$3.62 (72.4%)

My co-player has been very consistent and cooperative:

- As sender, they've sent generous amounts (\$3.00 and now \$3.62 - 72.4%!)
- As receiver, they've returned about 50% consistently

They've sent an even larger amount this round (\$3.62, which is 72.4% of their endowment).
This shows increasing trust.

I should continue to reciprocate fairly. If I return around 50-55% like they've been doing
:

- 50% would be \$5.43
- 55% would be \$5.97

Let me aim for about \$5.50 to \$6.00 to maintain the cooperative pattern and reward their increased generosity.

If I return \$5.50:

- Sender ends with: $\$1.38 + \$5.50 = \$6.88$ (profit of \$1.88)
- I keep: $\$10.86 - \$5.50 = \$5.36$

If I return \$6.00:

- Sender ends with: $\$1.38 + \$6.00 = \$7.38$ (profit of \$2.38)
- I keep: $\$10.86 - \$6.00 = \$4.86$

Given they sent 72.4% of their endowment (very high trust), I should return a bit more generously. ...

usage:

```
{
  "input_tokens": 2074,
  "output_tokens": 706,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.79 returned, visible payoff \$7.79.
- Round 4 against you, as RECEIVER: saw \$3.93 sent, it became \$11.790000000000001, returned \$5.9, visible payoff \$5.890000000000001.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.97 returned, visible payoff \$7.97.

Your total visible earnings across all rounds are \$29.189999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the visible history:

****Round 3:**** Co-player sent \$3.00, saw \$5.79 returned, payoff \$7.79
****Round 4:**** Co-player returned \$5.90 from \$11.79 (50%)
****Round 5:**** Co-player sent \$3.00, saw \$5.97 returned, payoff \$7.97

My co-player has been extremely consistent and cooperative:

- As sender: consistently sends \$3.00 (60% of endowment)
- As receiver: returns about 50% fairly
- They've received good returns, reinforcing mutual trust

This is a stable, mutually beneficial cooperative relationship. I should continue matching

their pattern by sending \$3.00.

```
{"send": 3.00}
```

reasoning:

Looking at the visible history from my co-player's perspective:

****Round 3:**** Co-player sent \$3.00, saw \$5.79 returned, payoff \$7.79

****Round 4:**** Co-player returned \$5.90 from \$11.79 (50%)

****Round 5:**** Co-player sent \$3.00, saw \$5.97 returned, payoff \$7.97

My co-player has been very consistent:

- As sender: consistently sends \$3.00 (60% of endowment)
- As receiver: returns about 50%
- They've seen good returns when sending (\$5.79 and \$5.97), leading to strong payoffs (~\$8)

This is a very cooperative and stable relationship. My co-player has maintained trust by consistently sending \$3.00 and returning fairly. I should continue this pattern of mutual cooperation by sending \$3.00 again.

The relationship is working well - we're both benefiting from the cooperation. No reason to deviate from what's working.

usage:

```
{
  "input_tokens": 2569,
  "output_tokens": 425,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$3.93 sent, it became \$11.790000000000001, returned \$5.9, visible payoff \$5.890000000000001.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.97 returned, visible payoff \$7.97.
- Round 6 against you, as RECEIVER: saw \$3.02 sent, it became \$9.06, returned \$4.5, visible payoff \$4.5600000000000005.

Your total visible earnings across all rounds are \$35.76.

This round, you are the RECEIVER against your current co-player. They sent you \$2.78, so you received \$8.34. How much do you return? (0-8.34)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at the visible history:

****Round 4:**** Co-player returned \$5.90 from \$11.79 (50%)

****Round 5:**** Co-player sent \$3.00, saw \$5.97 returned

****Round 6:**** Co-player returned \$4.50 from \$9.06 (49.7%)

****Round 7:**** Co-player sent \$2.78 (55.6%)

My co-player has been consistently cooperative, returning about 50% when receiver and sending generously as sender. This round they sent \$2.78, slightly less than their usual \$3.00, but still a majority of their endowment (55.6%).

I should maintain our cooperative relationship with a fair return. If I return \$4.50:

- Sender ends with: \$2.22 + \$4.50 = \$6.72 (profit of \$1.72)
- I keep: \$8.34 - \$4.50 = \$3.84

This returns about 54% of what I received, which is fair and maintains the mutual trust we've built.

```
{"return": 4.50}
```

reasoning:

Looking at the visible history:

****Round 4:**** Co-player returned \$5.90 from \$11.79 (50%)

****Round 5:**** Co-player sent \$3.00, saw \$5.97 returned, payoff \$7.97
****Round 6:**** Co-player returned \$4.50 from \$9.06 (49.7%)
****Round 7:**** Co-player sent \$2.78 (55.6%)

My co-player has been very consistent:

- As sender: sending around \$3.00 (though this round \$2.78, slightly less)
- As receiver: returning about 50% consistently

This round they sent \$2.78, which is a bit less than their usual \$3.00. Let me think about what to return.

If I return \$4.50:

- Sender ends with: $\$2.22 + \$4.50 = \$6.72$ (profit of \$1.72)
- I keep: $\$8.34 - \$4.50 = \$3.84$

This returns about 54% of what I received, which is fair and maintains our cooperative relationship.

Actually, let me reconsider. They sent slightly less this round (\$2.78 vs usual \$3.00). I should still be generous but maybe match their slight reduction in trust with a proportional return.

If I return \$4.50:

- That's 54% return rate
- Sender profits \$1.72
- I keep \$3.84

If I return \$4.00:

- That's 48% return rate
- Sender profits \$1.22
- I keep \$4.34

I think \$4.50 is still appropriate to maintain the good cooperative relationship, even though they sent slightly less. The relationship has been strong overall.

usage:

```
{
  "input_tokens": 2990,
  "output_tokens": 685,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.97 returned, visible payoff \$7.97.
- Round 6 against you, as RECEIVER: saw \$3.02 sent, it became \$9.06, returned \$4.5, visible payoff \$4.5600000000000005.
- Round 7 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.84 returned, visible payoff \$6.84.

Your total visible earnings across all rounds are \$39.599999999999994.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at the visible history:

****Round 5:**** Co-player sent \$3.00, saw \$5.97 returned, payoff \$7.97
****Round 6:**** Co-player returned \$4.50 from \$9.06 (~50%)
****Round 7:**** Co-player sent \$3.00, saw \$4.84 returned, payoff \$6.84

My co-player continues to be consistently cooperative:

- As sender: reliably sends \$3.00 (60% of endowment)
- As receiver: returns about 50%, maintaining fairness
- They continue cooperating despite variable returns (due to communication noise)

We've established a strong, mutually beneficial pattern over 7 rounds. I should continue matching their consistent trust by sending \$3.00.

```
{"send": 3.00}
```

reasoning:

Looking at the visible history:

```
**Round 5:** Co-player sent $3.00, saw $5.97 returned, payoff $7.97
**Round 6:** Co-player returned $4.50 from $9.06 (49.7%)
**Round 7:** Co-player sent $3.00, saw $4.84 returned, payoff $6.84
```

My co-player has been very consistent:

- As sender: consistently sends \$3.00 (60% of endowment)
- As receiver: returns about 50% fairly
- They continue to show trust despite some variation in returns (due to communication noise)

This is Round 8. We've built a strong cooperative relationship over 7 rounds. My co-player has been extremely consistent in sending \$3.00 when they're the sender. I should continue to match their consistent behavior.

I'll send \$3.00 to maintain our established pattern of cooperation.

usage:

```
{
  "input_tokens": 3240,
  "output_tokens": 415,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as RECEIVER: saw \$3.02 sent, it became \$9.06, returned \$4.5, visible payoff \$4.5600000000000005.
- Round 7 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.84 returned, visible payoff \$6.84.
- Round 8 against you, as RECEIVER: saw \$2.39 sent, it became \$7.17, returned \$3.6, visible payoff \$3.57.

Your total visible earnings across all rounds are \$44.55.

This round, you are the RECEIVER against your current co-player. They sent you \$3.77, so you received \$11.31. How much do you return? (0-11.31)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at the visible history:

```
**Round 6:** Co-player returned $4.50 from $9.06 (~50%)
**Round 7:** Co-player sent $3.00, saw $4.84 returned
**Round 8:** Co-player returned $3.60 from $7.17 (~50%)
**Round 9:** Co-player sent $3.77 (75.4%!) - very high trust!
```

My co-player continues to be consistently cooperative, returning about 50% when receiver.

This round they sent \$3.77, which is an even larger amount (75.4% of their endowment) - showing exceptional trust!

I should reward this very high level of trust. If I return \$6.00:

- Sender ends with: \$1.23 + \$6.00 = \$7.23 (profit of \$2.23)
- I keep: \$11.31 - \$6.00 = \$5.31

This returns about 53% of what I received, acknowledging their exceptional trust while maintaining fairness.

```
{"return": 6.00}
```

reasoning:

Looking at the visible history:

```
**Round 6:** Co-player returned $4.50 from $9.06 (~50%)
```

****Round 7:**** Co-player sent \$3.00, saw \$4.84 returned
****Round 8:**** Co-player returned \$3.60 from \$7.17 (~50%)
****Round 9:**** Co-player sent \$3.77 (75.4%!) - very high trust!

My co-player has been consistently cooperative, maintaining about 50% returns when receive r. As sender, they've been generous, and this round they sent an even larger amount (\$3.77, which is 75.4% of their endowment) - showing strong trust.

I should reciprocate this high trust. Looking at the pattern:

- They consistently return ~50% as receiver
- They send generously as sender
- This round shows particularly high trust with \$3.77

If I return \$6.00:

- Sender ends with: $\$1.23 + \$6.00 = \$7.23$ (profit of \$2.23)
- I keep: $\$11.31 - \$6.00 = \$5.31$

This returns about 53% of what I received, slightly more than their 50% pattern to acknowledge their increased trust.

Actually, looking more carefully at the amounts I've been returning based on the visible history:

- Round 3: I returned \$5.00 out of \$9.69 (51.6%)
- Round 5: I returned \$6.00 out of \$10.86 (55.2%)
- Round 7: I returned \$4.50 out of \$8.34 (54%)

I've been consistently returning around 50-55%. This round, with the high trust shown (\$3.77 sent), I should return around 53-55% which would be about \$6.00-\$6.20.

Let me go with \$6.00 to maintain consistency and reward their high trust.

usage:

```
{
  "input_tokens": 3356,
  "output_tokens": 723,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 7 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.84 returned, visible payoff \$6.84.
- Round 8 against you, as RECEIVER: saw \$2.39 sent, it became \$7.17, returned \$3.6, visible payoff \$3.57.
- Round 9 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.93 returned, visible payoff \$7.93.

Your total visible earnings across all rounds are \$49.86.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the visible history:

****Round 7:**** Co-player sent \$3.00, saw \$4.84 returned, payoff \$6.84
****Round 8:**** Co-player returned \$3.60 from \$7.17 (~50%)
****Round 9:**** Co-player sent \$3.00, saw \$5.93 returned, payoff \$7.93

My co-player has been remarkably consistent and cooperative throughout our 9 rounds together:

- As sender: reliably sends \$3.00 (60% of endowment)
- As receiver: consistently returns about 50%
- We've built strong mutual trust with stable, beneficial outcomes

This has been one of the most cooperative relationships in the game. I should continue matching their trustworthy pattern by sending \$3.00.

```
{"send": 3.00}
```

Reasoning:

Looking at the visible history:

****Round 7:**** Co-player sent \$3.00, saw \$4.84 returned, payoff \$6.84
****Round 8:**** Co-player returned \$3.60 from \$7.17 (~50%)
****Round 9:**** Co-player sent \$3.00, saw \$5.93 returned, payoff \$7.93

My co-player continues to be extremely consistent and cooperative:

- As sender: consistently sends \$3.00 (60% of endowment)
- As receiver: returns about 50% fairly
- They continue receiving good returns, reinforcing mutual trust
- We've had 9 rounds together with a stable, cooperative pattern

This is the 10th round. We've built strong mutual trust over many rounds. I should continue matching their consistent pattern by sending \$3.00.

This relationship has been extremely stable and cooperative. Both of us have been consistently trustworthy. I should maintain this by sending \$3.00.

```
Usage: {  
  "input_tokens": 3411,  
  "output_tokens": 427,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_5 - Receiver - dyad_3

Timestamp: 2026-08-12T08:18:20

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_3",  
  "move_index": 5,  
  "opponent_id": "Agent_6",  
  "pairing": {  
    "agents": [  
      "Agent_6",  
      "Agent_5"  
    ],  
    "investor": "Agent_6",  
    "roles": {  
      "Agent_5": "trustee",  
      "Agent_6": "investor"  
    },  
    "trustee": "Agent_5"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 10,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5

- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.66 sent, it became \$7.98, returned \$4.0, visible payoff \$3.9800000000000004.
- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.03 returned, visible payoff \$7.03.
- Round 3 against you, as RECEIVER: saw \$3.23 sent, it became \$9.69, returned \$5.0, visible payoff \$4.6899999999999995.

Your total visible earnings across all rounds are \$17.33.

This round, you are the RECEIVER against your current co-player. They sent you \$3.93, so you received \$11.790000000000001. How much do you return? (0-11.790000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at our continued partnership:

****Round 1:**** They returned \$4.0 from \$7.98 (?50%)

****Round 2:**** I returned \$4.50 from \$9.0 (?50%)

****Round 3:**** They returned \$5.0 from \$9.69 (?52%)

****Current Round 4:****

- They sent \$3.93 (79% of endowment!) - very high trust signal
- I received \$11.79
- They kept only \$1.07

****Analysis:****

This co-player has shown increasing trust and consistent 50% reciprocity. They're now sending nearly their entire endowment, demonstrating strong faith in our relationship.

I should maintain our established 50% return pattern to:

- Honor their trust
- Keep our cooperative equilibrium
- Ensure mutual benefit

Returning \$5.90:

- They end with: \$1.07 + \$5.90 = \$6.97
- I end with: \$11.79 - \$5.90 = \$5.89

```
'''json
{"return": 5.90}
'''
```

reasoning:

Let me analyze the history:

****Round 1:****

- I sent (saw \$2.66 with noise from my \$3)
- They received \$7.98, returned \$4.0 (about 50%)

****Round 2:****

- They sent \$3
- I received \$9, returned \$4.50 (they saw \$5.03 with noise)
- They got \$7.03 total

****Round 3:****

- I sent \$3 (they saw \$3.23 with noise)
- They received \$9.69, returned \$5.0 (about 52%)
- They kept \$4.69

****Round 4 (current):****

- They sent \$3.93 (about 79% of their endowment!) - very high trust
- I received \$11.79
- They kept \$1.07

This co-player is showing extremely high trust by sending \$3.93. They've consistently returned around 50% when they were receiver. I should reciprocate this trust and maintain our cooperative relationship.

If I return about 50% of what I received:

- Return \$5.90
- They end with: $\$1.07 + \$5.90 = \$6.97$
- I end with: $\$11.79 - \$5.90 = \$5.89$

But they're showing increasing trust (sending more and more). Maybe I should return a bit more to reward this and encourage continued cooperation. Let me think...

Actually, looking at the pattern:

- Round 1 receiver: returned \$4 from \$7.98 (50%)
- Round 3 receiver: returned \$5 from \$9.69 (52%)

They're consistently returning around 50%. I've been doing similar - returned \$4.50 from \$9 in Round 2 (50%).

Given they sent \$3.93 (very high trust), I should return around 50% or slightly more to maintain the relationship.

...

usage:

```
{
  "input_tokens": 1695,
  "output_tokens": 859,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.03 returned, visible payoff \$7.03.
- Round 3 against you, as RECEIVER: saw \$3.23 sent, it became \$9.69, returned \$5.0, visible payoff \$4.6899999999999995.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.63 returned, visible payoff \$8.629999999999999.

Your total visible earnings across all rounds are \$23.22.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at our established partnership:

****Round 2:**** They sent \$3.0, I returned generously
****Round 3:**** I sent \$3.23 (with noise), they returned \$5.0 (?52%)
****Round 4:**** They sent \$3.0, I returned well, they saw \$6.63

****Pattern Analysis:****

- They consistently send \$3 when sender (Rounds 2, 4)
- They reciprocate fairly at ~50% when receiver (Round 3)
- We've built a stable, mutually beneficial relationship
- Both earning well (\$23.22 total for me)

****My decision:****

Match their consistency by sending \$3 again. This:

- Maintains our established cooperative equilibrium
- Shows I trust the pattern we've developed
- Signals reliability and reciprocity

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at the visible history:

****Round 2:**** They sent \$3.0, I returned (they saw \$5.03) - I actually returned \$4.50 but they saw \$5.03 due to noise
****Round 3:**** They received \$9.69 (from my \$3 send), returned \$5.0
****Round 4:**** They sent \$3.0, I returned \$5.90 but they saw \$6.63 - generous due to noise

Pattern analysis:

- This co-player consistently sends \$3.0 when they're sender (Rounds 2 and 4)
- They return roughly 50% when receiver (Round 3: \$5.0 from \$9.69)
- They've been seeing higher returns from me due to noise (\$5.03 and \$6.63 vs what I actually sent \$4.50 and \$5.90)

Our cooperation has been strong and consistent. They've shown:

- Reliable sending patterns (\$3)
- Fair reciprocity (~50%)
- Building trust over rounds

I should continue with the same sending pattern to maintain this cooperative equilibrium. Sending \$3 has been working well for both of us.

My total visible earnings are \$23.22 which suggests we're both doing well.

I'll send \$3 again to maintain consistency and trust.

usage:

```
{
  "input_tokens": 2185,
  "output_tokens": 517,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$3.23 sent, it became \$9.69, returned \$5.0, visible payoff \$4.6899999999999995.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.63 returned, visible payoff \$8.629999999999999.
- Round 5 against you, as RECEIVER: saw \$3.62 sent, it became \$10.86, returned \$6.0, visible payoff \$4.859999999999999.

Your total visible earnings across all rounds are \$31.189999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$3.02, so you received \$9.06. How much do you return? (0-9.06)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at our continued partnership:

****Round 3:**** I sent, they returned \$5.0 from \$9.69 (?52%)

****Round 4:**** They sent \$3.0, I returned generously

****Round 5:**** I sent, they returned \$6.0 from \$10.86 (?55%)

****Current Round 6:****

- They sent \$3.02 (?60% of endowment) - continued trust

- I received \$9.06

- They kept \$1.98

****Analysis:****

Our partnership shows consistent mutual cooperation with ~50-55% return rates. They've reliably reciprocated in this range, and I should maintain this equilibrium.

Returning \$4.50:

- They end with: \$1.98 + \$4.50 = \$6.48

- I end with: \$9.06 - \$4.50 = \$4.56

This maintains our cooperative pattern.

```
'''json
{"return": 4.50}
'''
```

reasoning:

Looking at our history:

****Round 3:**** I sent (they saw \$3.23), they returned \$5.0 from \$9.69 (?52%)

****Round 4:**** They sent \$3.0, I returned, they saw \$6.63 (generous due to noise)

****Round 5:**** I sent (they saw \$3.62), they returned \$6.0 from \$10.86 (?55%)

****Current Round 6:****

- They sent \$3.02 (?60% of endowment)

- I received \$9.06

- They kept \$1.98

****Pattern:****

- They consistently send around \$3

- They consistently return around 50-55% when receiver

- We have a stable, cooperative relationship

I should maintain the ~50% return pattern:

- Returning \$4.50 would be about 50%

- This keeps our cooperative equilibrium

- They end with: \$1.98 + \$4.50 = \$6.48

- I end with: \$9.06 - \$4.50 = \$4.56

Both benefit reasonably, and this maintains the pattern.

usage:

```
{
  "input_tokens": 2634,
  "output_tokens": 514,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.63 returned, visible payoff \$8.629999999999999.

- Round 5 against you, as RECEIVER: saw \$3.62 sent, it became \$10.86, returned \$6.0, visible payoff \$4.859999999999999.

- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.57 returned, visible payoff \$6.57.

Your total visible earnings across all rounds are \$35.75.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at our established pattern:

****Round 4:**** They sent \$3.0, I returned generously
****Round 5:**** I sent, they returned \$6.0 (?55%)
****Round 6:**** They sent \$3.0, I returned (they saw \$4.57)

****Analysis:****

- They consistently send \$3 when in the sender role (Rounds 4, 6)
- They reciprocate fairly (50-55%) when receiver (Round 5)
- We've maintained a stable, mutually beneficial partnership
- Total earnings of \$35.75 shows this strategy is working well

****My decision:****

Continue the established pattern by sending \$3. This:

- Maintains our successful cooperative equilibrium
- Matches their consistent behavior
- Sustains mutual trust and benefit

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at the history:

****Round 4:**** They sent \$3.0, I returned well (they saw \$6.63)
****Round 5:**** I sent, they returned \$6.0
****Round 6:**** They sent \$3.0, I returned (they saw \$4.57 with noise)

****Pattern:****

- They consistently send \$3 when sender
- They reciprocate fairly when receiver
- We've maintained a strong cooperative relationship
- Total earnings are healthy at \$35.75

This is Round 7, and we've established a stable pattern where:

- Both of us send \$3 when sender
- Both of us return around 50% when receiver
- This creates mutual benefit

I should continue with \$3 to maintain this successful equilibrium.

usage:

```
{
  "input_tokens": 3095,
  "output_tokens": 378,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as RECEIVER: saw \$3.62 sent, it became \$10.86, returned \$6.0, visible payoff \$4.859999999999999.
- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.57 returned, visible payoff \$6.57.
- Round 7 against you, as RECEIVER: saw \$2.78 sent, it became \$8.34, returned \$4.5, visible payoff \$3.84.

Your total visible earnings across all rounds are \$42.59.

This round, you are the RECEIVER against your current co-player. They sent you \$2.39, so you received \$7.17. How much do you return? (0-7.17)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at our partnership progression:

****Round 5:**** I sent, they returned \$6.0 from \$10.86 (?55%)
****Round 6:**** They sent \$3.0, I returned (noise showed \$4.57)
****Round 7:**** I sent, they returned \$4.5 from \$8.34 (?54%)

****Current Round 8:****

- They sent \$2.39 (?48% of endowment) - slightly less than usual
- I received \$7.17
- They kept \$2.61

****Analysis:****

They've been consistently cooperative with ~50-55% returns. Their lower send amount this round might be:

- Natural variation
- Response to perceived lower returns (due to noise)

I should maintain our cooperative equilibrium with a ~50% return to reinforce trust.

Returning \$3.60:

- They end with: \$2.61 + \$3.60 = \$6.21
- I end with: \$7.17 - \$3.60 = \$3.57

```
'''json
{"return": 3.60}
'''
```

reasoning:

Looking at the recent history:

****Round 5:**** I sent, they returned \$6.0 from \$10.86 (?55%)
****Round 6:**** They sent \$3.0, I returned, they saw \$4.57 (with noise)
****Round 7:**** I sent, they returned \$4.5 from \$8.34 (?54%)

****Current Round 8:****

- They sent \$2.39 (?48% of endowment) - slightly lower than before
- I received \$7.17
- They kept \$2.61

****Analysis:****

Our partnership has been consistently cooperative. They've been returning around 50-55% when receiver. I've been returning around 50% as well to maintain the equilibrium.

They sent slightly less this round (\$2.39 vs previous \$3.0), which could be:

1. Natural variation
2. A slight reduction in trust after seeing lower returns in recent rounds (though that's due to noise)

I should maintain the cooperative pattern and return around 50%:

- Returning \$3.50-\$4.00 would be fair
- This keeps the relationship stable
- Shows I'm still trustworthy despite their lower send amount

Returning \$3.60:

- They end with: \$2.61 + \$3.60 = \$6.21
- I end with: \$7.17 - \$3.60 = \$3.57

This is around 50% of what I received, maintaining our established pattern.

usage:

```
{
  "input_tokens": 3328,
  "output_tokens": 616,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.57 returned, visible payoff \$6.57.
- Round 7 against you, as RECEIVER: saw \$2.78 sent, it became \$8.34, returned \$4.5, visible payoff \$3.84.
- Round 8 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.95 returned, visible payoff \$4.95.
Your total visible earnings across all rounds are \$46.160000000000004.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant
content:

Looking at the recent pattern:

****Round 6:**** They sent \$3.0

****Round 7:**** I sent, they returned \$4.5 (?54%)

****Round 8:**** They sent \$3.0, I returned \$3.60, but noise distorted it to \$2.95 visible

****Analysis:****

- They've maintained consistent \$3 sends (Rounds 6, 8)
- They reciprocate fairly at ~50-55% when receiver (Round 7)
- Unfortunately noise in Round 8 made my \$3.60 return look like only \$2.95
- This may have made them think I reduced cooperation

****My decision:****

I should continue sending \$3 to:

- Demonstrate continued commitment to cooperation
- Match their consistent behavior
- Help overcome the noise distortion from last round
- Maintain our mutually beneficial partnership

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at the recent history:

****Round 6:**** They sent \$3.0, I returned (they saw \$4.57 due to noise)

****Round 7:**** I sent, they returned \$4.5 from \$8.34 (?54%)

****Round 8:**** They sent \$3.0, I returned \$3.60, but they only saw \$2.95 (significant noise!)

****Analysis:****

- They continue to send \$3 consistently (Rounds 6, 8)
- They reciprocate fairly at ~50-55% (Round 7)
- In Round 8, noise significantly distorted what they saw - I returned \$3.60 but they only saw \$2.95
- This might make them think I'm reducing cooperation

The noise in Round 8 is unfortunate - they sent \$3 (60% of endowment) but saw only \$2.95 returned (about 33% return rate instead of the actual 40%).

****My decision:****

Despite the noise issue, I should maintain the cooperative pattern. Sending \$3 again:

- Shows I'm still committed to our partnership
- Matches their consistent behavior
- Helps rebuild trust after the unfortunate noise in Round 8

They've been a reliable partner, always sending \$3 and returning fairly when receiver. I should continue cooperating.

usage:

```
{
  "input_tokens": 3403,
  "output_tokens": 526,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 7 against you, as RECEIVER: saw \$2.78 sent, it became \$8.34, returned \$4.5, visible payoff \$3.84.
- Round 8 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$2.95 returned, visible payoff \$4.95.
- Round 9 against you, as RECEIVER: saw \$3.77 sent, it became \$11.31, returned \$6.0, visible payoff \$5.3100000000000005.

Your total visible earnings across all rounds are \$54.09.

This round, you are the RECEIVER against your current co-player. They sent you \$3.16, so you received \$9.48. How much do you return? (0-9.48)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at our partnership:

****Round 7:**** I sent, they returned \$4.5 from \$8.34 (?54%)

****Round 8:**** They sent \$3.0, but noise distorted my return to look low (\$2.95)

****Round 9:**** I sent, they returned \$6.0 from \$11.31 (?53%)

****Current Round 10:****

- They sent \$3.16 (?63% of endowment) - high trust signal
- I received \$9.48
- They kept only \$1.84

****Analysis:****

Despite noise making my Round 8 return look poor, they've maintained consistent cooperation at ~50-55% returns. They're showing remarkable trust by continuing to send substantial amounts

I should maintain our cooperative equilibrium with a ~50% return.

Returning \$4.75:

- They end with: \$1.84 + \$4.75 = \$6.59
- I end with: \$9.48 - \$4.75 = \$4.73

```
'''json
{"return": 4.75}
'''
```

Reasoning:

Looking at the recent pattern:

****Round 7:**** I sent (they saw \$2.78 due to noise), they returned \$4.5 from \$8.34 (?54%)

****Round 8:**** They sent \$3.0, I returned \$3.60 but noise showed \$2.95

****Round 9:**** I sent (they saw \$3.77 due to noise), they returned \$6.0 from \$11.31 (?53%)

****Current Round 10:****

- They sent \$3.16 (?63% of endowment)
- I received \$9.48
- They kept \$1.84

****Analysis:****

Despite the noise in Round 8 showing a low return from me (they saw \$2.95), they:

- Continued to cooperate (Round 9 returned \$6.0, about 53%)
- Are still sending substantial amounts (\$3.16 this round)
- Show strong trust and consistent cooperation

They've been consistently reciprocating at ~50-55% range. I should maintain this pattern.

Returning ~50%:

$\$9.48 * 0.50 = \4.74

Let me return \$4.75:

- They end with: \$1.84 + \$4.75 = \$6.59
- I end with: \$9.48 - \$4.75 = \$4.73

This maintains our cooperative equilibrium.

Usage: {
"input_tokens": 3472,

```
"output_tokens": 599,  
"reasoning_tokens": 0  
}
```

GAME - Agent_8 - Sender - dyad_4

Timestamp: 2026-08-12T08:18:25

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_4",  
  "move_index": 6,  
  "opponent_id": "Agent_7",  
  "pairing": {  
    "agents": [  
      "Agent_8",  
      "Agent_7"  
    ],  
    "investor": "Agent_8",  
    "roles": {  
      "Agent_7": "trustee",  
      "Agent_8": "investor"  
    },  
    "trustee": "Agent_7"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 10,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.37 returned, visible payoff \$7.37.
- Round 2 against you, as RECEIVER: saw \$2.35 sent, it became \$7.050000000000001, returned \$3.75, visible payoff \$3.3000000000000007.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.3 returned, visible payoff \$6.3.

Your total visible earnings across all rounds are \$13.549999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at the history with my current co-player:

- Round 1: They sent \$3.0 (strong trust)
- Round 2: I sent, they returned \$3.75 (cooperative)
- Round 3: They sent \$3.0 again (consistent trust)

My co-player has shown consistent cooperation, sending \$3.0 both times they were the sender. They also returned a fair amount when they were receiver. I should reciprocate this consistent trust.

```
{"send": 3.0}
```

reasoning:

Let me analyze the visible history with my current co-player:

Round 1: They were SENDER, sent \$3.0 (60%), I returned something, they saw \$5.37 returned
 Round 2: They were RECEIVER, they saw \$2.35 sent (but I actually sent \$3.0 - noise!), they returned \$3.75
 Round 3: They were SENDER, sent \$3.0 (60%), they saw \$4.3 returned (but I actually returned \$3.33 - noise showing higher!)

So my co-player has:

- Sent \$3.0 twice when they were sender (consistent trust)
- Returned \$3.75 when they received \$7.05

They're showing good cooperation. In Round 3, they saw a higher return (\$4.3) than I actually sent (\$3.33), which might encourage them.

I should continue cooperating. Sending \$3.0 again would be consistent and fair.

My total visible earnings: \$13.55

- Round 1 as RECEIVER: kept \$4.51
- Round 2 as SENDER: sent \$3.0, saw return, kept something
- Round 3 as RECEIVER: kept \$3.33

Actually, let me recalculate:

Round 1: I was RECEIVER, got \$9.51, returned \$5.00, kept \$4.51
 Round 2: I was SENDER, sent \$3.0, they returned \$3.75, I kept \$2 + \$3.75 = \$5.75
 Round 3: I was RECEIVER, got \$6.66, returned \$3.33, kept \$3.33

Total: \$4.51 + \$5.75 + \$3.33 = \$13.59 ? \$13.55 (matches with rounding)

My co-player has been consistent in sending \$3.0 when they're sender. ...

usage:

```
{
  "input_tokens": 1610,
  "output_tokens": 585,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.35 sent, it became \$7.050000000000001, returned \$3.75, visible payoff \$3.3000000000000007.
 - Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.3 returned, visible payoff \$6.3.
 - Round 4 against you, as RECEIVER: saw \$2.62 sent, it became \$7.86, returned \$4.2, visible payoff \$3.66.
- Your total visible earnings across all rounds are \$18.86.

This round, you are the RECEIVER against your current co-player. They sent you \$3.72, so you received \$11.16. How much do you return? (0-11.16)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the current situation:

- Sender sent \$3.72 (74.4% of their endowment - very high trust!)
- I received \$11.16
- Sender currently has: \$5 - \$3.72 = \$1.28

History shows this co-player has been consistently cooperative and trusting. Now they've increased their trust even more, sending nearly 75% of their endowment. I should reciprocate this strong trust generously.

Returning \$6.0:

- Sender would get: \$1.28 + \$6.0 = \$7.28 (profit of \$2.28)
- I keep: \$5.16

This rewards their exceptional trust while both of us profit from the cooperation.

{"return": 6.0}

reasoning:

Looking at the current situation:

- Sender sent \$3.72 (74.4% of their endowment - very high trust!)
- I received \$11.16 after 3x multiplier
- Sender currently has: \$5 - \$3.72 = \$1.28

Looking at history with this co-player:

- They've been consistently cooperative
- Round 3: They sent \$3.0
- Round 4: They saw me send \$2.62 (due to noise, I actually sent \$3.0), returned \$4.2
- Now Round 5: They sent \$3.72 (even more!)

This is a very trusting move - 74.4% of endowment. I should reciprocate generously.

If I return \$6.0:

- Sender gets: \$1.28 + \$6.0 = \$7.28 (profit of \$2.28)
- I keep: \$11.16 - \$6.0 = \$5.16

This rewards their high trust appropriately. Both of us profit well.

usage:

```
{
  "input_tokens": 1967,
  "output_tokens": 440,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.3 returned, visible payoff \$6.3.
- Round 4 against you, as RECEIVER: saw \$2.62 sent, it became \$7.86, returned \$4.2, visible payoff \$3.66.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.04 returned, visible payoff \$7.04.

Your total visible earnings across all rounds are \$24.02.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the history with my current co-player:

- Round 3: They sent \$3.0 (consistent trust)
- Round 4: They returned \$4.2 when they received \$7.86 (generous)
- Round 5: They sent \$3.0 again (consistently cooperative)

This co-player has been very reliable and cooperative throughout our interactions. They consistently send \$3.0 when they're the sender and return fair amounts when they're the receiver. I should continue matching this level of trust and cooperation.

{"send": 3.0}

reasoning:

Looking at the visible history with my current co-player:

- Round 3: They sent \$3.0 as sender
- Round 4: They returned \$4.2 as receiver (I sent \$2.62 visible to them)
- Round 5: They sent \$3.0 as sender

This co-player has been consistently cooperative, sending \$3.0 whenever they're the sender and returning fair amounts when they're the receiver (\$4.2 when they received \$7.86).

I've been sending \$3.0 consistently when I'm the sender. This has been working well for both of us. I should continue this pattern of cooperation.

{"send": 3.0}

usage:

```
{
  "input_tokens": 2364,
  "output_tokens": 295,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$2.62 sent, it became \$7.86, returned \$4.2, visible payoff \$3.66.
- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.04 returned, visible payoff \$7.04.
- Round 6 against you, as RECEIVER: saw \$3.23 sent, it became \$9.69, returned \$5.15, visible payoff \$4.539999999999999.

Your total visible earnings across all rounds are \$32.07.

This round, you are the RECEIVER against your current co-player. They sent you \$2.85, so you received \$8.55. How much do you return? (0-8.55)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at the current situation:

- Sender sent \$2.85 (57% of their endowment - good trust)
- I received \$8.55
- Sender currently has: \$5 - \$2.85 = \$2.15

History shows this co-player has been consistently cooperative:

- Round 4: Returned \$4.2 generously
- Round 5: Sent \$3.0 showing trust
- Round 6: Returned \$5.15 very generously (over 50% of what they received)

They've demonstrated strong cooperation throughout. Returning about 50% of what I received would be fair:

Returning \$4.5:

- Sender gets: $\$2.15 + \$4.5 = \$6.65$ (profit of \$1.80)
- I keep: \$4.05

This maintains our cooperative relationship.

{"return": 4.5}

reasoning:

Looking at the current situation:

- Sender sent \$2.85 (57% of their endowment)
- I received \$8.55
- Sender currently has: $\$5 - \$2.85 = \$2.15$

History with this co-player:

- Round 4: They returned \$4.2 when receiving \$7.86 (generous)
- Round 5: They sent \$3.0, saw \$5.04 returned
- Round 6: They returned \$5.15 when receiving \$9.69 (very generous!)

This co-player has been consistently cooperative and generous when in the receiver role. They sent \$2.85 this round, which is still a significant amount showing trust.

If I return around \$4.5-5.0:

- Returning \$4.5: Sender gets $\$2.15 + \$4.5 = \$6.65$ (profit of \$1.65), I keep \$4.05
- Returning \$5.0: Sender gets $\$2.15 + \$5.0 = \$7.15$ (profit of \$2.15), I keep \$3.55

Given their consistent generosity, I should return around \$4.5-5.0. Let me go with \$4.75 as a middle ground.

Actually, looking at proportions - they've been returning around 50-53% of what they receive. I should reciprocate similarly.

50% of \$8.55 = \$4.275

Let me return \$4.5 to be fair and generous.

{"return": 4.5}

usage:

```
{
  "input_tokens": 2727,
  "output_tokens": 589,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.04 returned, visible payoff \$7.04.
- Round 6 against you, as RECEIVER: saw \$3.23 sent, it became \$9.69, returned \$5.15, visible payoff \$4.539999999999999.
- Round 7 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.07 returned, visible payoff \$6.07.

Your total visible earnings across all rounds are \$36.120000000000005.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at the history with my current co-player:

- Round 5: They sent \$3.0 (strong trust)
- Round 6: They returned \$5.15 very generously (53% of what they received)
- Round 7: They sent \$3.0 again (consistently cooperative)

This co-player has been exceptionally reliable and cooperative throughout all our interactions. They consistently send \$3.0 when they're the sender and return generously when they're the receiver. I should continue matching this trust level.

```
{"send": 3.0}
```

reasoning:

Looking at the history with my current co-player:

- Round 5: They sent \$3.0 (consistent trust)
- Round 6: They returned \$5.15 when they received \$9.69 (very generous - over 50%)
- Round 7: They sent \$3.0 again (consistently cooperative)

This co-player has been extremely reliable and cooperative throughout all our interactions. They:

1. Consistently send \$3.0 as sender (60% of endowment)
2. Return generously as receiver (around 50%+ of what they receive)

This is Round 8, which might be the final round or near the end. They've been a model cooperative partner. I should continue reciprocating this high level of trust and cooperation.

Sending \$3.0 matches their consistent behavior and maintains our strong cooperative relationship.

usage:

```
{
  "input_tokens": 2929,
  "output_tokens": 330,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as RECEIVER: saw \$3.23 sent, it became \$9.69, returned \$5.15, visible payoff \$4.539999999999999.
- Round 7 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.07 returned, visible payoff \$6.07.
- Round 8 against you, as RECEIVER: saw \$3.87 sent, it became \$11.61, returned \$6.15, visible payoff \$5.459999999999999.

Your total visible earnings across all rounds are \$44.150000000000006.

This round, you are the RECEIVER against your current co-player. They sent you \$3.62, so you received \$10.86. How much do you return? (0-10.86)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at the current situation:

- Sender sent \$3.62 (72.4% of their endowment - very high trust)
- I received \$10.86
- Sender currently has: \$5 - \$3.62 = \$1.38

History shows this co-player has been exceptionally cooperative:

- Round 6: Returned \$5.15 very generously
- Round 7: Sent \$3.0 consistently
- Round 8: Returned \$6.15 exceptionally generously (53% of what they received)

They continue to show strong trust with this \$3.62 send. I should reciprocate their consistent generosity.

Returning \$5.5:

- Sender gets: \$1.38 + \$5.5 = \$6.88 (profit of \$1.88)

- I keep: \$5.36

This maintains our highly cooperative relationship.

```
{"return": 5.5}
```

reasoning:

Looking at the current situation:

- Sender sent \$3.62 (72.4% of their endowment - very high trust)
- I received \$10.86
- Sender currently has: \$5 - \$3.62 = \$1.38

History shows this co-player has been excellent:

- Round 6: Returned \$5.15 very generously
- Round 7: Sent \$3.0 showing consistent trust
- Round 8: Returned \$6.15 exceptionally generously (53% of what they received)

This co-player has been incredibly cooperative and generous throughout all our interactions. They're now sending \$3.62, showing even stronger trust. I should reciprocate generously.

Returning about 50-55% of what I received would be appropriate given their exceptional cooperation:

Returning \$5.5:

- Sender gets: \$1.38 + \$5.5 = \$6.88 (profit of \$1.88)
- I keep: \$5.36

This rewards their strong trust while maintaining our cooperative relationship.

usage:

```
{
  "input_tokens": 3069,
  "output_tokens": 477,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 7 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.07 returned, visible payoff \$6.07.
- Round 8 against you, as RECEIVER: saw \$3.87 sent, it became \$11.61, returned \$6.15, visible payoff \$5.4599999999999999.
- Round 9 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.03 returned, visible payoff \$7.03.

Your total visible earnings across all rounds are \$49.510000000000005.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the history with my current co-player:

- Round 7: They sent \$3.0 (consistent trust)
- Round 8: They returned \$6.15 very generously (53% of what they received)
- Round 9: They sent \$3.0 again (consistently cooperative)

This co-player has demonstrated exceptional reliability and cooperation throughout all our interactions. They consistently send \$3.0 when they're the sender and return very generously when they're the receiver. We've built a strong, mutually beneficial partnership. I should continue with the same level of trust.

```
{"send": 3.0}
```

Reasoning:

Looking at the history with my current co-player:

- Round 7: They sent \$3.0 (consistent trust)
- Round 8: They returned \$6.15 very generously (53% of what they received)

- Round 9: They sent \$3.0 again (consistently cooperative)

This co-player has been exceptionally reliable throughout all our interactions. They consistently send \$3.0 when they're the sender and return very generously when they're the receiver. We've established a strong cooperative relationship.

I should continue sending \$3.0 to maintain this highly successful partnership.

```
Usage: {  
  "input_tokens": 3068,  
  "output_tokens": 285,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_7 - Receiver - dyad_4

Timestamp: 2026-08-12T08:18:32

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_4",  
  "move_index": 7,  
  "opponent_id": "Agent_8",  
  "pairing": {  
    "agents": [  
      "Agent_8",  
      "Agent_7"  
    ],  
    "investor": "Agent_8",  
    "roles": {  
      "Agent_7": "trustee",  
      "Agent_8": "investor"  
    },  
    "trustee": "Agent_7"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 10,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.17 sent, it became \$9.51, returned \$5.00, visible payoff \$4.51.
- Round 2 against you, as SENDER: sent \$3.00, it became \$9.00, saw \$3.71 returned, visible payoff \$5.71.
- Round 3 against you, as RECEIVER: saw \$2.22 sent, it became \$6.66, returned \$3.33, visible payoff \$3.33.

Your total visible earnings across all rounds are \$16.970000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$2.62, so you received \$7.86. How much do you return? (0-7.86)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at our continued interaction:

****Round 1:**** They returned \$5.00 from \$9.51 (53%)

****Round 3:**** They returned \$3.33 from \$6.66 (50%)

****Current situation:****

- They sent \$2.62, I received \$7.86
- They've been consistently cooperative, returning around 50-53%

****My decision:****

I should match their cooperative approach. Returning about 53% would be:

- 53% of \$7.86 ? \$4.17

This ensures:

- They profit significantly (\$4.17 back vs \$2.62 sent = +\$1.55 net)
- I also benefit (\$3.69 kept from the tripled amount)
- We maintain positive reciprocity

```
'''json
{"return": 4.20}
'''
```

reasoning:

Looking at the history with this co-player:

****Round 1:****

- I sent \$3, they saw \$3.17 (noise), received \$9.51
- They returned \$5.00 (~53% return rate)

****Round 2:****

- They sent \$3.00, became \$9.00
- I returned \$3.75, they saw \$3.71
- Visible return rate for them: ~41%

****Round 3:****

- I sent \$3, they saw \$2.22 (significant noise), received \$6.66
- They returned \$3.33 (50% return rate)

****Current Round 4:****

- They sent \$2.62, I received \$7.86

****Analysis:****

- This co-player has been consistently cooperative
- Round 1: 53% return
- Round 3: 50% return
- Their sending amounts haven't decreased dramatically despite seeing noise

****My approach:****

I should maintain a similar return rate. They've been returning around 50-53%.

- 50% of \$7.86 = \$3.93
- 53% of \$7.86 = \$4.17

I'll return around \$4.00 to:

- Match their cooperative behavior
- Ensure they profit from sending (\$2.62 sent, \$4.00 back)
- Keep the relationship positive
- I keep \$3.86, they get \$4.00 back vs \$2.62 sent = net gain of \$1.38 for them

Actually, let me be a bit more generous to match their Round 1 behavior. I'll return \$4.20, which is about 53% of what I received.

Their final payoff: $\$5 - \$2.62 + \$4.20 = \6.58

My final payoff: $\$7.86 - \$4.20 = \$3.66$

...

usage:

```
{
  "input_tokens": 1649,
  "output_tokens": 650,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.71 returned, visible payoff \$5.71.
- Round 3 against you, as RECEIVER: saw \$2.22 sent, it became \$6.66, returned \$3.33, visible payoff \$3.33.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.31 returned, visible payoff \$5.3100000000000005.

Your total visible earnings across all rounds are \$20.630000000000003.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at our ongoing interaction:

****Pattern Analysis:****

- They've consistently sent \$3.00 in Rounds 2 and 4
- As receiver (Rounds 1 & 3), they returned generously (~50-53%)
- Unfortunately, noise has made my returns appear lower than actual:
 - Round 2: I returned \$3.75, they saw \$3.71
 - Round 4: I returned \$4.20, they saw only \$3.31

****Current Strategy:****

Despite the noise distorting communication, they've continued trusting me with \$3 sends. I should:

- Maintain consistency by sending \$3
- Signal continued cooperation
- Trust they'll recognize the pattern despite noise

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze our history:

```
**Round 1:** I sent $3, they received ~$9.51, returned $5.00
**Round 2:** They sent $2.35, I received $7.05, returned $3.75, but they saw $3.71
**Round 3:** I sent $3, they received ~$6.66, returned $3.33, but they saw I sent $2.22
**Round 4:** They sent $2.62, I received $7.86, returned $4.20, but they saw $3.31
```

Looking at what they're seeing from my perspective:

- Round 2: They sent \$3.00, saw \$3.71 returned (visible return rate ~41%)
- Round 4: They sent \$3.00, saw \$3.31 returned (visible return rate ~37%)

The noise is making it look like I'm returning less than I actually am.

From their rounds as receiver:

- Round 1: They returned \$5.00 from \$9.51 (53%)
- Round 3: They returned \$3.33 from \$6.66 (50%)

They've been consistently cooperative. However, the noise is making my returns look less generous than they actually are. Despite this, they've continued sending \$3 in rounds 2 and 4.

I should:

1. Continue showing trust and cooperation
2. Send a similar amount to maintain consistency
3. Hopefully they'll continue to be cooperative

I'll send \$3 again.

usage:

```
{
  "input_tokens": 2072,
  "output_tokens": 533,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$2.22 sent, it became \$6.66, returned \$3.33, visible payoff \$3.33.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.31 returned, visible payoff \$5.3100000000000005.
- Round 5 against you, as RECEIVER: saw \$3.72 sent, it became \$11.16, returned \$6.0, visible payoff \$5.16.

Your total visible earnings across all rounds are \$27.67.

This round, you are the RECEIVER against your current co-player. They sent you \$3.23, so you received \$9.69. How much do you return? (0-9.69)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at our continued interaction:

****Round 5 analysis:****

- They saw I sent \$3.72, received \$11.16
- They returned a very generous \$6.00 (53.8%)

****Current situation:****

- They sent \$3.23, I received \$9.69
- They're maintaining strong cooperation despite noise

****My decision:****

I should match their generosity. Returning about 53% would be:

- 53% of \$9.69 ? \$5.14

This ensures:

- They profit well (\$5.14 back vs \$3.23 sent = +\$1.91 net)
- I also benefit (\$4.55 kept)
- We maintain mutual cooperation

```
'''json
{"return": 5.15}
'''
```

reasoning:

Looking at the history:

****Round 5 analysis:****

- They saw I sent \$3.72 (noise from my actual \$3), received \$11.16
- They returned \$6.00, which is about 53.8% of what they received
- Very generous and cooperative again!

****Current situation:****

- They sent me \$3.23, I received \$9.69
- They've been consistently generous, especially in Round 5 where they returned \$6.00 from \$11.16

****My decision:****

They're showing strong cooperation. I should reciprocate at a similar level:

- Around 53-54% of \$9.69 would be approximately \$5.15-\$5.23

Let me calculate:

- 53% of \$9.69 = \$5.14
- 54% of \$9.69 = \$5.23

I'll return around \$5.15 to match their cooperative spirit and ensure they profit well from sending to me.

This gives them: \$5.15 back vs \$3.23 sent = +\$1.92 profit

And I keep: \$9.69 - \$5.15 = \$4.54

Both profit nicely.

usage:

```
{
  "input_tokens": 2482,
  "output_tokens": 468,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.31 returned, visible payoff \$5.3100000000000005.
- Round 5 against you, as RECEIVER: saw \$3.72 sent, it became \$11.16, returned \$6.0, visible payoff \$5.16.
- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.05 returned, visible payoff \$8.05.

Your total visible earnings across all rounds are \$32.21.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at our interaction pattern:

****Recent trends:****

- Round 5: I sent \$3, they returned a generous \$6.00 from \$11.16
- Round 6: They sent \$3, I returned \$5.15, they saw \$6.05 (noise helped!)
- They're consistently sending \$3 and showing strong cooperation

****Analysis:****

- They saw my returns improve from their perspective (\$3.31 ? \$6.05)

- They responded very generously in Round 5 (\$6.00 return)
- We've established a stable, mutually beneficial pattern

****My decision:****

Continue with \$3 to:

- Maintain the cooperative equilibrium we've built
- Show consistency and reliability
- Keep generating mutual gains

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at the history:

****Round 4:**** They sent \$3.00, I returned \$4.20, they saw \$3.31 (noise made it look lower)
****Round 5:**** I sent \$3.00, they received ~\$11.16, returned \$6.00 (very generous)
****Round 6:**** They sent \$3.00, I returned \$5.15, they saw \$6.05 (noise made it look higher!
)

****Analysis:****

- This co-player has been consistently cooperative
- They've been sending \$3.00 regularly
- In Round 5 they returned a very generous \$6.00 from ~\$11.16
- In Round 6, the noise actually worked in my favor - they saw \$6.05 when I returned \$5.15, which looks very generous
- They're seeing improved returns from me (from their perspective: \$3.31 ? \$6.05)

****Strategy:****

- They're maintaining trust and cooperation
- I should continue sending \$3 to maintain this productive relationship
- The pattern of mutual cooperation is working well for both of us

I'll send \$3 again.

usage:

```
{
  "input_tokens": 2885,
  "output_tokens": 455,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as RECEIVER: saw \$3.72 sent, it became \$11.16, returned \$6.0, visible payoff \$5.16.
- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.05 returned, visible payoff \$8.05.
- Round 7 against you, as RECEIVER: saw \$2.85 sent, it became \$8.55, returned \$4.5, visible payoff \$4.050000000000001.

Your total visible earnings across all rounds are \$38.28.

This round, you are the RECEIVER against your current co-player. They sent you \$3.87, so you received \$11.61. How much do you return? (0-11.61)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at our ongoing interaction:

****Round 7 analysis:****

- They saw I sent \$2.85, received \$8.55
- They returned \$4.50 (52.6%)

****Current situation:****

- They sent \$3.87, I received \$11.61
- This is their highest send yet - showing increased trust!

****My decision:****

I should reward their increased trust with a generous return. Using their ~53% return rate pattern:

- 53% of \$11.61 ? \$6.15

This ensures:

- They profit well (\$6.15 back vs \$3.87 sent = +\$2.28 net)
- I also benefit (\$5.46 kept)
- We reinforce our strong cooperative relationship

```
'''json
{"return": 6.15}
'''
```

reasoning:

Looking at our interaction:

****Round 7 analysis:****

- They saw I sent \$2.85, received \$8.55
- They returned \$4.50 (52.6%)

****Current situation:****

- They sent \$3.87, I received \$11.61
- This is their highest send yet!

****Analysis:****

- They've been very cooperative throughout
- Round 7: returned \$4.50 from \$8.55 (52.6%)
- Round 6: they sent \$3.00, saw me return \$6.05
- Round 5: they returned \$6.00 from \$11.16 (53.8%)

They're sending more (\$3.87) which shows increased trust. I should reciprocate with a generous return.

Using ~53% return rate:

- 53% of \$11.61 ? \$6.15

This would give them:

- Received back: \$6.15
- Sent: \$3.87
- Net profit: \$2.28

And I keep:

- \$11.61 - \$6.15 = \$5.46

This is fair and maintains our cooperative relationship.

usage:

```
{
  "input_tokens": 3073,
  "output_tokens": 469,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.05 returned, visible payoff \$8.05.
- Round 7 against you, as RECEIVER: saw \$2.85 sent, it became \$8.55, returned \$4.5, visible payoff \$4.050000000000001.
- Round 8 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.03 returned, visible payoff \$8.030000000000001.

Your total visible earnings across all rounds are \$43.74.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at our established pattern:

****Recent interaction:****

- Round 6: They sent \$3, saw ~\$6.05 back (payoff \$8.05)
- Round 7: They returned \$4.50 when they received \$8.55
- Round 8: They sent \$3, saw ~\$6.03 back (payoff \$8.03)

****Analysis:****

- We've built a very stable, mutually beneficial relationship
- They consistently send \$3 and return generously (~53%)
- I've been getting strong returns from them
- Both of us are profiting well from this cooperation

****My decision:****

Continue with \$3 to:

- Maintain our successful equilibrium
- Show reliability and consistency
- Keep generating mutual gains

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at our interaction history:

****Recent pattern:****

- Round 6: They sent \$3, I returned \$5.15, they saw \$6.05 (good payoff: \$8.05)
- Round 7: I sent \$3, they returned \$4.50
- Round 8: They sent \$3, I returned \$6.15, they saw \$6.03 (good payoff: \$8.03)

****Analysis:****

- They're consistently sending \$3
- They've seen good returns recently (around \$6)
- In Round 7, they returned \$4.50 from \$8.55 (52.6%)
- In Round 8, they increased their send amount to \$3.87, showing trust
- We've built a strong cooperative relationship

****My strategy:****

- They've been consistent with \$3 sends and generous returns
- I should continue sending \$3 to maintain our successful pattern
- This has been working well for both of us

Let me send \$3 again.

usage:

```
{
  "input_tokens": 3152,
  "output_tokens": 426,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 7 against you, as RECEIVER: saw \$2.85 sent, it became \$8.55, returned \$4.5, visible payoff \$4.050000000000001.
- Round 8 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.03 returned, visible payoff \$8.030000000000001.
- Round 9 against you, as RECEIVER: saw \$3.62 sent, it became \$10.86, returned \$5.5, visible payoff \$5.359999999999999.

Your total visible earnings across all rounds are \$50.77.

This round, you are the RECEIVER against your current co-player. They sent you \$3.88, so you received \$11.64. How much do you return? (0-11.64)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at our continued interaction:

****Round 9 analysis:****

- They saw I sent \$3.62, received \$10.86

- They returned \$5.50 (50.6%)

****Current situation:****

- They sent \$3.88, I received \$11.64
- Another high trust send, showing confidence in our relationship

****My decision:****

I should match their cooperative pattern. Returning about 53% would be:

- 53% of \$11.64 ? \$6.17

This ensures:

- They profit well (\$6.17 back vs \$3.88 sent = +\$2.29 net)
- I also benefit (\$5.47 kept)
- We maintain our successful cooperative equilibrium

```
'''json
{"return": 6.17}
'''
```

Reasoning:

Looking at the pattern:

****Round 9 analysis:****

- They saw I sent \$3.62, received \$10.86
- They returned \$5.50 (50.6%)

****Current situation:****

- They sent \$3.88, I received \$11.64
- This is again a high send amount, showing continued trust

****Pattern from their returns:****

- Round 5: \$6.00 from \$11.16 (53.8%)
- Round 7: \$4.50 from \$8.55 (52.6%)
- Round 9: \$5.50 from \$10.86 (50.6%)

They're consistently returning around 50-53%.

****My decision:****

I should continue matching their cooperative approach. Using their ~53% return rate:

- 53% of \$11.64 ? \$6.17

This ensures:

- They profit well (\$6.17 back vs \$3.88 sent = +\$2.29 net)
- I also benefit (\$5.47 kept)
- We maintain our strong cooperative relationship

Let me calculate more precisely:

$11.64 * 0.53 = 6.1692$? \$6.17

That seems right. I'll return \$6.17.

```
Usage: {
  "input_tokens": 3197,
  "output_tokens": 492,
  "reasoning_tokens": 0
}
```